SGA 7

Exposé I

Summary of the First Lectures by A. Grothendieck

written up by P. Deligne

Contents

0. Preliminaries

0.0. Terminology. Recall the following definitions:

(0.0.1) trait: the spectrum of a discrete valuation ring. The generic and closed points of a trait are often denoted by η and s.

(0.0.2) strictly local: for a ring, local henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.0.3) geometric point of S (in this exposé): a morphism $\bar{x} : \mathrm{Spec}(\bar{k}) \longrightarrow S$ with image $x \in S$, where $k(\bar{x}) \underset{\mathrm{dfn}}{=} \bar{k}$ is a separable closure of $k(x)$. For a henselian trait $S = \mathrm{Spec}(V)$, a generic geometric point of S (that is, one with image η) is often denoted by $\bar{\eta}$. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S centered at s.

(0.0.4) strict localization of X at the geometric point $\bar{x}$: SGA 4 VIII 4.4.

(0.0.5) essentially of finite type over S: for a local (respectively, henselian local; respectively, strictly local) S-scheme, the spectrum of a local ring (respectively, the henselization of a local ring; respectively, a strict localization) of a scheme of finite type over S.

(0.0.6) $F \longmapsto F(n)$: Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, that is, if there exist $N \geqslant 1$ and $M \geqslant 1$ such that $(T^N - 1)^M = 0$.

0.1. Review of ℓ-adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p, and let ℓ be a prime different from p.

The ℓ-adic cohomology groups of X were defined in SGA 4 and SGA 5.

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i-th cohomology group of the ètale site X_{et} of X with coefficients in $\mathbb{Z}/\ell^n$;

b) by definition,

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n);$$

c) by definition,

$$H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The ℓ-adic cohomology groups of X with proper supports are defined in the same way from the groups $H^i_c(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases, one knows how to prove that these groups $H^i(X)$ have reasonable finiteness properties:

(0.1.1) for cohomology with proper supports;

(0.1.2) when X is proper over k;

(0.1.3) when X is smooth over k (by (0.1.1) and Poincaré duality, SGA 4 XVIII);

(0.1.4) when Hironaka's resolution of singularities is available for schemes of finite type over k of dimension at most $\dim(X)$;

(0.1.5) when $i = 0, 1$.

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are finitely generated, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and (except perhaps in case (0.1.5)) there are split short exact sequences

$$0 \longrightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \longrightarrow H^i(X, \mathbb{Z}/\ell^n) \longrightarrow \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \longrightarrow 0.$$

Moreover, in each case the proof of the finiteness theorem also proves a generic constructibility theorem: if k is the field of fractions of an integral noetherian scheme S, and X is the generic fibre of a morphism $f_1 : X_1 \longrightarrow S$ of finite type, then there exists a nonempty open subset $U \subset S$ such that over U the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and their formation commutes with every base change.

0.2. We no longer suppose that k is algebraically closed. Let $\bar{k}$ be an algebraic closure of k (or a separable closure, which amounts to the same thing here). We consider the "geometric" cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extension of scalars from k to $\bar{k}$. By transport of structure, the Galois group $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$-cohomology, in any of the cases (0.1.1)–(0.1.5), this gives continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Review of traits (0.0.1)

For the proofs, we refer to [4]. Let S be a henselian trait, and let η, s, V, $\bar{V}$, $\bar{\eta}$, and $\bar{s}$ be as in (0.0.1) and (0.0.3). Let p denote the characteristic exponent of $k(s)$, and let $\mathrm{Gal}(\bar{\eta}/\eta)$ (respectively, $\mathrm{Gal}(\bar{s}/s)$) denote the Galois group $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ (respectively, the corresponding group for $\bar{s}/s$). By transport of structure, $\mathrm{Gal}(\bar{\eta}/\eta)$ acts on $k(\bar{s})$; hence there is a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ onto $\mathrm{Gal}(\bar{s}/s)$ whose kernel is the inertia group I. The subfield of $k(\bar{\eta})$ fixed by I is a maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro-p subgroup P. The quotient I/P is the tame inertia group. Canonically,

$$I/P \simeq \varprojlim_n \mu_n(k(\bar{s})) \underset{\text{dfn}}{=} \widehat{\mathbb{Z}}(1)(k(\bar{s})).$$

If ϕ_n is the canonical map from I/P to $\mu_n(\bar{k})$ and u is an element of $\bar{V}$ of valuation $1/n$, then, for $\sigma \in I$,

$$\sigma(u) \equiv \phi_n(\sigma)\, u \qquad \text{(modulo elements of valuation greater than } 1/n\text{)}.$$

In summary,

$$\begin{aligned} \mathrm{Gal}(\bar{\eta}/\eta) &\supset I \supset P, \\ \mathrm{Gal}(\bar{\eta}/\eta)/I &\simeq \mathrm{Gal}(\bar{s}/s), \\ I/P &\simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ P &: \text{ a pro-}p\text{ group} \quad (\{e\} \text{ if } p = 1), \end{aligned} \tag{0.3.1}$$

and the action (by inner automorphisms in $\mathrm{Gal}(\bar{\eta}/\eta)$) of $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.4. More generally, let S be strictly local and regular, let D be a regular divisor in S, and let p' be the characteristic exponent of the generic point of D. Let $\bar{\eta}$ be a geometric generic point of S, and let $\bar{s}$ be the resulting geometric point centered at the closed point s. By Abhyankar's lemma, the fundamental group $\pi_1(S - D, \bar{\eta})$ has a filtration analogous to (0.3.1):

$$\begin{aligned} \pi_1(S - D, \bar{\eta}) &\supset I \supset P, \\ \pi_1(S - D, \bar{\eta})/I &\simeq \mathrm{Gal}(\bar{s}/s), \\ I/P &\simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ P &: \text{ with no quotient of order prime to } p'. \end{aligned} \tag{0.4.1}$$

The action by inner automorphisms of $\pi_1(S-D,\bar{\eta})/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.5. Néron desingularization

The method of Néron desingularization gives the following result.

Lemma 0.5.1. Let $S \longrightarrow S_o$ be a morphism of henselian traits, with $S = \mathrm{Spec}(V)$ and $S_o = \mathrm{Spec}(V_o)$, and let π be a uniformizer of V_o. Suppose that π is a uniformizer of V, and that the extensions $k(s)/k(s_o)$ and $k(\eta)/k(\eta_o)$ are separable. Then V is a filtered direct limit of smooth henselian V_o-algebras B_i essentially of finite type over V_o.

To construct the desired V_o-algebras, one takes $V_o \subset B \subset V$, with B of finite type, and desingularizes $\mathrm{Spec}(B)$ by Néron's method ([1], § 4).

Let $S = \mathrm{Spec}(V)$ be a henselian trait.

(0.5.2) If S is of equal characteristic, with uniformizer π, one may apply (0.5.1) with $S_o = \mathbf{F}_p[\pi]_{(\pi)}$. The extension s/s_o is separable because $\mathbf{F}_p$ is perfect; the extension η/η_o is separable because π, being a uniformizer, is not a p-th power (if $p \neq 1$), and hence belongs to a p-basis. It follows that S is a limit of spectra of smooth henselian $\mathbf{F}_p[\pi]_{(\pi)}$-algebras essentially of finite type.

(0.5.3) If S is complete and of mixed characteristic, with perfect residue field k, then V is an Eisenstein extension of the ring of Witt vectors (that is, the Cohen ring) $W(k)$:

$$V \simeq W(k)[\pi]/\left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i\right).$$

If the a_i are perturbed slightly, one obtains an isomorphic extension. We may therefore suppose that the a_i belong to $W(k')$, where k' is of finite type over $\mathbf{F}_p$. Let k_o be the perfect closure of k' in k, and set

$$V_o = W(k_o)[\pi]/\left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i\right).$$

Lemma 0.5.1 applies to V/V_o, and the residue field of V_o is a radicial extension of a field of finite type over $\mathbf{F}_p$.

1. Arithmetic proof of the monodromy theorem

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime different from the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\mathrm{Gal}(\bar{\eta}/\eta)$ in $\mathrm{GL}(n,\mathbb{Q}_\ell)$. Suppose that the following condition holds:

$(*_\ell)$ No finite extension of $k(s)$ contains all roots of unity of ℓ-power order.

Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for every $\sigma \in I_1$.

Condition $(*_\ell)$ is automatically satisfied when $k(s)$ is of finite type over the prime field (or is radicial over such a field).

The following theorem follows immediately from 1.1.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η, and let H be a cohomology space of $X_{\bar{\eta}}$ with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If condition $(*_\ell)$ is satisfied, the action on H of every element of I is quasi-unipotent.

The limit arguments of 0.5 allow us to dispense with hypothesis $(*_\ell)$.

Variant 1.3. Condition $(*_\ell)$ in 1.2 is superfluous.

Let H'_o be a $\mathbb{Z}_\ell$-cohomology space of $X_{\bar{\eta}}$ giving rise to H, and put $H_o = H'_o/\mathrm{torsion}$. By hypothesis, $H = H_o \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and the action of $\mathrm{Gal}(\bar{\eta}/\eta)$ is induced by

$$\rho : \mathrm{Gal}(\bar{\eta}/\eta) \longrightarrow \mathrm{GL}(H_o).$$

Let $I_{\ell'}$ be the largest prime-to-ℓ subgroup of I. Its image in $\mathrm{GL}(H_o)$ is an ℓ-adic Lie group of order prime to ℓ, and is therefore finite.

Make an extension of traits $S' \longrightarrow S$. This does not change H_o (SGA 4 XVI 6.6), and $\rho(I')$ has finite index in $\rho(I)$, because I'/P' has finite index in I/P, by what precedes. It therefore suffices to prove the theorem over S'; we may suppose that

a) $\mathrm{Gal}(\bar\eta/\eta)$ acts trivially on $H_o/\ell H_o$;

b) if S is of mixed characteristic, then S is complete with perfect residue field (EGA O_{III} 10.3.1).

Put $S = \mathrm{Spec}(V)$, define V_o as in (0.5.2) and (0.5.3), and apply (0.5.1). We have

$$V = \varinjlim_i B_i.$$

Here $S_i = \mathrm{Spec}(B_i)$ is essentially of finite type and smooth over V_o. Let $D_i \subset S_i$ be defined by the equation $\pi = 0$. For sufficiently large i, H_o comes from a twisted constant $\mathbb{Z}_\ell$-sheaf $\mathcal{H}_i$ on $S_i - D_i$ (by "generic constructibility," (0.1)), and $\mathcal{H}_i/\ell\mathcal{H}_i$ is constant.

The filtrations (0.3.1) and (0.4.1) give rise to commutative diagrams:

$$\text{(1.3.1)}\qquad \begin{array}{ccccc} \mathrm{Gal}(\bar\eta/\eta) & \supset & I & \supset & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar\eta) & \supset & I_i & \supset & P_i \end{array} \qquad \text{and} \qquad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \\ \downarrow & & \| \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \end{array}$$

The representation on H_o of $\mathrm{Gal}(\bar\eta/\eta)$ is induced by an action of $\pi_1(S_i - D_i, \bar\eta)$ on H_o that is trivial on $H_o/\ell H_o$. The group

$$\mathrm{Ker}\big(\mathrm{GL}(H_o) \longrightarrow \mathrm{GL}(H_o/\ell H_o)\big)$$

is a pro-ℓ group. Thus P_i acts trivially, and we apply the following variant of 1.1.

Variant 1.4. Retain the notation of 0.4, and let ρ be a continuous representation of $\mathrm{Gal}(\bar\eta/\eta)$ in $\mathrm{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies $(*_\ell)$ and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

2.1. Let S be a strictly local trait, and retain the notation of 0.3. Thus, by hypothesis, $s = \bar s$. Let $f : X \longrightarrow S$ be a scheme of finite type over S. The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar\eta}$, as well as the action of the inertia group on the latter, using information about f that is local on X.

A key result of this kind is the specialization theorem (SGA 4 XVI 2.2), according to which X_s and $X_{\bar\eta}$ have the same cohomology when f is proper and smooth.

In the transcendental theory, if $f : X \longrightarrow D$ is a proper morphism from an analytic space X to the disk D, vanishing cycles arise as follows:

a) After shrinking D if necessary, one may suppose that X is a topological fibre bundle over the punctured disk D^* (Thom's isotopy theorem). In particular, the fibres

X_t ($t \in D^*$) are all isomorphic.

b) It is not unreasonable to picture the special fibre X_s as obtained from "the" general fibre X_t by contracting polyhedra in X_t; one would then have a map $\pi : X_t \longrightarrow X_s$. The method of vanishing cycles consists in studying the difference between the cohomology of X_s and that of X_t by means of the Leray spectral sequence for π.

The geometric theory explained below is very close to this transcendental theory: the general point $t \in D$ must be replaced by $\bar\eta$, the geometric generic point of S. Conversely, the transcendental theory can be modeled on the geometric theory by replacing t or $\bar\eta$ with the universal cover $\widetilde{D}^*$ of D^*, and X_t or $X_{\bar\eta}$ with $X \times_D \widetilde{D}^*$. This eliminates many ε's and η's, but sometimes loses information.

2.2. Fix a torsion ring Λ in which p is invertible (for example, $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ prime to p). Consider the morphisms

$$
\begin{tikzcd}
X_s \arrow[r, "i"] \arrow[d, "f_s"] & X \arrow[d, "f"] & & X_{\bar\eta} \arrow[ll, "\bar j"'] \arrow[d, "f_{\bar\eta}"] \\
s \arrow[r] & S & \eta \arrow[l] & \bar\eta \arrow[l]
\end{tikzcd}
$$

The sheaves of vanishing cycles ψ^i on X_s are

$$\psi^i = i^* R^i \bar j_* \Lambda. \tag{2.2.1}$$

If f is proper, then (SGA 4 XII 5.5)

$$i^* : H^p(X, R^q \bar j_* \Lambda) \xrightarrow{\sim} H^p(X_s, i^* R^q \bar j_* \Lambda), \tag{2.2.2}$$

and the Leray spectral sequence for $\bar j$ becomes

$$H^p(X_s, \psi^q) \Longrightarrow H^{p+q}(X_{\bar\eta}, \Lambda). \tag{2.2.3}$$

When nothing happens at infinity, this spectral sequence can sometimes be obtained without a properness hypothesis. That some hypothesis at infinity is necessary is already clear from the trivial case $X_s = \varnothing$.

The definitions immediately give:

Proposition 2.3. Let $\bar x$ be a geometric point of X_s (for example, a closed point), let $X_{(\bar x)}$ be the strict henselization of X at $\bar x$, and let $X_{(\bar x)\bar\eta}$ be the geometric generic fibre of the projection $X_{(\bar x)} \longrightarrow S$. Then

$$\psi^i_{\bar x} = H^i(X_{(\bar x)\bar\eta}, \Lambda).$$

Corollary 2.4. Wherever f is smooth, $\psi^i = 0$ for $i > 0$ and $\psi^0 = \Lambda$.

This is SGA 4 XV 2.1.

The group $\mathrm{Gal}(\bar\eta/\eta)$ (which, by hypothesis, equals the inertia group) acts on the ψ^i by transport of structure, and

Scholium 2.5. The spectral sequence (2.2.3) is $\mathrm{Gal}(\bar\eta/\eta)$-equivariant.

Variant 2.6. Put $\bar S = \mathrm{Spec}(\bar V)$ and $\bar X = X \times_S \bar S$. The scheme $X_{\bar\eta}$ is the complement of $X_s = X_{\bar s}$ in $\bar X$. Put

$$\phi^i = \underline{H}^{i+1}_{X_s}(\bar X, \Lambda)$$

(the local-cohomology sheaves). There is an exact sequence of sheaves

$$0 \longrightarrow \phi^{-1} \longrightarrow \Lambda \longrightarrow \psi^0 \longrightarrow \phi^0 \longrightarrow 0,$$

and, for $i > 0$, isomorphisms $\psi^i \xrightarrow{\sim} \phi^i$. If f is smooth, the ϕ^i vanish.

Put $\phi^i_{\mathrm{gl}} = H^i_{X_s}(\bar X, \Lambda)$, the global analogue of ϕ^i. There is a spectral sequence

$$H^p(X_s, \phi^q) \Longrightarrow \phi^{p+q}_{\mathrm{gl}}, \tag{2.6.1}$$

and a long exact cohomology sequence which, when f is proper, becomes

$$\cdots \longrightarrow H^i(X_s, \Lambda) \longrightarrow H^i(X_{\bar\eta}, \Lambda) \longrightarrow \phi^i_{\mathrm{gl}} \longrightarrow \cdots . \tag{2.6.2}$$

Variant 2.7. Let K_t be the maximal tamely ramified extension of K in $\bar{K}$. Put $X_t = X \times_S \mathrm{Spec}(K_t)$, and let $\bar{j}_t$ be the projection from X_t to X. Define tame variants of the ψ^i by

$$\psi_t^i = R^i \bar{j}_{t*} \Lambda.$$

Since P is a pro-p group and p is invertible in Λ, one obtains

$$H^*(X_t, \Lambda) = H^*(X_{\bar{\eta}}, \Lambda)^P, \tag{2.7.1}$$

$$\psi_t^i = (\psi^i)^P. \tag{2.7.2}$$

When f is proper, the Leray spectral sequence for $\bar{j}_t$ is obtained from (2.2.3) by passing to P-invariants:

$$H^p(X_s, \psi_t^q) \Longrightarrow H^{p+q}(X_t, \Lambda) = H^{p+q}(X_{\bar{\eta}}, \Lambda)^P. \tag{2.7.3}$$

3. Geometric proof of the monodromy theorem

Let Λ be as in 2.2.

Conjecture 3.1 (purity). *Let A be an excellent regular strictly local ring* (0.0.2) *and let D be a regular divisor in $\mathrm{Spec}(A)$ defined by a parameter x. Put $U = \mathrm{Spec}(A) - D = \mathrm{Spec}(A[1/x])$. Then*

$$H^q(U, \Lambda) = \begin{cases} \Lambda & \text{if } q = 0, \\ \Lambda(-1) & \text{if } q = 1 \quad \text{(notation (0.0.6))}, \\ 0 & \text{if } q > 1. \end{cases} \tag{3.1.1}$$

Here are some cases in which this conjecture is known (SGA 4 XIX): for $q = 0$ or 1; when A has characteristic 0; when A has equal characteristic p and resolution of singularities is available in characteristic p in dimensions $\leqslant \dim(A)$; or when $(\mathrm{Spec}(A), D)$ is a smooth pair (the relative purity theorem, SGA 4 XVI 3.7).

When purity is available and $D = \sum_{i \in B} D_i$ is a normal-crossings divisor in $\mathrm{Spec}(A)$ defined by part of a regular system of parameters, the cohomology of $U = \mathrm{Spec}(A) - D$ is given by

$$\begin{aligned} &H^0(U, \Lambda) = \Lambda, \\ &H^1(U, \Lambda) = \Lambda(-1)^B, \\ &H^q(U, \Lambda) = \textstyle\bigwedge^q H^1(U, \Lambda) \qquad \text{(via the cup product).} \end{aligned} \tag{3.1.2}$$

These formulas are identical to those which, over $\mathbf{C}$, give the cohomology of $D^{*B} \times D^k$.

3.2. Let S be a strictly local trait, and retain the notation of (0.0.1) and (0.0.3). Let $f : X \longrightarrow S$ be a morphism of finite type. Suppose that X is regular, that X_η is smooth, and that X_s is a normal-crossings divisor in X. We shall suppose that this divisor is, globally and not merely locally for the étale topology, a sum of regular divisors:

$$(X_s)_{\mathrm{red}} = \sum_{i \in B} D_i$$

(the general case can be recovered by localization). Denote their multiplicities by m_i, so that $X_s = \sum_{i \in B} m_i D_i$.

Let x be a point of X_s, let $\bar{x}$ be a geometric point of X lying over x (for example, x may be closed and equal to $\bar{x}$), put $C = \{i \mid x \in D_i\}$, and let K be the complex concentrated in degrees 0 and -1

$$K : \mathbf{Z}^C \xrightarrow{d} \mathbf{Z}, \qquad d((n_j)) = \sum_j m_j n_j.$$

Theorem 3.3 (uses purity). The groups of tame vanishing cycles $\psi^i_{t\bar{x}}$ are as follows:
(i) There is a canonical isomorphism of graded Λ-algebras

$$\psi^*_{t\bar{x}} \simeq \bigwedge^* \big(H_1(K) \otimes \Lambda(-1)\big) \otimes_\Lambda \psi^0_{t\bar{x}}.$$

(ii) $\psi^0_{t\bar{x}}$ is the Λ-algebra Λ^E, where E is a set on which the tame inertia group $I_t = I/P = \widehat{\mathbf{Z}}(1)(k(s))$ acts transitively, and whose cardinality is the largest prime-to-p divisor of $\# H_0(K)$.

Here is the method of proof.

a) Let $X_{(\bar{x})}$ be the strict localization of X at $\bar{x}$, let $U = X_{(\bar{x})} - X_{(\bar{x})s}$ (the complement of a normal-crossings divisor), let t_j be a local equation for D_j, put

$$X_n = X_{(\bar{x})}\big[(t_j^{1/n})\big] \qquad ((n,p)=1),$$

which is a regular scheme, and let U_n be the inverse image of U in X_n. By (3.1.2),

$$H^q(U_n, \Lambda) = \bigwedge^q \big(\Lambda(-1)^C\big).$$

For $m = nd$, the pullback map from $H^q(U_n,\Lambda)$ to $H^q(U_m,\Lambda)$ is multiplication by d^q. Putting $\widetilde{U} = \varprojlim_{(n,p)=1} U_n$, one therefore has

$$(3.3.1) \qquad \begin{cases} H^0(\widetilde{U}, \Lambda) = \Lambda, \\ H^q(\widetilde{U}, \Lambda) = \varinjlim_n H^q(U_n, \Lambda) = 0 \quad \text{if } q \neq 0. \end{cases}$$

b) $\widetilde{U}$ is a pro-covering of U with group $\widehat{\mathbf{Z}}(1)(k(s))^C$. By definition,

$$\psi^*_{t\bar{x}} = H^*(U_{\eta t}, \Lambda).$$

$\widetilde{U}$ is a pro-covering, with group $H_1(K) \otimes \widehat{\mathbf{Z}}(1)(k(s))$, of each connected component of $U_{\eta t}$, and these components are permuted transitively by the tame inertia group. Formula (3.3.1) and the Hochschild–Serre spectral sequence now give the formulas in 3.3.

Pictorially, one may say that $U_{\bar{\eta}}$ is the fibre of a morphism δ from the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s))^C, 1)$ represented by U to the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s)), 1)$ represented by η; the morphism δ induces

$$d : (n_j) \longmapsto \sum_j m_j n_j : \widehat{\mathbf{Z}}(1)(k(s))^C \longrightarrow \widehat{\mathbf{Z}}(1)(k(s)).$$

Corollary 3.4 (uses purity). Suppose that f is proper. Let N be the least common multiple of the multiplicities m_j ($j \in B$). Let $q \geqslant 0$, and let q' be the minimum of q and the greatest number of divisors D_j passing through one point. For T in the tame inertia group $I_t = I/P$, one has

$$(T^N - 1)^{q'+1} = 0 \quad \underline{\text{on}} \quad H^q(X_{\bar{\eta}}, \Lambda)^P.$$

This follows at once from 3.3, Scholium 2.5, and Variant 2.7.

The required purity results are available for H^1, in characteristic 0, or when S is essentially of finite type over a field.

Theorem 3.5. Let C be a smooth projective curve over the fraction field $k(\eta)$ of a discrete valuation ring V. The inertia group I acts on $H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell)$ by quasi-unipotent automorphisms of level 2: there exists N such that, for $T \in I$,

$$(T^N - 1)^2 = 0 \quad \text{on} \quad H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell).$$

We reduce to the case in which V is strictly local. The assertion is that there exists an open subgroup I_1 of I such that $(T-1)^2 = 0$ on H^1 for $T \in I_1$.

We may extend the scalars of $k(\eta)$ to a finite extension, replacing V by its normalization in that extension. Since the image of the pro-p group P in an ℓ-adic Lie group $\mathrm{GL}(n, \mathbf{Q}_\ell)$ is finite, we may in particular suppose that P acts trivially.

Let C_o be a minimal model of C over V (for the existence of C_o, which is based on Abhyankar's theorem, see the discussion in [3], § 2, p. 87; one could instead reduce to the case where V is excellent by passing to its completion and invoking SGA 4 XVI 1.6).

Let C_1 be a regular model of C, obtained from C_o by blowups, whose special fibre is a normal-crossings divisor. The arguments of 3.3 and 3.4 apply to C_1, ψ_t^0, ψ_t^1, and

$$H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n)^P = H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n),$$

with N independent of n; Theorem 3.5 follows.

Remark 3.6. Let A be an abelian variety over $k(\eta)$. Since A is a quotient of a Jacobian, it follows again that the action of I on $H^1(A_{\bar{\eta}}, \mathbb{Q}_\ell)$, or on $T_\ell(A)$, is quasi-unipotent of level 2.

Remark 3.7. The action of I on $H^1(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is in fact quasi-unipotent of level 2 for every scheme X of finite type over η.

Theorem 3.5 (or Remark 3.6) is the basis for the proof of the stable reduction theorem for abelian varieties that will be given in Exposé IX, § 3.

3.8. It is clear that whenever purity and resolution of singularities are available (for example, in characteristic 0), the preceding method proves the monodromy theorem and gives bounds for the nilpotence exponent. Thus, in equal characteristic 0, if X is proper and smooth over η and T belongs to an open subgroup I_1 of I, one has

$$(T-1)^{q+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \mathbb{Z}_\ell) \qquad (T \in I_1 \subset I).$$

4. Vanishing criteria for the sheaves of vanishing cycles

4.1. Let S be a strictly local trait and let $f : X \longrightarrow S$ be a scheme of finite type over S. Put $n = \dim(X_\eta)$.

If X' is the scheme-theoretic closure of X_η in X, then:

a) On $X_s - X'_s$, one has $\phi^{-1} = \Lambda$, and the other ϕ^i vanish;

b) on X'_s, the ϕ^i and ψ^i relative to X/S or to X'/S coincide, and $\phi^{-1} = 0$.

Since X' is flat over S, this often permits a reduction to the case in which X is flat over S.

Theorem 4.2.

(i) One has $\psi^i = 0$ for $i > n$.

(ii) More precisely, let x be a point of X_s whose closure has dimension k ($k = \mathrm{trdeg}_{k(s)}\, k(x)$), and let $\bar{x}$ be a geometric point lying over x. Then

$$\psi^i_{\bar{x}} = 0 \quad \underline{\text{for}} \quad i > n-k, \qquad \text{i.e.} \qquad d(\psi^i) \leqslant n - i \quad (\underline{\text{notation of SGA 4 XIV 2.1}}).$$

Indeed,

$$\psi^i_{\bar{x}} = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda),$$

and $X_{(\bar{x})\bar{\eta}}$ is an inverse limit of affine schemes of dimension at most n over $k(\bar{\eta})$. Assertion (i) therefore follows from the affine Lefschetz theorem (SGA 4 XIV 3.2) by passage to the limit.

To prove (ii), we may suppose that X is flat over S (4.1) and that there is a quasi-finite flat S-morphism

$$g : X \longrightarrow \mathbb{A}^k_S$$

such that $g(x)$ is the generic point of $\mathbb{A}^k_s$. Let S' be the strict localization of $\mathbb{A}^k_S$ at $\bar{x}$. Then:

a) $S'_{\bar{\eta}}$ is the spectrum of a field, and $\mathrm{Gal}(\bar{\eta}'/s'_{\bar{\eta}})$ is a p-group Q;

b) by (i), applied to g,

$$H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda) = 0 \qquad \text{for } i > n-k;$$

c) one has

$$H^i(X_{(\bar{x})\bar{\eta}}, \Lambda) = H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda)^Q,$$

which proves the theorem.

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension at most d, the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n+d+1$ and an epimorphism for $i = n+d+1$.

4.4. To prove, in certain cases, that the ϕ^i vanish for small i, we shall use local duality, either directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities à la Hironaka is available in the dimensions under consideration.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \longrightarrow S$ be a morphism of finite type and let x be a closed point of X_s. Suppose that:

(a) x is an isolated non-smooth point in its fibre;

(b) X has relative geometric depth at least n at x: in a neighbourhood of x, the scheme X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x, with $n \leqslant N-k$.

Then $\phi^i_x = 0$ for $i < n$.

Let X_x be the strict localization of X at x, and put $V = X_x - \{x\}$. For every trait S' finite over S, put similarly

$$X'_x = X_x \times_S S', \qquad V' = V \times_S S'.$$

Let $\bar{S}$ be the normalization of S in $\bar{\eta}$, and put

$$\bar{X}_x = X_x \times_S \bar{S}, \qquad \bar{V} = V \times_S \bar{S}.$$

We have:

(1) If $\phi^i = 0$ on $(X_x)_s$ (i.e. at every generization of x in X_s) for $i \leqslant m$, then

$$H^{i-1}_{\{x\}}(\bar{X}_x) = \widetilde{H}^{\cdot i}(\bar{V}) \xrightarrow{\sim} \widetilde{H}^{\cdot i}(\bar{V}_{\bar{\eta}}) = \phi^i \qquad \text{for } i \leqslant m.$$

The ϕ^i are indeed the cohomology sheaves of $\bar{V}$ with support in V_s; under the hypotheses of 4.5, condition (a) makes (1) applicable with $m = \infty$ (SGA 4 XV 2.1).

(2) Hypothesis (b) of 4.5 is stable under base change. By SGA 2 XIV 5.6, it implies

$$\widetilde{H}^{\cdot i}(V') = H^{i-1}_{\{x\}}(X'_x) = 0 \qquad \text{for } i < n.$$

We conclude by observing that $\widetilde{H}^{\cdot i}(\bar{V})$ is the filtered colimit of the $\widetilde{H}^{\cdot i}(V')$.

Combining 4.2 and 4.5 gives:

Corollary 4.6. Let S be a henselian trait of characteristic 0, let X/S be a relative complete intersection of relative dimension n, and let $x \in X_s$ be an isolated non-smooth point in its fibre. Then $\phi^i_x = 0$ for $i \neq n$.

Corollary 4.7. Under the hypotheses of 4.6, ϕ^n_x is a flat Λ-module.

Indeed, if Λ' is a finite Λ-algebra, a general nonsense (SGA 4 XVII 5.2.11) gives, for the $\phi^i_{\Lambda'}$,

$$\phi^{n-i}_{\Lambda'} = \mathrm{Tor}^{\Lambda}_i(\Lambda', \phi^n_\Lambda),$$

and we apply 4.6 to Λ'.

Variant 4.8. In 4.5, omit condition (a), and let Z be the non-smooth locus of f. One may still conclude that

$$\phi_x^i = 0 \qquad \text{for } i < n - \dim Z_s.$$

We proceed by descending induction on $\dim \overline{\{x\}}$. The induction hypothesis permits us to apply 4.5 (1) with $m = n - \dim Z_s$, and the proof is completed as before.

4.9. It is possible to obtain a vanishing criterion using only information about the geometric fibres of f. For complete intersections, however, the result is one degree weaker than 4.5.

Proposition 4.10. Under the general hypotheses of 4.5, suppose that:

(a) x is an isolated non-smooth point in its fibre;

(b) for every point y of $X_{\bar\eta}$ whose closure in $X_{\bar\eta}$ has dimension $d(y)$ and which is a generization of x, the étale depth satisfies

$$\operatorname{depth}_{\text{ét},y}(X_{\bar\eta}) \geqslant n - d(y), \quad \text{i.e.} \quad H^i_{\{y\}}(X_{\bar\eta}) = 0 \quad \text{for } i \leqslant n - d(y);$$

(c) $\operatorname{depth}_{\text{ét},x}(X_s) \geqslant n$.

Then $\phi_x^i = 0$ for $i < n - 1$.

Retain the notation of 4.5. We may apply 4.5 (1) with $m = \infty$. By the local Lefschetz theorem (SGA 2 XIV), hypothesis (b) gives

$$H^i(V') \xrightarrow{\sim} H^i(V_s) \qquad (\ i < n\text{-}1\ )$$

$$H^i(V') \hookrightarrow H^i(V_s) \qquad (\ i = n\text{-}1\ )$$

Condition (c) then gives

$$H^i(\bar V) = \varinjlim H^i(V') = 0 \qquad \text{for } i \leqslant n - 1,$$

which proves 4.10.

4.11. As in 4.8, one can give a variant of 4.10 in which (a) is not assumed.

5. Action of monodromy on π_1

In this section (Grothendieck's exposé of 19 November 1968), we shall use the semistable reduction theorem for curves. Artin and Winters [2] gave a direct proof of this theorem; it can also be deduced [3] from the analogous theorem for abelian varieties (IX § 3, using 3.6). We shall also use Schlessinger's theory (Rim's Exposé VI).

5.1. Let k be an algebraically closed field, let C be a projective curve over k, and let $x_1, \ldots, x_n$ be a finite set of closed points of C. The pointed curve

$$X = (C; x_1, \ldots, x_n)$$

is called stable if:

(a) C is reduced, and its only singularities are double points with distinct tangents;

(b) the x_i are distinct smooth points of C;

(c) if C_1 is an irreducible component of C, and E is the set of points of the normalization C_1' of C_1 lying above a singular point of C or above one of the x_i, then, if C_1' has genus 0 (respectively 1), the set E has at least 3 (respectively 1) elements.

Condition (c) is equivalent to each of the following:

(c') X has no nontrivial infinitesimal automorphisms.

(c") If ω is the invertible dualizing sheaf, the invertible sheaf

$$\omega' = \omega\Big(\sum_i x_i\Big)$$

is ample.

An S-scheme C equipped with sections $x_1, \ldots, x_n$ is a stable curve over S if its geometric fibres are stable curves. In [3], §§ 1–2, only the case $n = 0$ is considered. The extension of the results of loc. cit. to the general case is immediate:

(1) If X/k is stable, then $H^1(C, \omega'^{\otimes n}) = 0$ for $n \geqslant 2$, and $\omega'^{\otimes n}$ is very ample for $n \geqslant 3$ (cf. [3] 1.2).

(2) The formal moduli scheme M of X is smooth, and C remains singular over a normal-crossings divisor of M.

(3) If X and Y are stable over S, then $\mathrm{Isom}_S(X, Y)$ is finite and unramified over S (isomorphisms are required to respect the marked points).

(4) If X/k is stable, the map

$$\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k \mathrm{Pic}^\circ(C)$$

is injective.

Finally, we have:

Proposition 5.1.1. Let $X_\eta = (C_\eta; x_1, \ldots, x_n)$ be stable over the generic point η of a trait S, with C_η smooth. There exist a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X_\eta \otimes_\eta \eta'.$$

This follows from the usual semistable reduction theorem, which permits one to choose η' so that the geometric special fibre of the minimal model C'' of $C_{\eta'}$ over S' satisfies (a). One then blows up, iteratively if necessary, smooth points of C'' so that (b) is satisfied. Next one contracts the chains of smooth rational components of self-intersection -2 which contain none of the $\bar{x}_i$ in the special fibre; the result is C' (cf. [3], proof of 2.3, p. 88).

As in loc. cit., one can check that C' exists as soon as the Jacobian of C_η has semistable reduction. The smoothness hypothesis on C_η is moreover unnecessary.

5.2. Let X_η be a geometrically connected scheme of finite type over the fraction field of a strictly local trait S. For a geometric point ξ of $X_{\bar\eta}$, there is an exact sequence

$$0 \longrightarrow \pi_1(X_{\bar\eta}, \xi) \longrightarrow \pi_1(X, \xi) \longrightarrow \mathrm{Gal}(\bar\eta/\eta) \longrightarrow 0. \tag{5.2.1}$$

It induces a homomorphism from $\mathrm{Gal}(\bar\eta/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar\eta})$. Passing to the largest prime-to-p quotient $\pi_1^{(p)}(X_{\bar\eta})$ of $\pi_1(X_{\bar\eta})$, one obtains

$$\mathrm{Gal}(\bar\eta/\eta) \longrightarrow \mathrm{Aut}_{\mathrm{ext}}\big(\pi_1^{(p)}(X_{\bar\eta})\big). \tag{5.2.2}$$

Suppose that X_η has a rational point x, and let $\bar{x}$ be the geometric point

$$\bar\eta \longrightarrow \eta \xrightarrow{x} X_\eta.$$

For $\xi = \bar{x}$, the exact sequence (5.2.1) splits canonically, giving

$$[x] : \mathrm{Gal}(\bar\eta/\eta) \longrightarrow \mathrm{Aut}\big(\pi_1(X_{\bar\eta}, \bar{x})\big) \longrightarrow \mathrm{Aut}\big(\pi_1^{(p)}(X_{\bar\eta}, \bar{x})\big). \tag{5.2.3}$$

Theorem 5.3. The image of P under (5.2.2) (or under (5.2.3), which amounts to the same thing) is finite.

5.3.1. Reduction to the case in which X_η is geometrically normal and integral.

Let X'_η be the normalization of X_η, let $(X'_{i\eta})_i$ be the family of connected components of X'_η, put

$$X''_\eta = X'_\eta \times_{X_\eta} X'_\eta,$$

and let $(X''_{j\eta})_j$ be the family of connected components of X''_η. After a finite extension of $k(\eta)$, we may suppose that the $X'_{i\eta}$ are geometrically normal and integral and have rational points x_i, and that the $X''_{j\eta}$ are geometrically connected and have rational points y_j.

For $k = 1$ or 2, and whenever $p_k(y_j) \in X'_{i\eta}$, choose a class of paths ℓ_{kij} from $\mathrm{pr}_k(\bar{y}_j)$ to $\bar{x}_i$. It is an element of a torsor under $\pi_1(X'_{i\bar{\eta}}, \bar{x}_i)$. Let $\bar{\ell}_{kij}$ be the corresponding prime-to-p path (the same path modulo $\mathrm{Ker}(\pi \longrightarrow \pi^{(p)})$). We may choose $\bar{\ell}_{kij}$ to be invariant under P: the set of prime-to-p paths is an inverse limit of finite sets of cardinality prime to p on which the pro-p group P acts.

Consequently, if a finite-index subgroup P' of P acts trivially on the groups $\pi_1^{(p)}(X'_{i\bar{\eta}}, \bar{x}_i)$, the Van Kampen formulas—equivalently, the fact that the normalization map $X'_{\bar{\eta}} \longrightarrow X_{\bar{\eta}}$ is effective descent for étale coverings—show that P' acts trivially on $\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})$.

5.3.2. If $X_{\bar{\eta}}$ is normal and U is an open subset, then $\pi_1(U)$ maps onto $\pi_1(X_{\bar{\eta}})$; we may therefore suppose that $X_{\bar{\eta}}$ is smooth and affine. By cutting with generic hyperplane sections (which replaces S by its strict localization at the generic point of the special fibre of an $\mathbb{A}^n_S$) and applying Bertini, we reduce to the case in which X_η is smooth, geometrically connected, and of dimension one.

After a finite extension of $k(\eta)$, we may suppose that X_η is the complement in a smooth projective curve $\bar{X}_\eta$ of a finite set of rational points x_i, whose number we are free to increase. By the semistable reduction theorem (5.1.1), we reduce to the case in which there is a stable curve $X = (C; x_1, \ldots, x_n)$ over S such that

$$X_\eta = C_\eta - \{x_1, \ldots, x_n\}.$$

We then apply the following result.

Theorem 5.4. *Let S be a strictly local scheme, let $f : \bar{X} \longrightarrow S$ be a proper flat morphism whose special fibre is a curve with ordinary double points as its only non-smooth points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Let x be a rational point of $X = \bar{X} - Y$. Let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U. Then the image of $\pi_1(U, \xi)$ in $\mathrm{Aut}(\pi_1^{(p)}(X_\xi, x\xi))$ is abelian and prime to p.*

Observe that $\pi_1(U, \xi)$ can act on $\pi_1^{(p)}(X_\xi, x\xi)$ only because the groups $\pi_1^{(p)}$ satisfy the specialization theorem.

We reduce first to the case in which S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is the spectrum of a formal power-series ring over a Cohen ring:

$$S = \mathrm{Spec}\big(W[[t_1, \ldots, t_n]]\big),$$

and U is the complement of a normal-crossings divisor with equation $t_1 \cdots t_m = 0$. In this universal case, Abhyankar's lemma shows that $\pi_1(U, \xi)$ is abelian and prime to p, which proves the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by taking a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem

(by P. Deligne)

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \longrightarrow S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ obtained from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Néron model of $A_{\eta'}$ over S', let $\mathcal{A}_{S',s'}$ be its special fibre, and let $\mathcal{A}^\circ_{S',s'}$ be the identity component of $\mathcal{A}_{S',s'}$. The stable reduction theorem is as follows.

Theorem 6.1. *For a suitable S', $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}^\circ_{S',s'}$ is an extension of an abelian variety by a torus.*

It follows fairly readily from 6.1 that one may choose S' such that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field (the same argument would work if $k(s)$ were radicial over such a field).

Let ℓ be a prime different from the residual characteristic exponent p, let $T_\ell(A)$ be the Tate module, let $V_\ell(A) = T_\ell(A) \otimes \mathbf{Q}_\ell$, and let

$$\rho : \mathrm{Gal}(\bar\eta/\eta) \longrightarrow \mathrm{GL}(T_\ell(A))$$

be the natural representation. After a preliminary extension of scalars, we are reduced to the case in which the inertia group I acts unipotently (1.2), and hence through its largest pro-ℓ quotient $\mathbf{Z}_\ell(1)$ (0.3). Let T be the image in $\mathrm{GL}(T_\ell(A))$ of a generator of $\mathbf{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T-1)^r \neq 0$. We have:

Lemma 6.2. The map

$$(\sigma, v) \in I \times V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces the morphism and the isomorphism in the commutative diagram

(6.2.1)
$$\begin{tikzcd}
V_\ell(A)_I \otimes \mathbf{Q}_\ell(r) \arrow[r, "M"] \arrow[d] & V_\ell(A)^I \\
\big(V_\ell(A)/\operatorname{Ker}(T-1)^r\big) \otimes \mathbf{Q}_\ell(r) \arrow[r, "\sim"] & \operatorname{Im}(T-1)^r \arrow[u]
\end{tikzcd}$$

$V_\ell(A)_I$ and $V_\ell(A)^I$ are $\mathrm{Gal}(\bar s/s)$-modules, and M is Galois-equivariant.

6.3. An ℓ-adic representation

$$\tau : \mathrm{Gal}(\bar s/s) \longrightarrow \mathrm{GL}(V)$$

is said to be of weight n ($n \in \mathbf{Z}$) if the following condition is satisfied:

(*) For a suitable model X of $k(s)$ (that is, an irreducible variety of finite type over the prime field such that $k(s) = k(X)$), τ factors through $\pi_1(X, \bar s)$ and, for every closed point x of X, the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value $|k(x)|^{n/2}$ (here F_x denotes the geometric Frobenius ϕ_x^{-1}).

If V (respectively W) has weight n (respectively m) and $n \neq m$, then $\mathrm{Hom}_{\mathrm{Gal}}(V, W) = 0$.

6.4. It follows from the universal property of the Néron model that $V_\ell(A)^I = V_\ell(\mathcal{A}^\circ_{S,s})$. If a is the dimension of the largest abelian quotient of $\mathcal{A}^\circ_{S,s}$ and m is the dimension of its multiplicative part, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2.

6.5. Let A^* be the abelian variety dual to A. Recall that $V_\ell(A^*)$ and $V_\ell(A)$ (and hence $V_\ell(A^*)^I$ and $V_\ell(A)_I$) are in duality with values in $\mathbf{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m are for A, it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1.

6.6. The Galois module $\mathbf{Q}_\ell(r)$ has weight $-2r$. Since $\operatorname{Im}(T-1)^r \neq 0$, it follows from 6.4, 6.5, and the isomorphism in 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

so $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. It then follows from 6.2 that $\operatorname{Im}(T-1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0; these spaces have the same dimension $m_1 \leq m$.

We then have

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2 \dim \mathcal{A}_{S,s} \geq 2(a+m) \geq 2a + m + m_1 = 2d = 2 \dim \mathcal{A}_{S,s},$$

so A_η has stable reduction ($\dim \mathcal{A}_{S,s} = a + m$). At the same time, we have obtained $m_1 = m$; that is,

$$\mathrm{Im}(T-1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}^\circ_{S,s})$$

is the V_ℓ of the multiplicative part of $\mathcal{A}^\circ_{S,s}$.

6.7. To treat the general case, we shall use (0.5) (cf. 1.3). We reduce to the case in which:

(a) the conditions of (0.5.2) or (0.5.3) are satisfied;

(b) $\mathrm{Gal}(\bar\eta/\eta)$ acts trivially on A_ℓ;

(c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses, we shall prove that A_η has stable reduction. Hypothesis (a) allows us to apply (0.5.1) and write $S = \varprojlim S_i$ as in 1.3, whose notation we retain. Let s_i be the closed point of S_i.

For sufficiently large i, the identity component $\mathcal{A}^\circ$ of the Néron model $\mathcal{A}_S$ descends to a smooth group scheme $\mathcal{A}_i$ with connected fibres over S_i, and $(\mathcal{A}_i)_\ell$ is constant over $S_i - D_i$. The group $\pi_1(S_i - D_i, \bar\eta)$ acts on $T_\ell(A)$ and acts trivially on $T_\ell(A)/\ell T_\ell(A) \simeq A_\ell$. As in 1.3, one sees that P_i acts trivially. We then have, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \qquad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}^\circ_s) = T_\ell(A)^I = T_\ell(A)^{I_i},$$

and (6.2.1) is a diagram of $\mathrm{Gal}(\bar s_i/s_i)$-modules to which the arguments of 6.4–6.6 may be applied.

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Exposé II

Finiteness properties of the fundamental group

by Michèle Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k, and let $\Pi_1^{(p)}(X)$ be the largest "prime-to-p" quotient of $\Pi_1(X)$. In this exposé we prove that, when X is a surface, $\Pi_1^{(p)}(X)$ is topologically finitely presented, and that the same holds for arbitrary X if resolution of singularities is assumed.

For a surface, the proof method, due to J.-P. Murre, consists in reducing to a fibration over P^1 having properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k. Then there exists a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow P^1,$$

such that f has a section σ and the following conditions hold:

(i) f is smooth at the points of $\sigma(P^1)$.

(ii) The fibres of f are geometrically integral curves having only ordinary singularities.

(iii) There exists a nonempty open subset U of P^1 such that the fibres of f over the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \longrightarrow P^r$ for which a Lefschetz pencil exists (XVII 2.5). Indeed, let $D = \{H_t\}_{t \in P^1}$ be such a pencil of hyperplanes. Recall that it has the following properties (loc. cit. 2.2):

(a) The axis Δ of D meets S transversely; the closed subscheme $\Delta \cap S$ is therefore reduced and consists of finitely many closed points $x_1, \ldots, x_n$ of S.

(b) There exists a nonempty open subset U of P^1 such that, for every $t \in U$, H_t meets S transversely.

(c) For every $t \in P^1 - U$, H_t meets S transversely except at one point, which is an ordinary quadratic singularity.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t. The map assigning $t \in P^1$ to x is a rational map $a : S \dashrightarrow P^1$ with domain of definition $S - (\Delta \cap S)$.

Consider the blow-up $g : S' \longrightarrow S$ of the closed subscheme $S \cap \Delta$. The scheme S' is regular, and the fibre S'_{x_i} over x_i is the projective bundle defined by the tangent space of S at x_i, hence is isomorphic to P^1 (EGA IV 19.4.2, 19.4.4). Moreover, the rational map $f = ag$ is everywhere defined (XVIII 3.1). Finally, to each x_i one can associate the section of f that sends $t \in P^1$ to the tangent vector at x_i of $H_t \cap S$.

Let us verify conditions (i), (ii), and (iii). For each $t \in P^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic. Indeed, the morphism $S'_t \longrightarrow S_t$ is an isomorphism except possibly at the points x_i, where S_t is regular; moreover, since S'_t is a complete intersection in S', it is integral, and is therefore isomorphic to S_t.

The asserted properties of the fibres follow from (a)–(c). Finally, f is smooth at the points of σ, since f is flat and the fibres S'_t are geometrically regular at $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S, and let $Y' = Y \times_S S'$. Then f, g, and U may be chosen so that $Y' \times_{P^1} U$ is a divisor with normal crossings relative to U.

It is known that, if a Lefschetz pencil exists for an embedding $S \longrightarrow P^r$, the Lefschetz pencils form an open subset of the variety of lines in P^r (XVII 3.2.1). The same is true of the set of

Lefschetz pencils whose axis does not meet Y and for which there is a nonempty open subset U_1 of P^1 such that $Y \cap H_t$ is smooth for $t \in U_1$. It is then enough to construct f and g from a pencil satisfying these two conditions and replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is said to be topologically finitely generated if there is an integer n such that G is isomorphic to a quotient of the free pro-group $F(n)$ on n generators. Put $K = \mathrm{Ker}(F(n) \longrightarrow G)$. The group G is said to be topologically finitely presented if, in addition, there are finitely many elements $k_1, \ldots, k_r$ of K such that the smallest closed normal subgroup of K containing $k_1, \ldots, k_r$ is K.

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k. Suppose that schemes of finite type and dimension at most $\dim X$ over an algebraic closure of k admit strong resolution of singularities (SGA 5 I 3.1.5). This condition is satisfied when $\dim X = 2$, by [1], or when $\mathrm{char}\, k = 0$, by [2]. Then the fundamental group $\Pi_1^{(p)}(X)$ is topologically finitely presented.

By SGA 1 IX 4.10, we may suppose that k is algebraically closed.

1) Reduction to the case in which X is a regular surface.

If $f : X' \longrightarrow X$ is a morphism of finite type, put $X'' = X' \times_X X'$, and let $(X'_a)_{a \in A}$ and $(X''_b)_{b \in B}$ be the families of connected components of X' and X'', respectively. When f is an effective descent morphism for the category of finite étale coverings, the fundamental group $\Pi_1(X)$ can be computed explicitly

in terms of the fundamental groups $\Pi_1(X'_a)$ and $\Pi_1(X''_b)$. It follows (SGA 1 IX 5.2, 5.3) that if the $\Pi_1(X'_a)$ are topologically finitely generated, then so is $\Pi_1(X)$; and if the $\Pi_1(X'_a)$ are topologically finitely presented while the $\Pi_1(X''_b)$ are topologically finitely generated, then $\Pi_1(X)$ is topologically finitely presented.

Since X admits resolution of singularities, there is a regular scheme X' and a proper birational morphism $f : X' \longrightarrow X$. Since f is an effective descent morphism for finite étale coverings, it suffices to prove the theorem under the additional assumption that X is regular. One first deduces that $\Pi_1(X)$ is topologically finitely generated; knowing that $\Pi_1(X')$ is topologically finitely presented and that the $\Pi_1(X''_b)$ are topologically finitely generated then shows that $\Pi_1(X)$ is topologically finitely presented.

Now let $(U_i)_{i \in I}$ be a finite affine open covering of X, and let $f : \coprod U_i \longrightarrow X$ be the canonical morphism. Since f is an effective descent morphism for finite étale coverings, it is enough to prove the theorem for the U_i. We are thus reduced to the case in which X is affine and smooth.

Assume henceforth that X is affine and smooth over k, with $\dim X > 2$. The Lefschetz theorem for quasi-projective schemes ([3] V 7.4) applies: if Y is a sufficiently general hyperplane section of X, then

$$\Pi_1(Y) \simeq \Pi_1(X).$$

It therefore suffices to prove the theorem for Y, reducing us to the case $\dim X = 2$.

2) The case in which X is a smooth surface over k.

By the strong resolution theorem for surfaces, there exist a smooth, integral, projective surface S and an open immersion $i : X \longrightarrow S$ such that $Y = S - X$ is a normal-crossings divisor. We apply Corollary 2.2 and retain its notation.

We show that it is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically finitely presented. Since X and X' are regular and

$$\mathrm{codim}\left(\bigcup_{1 \leq i \leq n} \{x_i\}, X \right) \geq 2,$$

every étale covering of X' induces on $X - \bigcup_{1 \leq i \leq n} \{x_i\}$ an étale covering which extends uniquely to an étale covering of X (SGA 2 1.11). This gives an isomorphism

$$\Pi_1(X') \simeq \Pi_1(X).$$

Consider the cartesian diagram

$$\begin{tikzcd} X' \arrow[d, "h"'] & X'_U \arrow[l] \arrow[d, "h_U"] \\ P^1 & U \arrow[l] \end{tikzcd}$$

where $h = f|_{X'}$. Let L be the set of primes different from p, and let $\bar{\eta}$ be a geometric point over the generic point η of P^1. We verify the hypotheses of SGA 1 XIII 4.8. The morphism h has a section σ, because f does. To prove condition (a) of SGA 1 XIII 4.7, note first that, since the fibres of h are geometrically integral, h is 0-acyclic and locally 0-acyclic (SGA 4 XV 4.1, 1.16). It remains to show that, for every locally constant sheaf of sets F on X', (F, h) is cohomologically proper in dimensions ≤ 0. If ξ is a closed point of P^1 and Z denotes the integral closure of $\operatorname{Spec} \mathcal{O}_{P^1,\xi}$ in $\bar{\eta}$, then $X'_Z = X' \times_{P^1} Z$ satisfies condition (S_2) at its closed points and is regular at every other point, hence is normal. Consequently, an étale covering of X'_Z whose restriction to $X'_{\bar{\eta}}$ is connected is necessarily connected. This proves the assertion, and hence condition (a).

Since X'_U is the complement in S'_U of a divisor with normal crossings relative to U, h_U is locally 1-aspherical for L (SGA 4 XV 2.1); and, for every finite constant sheaf of L-groups F on X'_U, (F, h_U) is cohomologically proper in dimensions ≤ 1 (SGA 1 XIII 2.4), which gives condition (b). Since P^1 is normal and $T = P^1 - U$ has codimension 1, we have

$$\operatorname{prof} \text{ét}_T(P^1) \geq 2,$$

which gives condition (c). The condition

$$\operatorname{prof} \operatorname{hop}_x(X') \geq 3$$

holds at every closed point of X' by the purity theorem, which gives condition (e). Finally, for every $t \in P^1$, h is smooth at $\sigma(t)$, and hence is locally 1-aspherical for L at $\sigma(t)$, which gives condition (d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, there is an exact sequence

$$1 \longrightarrow \Pi_1^{(p)}(X'_{\bar{\eta}}, a) \longrightarrow \Pi'_1(X'_U, a) \longrightarrow \Pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K be the image of $\Pi_1(U, a)$ in $\operatorname{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$. The group of coinvariants under K,

$$\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ by the closed normal subgroup generated by the elements $x^{-1}q(x)$, where $x \in \Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. (4.7.4), there is an isomorphism

$$\Pi_1^{(p)}(X', a) \simeq \Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K.$$

Write $P^1 - U = \coprod_{1 \leq i \leq n} \{t_i\}$. For each i, let $\bar{T}_i$ be the strict localization of P^1 at a geometric point $\bar{t}_i$ over t_i, and let s_i be the generic point of $\bar{T}_i$. An étale covering of U whose restriction to every s_i is trivial extends to an étale covering of P^1, and is therefore trivial. For each i, let $\overline{k(s_i)}$ be an algebraic closure of $k(s_i)$. The preceding assertion says that $\Pi_1(U, a)$ is the smallest closed normal subgroup containing the images of the Galois groups $\operatorname{Gal}(\overline{k(s_i)}/k(s_i))$ in $\Pi_1(U, a)$. By I 5.3, the image K_i of $\operatorname{Gal}(\overline{k(s_i)}/k(s_i))$ in $\operatorname{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed normal subgroup containing all the K_i, and since $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ is topologically finitely presented (SGA 1 XIII 3.2), this proves that $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K$, and hence also the isomorphic group $\Pi_1^{(p)}(X)$, is topologically finitely presented.

Remark 2.3.2. Resolution of singularities is not needed to prove Theorem 2.3.1 when X is the open complement of a normal-crossings divisor in a projective scheme smooth over k.

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Exposé VI

FORMAL DEFORMATION THEORY

by D. S. Rim

1. Formal existence theorem

We recall the notion of a cofibered category (S.G.A. 1 IV). Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \longrightarrow \underline{C}$ such that:

Ax. 1. For every map f in $\underline{C}$ and every object a in $\underline{A}$ for which $P(a)$ is the source of f, there is a cocartesian f-morphism $\alpha : a \longrightarrow b$. Thus, for every f-morphism $\alpha' : a \longrightarrow b'$, there is a unique T-morphism $\tau : b \longrightarrow b'$ in $\underline{A}$ such that $\alpha' = \tau\alpha$, where T is the target of f.

Ax. 2. A composite of cocartesian morphisms in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \longrightarrow \underline{C}$ follow immediately from the definitions.

(Cancellation) Let $a \underset{\beta}{\overset{\alpha}{\rightrightarrows}} a'$ be two cocartesian maps such that $P(\alpha) = P(\beta)$. If there is a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(Factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$ and $a \xrightarrow{\beta} a''$, there is a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there is a map $f : P(a') \longrightarrow P(a'')$ in $\underline{C}$ such that $fP(\alpha) = P(\beta)$. Moreover, γ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object S of $\underline{C}$, denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects a for which $P(a) = S$, and in which

$$\mathrm{Hom}_{\underline{A}(S)}(a, a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a, a') \mid P(\alpha) = \mathrm{id}_S\}.$$

A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$ is a groupoid for every object S of $\underline{C}$. Recall that a groupoid is a category in which every morphism is an isomorphism. Thus a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \longrightarrow \underline{C}$ in which every map of $\underline{A}$ is cocartesian. Furthermore, for every $\alpha \in \mathrm{Fl}(\underline{A})$, α is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, fix a complete noetherian local ring Λ with residue field k, and let $\underline{C}_\Lambda$ be the category of artinian local Λ-algebras with residue field k, whose maps are local homomorphisms over Λ. From 1.2 onward, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to satisfy $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of exactly one element for every two objects a, b of $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will simply be called a $\underline{C}_\Lambda$-map if $P(a') = P(a)$ and $P(\alpha) = 1_{P(a)}$. By abuse of language, a map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called surjective if $P(\alpha) : P(a') \longrightarrow P(a)$ is surjective in $\underline{C}_\Lambda$.

Ex. 1.1(a). Let $F : \underline{C} \longrightarrow$ (categories) be a functor. Define the category $\underline{F}$ as follows. An object of $\underline{F}$ is a pair (S, a) with $S \in \mathrm{ob}(\underline{C})$ and $a \in \mathrm{ob}F(S)$; a map $(S, a) \longrightarrow (S', a')$ is a pair (f, u'), where $f : S \longrightarrow S'$ is a map in $\underline{C}$ and $u' : F(f)a \longrightarrow a'$ is a map in $F(S')$. The composite of

$$(S, a) \xrightarrow{(f,u')} (S', a') \xrightarrow{(g,u'')} (S'', a'')$$

is

$$(gf,\ u'' \cdot F(g)u') : (S, a) \longrightarrow (S'', a'').$$

With respect to the functor $P : \underline{F} \longrightarrow \underline{C}$ given by $(S, a) \longmapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \longrightarrow$ (groups) yields a cofibered group, and a functor $F : \underline{C} \longrightarrow$ (Ens) yields a cofibered discrete groupoid. A natural transformation $\varphi : F \longrightarrow G$, where $F, G : \underline{C} \longrightarrow$ (categories), induces a cocartesian functor $\varphi : \underline{F} \longrightarrow \underline{G}$ over $\underline{C}$.

Ex. 1.1(b). Let X and Y be schemes over Λ together with a fixed isomorphism

$$X \otimes_\Lambda k \overset{\tau}{\longleftrightarrow} Y \otimes_\Lambda k.$$

Then the functor $\mathrm{Isom}_{X,Y} : \underline{C}_\Lambda \longrightarrow$ (groupoids) given by

$$\mathrm{Isom}_{X,Y}(S) = \left\{ X \otimes_\Lambda S \overset{\sigma}{\longleftrightarrow} Y \otimes_\Lambda S \;\middle|\; \sigma \otimes_\Lambda k = \tau \right\}$$

yields a cofibered groupoid $P : \underline{\mathrm{Isom}}_{X,Y} \longrightarrow \underline{C}_\Lambda$.

Ex. 1.1(c). Let X_0 be a fixed scheme over k, and let $\underline{M}_{X_0}$ be the category of deformations of X_0 over $\underline{C}_\Lambda$ (see §4). Thus an object of $\underline{M}_{X_0}$ is a pair (R, X), where $R \in \underline{C}_\Lambda$ and X is a deformation of X_0 to R. Then the functor $P : \underline{M}_{X_0} \longrightarrow \underline{C}_\Lambda$ given by $(R, X) \longmapsto R$ defines a cofibered groupoid over $\underline{C}_\Lambda$.

Ex. 1.1(d). For any groupoid X, denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects of X. If $P : \underline{F} \longrightarrow \underline{C}_\Lambda$ is a cofibered groupoid, we obtain a functor $[\underline{F}] : \underline{C}_\Lambda \longrightarrow$ (Ens) given by $[\underline{F}](S) = [\underline{F}(S)]$, and hence a cofibered discrete groupoid which, by abuse of notation, is again denoted by $[\underline{F}]$. Definition 1.2. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called semi-homogeneous if the following properties hold:

(1) Every diagram

$$\begin{tikzcd}
a'' \arrow[r, dashed] \arrow[d, dotted] & a' \arrow[d] \\
a_1 \arrow[r] & a
\end{tikzcd}$$

of solid arrows in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(2) Let

$$\begin{tikzcd}
 & R \\
R \underset{k}{\times} k[\varepsilon] \arrow[ur, "\pi_1"] \arrow[dr, "\pi_2"'] & \\
 & k[\varepsilon]
\end{tikzcd}$$

be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k, and let

$$\begin{tikzcd}
a' \arrow[dr, "\alpha"] & \\
 & a \\
b' \arrow[ur, "\beta"'] &
\end{tikzcd}$$

be π_1-maps in $\underline{A}$. Then there is a $\underline{C}_\Lambda$-map $\gamma : a' \longrightarrow b'$ such that $\alpha = \beta\gamma$ if and only if $(\pi_2)_*(a') \simeq (\pi_2)_*(b')$.

Remark 1.3(a). A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. In particular, 1.2(2) implies that the canonical map

$$[\underline{A}(R \times_k k[\varepsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\varepsilon])]$$

is bijective.

Remark 1.3(b). Consider a commutative diagram

$$\begin{tikzcd} & a' \arrow[dl, "v_1"'] \arrow[dr, "v_2"] & \\ a_1 \arrow[dr, "u_1"'] & & a_2 \arrow[dl, "u_2"] \\ & a & \end{tikzcd}$$

in a cofibered category $\underline{A}$ over $\underline{C}_\Lambda$. Put $R = P(a)$, $R' = P(a')$, and $R_i = P(a_i)$ for $i = 1, 2$. If a'' is the direct image under the map

$$(P(v_1), P(v_2)) : R' \longrightarrow R_1 \times_R R_2,$$

we obtain a commutative diagram

$$\begin{tikzcd} & a' \arrow[d] \arrow[ddl, "v_1"'] \arrow[ddr, "v_2"] & \\ & a'' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr, "u_1"'] & & a_2 \arrow[dl, "u_2"] \\ & a & \end{tikzcd}$$

Thus 1.2(1) may be replaced by:

(1)$'$ Every diagram

$$\begin{tikzcd} a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd} \quad \text{in } \underline{A} \text{ above} \quad \begin{tikzcd} R_1 \arrow[dr] & & R_2 \arrow[dl] \\ & R & \end{tikzcd} \quad \text{in } \underline{C}_\Lambda,$$

where $R_2 \longrightarrow R$ is surjective, can be completed to a commutative diagram

$$\begin{tikzcd} & a' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd} \quad \text{above} \quad \begin{tikzcd} & R_1 \times_R R_2 \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ R_1 \arrow[dr] & & R_2 \arrow[dl] \\ & R & \end{tikzcd}$$

where π_i $(i = 1, 2)$ are the projections.

Remark 1.3(c). If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[\varepsilon])]$ is canonically endowed with a vector space structure over k, where $k[\varepsilon]$ is the ring of dual numbers over k. Indeed, let $\underline{\underline{V}}$ be the category of

finite-dimensional vector spaces over k. Every product-preserving functor $H : \underline{\underline{V}} \longrightarrow$ (Ens) factors canonically as

$$\begin{tikzcd} \underline{\underline{V}} \arrow[rr, "H"] \arrow[dr, "\overline{H}"'] & & (\mathrm{Ens}) \\ & \underline{\underline{V}} \arrow[ur] & \end{tikzcd}$$

where $\underline{\underline{V}} \longrightarrow (\text{Ens})$ is the forgetful functor. On the other hand, the functor

$$G : \underline{\underline{V}} \longrightarrow \underline{C}_\Lambda, \qquad W \longmapsto k[W] = k \oplus W, \qquad W^2 = 0,$$

takes products in $\underline{\underline{V}}$ to fibre products over k in $\underline{C}_\Lambda$. If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then

$$[\underline{A}(k[V] \times_k k[W])] \longrightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$$

is bijective by 1.3(a). Hence the functor $V \longmapsto [\underline{A}(k[V])]$ commutes with products, proving the assertion.

Definition 1.4. (1) Let R be an object of $\underline{C}_\Lambda$. A small extension of R in $\underline{C}_\Lambda$ is a surjective map $R' \xrightarrow{f} R$ in $\underline{C}_\Lambda$ such that $(\ker f)^2 = 0$ and $\ker f \simeq k$. In other words, $R' \xrightarrow{f} R$ is a small extension of R if

$$0 \longrightarrow k \xrightarrow{t} R' \xrightarrow{f} R \longrightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

(2) Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let a be an object of $\underline{A}$. A small prolongation of a is an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ such that $P(\alpha)$ is a small extension in $\underline{C}_\Lambda$. An arrow $a' \xrightarrow{\alpha} a$ will be called versal for small prolongations of a if, for every small prolongation $a'_1 \longrightarrow a$, there exists a commutative diagram

$$\begin{tikzcd} a' \arrow[rr] \arrow[dr, "\alpha"'] & & a'_1 \arrow[dl] \\ & a & \end{tikzcd}$$

Proposition 1.5. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$.
(1) Let

$$\begin{tikzcd} & R \\ R \times_k k[\varepsilon] \arrow[ur, "\pi_1"] \arrow[dr, "\pi_2"'] & \\ & k[\varepsilon] \end{tikzcd}$$

be the projection maps in $\underline{C}_\Lambda$, and let $a' \xrightarrow{\alpha} a$ be a π_1-map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \longrightarrow (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{C}_\Lambda$, and consider a commutative diagram

$$\begin{tikzcd} & a'' \arrow[dl, "\alpha'"'] \arrow[dr, "\alpha'_1"] & \\ a' \arrow[dr, "\alpha"'] & & a' \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd} \quad \text{above} \quad \begin{tikzcd} & R' \times_R R' \arrow[dl] \arrow[dr] & \\ R' \arrow[dr] & & R' \arrow[dl] \\ & R & \end{tikzcd}$$

Define the canonical map $\widetilde{f} : R' \times_R R' \longrightarrow k[\ker f]$ by

$$\widetilde{f}(r'_1, r'_2) = r'_1(0) + (r'_1 - r'_2).$$

If there exists a map $a' \longrightarrow (\widetilde{f})_*(a'')$, then there exists a map $\gamma : a' \longrightarrow a'$ such that $\alpha = \alpha_1\gamma$.

Proof. (1) The "only if" part is clear. Assume conversely that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$. Set

$$h : R \longrightarrow R \times_k k[\varepsilon], \qquad h = 1 \times P(\varphi).$$

The composite $\pi_1 h$ is the identity. We therefore obtain maps

$$a \underset{\alpha_1}{\overset{\gamma}{\rightleftarrows}} h_* a, \qquad \alpha_1 \gamma = 1_a, \qquad P(\gamma) = h, \qquad P(\alpha_1) = \pi_1.$$

On the other hand, $\pi_2 h = P(\varphi)$, so there exists a π_2-map

$$h_* a \longrightarrow (\pi_2)_* a'.$$

Thus $(\pi_2)_*(h_* a) = (\pi_2)_* a'$, and 1.2(2) gives a $\underline{C}_\Lambda$-map $u : h_* a \longrightarrow a'$ such that $\alpha u = \alpha_1$. Define $\beta = u\gamma$. Then

$$\alpha\beta = \alpha u \gamma = \alpha_1 \gamma = 1_a.$$

(2) The map

$$1 \times \widetilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\ker f]$$

is an isomorphism compatible with the projections onto the first factor, and $\pi_2(1 \times \widetilde{f}) = \widetilde{f}$, where $\pi_2 : R' \times_k k[\ker f] \longrightarrow k[\ker f]$ is the second projection. Since $R' \xrightarrow{f} R$ is a small extension, $k[\ker f] \simeq k[\varepsilon]$. Hence, if there exists a map

$$a' \longrightarrow (\widetilde{f})_* a'' = (\pi_2)_* \big((1 \times \widetilde{f})_* a''\big),$$

part (1) yields a map $\gamma' : a' \longrightarrow a''$ such that $\alpha' \gamma' = 1_{a'}$. Define $\gamma = \alpha_1' \gamma' : a' \longrightarrow a'$. Then $\alpha_1 \gamma = \alpha$.

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$\begin{tikzcd} a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd}$$

in $\underline{A}$, where α_1 is a small prolongation. Assume that a is versal for small prolongations of e. Then there exists a map $\rho : a' \longrightarrow a_1$ such that $\alpha = \alpha_1 \rho$ if and only if there exists a map $f : P(a') \longrightarrow P(a_1)$ such that $P(\alpha) = P(\alpha_1) f$ in $\underline{C}_\Lambda$.

Proof. Assume that $P(\alpha) = P(\alpha_1) f$, where $f : P(a') \longrightarrow P(a_1)$. Replacing a' by its direct image under f, we may and shall assume that $P(\alpha) = P(\alpha_1)$. Put $R' = P(a_1)$ and $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we obtain a commutative diagram

$$\begin{tikzcd} & a'' \arrow[dl, "\beta"'] \arrow[dr, "\beta_1"] & \\ a' \arrow[dr, "\alpha"'] & & a_1 \arrow[dl, "\alpha_1"] \\ & a & \end{tikzcd} \quad \text{above} \quad \begin{tikzcd} & R' \times_R R' \arrow[dl] \arrow[dr] & \\ R' \arrow[dr] & & R' \arrow[dl] \\ & R & \end{tikzcd}$$

Since a is versal for small prolongations of e, so is a'; the assertion therefore follows from 1.5(2).

Definition 1.7. For each object R of $\underline{C}_\Lambda$, put

$$\overline{\Omega}_R = M/(m_\Lambda R + M^2),$$

where m_Λ and M are the maximal ideals of Λ and R, respectively. This is a finite-dimensional vector space over k, and $R \longmapsto \overline{\Omega}_R$ defines a functor

$$\overline{\Omega} : \underline{C}_\Lambda \longrightarrow \underline{\underline{V}},$$

where $\underline{\underline{V}}$ is the category of finite-dimensional vector spaces over k. If $\underline{A}$ is a cofibered category over $\underline{C}_\Lambda$, the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\Lambda \longrightarrow \underline{\underline{V}}$ is again denoted $\overline{\Omega}$, by abuse of notation.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$. Given an object a of $\underline{A}$ that is versal for small prolongations of e, there exists an arrow $a' \longrightarrow a$ such that:

(i) a' is versal for small prolongations of a;

(ii) $P(a') \longrightarrow P(a)$ is surjective, and its kernel is annihilated by the maximal ideal of $P(a)$;

(iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

Proof. Put $R = P(a)$, and consider an exact sequence

$$0 \longrightarrow J \longrightarrow S \longrightarrow R \longrightarrow 0,$$

where S is a formal power-series ring over Λ in finitely many variables and $J \subset M^2$, with M the maximal ideal of S. Let Σ be the set of ideals J_1 of S such that

(i) $J \supset J_1 \supset MJ$, and

(ii) there exists a π_1-morphism $a_1 \longrightarrow a$ in $\underline{A}$, where $\pi_1 : S/J_1 \longrightarrow R$ is the canonical map.

Property 1.2(1) of $\underline{A}$, together with the fact that J/MJ is a finite-dimensional vector space over k, implies that Σ has a smallest ideal. Denote it by J', and choose a π'-morphism $a' \xrightarrow{g} a$, where $\pi' : S/J' \longrightarrow R$ is the canonical map. Since $J' \subset J \subset M^2$, the map $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

We claim that $a' \xrightarrow{g} a$ is versal for small prolongations of a. Indeed, let $a_1 \longrightarrow a$ be a small prolongation, and put $R_1 = P(a_1)$. We obtain an exact commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & J/MJ \arrow[r] \arrow[d, "h"] & S/MJ \arrow[r] \arrow[d, "h_1"] & R \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & k \arrow[r] & R_1 \arrow[r] & R \arrow[r] & 0.
\end{tikzcd}$$

If $h = 0$, then $\pi_1 : R_1 \longrightarrow R$ admits a cross-section, so h_1 induces a map $f : S/J' \longrightarrow R_1$. If $h \neq 0$, then $h_1 : S/MJ \longrightarrow R_1$ is surjective and hence, by the definition of J', again induces $f : S/J' \longrightarrow R_1$. In either case we obtain a commutative diagram

$$\begin{tikzcd}
S/J' \arrow[rr, "f"] \arrow[dr, "\pi'"'] & & R_1 \arrow[dl, "\pi_1"] \\
& R &
\end{tikzcd}$$

Since a is versal for small prolongations of e, Corollary 1.6 gives a commutative diagram

$$\begin{tikzcd}
a' \arrow[rr, "\rho"] \arrow[dr] & & a_1 \arrow[dl] \\
& a &
\end{tikzcd}$$

as required.

Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category in groupoids. It then gives a cofibered category in groupoids

$$\mathrm{pro}(P) : \mathrm{pro}(\underline{A}) \longrightarrow \mathrm{pro}(\underline{C}_\Lambda)$$

(for the definition of the category of pro-objects $\mathrm{pro}(\underline{A})$, see A. Grothendieck, Séminaire Bourbaki 195), fitting into the commutative diagram

$$\begin{tikzcd} \underline{A} \arrow[r] \arrow[d, "P"'] & \mathrm{pro}(\underline{A}) \arrow[d, "\mathrm{pro}(P)"] \\ \underline{C}_\Lambda \arrow[r] & \mathrm{pro}(\underline{C}_\Lambda). \end{tikzcd}$$

Let $\widehat{\underline{C}}_\Lambda$ be the full subcategory of $\mathrm{pro}(\underline{C}_\Lambda)$ whose objects are systems $(R_i)_{i\in\mathbb{N}}$ with $R_i \in \underline{C}_\Lambda$ such that each map $\varphi_i : R_i \longrightarrow R_{i-1}$ is surjective and, for all sufficiently large i, induces an isomorphism

$$\frac{m_i}{m_\Lambda R_i + m_i^2} \xrightarrow{\ \sim\ } \frac{m_{i-1}}{m_\Lambda R_{i-1} + m_{i-1}^2},$$

where $\mathbb{N}$ denotes the ordered category of natural numbers and m_i is the maximal ideal of R_i.

Let $\widehat{\underline{A}}$ be the full subcategory of $\mathrm{pro}(\underline{A})$ defined by

$$\mathrm{ob}(\widehat{\underline{A}}) = \left\{ a \in \mathrm{ob}(\mathrm{pro}(\underline{A})) \,\middle|\, \mathrm{pro}(P)(a) \in \mathrm{ob}(\widehat{\underline{C}}_\Lambda) \right\}.$$

We obtain a cofibered category in groupoids $\widehat{P} : \widehat{\underline{A}} \longrightarrow \widehat{\underline{C}}_\Lambda$ and a commutative diagram

$$\begin{tikzcd} \underline{A} \arrow[r, hook] \arrow[d, "P"'] & \widehat{\underline{A}} \arrow[d, "\widehat{P}"] \\ \underline{C}_\Lambda \arrow[r, hook] & \widehat{\underline{C}}_\Lambda. \end{tikzcd}$$

The category $\widehat{\underline{C}}_\Lambda$ is canonically equivalent to the category of complete local Λ-algebras of formally finite type over Λ with fixed residue field k; we henceforth make this identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. An object ξ of $\widehat{\underline{A}}$ is called quasi-initial if, for every object a of $\widehat{\underline{A}}$, there exists a map $\xi \to a$ in $\widehat{\underline{A}}$. A quasi-initial object ξ is called minimal if the morphism $\xi : h_R \longrightarrow [\widehat{\underline{A}}]$ induces a bijection

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\ \sim\ } [\underline{A}(k[\varepsilon])], \qquad R = P(\xi).$$

An object ξ of $\widehat{\underline{A}}$ is called a versal formal object of $\underline{A}$ if every surjective map $\alpha : a' \to a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

A versal formal object ξ is called minimal if $\xi : h_R \longrightarrow [\widehat{\underline{A}}]$ induces the above bijection on $k[\varepsilon]$, where $R = P(\xi)$.

Proposition 1.10. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$.

(i) Every (minimally) versal formal object of $\underline{A}$ is a quasi-initial (minimal) object of $\widehat{\underline{A}}$.

(ii) Suppose that ξ is an object of $\widehat{\underline{A}}$ for which

$$\widetilde{\xi}(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow [\underline{A}(k[\varepsilon])]$$

is injective, where $P(\xi) = R$. Then every endomorphism of ξ in $\widehat{\underline{A}}$ is an automorphism. In particular, a quasi-initial minimal object of $\widehat{\underline{A}}$, if it exists, is unique up to noncanonical isomorphism.

Proof. (i) Let ξ be a versal formal object of $\underline{A}$. For an object a of $\widehat{\underline{A}}$, choose a presentation by surjective maps in $\underline{A}$,

$$\cdots \longrightarrow a_n \xrightarrow{\alpha_n} a_{n-1} \longrightarrow \cdots \longrightarrow a_1 \xrightarrow{\alpha_1} a_0, \qquad a = \varprojlim_n a_n.$$

Suppose maps $\varphi_i : \xi \longrightarrow a_i$ have been chosen with $\alpha_i \varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is versal and α_n is surjective, there is a map $\varphi_n : \xi \longrightarrow a_n$ with $\alpha_n \varphi_n = \varphi_{n-1}$. Thus $\varphi = \varprojlim_n \varphi_n$ defines a map $\xi \longrightarrow a$.

(ii) Let $\alpha : \xi \longrightarrow \xi$ be an endomorphism in $\widehat{\underline{A}}$. For every $\lambda \in \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon])$, we have

$$\lambda_* \xi \simeq (\lambda \circ P(\alpha))_* \xi.$$

Injectivity therefore gives $\lambda = \lambda \circ P(\alpha)$ for every λ. It follows that $P(\alpha) : R \longrightarrow R$ is surjective. Every surjective endomorphism of a noetherian ring is an automorphism, so $P(\alpha)$, and hence α, is an automorphism.

Theorem 1.11 (Schlessinger). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then $\underline{A}$ admits a minimally versal formal object if and only if:

(i) $\underline{A}$ is semi-homogeneous;

(ii) $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof. Necessity. Let $\xi \in \widehat{\underline{A}}$ be a minimally versal formal object of $\underline{A}$, and consider a diagram

$$\begin{tikzcd} a_1 \arrow[dr] & & a_2 \arrow[dl, "\alpha"] \\ & a & \end{tikzcd}$$

in $\underline{A}$, where α is surjective. Since ξ is quasi-initial, there is a map $\xi \longrightarrow a_1$. Since α is surjective, the composite $\xi \longrightarrow a_1 \longrightarrow a$ lifts to a_2; hence we obtain a commutative diagram

$$\begin{tikzcd} & \xi \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl, "\alpha"] \\ & a & \end{tikzcd}$$

which proves condition 1.2(1).

We now prove condition 1.2(2). Let

$$\begin{tikzcd} & S \\ S \times_k k[\varepsilon] \arrow[ur, "\pi_1"] \arrow[dr, "\pi_2"'] & \\ & k[\varepsilon] \end{tikzcd}$$

be the projections, and let

$$\begin{tikzcd} a' \arrow[dr, "\alpha"'] & & b' \arrow[dl, "\beta"] \\ & a & \end{tikzcd}$$

be π_1-maps in $\underline{A}$. By the definition of ξ, there is a commutative diagram

$$\begin{tikzcd} & \xi \arrow[dl, "u"'] \arrow[dr, "v"] & \\ a' \arrow[dr, "\alpha"'] & & b' \arrow[dl, "\beta"] \\ & a & \end{tikzcd}$$

and therefore $\pi_1 P(u) = \pi_1 P(v)$. Suppose now that $(\pi_2)_* a' \simeq (\pi_2)_* b'$. Minimality of ξ gives $\pi_2 P(u) = \pi_2 P(v)$, whence

$$P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v).$$

Since $\underline{A}$ is a cofibered groupoid, there is a unique isomorphism $\sigma : a' \xrightarrow{\sim} b'$ in $\underline{A}(S \times_k k[\varepsilon])$ such that $\sigma u = v$. The equality $\beta\sigma u = \alpha u$ then implies $\beta\sigma = \alpha$.

Sufficiency. Let $c_1, \dots, c_n$ be objects of $\underline{A}(k[\varepsilon])$ whose classes form a k-basis of $[\underline{A}(k[\varepsilon])]$. Choose

$$a_1 \in \underline{A}(k[\varepsilon] \times_k \cdots \times_k k[\varepsilon])$$

such that $(\pi_j)_* a_1 \simeq c_j$. Then

$$\widetilde{a}_1(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \longrightarrow [\underline{A}(k[\varepsilon])]$$

is bijective, where $R_1 = P(a_1)$. In particular, $a_1 \longrightarrow e$ is versal for small prolongations of e. Choose $a_2 \longrightarrow a_1$ versal for small prolongations of a_1 such that $\overline{\Omega}_{a_2} \longrightarrow \overline{\Omega}_{a_1}$ is bijective. Continue inductively, choosing $a_{n+1} \longrightarrow a_n$ versal for small prolongations and inducing a bijection on $\overline{\Omega}$, and set

$$\xi = \varprojlim_n a_n.$$

Then $\overline{\Omega}_\xi \longrightarrow \overline{\Omega}_{a_1}$ is bijective, so

$$\xi(k[\varepsilon]) : \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])]$$

is bijective.

It remains to prove versality. Consider a diagram

$$\begin{tikzcd} & a' \arrow[d, "\alpha"] \\ \xi \arrow[r, "h"] & a \end{tikzcd}$$

where α is surjective. We must lift h to a'. By induction we may assume that a' is a small prolongation of a. Since $P(a)$ is artinian, h factors through some a_q,

$$\xi \longrightarrow a_q \longrightarrow a.$$

Semi-homogeneity gives a commutative diagram

$$\begin{tikzcd} & a'' \arrow[r] \arrow[d, "\alpha'"] & a' \arrow[d, "\alpha"] \\ \xi \arrow[r] & a_q \arrow[r] & a, \end{tikzcd}$$

where α' is also a small prolongation. Since $a_{q+1} \longrightarrow a_q$ is versal for small prolongations of a_q, there is a commutative diagram

$$\begin{tikzcd} & & a'' \arrow[d, "\alpha'"] \\ \xi \arrow[r] & a_{q+1} \arrow[r] \arrow[ur] & a_q, \end{tikzcd}$$

which supplies the required lift.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_0 = e$$

be an infinite sequence of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1}, \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective}, \\ \overline{\Omega}_{a_i} \longrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i \gg 0. \end{cases}$$

The preceding proof shows that $\varprojlim_i a_i$ is a versal formal object of $\underline{A}$.

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. If $\underline{A}$ satisfies 1.11(i) and (ii), then so does $[\underline{A}]$; moreover, $\xi \in \mathrm{ob}(\widehat{\underline{A}})$ is a minimally versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$, by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ. Any invariant of $R = P(\xi)$, a complete local Λ-algebra of formally finite type over Λ, is automatically an invariant of $\underline{A}$ over $\underline{C}_\Lambda$. Thus, for example, we say that $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ if R is (formally) smooth over Λ, that is, a formal power-series algebra over Λ; and we say that $\underline{A}$ is irreducible over $\underline{C}_\Lambda$ if $\mathrm{Spec}(R)$ is irreducible.

In the next section we introduce smoothness for a functor $\underline{A} \longrightarrow \underline{B}$ over $\underline{C}_\Lambda$, where $\underline{A}$ and $\underline{B}$ are cofibered categories in groupoids over $\underline{C}_\Lambda$. It follows from 2.4 that $\underline{A}$ being (formally) smooth over $\underline{C}_\Lambda$ is equivalent to the functor $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ being smooth, so no ambiguity arises. We define

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim(k \otimes_\Lambda R).$$

For this, let $\underline{C}_k$ be the category of artinian local k-algebras with residue field k. The functor $\underline{C}_k \longrightarrow \underline{C}_\Lambda$ is fully faithful, and its left adjoint is

$$\underline{C}_\Lambda \longrightarrow \underline{C}_k, \qquad R \longmapsto k \otimes_\Lambda R.$$

Let $\underline{A}|_{\underline{C}_k}$ be the pullback of $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ along $\underline{C}_k \longrightarrow \underline{C}_\Lambda$. If ξ is a minimally versal formal object for $\underline{A}$ over $\underline{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is minimally versal for $\underline{A}|_{\underline{C}_k}$ over $\underline{C}_k$; the arrow $\xi \longrightarrow k \otimes_\Lambda \xi$ is cocartesian over $R \longrightarrow k \otimes_\Lambda R$. Consequently,

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim_{\underline{C}_k}(\underline{A}|_{\underline{C}_k}).$$

1.15. In the remainder of this paragraph, we briefly explain how to modify the preceding theory to allow residue-field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ-algebra. Assume that:

(a) the natural map $\Lambda \longrightarrow K$ is a local homomorphism;

(b) the residue field K' of K is an extension of finite type of the residue field k of Λ.

The most interesting case is $K = K'$ with K a finite inseparable extension of k.

Denote by $\widehat{C}_{\Lambda,K}$ the category whose objects are complete noetherian local rings R equipped with a Λ-algebra structure and a Λ-epimorphism $s : R \longrightarrow K$. Let $C_{\Lambda,K}$ be its full subcategory consisting of pairs (R, s) for which $\ker(s)$ has finite length as an R-module. The category $\widehat{C}_{\Lambda,K}$ identifies with a full subcategory of $\mathrm{Pro}(C_{\Lambda,K})$. If K is artinian, then every object of $C_{\Lambda,K}$ is artinian.

For a finite-dimensional vector space V over K', write $D_V(K)$ for the algebra of dual numbers over K defined by V. Its elements have the form $x + v\varepsilon$, where $x \in K$, $v \in V$, and $\varepsilon^2 = 0$, with multiplication

$$(x + v\varepsilon)(x' + v'\varepsilon) = xx' + (xv' + x'v)\varepsilon.$$

The map

$$s : D_V(K) \longrightarrow K, \qquad s(x + v\varepsilon) = x,$$

makes $D_V(K)$ an object of $C_{\Lambda,K}$. The role previously played by $k[\varepsilon]$ is now played by $D_{K'}(K)$.

As in 1.8, every cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ extends naturally to a cofibered category $\widehat{\underline{A}}$ over $\widehat{C}_{\Lambda,K}$. A map in $\underline{A}$ or $\widehat{\underline{A}}$ is called *surjective* when the corresponding algebra homomorphism in $C_{\Lambda,K}$ or $\widehat{C}_{\Lambda,K}$ is surjective.

We assume that $\underline{A}(K) = \{e\}$.

Definition 1.16. A cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ is called semi-homogeneous if the following properties hold:

(I) *Every diagram of solid arrows*

$$\begin{tikzcd} a'' \arrow[r, dashed] \arrow[d, dashed] & a' \arrow[d] \\ a_1 \arrow[r] & a \end{tikzcd}$$

in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(II) *For every R, the natural map*

$$[\underline{A}(R \times_K D_{K'}(K))] \longrightarrow [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$.

1.17. First restrict $\underline{A}$ to the full subcategory of $C_{\Lambda,K}$ whose objects are the $D_V(K)$. Put

$$\Omega = \Omega^1_{K/\Lambda} \otimes_K K', \qquad d : K \longrightarrow \Omega.$$

A map $f : D_V(K) \longrightarrow D_W(K)$ can be written

$$f(x + v\varepsilon) = x + \big(u(v) + D(dx)\big)\varepsilon,$$

with $u \in \mathrm{Hom}_{K'}(V, W)$ and $D \in \mathrm{Hom}_{K'}(\Omega, W)$. Identifying f with (u, D), composition is given by

$$(u, D) \circ (u', D') = (uu', D + uD').$$

Condition (II) implies:

(II$'$) the functor from K'-vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 1.18. Assume that condition (II$'$) is satisfied by $\underline{A}$. Then:

(1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector-space structure such that, functorially in V,

$$[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))].$$

(2) There is a linear map

$$\gamma : \mathrm{Hom}(\Omega, K') \longrightarrow [\underline{A}(D_{K'}(K))]$$

such that, for every $D \in \mathrm{Hom}(\Omega, K')$, the map

$$[\underline{A}((1, D))] : [\underline{A}(D_{K'}(K))] \longrightarrow [\underline{A}(D_{K'}(K))]$$

is $x \longmapsto x + \gamma(D)$.

(3) For every $(u, D) : D_V(K) \longrightarrow D_W(K)$, the induced map

$$(u, D) : [\underline{A}(D_V(K))] \longrightarrow [\underline{A}(D_W(K))],$$

that is,

$$(u,D) : V \otimes [\underline{A}(D_{K'}(K))] \longrightarrow W \otimes [\underline{A}(D_{K'}(K))],$$

is given by

$$x \longmapsto u(x) + \gamma(D).$$

Introduce the notation

$$\begin{aligned} F(V) &= [\underline{A}(D_V(K))], \\ A &= F(K'), \\ F(u,D) : F(V) &\longrightarrow F(W) \qquad \text{for the map induced by } (u,D) : D_V(K) \longrightarrow D_W(K), \\ F(D) : F(V) &\longrightarrow F(V), \qquad F(D) = F(1,D). \end{aligned}$$

By (II$'$), and because K' is a K'-vector-space object in the category of K'-vector spaces, $F(K')$ is a K'-vector-space object in the category of sets, hence a K'-vector space. This proves (1). We henceforth identify $F(V)$ with $V \otimes A$.

The identity

$$(u,D) = (1,D) \circ (u,0)$$

gives

$$F(u,D) = F(D) \circ (u \otimes 1_A).$$

The functoriality of F amounts to the identities

$$\begin{aligned} F(0) &= \mathrm{Id}, \\ F(D) \circ F(D') &= F(D + D'), \\ (u \otimes 1) \circ F(D) &= F(uD) \circ (u \otimes 1_A). \end{aligned}$$

For any $D : \Omega \longrightarrow V$ and any $n \in \mathbb{N}$, let

$$i : V \hookrightarrow V \oplus K'^{,n}, \qquad p : V \oplus K'^{,n} \longrightarrow K'^{,n}$$

be the canonical inclusion and projection. Then $piD = 0$, and the following diagram is commutative:

$$\begin{tikzcd} V \otimes A \arrow[r] \arrow[d, "F(D)"] & (V \oplus K'^{,n}) \otimes A \arrow[r] \arrow[d, "F(i,D)"] & K'^{,n} \otimes A \arrow[d, equal] \\ V \otimes A \arrow[r] & (V \oplus K'^{,n}) \otimes A \arrow[r] & K'^{,n} \otimes A. \end{tikzcd}$$

Let $x : K'^{,n} \longrightarrow V$ be a map, put $x_i = x(e_i)$, and let (a_i) be a family of elements of A. We have

$$(1_V + x) \circ i \circ D = D.$$

Consequently, if

$$F(i,D)\Big(\sum_i e_i a_i\Big) = \sum_i e_i a_i + a, \qquad a \in V \otimes A,$$

then

$$F(D)\Big(\sum_i x_i a_i\Big) = \sum_i x_i a_i + a,$$

and a is independent of the x_i. Hence

$$\begin{aligned} F(D)(y) &= y + G(D), \\ G(D + D') &= G(D) + G(D'), \\ uG(D) &= G(uD) \qquad \text{for every } u : V \longrightarrow W. \end{aligned}$$

Every such G is given, for some $\gamma \in \Omega \otimes A$, by

$$G(D) = (D \otimes 1_A)(\gamma).$$

This proves (2) and (3).

Definition 1.19. (1) An object $\xi \in \widehat{\underline{A}}$ is called a versal formal object if every surjective map $\alpha : a' \longrightarrow a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \longrightarrow \mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

(2) A versal formal object $\xi \in \widehat{\underline{A}}$, with image (R, s) in $\widehat{C}_{\Lambda,K}$, is called minimal if the following condition holds. For any two maps

$$\varphi_1, \varphi_2 : R \rightrightarrows D_{K'}(K),$$

if $\varphi_1(\xi)$ is isomorphic to $\varphi_2(\xi)$, then there is a derivation $D : K \longrightarrow K'$ such that

$$(1, D)(\varphi_1) = \varphi_2, \qquad \text{that is,} \qquad \varphi_2 - \varphi_1 = Ds.$$

Theorem 1.20. If $\underline{A}$ is semi-homogeneous and

$$\dim_{K'}[\underline{A}(D_{K'}(K))] < \infty,$$

then $\underline{A}$ admits a minimally versal formal object, and any two such objects are isomorphic.

With the notation of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$ complementary to the image of γ, and let H be the dual of H^*. Set

$$R_0 = D_H(K).$$

Choose $\xi_0 \in \underline{A}(R_0)$ whose class, under the identifications

$$[\underline{A}(R_0)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}\big(H^*, [\underline{A}(D_{K'}(K))]\big),$$

is the inclusion $H^* \hookrightarrow [\underline{A}(D_{K'}(K))]$. Then:

(1) for every V, the maps

$$\mathrm{Hom}_{C_{\Lambda,K}}\big(D_H(K), D_V(K)\big) \longrightarrow [\underline{A}(D_V(K))]$$

are surjective;

(2) the minimality condition of 1.19(2) holds for (R_0, ξ_0).

Choose $S \in \widehat{C}_{\Lambda,K}$ formally smooth over Λ, together with an epimorphism $t : S \longrightarrow R_0$, such that

$$\Omega_{R_0/\Lambda} \otimes K' \xrightarrow{\sim} \Omega_{S/\Lambda} \otimes K'.$$

If M is the maximal ideal of S, define inductively, for $n \geq 1$,

$$R_n = S/I_n, \qquad \xi_n \in \mathrm{Ob}(\underline{A}(R_n)),$$

by the following conditions:

(a) I_{n+1} is the intersection of the ideals I such that

$$I_n M \subset I \subset I_n$$

and ξ_n extends to an object $\xi \in \underline{A}(S/I)$;

(b) ξ_{n+1} is an extension of ξ_n.

If

$$R = \varprojlim_n R_n, \qquad \xi = \varprojlim_n \xi_n,$$

then (R, ξ) is a minimally versal formal object, and it is unique up to isomorphism.

2. Prorepresentable cofibered groupoids

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ. In general, ξ alone does not determine $\underline{A}$ up to $\underline{C}_\Lambda$-equivalence. One must also know when two maps

$$R \underset{g}{\overset{f}{\rightrightarrows}} S$$

give rise to a $\underline{C}_\Lambda$-isomorphism $\xi(f) \longrightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12, we now consider the problem of recovering $\underline{A}$, up to $\underline{C}_\Lambda$-equivalence. We obtain a satisfactory answer for a broad class of cofibered groupoids.

Many cofibered groupoids that arise in practice satisfy a property slightly stronger than semi-homogeneity; it will be defined explicitly in 2.5. To investigate them, it is convenient to recast the preceding discussion in terms of "smooth functors." All the notation and terminology of the preceding section remain in force throughout this section.

Definition 2.1. Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that φ is smooth if every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

$$(*) \qquad \begin{tikzcd} \varphi(a') \arrow[r] \arrow[dr, "\varphi(\alpha)"'] & b' \arrow[d, "\beta"] \\ & \varphi(a) \end{tikzcd}$$

for some $\alpha : a' \longrightarrow a$ in $\underline{A}$. We say that φ is minimally smooth if it is smooth and

$$\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \longrightarrow [\underline{B}(k[\varepsilon])]$$

is bijective.

Remark 2.2(a). Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. The following conditions are equivalent:

(i) φ is smooth;

(ii) every surjective map $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a diagram $(*)$ in which $\varphi(a') \longrightarrow b'$ is a $\underline{C}_\Lambda$-isomorphism;

(iii) every small prolongation $\beta : b' \longrightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram $(*)$.

Remark 2.2(b). For every cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor

$$\underline{A} \longrightarrow [\underline{A}]$$

is minimally smooth.

Remark 2.2(c). For each $R \in \widehat{\underline{C}}_\Lambda$, let

$$h_R : \underline{C}_\Lambda \longrightarrow (\mathrm{Ens})$$

be the functor defined by

$$h_R(S) = \mathrm{Hom}_{\Lambda\text{-alg}}(R, S) \qquad (S \in \underline{C}_\Lambda).$$

It determines a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$. A map $R \longrightarrow S$ in $\widehat{\underline{C}}_\Lambda$ induces a functor

$$\underline{h}_S \longrightarrow \underline{h}_R.$$

This functor is smooth if and only if S is a formal power-series ring over R.

Remark 2.2(d). Given $R \in \widehat{\underline{C}}_\Lambda$, denote by $(R, \underline{C}_\Lambda)$ the category of arrows in $\widehat{\underline{C}}_\Lambda$ with source R and target in $\underline{C}_\Lambda$. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Given $\xi \in \mathrm{Ob}\,\widehat{\underline{A}}$ with $P(\xi) = R$, let

$$\underline{A}' \longrightarrow (R, \underline{C}_\Lambda)$$

be the pullback of $\underline{A} \longrightarrow \underline{C}_\Lambda$ along the canonical functor $(R, \underline{C}_\Lambda) \longrightarrow \underline{C}_\Lambda$. Choose a section $\xi^\#$ such that $\xi^\#(1_R) = \xi$. Thus, for every map $f : R \longrightarrow S$, we choose an f-map $\xi \longrightarrow \xi(f)$.

We obtain a functor

$$\xi^\# : \underline{h}_R \longrightarrow \underline{A}$$

over $\underline{C}_\Lambda$ by sending $(R \xrightarrow{f} S)$ to $\xi(f)$. Up to natural equivalence, this functor does not depend on the choice of $\xi^\#$.

The assertion that $\xi^\# : \underline{h}_R \longrightarrow \underline{A}$ is (minimally) smooth means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed to a commutative diagram

$$\begin{tikzcd} \xi(f') \arrow[r] \arrow[dr, "\xi()"'] & a' \arrow[d, "\alpha"] \\ & \xi(f) \end{tikzcd}$$

(and, in the minimal case, that $\underline{h}_R(k[\varepsilon]) \longrightarrow \underline{A}(k[\varepsilon])$ is bijective). This means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\Lambda$.

Thus the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor

$$\underline{h}_R \longrightarrow \underline{A}$$

over $\underline{C}_\Lambda$ for some $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$.

The following general properties of smoothness are immediate.

Proposition 2.3. Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$.

(a) If φ and ψ are both smooth, then so is $\psi \circ \varphi$.

(b) If $\psi \circ \varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a versal formal object ξ. By 2.2(d), this means that

$$\xi : \underline{h}_R \longrightarrow \underline{A}$$

is smooth, where $R = P(\xi)$. Therefore the following assertions are equivalent:

— $\underline{A}$ is formally smooth over $\underline{C}_\Lambda$ (cf. 1.13), that is, R is a formal power-series algebra over Λ;

— the composite functor

$$\underline{h}_R \xrightarrow{\xi} \underline{A} \xrightarrow{P} \underline{C}_\Lambda \ (= \underline{h}_\Lambda)$$

is smooth;

— $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ is smooth.

The last equivalence follows from Proposition 2.3(b).

We now consider a stronger property than semi-homogeneity.

Definition 2.5. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called homogeneous if every diagram

$$\begin{tikzcd} & a' \arrow[d] \\ b' \arrow[r] & a \end{tikzcd}$$

in which $a' \longrightarrow a$ is surjective admits a fibre product $a' \times_a b'$ in $\underline{A}$ such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b').$$

Remark 2.6(a). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$(*) \qquad \begin{tikzcd} & a' \arrow[dl] \arrow[dr] & \\ a_1 \arrow[dr] & & a_2 \arrow[dl] \\ & a & \end{tikzcd} \qquad \text{in } \underline{A} \text{ above the diagram} \qquad \begin{tikzcd} & R_1 \times_R R_2 \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ R_1 \arrow[dr] & & R_2 \arrow[dl] \\ & R & \end{tikzcd}$$

in $\underline{C}_\Lambda$, where the π_i are the projections. If a fibre product $a_1 \times_a a_2$ exists in $\underline{A}$, there is a unique map

$$\varphi : a' \longrightarrow a_1 \times_a a_2$$

compatible with the projections. If

$$P(a_1 \times_a a_2) = R_1 \times_R R_2,$$

then $P(\varphi)$ is the identity map of $R_1 \times_R R_2$; hence φ is a $\underline{C}_\Lambda$-isomorphism, that is, $a' = a_1 \times_a a_2$. Thus, if $\underline{A}$ is homogeneous, every diagram $(*)$ defines a fibre product $a_1 \times_a a_2$ whenever $R_1 \longrightarrow R$ is surjective. It follows at once

that every homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

Remark 2.6(b). Consider a diagram of categories and functors

$$\begin{tikzcd} X \arrow[dr, "\varphi"'] & & Y \arrow[dl, "\psi"] \\ & Z & \end{tikzcd}$$

The ordinary fibre product $X \times_Z Y$ is the category whose objects are pairs (x, y) satisfying $\varphi(x) = \psi(y)$ and whose maps $(x, y) \longrightarrow (x', y')$ are pairs

$$\alpha : x \longrightarrow x', \qquad \beta : y \longrightarrow y'$$

such that $\varphi(\alpha) = \psi(\beta)$. The fibre product $X \times^2_Z Y$ in the sense of 2-categories is the category whose objects are triples $(x, y; \gamma)$, where

$$\varphi(x) \xrightarrow{\gamma} \psi(y)$$

is an isomorphism. A map $(x, y; \gamma) \longrightarrow (x', y'; \gamma')$ is a pair $\alpha : x \longrightarrow x'$ and $\beta : y \longrightarrow y'$ for which the diagram

$$\begin{tikzcd} \varphi(x) \arrow[r, "\varphi(\alpha)"] \arrow[d, "\gamma"'] & \varphi(x') \arrow[d, "\gamma'"] \\ \psi(y) \arrow[r, "\psi(\beta)"'] & \psi(y') \end{tikzcd}$$

commutes.

The notions of a (semi-)homogeneous cofibered groupoid and of a smooth functor can be rephrased in terms of fibre products of categories in this 2-categorical sense. Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category, and consider $R_1 \times_R R_2$ in $\underline{C}_\Lambda$. Choose direct-image functors

$$(\pi_i)_* : \underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_i), \qquad i = 1, 2.$$

They induce a functor

$$\underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_1) \times^2_{\underline{A}(R)} \underline{A}(R_2).$$

Up to canonical natural equivalence, it is independent of the chosen direct-image functors. The groupoid $\underline{A}$ is homogeneous if and only if this functor is an equivalence whenever $R_1 \longrightarrow R$ is surjective; it is semi-homogeneous if and only if

$$[\underline{A}(R_1 \times_R R_2)] \longrightarrow [\underline{A}(R_1)] \times_{[\underline{A}(R)]} [\underline{A}(R_2)]$$

is surjective whenever $R_1 \longrightarrow R$ is surjective, and is bijective whenever $R = k$ and $R_1 = k[\varepsilon]$.

Now let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For every map $R' \longrightarrow R$ in $\underline{C}_\Lambda$, φ induces, up to canonical natural equivalence, a functor

$$\underline{A}(R') \longrightarrow \underline{A}(R) \times^2_{\underline{B}(R)} \underline{B}(R').$$

The functor φ is smooth if and only if this functor is surjective on isomorphism classes of objects whenever $R' \longrightarrow R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is homogeneous, then for any finite-dimensional k-vector spaces V and W the functor

$$\underline{A}(k[V \times W]) \longrightarrow \underline{A}(k[V]) \times \underline{A}(k[W])$$

is an equivalence of categories. Recall our convention $\underline{A}(k) = \{e\}$. In particular,

$$\mathrm{Aut}_{\underline{A}(k[V\times W])}(1_{V\times W}) \xrightarrow{\sim} \mathrm{Aut}_{\underline{A}(k[V])}(1_V) \times \mathrm{Aut}_{\underline{A}(k[W])}(1_W),$$

where 1_V is the direct image of e under the unique map $k \longrightarrow k[V]$ in $\underline{C}_\Lambda$. Consequently, if $\underline{A}$ is homogeneous, both $\mathrm{Aut}(1_\varepsilon)$ and $[\underline{A}(k[\varepsilon])]$ carry canonical k-vector-space structures.

Remark 2.6(d). For every $R \in \widehat{\underline{C}}_\Lambda$, the cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. Homogeneity of $\underline{A}$, however, does not in general imply homogeneity of $[\underline{A}]$.

Proposition 2.7. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following conditions are equivalent.

(i) $[\underline{A}]$ is homogeneous; that is,

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is bijective for every small extension $R' \longrightarrow R$.

(ii) For every small prolongation $a' \longrightarrow a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{\underline{A}(R')}(a') \longrightarrow \mathrm{Aut}_{\underline{A}(R)}(a)$$

is surjective, where $R' = P(a')$ and $R = P(a)$.

Proof. (i)⇒(ii). Let $f : R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $a' \longrightarrow a$ be an f-map in $\underline{A}$. Given $\tau \in \mathrm{Aut}_{\underline{A}(R)}(a)$, consider $a' \times_a a'$ and $a' \times_a a'_\tau$, where $a'_\tau \longrightarrow a$ is the composite

$$a' \longrightarrow a \xrightarrow{\tau} a.$$

Since

$$[\underline{A}(R' \times_R R')] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$$

is bijective, there is an isomorphism

$$a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau.$$

Projection onto the second factor then gives an isomorphism $\tau' : a' \longrightarrow a'$ lifting τ.

(ii)⇒(i). Let a''_i $(i = 1, 2)$ be objects of $\underline{A}(R' \times_R S)$ with the same image under

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)].$$

We obtain a diagram of solid arrows

$$
\begin{tikzcd}
 & & a_2'' \arrow[dll] \arrow[drr] & & \\
 & & a_1'' \arrow[u, dashed] \arrow[dl] \arrow[dr] & & \\
a_2' \arrow[drr] & a_1' \arrow[l, "\sim"'] \arrow[dr] & & b_1 \arrow[r, "\tau"] \arrow[dl] & b_2 \arrow[dll] \\
 & & a_1 \arrow[d, "\bar{\tau}"] & & \\
 & & a_2 & &
\end{tikzcd}
\qquad \text{above} \qquad
\begin{tikzcd}
 & R' \times_R S \arrow[dl] \arrow[dr] & \\
R' \arrow[dr] & & S \arrow[dl] \\
 & R &
\end{tikzcd}
$$

where $\bar{\tau} : a_1 \longrightarrow a_2$ is the isomorphism induced by τ, so the lower-right square commutes. Under condition (ii), the given isomorphism $a_1' \xrightarrow{\sim} a_2'$ can be replaced so that the left square also commutes. Since

$$a_i'' = a_i' \times_{a_i} b_i \qquad (i = 1, 2)$$

by Remark 2.6(a), we obtain an isomorphism $a_1'' \xrightarrow{\sim} a_2''$. Hence

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is injective, and therefore bijective because $\underline{A}$ is homogeneous by hypothesis. □

Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $f : R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$. There is a canonical map

$$\widetilde{f} : R' \times_R R' \longrightarrow k[\operatorname{Ker} f], \qquad (r_1', r_2') \longmapsto r_1'(0) + (r_1' - r_2'),$$

and

$$1 \times \widetilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\operatorname{Ker} f]$$

is an isomorphism. Given two f-maps $a_i \longrightarrow a$ $(i = 1, 2)$ in $\underline{A}$, the fibre product $a_1 \times_a a_2$ is an object over $R' \times_R R'$. We may therefore form

$$\tau(a_1, a_2) = (\widetilde{f})_*(a_1 \times_a a_2) \in \underline{A}(k[\operatorname{Ker} f]).$$

Since $1 \times \widetilde{f}$ is an isomorphism, there is an isomorphism

$$a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2)$$

compatible with projection onto the first factor. Statement 1.5(2) now takes the following more definite form.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let

$$a_i \xrightarrow{\alpha_i} a \qquad (i = 1, 2)$$

be f-maps in $\underline{A}$.

(1) There exists $\rho : a_1 \longrightarrow a_2$ in $\underline{A}$ with $\alpha_2 \circ \rho = \alpha_1$ if and only if there is an arrow $a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$.

(2) There exists such a ρ with $P(\rho) = \mathrm{id}_{R'}$ if and only if

$$\tau(a_1, a_2) = 0 \quad \text{in} \quad [\underline{A}(k[\operatorname{Ker} f])].$$

Proof. We prove both assertions simultaneously. The existence of $\rho : a_1 \longrightarrow a_2$ satisfying $\alpha_2 \circ \rho = \alpha_1$ (and $P(\rho) = \mathrm{id}_{R'}$) is equivalent successively to the existence of

$$h : a_1 \longrightarrow a_1 \times_a a_2, \qquad \mathrm{pr}_1 \circ h = \mathrm{id}_{a_1},$$

with $P(h)$ the diagonal map; to the existence of

$$g : a_1 \longrightarrow a_1 \times \tau(a_1, a_2), \qquad \mathrm{pr}_1 \circ g = \mathrm{id}_{a_1},$$

with

$$P(g) : R' \longrightarrow R' \times_k k[\operatorname{Ker} f], \qquad r' \longmapsto (r', r'(0));$$

and to the existence of an arrow $\gamma : a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$, with

$$P(\gamma) : R' \longrightarrow k[\operatorname{Ker} f], \qquad r' \longmapsto r'(0).$$

Such an arrow factors through $R' \longrightarrow k \longrightarrow k[\operatorname{Ker} f]$ if and only if $\tau(a_1, a_2) = 0$. $\square$

Proposition 2.9. (a) Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that

(i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous;

(ii) the map

$$\psi(k[\varepsilon]) : [\underline{B}(k[\varepsilon])] \longrightarrow [\underline{C}(k[\varepsilon])]$$

is injective.

Then $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth.

(b) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then

$$\widehat{\varphi} : \widehat{\underline{A}} \longrightarrow \widehat{\underline{B}}$$

is smooth. Explicitly, for every surjective map $b' \xrightarrow{\beta} \widehat{\varphi}(a)$ in $\widehat{\underline{B}}$, there are a map $\alpha : a' \longrightarrow a$ in $\widehat{\underline{A}}$ and a $\widehat{\underline{C}}_\Lambda$-isomorphism

$$\rho : \widehat{\varphi}(a') \xrightarrow{\sim} b'$$

such that $\beta \circ \rho = \varphi(\alpha)$.

Proof. (a) Suppose $\psi \circ \varphi$ is smooth. If φ is smooth, then ψ is smooth by Proposition 2.3(b), so it remains to prove that φ is smooth. Let $b' \xrightarrow{\beta} \varphi(a)$ be a small prolongation in $\underline{B}$. Since $\psi \circ \varphi$ is smooth, there are an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism

$$\rho : \psi(b') \longrightarrow \psi\varphi(a')$$

for which

$$\begin{tikzcd} \psi(b') \arrow[rr, "\rho"] \arrow[dr, "\psi(\beta)"'] & & \psi\varphi(a') \arrow[dl, "\psi\varphi(\alpha)"] \\ & \psi\varphi(a) & \end{tikzcd}$$

commutes. Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

$$\begin{tikzcd} & b'' \arrow[dl] \arrow[dr] & \\ b' \arrow[dr, "\beta"'] & & \varphi(a') \arrow[dl, "\varphi(\alpha)"] \\ & \varphi(a) & \end{tikzcd} \quad \text{above} \quad \begin{tikzcd} & R' \times_R R' \arrow[dl] \arrow[dr] & \\ R' \arrow[dr] & & R' \arrow[dl] \\ & R & \end{tikzcd}$$

where $R = P(a)$ and $R' = P(b') = P(a')$. Homogeneity of $\underline{C}$ gives

$$\psi(b'') = \psi(b') \times_{\psi\varphi(a)} \psi\varphi(a')$$

by Remark 2.6(a). Lemma 2.8(2) then gives

$$0 = \tau(\psi\varphi(a'), \psi(b')) = (\widetilde{f})_* \psi(b'').$$

By hypothesis (ii), it follows that $(\widetilde{f})_*(b'') = 0$. Statement 1.5(2) now provides a map $\gamma : b' \longrightarrow \varphi(a')$ satisfying $\beta = \varphi(\alpha) \circ \gamma$. Thus φ is smooth, and then so is ψ. The converse follows from Proposition 2.3(a).

(b) Let a surjective map $\beta : b' \longrightarrow \widehat{\varphi}(a)$ in $\underline{\widehat{B}}$ be given. It is represented by a sequence of commutative diagrams

$$\begin{tikzcd}
\cdots \arrow[r, dashed] & b'_3 \arrow[r, "\beta'_2"] \arrow[d, "\beta_3"] & b'_2 \arrow[r, "\beta'_1"] \arrow[d, "\beta_2"] & b'_1 \arrow[d, "\beta_1"] \\
\cdots \arrow[r, dashed] & \varphi(a_3) \arrow[r, "\varphi(\alpha_2)"'] & \varphi(a_2) \arrow[r, "\varphi(\alpha_1)"'] & \varphi(a_1)
\end{tikzcd}$$

in $\underline{B}$, where every α_v, β_v, and β'_v is surjective and $\beta = \varprojlim_v \beta_v$. It is enough to show inductively that every diagram of solid arrows below, with α_n, β_n, β_{n+1}, and β'_n surjective, can be completed by the dotted arrows:

$$\begin{tikzcd}
\varphi(a'_{n+1}) \arrow[rrr, dashed, "\varphi()"] \arrow[dr, dashed, leftrightarrow] \arrow[ddr, dashed, "\varphi()"'] & & & \varphi(a'_n) \arrow[dl] \arrow[ddl, "\varphi(\alpha'_n)"] \\
& b'_{n+1} \arrow[r, "\beta'_n"] \arrow[d, "\beta_{n+1}"] & b'_n \arrow[d, "\beta_n"] & \\
& \varphi(a_{n+1}) \arrow[r, "\varphi(\alpha_n)"'] & \varphi(a_n) &
\end{tikzcd}$$

By hypothesis $\underline{A}$ is semi-homogeneous and $\alpha_n : a'_n \longrightarrow a_n$ is surjective. Hence there is a commutative diagram

$$\begin{tikzcd}
& a' \arrow[dl, "\gamma"'] \arrow[dr, "\gamma'"] & \\
a_{n+1} \arrow[dr, "\alpha_n"'] & & a'_n \arrow[dl, "\alpha'_n"] \\
& a_n &
\end{tikzcd}
\quad \text{in } \underline{A} \text{ over} \quad
\begin{tikzcd}
& R_{n+1} \times_{R_n} R'_n \arrow[dl] \arrow[dr] & \\
R_{n+1} \arrow[dr] & & R'_n \arrow[dl] \\
& R_n &
\end{tikzcd}
\quad \text{in } \underline{C}_\Lambda$$

Homogeneity of $\underline{B}$ implies that

$$\begin{tikzcd}
& \varphi(a') \arrow[dl, "\varphi(\gamma)"'] \arrow[dr, "\varphi(\gamma')"] & \\
\varphi(a_{n+1}) \arrow[dr, "\varphi(\alpha_n)"'] & & \varphi(a'_n) \arrow[dl, "\varphi(\alpha'_n)"] \\
& \varphi(a_n) &
\end{tikzcd}$$

is a fibre-product diagram in $\underline{B}$. Thus there is a unique map $\delta : b'_{n+1} \longrightarrow \varphi(a')$ with the required commutativity. Since β_{n+1} and β'_n are surjective, so is δ. Smoothness of φ therefore gives a map $a'_{n+1} \xrightarrow{\alpha'} a'$ in $\underline{A}$ and a $\underline{C}_\Lambda$-isomorphism

$$\rho : \varphi(a'_{n+1}) \xrightarrow{\sim} b'_{n+1}$$

such that $\delta \circ \rho = \varphi(\alpha')$. We obtain the commutative diagram

$$\begin{tikzcd}
\varphi(a'_{n+1}) \arrow[rrr, "\varphi(\gamma' \circ \alpha')"] \arrow[dr, "\rho"] \arrow[ddr, "\varphi(\gamma \circ \alpha')"'] & & & \varphi(a'_n) \arrow[dl, "\sim"'] \arrow[ddl, "\varphi(\alpha'_n)"] \\
& b'_{n+1} \arrow[r] \arrow[d] & b'_n \arrow[d] & \\
& \varphi(a_{n+1}) \arrow[r] & \varphi(a_n) &
\end{tikzcd}$$

which completes the induction and proves (b). $\square$

Corollary 2.10. (a) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. Let ξ be a versal formal object of $\underline{A}$, and let η be a minimally versal formal object of $\underline{B}$. If $\underline{B}$ is homogeneous, then for every map $\beta : \eta \longrightarrow \varphi(\xi)$ in $\widehat{\underline{B}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\beta) : P(\eta) \longrightarrow P(\varphi(\xi)) = P(\xi).$$

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let ξ and η be versal formal objects of $\underline{A}$. If η is minimal, then for every map $\alpha : \eta \longrightarrow \xi$ in $\widehat{\underline{A}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\alpha) : P(\eta) \longrightarrow P(\xi).$$

Proof. (a) A map $\beta : \eta \longrightarrow \varphi(\xi)$ induces a commutative diagram

$$\begin{tikzcd}
\underline{h}_{P(\xi)} \arrow[r, "\widetilde{\beta}"] \arrow[d, "\widetilde{\xi}"'] & \underline{h}_{P(\eta)} \arrow[d, "\widetilde{\eta}"] \\
\underline{A} \arrow[r, "\varphi"'] & \underline{B}.
\end{tikzcd}$$

Since $\widetilde{\xi}$ and φ are smooth, the composite $\widetilde{\eta} \circ \widetilde{\beta} = \varphi \circ \widetilde{\xi}$ is smooth. On the other hand,

$$\widetilde{\eta}(k[\varepsilon]) : \underline{h}_{P(\eta)}(k[\varepsilon]) \longrightarrow [\underline{B}(k[\varepsilon])]$$

is an isomorphism. Proposition 2.9(a) therefore shows that $\widetilde{\beta}$ is smooth, which means precisely that $P(\xi)$ is a formal power-series ring over $P(\eta)$.

(b) Apply (a) with $\underline{A} = \underline{B}$. $\square$ We conclude by considering the "representability" of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Every object R of $\underline{C}$ defines a functor

$$h_R : \underline{C} \longrightarrow (\text{Sets}), \qquad h_R(S) = \operatorname{Hom}_{\underline{C}}(R, S),$$

and hence a fully faithful functor

$$h : \underline{C}^{\text{op}} \longrightarrow \underline{\operatorname{Hom}}(\underline{C}, (\text{Sets})).$$

A *cocategory in* $\underline{C}$ is a diagram

$$\begin{tikzcd}
h_{R_1} \times_{h_{R_0}} h_{R_1} \arrow[r] & h_{R_1} & h_{R_0} \arrow[l, "\epsilon"'] \arrow[from=1-2, to=1-3, shift left=2, "s_1"] \arrow[from=1-2, to=1-3, shift right=2, "s_2"']
\end{tikzcd} \qquad \text{with } s_i \circ \epsilon = \text{id} \quad (i = 1, 2),$$

where the first arrow is denoted by c and

$$h_{R_1} \times_{h_{R_0}} h_{R_1} = (h_{R_1})_{s_1} \times_{h_{R_0}} {}_{s_2}(h_{R_1}),$$

such that

(i) c is associative and compatible with s_i for $i = 1, 2$;

(ii) both composites

$$h_{R_1} \xrightarrow{\sim} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\epsilon\times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1},$$

$$h_{R_1} \xrightarrow{\sim} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1\times\epsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}$$

are identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$, then for every $S \in \underline{C}$ we define a category $h_{(R_0,R_1,\ldots)}(S)$ by

$$\operatorname{ob} h_{(R_0,R_1,\ldots)}(S) = \operatorname{Hom}_{\underline{C}}(R_0, S), \qquad \operatorname{Fl} h_{(R_0,R_1,\ldots)}(S) = \operatorname{Hom}_{\underline{C}}(R_1, S),$$

with composition supplied by

$$c : h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1}.$$

The categories so obtained depend functorially on S and define a cofibered category $\underline{h}_{(R_0,R_1,\ldots)}$ over $\underline{C}$. For example, every object R of $\underline{C}$ defines a discrete cocategory

$$\underline{h}_R = \underline{h}_{(R,R,s_1,s_2,\epsilon,c)}$$

by taking s_1, s_2, ϵ, and c all to be identity maps. A homomorphism

$$(R_0, R_1, \ldots) \longrightarrow (R_0', R_1', \ldots)$$

of cocategories is a pair of maps of represented functors $h_{R_0} \longrightarrow h_{R_0'}$ and $h_{R_1} \longrightarrow h_{R_1'}$ whose induced map

$$h_{(R_0,R_1,\ldots)}(S) \longrightarrow h_{(R_0',R_1',\ldots)}(S)$$

is a functor for every $S \in \underline{C}$. Every cocategory $(R_0, R_1, \ldots)$ has, in particular, a homomorphism

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}.$$

A *cogroupoid in* $\underline{C}$ is a cocategory $(R_0, R_1, \ldots)$ for which $h_{(R_0,R_1,\ldots)}(S)$ is a groupoid for every $S \in \underline{C}$; equivalently, $\underline{h}_{(R_0,R_1,\ldots)}$ is a category cofibered in groupoids over $\underline{C}$.

In practice, cocategories and cogroupoids arise as follows. Let $P : \underline{F} \longrightarrow \underline{C}$ be a cofibered category, and fix an object ξ of $\underline{F}$. Put $R_0 = P(\xi)$, and let $(R_0, \underline{C})$ denote the category of arrows in $\underline{C}$ with source R_0. Let $P : \underline{F}' \longrightarrow (R_0, \underline{C})$ be the pullback of $\underline{F}$ along the canonical functor $(R_0, \underline{C}) \longrightarrow \underline{C}$. Choose a cocartesian section $\xi^\#$ of $\underline{F}'$ over $(R_0, \underline{C})$ such that $\xi^\#(\mathrm{id}_{R_0}) = \xi$. Thus, for every arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$, we choose a cocartesian f-map $\xi \longrightarrow \xi^\#(f)$.

Define

$$\operatorname{Hom}_{\xi^\#} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\text{Sets})$$

by

$$X = (f_1, f_2) \longmapsto \operatorname{Hom}_{\underline{F}(S)}(\xi^\#(f_1), \xi^\#(f_2)),$$

where S is the common target of f_1 and f_2. Up to canonical natural equivalence, this functor is independent of the choice of $\xi^\#$; henceforth we write simply ξ for $\xi^\#$.

Suppose the functor is represented by an object

$$X_0 = \left(R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_1 \right)$$

of $(R_0 \amalg R_0, \underline{C})$. Thus there is an R_0-map $\varphi : \xi(s_1) \longrightarrow \xi(s_2)$ in $\underline{F}$ such that

$$\widetilde{\varphi} : \mathrm{Hom}_{(R_0 \amalg R_0, \underline{C})}(X_0, X) \longrightarrow \mathrm{Hom}_{\xi}(X)$$

is bijective for every X in $(R_0 \amalg R_0, \underline{C})$. Then X_0 naturally defines a cocategory $(R_0, R_1, \ldots)$ in $\underline{C}$. Indeed, a triple of arrows

$$\begin{tikzcd} R_0 \arrow[r, shift left=2, "u"] \arrow[r, "v"] \arrow[r, shift right=2, "w"'] & S \end{tikzcd}$$

in $\underline{C}$ defines a composition

$$\mathrm{Hom}_{\underline{F}(S)}(\xi(v), \xi(w)) \times \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(v)) \longrightarrow \mathrm{Hom}_{\underline{F}(S)}(\xi(u), \xi(w)),$$

and hence a composition rule

$$h_{R_1}(S) \times_{h_{R_0}(S)} h_{R_1}(S) \longrightarrow h_{R_1}(S),$$

functorial in S. Consequently, Hom_{ξ} defines the cocategory $(R_0, R_1, \ldots)$ in $\underline{C}$ and the cofibered category $\underline{h}_{(R_0, R_1, \ldots)}$ over $\underline{C}$. Explicitly,

$$\mathrm{ob}\, \underline{h}_{(R_0, R_1, \ldots)} = \coprod_{S \in \mathrm{ob}(\underline{C})} \mathrm{Hom}_{\underline{C}}(R_0, S).$$

An arrow

$$(R_0 \xrightarrow{f} S) \longrightarrow (R_0 \xrightarrow{f'} S')$$

is a pair of arrows $S \xrightarrow{h} S'$ and $R_1 \xrightarrow{u} S'$ in $\underline{C}$ such that

$$u \circ s_1 = h \circ f, \qquad u \circ s_2 = f'.$$

The structural functor $P : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{C}$ sends $(R_0 \xrightarrow{f} S)$ to S.

The pair (ξ, φ) now defines a functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{F} \quad \text{over } \underline{C}$$

by

$$(R_0 \xrightarrow{f} S) \longmapsto \xi(f)$$

on objects and by

$$\left[(R_0 \xrightarrow{f} S) \xrightarrow{(h,u)} (R_0 \xrightarrow{f'} S')\right] \longmapsto \left[\xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f')\right]$$

on arrows. For every $S \in \underline{C}$, the functor

$$(\xi, \varphi)(S) : \underline{h}_{(R_0, R_1, \ldots)}(S) \longrightarrow \underline{F}(S)$$

is fully faithful by the representability of Hom_{ξ}. It follows that

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{F}$$

is a $\underline{C}$-equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$, that is, if every object a of $\underline{F}$ admits a cocartesian arrow $\xi \longrightarrow a$.

One may instead consider the functor

$$\mathrm{Isom}_{\xi} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\mathrm{Sets}),$$

given by

$$X = (f_1, f_2) \longmapsto \mathrm{Isom}_{\underline{F}(S)}(\xi(f_1), \xi(f_2)),$$

where S is the common target of f_1 and f_2. If this functor is represented by an object $(R_0 \rightrightarrows R_1)$ of $(R_0 \amalg R_0, \underline{C})$, then it defines a cogroupoid $(R_0, R_1, \ldots)$ in $\underline{C}$, together with a functor

$$(\xi, \varphi) : \underline{h}_{(R_0,R_1,\ldots)} \longrightarrow \underline{F}.$$

When $\underline{F}$ is a cofibered groupoid, this functor is a $\underline{C}$-equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$. Definition and Proposition 2.11. A cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$ is called smooth if either, and hence both, of the following equivalent conditions holds:

(i) R_1 is a formal power-series ring over R_0 via s_1 (and via s_2);

(ii) the functor

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$$

is smooth over $\underline{C}_\Lambda$.

A minimally smooth cogroupoid, also called a normalized smooth cogroupoid, in $\widehat{\underline{C}}_\Lambda$ is a smooth cogroupoid $(R_0, R_1, \ldots)$ for which $\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])$ is totally disconnected; equivalently,

$$s_1(k[\varepsilon]) = s_2(k[\varepsilon]).$$

Proof. Smoothness of $\epsilon : \underline{h}_{R_0} \to \underline{h}_{(R_0,R_1,\ldots)}$ means that, for every surjective map

$$(S', f') \xrightarrow{(h,u)} (S, f)$$

in $\underline{h}_{(R_0,R_1,\ldots)}$, there are a lift f'' of f to S' and an isomorphism

$$(S', f'') \xrightarrow{(1,u')} (S', f').$$

Equivalently, given $u \in h_{R_1}(S)$ and a lift f' of $s_1(u)$ to S', there is a lift u' of u to S' such that $s_1(u') = f'$, for every surjection $S' \to S$ in $\underline{C}_\Lambda$. This is precisely the smoothness of

$$s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$$

over $\underline{C}_\Lambda$, and that in turn says that R_1 is a formal power-series ring over R_0 via s_1. □

Proposition 2.12. Let $(R_0, R_1, \ldots)$ be a cogroupoid in $\widehat{\underline{C}}_\Lambda$.

(a) If $(R_0, R_1, \ldots)$ is smooth, then $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous.

(b) If $s_1(k[\varepsilon]) = s_2(k[\varepsilon])$, then the following conditions are equivalent:

(i) $(R_0, R_1, \ldots)$ is smooth;

(ii) $\underline{h}_{(R_0,R_1,\ldots)}$ is homogeneous;

(iii) $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous.

(c) If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0,S_1,\ldots)}$ for some minimally smooth cogroupoid S_*.

Proof. (a) Consider a diagram

$$\begin{tikzcd}
(S_1, f_1) \arrow[dr, "(h_1{,}u_1)"'] & & (S_2, f_2) \arrow[dl, "(h_2{,}u_2)"] \\
& (S, f) &
\end{tikzcd}$$

in $\underline{h}_{(R_0,R_1,\ldots)}$, where $h_2 : S_2 \longrightarrow S$ is surjective in $\underline{C}_\Lambda$. Since $s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$ is smooth, $u_1^{-1}u_2$ lifts to an element $v \in h_{R_1}(S_2)$ with $s_1(v) = f_2$. We have

$$h_2 \circ s_2(v) = h_1 \circ f_1,$$

so $f_1 \times s_2(v)$ is an element of $h_{R_0}(S_1 \times_S S_2)$ and gives a commutative diagram

$$\begin{tikzcd}
 & (S_1 \times_S S_2,, f_1 \times s_2(v)) \arrow[dl, "\pi_1"'] \arrow[dr, "(\pi_2, v^{-1})"] & \\
(S_1, f_1) \arrow[dr, "(h_1,u_1)"'] & & (S_2, f_2) \arrow[dl, "(h_2,u_2)"] \\
 & (S, f) &
\end{tikzcd}$$

We claim that its top object is the fibre product

$$(S_1, f_1) \times_{(S,f)} (S_2, f_2)$$

in $\underline{h}_{(R_0,R_1,\ldots)}$. Indeed, for every object (T, g), a map to the top object is equivalently a tuple $(h_1' \times h_2', w_1, w_2)$ satisfying

$$\begin{aligned}
h_1 w_1 &= h_2 w_2, & s_1(w_1) &= h_1' g, & s_2(w_1) &= f_1, \\
s_1(w_2) &= h_2' g, & s_2(w_2) &= s_2(v).
\end{aligned}$$

Putting $w_2' = v^{-1} w_2$ gives a bijection with the tuples $(h_1' \times h_2', w_1, w_2')$ satisfying

$$\begin{aligned}
u_2 h_2 w_2' &= u_1 h_1 w_1, & s_1(w_1) &= h_1' g, & s_2(w_1) &= f_1, \\
s_1(w_2') &= h_2' g, & s_2(w_2') &= f_2.
\end{aligned}$$

These are exactly the pairs of arrows from (T, g) to (S_1, f_1) and (S_2, f_2) whose composites to (S, f) coincide. Thus

$$\begin{aligned}
&\mathrm{Hom}((T, g), (S_1 \times_S S_2, f_1 \times s_2(v))) \\
&\qquad\qquad \simeq \mathrm{Hom}((T, g), (S_1, f_1)) \times_{\mathrm{Hom}((T,g),(S,f))} \mathrm{Hom}((T, g), (S_2, f_2)),
\end{aligned}$$

which proves (a).

(b) It remains only to prove (iii)⇒(i). If $\underline{h}_{(R_0,R_1,\ldots)}$ is semi-homogeneous, then

$$\dim[\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])] = \dim h_{R_0}(k[\varepsilon]) < \infty,$$

so it admits a minimally versal formal object. On the other hand, $(R_0, 1_{R_0})$ is a quasi-initial minimal object of $\widehat{\underline{h}}_{(R_0,R_1,\ldots)}$, because

$$h_{R_0}(k[\varepsilon]) \longrightarrow [\underline{h}_{(R_0,R_1,\ldots)}(k[\varepsilon])]$$

is bijective. Proposition 1.10(ii) therefore makes $(R_0, 1_{R_0})$ a minimally versal formal object of $\underline{h}_{(R_0,R_1,\ldots)}$. Equivalently,

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0,R_1,\ldots)}$$

is minimally smooth. This proves (i). (c) Let (S_0, ξ_0) be a minimally versal formal object of $\underline{h}_{(R_0,R_1,\ldots)}$ (Theorem 1.11). For any object

$$\xi : R_0 \longrightarrow T \qquad \text{of } \underline{h}_{(R_0,R_1,\ldots)}(T),$$

consider the functor

$$\mathrm{Isom}(\xi, \xi) : \underline{C}_\Lambda / (T \widehat{\otimes}_\Lambda T) \longrightarrow (\mathrm{Sets}).$$

It assigns to R the set of isomorphisms between the two images of ξ on R obtained from

$$R_0 \xrightarrow{\xi} T \rightrightarrows T \widehat{\otimes}_\Lambda T \longrightarrow R.$$

This functor is represented by

$$R_1 \otimes_{R_0 \widehat{\otimes}_\Lambda R_0} (T \widehat{\otimes}_\Lambda T).$$

Passing to the inverse limit shows that $\mathrm{Isom}(\xi_0, \xi_0)$ is prorepresented by an object S_1, and that

$$\underline{h}_{(R_0, R_1, \ldots)} \simeq \underline{h}_{(S_0, S_1, \ldots)}.$$

By (b), the cogroupoid S_* is minimally smooth. □

Definition 2.13. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called (minimally) prorepresentable if it is $\underline{C}_\Lambda$-equivalent to $\underline{h}_{(R_0, R_1, \ldots)}$ for some (normalized) cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$.

Proposition 2.14. If $(R_0, R_1, \ldots)$ is a normalized cogroupoid in $\widehat{\underline{C}}_\Lambda$, then every homomorphism

$$(\alpha_0, \alpha_1) : (R_0, R_1, \ldots) \longrightarrow (R_0, R_1, \ldots)$$

which induces a $\underline{C}_\Lambda$-equivalence

$$\underline{h}_{(\alpha_0, \alpha_1)} : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{h}_{(R_0, R_1, \ldots)}$$

is an isomorphism. In

particular, a normalized prorepresentation of a cofibered groupoid over $\underline{C}_\Lambda$, if it exists, is unique up to isomorphism.

Proof. Consider the commutative diagram

$$\begin{array}{ccc}
\underline{h}_{(R_0, R_1, \ldots)} & \xrightarrow{\underline{h}_{(\alpha_0, \alpha_1)}} & \underline{h}_{(R_0, R_1, \ldots)} \\
\uparrow \epsilon & & \uparrow \epsilon \\
\underline{h}_{R_0} & \xrightarrow[\underline{h}_{\alpha_0}]{} & \underline{h}_{R_0}
\end{array}$$

in which $\underline{h}_{(\alpha_0, \alpha_1)}$ is a $\underline{C}_\Lambda$-equivalence. Since $\underline{h}_{(R_0, R_1, \ldots)}(k[\varepsilon])$ is totally disconnected, both

$$[\underline{h}_{(\alpha_0, \alpha_1)}(k[\varepsilon])] \quad \text{and} \quad [\epsilon(k[\varepsilon])]$$

are bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$. Thus $\alpha_0 : R_0 \longrightarrow R_0$ is a surjective endomorphism and therefore an automorphism. It follows that α_1 is also an automorphism of R_1. □ Lemma 2.15. Let A and B be discrete groupoids over $\underline{C}_\Lambda$, and let $\varphi : A \longrightarrow B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\underline{C}_\Lambda$-equivalence.

Proof. We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for every $S \in \mathrm{ob}(\underline{C}_\Lambda)$. It is surjective for every S and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S. Let $S' \xrightarrow{f} S$ be a small extension in $\underline{C}_\Lambda$. Since A and B are homogeneous discrete groupoids, the canonical isomorphism

$$1 \times \widetilde{f} : S' \times_S S' \xrightarrow{\sim} S' \times_k k[\mathrm{Ker}\, f]$$

induces a commutative diagram

$$\begin{array}{ccc}
A(S') \times_{A(S)} A(S') & \xrightarrow{\sim} & A(S') \times A(k[\mathrm{Ker}\, f]) \\
\downarrow & & \downarrow \\
B(S') \times_{B(S)} B(S') & \xrightarrow[\sim]{} & B(S') \times B(k[\mathrm{Ker}\, f]).
\end{array}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets over $A(S)$ and $B(S)$, with structural groups identified by

$$A(k[\mathrm{Ker}\, f]) \xrightarrow{\sim} B(k[\mathrm{Ker}\, f]).$$

If $\varphi(S)$ is bijective, it follows that $\varphi(S')$ is injective, and hence bijective. This completes the induction. $\square$

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\Lambda$ such that

$$\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$$

is finite-dimensional. For every $\xi \in \mathrm{ob}(\widehat{\underline{A}})$, the functor

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R\widehat{\otimes}_\Lambda R} \longrightarrow (\mathrm{Sets})$$

is prorepresentable, where $R = P(\xi)$.

Proof. If Isom_ξ is homogeneous, Theorem 1.11 supplies a minimally smooth functor

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}_\xi}.$$

Lemma 2.15 makes Φ a $\widehat{\underline{C}}_{R\widehat{\otimes}_\Lambda R}$-equivalence, and hence Isom_ξ is prorepresentable. It remains to prove homogeneity. Consider a diagram

$$\begin{tikzcd} T_1 \arrow[dr] & & T_2 \arrow[dl] \\ & T & \end{tikzcd}$$

in $\widehat{\underline{C}}_{R\widehat{\otimes}_\Lambda R}$, with $T_2 \longrightarrow T$ surjective. Since $\underline{A}$ is homogeneous, the fibre products

$$\xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$$

exist in $\underline{A}$, and there are canonical isomorphisms

$$\xi_i(T_1 \times_T T_2) \xrightarrow{\sim} \xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2) \qquad (i = 1, 2)$$

(cf. 2.6(a)). Therefore

$$\mathrm{Isom}_\xi(T_1 \times_T T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \times_{\mathrm{Isom}_\xi(T)} \mathrm{Isom}_\xi(T_2)$$

is an isomorphism. $\square$

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

(1) $\underline{A}$ is homogeneous;

(2) $[\underline{A}(k[\varepsilon])]$ and $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$ are finite-dimensional vector spaces over k.

Proof. Suppose first that $\underline{A}$ is prorepresented by a smooth cogroupoid. Thus $\underline{A}$ is $\underline{C}_\Lambda$-equivalent to

$$\underline{h} = \underline{h}_{(R_0, R_1, \ldots)}$$

for a smooth cogroupoid $(R_0, R_1, \ldots)$ in $\widehat{\underline{C}}_\Lambda$. Proposition 2.12(a) shows that $\underline{h}$ is homogeneous. Moreover,

$$[\underline{h}(k[\varepsilon])] = \mathrm{Coker}\left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \overset{s_1}{\underset{s_2}{\rightrightarrows}} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon])\right),$$

$$\begin{aligned} \mathrm{Aut}_{\underline{h}}(1_\varepsilon) &= \mathrm{Ker}\left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon])\right) \\ &= \mathrm{Ker}\left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon])\right). \end{aligned}$$

Both are finite-dimensional vector spaces over k, so the same is true for $\underline{A}$.

Conversely, assume (1) and (2). Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\varepsilon])]$ is finite-dimensional, it has a minimally versal formal object ξ. Put $P(\xi) = R_0$ and consider

$$\mathrm{Isom}_\xi : \underline{\widehat{C}}_{R_0 \widehat{\otimes}_\Lambda R_0} \longrightarrow (\mathrm{Sets}),$$

given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T), \xi_2(T)),$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \longrightarrow T$ are the composite maps in

$$R_0 \underset{s_2}{\overset{s_1}{\rightrightarrows}} R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow T.$$

Corollary 2.16 prorepresents this functor by an object

$$R_1 \in \mathrm{ob}(\underline{\widehat{C}}_{R_0 \widehat{\otimes}_\Lambda R_0})$$

and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ for which

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}_\xi}$$

is an isomorphism over $\underline{\widehat{C}}_{R_0 \widehat{\otimes}_\Lambda R_0}$. The pair (R_0, R_1) therefore defines a cogroupoid, and

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \ldots)} \longrightarrow \underline{A}$$

is a $\underline{C}_\Lambda$-equivalence. Because ξ is minimally versal, the composite

$$\underline{h}_{(R_0, R_0, \ldots)} \xrightarrow{\epsilon} \underline{h}_{(R_0, R_1, \ldots)} \xrightarrow{(\xi,\varphi)} \underline{A}$$

is minimally smooth. Since (ξ, φ) is an equivalence, ϵ is minimally smooth as well. Thus $(R_0, R_1, \ldots)$ is a normalized smooth cogroupoid in $\underline{\widehat{C}}_\Lambda$. $\square$

Corollary 2.18. A functor $F : \underline{C}_\Lambda \longrightarrow (\mathrm{Sets})$ is prorepresentable if and only if it is homogeneous and $\dim F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\underline{\widehat{C}}_\Lambda$ can be normalized up to isomorphism.

Given a cogroupoid $(R_0, R_1, \ldots)$ in $\underline{\widehat{C}}_\Lambda$, set $\bar{R}_1 = R_1/J$, where J is the ideal of R_1 generated by

$$\{s_1(r) - s_2(r) \mid r \in R_0\}.$$

Then $(R_0, \bar{R}_1, \ldots)$ defines a formal group scheme over R_0 with the induced composition law.

Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$ via (ξ, φ). Let

$$\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$$

be the embeddings corresponding to the surjections

$$R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k,$$

and let Aut_ξ and Aut^o be the restrictions of Isom_ξ to $\underline{C}_{R_0}$ and $\underline{C}_k$, respectively. Thus $\mathrm{Aut}_\xi(S)$ is the automorphism group of $\xi(S)$ for $S \in \mathrm{ob}(\underline{C}_{R_0})$, while $\mathrm{Aut}^o(S)$ is the automorphism group of 1_S for $S \in \mathrm{ob}(\underline{C}_k)$, where 1_S is the direct image of $1 \in \underline{A}(k)$ under $k \longrightarrow S$.

Since R_1 prorepresents Isom_ξ on $\underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents Aut_ξ on $\underline{C}_{R_0}$, and that $k \otimes_{R_0} \bar{R}_1$ prorepresents Aut^o on $\underline{C}_k$. Although $\underline{A}$ is prorepresented by the smooth cogroupoid $(R_0, R_1, \ldots)$, neither $(R_0, \bar{R}_1, \ldots)$ nor $(k, k \otimes_{R_0} \bar{R}_1, \ldots)$ is smooth in general. More precisely:

Corollary 2.20. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$ via (ξ, φ).

(a) $(R_0, \bar{R}_1, \ldots)$ and $(k, k \otimes_{R_0} \bar{R}_1, \ldots)$ prorepresent Aut_ξ and Aut^o, respectively.

(b) The following statements are equivalent:

(i) the formal group scheme $\underline{h}_{(R_0,\bar{R}_1,\ldots)}$ is formally smooth over R_0;

(ii) for every surjective arrow $a' \longrightarrow a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{P(a')}(a') \longrightarrow \mathrm{Aut}_{P(a)}(a)$$

is surjective;

(iii) R_0 prorepresents the functor $[\underline{A}]$ over $\underline{C}_\Lambda$.

Proof. We have already noted (a). For (b), the equivalence (i)⇔(ii) is clear, whereas (ii)⇔(iii) follows from 2.7 and 2.15, since

$$\widetilde{\xi} : \underline{h}_{R_0} \longrightarrow [\underline{A}]$$

is minimally smooth. □

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|\underline{C}_k$ be the pullback of $p : \underline{A} \longrightarrow \underline{C}_\Lambda$ along $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. If $\underline{A}$ is prorepresented by $(R_0, R_1, \ldots)$ via (ξ, φ), then $\underline{A}|\underline{C}_k$ is prorepresented by

$$(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \ldots)$$

via $(k \otimes_\Lambda \xi, k \otimes_\Lambda \varphi)$. Here $k \otimes_\Lambda \xi$ is the direct image of ξ under $R_0 \longrightarrow k \otimes_\Lambda R_0$, and $k \otimes_\Lambda \varphi$ is the image of φ under

$$\mathrm{Isom}_{\underline{A}(R_0)}\big(\xi(s_1), \xi(s_2)\big) \longrightarrow \mathrm{Isom}_{\underline{A}(k \otimes_\Lambda R_1)}\big(k \otimes_\Lambda \xi(s_1), k \otimes_\Lambda \xi(s_2)\big).$$

If $(R_0, R_1, \ldots)$ is normalized, then so is $(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \ldots)$; moreover, the relative dimension of R_1 over R_0 equals that of $k \otimes_\Lambda R_1$ over $k \otimes_\Lambda R_0$, since R_1 is smooth over R_0.

Remark 2.22. Suppose that $\underline{A}$ is prorepresented by smooth cogroupoids $(R_0, R_1, \ldots)$ and $(R'_0, R'_1, \ldots)$, with $(R_0, R_1, \ldots)$ normalized. We obtain a homomorphism

$$(i_0, i_1) : (R_0, R_1, \ldots) \longrightarrow (R'_0, R'_1, \ldots).$$

By 2.10(b), R'_0 is a formal power-series ring over R_0, and (i_0, i_1) induces an isomorphism

$$R'_0 \otimes_{R_0} (R_0, R_1, \ldots) \xrightarrow{\sim} (R'_0, R'_1, \ldots).$$

3. The coherent sheaves $\underline{D}^i_X$

One important functor which is usually not prorepresentable, but nevertheless admits a formal versal object, is the deformation functor. With this in view, we recall some basic facts about extensions of commutative rings and their obstruction theory. This section contains nothing new, except perhaps for its systematic use of Grothendieck cohomology on a site. Its purpose is to make this exposé as self-contained as possible; in fact, we prove somewhat more than will be needed later. The basic results are stated in Theorem 3.12. Throughout, all rings are commutative and have a unit.

Given a ring R, let $\underline{C}_R$ denote the category of R-algebras. For every $A \in \mathrm{Ob}(\underline{C}_R)$, put

$$\mathrm{Cov}(A) = \{\text{surjective arrows } \varphi : A' \longrightarrow A \text{ in } \underline{C}_R\}.$$

This defines a pretopology, and hence a site, on $\underline{C}_R$. For every $A \in \mathrm{Ob}(\underline{C}_R)$, let $\underline{C}_{A|R}$ be the category of arrows in $\underline{C}_R$ with target A. We endow $\underline{C}_{A|R}$ with the topology induced by the forgetful functor

$$j_A : \underline{C}_{A|R} \longrightarrow \underline{C}_R.$$

The category $\underline{C}_{A|R}$ has a final object $(1_A : A \longrightarrow A)$ and an initial object $(i_A : R \longrightarrow A)$, and it admits both fibre products and coproducts. When there is no risk of confusion, we write simply B for an object $(B \longrightarrow A)$ of $\underline{C}_{A|R}$.

Let $\mathscr{C}$ be an abelian category. A $\mathscr{C}$-valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor

$$F : \underline{C}_{A|R}^{\mathrm{op}} \longrightarrow \mathscr{C}$$

such that, for every surjective arrow $B' \longrightarrow B$ in $\underline{C}_{A|R}$, the sequence

$$0 \longrightarrow F(B) \longrightarrow F(B') \rightrightarrows F(B' \times_B B')$$

is exact. The <u>additive</u> presheaves are those which preserve coproducts; thus

$$F(B_1 \otimes_R B_2) = F(B_1) \oplus F(B_2).$$

For example, an A-module M defines the functor

$$\mathrm{Der}_R(-, M) : \underline{C}_{A|R}^{\mathrm{op}} \longrightarrow (A\text{-mod}),$$

which sends B to the R-linear derivations $\mathrm{Der}_R(B, M)$, with M regarded as a B-module via the structural map $B \longrightarrow A$. This is plainly an additive sheaf on $\underline{C}_{A|R}$. Here (A-mod) denotes the category of A-modules.

A presheaf F on $\underline{C}_{A|R}$ with values in (A-mod) is said to be of <u>finite type</u> if $F(B)$ is an A-module of finite type whenever B is an R-algebra of finite type. For example, if A is noetherian and M is an A-module of finite type, then $\mathrm{Der}_R(-, M)$ is an additive sheaf of finite type on $\underline{C}_{A|R}$.

We record a little more notation and a few basic facts from homological algebra.

<u>3.1(a).</u> Throughout the remainder of this section, $\widetilde{\underline{C}}_{A|R}$ and $\overset{\vee}{\underline{C}}_{A|R}$ denote, respectively, the categories of sheaves and presheaves with values in (A-mod). There is a left-exact inclusion functor

$$i_{A|R} : \widetilde{\underline{C}}_{A|R} \longrightarrow \overset{\vee}{\underline{C}}_{A|R}$$

and an exact associated-sheaf functor

$$G_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \longrightarrow \widetilde{\underline{C}}_{A|R},$$

left adjoint to $i_{A|R}$.

An object $B \in \mathrm{Ob}(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \longrightarrow \underline{C}_{A|R}$ and hence a commutative diagram

(*)
$$\begin{tikzcd}
\widetilde{\underline{C}}_{A|R} \arrow[r, "\widetilde{j}"] \arrow[d, "i_{A|R}"'] & \widetilde{\underline{C}}_{B|R} \arrow[d] \\
\overset{\vee}{\underline{C}}_{A|R} \arrow[r, "j^{\vee}"'] & \overset{\vee}{\underline{C}}_{B|R}.
\end{tikzcd}$$

The restriction functor $\widetilde{j}$, as well as $j^{\vee}$, is exact, because the site $\underline{C}_{B|R}$ is simply the category of arrows in $\underline{C}_{A|R}$ with target B, endowed with the induced topology.

A ring homomorphism $S \longrightarrow R$ induces a continuous functor $\beta : \underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$ and hence a commutative diagram

(**)
$$\begin{tikzcd}
\widetilde{\underline{C}}_{A|S} \arrow[r, "\widetilde{\beta}"] \arrow[d, "i_{A|S}"'] & \widetilde{\underline{C}}_{A|R} \arrow[d] \\
\overset{\vee}{\underline{C}}_{A|S} \arrow[r, "\beta^{\vee}"'] & \overset{\vee}{\underline{C}}_{A|R}.
\end{tikzcd}$$

Notice that $\widetilde{\beta}$ need not be exact.

3.1(b). The right-derived functor of the inclusion

$$i_{A|R} : \underline{\widetilde{C}}_{A|R} \longrightarrow \underline{\overset{\vee}{C}}_{A|R}$$

will be denoted by $\underline{H}^{\bullet}_{A|R}$. For every sheaf F on $\underline{C}_{A|R}$, put

$$H^{\bullet}(A|R, F) = [\underline{H}^{\bullet}_{A|R}(F)](A).$$

For $B' \longrightarrow B$ in $\mathrm{Cov}(B)$, define

$$H^0(\{B' \longrightarrow B\}, F) = \ker\bigl(F(B') \rightrightarrows F(B' \times_B B')\bigr) .$$

We denote the right-derived functors of $H^0(\{B' \longrightarrow B\}, -)$ by $H^{\bullet}(\{B' \longrightarrow B\}, -)$. These are the cohomology groups of the Čech semisimplicial module

$$F(B') \;\substack{\longrightarrow\\ \longrightarrow}\; F(B' \times_B B') \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; F(B' \times_B B' \times_B B') \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \cdots .$$

The right-derived functor of

$$\underline{\check{H}}^0_{A|R} : \underline{\overset{\vee}{C}}_{A|R} \longrightarrow \underline{\overset{\vee}{C}}_{A|R}$$

will be denoted by $\underline{\check{H}}^{\bullet}_{A|R}$, where

$$[\underline{\check{H}}^0_{A|R}(F)](B) = \varinjlim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \longrightarrow B\}, F).$$

For every presheaf F on $\underline{C}_{A|R}$, put

$$\check{H}^{\bullet}(A|R, F) = [\underline{\check{H}}^{\bullet}_{A|R}(F)](A) = \varinjlim_{\{A' \to A\} \in \mathrm{Cov}(A)} H^{\bullet}(\{A' \longrightarrow A\}, F).$$

For every sheaf F on $\underline{C}_{A|R}$ there is the usual spectral sequence

$$\check{H}^p\bigl(A|R, \underline{H}^q_{A|R}(F)\bigr) \Longrightarrow H^{p+q}(A|R, F).$$

Here $\check{H}^0(A|R, \underline{H}^q_{A|R}(F)) = 0$ for every $q > 0$. Consequently,

$$\check{H}^1(A|R, F) = H^1(A|R, F),$$

and the sequence

$$0 \longrightarrow \check{H}^2(A|R, F) \longrightarrow H^2(A|R, F) \longrightarrow \check{H}^1\bigl(A|R, \underline{H}^1_{A|R}(F)\bigr) \longrightarrow \check{H}^3(A|R, F)$$

is exact. 3.1(c). If B is an object of $\underline{C}_{A|R}$, diagram $(*)$ in 3.1(a), together with the exactness of $\widetilde{j}$ and $j^{\vee}$, gives a canonical isomorphism

$$H^{\bullet}(B|R, \widetilde{j}F) = [\underline{H}^{\bullet}_{A|R}(F)](B).$$

On the other hand, a ring homomorphism $S \longrightarrow R$ induces diagram $(**)$ in 3.1(a). The functor $\widetilde{\beta}$ need not be exact; however, β preserves fibre products, and we obtain a spectral sequence

$$\underline{H}^p_{A|R}(R^q\widetilde{\beta}F) \Longrightarrow \beta^{\vee}\underline{H}^{p+q}_{A|S}(F),$$

or, on evaluating at A,

$$H^p(A|R, R^q\widetilde{\beta}F) \Longrightarrow H^{p+q}(A|S, F).$$

Since there is a canonical isomorphism

$$\widetilde{\beta} = G_{A|R} \circ \beta^{\vee} \circ i_{A|S}$$

and the functor $\beta^{\#} = G_{A|R} \circ \beta^{\vee}$ is exact, we have

$$R^{\bullet}\widetilde{\beta}F = \beta^{\#}\underline{H}^{\bullet}_{A|S}(F).$$

Thus the preceding spectral sequence may be written

$$H^p\big(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)\big) \Longrightarrow H^{p+q}(A|S, F).$$

Lemma 3.2. Let $B \in \mathrm{Ob}(\underline{C}_{A|R})$, and choose a surjective arrow $P \longrightarrow B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R. Then, for every $F \in \mathrm{Ob}(\check{\underline{C}}_{A|R})$, the natural map

$$H^{\bullet}(\{P \longrightarrow B\}, F) \longrightarrow [\check{\underline{H}}^{\bullet}(F)](B)$$

is an isomorphism.

Proof. It suffices to prove that the natural map

$$H^0(\{P \longrightarrow B\}, F) \longrightarrow [\check{\underline{H}}^{0}(F)](B) = \varinjlim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \longrightarrow B\}, F)$$

is an isomorphism. This is clear because, for every $\{B' \longrightarrow B\} \in \mathrm{Cov}(B)$, there is a map of covers

$$\{P \longrightarrow B\} \longrightarrow \{B' \longrightarrow B\}.$$

$\square$

Corollary 3.3.

(a) Suppose that A is noetherian. If $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for every n.

(b) If P is a polynomial algebra over R, then

$$H^n(P|R, F) = 0$$

for every $n > 0$ and every $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{P|R})$.

(c) A sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

in $\widetilde{\underline{C}}_{A|R}$ is exact if and only if

$$0 \longrightarrow F'(P) \longrightarrow F(P) \longrightarrow F''(P) \longrightarrow 0$$

is exact for every polynomial R-algebra $P \in \mathrm{Ob}(\underline{C}_{A|R})$.

Proof.
(a) In view of the spectral sequence

$$\check{H}^p\big(B|R, \underline{H}^q(F)\big) \Longrightarrow H^{p+q}(B|R, F)$$

and the equality $\check{H}^0(\underline{H}^q(F)) = 0$ for $q > 0$, it is enough to show that $\check{\underline{H}}^n(G)$ is of finite type for every n and every presheaf G of finite type. By Lemma 3.2,

$$\check{H}^{\bullet}(B|R, G) = H^{\bullet}(\{P \longrightarrow B\}, G),$$

where P is a polynomial algebra over R and $\{P \longrightarrow B\} \in \mathrm{Cov}(B)$. The latter is the cohomology of the complex

$$\cdots G(P \times_B P \times_B P) \Lleftarrow G(P \times_B P) \Leftarrow G(P).$$

If B is an R-algebra of finite type, P may be chosen of finite type over R. All the iterated fibre products $P \times_B \cdots \times_B P$ are then R-algebras of finite type. If G is of finite type, their images under G are A-modules of finite type. Hence $H^\bullet(\{P \longrightarrow B\}, G)$ is an A-module of finite type because A is noetherian.

(b) Let P be a polynomial algebra over R. Every surjective R-algebra map $B \longrightarrow P$ admits a section, so

$$H^i(\{B \longrightarrow P\}, G) = 0 \qquad (i > 0).$$

It follows that $\check{H}^i(P|R, G) = 0$ for every $i > 0$ and every presheaf G on $\underline{C}_{P|R}$. Consequently the spectral sequence

$$\check{H}^p\big(P|R, \underline{H}^q_{P|R}(F)\big) \Longrightarrow H^{p+q}(P|R, F)$$

degenerates, and

$$\check{H}^0\big(P|R, \underline{H}^n(F)\big) \simeq H^n(P|R, F)$$

for every n. For a sheaf F, however, $\check{H}^0(P|R, \underline{H}^n(F)) = 0$ whenever $n > 0$. Thus $H^n(P|R, F) = 0$ for every $n > 0$.

(c) If $0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$ is exact in $\widetilde{\underline{C}}_{A|R}$, then the sequence

$$0 \longrightarrow F'(B) \longrightarrow F(B) \longrightarrow F''(B) \longrightarrow H^1(B|R, F') \longrightarrow H^1(B|R, F) \longrightarrow \cdots$$

is exact. If $P \in \mathrm{Ob}(\underline{C}_{A|R})$ is a polynomial algebra over R, then $H^1(P|R, F') = 0$ by (b), and hence

$$0 \longrightarrow F'(P) \longrightarrow F(P) \longrightarrow F''(P) \longrightarrow 0$$

is exact.

Conversely, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ with

$$\mathrm{Ob}(\underline{P}_{A|R}) = \{P \in \mathrm{Ob}(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\},$$

endowed with the induced topology, and let $p : \underline{P}_{A|R} \longrightarrow \underline{C}_{A|R}$ be the inclusion. Every object of $\underline{C}_{A|R}$ is covered by an object of $\underline{P}_{A|R}$; the comparison lemma therefore shows that

$$\widetilde{p} : \widetilde{\underline{C}}_{A|R} \longrightarrow \widetilde{\underline{P}}_{A|R}$$

is an equivalence of categories. If the sequence obtained by evaluating $F' \longrightarrow F \longrightarrow F''$ at every $P \in \mathrm{Ob}(\underline{P}_{A|R})$ is exact, then

$$0 \longrightarrow \widetilde{p}F' \longrightarrow \widetilde{p}F \longrightarrow \widetilde{p}F'' \longrightarrow 0$$

is exact in $\widetilde{\underline{P}}_{A|R}$, and therefore the original sequence is exact in $\widetilde{\underline{C}}_{A|R}$. $\square$

<u>Remark 3.4.</u> Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows at once that every presheaf on $\underline{P}_{A|R}$ is a sheaf; that is,

$$\widetilde{\underline{P}}_{A|R} = \overset{\vee}{\underline{P}}_{A|R}.$$

Consequently, the restriction functor

$$\widetilde{\underline{C}}_{A|R} \longrightarrow \overset{\vee}{\underline{P}}_{A|R}$$

is an equivalence of categories. <u>Lemma 3.5.</u> Consider a diagram of R-algebras

$$\begin{tikzcd} & C \arrow[d] \\ B' \arrow[r, "\text{surjective}"'] & B \end{tikzcd}$$

and let S be an R-algebra.

(i) If $\mathrm{Tor}_1^R(B,S)=0$, then the canonical map

$$(B' \times_B C) \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S)$$

is an isomorphism.

(ii) Suppose that B' is flat over R. If

$$\mathrm{Tor}_i^R(C,S)=0 \quad (0<i<n) \qquad \text{and} \qquad \mathrm{Tor}_i^R(B,S)=0 \quad (0<i\le n),$$

then

$$\mathrm{Tor}_i^R(B' \times_B C, S)=0 \qquad (0<i<n).$$

Proof. Let $0 \longrightarrow J \longrightarrow B' \longrightarrow B \longrightarrow 0$ be exact.
(i) If $\mathrm{Tor}_1^R(B,S)=0$, then

$$0 \longrightarrow J \otimes_R S \longrightarrow B' \otimes_R S \longrightarrow B \otimes_R S \longrightarrow 0$$

is exact, and hence so is

$$0 \longrightarrow J \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) \longrightarrow C \otimes_R S \longrightarrow 0.$$

We therefore obtain a commutative diagram with exact rows:

$$\begin{tikzcd}
 & J \otimes_R S \arrow[r] \arrow[d, equal] & (B' \times_B C) \otimes_R S \arrow[r] \arrow[d] & C \otimes_R S \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & J \otimes_R S \arrow[r] & (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) \arrow[r] & C \otimes_R S \arrow[r] & 0.
\end{tikzcd}$$

It follows that

$$(B' \times_B C) \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S)$$

is an isomorphism.

(ii) Since B' is flat over R and $\mathrm{Tor}_i^R(B,S)=0$ for $0<i\le n$, the long exact Tor sequence gives

$$\mathrm{Tor}_i^R(J,S)=0 \qquad (0<i<n).$$

Applying the same sequence to

$$0 \longrightarrow J \longrightarrow B' \times_B C \longrightarrow C \longrightarrow 0$$

and using the hypothesis on C gives

$$\mathrm{Tor}_i^R(B' \times_B C, S)=0 \qquad (0<i<n).$$

$\square$

Corollary 3.6.

(a) Let A and S be R-algebras. If $\mathrm{Tor}_1^R(A,S)=0$, then for every $G \in \mathrm{Ob}(\check{\underline{C}}_{A \otimes_R S|S})$ the canonical map

$$\check{H}^\bullet(A \otimes_R S|S, G) \longrightarrow \check{H}^\bullet(A|R, \check{S}G)$$

is an isomorphism. Here

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \qquad X \longmapsto X \otimes_R S.$$

In particular, if S is flat over R, then

$$\check{S}\, \underline{\check{H}}^\bullet(G) \longrightarrow \underline{\check{H}}^\bullet(\check{S}G)$$

is an isomorphism for every such G.

(b) Let $P \longrightarrow B$ be a surjective map of R-algebras, where P is a polynomial algebra over R. If

$$\operatorname{Tor}_i^R(B,S) = 0 \qquad (0 < i \le n),$$

then, for every iterated fibre product of copies of P over B,

$$\operatorname{Tor}_i^R(P \times_B P \times_B \cdots \times_B P, S) = 0 \qquad (0 < i < n).$$

Proof.
(a) Given $B \in \operatorname{Ob}(\underline{C}_{A|R})$, choose a cover $\{P \longrightarrow B\} \in \operatorname{Cov}(B)$ with P a polynomial algebra over R. If $\operatorname{Tor}_1^R(B,S) = 0$, Lemma 3.5(i) shows that the natural map

$$\left(\overbrace{P \times_B \cdots \times_B P}^{n}\right) \otimes_R S \longrightarrow \overbrace{(P \otimes_R S) \times_{B\otimes_R S} \cdots \times_{B \otimes_R S} (P \otimes_R S)}^{n}$$

is an isomorphism for every n. Consequently,

$$\begin{aligned}\check{H}^\bullet(B|R, \check{S}G) &= H^\bullet(\{P \longrightarrow B\}, \check{S}G)\\ &= H^\bullet(\{P \otimes_R S \longrightarrow B \otimes_R S\}, G)\\ &= \check{H}^\bullet(B \otimes_R S|S, G).\end{aligned}$$

If S is flat over R, then $\operatorname{Tor}_1^R(X,S) = 0$ for every $X \in \operatorname{Ob}(\underline{C}_{A|R})$. The preceding isomorphism is therefore natural in X, and gives

$$\check{S}\,\underline{\check{H}}^\bullet(G) \xrightarrow{\sim} \underline{\check{H}}^\bullet(\check{S}G).$$

(b) Put

$$P^{(m)} = \overbrace{P \times_B \cdots \times_B P}^{m+1}.$$

For $m = 0$ there is nothing to prove, since P is flat over R. Suppose inductively that $\operatorname{Tor}_i^R(P^{(m-1)}, S) = 0$ for $0 < i < n$. Since $P \longrightarrow B$ is surjective and $P^{(m)} = P \times_B P^{(m-1)}$, Lemma 3.5(ii) gives

$$\operatorname{Tor}_i^R(P^{(m)}, S) = 0 \qquad (0 < i < n).$$

□

Proposition 3.7.

(a) Let A and S be R-algebras. If

$$\operatorname{Tor}_i^R(A,S) = 0 \qquad (0 < i \le n),$$

then, for every sheaf F on $\underline{C}_{A \otimes_R S|S}$, the canonical map

$$H^i(A \otimes_R S|S, F) \longrightarrow H^i(A|R, \widetilde{S}F)$$

is an isomorphism for every $i \le n$. Here

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \qquad X \longmapsto X \otimes_R S.$$

(b) Let P be a polynomial algebra over R. Then, for every additive sheaf F on $\underline{C}_{A \otimes_R P|R}$, the canonical map

$$H^n(A \otimes_R P|R, F) \longrightarrow H^n(A|R, F)$$

is an isomorphism for every $n > 0$.

(c) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every additive sheaf F on $\underline{C}_{A|S}$, there is an exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(A|R, F_R) \longrightarrow H^0(A|S, F) \longrightarrow H^0(R|S, F) \longrightarrow H^1(A|R, F_R) \\ \longrightarrow H^1(A|S, F) \longrightarrow H^1(R|S, F) \longrightarrow \cdots \longrightarrow H^n(A|R, F_R) \\ \longrightarrow H^n(A|S, F) \longrightarrow H^n(R|S, F) \longrightarrow H^{n+1}(A|R, F_R) \longrightarrow \cdots, \end{aligned}$$

where F_R is the sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \ker\big(F(X) \longrightarrow F(R)\big)$$

for every $X \in \mathrm{Ob}(\underline{C}_{A|R})$, via the forgetful functor $\underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$.

Proof.

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A\otimes_R S|S}, \qquad S(X) = X \otimes_R S.$$

Since S sends polynomial algebras over R to polynomial algebras over S, the functor

$$\widetilde{S} : \widetilde{\underline{C}}_{A\otimes_R S|S} \longrightarrow \widetilde{\underline{C}}_{A|R}$$

is exact by 3.3(c). We therefore obtain the canonical map

$$\check{S}\, \underline{H}^{\bullet}_{A\otimes_R S|S}(F) \longrightarrow \underline{H}^{\bullet}_{A|R}(\widetilde{S}F)$$

and the following commutative diagram of spectral sequences:

$$\begin{array}{ccc} E_S^{p,q}(S(X)) = \check{H}^p\big(X \otimes_R S|S, \underline{H}^q_{A\otimes_R S|S}(F)\big) & \Longrightarrow & H^{p+q}(X \otimes_R S|S, F) \\ \downarrow & & \\ \check{H}^p\big(X|R, \check{S}\, \underline{H}^q_{A\otimes_R S|S}(F)\big) & & \downarrow \\ \downarrow & & \\ E_R^{p,q}(X) = \check{H}^p\big(X|R, \underline{H}^q_{A|R}(\widetilde{S}F)\big) & \Longrightarrow & H^{p+q}(X|R, \widetilde{S}F). \end{array}$$

By 3.6(a), the map

$$E_S^{p,q}(S(X)) \longrightarrow \check{H}^p\big(X|R, \check{S}\, \underline{H}^q_{A\otimes_R S|S}(F)\big)$$

is an isomorphism whenever $\mathrm{Tor}_1^R(X, S) = 0$. Suppose inductively that

$$[\check{S}\, \underline{H}^q_{A\otimes_R S|S}(F)](X) \longrightarrow [\underline{H}^q_{A|R}(\widetilde{S}F)](X)$$

is an isomorphism for every $q < n$ and every $X \in \mathrm{Ob}(\underline{C}_{A|R})$ satisfying

$$\mathrm{Tor}_i^R(X, S) = 0 \qquad (0 < i \leq n).$$

It follows from 3.6(b) that $E_S^{p,q}(S(X)) \longrightarrow E_R^{p,q}(X)$ is an isomorphism for every $q < n$ and every p, under the same Tor hypothesis. Moreover, $E_S^{0,n}(S(X)) \longrightarrow E_R^{0,n}(X)$ is an isomorphism, since both sides are zero. Hence

$$H^n(X \otimes_R S|S, F) \longrightarrow H^n(X|R, \widetilde{S}F)$$

is an isomorphism whenever $\mathrm{Tor}_i^R(X, S) = 0$ for $0 < i \leq n$. Equivalently,

$$[\check{S}\, \underline{H}^i_{A\otimes_R S|S}(F)](X) \longrightarrow [\underline{H}^i_{A|R}(\widetilde{S}F)](X)$$

is an isomorphism for every $i \leq n$ under that hypothesis.

(b) Let

$$f : \underline{C}_{A|R} \longrightarrow \underline{C}_{A\otimes_R P|R}, \qquad f(X) = X \otimes_R P.$$

Choose a surjective arrow $P' \longrightarrow X$ in $\underline{C}_{A|R}$, with P' polynomial over R. Then

$$P' \otimes_R P \longrightarrow X \otimes_R P$$

is a surjective arrow in $\underline{C}_{A\otimes_R P|R}$, and $P' \otimes_R P$ is again polynomial over R. Since P is flat over R, 3.5(i) gives

$$\begin{aligned}\check{H}^\bullet(X|R, \check{f}G) &= H^\bullet(\{P' \longrightarrow X\}, \check{f}G)\\ &= H^\bullet(\{P' \otimes_R P \longrightarrow X \otimes_R P\}, G)\\ &= \check{H}^\bullet(X \otimes_R P|R, G)\end{aligned}$$

for every $G \in \mathrm{Ob}(\overset{\vee}{\underline{C}}_{A\otimes_R P|R})$. Thus $\widetilde{f}$ sends acyclic sheaves to acyclic sheaves. On the other hand, f sends polynomial algebras over R to polynomial algebras over R, so

$$\widetilde{f} : \widetilde{\underline{C}}_{A\otimes_R P|R} \longrightarrow \widetilde{\underline{C}}_{A|R}$$

is exact by 3.3(c). It follows that

$$H^\bullet(A \otimes_R P|R, F) \longrightarrow H^\bullet(A|R, \widetilde{f}F)$$

is an isomorphism for every $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A\otimes_R P|R})$.

For each $X \in \mathrm{Ob}(\underline{C}_{A|R})$ there is a canonical map

$$(\widetilde{f}F)(X) = F(X \otimes_R P) \longrightarrow F(X) \oplus F(P),$$

and hence a canonical map $\widetilde{f}F \longrightarrow F \oplus F(P)$, where $F(P)$ denotes the constant sheaf and F on the right denotes the restriction of F to $\underline{C}_{A|R}$. If F is additive, this map is an isomorphism. Therefore, for every $n > 0$,

$$H^n(A|R, \widetilde{f}F) \xrightarrow{\sim} H^n(A|R, F \oplus F(P)) = H^n(A|R, F),$$

because the constant sheaf $F(P)$ is flasque. This proves (b).

(c) This argument was communicated by Rinehart. Consider the forgetful functor $\beta : \underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$. There is a spectral sequence

$$E^{p,q} = H^p\big(A|R, \beta^\# \underline{H}^q_{A|S}(F)\big) \Longrightarrow H^{p+q}(A|S, F),$$

where

$$\beta^\# = G_{A|R} \circ \beta^\vee, \qquad G_{A|R} : \overset{\vee}{\underline{C}}_{A|R} \longrightarrow \widetilde{\underline{C}}_{A|R}$$

is the associated-sheaf functor.

Suppose now that F is additive. We claim that $\beta^\# \underline{H}^q_{A|S}(F)$ is a constant sheaf for every $q > 0$. Indeed, if P is a polynomial algebra over R, then $P = R \otimes_S P'$ for a polynomial algebra P' over S, and

$$[\beta^\# \underline{H}^q_{A|S}(F)](P) = H^q(P|S, F) = H^q(R \otimes_S P'|S, F) \longrightarrow H^q(R|S, F)$$

is an isomorphism for $q > 0$ by (b), and is surjective for $q = 0$. By 3.3(c), the canonical map

$$\beta^\# \underline{H}^q_{A|S}(F) \longrightarrow H^q(R|S, F)$$

to the corresponding constant sheaf is therefore an isomorphism for $q > 0$ and a surjection for $q = 0$. Hence $E^{p,q} = 0$ whenever $p, q > 0$, and the spectral sequence yields the exact sequence

$$(*) \quad 0 \longrightarrow E^{1,0} \longrightarrow H^1 \longrightarrow E^{0,1} \longrightarrow E^{2,0} \longrightarrow H^2 \longrightarrow E^{0,2} \longrightarrow E^{3,0} \longrightarrow H^3 \longrightarrow \cdots .$$

Here

$$E^{0,q} = [\beta^\# \underline{H}^q_{A|S}(F)](A) \xrightarrow{\sim} H^q(R|S,F) \quad (q>0), \qquad E^{p,0} = H^p(A|R,\widetilde{\beta}F).$$

On the other hand, the exact sequence

$$0 \longrightarrow F_R \longrightarrow \widetilde{\beta}F \longrightarrow F(R) \longrightarrow 0$$

gives the exact diagram

$$\begin{array}{ccccccccccc} 0 \to H^0(A|R,F_R) & \to & H^0(A|R,\widetilde{\beta}F) & \to & H^0(A|R,F(R)) & \to & H^1(A|R,F_R) & \to & H^1(A|R,\widetilde{\beta}F) \to 0 \\ & & \| & & \| & & & & \\ & & H^0(A|S,F) & & H^0(R|S,F) & & & & \end{array}$$

labelled $(**)$ in the source. Combining $(*)$ and $(**)$ gives the desired exact sequence. □

Corollary 3.8.

(a) Let A and B be R-algebras such that

$$\mathrm{Tor}_i^R(A,B) = 0 \qquad (0 < i \leq n).$$

Then, for every additive sheaf $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$, the canonical map

$$H^i(A\otimes_R B|R,F) \longrightarrow H^i(A|R,F) \oplus H^i(B|R,F)$$

is an isomorphism for every $i \leq n$.

(b) Let A be an R-algebra such that $\mathrm{Tor}_i^R(A,A) = 0$ for every $i > 0$. If the map $A\otimes_R A \longrightarrow A$ is an isomorphism, then, for every additive sheaf $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A|R})$, one has

$$H^i(A|R,F) = 0$$

for every i.

Proof.

(a) Let $F \in \mathrm{Ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$ be additive. Applied to $R \longrightarrow A \longrightarrow A\otimes_R B$, Proposition 3.7(c) gives the exact sequence

$$(*) \qquad \cdots \longrightarrow H^i(A\otimes_R B|A,F_A) \longrightarrow H^i(A\otimes_R B|R,F) \longrightarrow H^i(A|R,F) \longrightarrow \cdots.$$

Consider the commutative diagram

$$\begin{array}{ccc} H^i(A\otimes_R B|A,F_A) & \longrightarrow & H^i(A\otimes_R B|R,F) \\ \downarrow & & \downarrow \\ H^i(B|R,\widetilde{A}F_A) & \xrightarrow{\approx} & H^i(B|R,F). \end{array}$$

Here

$$A : \underline{C}_{B|R} \longrightarrow \underline{C}_{A\otimes_R B|A}, \qquad X \longmapsto A\otimes_R X,$$

and the bottom map is induced by $\widetilde{A}F_A \longrightarrow F$, which is an isomorphism because F is additive. By functoriality, the commutativity of the diagram reduces to the case $i=0$.

If $\mathrm{Tor}_i^R(A,B) = 0$ for $0 < i \leq n$, then Proposition 3.7(a) shows that the left vertical map is an isomorphism for $i \leq n$. Consequently, both composites

$$H^i(A\otimes_R B|A,F_A) \longrightarrow H^i(A\otimes_R B|R,F) \longrightarrow H^i(B|R,F)$$

and, by symmetry,

$$H^i(A \otimes_R B|B, F_B) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F)$$

are isomorphisms for $i \leq n$. It follows that $(*)$ breaks into split short exact sequences

$$0 \longrightarrow H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \longrightarrow 0$$

for $i \leq n$, and hence that

$$H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism in that range.

(b) By (a), the map

$$H^i(A \otimes_R A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F)$$

is an isomorphism for every i. On the other hand, the assumed isomorphism $A \otimes_R A \longrightarrow A$ induces an isomorphism

$$H^i(A|R, F) \longrightarrow H^i(A \otimes_R A|R, F).$$

Their composite is the diagonal map

$$H^i(A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F).$$

Since it is an isomorphism, $H^i(A|R, F) = 0$ for every i. $\square$

Corollary 3.9. Let S be a multiplicative subset of an R-algebra A. Suppose that

$$F \in \mathrm{Ob}(\widetilde{\underline{C}}_{S^{-1}A|R})$$

is additive and that F_A is also additive. Then the canonical map

$$H^i(S^{-1}A|R, F) \longrightarrow H^i(A|R, F)$$

is an isomorphism for every i. Here $F_A \in \mathrm{Ob}(\widetilde{\underline{C}}_{S^{-1}A|A})$ is defined by

$$F_A(X) = \ker\big(F(X) \longrightarrow F(A)\big)$$

for every $X \in \mathrm{Ob}(\underline{C}_{S^{-1}A|A})$.

Proof. Applied to $R \longrightarrow A \longrightarrow S^{-1}A$, Proposition 3.7(c) gives the exact sequence

$$\cdots \longrightarrow H^n(S^{-1}A|A, F_A) \longrightarrow H^n(S^{-1}A|R, F) \longrightarrow H^n(A|R, F) \longrightarrow \cdots .$$

The map

$$S^{-1}A \otimes_A S^{-1}A \longrightarrow S^{-1}A$$

is an isomorphism, and F_A is additive by hypothesis. Corollary 3.8(b) therefore gives $H^n(S^{-1}A|A, F_A) = 0$ for every n. The middle map in the exact sequence is consequently an isomorphism for every n. $\square$

Let A be an R-algebra. An A-module M defines an additive sheaf

$$\mathrm{Der}_R(-, M) \in \mathrm{Ob}(\widetilde{\underline{C}}_{A|R}), \qquad X \longmapsto \mathrm{Der}_R(X, M).$$

We make the following definition.

Definition 3.10. Let A be an R-algebra. For every A-module M, set

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)).$$

We now compute the groups of low degree, namely $D^i(A|R, M)$ for $i = 1, 2$. Choose a surjective arrow $P \longrightarrow A$ in $\underline{C}_{A|R}$ with P a polynomial algebra over R, and write

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0.$$

Lemma 3.10.

$$\begin{aligned} D^1(A|R, M) &= \operatorname{Coker}\big(\operatorname{Der}_R(P, M) \longrightarrow \operatorname{Hom}_P(I, M)\big), \\ D^2(A|R, M) &= H^1\big(A|R, \underline{H}^1_{A|R}(\operatorname{Der}_R(-, M))\big). \end{aligned}$$

Proof. Consider the simplicial algebra

$$(*) \qquad \cdots\ P_A^{(2)} \mathrel{\substack{\longrightarrow\\[-0.3em]\longrightarrow\\[-0.3em]\longrightarrow}} P_A^{(1)} \rightrightarrows P_A^{(0)},$$

where

$$P_A^{(n)} = \overbrace{P \times_A \cdots \times_A P}^{n+1}.$$

For every n there is an exact sequence

$$0 \longrightarrow I^{(n)} \longrightarrow P_A^{(n)} \longrightarrow A \longrightarrow 0, \qquad I^{(n)} = \overbrace{I \times \cdots \times I}^{n+1}.$$

Put $\overline{I} = I/I^2$, and let Δ denote the diagonal map. It is straightforward to verify that, for every n, the sequence

$$\text{(a)} \qquad 0 \longrightarrow \operatorname{Hom}_A\big(\overline{I}^{(n)}/\Delta(\overline{I}), M\big) \longrightarrow \operatorname{Der}_R(P_A^{(n)}, M) \xrightarrow{\operatorname{Der}(\Delta, M)} \operatorname{Der}_R(P, M) \longrightarrow 0$$

is exact. We also have the exact sequence

$$\text{(b)} \qquad 0 \longrightarrow \operatorname{Hom}_A\big(\overline{I}^{(n)}/\Delta(\overline{I}), M\big) \longrightarrow \operatorname{Hom}_A(\overline{I}^{(n)}, M) \xrightarrow{\operatorname{Hom}(\Delta, M)} \operatorname{Hom}_A(\overline{I}, M) \longrightarrow 0.$$

The sequences (a) and (b) yield the corresponding exact sequences of simplicial A-modules. Since $\{I \longrightarrow 0\}$ and $\{I \longrightarrow I\}$ admit sections,

$$H^i(\{I \longrightarrow 0\}, \operatorname{Hom}_A(-, M)) = H^i(\{I \longrightarrow I\}, \operatorname{Hom}_A(-, M)) = 0 \qquad (i > 0).$$

It follows that

$$\begin{cases} \quad 0 \longrightarrow \operatorname{Der}_R(A, M) \longrightarrow \operatorname{Der}_R(P, M) \longrightarrow \operatorname{Hom}_A(\overline{I}, M) & \\ \qquad\qquad \longrightarrow H^1(\{P \longrightarrow A\}, \operatorname{Der}_R(-, M)) \longrightarrow 0 & \text{is exact,} \\ H^i(\{P \longrightarrow A\}, \operatorname{Der}_R(-, M)) = 0 & (i > 1). \end{cases}$$

By 3.1(b), therefore,

$$\begin{aligned} D^1(A|R, M) &= \check{H}^1(A|R, \operatorname{Der}_R(-, M)) \\ &= H^1(\{P \longrightarrow A\}, \operatorname{Der}_R(-, M)) \\ &= \operatorname{Coker}\big(\operatorname{Der}_R(P, M) \longrightarrow \operatorname{Hom}_A(\overline{I}, M)\big) \\ &= \operatorname{Coker}\big(\operatorname{Der}_R(P, M) \longrightarrow \operatorname{Hom}_P(I, M)\big), \\ D^2(A|R, M) &= H^1\big(A|R, \underline{H}^1_{A|R}(\operatorname{Der}_R(-, M))\big). \end{aligned}$$

□

Choose a free P-module F and a P-linear map $\gamma : F \longrightarrow P$ such that $\gamma(F) = I$. Associated with γ is a Koszul complex, denoted by K_I by abuse of notation. Lemma 3.11. There is an isomorphism

$$D^2(A|R, M) \simeq \operatorname{Coker}\big(H^1(K_I, M) \longrightarrow \operatorname{Hom}_A(H_1(K_I), M)\big).$$

Proof. We have

$$\begin{aligned} D^2(A|R,M) &= H^1\big(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))\big) \\ &= \ker\Big(H^1(P_A^{(1)}|R,M) \rightrightarrows H^1(P_A^{(2)}|R,M)\Big), \end{aligned}$$

because

$$[\underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))](P_A^{(0)}) = [\underline{H}^1_{A|R}(\mathrm{Der}_R(-,M))](P) = 0.$$

Let

$$0 \longrightarrow K \longrightarrow F \xrightarrow{\gamma} I \longrightarrow 0$$

be exact. Write $P[F]$ for the symmetric algebra over P generated by F. Then $P[F]$ and $P[F] \otimes_P P[F]$ are polynomial algebras over R, and there are surjective maps

$$\begin{aligned} P[F] &\longrightarrow P_A^{(1)} = P \times_A P, & X &\longmapsto (0, \gamma(X)), \\ P[F] \otimes_P P[F] &\longrightarrow P_A^{(2)} = P \times_A P \times_A P, & X \otimes 1 &\longmapsto (0, \gamma(X), 0), \\ & & 1 \otimes X &\longmapsto (0, 0, \gamma(X)), \end{aligned}$$

for $X \in F$. Put

$$J = \ker(P[F] \longrightarrow P_A^{(1)}), \qquad J' = \ker(P[F] \otimes_P P[F] \longrightarrow P_A^{(2)}).$$

If F_γ denotes the ideal generated by $\{X - \gamma(X) \mid X \in F\}$, then

$$\begin{aligned} J &= (F) \cap (F_\gamma) = (K) + (F \cdot F_\gamma), \\ J' &= (F \otimes 1 + 1 \otimes F) \cap (F_\gamma \otimes 1 + 1 \otimes F) \cap (F \otimes 1 + 1 \otimes F_\gamma) \\ &= (K \otimes 1 + 1 \otimes K) + (F \cdot F_\gamma \otimes 1 + 1 \otimes F \cdot F_\gamma) + (F \otimes F). \end{aligned}$$

The three maps $\pi_i : P_A^{(2)} \longrightarrow P_A^{(1)}$ admit lifts $\widehat{\pi}_i : P[F] \otimes_P P[F] \longrightarrow P[F]$ given, for $X \in F$, by

$$\widehat{\pi}_0 : \begin{cases} X \otimes 1 \longmapsto \gamma(X) - X, \\ 1 \otimes X \longmapsto X, \end{cases} \qquad \widehat{\pi}_1 : \begin{cases} X \otimes 1 \longmapsto 0, \\ 1 \otimes X \longmapsto X, \end{cases} \qquad \widehat{\pi}_2 : \begin{cases} X \otimes 1 \longmapsto X, \\ 1 \otimes X \longmapsto 0. \end{cases}$$

A direct computation shows that $\widehat{\pi}_0 - \widehat{\pi}_1 + \widehat{\pi}_2$ annihilates

$$K \otimes 1 + 1 \otimes K + 1 \otimes F \cdot F_\gamma$$

and maps $F \cdot F_\gamma \otimes 1 + F \otimes F$ onto $F \cdot F_\gamma$. For every

$$D \in \mathrm{Der}_R(P[F] \otimes_P P[F], M)$$

one has $D(F \cdot F_\gamma \otimes 1 + F \otimes F) = 0$, because $I \cdot M = 0$. Consequently,

$$\begin{aligned} D^2(A|R,M) &= \ker\Big(D^1(P_A^{(1)}|R,M) \rightrightarrows D^1(P_A^{(2)}|R,M)\Big) \\ &= \{[f] \in D^1(P_A^{(1)}|R,M) \mid f(F \cdot F_\gamma) = 0\} \\ &= \mathrm{Coker}\left(\mathrm{Der}_R(P[F],M) \longrightarrow \mathrm{Hom}_{P[F]}\left(\frac{(F) \cap (F_\gamma)}{F \cdot F_\gamma}, M\right)\right) \\ &= \mathrm{Coker}\left(\mathrm{Der}_P(P[F],M) \longrightarrow \mathrm{Hom}_{P[F]}\left(\frac{(F) \cap (F_\gamma)}{(F)(F_\gamma)}, M\right)\right). \end{aligned}$$

The sequences

$$0 \longrightarrow (F) \longrightarrow P[F] \xrightarrow{P[0]} P \longrightarrow 0, \qquad 0 \longrightarrow (F_\gamma) \longrightarrow P[F] \xrightarrow{P[\gamma]} P_\gamma \longrightarrow 0$$

are exact. Hence
$$\frac{(F)\cap(F_\gamma)}{(F)(F_\gamma)} = \mathrm{Tor}_1^{P[F]}(P,P_\gamma) = H_1(K_I),$$
where K_I is the Koszul complex associated with $F \xrightarrow{\gamma} I \subset P$. We therefore obtain
$$\begin{aligned} D^2(A|R,M) &= \mathrm{Coker}\big(\mathrm{Hom}_P(F,M) \longrightarrow \mathrm{Hom}_P(H_1(K_I),M)\big) \\ &= \mathrm{Coker}\big(H^1(K_I,M) \longrightarrow \mathrm{Hom}_A(H_1(K_I),M)\big). \end{aligned}$$

□

We can now state the basic facts needed below, summarizing the results already established.

Theorem 3.12.

(1) If
$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$
is an exact sequence of A-modules, then the sequence
$$\begin{aligned} 0 \longrightarrow D^0(A|R,M') \longrightarrow D^0(A|R,M) \longrightarrow D^0(A|R,M'') \longrightarrow D^1(A|R,M') \\ \longrightarrow D^1(A|R,M) \longrightarrow D^1(A|R,M'') \longrightarrow \cdots \longrightarrow D^n(A|R,M') \\ \longrightarrow D^n(A|R,M) \longrightarrow D^n(A|R,M'') \longrightarrow \cdots \end{aligned}$$
is exact.

(2) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every A-module M, the sequence
$$\begin{aligned} 0 \longrightarrow D^0(A|R,M) \longrightarrow D^0(A|S,M) \longrightarrow D^0(R|S,M) \longrightarrow D^1(A|R,M) \\ \longrightarrow D^1(A|S,M) \longrightarrow D^1(R|S,M) \longrightarrow \cdots \longrightarrow D^n(A|R,M) \\ \longrightarrow D^n(A|S,M) \longrightarrow D^n(R|S,M) \longrightarrow \cdots \end{aligned}$$
is exact.

(3) Let A and S be R-algebras such that $\mathrm{Tor}_i^R(A,S) = 0$ for every $i > 0$. Then, for every $A \otimes_R S$-module M, the canonical map
$$D^i(A \otimes_R S|S,M) \longrightarrow D^i(A|R,M)$$
is an isomorphism for every i. In particular, if A is flat over R, then
$$D^i(A \otimes_R S|S,M) \xrightarrow{\sim} D^i(A|R,M)$$
for every i, every R-algebra S, and every $A \otimes_R S$-module M.

(4) Let S be a multiplicative subset of A. For every $S^{-1}A$-module N, the canonical map
$$D^i(S^{-1}A|R,N) \longrightarrow D^i(A|R,N)$$
is an isomorphism for every i. Consequently, for every A-module M, there is a canonical isomorphism
$$D^i(S^{-1}A|R,S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R,M)$$
for every i.

(5) Let A and B be R-algebras such that $\mathrm{Tor}_i^R(A,B) = 0$ for $0 < i \leq n$. Then, for every $A \otimes_R B$-module M, the canonical map
$$D^i(A \otimes_R B|R,M) \longrightarrow D^i(A|R,M) \oplus D^i(B|R,M)$$

is an isomorphism for $i \le n$. In particular, suppose that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that the diagonal

$$\Delta : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion—for example, suppose that A is *étale* over R. Then

$$D^i(A|R, M) = 0$$

for every $i > 0$ and every A-module M.

(6) Let R be noetherian, let A be an R-algebra essentially of finite type, and let E be a flat A-module. Then the canonical map

$$E \otimes_A D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \otimes_A M)$$

is an isomorphism for every $i \ge 0$ and every A-module M.

(7) If R is noetherian, A is an R-algebra essentially of finite type, and M is an A-module of finite type, then $D^i(A|R, M)$ is an A-module of finite type for every i.

(8) Let P be a polynomial algebra over R, and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. Then

$$D^0(A|R, M) = \mathrm{Der}_R(A, M),$$

$$\begin{aligned} D^1(A|R, M) &= \mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big) \\ &= \big\{ \text{ isomorphism classes of commutative } R\text{-algebra extensions of } A \text{ by } M \ \big\}, \\ D^2(A|R, M) &= \mathrm{Coker}\big(H^1(K_I, M) \longrightarrow \mathrm{Hom}_A(H_1(K_I), M)\big). \end{aligned}$$

Proof.

(1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A-modules, then, for every polynomial algebra P over R, the sequence

$$0 \longrightarrow \mathrm{Der}_R(P, M') \longrightarrow \mathrm{Der}_R(P, M) \longrightarrow \mathrm{Der}_R(P, M'') \longrightarrow 0$$

is exact. Hence, by 3.3(c),

$$0 \longrightarrow \mathrm{Der}_R(-, M') \longrightarrow \mathrm{Der}_R(-, M) \longrightarrow \mathrm{Der}_R(-, M'') \longrightarrow 0$$

is an exact sequence of sheaves.

(2) The functor $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{C}_{A|S}$, and

$$\mathrm{Der}_R(B, M) = \ker\big(\mathrm{Der}_S(B, M) \longrightarrow \mathrm{Der}_S(R, M)\big)$$

for every $B \in \mathrm{Ob}(\underline{C}_{A|R})$. The assertion therefore follows from 3.7(c).

(3) Let

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A\otimes_R S|S}, \qquad X \longmapsto X \otimes_R S.$$

Then $\widetilde{S}\,\mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$, so the assertion follows from 3.7(a).

(4) This follows from 3.9. Indeed, $F = \mathrm{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \mathrm{Ob}(\underline{\widetilde{C}}_{S^{-1}A|A})$ is also additive, since

$$\begin{aligned} F_A(X) &= \ker\bigl(F(X) \longrightarrow F(A)\bigr) \\ &= \ker\bigl(\mathrm{Der}_R(X, N) \longrightarrow \mathrm{Der}_R(A, N)\bigr) \\ &= \mathrm{Der}_A(X, N). \end{aligned}$$

Thus $F_A = \mathrm{Der}_A(-, N)$.

(5) The first assertion follows from 3.8(a). Now suppose that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that

$$\Delta : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion. Let $\varphi : A \otimes_R A \longrightarrow A$ be the multiplication map. For every maximal ideal $\mathfrak{m}$ of A, the induced map

$$(A \otimes_R A)_{\varphi^{-1}(\mathfrak{m})} \longrightarrow A_{\mathfrak{m}}$$

is an isomorphism. The first assertion, together with 3.12(4), therefore implies that the diagonal map

$$\Delta_{\mathfrak{m}} : D^i(A|R, M)_{\mathfrak{m}} \longrightarrow D^i(A|R, M)_{\mathfrak{m}} \oplus D^i(A|R, M)_{\mathfrak{m}}$$

is an isomorphism for every maximal ideal $\mathfrak{m}$ of A. Consequently,

$$\Delta : D^i(A|R, M) \longrightarrow D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, and hence $D^i(A|R, M) = 0$ for every $i > 0$.

(6) In view of 3.12(4), we may assume that A is an R-algebra of finite type. Consider the morphism of sheaves

$$E \otimes_A \mathrm{Der}_R(-, M) \longrightarrow \mathrm{Der}_R(-, E \otimes_A M)$$

on $\underline{\widetilde{C}}_{A|R}$. Since E is flat over A, the map

$$E \otimes_A \mathrm{Der}_R(B, M) \xrightarrow{\sim} \mathrm{Der}_R(B, E \otimes_A M)$$

is an isomorphism whenever $B \in \mathrm{Ob}(\underline{C}_{A|R})$ is of finite type over R. By 3.3(b), it follows that

$$D^i\bigl(A|R, E \otimes_A \mathrm{Der}_R(-, M)\bigr) \longrightarrow D^i\bigl(A|R, \mathrm{Der}_R(-, E \otimes_A M)\bigr)$$

is an isomorphism. On the other hand, $E \otimes_A -$ is exact, and therefore

$$E \otimes_A D^i(A|R, M) = D^i\bigl(A|R, E \otimes_A \mathrm{Der}_R(-, M)\bigr).$$

This proves the assertion.

(7) This follows from 3.3(a).

(8) In view of 3.10 and 3.11, it suffices to show that

$$\mathrm{Coker}\bigl(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\bigr) = \mathrm{Exalcomm}(A|R, M),$$

where $\mathrm{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R-algebra extensions of A by M. Let

$$0 \longrightarrow M \longrightarrow A' \longrightarrow A \longrightarrow 0$$

be such an extension. Since P is a polynomial algebra over R, we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0. \end{array}$$

The map $I \longrightarrow M$ is determined by the extension A', up to the action of $\mathrm{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \mathrm{Exalcomm}(A|R, M) \longrightarrow \mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big).$$

Conversely, a P-linear map $I \longrightarrow M$ yields, by pushout, an extension fitting into a diagram

$$\begin{tikzcd}
0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0.
\end{tikzcd}$$

Two P-linear maps $I \rightrightarrows M$ that differ by an element of $\mathrm{Der}_R(P, M)$ yield isomorphic extensions. Hence there is a map

$$\Psi : \mathrm{Coker}\big(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)\big) \longrightarrow \mathrm{Exalcomm}(A|R, M).$$

It is straightforward to verify that Φ and Ψ are inverse to each other.

□

Corollary 3.13.

(a) The R-algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A-module M.

(b) Let R be noetherian and let A be an R-algebra of finite type. Then A is a relative complete intersection over R if and only if $D^2(A|R, M) = 0$ for every A-module M.

(c) Let R be noetherian and let A be an R-algebra of finite type. For every multiplicative subset S of A, there is a canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M) \qquad (i \geq 0).$$

(d) Let R be noetherian and let A be a local R-algebra essentially of finite type. Write $\widehat{A}$ for the $\mathfrak{m}$-adic completion of A, where $\mathfrak{m}$ is the maximal ideal of A. Then, for every A-module E of finite type, there is a canonical isomorphism

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \xrightarrow{\sim} \widehat{A} \otimes_A D^1(A|R, E).$$

Proof.
Part (a) follows immediately from $D^1(A|R, M) = \mathrm{Exalcomm}(A|R, M)$.
For (b), let P be a polynomial algebra over R in finitely many variables, and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. After localization, we may suppose that I is generated by a P-regular sequence. Then $H_1(K_I) = 0$, and hence

$$D^2(A|R, M) = \mathrm{Coker}\big(H^1(K_I, M) \longrightarrow \mathrm{Hom}_A(H_1(K_I), M)\big) = 0.$$

Conversely, suppose that $D^2(A|R, M) = 0$ for every A-module M. Choose an exact sequence

$$0 \longrightarrow K \longrightarrow F \longrightarrow I \longrightarrow 0,$$

where F is a free P-module of finite type. If

$$K_0 = \mathrm{Im}\big(\textstyle\bigwedge^2 F \longrightarrow \bigwedge^1 F\big),$$

then $H_1(K_I) = K/K_0$. Since $D^2(A|R, H_1(K_I)) = 0$, the map

$$H^1(K_I, H_1(K_I)) \longrightarrow \mathrm{Hom}_A(H_1(K_I), H_1(K_I))$$

is surjective. Consequently, the exact sequence

$$0 \longrightarrow K/K_0 \longrightarrow F/IF \longrightarrow I/I^2 \longrightarrow 0$$

splits as a sequence of A-modules. In particular, I/I^2 is projective. After localizing A once more, we may suppose that I/I^2 is free and that $F/IF \longrightarrow I/I^2$ is an isomorphism. The split exact sequence then gives $H_1(K_I) = K/K_0 = 0$. Thus I is generated by a P-regular sequence, so A is a relative complete intersection over R.

Part (c) follows immediately from 3.12(4) and (6).

For (d), 3.12(6) first gives canonical isomorphisms

$$D^i(A|R, \widehat{A} \otimes_A E) \xrightarrow{\sim} \widehat{A} \otimes_A D^i(A|R, E)$$

for every i. It therefore remains to prove that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is an isomorphism. We use the description of D^1 in terms of algebra extensions.

Let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow A' \xrightarrow{\varphi} \widehat{A} \longrightarrow 0$$

be an R-algebra extension whose restriction to A splits. Since $\widehat{A} \otimes_A E$ is an $\widehat{A}$-module of finite type, A' is a local R-algebra complete for its $\mathfrak{m}'$-adic topology, where $\mathfrak{m}'$ is the maximal ideal of A'. If $\psi : A \longrightarrow A'$ is an R-algebra map lifting the canonical map $A \longrightarrow \widehat{A}$, then ψ extends uniquely to a map $\widehat{\psi} : \widehat{A} \longrightarrow A'$ satisfying $\varphi \circ \widehat{\psi} = \mathrm{id}_{\widehat{A}}$. This proves that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is injective.

Now let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow B \xrightarrow{\varphi} A \longrightarrow 0$$

be an R-algebra extension, and put $\widehat{E} = \widehat{A} \otimes_A E$. We first claim that the topology on $\widehat{E}$ induced by the $\mathfrak{m}_B$-adic topology on B coincides with its topology as an $\widehat{A}$-module. Indeed, since A is essentially of finite type over R, there is a subalgebra $B_0 \subset B$ such that B_0 is a local R-algebra essentially of finite type and $\varphi(B_0) = A$. Put $E_0 = \widehat{E} \cap B_0$, and denote the maximal ideals of B, B_0, and A by $\mathfrak{m}_B$, $\mathfrak{m}_0$, and $\mathfrak{m}$, respectively. Then

$$\mathfrak{m}_B = \mathfrak{m}_0 + \widehat{E}, \qquad \mathfrak{m}_B^{n+1} = \mathfrak{m}_0^{n+1} + \mathfrak{m}^n \widehat{E}$$

for every n. Since B_0 is noetherian, the Artin–Rees theorem gives an integer $r > 0$ such that, for every $n \geq r$,

$$\begin{aligned} \widehat{E} \cap \mathfrak{m}_B^{n+1} &= \mathfrak{m}^n \widehat{E} + \widehat{E} \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + E_0 \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + \mathfrak{m}^{n+1-r}(E_0 \cap \mathfrak{m}_0^r) \subset \mathfrak{m}^{n+1-r} \widehat{E}. \end{aligned}$$

This proves the assertion about the two topologies. In particular, $\widehat{E}$ is complete for the topology induced by the $\mathfrak{m}_B$-adic topology on B, and hence completion gives an exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \widehat{E} & \longrightarrow & B & \longrightarrow & A & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \widehat{E} & \longrightarrow & \widehat{B} & \longrightarrow & \widehat{A} & \longrightarrow & 0. \end{array}$$

Here $\widehat{B}$ denotes the $\mathfrak{m}_B$-adic completion of B. Thus

$$D^1(\widehat{A}|R, \widehat{E}) \longrightarrow D^1(A|R, \widehat{E})$$

is surjective, and the proof is complete. $\square$

Corollary 3.14. Let R be noetherian, and let A be an R-algebra of finite type that is smooth everywhere except at one closed point $x \in \operatorname{Spec}(A)$. Then there are isomorphisms

$$D^1(A|R, A) \simeq D^1(A_x|R, A_x) \simeq D^1(\widehat{A}_x|R, \widehat{A}_x).$$

Proof. Since A is smooth over R away from x, $D^1(A|R, A)$ is an A-module of finite type supported on $\{x\}$. Hence

$$D^1(A|R, A) = D^1(A|R, A)_x = D^1\widehat{(A|R, A)}_x.$$

The assertion now follows from 3.13. $\square$

Remark 3.15. Let R be a topological ring and A a topological R-algebra. For every A-module E, define

$$D^i_{\text{top}}(A|R, E) = \varinjlim_{\alpha} D^i(A_\alpha|R_\alpha, A_\alpha \otimes_A E),$$

where $A_\alpha = A/J_\alpha$, $R_\alpha = R/I_\alpha$, and J_α and I_α are open in A and R, respectively. Thus A is formally smooth as a topological R-algebra precisely when $D^1_{\text{top}}(A|R, E) = 0$ for every A-module E.

Now let X be a scheme over a ring R, and let $\underline{E}$ be a quasi-coherent $\mathcal{O}_X$-module. We define sheaves

$$\underline{D}^i_{X|R}(\underline{E}) = \underline{D}^i(X|R, \underline{E})$$

on X from the presheaves on affine open subsets $U \subset X$ given by

$$\big[\underline{D}^i_{X|R}(\underline{E})\big](U) = D^i\big(\Gamma(U, \mathcal{O}_X)|R, \Gamma(U, \underline{E})\big).$$

If R is noetherian and X is of finite type over R, the localization property 3.13(c) shows that these presheaves canonically form quasi-coherent $\mathcal{O}_X$-modules.

When no confusion is possible, we abbreviate

$$\underline{D}^i_{X|R} = \underline{D}^i_{X|R}(\mathcal{O}_X).$$

All the assertions of 3.13 have corresponding formulations for the quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$. We leave those formulations to the reader and record below the properties most relevant to our purpose.

Corollary 3.15. Let R be noetherian and let X be an R-scheme of finite type.

(a) For every coherent $\mathcal{O}_X$-module $\underline{E}$, $\underline{D}^i_{X|R}(\underline{E})$ is again coherent for every i. Moreover,

$$\operatorname{Supp} \underline{D}^i_{X|R}(\underline{E}) \subset X^{\text{Sing}} \qquad (i > 0),$$

where

$$X^{\text{Sing}} = \{x \in X \mid X \text{ is not smooth over } R \text{ at } x\}.$$

(b) If X is flat over R, then for every R-algebra S there is a canonical isomorphism

$$\underline{D}^i_{X \otimes_R S\,|\,S}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$$

for every i.

(c) Suppose that X is a relative complete intersection over R. For every surjective ring homomorphism $S \longrightarrow R$, there is an exact sequence

$$0 \longrightarrow \underline{D}^1_{X|R}(\underline{E}) \longrightarrow \underline{D}^1_{X|S}(\underline{E}) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_X}\big((I/I^2)\otimes_R \mathcal{O}_X, \underline{E}\big) \longrightarrow 0,$$

where $I = \ker(S \longrightarrow R)$. If, moreover, R is local with residue field k and X is flat over R, then

$$0 \longrightarrow \underline{D}^1_{X_0|k}(\mathcal{O}_{X_0}) \longrightarrow \underline{D}^1_{X|S}(\mathcal{O}_{X_0}) \longrightarrow D^1(R|S,k)\otimes_k \mathcal{O}_{X_0} \longrightarrow 0,$$

where $X_0 = X \otimes_R k$.

Proof. Part (a) follows from the finiteness and low-degree descriptions in 3.12, while (b) follows from 3.12(3).

For (c), because X is a relative complete intersection over R, we have $\underline{D}^i_{X|R}(\underline{E}) = 0$ for every $i \geq 2$. The transitivity sequence 3.12(2) therefore gives an exact sequence

$$\begin{aligned} 0 \longrightarrow \underline{D}^0_{X|R}(\underline{E}) \longrightarrow \underline{D}^0_{X|S}(\underline{E}) \longrightarrow \underline{D}^0_{R|S}(\underline{E}) \longrightarrow \underline{D}^1_{X|R}(\underline{E}) \\ \longrightarrow \underline{D}^1_{X|S}(\underline{E}) \longrightarrow \underline{D}^1_{R|S}(\underline{E}) \longrightarrow 0, \end{aligned}$$

where the quasi-coherent sheaves $\underline{D}^i_{R|S}(\underline{E})$ on X are defined on every affine open subset $U \subset X$ by

$$\big[\underline{D}^i_{R|S}(\underline{E})\big](U) = D^i(R|S, \underline{E}(U)).$$

Since $S \longrightarrow R$ is surjective, $\underline{D}^0_{R|S}(\underline{E}) = 0$; and the low-degree description in 3.12(8) gives

$$\underline{D}^1_{R|S}(\underline{E}) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}\big((I/I^2)\otimes_R \mathcal{O}_X, \underline{E}\big).$$

This proves the first exact sequence.

If X is flat over R, then, for every quasi-coherent $\mathcal{O}_{X_0}$-module $\underline{E}$, there are canonical isomorphisms

$$\underline{D}^i_{X_0|k}(\underline{E}) \xrightarrow{\sim} \underline{D}^i_{X|R}(\underline{E})$$

for every i. In particular,

$$\underline{D}^1_{X_0|k}(\mathcal{O}_{X_0}) \xrightarrow{\sim} \underline{D}^1_{X|R}(\mathcal{O}_{X_0}),$$

from which the second exact sequence follows. □

Remark 3.16. The quasi-coherent sheaves $\underline{D}^i_{X|R}(\underline{E})$ defined above should occur as the terms $E_2^{0,i}$ of a spectral sequence relating the local theory to the global theory. L. Illusie has obtained such a theory in *Lecture Notes in Mathematics* 239. His method uses "homotopical algebra," generalizing the methods of M. André and D. Quillen, rather than "cohomological algebra." It seems that the approach taken here can be globalized by means of Grothendieck cohomology on the category of schemes, equipped with an appropriate topology generalizing the one on the affine category.

4. Formal moduli of deformations

Among the natural cofibered categories whose isomorphism classes of objects are generally not prorepresentable but nevertheless admit a formal versal object is the cofibered category of deformations. We return to the situation and notation of section 1. Thus, throughout this section, Λ is a fixed complete noetherian local ring with residue field k, and $\underline{C}_\Lambda$ is the category of artinian local Λ-algebras with residue field k.

Let X_0 be a fixed scheme over k. By an (infinitesimal) deformation of X_0 we mean a pair (R, X), where $R \in \mathrm{Ob}(\underline{C}_\Lambda)$ and X is a flat R-scheme, together with a commutative diagram

$$\begin{tikzcd}
X_0 \arrow[r, "i_X"] \arrow[d] & X \arrow[d, "\text{flat}"] \\
\mathrm{Spec}(k) \arrow[r] & \mathrm{Spec}(R)
\end{tikzcd}$$

such that the induced morphism

$$X_0 \longrightarrow X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$$

is an isomorphism. Since R is artinian local, $i_X : X_0 \longrightarrow X$ is necessarily a closed immersion and a homeomorphism on the underlying topological spaces.

If (R, X) and (R', X') are deformations of X_0 over R and R', respectively, an arrow

$$(R', X') \longrightarrow (R, X)$$

consists of a map $\varphi : R' \longrightarrow R$ in $\underline{C}_\Lambda$ and a morphism of schemes $\Phi : X \longrightarrow X'$ fitting into a commutative diagram

$$\begin{tikzcd}
X \arrow[rr, "\Phi"] \arrow[ddd] & & X' \arrow[ddd] \\
& X_0 \arrow[ul, "i_X"'] \arrow[ur, "i_{X'}"] \arrow[d] & \\
& \mathrm{Spec}(k) \arrow[dl] \arrow[dr] & \\
\mathrm{Spec}(R) \arrow[rr, "\mathrm{Spec}(\varphi)"'] & & \mathrm{Spec}(R').
\end{tikzcd}$$

This defines the category $\underline{M}_{X_0}$ of infinitesimal deformations of X_0. It is equipped with the functor

$$p : \underline{M}_{X_0} \longrightarrow \underline{C}_\Lambda, \qquad (R, X) \longmapsto R.$$

Let $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$ and let $\varphi : R \longrightarrow S$ be a map in $\underline{C}_\Lambda$. There is a commutative diagram

$$\begin{tikzcd}
X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S) \arrow[r, "p_1"] \arrow[d, "\text{flat}"'] & X \arrow[d, "\text{flat}"] \\
\mathrm{Spec}(S) \arrow[r, "\mathrm{Spec}(\varphi)"'] & \mathrm{Spec}(R),
\end{tikzcd}$$

and the φ-arrow

$$(\varphi, p_1) : (R, X) \longrightarrow \big(S, X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(S)\big)$$

is cocartesian in $\underline{M}_{X_0}$. Hence $\underline{M}_{X_0}$ is a cofibered category over $\underline{C}_\Lambda$. We record the following facts.

4.1(a). Let

$$(1_R, \Phi) : (R, X_1) \longrightarrow (R, X_2)$$

be an arrow in $\underline{M}_{X_0}$. Thus $\Phi : X_2 \longrightarrow X_1$ is an R-morphism whose reduction

$$X_2 \otimes_R k \longrightarrow X_1 \otimes_R k$$

is an isomorphism. Since R is artinian and the X_i are flat over R, Φ is an isomorphism. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b). Suppose that X_0 is affine, say $X_0 = \mathrm{Spec}(A_0)$. Then every $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$ has X affine as well. Consequently, for every $R \in \mathrm{Ob}(\underline{C}_\Lambda)$, taking global sections identifies $\underline{M}_{X_0}(R)$ with the following category. An object is a flat R-algebra A together with an R-algebra map $j_A : A \longrightarrow A_0$ inducing an isomorphism

$$A \otimes_R k \xrightarrow{\sim} A_0.$$

A morphism $\Phi : A_1 \longrightarrow A_2$ is an R-algebra map for which the diagram

$$\begin{tikzcd} A_1 \arrow[r, "\Phi"] \arrow[d, "j_{A_1}"'] & A_2 \arrow[d, "j_{A_2}"] \\ A_0 \arrow[r, equal] & A_0 \end{tikzcd}$$

commutes.

4.1(c). Consider the cofibered category $\widehat{\underline{M}}_{X_0}$ in groupoids over $\widehat{\underline{C}}_\Lambda$. By definition, for every $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$, the category $\widehat{\underline{M}}_{X_0}(R)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over $\mathrm{Spf}(R)$, such that

$$\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\mathfrak{m}^{n+1}$$

is flat over $R/\mathfrak{m}^{n+1}$ for every $n \geq 0$. If X_0 is noetherian, this is equivalent to requiring $\mathfrak{X}$ to be flat over $\mathrm{Spf}(R)$. Thus, in that case, $\widehat{\underline{M}}_{X_0}$ is canonically equivalent over $\widehat{\underline{C}}_\Lambda$ to the category of deformations of X_0 over the objects of $\widehat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d). Let $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$, so that there is a commutative diagram

$$\begin{tikzcd} X_0 \arrow[r, "i_X"] \arrow[d] & X \arrow[d, "\text{flat}"] \\ \mathrm{Spec}(k) \arrow[r] & \mathrm{Spec}(R) \end{tikzcd}$$

whose induced map $X_0 \longrightarrow X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k)$ is an isomorphism.

Let $U_0 \subset X_0$ be an open subscheme. Since $i_X : X_0 \longrightarrow X$ is a closed immersion and a homeomorphism on underlying spaces, restriction gives a commutative diagram

$$\begin{tikzcd} U_0 \arrow[r, "i_X|_{U_0}"] \arrow[d] & X|_{U_0} \arrow[d, "\text{flat}"] \\ \mathrm{Spec}(k) \arrow[r] & \mathrm{Spec}(R). \end{tikzcd}$$

The induced map

$$U_0 \longrightarrow (X|_{U_0}) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(k) = (X \otimes_R k)|_{U_0}$$

is an isomorphism, so $(R, X|_{U_0})$ is a deformation of U_0 over R. Moreover, every arrow $(\varphi, \Phi) : (R, X) \longrightarrow (R', X')$ in $\underline{M}_{X_0}$ restricts to an arrow

$$(\varphi, \Phi|_{U_0}) : (R, X|_{U_0}) \longrightarrow (R', X'|_{U_0})$$

in $\underline{M}_{U_0}$. We therefore have a restriction functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0}$$

over $\underline{C}_\Lambda$.

One may likewise localize and complete at a point. Let $x_0 \in X_0$. If $(R, X) \in \mathrm{Ob}(\underline{M}_{X_0})$, then $\mathrm{Spec}(\mathcal{O}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\mathcal{O}_{X_0,x_0})$ over R, while $\mathrm{Spec}(\widehat{\mathcal{O}}_{X,x_0})$ is a deformation of $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})$ over R; the hats denote completion for the topology defined by powers of the maximal ideal. Thus there are functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})}.$$

We first prove that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. Consider a diagram

$$\begin{tikzcd} & R' \arrow[d] \\ R \arrow[r, "\text{surjective}"'] & \bar{R} \end{tikzcd}$$

in $\underline{C}_\Lambda$, and let E and E' be flat modules over R and R', respectively. Then, for every R'-homomorphism

$$E' \longrightarrow \bar{E} = E \otimes_R \bar{R},$$

the fiber product $E \times_{\bar{E}} E'$ is flat as a module over $R \times_{\bar{R}} R'$.

Proof. Let

$$0 \longrightarrow I \longrightarrow R \longrightarrow \bar{R} \longrightarrow 0$$

be exact, so that

$$0 \longrightarrow I \longrightarrow R \times_{\bar{R}} R' \longrightarrow R' \longrightarrow 0$$

is exact. Since E is flat over R,

$$0 \longrightarrow I \otimes_R E \longrightarrow E \longrightarrow \bar{E} \longrightarrow 0$$

is exact, and hence we obtain the exact sequence

$$(*) \qquad 0 \longrightarrow I \otimes_R E \longrightarrow E \times_{\bar{E}} E' \longrightarrow E' \longrightarrow 0.$$

Set $S = R \times_{\bar{R}} R'$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S-module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R'-module and

$$\mathrm{Tor}_1^S(E \times_{\bar{E}} E', R') = 0.$$

But

$$I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E'), \qquad E' = R' \otimes_S (E \times_{\bar{E}} E'),$$

and therefore the exact sequence $(*)$ may be rewritten as

$$0 \longrightarrow I \otimes_S (E \times_{\bar{E}} E') \longrightarrow E \times_{\bar{E}} E' \longrightarrow R' \otimes_S (E \times_{\bar{E}} E') \longrightarrow 0.$$

Thus

$$\mathrm{Tor}_1^S(R', E \times_{\bar{E}} E') = 0, \qquad R' \otimes_S (E \times_{\bar{E}} E') = E',$$

and E' is flat over R'. It follows that $E \times_{\bar{E}} E'$ is flat over S. $\square$

Corollary 4.3. Given a diagram

$$\begin{tikzcd}[column sep=small] (R,X) \arrow[dr, "(\varphi,\Phi)"'] & & (R',X') \arrow[dl, "(\varphi',\Phi')"] \\ & (\bar{R},\bar{X}) & \end{tikzcd}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \amalg_{\bar{X}} X'$ (in the category of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided that $\varphi : R \longrightarrow \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof. For every affine open $U_0 \subset X_0$, by definition we have

$$(X \amalg_{\bar{X}} X')|_{U_0} \simeq \operatorname{Spec}\Big(\Gamma(U_0, X) \times_{\Gamma(U_0,\bar{X})} \Gamma(U_0, X')\Big).$$

Consequently, if $\varphi : R \longrightarrow \bar{R}$ is surjective, $X \amalg_{\bar{X}} X'$ is flat over $R \times_{\bar{R}} R'$ by Lemma 4.2. Moreover, the morphism

$$i_X \amalg_{i_{\bar{X}}} i_{X'} : X_0 \longrightarrow X \amalg_{\bar{X}} X'$$

induces an isomorphism

$$X_0 \longrightarrow (X \amalg_{\bar{X}} X') \otimes_S k, \qquad S = R \times_{\bar{R}} R'.$$

Therefore

$$\big(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X'\big) \in \operatorname{Ob}(\underline{M}_{X_0}),$$

and, in $\underline{M}_{X_0}$, we have

$$\big(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X'\big) = (R, X) \times_{(\bar{R},\bar{X})} (R', X').$$

Thus $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$. $\square$

Lemma 4.4. For every scheme X_0 over k, there is a canonical exact sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}),$$

where the $\underline{D}^i_{X_0}$ are as defined in section 3.

Proof. First suppose that X_0 is affine, and put $\Gamma(X_0, \mathcal{O}_{X_0}) = A_0$. As noted in 4.1(b), the category $\underline{M}_{X_0}(k[\varepsilon])$ is canonically equivalent to a category $\underline{E}$ whose objects are flat $k[\varepsilon]$-algebras A equipped with a $k[\varepsilon]$-algebra map $j_A : A \longrightarrow A_0$ inducing an isomorphism

$$k \otimes_{k[\varepsilon]} A \xrightarrow{\sim} A_0.$$

Equivalently, an object of $\underline{E}$ is a k-algebra A equipped with an exact sequence

$$0 \longrightarrow A_0 \xrightarrow{\varepsilon} A \xrightarrow{j_A} A_0 \longrightarrow 0$$

in which $\varepsilon(A_0)^2 = 0$ as an ideal of A.

An arrow $A_1 \longrightarrow A_2$ in $\underline{E}$, in the first description, is a $k[\varepsilon]$-algebra map $\Phi : A_1 \longrightarrow A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. In the second description it is a k-algebra map $\Phi : A_1 \longrightarrow A_2$ yielding a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & A_0 \arrow[r, "\varepsilon"] \arrow[d, equal] & A_1 \arrow[r, "j_{A_1}"] \arrow[d, "\Phi"] & A_0 \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & A_0 \arrow[r] & A_2 \arrow[r, "j_{A_2}"] & A_0 \arrow[r] & 0. \end{tikzcd}$$

It follows that, via $\Gamma(X_0, -)$, $[\underline{M}_{X_0}(k[\varepsilon])]$ is canonically isomorphic to the set of isomorphism classes of k-algebra extensions of A_0 by A_0. In other words, there is a canonical bijection

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}^1_{X_0})$$

which plainly respects the vector-space structures on both sides.

In general, for every affine open $U_0 \subset X_0$, restriction and the affine case give the composite map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{U_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(U_0, \underline{D}^1_{X_0}).$$

These maps give a canonical map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}).$$

Its kernel consists of the isomorphism classes of locally trivial deformations of X_0 over $k[\varepsilon]$, that is, of deformations locally isomorphic to $X_0 \otimes_k k[\varepsilon]$. The sheaf of automorphisms of $X_0 \otimes_k k[\varepsilon]$ inducing the identity on X_0 is

$$\underline{D}^0_{X_0} = \underline{\mathrm{Der}}_k(\mathcal{O}_{X_0}, \mathcal{O}_{X_0}).$$

The isomorphism classes of such locally trivial deformations are therefore canonically identified with $H^1(X_0, \underline{D}^0_{X_0})$, which gives the asserted exact sequence. $\square$

Theorem 4.5. Let X_0 be a scheme of finite type over k for which either

(i) X_0 is proper over k, or

(ii) X_0 is affine and

$$X_0^{\mathrm{sing}} = \{x \in X_0 \mid X_0 \text{ is not smooth over } k \text{ at } x\}$$

is finite.

(a) The category $\underline{M}_{X_0}$ admits a formal versal object over $\underline{C}_\Lambda$, called a formal versal deformation of X_0. If $(R, \mathcal{X})$ is a formal versal deformation of X_0, then for every $(S, \mathcal{Y}) \in \mathrm{Ob}(\widehat{\underline{M}}_{X_0})$ there is a map

$$(\varphi, \Phi) : (R, \mathcal{X}) \longrightarrow (S, \mathcal{Y})$$

in $\widehat{\underline{M}}_{X_0}$ such that the following diagram is cartesian:

$$\begin{tikzcd} \mathcal{Y} \arrow[r, "\Phi"] \arrow[d] & \mathcal{X} \arrow[d] \\ \mathrm{Spf}(S) \arrow[r, "\mathrm{Spf}(\varphi)"'] & \mathrm{Spf}(R). \end{tikzcd}$$

Furthermore, a formal versal deformation $(R, \mathcal{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \longrightarrow R$ in $\underline{C}_\Lambda$ and every deformation (R', X') of X_0 over R', the homomorphism

$$\mathrm{Aut}_{R'}(X' \mid X_0) \longrightarrow \mathrm{Aut}_R(X' \otimes_{R'} R \mid X_0)$$

is surjective.

(b) If X_0 is proper over k, then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$.

Proof. **(a)** By Corollary 4.3, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$. By 1.11 and 2.9(b), it therefore suffices to prove that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k.

Consider the exact sequence

$$(*) \qquad 0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}).$$

If X_0 is proper over k, both $H^1(X_0, \underline{D}^0_{X_0})$ and $H^0(X_0, \underline{D}^1_{X_0})$ are finite-dimensional, since the $\underline{D}^i_{X_0}$ are coherent for all $i \geq 0$ by 2.13(a).

Now suppose that X_0 is affine and X_0^{sing} is finite. The sheaf $\underline{D}^1_{X_0}$ is coherent and

$$\operatorname{Supp} \underline{D}^1_{X_0} \subset X_0^{\text{sing}}$$

by 2.13(a). Hence $H^0(X_0, \underline{D}^1_{X_0})$ is finite-dimensional over k, whereas $H^1(X_0, \underline{D}^0_{X_0}) = 0$ because X_0 is affine. In either case, the exact sequence $(*)$ shows that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k. The second assertion of part (a) follows from 2.7.

(b) If X_0 is proper, then

$$\operatorname{Aut}(X_0 \otimes_k k[\varepsilon] \mid X_0) = H^0(X_0, \underline{D}^0_{X_0})$$

is finite-dimensional, and the assertion follows from 2.13. $\square$

Remark 4.6(a). Let X_0 be affine. A formal versal deformation of X_0 exists if and only if the support of $\underline{D}^1_{X_0}$ is a finite set. This does not imply, however, that all nonsmooth points of X_0 are isolated: there are affine schemes X_0 over k having a nonisolated nonsmooth point for which $\operatorname{Supp} \underline{D}^1_{X_0}$ is finite.

Remark 4.6(b). Henceforth, a deformation of X_0 means an object of $\underline{M}_{X_0}$. Where confusion is possible, such an object will be called an infinitesimal deformation of X_0.

Remark 4.6(c). Let X_0 be a scheme over k. We have fixed a complete noetherian local ring Λ with residue field k and considered deformations of X_0 over $\underline{C}_\Lambda$.

In practice one usually takes $\Lambda = k$, or $\Lambda = W(k)$ when k is a perfect field of positive characteristic. The inclusion $\underline{C}_k \subset \underline{C}_\Lambda$ is fully faithful, and $\underline{M}_{X_0}|_{\underline{C}_k}$ is simply the category of deformations of X_0 over the objects of $\underline{C}_k$. Consequently, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_\Lambda$, then

$$(k \otimes_\Lambda R,\ k \otimes_\Lambda \mathfrak{X})$$

is a formal versal deformation of X_0 over $\underline{C}_k$. It follows, for example, that

$$\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim(k \otimes_\Lambda R)$$

does not depend on the choice of Λ, but only on X_0.

The preceding theorem admits many generalizations. For example, consider a pair $Y_0 \longrightarrow X_0$, where X_0 is a scheme of finite type over k and Y_0 is a fixed closed subscheme of X_0. For $R \in \operatorname{Ob}(\underline{C}_\Lambda)$, a deformation of (X_0, Y_0) over R is a commutative diagram

$$\begin{tikzcd}
Y_0 \arrow[rrr] \arrow[dr] \arrow[ddr] & & & Y \arrow[dl] \arrow[ddl, "\text{flat}"] \\
& X_0 \arrow[r] \arrow[d] & X \arrow[d, "\text{flat}"'] & \\
& \operatorname{Spec}(k) \arrow[r] & \operatorname{Spec}(R) &
\end{tikzcd}$$

such that the induced map

$$(X_0, Y_0) \longrightarrow \big(X \times_{\operatorname{Spec}(R)} \operatorname{Spec}(k), Y \times_{\operatorname{Spec}(R)} \operatorname{Spec}(k)\big)$$

is an isomorphism. Defining morphisms

$$(R'; X', Y') \longrightarrow (R; X, Y)$$

of deformations of (X_0, Y_0) in the evident way gives a cofibered category $\underline{M}_{X_0,Y_0}$ in groupoids over $\underline{C}_\Lambda$. This construction is frequently used to rigidify automorphisms in deformation-moduli problems. The argument of Corollary 4.3, based on Lemma 4.2, immediately shows that $\underline{M}_{X_0,Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Lemma 4.7. The sequence

$$0 \longrightarrow H^0\Big(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\Big) \longrightarrow [\underline{M}_{X_0,Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]$$

is exact, where $\underline{I}$ is the ideal sheaf defining Y_0 in X_0; thus

$$0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$$

is exact.

Proof. Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0,Y_0}(k[\varepsilon])$ whose objects have the form

$$\big(k[\varepsilon]; X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]), Y\big).$$

Then

$$[\underline{B}] = \ker\big([\underline{M}_{X_0,Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]\big)\,.$$

The category $\underline{B}$ is the groupoid of deformations of Y_0 over $k[\varepsilon]$ inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon])$. Thus an object is a commutative diagram

$$\begin{tikzcd} Y \arrow[rr, "i_Y"] \arrow[dr, "\text{flat}"'] & & X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]) \arrow[dl] \\ & \mathrm{Spec}(k[\varepsilon]) & \end{tikzcd}$$

whose special fiber is the fixed closed immersion i_{Y_0}. An arrow $(Y_1, i_{Y_1}) \longrightarrow (Y_2, i_{Y_2})$ is a $k[\varepsilon]$-morphism $\Phi : Y_1 \longrightarrow Y_2$ (necessarily an isomorphism) whose special fiber is id_{Y_0} and for which $i_{Y_2} \circ \Phi = i_{Y_1}$.

Equivalently, an object of $\underline{B}$ is a pair (Y, j_Y), where Y is a deformation of Y_0 over $k[\varepsilon]$ and $j_Y : Y \longrightarrow X_0$ is a k-morphism whose composite $Y_0 \longrightarrow Y \xrightarrow{j_Y} X_0$ is i_{Y_0}. An arrow $(Y_1, j_{Y_1}) \longrightarrow (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \longrightarrow Y_2$ whose special fiber is id_{Y_0} and for which $j_{Y_2} \circ \Phi = j_{Y_1}$.

The assignment sending (Y, j_Y) to the diagram

$$\begin{tikzcd} 0 \arrow[r] & \underline{I} \arrow[r] \arrow[d] & \mathcal{O}_{X_0} \arrow[r] \arrow[d, "j_Y"] & \mathcal{O}_{Y_0} \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & \mathcal{O}_{Y_0} \arrow[r, "\varepsilon"] & \mathcal{O}_Y \arrow[r] & \mathcal{O}_{Y_0} \arrow[r] & 0 \end{tikzcd}$$

gives an isomorphism

$$\begin{aligned} [\underline{B}] &\xrightarrow{j_Y|_{\underline{I}}} \mathrm{Hom}_{\mathcal{O}_{X_0}}(\underline{I}, \mathcal{O}_{Y_0}) \\ &= \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0}) \\ &= H^0\Big(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\Big)\,. \end{aligned}$$

This proves exactness. ∎

Remark 4.8. For a more general formula implying Lemma 4.7, see SGA 3, III, 4.4. One may also note the following. For every S-scheme X and every quasi-coherent $\mathcal{O}_X$-module E, one can define quasi-coherent $\mathcal{O}_X$-modules $\underline{D}^i_{X|S}(E)$ for all $i \geq 0$, generalizing the case in which S is affine. As usual, we write

$$\underline{D}^i_{X|S}(\mathcal{O}_X) = \underline{D}^i_{X|S}.$$

For the closed immersion $Y_0 \hookrightarrow X_0$, we then have

$$H^0\Big(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\Big) = H^0(Y_0, \underline{D}^1_{Y_0|X_0}),$$

where $0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$ is exact.

Theorem 4.9. Let X_0 be a k-scheme of finite type and Y_0 a closed subscheme of X_0. Suppose either

(i) X_0 is proper over k, or

(ii) X_0 is affine and smooth over k outside finitely many points, and Y_0 is proper over k.

(a) The category $\underline{M}_{X_0,Y_0}$ admits a formal versal deformation over $\underline{C}_\Lambda$. It prorepresents the functor $[\underline{M}_{X_0,Y_0}]$ if and only if, for every surjective map $R' \longrightarrow R$ in $\underline{C}_\Lambda$ and every deformation $(R'; X', Y')$, the map

$$\mathrm{Aut}_{R'}((X',Y') \mid X_0) \longrightarrow \mathrm{Aut}_R\big((X' \otimes_{R'} R, Y' \otimes_{R'} R) \mid X_0\big)$$

is surjective.

(b) If, in addition,

$$H^0\Big(X_0, \underline{D}^0_{Y_0|k} \times_{\underline{D}^0_{X_0|k}(\mathcal{O}_{Y_0})} \underline{D}^0_{X_0|k}\Big)$$

is finite-dimensional, then $\underline{M}_{X_0,Y_0}$ is smoothly pseudo-representable.

Proof. This follows immediately from Lemmas 4.4 and 4.7 together with 1.11, 2.7, and 2.13. $\square$

We now consider localization and completion in the deformation theory of schemes. We first fix notation. Let X_0 be a k-scheme of finite type and let $x_0 \in X_0$ be a fixed point. There are natural functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})},$$

and hence functors

$$\widehat{\underline{M}}_{X_0} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\mathcal{O}_{X_0,x_0})} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_0})}.$$

As observed in 4.1(c), $\widehat{\underline{M}}_{X_0}$ is canonically equivalent to the category whose objects are pairs $(R, \mathfrak{X})$, where $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$ and $\mathfrak{X}$ is a flat R-adic formal scheme equipped with a fixed isomorphism $\mathfrak{X}_0 \xrightarrow{\sim} X_0$.

When there is no danger of confusion, we denote the images of $(R, \mathfrak{X})$ under localization and completion by $(R, \mathfrak{X}_{(x_0)})$ and $(R, \mathfrak{X}_{(\widehat{x}_0)})$, respectively. Thus, if

$$\mathfrak{X} = \varinjlim_{\nu} \mathfrak{X}_\nu, \qquad \mathfrak{X}_\nu = \mathfrak{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\mathfrak{m}^{\nu+1}),$$

then

$$\mathfrak{X}_{(x_0)} = \varinjlim_{\nu} \mathrm{Spec}(\mathcal{O}_{\mathfrak{X}_\nu, x_0})$$

and

$$\mathfrak{X}_{(\widehat{x}_0)} = \varinjlim_{\nu} \mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X}_\nu, x_0}).$$

Theorem 4.10. Let X_0 be an affine scheme of finite type over k.

(a) If X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$.

(b) Suppose that, for a fixed closed point $x \in X_0$, the open $X_0 - \{x\}$ is smooth over k. Then both functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}$$

are minimally smooth. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0, then $(R, \mathfrak{X}_{(x)})$ and $(R, \mathfrak{X}_{(\widehat{x})})$ are formal versal deformations of $\mathrm{Spec}(\mathcal{O}_{X_0,x})$ and $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})$, respectively.

Proof. We use the notation for the cohomology of commutative algebras from section 3, and write $X_0 = \mathrm{Spec}(A_0)$.

(a) It suffices to show that $P : [\underline{M}_{X_0}] \longrightarrow \underline{C}_\Lambda$ is smooth. Let $(R, \mathrm{Spec}(A))$ be a deformation of $\mathrm{Spec}(A_0)$ over $R \in \mathrm{Ob}(\underline{C}_\Lambda)$. Given a small extension $R' \longrightarrow R$ in $\underline{C}_\Lambda$, so that

$$0 \longrightarrow k \longrightarrow R' \longrightarrow R \longrightarrow 0$$

is exact, we must lift $(R, \mathrm{Spec}(A))$ to R'. The transitivity sequence for D^i with coefficients in A_0, applied to $R' \longrightarrow R \longrightarrow A$, gives

$$\cdots \longrightarrow D^1(A|R, A_0) \longrightarrow D^1(A|R', A_0) \longrightarrow D^1(R|R', k) \otimes_k A_0 \longrightarrow D^2(A|R, A_0) \longrightarrow \cdots.$$

Because $R' \longrightarrow R$ is a small extension, $D^1(R|R', k)$ is one-dimensional over k, generated by the class $[R']$ represented by $0 \longrightarrow k \longrightarrow R' \longrightarrow R \longrightarrow 0$. Since A is flat over R,

$$D^i(A|R, A_0) = D^i(A_0|k, A_0) \qquad (i > 0).$$

In particular, $D^2(A|R, A_0) = 0$ because A_0 is locally a complete intersection, by the low-degree description in 3.12(8). We therefore obtain an exact sequence

$$\longrightarrow D^1(A_0|k, A_0) \longrightarrow D^1(A|R', A_0) \longrightarrow D^1(R|R', k) \otimes_k A_0 \longrightarrow 0.$$

Hence there is an R'-algebra extension A' of A by A_0 whose class maps to $[R'] \otimes 1$. Equivalently, $(R, \mathrm{Spec}(A))$ lifts to the deformation $(R', \mathrm{Spec}(A'))$.

(b) By 3.14 there are isomorphisms

$$D^1(A_0|k, A_0) \xrightarrow{\sim} D^1((A_0)_x|k, (A_0)_x) \xrightarrow{\sim} D^1(\widehat{(A_0)_x}|k, \widehat{(A_0)_x}),$$

and hence

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}(k[\varepsilon])].$$

By 2.7(a), it remains to show that

$$[\underline{M}_{X_0}] \longrightarrow [\underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}] \longrightarrow [\underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})}]$$

are smooth functors.

Let $(R, \mathrm{Spec}(A))$ be a deformation of $X_0 = \mathrm{Spec}(A_0)$, and let $0 \longrightarrow k \longrightarrow R' \longrightarrow R \longrightarrow 0$ be a small extension. As above, the homomorphisms $R' \longrightarrow R \longrightarrow A$ give an exact commutative diagram

$$\begin{tikzcd}
{} \arrow[r] & D^1(A_0|k, A_0) \arrow[r] \arrow[d, "\sim"'] & D^1(A|R', A_0) \arrow[r, "p"] \arrow[d] & D^1(R|R', k) \otimes_k A_0 \arrow[r] \arrow[d] & D^2(A_0|k, A_0) \arrow[r] \arrow[d, "\sim"] & \cdots \\
{} \arrow[r] & D^1(A_0|k, (A_0)_x) \arrow[r] & D^1(A|R', (A_0)_x) \arrow[r] & D^1(R|R', k) \otimes_k (A_0)_x \arrow[r] & D^2(A_0|k, (A_0)_x) \arrow[r] & \cdots \\
{} \arrow[r] & D^1((A_0)_x|k, (A_0)_x) \arrow[r] \arrow[u] & D^1(A_x|R', (A_0)_x) \arrow[r, "p_x"] \arrow[u] & D^1(R|R', k) \otimes_k (A_0)_x \arrow[r] \arrow[u] & D^2((A_0)_x|k, (A_0)_x) \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

The indicated isomorphisms follow from the fact that A_0 is smooth over k away from the single closed point x, so that

$$D^i(A_0|k, (A_0)_x) = D^i(A_0|k, A_0)_x = D^i(A_0|k, A_0) \qquad (i > 0).$$

Suppose we are given an arrow

$$(R', \mathrm{Spec}(B)) \longrightarrow (R, \mathrm{Spec}(A_x))$$

in $[\underline{M}_{\mathrm{Spec}((A_0)_x)}]$. This amounts to a class $[B] \in D^1(A_x|R',(A_0)_x)$ satisfying $p_x([B]) = [R']$. A diagram chase gives a class $[A'] \in D^1(A|R',A_0)$ such that $p([A']) = [R']$ and $A'_x = B$. Thus $(R', \mathrm{Spec}(A'))$ is a deformation of $\mathrm{Spec}(A_0)$ over R' whose localization at x is $\mathrm{Spec}(B)$. This proves that

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0,x})}$$

is smooth.

The smoothness of

$$\underline{M}_{\mathrm{Spec}((A_0)_x)} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{(A_0)_x})}$$

follows from the analogous diagram and the identities

$$D^i((A_0)_x|k,\widehat{(A_0)_x}) = D^i(\widehat{(A_0)_x|k},(A_0)_x) = D^i((A_0)_x|k,(A_0)_x) \qquad (i>0),$$

since x is the only possible nonsmooth point. □

Theorem 4.11. Let X_0 be a separated scheme of finite type over k, smooth away from a finite closed subset

$$X_0^{\mathrm{Sing}} = \{x_1,\dots,x_r\}.$$

Let $U_0^1,\dots,U_0^r$ be affine open neighbourhoods of $x_1,\dots,x_r$ such that $U_0^i \cap X_0^{\mathrm{Sing}} = \{x_i\}$. If $H^2(X_0,\underline{D}^0_{X_0}) = 0$, then the natural functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0^1} \times \dots \times \underline{M}_{U_0^r}$$

is smooth.

Proof. Complete $U_0^1,\dots,U_0^r$ to an affine open covering of X_0 by choosing $U_0^{r+1},\dots,U_0^N$ such that $U_0^j \cap X_0^{\mathrm{Sing}} = \varnothing$ for every $j>r$. Let $R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $(R',U'^1),\dots,(R',U'^r)$ be deformations of $U_0^1,\dots,U_0^r$ over R' such that

$$U'^i \otimes_{R'} R = X|_{U_0^i}, \qquad 1 \le i \le r,$$

for some deformation (R,X) of X_0 over R. We must find a deformation (R',X') of X_0 over R' such that

$$X' \otimes_{R'} R = X, \qquad X'|_{U_0^i} = U'^i \quad (1 \le i \le r).$$

For $j>r$, the scheme U_0^j is smooth over k; hence, by 4.9(a), $X|_{U_0^j}$ lifts to a deformation (R',U'^j) of U_0^j over R'. Thus

$$U'^j \otimes_{R'} R = X|_{U_0^j} \qquad (j>r).$$

For every pair $i \neq j$, the intersection $U_0^i \cap U_0^j$ is affine and smooth over k. We therefore obtain an isomorphism

$$\tau_{ij} : U'^i|_{U_0^i\cap U_0^j} \xrightarrow{\sim} U'^j|_{U_0^i \cap U_0^j}$$

such that $\tau_{ij} \otimes_{R'} R = \mathrm{id}_{X|_{U_0^i\cap U_0^j}}$. On triple intersections, the automorphisms

$$\tau_{ij}\tau_{jk}\tau_{ik}^{-1}$$

define a cohomology class in $H^2(X_0,\underline{D}^0_{X_0})$. This class is the obstruction to gluing the U'^i to a global deformation. It vanishes by hypothesis, so the U'^i glue to a deformation (R',X') with $X' \otimes_{R'} R = X$ and $X'|_{U_0^i} \simeq U'^i$. □

Remark 4.12. The preceding proof assumes that X_0 is separated, but this assumption is unnecessary. Indeed, fix a deformation (R,X) of X_0 and deformations (R',U'^i) of the members U_0^i of an open covering. The deformations (R', V') of open subsets V of X over R' that are compatible with the already-given deformations (R', U'^i) form a "gerbe" (in Giraud's terminology) whose structure sheaf is $\underline{D}^0_{X_0}$. The obstruction to the existence of a section of this gerbe—that is, to a global deformation (R', X') of X compatible with the given data (R', U'^i)—lies in $H^2(X_0, \underline{D}^0_{X_0})$, and hence vanishes under the hypothesis of the theorem.

Corollary 4.13. Let X_0 be a scheme of finite type over k, smooth away from a finite subset

$$X_0^{\mathrm{Sing}} = \{x_1, \dots, x_r\},$$

and suppose that $H^2(X_0, \underline{D}^0_{X_0}) = 0$. Then:
(a) The functor

$$\underline{M}_{X_0} \longrightarrow \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}$$

is smooth. In particular,

$$\dim \underline{M}_{X_0} = \dim_k H^1(X_0, \underline{D}^0_{X_0}) + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}.$$

(b) If, in addition, X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth.

Proof. **(a)** The first assertion follows from 4.10 and 4.9(b). Let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1), \dots, (R_r, \mathfrak{X}_r)$ be formal versal deformations of X_0 and of

$$\mathrm{Spec}(\mathcal{O}_{X_0,x_1}), \dots, \mathrm{Spec}(\mathcal{O}_{X_0,x_r}),$$

respectively. We obtain a commutative diagram

$$\begin{array}{ccc} \underline{h}_R & \longrightarrow & \underline{h}_{R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r} \\ \downarrow & & \downarrow \\ \underline{M}_{X_0} & \longrightarrow & \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}. \end{array}$$

The composite functor

$$\underline{h}_R \longrightarrow \underline{M}_{X_0} \longrightarrow \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}$$

is smooth, whereas

$$\underline{h}_{R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r} \longrightarrow \prod_{i=1}^{r} \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}$$

is minimally smooth. It follows from 2.7(a) that

$$\underline{h}_R \longrightarrow \underline{h}_{R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r}$$

is smooth; equivalently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r$.

By 4.4, the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow \prod_{i=1}^{r} [\underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}(k[\varepsilon])]$$

is exact. Equivalently,

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow \mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow \mathrm{Hom}_{\Lambda\text{-alg}}(R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r, k[\varepsilon])$$

is exact. Consequently, R is a formal power-series ring over $R_1\widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r$ in d variables, where

$$d = \dim_k H^1(X_0, \underline{D}^0_{X_0}).$$

Therefore

$$\begin{aligned}
\dim \underline{M}_{X_0} &= \dim(k \otimes_\Lambda R) \\
&= d + \dim\big(k \otimes_\Lambda (R_1\widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r)\big) \\
&= d + \dim\big((k \otimes_\Lambda R_1)\widehat{\otimes}_k \cdots \widehat{\otimes}_k (k \otimes_\Lambda R_r)\big) \\
&= d + \sum_{i=1}^{r} \dim(k \otimes_\Lambda R_i) \\
&= d + \sum_{i=1}^{r} \dim \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x_i})}.
\end{aligned}$$

(b) This follows from 4.10 and 4.9(a). □

4.14. We close this section by constructing explicitly a formal versal deformation in a simple case. When X_0 is affine and a complete intersection, such a deformation is immediate to write down. Let

$$A_0 = k[x_1, \dots, x_n]/(f_1, \dots, f_r) = S/J,$$

where $S = k[x_1, \dots, x_n]$, $J = (f_1, \dots, f_r)$, and $f_1, \dots, f_r$ is an S-regular sequence. We have an exact sequence

$$\cdots \longrightarrow \mathrm{Der}_k(S, A_0) \longrightarrow \mathrm{Hom}_{A_0}(J/J^2, A_0) \longrightarrow D^1(A_0|k, A_0) \longrightarrow 0.$$

The A_0-module J/J^2 is free of rank r, generated by the classes of $f_1, \dots, f_r$, because these elements form an S-regular sequence.

Assume now that $D^1(A_0|k, A_0)$ is finite-dimensional over k; for example, assume that A_0 has only isolated nonsmooth points. Put

$$d = \dim_k D^1(A_0|k, A_0),$$

and choose maps

$$p_i : J/J^2 \longrightarrow A_0 \qquad (1 \le i \le d)$$

whose classes form a k-basis of $D^1(A_0|k, A_0)$. For $1 \le i \le d$ and $1 \le j \le r$, choose $P_{ij} \in \Lambda[x_1, \dots, x_n]$ such that

$$P_{ij} \equiv p_i(f_j) \pmod{\mathfrak{m}_\Lambda + J},$$

where $\mathfrak{m}_\Lambda$ is the maximal ideal of Λ. Set

$$R = \Lambda[[t_1, \dots, t_d]], \qquad B = R[x_1, \dots, x_n]/(G_1, \dots, G_r),$$

where

$$G_j = F_j - \sum_{i=1}^{d} t_i P_{ij},$$

and $F_j \in \Lambda[x_1, \dots, x_n]$ is obtained from f_j by lifting its coefficients from k to Λ.

The algebra $R[x_1, \dots, x_n]$ is flat over R, and the images of $G_1, \dots, G_r$ modulo $\mathfrak{m}_R$ form the regular sequence $f_1, \dots, f_r$ in $k \otimes_R R[x_1, \dots, x_n]$. Hence B is R-flat and $k \otimes_R B = A_0$. By construction, the map

$$\mathrm{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow D^1(A_0|k, A_0) = [\underline{M}_{\mathrm{Spec}(A_0)}(k[\varepsilon])]$$

is bijective. Thus, if

$$\operatorname{Spf}(B) = \varinjlim_{\nu} \operatorname{Spec}\big((R/\mathfrak{m}_R^{\nu+1}) \otimes_R B\big),$$

then $(R, \operatorname{Spf}(B))$ is an object of $\widehat{\underline{M}}_{\operatorname{Spec}(A_0)}$. The functor

$$\underline{h}_R \longrightarrow \underline{M}_{\operatorname{Spec}(A_0)}$$

that it induces is smooth by 1.6, since R is a formal power-series ring over Λ. In other words, $(R, \operatorname{Spf}(B))$ is a formal versal object of $\underline{M}_{\operatorname{Spec}(A_0)}$.

By 4.10(b), if instead we put

$$B = R[[x_1, \ldots, x_n]]/(G_1, \ldots, G_r),$$

then $(R, \operatorname{Spf}(B))$ is a formal versal deformation of $\operatorname{Spec}(A_0)$ for

$$A_0 = k[[x_1, \ldots, x_n]]/(f_1, \ldots, f_r).$$

Remark 4.15. Let X_0 be a scheme of finite type over k with only isolated nonsmooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. The preceding construction shows that, when X_0 is affine and locally a complete intersection, $\mathfrak{X}$ is the formalization of an R-scheme of finite type. The same fact extends readily to the nonaffine case provided that $H^2(X_0, \underline{D}^0_{X_0}) = 0$.

Remark 4.16. The global theory of the relative cotangent complex—here reduced to the truncated complex of length one, which nevertheless suffices for many purposes in deformation theory—generalizes and simplifies results such as 4.10. Let S be a scheme, and let $S_0 \subset S' \subset S$ be subschemes defined by ideals $J \supset I$ with $JI = 0$. Let X' be flat over S', and write

$$L_\bullet = L^{X_0/S_0}$$

for the relative cotangent complex of X_0 over S_0, regarded as an object of $D(X_0, \mathcal{O}_{X_0})$. Let $\underline{F}$ be the category of flat S-schemes X extending X'/S'; thus an object is equipped with an S'-isomorphism

$$X \times_S S' \xrightarrow{\sim} X'.$$

Then:

(a) For an object X of $\underline{F}$, its group of automorphisms inducing the identity on X' is canonically isomorphic to

$$\operatorname{Ext}^0(X_0; L_\bullet, I_{X_0}),$$

where I_{X_0} is the inverse image on X_0 of the $\mathcal{O}_{S_0}$-module I.

(b) The set of isomorphism classes of objects of $\underline{F}$ is either empty or a torsor (a principal homogeneous space) under $\operatorname{Ext}^1(X_0; L_\bullet, I_{X_0})$.

(c) There is a canonical element

$$c \in \operatorname{Ext}^2(X_0; L_\bullet, I_{X_0})$$

whose vanishing is necessary and sufficient for $\underline{F}$ to be nonempty.

To study these groups, it is often convenient to write

$$\operatorname{Ext}^i(X_0; L_\bullet, I_{X_0}) = H^i\big(X_0, R\underline{\operatorname{Hom}}(L_\bullet, I_{X_0})\big).$$

The cohomology sheaves of $R\underline{\operatorname{Hom}}(L_\bullet, I_{X_0})$ are precisely the sheaves $\underline{D}^i_{X_0}(I_{X_0})$ introduced in section 3. For $i = 0, 1, 2$, they may be viewed respectively as the sheaf of automorphisms of an object X of $\underline{F}$ (automorphisms that induce the identity on X'), the sheaf measuring the indeterminacy of isomorphism classes of local solutions to the flat-deformation problem, and the sheaf of obstructions to the existence of local solutions.

The standard spectral sequence is

$$E_2^{p,q} = H^p\big(X_0, \underline{D}^q_{X_0}(I_{X_0})\big) \Longrightarrow \mathrm{Ext}^{p+q}(X_0; L_\bullet, I_{X_0}).$$

It gives the familiar isomorphisms

$$\begin{aligned}\mathrm{Aut}(X/X') &\simeq H^0\big(X_0, \underline{D}^0_{X_0}(I_{X_0})\big)\\ &\simeq H^0\big(X_0, \underline{\mathrm{Hom}}(\Omega^1_{X_0/S_0}, I_{X_0})\big)\\ &= \mathrm{Hom}(\Omega^1_{X_0/S_0}, I_{X_0}),\end{aligned}$$

and the low-degree exact sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}(I_{X_0})) \longrightarrow \mathrm{Ext}^1 \longrightarrow H^0(X_0, \underline{D}^1_{X_0}(I_{X_0})) \longrightarrow H^2(X_0, \underline{D}^0_{X_0}(I_{X_0})) \longrightarrow \mathrm{Ext}^2.$$

In particular, if $H^2(X_0, \underline{D}^0_{X_0}(I_{X_0})) = 0$ and local flat deformations of X'/S' exist, then compatible local deformations glue to a global flat deformation. More generally, for a prescribed section of the sheaf of isomorphism classes of local flat deformations, the argument of 4.10 places the obstruction to a compatible global deformation in $H^2(X_0, \underline{D}^0_{X_0}(I_{X_0}))$.

The same spectral sequence shows that Ext^2 vanishes whenever

$$(*) \qquad H^0(X_0, \underline{D}^2_{X_0}) = 0, \qquad H^1(X_0, \underline{D}^1_{X_0}) = 0, \qquad H^2(X_0, \underline{D}^0_{X_0}) = 0.$$

The first equality always holds when X_0 is a relative complete intersection over S_0, since then $\underline{D}^2_{X_0}(M) = 0$ for every $\mathcal{O}_{X_0}$-module M. The second holds when the support of $\underline{D}^1_{X_0}$ is discrete, as happens when X_0 is smooth over S_0 away from a discrete subset. If all three equalities in $(*)$ hold, then $\underline{F}$ is nonempty by (c), which generalizes 4.13(b).

There is also a geometric interpretation, requiring only the globalized functors $\underline{D}^i_{X_0}$ rather than the global theory of the relative cotangent complex. The three successive obstructions to extending X'/S' flatly over S lie in the three groups displayed in $(*)$. The local theory defines the first obstruction

$$c_1 \in H^0(X_0, \underline{D}^2_{X_0})$$

as the obstruction to the existence of local solutions. If $c_1 = 0$, the sheaf $\underline{F}$ of isomorphism classes of local solutions is canonically a torsor under $\underline{D}^1_{X_0}$. Its obstruction to having a section is

$$c_2 \in H^1(X_0, \underline{D}^1_{X_0});$$

such a section is a coherent system of isomorphism classes of local solutions. If $c_2 = 0$, choose a section of $\underline{F}$. The obstruction to a global solution realizing the chosen local classes is then

$$c_3 \in H^2(X_0, \underline{D}^0_{X_0}).$$

Changing the section of $\underline{F}$ by a section of $\underline{D}^1_{X_0}$ gives a map

$$d : H^0(X_0, \underline{D}^1_{X_0}) \longrightarrow H^2(X_0, \underline{D}^0_{X_0}).$$

A solution exists if and only if $c_1 = 0$, $c_2 = 0$, and the class of c_3 in $\mathrm{Coker}\, d$ is zero (each c_i being defined only after the preceding obstructions vanish).

The three obstruction groups

$$H^0(X_0, \underline{D}^2_{X_0}), \qquad H^1(X_0, \underline{D}^1_{X_0}), \qquad H^2(X_0, \underline{D}^0_{X_0})/\,\mathrm{Im}\, d$$

are the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the preceding spectral sequence; up to sign, d is its differential

$$d_2 : E_2^{0,1} \longrightarrow E_2^{2,0}.$$

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. The space $\mathfrak{X}$ is not an ordinary R-scheme, but an R-adic formal scheme. The purpose of this section is to clarify the meaning of the "nonsmooth locus of $\mathfrak{X}$ over R."

We first recall the Fitting ideals of a finite-type module. Let A be a commutative ring and let E be a finite-type A-module. Choose a presentation

$$K \longrightarrow F \longrightarrow E \longrightarrow 0,$$

where F is locally free of constant rank n. For $p = 0, 1, 2, \ldots$, define

$$I_A^p(E) = \operatorname{Im}\left(\bigwedge^{n-p} K \otimes_A \bigwedge^{n-p} F^* \longrightarrow A\right).$$

These ideals are independent of the chosen presentation. They are invariants of the A-module E, called its *Fitting ideals*. The following properties follow directly from the definition.

5.1. (a) For every ring homomorphism $A \longrightarrow B$, the canonical map

$$B \otimes_A A/I_A^p(E) \xrightarrow{\sim} B/I_B^p(B \otimes_A E)$$

is an isomorphism for every p. In particular,

$$I_{S^{-1}A}^p(S^{-1}E) = S^{-1}I_A^p(E)$$

for every multiplicative subset $S \subset A$.

(b) There are inclusions

$$I_A^0(E) \subset I_A^1(E) \subset I_A^2(E) \subset \cdots.$$

For every $x \in \operatorname{Spec}(A)$,

$$I_A^p(E)_x = A_x \qquad \text{for all } p \geq \dim_{\kappa(x)} E(x),$$

where $E(x) = \kappa(x) \otimes_A E$.

(c) A point $x \in \operatorname{Spec}(A) - \operatorname{Supp}(A/I_A^p(E))$ if and only if, for any (equivalently, for some) presentation

$$K \longrightarrow F \longrightarrow E \longrightarrow 0, \qquad F \simeq A^n,$$

the module K_x contains a copy of A_x^{n-p} that is a direct summand of F_x. In particular,

$$\operatorname{Supp}(A/I_A^0(E)) = \operatorname{Supp}(E).$$

(d) For any two finite-type A-modules,

$$I_A^r(E_1 \oplus E_2) = \sum_{p+q=r} I_A^p(E_1)I_A^q(E_2).$$

(e) If A is a noetherian adic ring, then

$$A/I_A^p(E) = \varprojlim_{\nu} A_\nu/I_{A_\nu}^p(E_\nu),$$

where $A = \varprojlim_\nu A_\nu$ and $E_\nu = E \otimes_A A_\nu$.

Property 5.1(a) globalizes the definition. If X is a scheme and $\underline{E}$ is a quasi-coherent $\mathcal{O}_X$-module of finite presentation, there are quasi-coherent ideal sheaves $\underline{I}_X^p(\underline{E})$, $p \geq 0$, characterized on every affine open $U \subset X$ by

$$\Gamma(U, \underline{I}_X^p(\underline{E})) \xrightarrow{\sim} I_{\Gamma(U,\mathcal{O}_X)}^p(\Gamma(U, \underline{E})).$$

For a morphism of schemes $X \longrightarrow Y$, the relative module of Kähler differentials $\Omega_{X|Y} = \Omega^1_{X|Y}$ is quasi-coherent. If the morphism is essentially of finite presentation, then $\Omega_{X|Y}$ is of finite presentation, so the ideal sheaves $\underline{I}^p_X(\Omega_{X|Y})$ are defined.

Definition 5.2. Let $X \longrightarrow Y$ be a morphism essentially of finite presentation. For every integer $p \geq 0$, define $J^p_X(X|Y)$ to be the closed subscheme of X
defined by the ideal sheaf

$$\underline{I}^p_X(X|Y) = \underline{I}^p_X(\Omega_{X|Y}).$$

It is called the pth *Jacobian subscheme* of X over Y. For every $x \in X$, put

$$X_x = \mathrm{Spec}(\mathcal{O}_{X,x}).$$

Theorem 5.3. Let $X \longrightarrow Y$ be a morphism essentially of finite presentation. Then:
(1) For every base change $Y' \longrightarrow Y$, the canonical map

$$J^p_{X'}(X'|Y') \xrightarrow{\sim} J^p_X(X|Y) \times_Y Y'$$

is an isomorphism for every p, where $X' = X \times_Y Y'$.
(2) For every $x \in X$, there is a canonical isomorphism

$$J^p_{X_x}(X_x|Y) \xrightarrow{\sim} J^p_X(X|Y)_x.$$

(3) There are inclusions

$$J^0_X(X|Y) \supset J^1_X(X|Y) \supset J^2_X(X|Y) \supset \cdots,$$

and

$$x \notin J^p_X(X|Y) \qquad \text{for every } p \geq \dim_{\kappa(x)} \Omega_{X|Y}(x).$$

(4) A point x does not belong to $J^p_X(X|Y)$ if and only if it has an open neighbourhood U that admits a closed immersion $i : U \hookrightarrow U'$ into a Y-scheme U' such that U' is smooth over Y at $x' = i(x)$ and

$$\dim_{x'} U'_{f(x)} = p.$$

(5) For the projections $\pi_i : X_1 \times_Y X_2 \longrightarrow X_i$,

$$\underline{I}^r_{X_1 \times_Y X_2}(X_1 \times_Y X_2|Y) = \sum_{p+q=r} \pi_1^* \underline{I}^p_{X_1}(X_1|Y)\, \pi_2^* \underline{I}^q_{X_2}(X_2|Y).$$

Proof. Assertion (3) follows from 5.1(b). Assertion (1) follows from 5.1(a) and the canonical isomorphism

$$P^*\Omega_{X|Y} \xrightarrow{\sim} \Omega_{X'|Y'}, \qquad P : X \times_Y Y' \longrightarrow X.$$

Likewise, (2) follows from 5.1(a) and

$$\mathcal{O}_{X,x} \otimes_{\mathcal{O}_X} \Omega_{X|Y} \xrightarrow{\sim} \Omega_{X_x|Y}.$$

Finally, (5) follows from 5.1(d) and

$$\pi_1^*\Omega_{X_1|Y} \oplus \pi_2^*\Omega_{X_2|Y} \xrightarrow{\sim} \Omega_{X_1 \times_Y X_2|Y}.$$

For (4), first suppose that there is a diagram

$$\begin{tikzcd} U \arrow[r, hook, "i"] \arrow[d, "f"'] & U' \arrow[dl] \\ Y & \end{tikzcd}$$

where U' is smooth over Y at $x' = i(x)$ and $\dim_{x'} U'_{f(x)} = p$. Then

$$\dim_{\kappa(x')} \Omega_{U'|Y}(x') = p,$$

and the sequence

$$i^*\Omega_{U'|Y} \longrightarrow \Omega_{U|Y} \longrightarrow 0$$

is exact. Hence $\dim_{\kappa(x)} \Omega_{U|Y}(x) \leq p$, so $x \notin J_U^p(U|Y)$ by (3). Since

$$J_U^p(U|Y) = J_X^p(X|Y)|_U,$$

we have $x \notin J_X^p(X|Y)$.

Conversely, suppose that $x \notin J_X^p(X|Y)$. The question is local, so write $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$. Choose an A-presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow B \longrightarrow 0, \qquad P = A[x_1, \ldots, x_n].$$

There is an exact sequence

$$J/J^2 \xrightarrow{d} B \otimes_P \Omega_{P|A} \longrightarrow \Omega_{B|A} \longrightarrow 0,$$

and $B \otimes_P \Omega_{P|A}$ is free of rank n over B. Thus

$$I_B^p(\Omega_{B|A}) = \operatorname{Im}\left(\bigwedge^{n-p}(J/J^2) \otimes_B (B \otimes_P \bigwedge^{n-p} \Omega_{P|A})^* \longrightarrow B\right).$$

Consequently, $x \notin J_X^p(X|Y)$ precisely when there are $f_1, \ldots, f_{n-p} \in J$ and a choice of $n-p$ among the variables $x_1, \ldots, x_n$ for which the corresponding Jacobian minor

$$\det\left(\frac{\partial f_i}{\partial x_j}\right)(x)$$

is nonzero. Equivalently,

$$\operatorname{Spec}\bigl(P/(f_1, \ldots, f_{n-p})\bigr) \longrightarrow \operatorname{Spec}(A)$$

is smooth at the image x' of x. Taking

$$U' = \operatorname{Spec}\bigl(P/(f_1, \ldots, f_{n-p})\bigr)$$

gives the required closed immersion locally. □

Theorem 5.4. Let $X \xrightarrow{f} Y$ be a morphism essentially of finite presentation. Suppose that there is a fixed integer $n \geq 0$ for which one of the following conditions holds:

(1) f is flat and $\dim_x X_{f(x)} = n$ for every $x \in X$;

(2) f is a relative complete intersection of virtual dimension n (SGA 6, VIII, 1.9).

Then $x \notin J_X^n(X|Y)$ if and only if f is smooth at x.

Proof. By 5.3(4), $x \notin J_X^n(X|Y)$ if and only if there is a diagram

$$\begin{tikzcd} U \arrow[r, hook, "i"] \arrow[d, "f"'] & U' \arrow[dl, "f'"] \\ Y & \end{tikzcd}$$

such that f' is smooth at x', with $\dim_{x'} U'_{f(x)} = n$, while i is a closed immersion and $x' = i(x)$. It therefore suffices to show that under either (1) or (2), i is an isomorphism at x.

Set

$$A = \mathcal{O}_{Y,f(x)}, \qquad B' = \mathcal{O}_{U',x'}, \qquad B = \mathcal{O}_{U,x},$$

and let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be exact. We must show that $I = 0$.

Assume (1). Since B is flat over A, the sequence

$$0 \longrightarrow I/\mathfrak{m}I \longrightarrow B'/\mathfrak{m}B' \longrightarrow B/\mathfrak{m}B \longrightarrow 0$$

is exact, where $\mathfrak{m}$ is the maximal ideal of A. By hypothesis,

$$\dim_x U_{f(x)} = \dim_{x'} U'_{f(x)},$$

and hence $\dim(B'/\mathfrak{m}B') = \dim(B/\mathfrak{m}B)$. Since $B'/\mathfrak{m}B'$ is a regular local ring, it follows that $I/\mathfrak{m}I = 0$, and therefore $I = 0$.

Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \otimes_{B'} \Omega_{B'|A} \longrightarrow \Omega_{B|A} \longrightarrow 0.$$

Since X is a relative complete intersection over Y and $U' \longrightarrow Y$ is smooth at x', the map

$$d : I/I^2 \longrightarrow B \otimes_{B'} \Omega_{B'|A}$$

is injective at x, and both I/I^2 and $B \otimes_{B'} \Omega_{B'|A}$ are free at x. By the definition of the virtual dimension of a relative complete intersection, we must therefore have

$$n = \dim_{\kappa(x)} \Omega_{B'|A}(x') - \dim_{\kappa(x)}(I/I^2)(x) = n - \dim_{\kappa(x)}(I/I^2)(x).$$

Thus $I/I^2 = 0$, and hence $I = 0$. $\square$ **Definition 5.5.** Let $X \xrightarrow{f} Y$ be a morphism essentially of finite presentation. If there is a fixed integer n satisfying (1) or (2) of 5.4, set

$$\underline{I}_X(X|Y) = \underline{I}_X^n(X|Y), \qquad J_X(X|Y) = J_X^n(X|Y).$$

This notion extends naturally to formal schemes. Let $\mathfrak{X}$ be a noetherian formal scheme and $\mathcal{E}$ a coherent $\mathfrak{X}$-module. For every affine open $U \subset \mathfrak{X}$, define

$$\underline{I}_{\mathfrak{X}}^p(\mathcal{E})(U) = I_{\mathfrak{X}(U)}^p\big(\mathcal{E}(U)\big).$$

If $\mathfrak{X} = \varinjlim_\nu X_\nu$, then $\mathcal{E} = \varprojlim_\nu \mathcal{E}_\nu$, where

$$\mathcal{E}_\nu = \mathcal{E} \otimes_{\mathcal{O}_{\mathfrak{X}}} \mathcal{O}_{X_\nu},$$

and noetherianness gives

$$\underline{I}_{\mathfrak{X}}^p(\mathcal{E}) = \varprojlim_\nu \underline{I}_{X_\nu}^p(\mathcal{E}_\nu).$$

Now let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme essentially of finite type. Let $\underline{I}_{\mathfrak{X}}$ be the ideal sheaf defining the closed diagonal immersion

$$\Delta : \mathfrak{X} \longrightarrow \mathfrak{X} \times_S \mathfrak{X}$$

(where the fibre product is taken in the category of formal schemes), so that

$$0 \longrightarrow \underline{I}_{\mathfrak{X}} \longrightarrow \mathcal{O}_{\mathfrak{X}\times_S\mathfrak{X}} \longrightarrow \mathcal{O}_{\mathfrak{X}} \longrightarrow 0$$

is exact. Define

$$\Omega_{\mathfrak{X}|S} = \underline{I}_{\mathfrak{X}}/\underline{I}_{\mathfrak{X}}^2.$$

Writing

$$\mathfrak{X} = \varinjlim_{\nu} X_\nu, \qquad X_\nu = \mathfrak{X} \times_S S_\nu,$$

we have

$$\Omega_{\mathfrak{X}|S} \simeq \varprojlim_{\nu} \Omega_{X_\nu|S_\nu},$$

so $\Omega_{\mathfrak{X}|S}$ is a coherent $\mathfrak{X}$-module. For every formal S-scheme S' of finite type there is a canonical isomorphism

$$\Omega_{\mathfrak{X}\times_S S'|S'} \simeq \mathcal{O}_{S'} \otimes_{\mathcal{O}_S} \Omega_{\mathfrak{X}|S}.$$

Definition 5.6. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme essentially of finite type. Define

$$\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S) = \underline{I}_{\mathfrak{X}}^p(\Omega_{\mathfrak{X}|S}),$$

and let $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ be the closed formal subscheme of $\mathfrak{X}$ defined by this ideal sheaf.

Since $\Omega_{\mathfrak{X}|S} = \varprojlim_\nu \Omega_{X_\nu|S_\nu}$, we have

$$\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S) = \varprojlim_{\nu} \underline{I}_{X_\nu}^p(X_\nu|S_\nu)$$

and

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S) = \varinjlim_{\nu} J_{X_\nu}^p(X_\nu|S_\nu).$$

The formal scheme $\mathfrak{X}_{(x)}$ is again S-adic and essentially of finite type, and there is a commutative diagram

$$\begin{tikzcd}
\mathfrak{X}_{(\widehat{x})} \arrow[rr, "i_{(\widehat{x})}"] \arrow[dr] & & \mathfrak{X} \\
& \mathfrak{X}_{(x)} \arrow[ur, "i_{(x)}"'] &
\end{tikzcd}$$

If x is an isolated point of $\mathfrak{X}$, then $i_{(x)}$ and $i_{(\widehat{x})}$ are both isomorphisms onto the connected component of $\mathfrak{X}$ containing x.

In particular, if

$$|\mathfrak{X}| = \{x_1, \ldots, x_m\}$$

is a finite discrete space, then there is a canonical identification

$$\mathfrak{X} = \coprod_{1\le i\le m} \mathfrak{X}_{(x_i)} \qquad \text{(disjoint union)}.$$

Corollary 5.7. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S-adic formal scheme essentially of finite type. Then:

(1) Formation of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ commutes with base change. More precisely, if

$$\mathfrak{X}' = \mathfrak{X} \times_S S',$$

then

$$J_{\mathfrak{X}'}^p(\mathfrak{X}'|S') = J_{\mathfrak{X}}^p(\mathfrak{X}|S) \times_S S'.$$

(2) The connected components of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ are in bijective correspondence with those of $J_{X_0}^p(X_0|S_0)$. For every

$$x \in \left|J_{X_0}^p(X_0|S_0)\right|$$

there is a canonical isomorphism

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S)_{(x)} \simeq J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S).$$

In particular, if

$$\left|J^p_{X_0}(X_0|S_0)\right| = \{x_1, \dots, x_m\}$$

is discrete, then

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1\le i\le m} J^p_{\mathfrak{X}_{(x_i)}}(\mathfrak{X}_{(x_i)}|S) \qquad \text{(disjoint union)}.$$

(3) Let S be an ordinary noetherian scheme and X an S-scheme essentially of finite type. Given a closed subscheme $S_0 \subset S$, let $\widehat{S}$ be the formal completion of S along S_0. If

$$\widehat{X} = X \times_S \widehat{S}$$

is the formal completion of X along $X_0 = X \times_S S_0$, then

$$J^p_{\widehat{X}}(\widehat{X}|\widehat{S}) = J^p_X(X|S) \times_S \widehat{S}.$$

Equivalently, it is the formal completion of $J^p_X(X|S)$ along $J^p_{X_0}(X_0|S_0)$.
(4) Suppose that a fixed integer $n \ge 0$ satisfies condition 5.4(1) or 5.4(2) for every $X_\nu \longrightarrow S_\nu$. Then

$$x \notin J^n_{\mathfrak{X}}(\mathfrak{X}|S)$$

if and only if $\mathfrak{X}$ is formally smooth over S at x.
Proof. Assertion (1) follows from

$$\Omega_{\mathfrak{X}'|S'} \simeq \mathcal{O}_{S'} \widehat{\otimes}_{\mathcal{O}_S} \Omega_{\mathfrak{X}|S},$$

whereas (2) follows from

$$\mathcal{O}_{\mathfrak{X}_{(x)}} \widehat{\otimes}_{\mathcal{O}_{\mathfrak{X}}} \Omega_{\mathfrak{X}|S} \simeq \Omega_{\mathfrak{X}_{(x)}|S}.$$

For (3),

$$\begin{aligned} J^p_{\widehat{X}}(\widehat{X}|\widehat{S}) &= \varinjlim_{\nu} J^p_{X\times_S S_\nu}(X \times_S S_\nu|S_\nu) \\ &= \varinjlim_{\nu} \big(J^p_X(X|S) \times_S S_\nu\big) = J^p_X(X|S) \times_S \widehat{S}. \end{aligned}$$

Finally, since $\mathfrak{X}$ is S-adic,

$$\begin{aligned} \mathfrak{X} \to S \text{ is formally smooth at } x &\iff X_\nu \to S_\nu \text{ is formally smooth at } x \text{ for every } \nu \\ &\iff x \notin J^n_{X_\nu}(X_\nu|S_\nu) \text{ for every } \nu \\ &\iff x \notin \varinjlim_{\nu} J^n_{X_\nu}(X_\nu|S_\nu) = J^n_{\mathfrak{X}}(\mathfrak{X}|S). \end{aligned}$$

□ **Remark 5.8.** Let $\mathfrak{X}$ be an S-adic formal scheme essentially of finite type. For a point $x \in \mathfrak{X}$, consider also the S-adic formal scheme

$$\mathfrak{X}_{(\widehat{x})} = \varinjlim_{\nu} \operatorname{Spec}\big(\widehat{\mathcal{O}_{X_\nu,x}}\big).$$

There are natural morphisms

$$\mathfrak{X}_{(\widehat{x})} \longrightarrow \mathfrak{X}_{(x)} \longrightarrow \mathfrak{X}.$$

In general, $\widehat{\mathcal{O}_{X_0,x}}$ is not essentially of finite type over $S_{(y)}$, where y is the image of x under $\mathfrak{X} \longrightarrow S$.
For simplicity, put

$$A_\nu = \mathcal{O}_{X_\nu,x}, \qquad R_\nu = \mathcal{O}_{S_\nu,y}.$$

The natural map

$$\Omega_{A_\nu|R_\nu} \longrightarrow \widehat{\Omega}_{A_\nu|R_\nu}$$

induces an isomorphism

$$\widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \widehat{\widehat{\Omega}_{A_\nu|R_\nu}},$$

and therefore

$$\varprojlim_\nu \widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \varprojlim_\nu \widehat{\widehat{\Omega}_{A_\nu|R_\nu}}.$$

Here the outer hat denotes the separated completion of a module for the topology induced by the ring. Consequently, if we define

$$\widehat{\Omega}_{\mathfrak{X}_{(\widehat{x})}|S} = \varprojlim_\nu \widehat{\widehat{\Omega}_{\mathcal{O}_{X_\nu,x}|\mathcal{O}_{S_\nu,y}}},$$

then $\widehat{\Omega}_{\mathfrak{X}_{(\widehat{x})}|S}$ is a coherent $\mathfrak{X}_{(\widehat{x})}$-module.

Define $J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S)$ to be the closed formal subscheme of $\mathfrak{X}_{(\widehat{x})}$ defined by

$$\underline{I}^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S) = \underline{I}^p_{\mathfrak{X}_{(\widehat{x})}}(\widehat{\Omega}_{\mathfrak{X}_{(\widehat{x})}|S}).$$

Then

$$J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)_{(\widehat{x})} \simeq J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S).$$

If x is an isolated point of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$, the natural morphism

$$J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S) \longrightarrow J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S)$$

is an isomorphism.

To form the "image" of $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ under $f : \mathfrak{X} \longrightarrow S$, one could simply take the formal subscheme of S defined by the ideal sheaf

$$\ker\left(\mathcal{O}_S \longrightarrow \mathcal{O}_{\mathfrak{X}}/\underline{I}^p_{\mathfrak{X}}(\mathfrak{X}|S)\right).$$

For our purposes, however, it is more convenient to use the Fitting ideal once again. **Definition 5.9.** Let S be a noetherian formal scheme and $f : \mathfrak{X} \longrightarrow S$ an S-adic formal scheme essentially of finite type. Suppose that $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ is proper over S, so that

$$f_*\mathcal{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)}$$

is a coherent $\mathcal{O}_S$-module. Define

$$\underline{I}^p_S(\mathfrak{X}|S) = \underline{I}^0_S\Big(f_*\mathcal{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)}\Big),$$

and let $J^p_S(\mathfrak{X}|S)$ be the closed formal subscheme of S defined by $\underline{I}^p_S(\mathfrak{X}|S)$.

If there is a fixed integer n satisfying condition 5.4(1) or 5.4(2) for every $X_\nu \longrightarrow S_\nu$, set

$$\begin{aligned} \underline{I}_{\mathfrak{X}}(\mathfrak{X}|S) &= \underline{I}^n_{\mathfrak{X}}(\mathfrak{X}|S), \qquad & J_{\mathfrak{X}}(\mathfrak{X}|S) &= J^n_{\mathfrak{X}}(\mathfrak{X}|S), \\ \underline{I}_S(\mathfrak{X}|S) &= \underline{I}^n_S(\mathfrak{X}|S), \qquad & J_S(\mathfrak{X}|S) &= J^n_S(\mathfrak{X}|S). \end{aligned}$$

Corollary 5.10. Let S be a noetherian formal scheme and $f : \mathfrak{X} \longrightarrow S$ an S-adic formal scheme essentially of finite type such that $J^p_{\mathfrak{X}}(\mathfrak{X}|S)$ is proper over S. Then:

(1)

$$\left|J^p_S(\mathfrak{X}|S)\right| = f\big(\,|J^p_{\mathfrak{X}}(\mathfrak{X}|S)|\,\big).$$

(2) Formation of $J^p_S(\mathfrak{X}|S)$ commutes with base change: if $\mathfrak{X}' = \mathfrak{X} \times_S S'$, then

$$J^p_{S'}(\mathfrak{X}'|S') = J^p_S(\mathfrak{X}|S) \times_S S'.$$

(3) If

$$J^p_{\mathfrak{X}}(\mathfrak{X}|S) = \coprod_{1 \le i \le m} \mathfrak{Y}^{(i)}$$

is a disjoint union, then

$$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1 \le i \le m} \underline{I}^0_S\big(f_* \mathcal{O}_{\mathfrak{Y}^{(i)}}\big).$$

In particular, if

$$\big|J^p_{X_0}(X_0|S_0)\big| = \{x_1, \dots, x_m\}$$

is discrete, then

$$\underline{I}^p_S(\mathfrak{X}|S) = \prod_{1 \le i \le m} \underline{I}^p_S(\mathfrak{X}_{(x_i)}|S) = \prod_{1 \le i \le m} \underline{I}^p_S(\mathfrak{X}_{(\widehat{x}_i)}|S).$$

(4) Let S be an ordinary noetherian scheme, X an S-scheme essentially of finite type, and $S_0 \hookrightarrow S$ a closed subscheme. Let $\widehat{S}$ be the formal completion of S along S_0. If

$$\widehat{X} = X \times_S \widehat{S}$$

is the formal completion of X along $X_0 = X \times_S S_0$, then

$$J^p_{\widehat{S}}(\widehat{X}|\widehat{S}) = J^p_S(X|S) \times_S \widehat{S}.$$

Thus it is the formal completion of $J^p_S(X|S)$ along $J^p_{S_0}(X_0|S_0)$.

Proof. By 5.1(c),

$$\operatorname{Supp} f_* \mathcal{O}_{J^p_{\mathfrak{X}}(\mathfrak{X}|S)} = \big|J^p_S(\mathfrak{X}|S)\big|,$$

which proves (1). Assertions (2) and (4) follow from 5.1(a), while (3) follows from 5.1(d) and, for an isolated point $x \in J^p_{\mathfrak{X}}(\mathfrak{X}|S)$, the isomorphism

$$J^p_{\mathfrak{X}_{(\widehat{x})}}(\mathfrak{X}_{(\widehat{x})}|S) \xrightarrow{\sim} J^p_{\mathfrak{X}_{(x)}}(\mathfrak{X}_{(x)}|S).$$

$\square$

Remark 5.11. Our definition of the Jacobian ideal, adopted from the classical definition over a base field, commutes with base change and, under the mild hypothesis of 5.4, gives a smoothness criterion. In a general setting, however, one can probably define in many ways a canonical subscheme whose support is the nonsmooth locus. We outline one such definition.

For an A-algebra B, put

$$I_{B|A} = \bigcap_E \operatorname{Ann} D^1(B|A, E),$$

where E ranges over all finite-type B-modules. By 3.13(c), the ideal $I_{B|A}$ commutes with localization. Thus a morphism $f : X \longrightarrow Y$ determines an ideal sheaf $\underline{I}_{X|Y}$. If f is essentially of finite type, then by 3.13(a), f is smooth at x if and only if

$$x \notin \operatorname{Supp}\big(\mathcal{O}_X / \underline{I}_{X|Y}\big).$$

This construction carries over naturally to adic formal schemes essentially of finite type, and is therefore equally suitable for deformation theory.

6. Non-degenerate quadratic singularities

The purpose of this section is to study deformations of a k-scheme whose only singularities are non-degenerate quadratic singularities. In view of 4.13, this becomes, in a suitable situation, a purely local problem.

We return to the notation of §4. Thus Λ is a fixed complete noetherian local ring with residue field k, and $\underline{C}_\Lambda$ denotes the category of local artinian Λ-algebras having residue field k.

Let X be a k-scheme of finite type, and let x be a closed point of X. Recall that x is called a *non-degenerate quadratic singularity rational over* k if

$$\widehat{\mathcal{O}}_{X,x} \simeq k[[x_1, \ldots, x_n]]/(f)$$

and

$$\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = (x_1, \ldots, x_n).$$

One may always take f to be a quadratic form. Indeed:

Lemma 6.1. Let $f \in k[[x_1, \ldots, x_n]]$, and put $M = (x_1, \ldots, x_n)$. If

$$\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = M,$$

then there are elements $x'_1, \ldots, x'_n$ satisfying $x'_i \equiv x_i \pmod{M^2}$ such that

$$f = q(x'_1, \ldots, x'_n)$$

for a quadratic form q.

Proof. We use successive approximation. Write

$$f = f_2 + h_3,$$

where f_2 is quadratic and $h_3 \in M^3$. Since the partial derivatives of f generate M, there are $\varepsilon_1^{(1)}, \ldots, \varepsilon_n^{(1)} \in M^2$ such that

$$\sum_i \varepsilon_i^{(1)} \frac{\partial f}{\partial x_i} \equiv h_3 \pmod{M^4}.$$

Put $x_i^{(1)} = x_i - \varepsilon_i^{(1)}$. Then

$$\begin{aligned} f(x_1^{(1)}, \ldots, x_n^{(1)}) &= f(x_1, \ldots, x_n) - \sum_i \varepsilon_i^{(1)} \frac{\partial f}{\partial x_i} + \cdots \\ &\equiv f_2(x_1, \ldots, x_n) \pmod{M^4}. \end{aligned}$$

Thus

$$f(x_1^{(1)}, \ldots, x_n^{(1)}) = f_2(x_1, \ldots, x_n) + h_4, \qquad h_4 \in M^4.$$

Moreover, the derivatives with respect to the new variables still generate M. Hence one can choose $\varepsilon_1^{(2)}, \ldots, \varepsilon_n^{(2)} \in M^3$ such that

$$\sum_i \varepsilon_i^{(2)} \frac{\partial f}{\partial x_i^{(1)}} \equiv h_4 \pmod{M^5}.$$

It follows that

$$\begin{aligned} &f(x_1^{(1)} - \varepsilon_1^{(2)}, \ldots, x_n^{(1)} - \varepsilon_n^{(2)}) \\ &\quad = f(x_1^{(1)}, \ldots, x_n^{(1)}) - \sum_i \varepsilon_i^{(2)} \frac{\partial f}{\partial x_i^{(1)}} + \cdots \\ &\quad \equiv f_2(x_1, \ldots, x_n) \pmod{M^5}. \end{aligned}$$

Continuing in this way, one finds elements $\varepsilon_i^{(\nu)} \in M^{\nu+1}$ such that, on setting

$$x_i^{(\nu)} = x_i - \big(\varepsilon_i^{(1)} + \cdots + \varepsilon_i^{(\nu)}\big),$$

one has

$$f(x_1^{(\nu)}, \ldots, x_n^{(\nu)}) \equiv f_2(x_1, \ldots, x_n) \pmod{M^{\nu+3}}.$$

Let $\varepsilon_i = \sum_\nu \varepsilon_i^{(\nu)}$. Completeness then gives

$$f(x_1 - \varepsilon_1, \ldots, x_n - \varepsilon_n) = f_2(x_1, \ldots, x_n),$$

or, equivalently,

$$f(x_1, \ldots, x_n) = f_2(x_1 + \varepsilon_1, \ldots, x_n + \varepsilon_n).$$

Taking $x_i' = x_i + \varepsilon_i$ proves the assertion. □

Lemma 6.2. Let A_0 be a local k-algebra essentially of finite type such that

$$\widehat{A}_0 \simeq k[[x_1, \ldots, x_n]]/(f),$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \mathrm{Spec}(A_0)$. Then:

(1) The canonical maps

$$J_{\mathfrak{X}}(\mathfrak{X}|R) \longrightarrow J_R(\mathfrak{X}|R) \longrightarrow \mathrm{Spec}(\Lambda)$$

are both isomorphisms.

(2) The ideal $I_R(\mathfrak{X}|R)$ is principal and generated by a free formal generator of R over Λ; in other words,

$$I_R(\mathfrak{X}|R) = (t), \qquad R = \Lambda[[t]].$$

(3) The morphism $\mathfrak{X} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth.

Proof. Write $\mathfrak{X} = \mathrm{Spf}(A)$. By 4.10(b), $(R, \mathrm{Spf}(\widehat{A}))$ is a formal versal deformation of $\widehat{X}_0 = \mathrm{Spec}(\widehat{A}_0)$, where

$$\widehat{A} = \varprojlim_{\nu} A \otimes_R R_\nu.$$

Now

$$\widehat{A}_0 \simeq k[[x_1, \ldots, x_n]]/(f), \qquad \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) = (x_1, \ldots, x_n).$$

It follows immediately that

$$\big[\underline{M}_{\widehat{X}_0}(k[\varepsilon])\big] \simeq k.$$

Consequently, by 4.14,

$$R \simeq \Lambda[[t]], \qquad \widehat{A} \simeq R[[x_1, \ldots, x_n]]/(F - t),$$

where F is the quadratic polynomial in $\Lambda[x_1, \ldots, x_n] \subset R[[x_1, \ldots, x_n]]$ obtained from f by lifting its coefficients to Λ.

The sequence

$$0 \longrightarrow \widehat{A} \xrightarrow{dF} \widehat{A}\, dx_1 \oplus \cdots \oplus \widehat{A}\, dx_n \longrightarrow \Omega_{\widehat{A}|R} \longrightarrow 0$$

is exact, where

$$1 \longmapsto \frac{\partial F}{\partial x_1} dx_1 + \cdots + \frac{\partial F}{\partial x_n} dx_n.$$

Therefore

$$I_{\widehat{A}}(\widehat{A}|R) = \left(\frac{\partial F}{\partial x_1}, \ldots, \frac{\partial F}{\partial x_n}\right) = (x_1, \ldots, x_n),$$

and hence

$$\begin{aligned}\widehat{A}/I_{\widehat{A}}(\widehat{A}|R) &= \widehat{A}/(x_1,\dots,x_n)\\ &= R[[x_1,\dots,x_n]]/(x_1,\dots,x_n,F-t) \simeq \Lambda.\end{aligned}$$

Thus the canonical map $\Lambda \longrightarrow \widehat{A}/I_{\widehat{A}}(\widehat{A}|R)$ is an isomorphism. The completion comparison identifies this quotient with $A/I_A(A|R)$, and consequently

$$\Lambda \xrightarrow{\sim} R/I_R(A|R).$$

This proves (1) and (2). Finally, (3) follows from the isomorphisms

$$\begin{aligned}\widehat{A} &\simeq R[[x_1,\dots,x_n]]/(F-t)\\ &\simeq \Lambda[[t,x_1,\dots,x_n]]/(F-t) \simeq \Lambda[[x_1,\dots,x_n]]\end{aligned}$$

of Λ-algebras. □

Corollary 6.3. Let X_0 be a k-scheme of finite type, and let $x \in X_0$ be a non-degenerate quadratic singularity rational over k.

(1) Let $(R,\mathfrak{X})$ and $(R,\mathfrak{Y})$ be two deformations of X_0, with $R \in \mathrm{ob}(\widehat{\underline{C}}_\Lambda)$. Assume that $k = k^2$. Then

$$\mathfrak{X}_{(\widehat{x})} \simeq \mathfrak{Y}_{(\widehat{x})}$$

as R-adic formal schemes if and only if

$$I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R).$$

(2) Let $R \in \mathrm{ob}(\widehat{\underline{C}}_\Lambda)$ be a regular local ring, and let $(R,\mathfrak{X})$ be a deformation of X_0. Then $\mathcal{O}_{\mathfrak{X},x}$ is regular if and only if

$$I_R(\mathfrak{X}_{(x)}|R) \not\subset \mathfrak{m}_R^2;$$

equivalently, $I_R(\mathfrak{X}_{(x)}|R)$ is a regular divisor in $\mathrm{Spf}(R)$.

Proof. Let (R_0,Z) be a formal versal deformation of $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0,x})$. Since x is a non-degenerate quadratic singularity rational over k, Lemma 6.2 allows us to take

$$R_0 \simeq \Lambda[[t]], \qquad Z \simeq \mathrm{Spf}\big(R_0[[X_1,\dots,X_n]]/(F-t)\big).$$

For (1), the deformations $(R,\mathfrak{X}_{(\widehat{x})})$ and $(R,\mathfrak{Y}_{(\widehat{x})})$ are induced from (R_0,Z) by continuous local Λ-algebra homomorphisms

$$\varphi,\psi : R_0 \longrightarrow R.$$

Put $a=\varphi(t)$ and $b=\psi(t)$. Then

$$\begin{aligned}\mathfrak{X}_{(\widehat{x})} &\simeq \mathrm{Spf}\big(R[[X_1,\dots,X_n]]/(F-a)\big),\\ \mathfrak{Y}_{(\widehat{x})} &\simeq \mathrm{Spf}\big(R[[X_1,\dots,X_n]]/(F-b)\big).\end{aligned}$$

Suppose that $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R)$; equivalently, $(a)=(b)$ in R. Then $a = ub$ for some unit $u \in R$. Since $k=k^2$ and R is complete, $u = v^2$ for some unit $v \in R$, and hence $a = v^2 b$. Thus

$$(F-a) \longmapsto v^2(F-b)$$

under $X_i \mapsto vX_i$, and this substitution induces an R-algebra isomorphism

$$R[[X_1,\dots,X_n]]/(F-a) \xrightarrow{\sim} R[[X_1,\dots,X_n]]/(F-b),$$

because F is quadratic. The converse is immediate.

For (2), there is a continuous local Λ-algebra homomorphism

$$\varphi : R_0 = \Lambda[[t]] \longrightarrow R$$

such that

$$\mathfrak{X}_{(\widehat{x})} \simeq \operatorname{Spf}\bigl(R[[X_1, \ldots, X_n]]/(F - \varphi(t))\bigr).$$

Moreover,

$$I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{X}_{(\widehat{x})}|R) = (\varphi(t)),$$

and

$$R[[X_1, \ldots, X_n]]/(F - \varphi(t)) = \widehat{\mathcal{O}}_{\mathfrak{X},x}$$

is regular if and only if $\mathcal{O}_{\mathfrak{X},x}$ is regular. Since R is regular, $R[[X_1, \ldots, X_n]]$ is a regular local ring. If M denotes its maximal ideal, then its quotient by $F - \varphi(t)$ is regular if and only if

$$F - \varphi(t) \notin M^2,$$

which is equivalent to $\varphi(t) \notin \mathfrak{m}_R^2$, and hence to

$$I_R(\mathfrak{X}_{(x)}|R) \not\subset \mathfrak{m}_R^2.$$

□

Theorem 6.4. Let X_0 be a k-scheme of finite type whose nonsmooth points are isolated, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Denote the set of nonsmooth points of X_0 by $\{z_1, \ldots, z_m\}$. Assume that

$$H^2(X_0, \underline{D}^0_{X_0}) = 0$$

and that every z_i is a non-degenerate quadratic singularity rational over k. Then:

(1) R is formally smooth over Λ; that is, R is a formal power-series ring over Λ.

(2) There is a disjoint-union decomposition

$$J_{\mathfrak{X}}(\mathfrak{X}|R) = \coprod_{1 \le i \le m} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R),$$

and the canonical morphism

$$J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \xrightarrow{\sim} J_R(\mathfrak{X}_{(z_i)}|R)$$

is an isomorphism.

(3) Each ideal $I_R(\mathfrak{X}_{(z_i)}|R)$ is principal, as is

$$I_R(\mathfrak{X}|R) = \prod_{1 \le i \le m} I_R(\mathfrak{X}_{(z_i)}|R).$$

If $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i)$, then $t_1, \ldots, t_m$ can be extended to a system of free formal generators of R over Λ.

(4) The morphism $\mathfrak{X} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth; equivalently,

$$J_{\mathfrak{X}}(\mathfrak{X}|\Lambda) = \varnothing.$$

(5) If X_0 is a complete irreducible curve, then

$$\dim \underline{M}_{X_0} = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, \underline{D}^0_{X_0}).$$

Proof. Assertion (1) follows from (2) and (3). The first assertion in (2) follows from 5.7(2). For each i, let $(R_i, \mathfrak{Y}_i)$ be a formal versal deformation of $\operatorname{Spec}(\widehat{\mathcal{O}}_{X_0,z_i})$. Since $(R, \mathfrak{X}_{(\widehat{z}_i)})$ is such a deformation, there is a continuous local Λ-algebra homomorphism

$$\varphi_i : R_i \longrightarrow R$$

and a cartesian diagram

$$\begin{tikzcd}
\mathfrak{X}_{(\widehat{z}_i)} \arrow[r] \arrow[d] & \mathfrak{Y}_i \arrow[d] \\
\operatorname{Spf}(R) \arrow[r, "\operatorname{Spf}(\varphi_i)"'] & \operatorname{Spf}(R_i).
\end{tikzcd}$$

Consequently,

$$\begin{aligned} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) &= J_{\mathfrak{X}_{(\widehat{z}_i)}}(\mathfrak{X}_{(\widehat{z}_i)}|R) \\ &\simeq J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \times_{\operatorname{Spf}(R_i)} \operatorname{Spf}(R), \end{aligned}$$

and

$$\begin{aligned} J_R(\mathfrak{X}_{(z_i)}|R) &= J_R(\mathfrak{X}_{(\widehat{z}_i)}|R) \\ &\simeq J_{R_i}(\mathfrak{Y}_i|R_i) \times_{\operatorname{Spf}(R_i)} \operatorname{Spf}(R). \end{aligned}$$

By Lemma 6.2(1),

$$J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \longrightarrow J_{R_i}(\mathfrak{Y}_i|R_i)$$

is an isomorphism, proving the remaining assertion in (2).

By Lemma 6.2(2),

$$I_{R_i}(\mathfrak{Y}_i|R_i) = (u_i), \qquad R_i = \Lambda[[u_i]].$$

It follows that

$$I_R(\mathfrak{X}_{(z_i)}|R) = I_R(\mathfrak{X}_{(\widehat{z}_i)}|R) = (t_i), \qquad t_i = \varphi_i(u_i),$$

and therefore

$$I_R(\mathfrak{X}|R) = \prod_{1\le i\le m} I_R(\mathfrak{X}_{(z_i)}|R) = \left(\prod_{i=1}^m t_i\right).$$

By 4.13, the natural morphism

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{X_0,(\widehat{z}_1)} \times \cdots \times \underline{M}_{X_0,(\widehat{z}_m)}$$

is smooth. Hence

$$R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_m \longrightarrow R$$

is formally smooth. Thus $t_1, \ldots, t_m$ extend to a system of free formal generators of R over Λ, proving (1) and (3).

For (4), the projection

$$\mathfrak{X}_{(\widehat{z}_i)} \xrightarrow{\sim} \mathfrak{Y}_i \times_{\operatorname{Spf}(R_i)} \operatorname{Spf}(R) \longrightarrow \mathfrak{Y}_i$$

is formally smooth because $\operatorname{Spf}(R) \longrightarrow \operatorname{Spf}(R_i)$ is formally smooth. Lemma 6.2(3) shows that $\mathfrak{Y}_i \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth. Hence every $\mathfrak{X}_{(\widehat{z}_i)} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth. Since the z_i are precisely the nonsmooth points of X_0, it follows that $\mathfrak{X} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth. For (5), R is formally smooth over Λ, and by 4.4 and 4.11 the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}^0_{X_0}) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}^1_{X_0}) \longrightarrow 0$$

is exact. Thus

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim_k H^1(X_0, \underline{D}^0_{X_0}) + \dim_k H^0(X_0, \underline{D}^1_{X_0}) \\ &= \dim_k H^1(X_0, \underline{D}^0_{X_0}) + m. \end{aligned}$$

It is therefore enough to show that

$$\dim_k H^1(X_0, \underline{D}^0_{X_0}) + m = 3g - 3 + \dim_k H^0(X_0, \underline{D}^0_{X_0}),$$

or equivalently

$$\chi(\underline{D}^0_{X_0}) = 3 - 3g + m,$$

where, for a coherent $\mathcal{O}_{X_0}$-module $\mathcal{E}$, we put

$$\chi(\mathcal{E}) = \dim_k H^0(X_0, \mathcal{E}) - \dim_k H^1(X_0, \mathcal{E}).$$

The Euler characteristic of $\underline{D}^0_{X_0}$ is unchanged by extension of the base field to its algebraic closure, so we may assume that k is algebraically closed. Let z be a nonsmooth point of X_0, put

$$A = \widehat{\mathcal{O}}_{X_0,z},$$

and let A' be the normalization of A. By hypothesis,

$$A \simeq k[[x,y]]/(f),$$

where f is a non-degenerate quadratic form. Over the algebraically closed field k we may take $f = xy$, so that

$$A \simeq k[[x,y]]/(xy).$$

It follows at once that

$$\ell(A'/A) = 1, \qquad \mathrm{Der}_k(A, \mathfrak{m}_A) = \mathrm{Der}_k(A, A).$$

Let C be the conductor of A in A'. Since A is a complete intersection,

$$\ell(A/C) = \ell(A'/A) = 1,$$

and hence $C = \mathfrak{m}_A$. Therefore

$$A' = \{a' \in K \mid a'\mathfrak{m}_A \subset \mathfrak{m}_A\},$$

where K is the total fraction ring of A.

We claim that

$$\mathrm{Der}_k(A, A) \subset \mathrm{Der}_k(A', A')$$

inside $\mathrm{Der}_k(K, K)$. Indeed, let $D \in \mathrm{Der}_k(A, A)$. For $a' \in A'$ and $x \in \mathfrak{m}_A$, one has $xa' \in \mathfrak{m}_A$, so $D(xa') \in \mathfrak{m}_A$ and $a'D(x) \in \mathfrak{m}_A$. The identity

$$D(xa') = a'D(x) + xD(a')$$

therefore gives $xD(a') \in \mathfrak{m}_A$ for every $x \in \mathfrak{m}_A$, whence $D(a') \in A'$.

Moreover, the exact sequence

$$0 \longrightarrow \mathrm{Der}_k(A, A) \longrightarrow \mathrm{Der}_k(A', A') \longrightarrow A'/\mathfrak{m}_A A' \longrightarrow 0$$

may be written

$$0 \longrightarrow \mathrm{Der}_k(A, A) \longrightarrow \mathrm{Der}_k(A', A') \longrightarrow A'/C \longrightarrow 0.$$

If X_0' is the normalization of X_0, these local sequences give an exact sequence

$$0 \longrightarrow \underline{D}^0_{X_0} \longrightarrow \underline{D}^0_{X_0'} \longrightarrow \mathcal{O}_{X_0'}/C \longrightarrow 0.$$

Since X_0 is locally a complete intersection,

$$\ell(\mathcal{O}_{X_0'}/C) = 2\ell(\mathcal{O}_{X_0'}/\mathcal{O}_{X_0}) = 2m,$$

whereas the Riemann–Roch theorem gives

$$\chi(\underline{D}^0_{X_0'}) = 3 - 3(g - m).$$

Consequently,

$$\begin{aligned}\chi(\underline{D}^0_{X_0}) &= \chi(\underline{D}^0_{X_0'}) - \ell(\mathcal{O}_{X_0'}/C)\\ &= 3 - 3(g-m) - 2m = 3 - 3g + m.\end{aligned}$$

This completes the proof. □

Remark 6.5. In the formula of Theorem 6.4(5), the number $a = \dim_k H^0(X_0, \underline{D}^0_{X_0})$ is easily computed by the Riemann–Roch theorem:

$$a = \begin{cases} 0, & g \geq 2,\\ 1, & g = 1,\\ 3, & g = 0.\end{cases}$$

We finish with a remark on characteristic 2. If $\operatorname{char} k = 2$, there is no non-degenerate quadratic form in an odd number of variables. Equivalently, there is no non-degenerate quadratic singularity of even dimension in characteristic 2. In this case one may instead consider a quadratic form of maximal non-degeneracy.

Definition 6.6. Let X be a k-scheme of finite type. A closed point $z \in X$ is called an *ordinary quadratic singularity rational over* k if the following conditions hold.

(i) If either $\operatorname{char} k \neq 2$, or $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd, then z is a non-degenerate quadratic singularity rational over k.

(ii) If $\operatorname{char} k = 2$ and $\dim \mathcal{O}_{X,z}$ is even, then

$$\overline{k} \otimes_k \widehat{\mathcal{O}}_{X,z} \simeq \overline{k}[[X_0, X_1, \ldots, X_{2n}]]/(f),$$

where $\overline{k}$ is an algebraic closure of k and

$$f = X_0^2 + X_1X_2 + X_3X_4 + \cdots + X_{2n-1}X_{2n}.$$

Lemma 6.7. Let k be a field of characteristic 2, and put

$$A_0 = k[[X_0, X_1, \ldots, X_{2n}]]/(f), \qquad f = X_0^2 + X_1X_2 + X_3X_4 + \cdots + X_{2n-1}X_{2n}.$$

Let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \operatorname{Spec}(A_0)$. Then

$$R = \Lambda[[t, t_1]],$$

where t, t_1 are free formal generators of R over Λ, and

$$I_R(\mathfrak{X}|R) = (t^2).$$

Moreover, $\mathfrak{X} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth.

Proof. Write $\mathfrak{X} = \operatorname{Spf}(A)$. Since

$$\left(f, \frac{\partial f}{\partial X_0}, \ldots, \frac{\partial f}{\partial X_{2n}}\right) = (X_0^2, X_1, \ldots, X_{2n}),$$

we have

$$D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2).$$

Thus $[\underline{M}_{X_0}(k[\varepsilon])]$ is a two-dimensional k-vector space. By 4.14,

$$R \simeq \Lambda[[t, t_1]], \qquad A \simeq R[[X_0, X_1, \ldots, X_{2n}]]/(F),$$

where

$$F = f + t_1 + tX_0.$$

The exact sequence

$$0 \longrightarrow A \xrightarrow{dF} A\,dX_0 \oplus \cdots \oplus A\,dX_{2n} \longrightarrow \widehat{\Omega}_{A|R} \longrightarrow 0,$$

in which

$$dF:\ 1 \longmapsto \sum_{0 \le i \le 2n} \frac{\partial f}{\partial X_i}\,dX_i + t\,dX_0,$$

gives

$$I_A(A|R) = (t, X_1, \ldots, X_{2n}) = (X_0^2 + t_1, t, X_1, \ldots, X_{2n}).$$

Hence

$$\begin{aligned} A/I_A(A|R) &\simeq A/(X_0^2 + t_1, t, X_1, \ldots, X_{2n}) \\ &\simeq \Lambda[[t_1, X_0]]/(X_0^2 + t_1). \end{aligned}$$

As an R-module this last module is isomorphic to $R/(t) \oplus R/(t)$. It follows that

$$I_R(A|R) = (t^2),$$

which proves the first assertion. On the other hand,

$$\begin{aligned} A &\simeq \Lambda[[t, t_1, X_0, X_1, \ldots, X_{2n}]]/(X_0^2 + X_1X_2 + \cdots + X_{2n-1}X_{2n} + tX_0 + t_1) \\ &\simeq \Lambda[[t, X_0, X_1, \ldots, X_{2n}]], \end{aligned}$$

so A is formally smooth over Λ. □

Lemma 6.7 gives the following analogue of Theorem 6.4; we leave its proof to the reader.

Corollary 6.8. Let X_0 be a k-scheme of finite type whose nonsmooth points are isolated, assume that

$$H^2(X_0, \underline{D}^0_{X_0}) = 0,$$

and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0. Denote the set of nonsmooth points by $\{z_1, \ldots, z_m\}$. Suppose that $\operatorname{char} k = 2$, that $\dim X_0$ is even, and that all the z_i are ordinary quadratic singularities. Then:

(1) R is formally smooth over Λ.

(2) Each ideal $I_R(\mathfrak{X}_{(z_i)}|R)$ is principal, as is

$$I_R(\mathfrak{X}|R) = \prod_{1 \le i \le m} I_R(\mathfrak{X}_{(z_i)}|R).$$

More precisely,

$$I_R(\mathfrak{X}_{(z_i)}|R) = (t_i^2),$$

and $t_1, \ldots, t_m$ can be extended to a system of free formal generators of R over Λ.

(3) The morphism $\mathfrak{X} \longrightarrow \operatorname{Spec}(\Lambda)$ is formally smooth.

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then

$$\dim \underline{M}_{X_0} = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \qquad a = \dim_k H^0(X_0, \underline{D}^0_{X_0}).$$

SGA 7

Exposé VII

BIEXTENSIONS OF SHEAVES OF GROUPS

by A. Grothendieck

0. Introduction

The notion of a biextension was introduced by D. Mumford [6], in the context of formal groups, as a convenient way of expressing the relations between the formal groups associated with an abelian scheme and the dual abelian scheme. In the present exposé we undertake a systematic study of this notion in the general context of sheaves of groups on a given site. In particular, we show that it fits remarkably simply into the classical cohomological formalism of $\overset{\mathrm{L}}{\otimes}$ and $\underline{\mathrm{RHom}}$ (3.6.4 and 3.6.5). In the following exposé we shall make explicit the most interesting special cases, notably that of biextensions of group schemes by the multiplicative group $\underline{G}_m$ (working with such sites as the Zariski site, the fpqc site, or an intermediate site). We shall see, in particular, that if A and B are two abelian schemes over a scheme S, biextensions of (A, B) by $\underline{G}_m$ admit a remarkable interpretation as, essentially, the classical divisorial correspondences on A, B. This gives a particularly convenient explicit cohomological meaning to the notion of divisorial correspondence (already implicit in Tate duality for abelian varieties over p-adic fields). This interpretation of divisorial correspondences is also convenient for following the fate of such a correspondence when A and B (assumed to be defined over the fraction field of a discrete valuation ring) are made to degenerate by means of Néron theory.

The results of Exposé VIII will show, for example, that one obtains a biextension of the pair of identity components of the Néron models, hence a biextension of their connected special fibres, which plays the role of a specialization of the original divisorial correspondence. These results will be used in Exposé IX, in conjunction with the monodromy theorem (I 3), to prove certain results about the Néron models of abelian varieties, the most important being the "semistable reduction theorem" for the Néron model.

Throughout what follows, unless expressly stated otherwise, we work in a fixed $\underline{U}$-topos denoted by $\underline{T}$; that is (SGA 4, Exposé IV), $\underline{T}$ is a category equivalent to the category of sheaves on a suitable $\underline{U}$-site (which may, if desired, be taken to be $\underline{T}$ itself, endowed with its canonical topology). In our terminology we shall often identify the objects of $\underline{T}$ with the sheaves of sets on $\underline{T}$. All groups, extensions of groups, and so forth considered below are relative to this topos $\underline{T}$.

1. Complements on extensions of groups

1.1. Let P and G be two groups of $\underline{T}$, and let

$$(1.1.1) \qquad 1 \longrightarrow G \xrightarrow{\ i\ } E \xrightarrow{\ j\ } P \longrightarrow 1$$

be an extension of P by G. Thus E is a group, $j : E \longrightarrow P$ is an epimorphism of groups, and i induces an isomorphism of G with $\ker j$. We shall reinterpret the datum of such an extension in different terms, which will be convenient for the definition and study of biextensions.

1.1.2. First, the homomorphism i makes G act on E in two ways, on the left and on the right, by setting

$$gxg' = i(g)xi(g')$$

for $g, g' \in G(S)$ and $x \in E(S)$, where S is any object of $\underline{T}$. If, by means of j, we regard E as an object "over P", that is, as an object of the induced topos $\underline{T}_{/P}$, this amounts to saying that the group $G_P = G \times P$ induced by G in $\underline{T}_{/P}$ acts on E on the left and on the right. Moreover, it is well known that each of these actions makes E a torsor (a left torsor and a right torsor, respectively) over P, with group G_P; more precisely still, it makes E a bitorsor under (G_P, G_P). Recall [2] that this means that E is a right torsor E_d under the right action of G_P, and that the left action induces an isomorphism of G_P with the group of automorphisms of E_d (the sheaf of groups of automorphisms of E that commute with the right action of G_P on E). This condition is equivalent to the symmetric condition: the left action of G_P on E makes E a left torsor E_s, and the right action induces an isomorphism of the opposite group G_P^{o} with the group of automorphisms of E_s.

1.1.2.1. When $\underline{T}$ is the "point topos", that is, the category of sets (equivalent to the category of sheaves on the one-point topological space), so that P and G are ordinary groups, the datum of a bitorsor E under (G_P, G_P) is plainly equivalent to the datum, for each $p \in P$, of a bitorsor E_p under (G, G), namely the fibre of E at p. An analogous interpretation is available in the general case. Indeed, giving a bitorsor E under (G_P, G_P) is equivalent to giving, for every "point" $p \in P(S)$, where S is any object of $\underline{T}$, a bitorsor E_p under (G_S, G_S) over S, in such a way that the formation of E_p is "compatible with base change along every morphism $S' \longrightarrow S$". Thus, if $p' : S' \longrightarrow P$ is the composite morphism, one is also given an isomorphism of bitorsors from $E_{p'}$ to the inverse image of E_p under $S' \longrightarrow S$, and these isomorphisms satisfy the usual transitivity condition for a composite $S'' \longrightarrow S' \longrightarrow S$.

When stated in canonical language (SGA 1, Exposé VI), this reduces to the following tautology: giving a bitorsor under (G_P, G_P) over P is equivalent to giving a cartesian section, over the category $\underline{T}_{/P}$, of the fibred category of bitorsors under (G_S, G_S) over a variable base $S \in \mathrm{ob}\,\underline{T}_{/P}$. More precisely, the "value at P" functor induces an equivalence between the category of the cartesian sections in question and the category of bitorsors under (G_P, G_P). This follows trivially from the fact that P is a final object of the base category $\underline{T}_{/P}$ of our fibred category: for <u>every</u> category fibred over $\underline{T}_{/P}$, the "value at P" functor is an equivalence between the category of cartesian sections of that fibred category and its fibre category over P. Despite its tautological nature, however, this interpretation will be useful for expressing certain compatibilities in language close to the set-theoretic case, using points with values in an object S in place of ordinary points.

1.1.3. In terms of the extension structure (1.1.1), the bitorsor structure of 1.1.2 on E under (G_P, G_P) used only part of the multiplicative structure of E: namely, multiplication of pairs of "points" of which at least one comes from a point of G (via i). We shall now interpret the multiplicative structure of E as an additional structure on the bitorsor under consideration.

For this purpose, first consider two sections p, p' of P. With the notation of 1.1.2.1, these give two bitorsors $E_p, E_{p'}$ under (G, G), namely the inverse images in E of the sections $p, p' : e \longrightarrow P$ (where e denotes the final object of $\underline{T}$). It is then clear that multiplication in E induces a morphism of objects of $\underline{T}$

$$\varphi : E_p \times E_{p'} \longrightarrow E_{pp'},$$

which, in terms of the left and right structures of the three bitorsors involved (and, as usual, using points with values in an arbitrary object $S \in \mathrm{ob}\,\underline{T}$), satisfies

$$\begin{cases} \varphi(gx, x') = g\varphi(x, x'), & \varphi(x, x'g') = \varphi(x, x')g', \\ \varphi(xg, x') = \varphi(x, gx'). \end{cases} \tag{1.1.3.0}$$

Here $g, g' \in G(S)$, $x \in E_p(S)$, and $x' \in E_{p'}(S)$. In terms of the contracted product [2]

$$E_p \overset{G}{\times} E_{p'},$$

the third relation says that φ factors as

$$E_p \times E_{p'} \longrightarrow E_p \overset{G}{\times} E_{p'} \xrightarrow{\varphi_{p,p'}} E_{pp'},$$

where the first arrow is the canonical quotient morphism. The first two relations, on the other hand, say that the morphism thus obtained,

$$\varphi_{p,p'} : E_p \overset{G}{\times} E_{p'} \longrightarrow E_{pp'}, \tag{1.1.3.1}$$

is a homomorphism of bitorsors, where the bitorsor structure on the source is the one for which the left (respectively, right) operations of G come from the left (respectively, right) operations of G on the first factor E_p (respectively, on the second factor $E_{p'}$).

Notice that, like every homomorphism of bitorsors under a fixed pair of groups, $\varphi_{p,p'}$ is necessarily an isomorphism.

The definition of the preceding isomorphism (1.1.3.1) immediately extends to the case in which p, p' are points of P with values in an arbitrary object S of $\underline{T}$, rather than sections of P: one simply applies the preceding discussion to the induced topos $\underline{T}_{/S}$ and to the extension of groups over S obtained from (1.1.1) by base change. The formation of the isomorphisms $\varphi_{p,p'}$ is then compatible with every base change $S' \longrightarrow S$. In other words, it corresponds to a homomorphism of cartesian sections of the fibred category over the category $\underline{T}_{/P\times P}$ of pairs of arrows $p, p' : S \rightrightarrows P$ with target P.

Giving a system of $\varphi_{p,p'}$ satisfying this condition is, of course, equivalent to giving $\varphi_{p,p'}$ in the "universal" case $S = P \times P$, where p, p' are the two projections pr_i $(i = 1, 2)$ from $P \times P$ to P. It is then an isomorphism of bitorsors under $(G_{P\times P}, G_{P\times P})$:

$$\varphi : \mathrm{pr}_1^*(E)\ \mathrm{pr}_2^*(E) \xrightarrow{\sim} \pi^*(E), \tag{1.1.3.2}$$

where

$$\pi : P \times P \longrightarrow P$$

is the morphism of objects of $\underline{T}$ defining the multiplicative structure of the group P. The compatibility properties below are most conveniently stated, however, in terms of the data (1.1.3.1), avoiding diagrams that are less close to intuition.

1.1.4. In writing the relations (1.1.3.0), which enabled us to express the given composition law on E in terms of isomorphisms of bitorsors (1.1.3.1), we used associativity of this composition law only for a product of three "points" of E of which at least one comes from a point of G. To express full associativity in terms of the $\varphi_{p,p'}$, one must require, for three points p, p', p'' of P with values in an object S of $\underline{T}$, that the following diagram be commutative:

$$\begin{tikzcd}
& E_{pp'}E_{p''} \arrow[dr, "\varphi_{pp',p''}"] & \\
E_pE_{p'}E_{p''} \arrow[ur, "\varphi_{p,p'}\wedge \mathrm{id}"] \arrow[dr, "\mathrm{id}\wedge\varphi_{p',p''}"'] & & E_{pp'p''} \\
& E_pE_{p'p''} \arrow[ur, "\varphi_{p,p'p''}"'] &
\end{tikzcd} \tag{1.1.4.1}$$

In this diagram we have written the contracted product of bitorsors without the signs $\overset{G_S}{\times}$. It is understood that the contracted product $Q.R$ of two bitorsors is formed using, in passing to the quotient in $Q \times R$, the action of the structural group G on the right of Q and on the left of R. This contracted-product operation is associative up to a canonical isomorphism [2], and we make the identification

$$(E_pE_{p'})E_{p''} \simeq E_p(E_{p'}E_{p''}),$$

writing both sides simply as $E_pE_{p'}E_{p''}$. Obvious conventions of this kind will be used throughout the rest of this exposé without being expressly noted each time.

1.1.4.2. Just as the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) are determined by the "universal" case $S = P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$ considered in (1.1.3.2), so the associativity condition (1.1.4.1) may likewise be checked only in the universal case $S = P \times P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, $p'' = \mathrm{pr}_3$. We leave it to the reader to write the diagram in whatever notation is preferred.

1.1.5. Conversely, suppose that the groups P and G of $\underline{T}$ are given, together with a bitorsor

$$j : E \longrightarrow P \tag{1.1.5.1}$$

under (G_P, G_P) (the two actions of G_P on E, or equivalently of G on E, being understood in the notation), and a system of isomorphisms of bitorsors over a variable base S,

$$\varphi_{p,p'} : E_p E_{p'} \xrightarrow{\sim} E_{pp'} \qquad (p, p' \in P(S)), \tag{1.1.5.2}$$

compatible with base change $S' \longrightarrow S$; in other words, an isomorphism of bitorsors (1.1.3.2). Suppose that the corresponding diagrams (1.1.4.1) commute. I claim that <u>there is a unique extension structure on E of P by G such that the preceding data are obtained from it by the constructions of 1.1.2 and 1.1.3</u>.

First observe that, on taking p, p' in (1.1.5.2) to be the identity section 1 of P, one obtains an isomorphism

$$\varphi_{1,1} : E_1 \xleftarrow{\sim} E_1 \overset{G}{\times} E_1. \tag{1.1.5.3}$$

Taking the contracted product, on the left say, of both sides with the inverse bitorsor E_1^{-1} of E_1 [2] therefore gives a canonical isomorphism

$$E_1 \simeq \text{the unit bitorsor } {}_sG_d \tag{1.1.5.4}$$

(that is, G acting on itself by left and right translations), in view of the canonical isomorphism

$$E_1^{-1}.E_1 \simeq \text{the unit bitorsor } {}_sG_d$$

of loc. cit., valid for every bitorsor. Equivalently, one has a section 1_E of E_1, hence of E,

$$1_E \in \Gamma(E_1) = \mathrm{Hom}(e, E_1) \subset \Gamma(E), \tag{1.1.5.5}$$

namely the image under (1.1.5.4) of the distinguished section of ${}_sG_d$, satisfying

$$g.1_E = 1_E.g \qquad (g \in G(S)).$$

By construction this section is characterized by

$$\varphi_{1,1}(1_E, 1_E) = 1_E. \tag{1.1.5.6}$$

On the other hand, giving the morphism of torsors φ of (1.1.3.2) is plainly equivalent to giving a morphism

$$\pi_E : E \times E \longrightarrow E, \qquad \text{also denoted } (x, y) \longmapsto xy, \tag{1.1.5.7}$$

satisfying

$$j(xy) = j(x)j(y) \qquad (x, y \in E(S)), \tag{1.1.5.8}$$

so that j is compatible with the multiplicative structures π_E on E and π on P, together with the three relations (1.1.3.0) (preferably beginning with the last of these). Finally define a morphism of objects of $\underline{T}$,

$$i : G \longrightarrow E,$$

by the formula

$$i(g) = g.1_E = 1_E.g. \tag{1.1.5.9}$$

The equality of the last two terms was already observed in constructing 1_E. I now claim that the multiplication law π_E on E makes E a group with identity section 1_E, and that $j : E \longrightarrow P$ and $i : G \longrightarrow E$ are homomorphisms of groups making E an extension of P by G.

Indeed, associativity of the composition law on E is precisely the commutativity of the diagrams (1.1.4.1). The construction of π_E by means of the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) also shows that, for every $x \in E(S)$, left and right translation by x in $E(S)$ are injective. Since $1_E 1_E = 1_E$ by (1.1.5.6), it follows at once that 1_E is a two-sided identity of E. We leave to the reader the verification that an $x \in E(S)$ lying over $p \in P(S)$ has an inverse; one uses the canonical isomorphism deduced from $\varphi_{p,p^{-1}}$,

(1.1.5.10) $$E_{p^{-1}} \simeq (E_p)^{\circ}.$$

Thus E is indeed a group. We have already observed that j is compatible with the composition laws (1.1.5.8), and the same is true of i, since, for $g, g' \in G(S)$,

$$\begin{aligned}
i(g)i(g') &= (g.1_E)(g'.1_E) \\
&= (g.1_E.g')1_E && \text{(the last relation of (1.1.3.0))} \\
&= g.1_E.g' && \text{(since 1_E is an identity)} \\
&= g.(1_E.g') = g.(g'.1_E) \\
&= (gg').1_E && \text{by (1.1.5.9)} \\
&= i(gg').
\end{aligned}$$

Since j is the structural projection of a torsor, it is an epimorphism of objects of $\underline{T}$. It remains only to see that i induces an isomorphism of G with $\ker j = E_1$, which is clear from the definition (1.1.5.9) and the construction of 1_E.

1.1.6. We shall accept that the preceding arguments prove the following. For a fixed topos $\underline{T}$, the preceding constructions give an equivalence between the category of extensions of groups of $\underline{T}$ as in (1.1.1), where P, G, E are all variable, and the category of systems (P, G, E, φ) in which P and G are groups of $\underline{T}$, E is a bitorsor under (G_P, G_P) over P, and φ is an isomorphism of bitorsors (1.1.3.2) (or, equivalently, a system of base-change-compatible isomorphisms of bitorsors (1.1.3.1)), satisfying the associativity condition expressed by commutativity of diagram (1.1.4.2) (or, equivalently, of the diagrams (1.1.4.1)).

We leave it to the reader to spell out, using the preceding dictionary, the evident notion of inverse image (respectively, direct image) of an extension (1.1.1) under a homomorphism of groups

$$P' \longrightarrow P \qquad \big(\text{respectively, } G \longrightarrow G'\big),$$

or under both operations simultaneously. We likewise leave to the reader the compatibility of the preceding equivalence of categories with the inverse-image functor f^* associated with a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

For this one should use the datum φ in the form (1.1.3.2), in preference to the $\varphi_{p,p'}$ of (1.1.3.1); it then defines the corresponding datum $f^*(\varphi)$ on the topos $\underline{T}'$.

1.2. The commutative case. From 1.4 onward we shall be concerned only with commutative extensions of groups. We therefore assume that P and G are commutative, and spell out here, in terms of the bitorsor description (P, G, E, φ) of an extension, the condition that E be a commutative group.

An obviously necessary first condition—which in fact expresses that $i(G)$ be a central subgroup of E—is that the bitorsors E_p, for $p \in P(S)$, be ordinary torsors; that is, the left and right actions of G on E_p must coincide. Equivalently, E over P is an ordinary torsor under G_P. (Thus there will henceforth be no need to speak of bitorsors, and we shall consider only ordinary torsors.)

Assuming this preliminary condition, commutativity of E is expressed by commutativity of the following diagrams, for $p, p' \in P(S)$:

$$
\begin{tikzcd}
E_p E_{p'} \arrow[r, "\varphi_{p,p'}"] \arrow[d, "\mathrm{sym}"'] & E_{pp'} \arrow[d, "\mathrm{id}"] \\
E_{p'} E_p \arrow[r, "\varphi_{p',p}"] & E_{p'p}.
\end{tikzcd}
\tag{1.2.1}
$$

Here the first vertical arrow is the canonical commutativity isomorphism (defined because G is now commutative); moreover $pp' = p'p$ because P is commutative. This condition is plainly equivalent to the corresponding condition in the universal situation $S = P \times P$, $p = \mathrm{pr}_1$, $p' = \mathrm{pr}_2$, which we need not write out again.

1.3. Relations with rigidified or birigidified torsors

1.3.1.0. Let P be an object of $\underline{T}$ equipped with a section e_P, let G be a group of $\underline{T}$, and let E be a torsor under G_P. A rigidification of E, relative to the given section e_P of P, is a trivialization of the torsor $E_1 = e_P^*(E)$ under G;

that is, a section of this torsor. This terminology is inspired by the case where G is commutative and

$$\mathrm{Hom}_{\mathrm{pt}}(P, G) \simeq (e), \tag{1.3.1.1}$$

where the left-hand side denotes the set of morphisms carrying e_P to the identity section e_G of G. This relation is equivalent to saying that the natural homomorphism

$$\Gamma(G) \xrightarrow{\sim} \mathrm{Hom}(P, G), \tag{1.3.1.2}$$

which associates with a section g of G the constant morphism $P \longrightarrow G$ with value g, is an isomorphism. It is immediate that these equivalent conditions are also equivalent to saying that every automorphism of the torsor E preserving the given rigidification is the identity automorphism (such automorphisms correspond precisely to the elements of the left-hand side of (1.3.1.1)). In particular, under condition (1.3.1.1), the category of torsors under G_P rigidified relative to e_P is "rigid".

1.3.2. Let, in addition, Q be an object of $\underline{T}$ equipped with a section e_Q, and now let E denote a torsor under $G_{P \times Q}$. Since e_Q defines a section $e_{P\times Q/P}$ of $P \times Q$ over P, one may consider rigidifications α of E relative to this section, working in the induced topos $\underline{T}_{/P}$; explicitly, these are sections of the torsor induced by E on $P \times e_Q$. Interchanging the roles of P and Q, one may likewise consider rigidifications β of E relative to the section $e_{P\times Q/Q}$ of $P \times Q$ over Q defined by e_P. Two such partial rigidifications are said to be compatible if the sections α and β agree on $e_P \times e_Q$, that is, if they define the same section of the torsor $E_{1,1}$ under G, the inverse image

of E under the section (e_P, e_Q) of $P \times Q$. A birigidification of the torsor E relative to the pair of sections (e_P, e_Q) is a pair of compatible partial rigidifications.

If G is commutative, then E admits a birigidification (α, β) if and only if it admits partial rigidifications α and β; moreover one may find a birigidification in which α (or β) is prescribed in advance. Indeed, it suffices to correct β by the morphism $Q \longrightarrow G$ obtained as the composite

$$Q \longrightarrow e \xrightarrow{\lambda} G,$$

where λ is defined by $\beta e_Q = (\alpha e_P)\lambda$.

Still in the commutative case, if E, E' are two birigidified torsors under $G_{P\times Q}$, saying that they are isomorphic merely as $G_{P\times Q}$-torsors is equivalent to saying that they are isomorphic as birigidified $G_{P\times Q}$-torsors. In other words, if there is an isomorphism $u : E \longrightarrow E'$ of $G_{P\times Q}$-torsors, then there is one $v : E \longrightarrow E'$ that preserves the birigidifications. Indeed, measuring the discrepancy of u from the given birigidifications gives morphisms $f : P \longrightarrow G$ and $g : Q \longrightarrow G$

such that $f(e_P) = g(e_Q) = a \in G(e)$. There is then a morphism $P \times Q \longrightarrow G$ whose restrictions along $e_{P\times Q/P}$ and $e_{P\times Q/Q}$ are respectively f and g, namely

$$h(p,q) = f(p)g(q)a^{-1},$$

and one may take $v = uh$.

1.3.3. The terminology of 1.3.1 extends immediately to the case of a bitorsor E under (G_P, G_P). A rigidification of the bitorsor E is then required to be an isomorphism from $e_P^*(E)$ to the trivial bitorsor ${}_sG_d$; equivalently, it is defined by a central section e_E of $e_P^*(E)$, one satisfying

$$g.e_E = e_E.g \qquad (g \in G(S)).$$

The terminology of birigidifications extends in the same way to bitorsors.

1.3.4. Return to the case in which two groups P and G are given, and choose the identity section e_P in order to define rigidifications of bitorsors over P and birigidifications of bitorsors over $P \times P$. There are natural functors

$$\textbf{EXT}(P;G) \xrightarrow{i} \mathrm{BITORSRIG}(P,G) \xrightarrow{\delta} \textbf{BITORSBIRIG}(P,P;G), \tag{1.3.4.1}$$

where the three expressions denote, respectively, the category of extensions of the group P by G, the category of bitorsors under (G_P, G_P) rigidified at e_P, and the category of bitorsors under $(G_{P\times P}, G_{P\times P})$ birigidified at (e_P, e_P). The functor i associates with an extension E of P by G the corresponding bitorsor over P (1.1.6), rigidified by the identity section of E. The functor δ is defined by

$$\delta(E) = \pi^*(E)\ \mathrm{pr}_1^*(E)^{-1}\ \mathrm{pr}_2^*(E)^{-1}, \tag{1.3.4.2}$$

where $\pi : P \times P \longrightarrow P$ is the composition law of P, and $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections. There is a canonical birigidification of $\delta(E)$. Indeed, the restriction of $\delta(E)$ to $P \times e_P$ is canonically isomorphic to

$$\mathrm{id}_P^*(E)\,\mathrm{id}_P^*(E)^{-1}\,e^*(E)^{-1},$$

where $e = \mathrm{pr}_2\,\mathrm{inj}_1$ is the constant morphism $P \longrightarrow P$ with value e_P. The given rigidification of E provides an isomorphism from $e^*(E)$ to the trivial bitorsor, and hence an isomorphism of the restriction of $\delta(E)$ to $P \times e_P$ with the trivial bitorsor. Symmetrically one defines a partial rigidification along $e_P \times P$, and the two partial rigidifications are immediately seen to be compatible.

1.3.4.3. Observe that the composite δi of the functors in (1.3.4.1) is canonically isomorphic to the trivial functor, that is, to the constant functor whose value is the trivial birigidified bitorsor over $P \times P$. Indeed, for every extension E of P by G, the isomorphism φ of (1.1.3.2) plainly defines a trivialization of $\delta i(E)$ compatible with its canonical birigidification.

Proposition 1.3.5. Let P and G be two groups of $\underline{T}$.

a) Suppose that every morphism of sheaves of sets $P \times P \longrightarrow G$ which is trivial on $P \times e_P$ and on $e_P \times P$ is trivial. Then the functor i of (1.3.4.1) is fully faithful. In particular, on a bitorsor E under (G_P, G_P) rigidified relative to e_P, that is, equipped with a central section e_E of $E_1 = e_P^*(E)$, there is at most one extension structure of P by G compatible with its bitorsor structure and having e_E as identity section.

b) Suppose in addition that every morphism of sheaves of sets $P \times P \times P \longrightarrow G$ which is trivial on the partial products $P \times P \times e_P$, $P \times e_P \times P$, and $e_P \times P \times P$ is trivial. Let E be a bitorsor under (G_P, G_P). In order that E admit an extension structure of P by G compatible with its bitorsor structure, it is necessary and sufficient that the bitorsor under $(G_{P\times P}, G_{P\times P})$ underlying

$$\delta(E) = \pi^*(E)\ \mathrm{pr}_1^*(E)^{-1}\ \mathrm{pr}_2^*(E)^{-1}$$

be trivial. When this is so, for every central section e_E of $E_1 = e_P^*(E)$, that is, every rigidification of E relative to e_P, there is a unique extension structure on E compatible with its bitorsor structure and having e_E as identity section. In other words, E, regarded as a bitorsor rigidified relative to e_P, lies in the essential image of i. Moreover, the birigidified bitorsor $\delta(E)$ has one and only one trivialization.

Thus, in terms of diagram (1.3.4.1), the essential image of the fully faithful functor i is "the kernel" of δ. More picturesquely, diagram (1.3.4.1) is an "exact sequence of pointed categories," and even of categories endowed with pseudo-group structures (in the terminology introduced in 2.5.1 below).

Let us prove 1.3.5. For a), let E, E' be two extensions of P by G, and let $f : E \longrightarrow E'$ be a morphism of the underlying rigidified bitorsors. We show that f is a morphism of extensions, or, what plainly amounts to the same thing, a homomorphism of groups. We must prove $f(xy) = f(x)f(y)$. If $j' : E' \longrightarrow P$ denotes the structural morphism of the torsor E' over P, then $j'f(xy) = j'(f(x)f(y))$: the two sides are respectively $j(xy)$ and $j(x)j(y)$, which are equal because j is multiplicative. Hence

$$f(xy) = u(j(x), j(y))f(x)f(y),$$

where

$$u : P \times P \longrightarrow G$$

is a uniquely determined morphism. Setting first $x = 1$ and then $y = 1$ in the preceding relation, and using $f(1) = 1$, gives

$$u(p, 1) = u(1, p) = 1.$$

Thus u is trivial on $P \times e_P$ and on $e_P \times P$; by hypothesis it is trivial, and therefore $f(xy) = f(x)f(y)$, as required.

For b), necessity of the stated condition for E to admit an extension structure compatible with its bitorsor structure is immediate from 1.3.4.3. It remains to prove sufficiency for the first assertion. Pulling a trivialization φ of $\delta(E)$ back along the identity section of $P \times P$ gives a trivialization of $E_1 = e_P^*(E)$ (compare 1.5.1), that is, a rigidification of E relative to e_P. Fix this rigidification. A given trivialization φ of $\delta(E)$ need not be compatible with the corresponding birigidification. Relative to the natural partial rigidifications of $\delta(E)$, its restrictions to $P \times e_P$ and $e_P \times P$ are described by central morphisms

$$f : P \longrightarrow G, qquadg : P \longrightarrow G,$$

whose values at the identity section of P coincide: $f(1) = g(1) = a \in \Gamma G$. The central morphism

$$h : P \times P \longrightarrow G, qquadh(p, q) = f(p)g(q)a^{-1},$$

has restrictions f and g to $P \times e_P$ and $e_P \times P$. Correcting φ by h, that is, replacing it by $\varphi(p, q)h(p, q)^{-1}$, therefore gives a trivialization of $\delta(E)$ compatible with its birigidification. Such a trivialization is unique: two of them differ by a central morphism $P \times P \longrightarrow G$ trivial on $P \times e_P$ and $e_P \times P$, hence trivial by hypothesis. This proves the last assertion of 1.3.5 b).

It remains to prove that φ, viewed as an isomorphism of the form (1.1.3.2), satisfies the associativity condition (1.1.4.1); by our choice of φ, e_E is already a two-sided identity for the composition law defined by φ. The two composites in (1.1.4.1) are isomorphisms of bitorsors under $(G_{P \times P \times P}, G_{P \times P \times P})$, so they differ by a central morphism

$$u : P \times P \times P \longrightarrow G$$

through the formula

$$(xy)z = u(jx, jy, jz)\, x(yz).$$

Setting successively $z = 1$, $x = 1$, and $y = 1$, and using the fact that 1 is a two-sided identity for the composition law, shows that u is trivial on the three partial products $P \times P \times e_P$, $e_P \times P \times P$, and $P \times e_P \times P$. The hypothesis then implies that u is trivial, so $(xy)z = x(yz)$. This completes the proof.

Corollary 1.3.6. Suppose that every morphism of sheaves of sets $P \times G \longrightarrow G$ which is trivial on $P \times e_G$ and on $e_P \times G$ is trivial. Then, if G is commutative, G is central in every extension of P by G. If P is also commutative and the hypothesis of 1.3.5 a) holds, then every extension E of P by G is commutative.

For the first assertion, we must prove that the action of P on G defined by the extension E is trivial. This action is expressed by a morphism $u : P \times G \longrightarrow G$. The restrictions to $P \times e_G$ and $e_P \times G$ of the corresponding morphism

$$(p, g) \longmapsto u(p, g)g^{-1}$$

are plainly trivial. Hence this last morphism is trivial, so $u(p, g) = g$, as required. Now, if E is an extension of P by G, its opposite group E^{o} is an extension of P^{o} by G^{o}, and the bitorsor structure underlying $i(E^{\mathrm{o}})$ is obtained from $i(E)$ by interchanging the left and right actions. When E is a central extension, one therefore has $i(E) = i(E^{\mathrm{o}})$; if, in addition, P is commutative, so that $P = P^{\mathrm{o}}$, the uniqueness assertion 1.3.5 a) gives $E = E^{\mathrm{o}}$. Thus E is commutative, proving the corollary.

1.3.7. One interesting special case in which all the hypotheses of 1.3.5 and 1.3.6 hold is

$$\underline{\mathrm{Hom}}_{\mathrm{pt}}(P, G) = e \quad \text{(the final sheaf)}, \tag{1.3.7.1}$$

where the left-hand side denotes the sheaf of morphisms of pointed sheaves of sets from P to G. Independently of the marked sections, this condition may also be written in the form, analogous to (1.3.1.2),

$$G \simeq \underline{\mathrm{Hom}}_{\mathrm{sets}}(P, G), \tag{1.3.7.2}$$

where the second member denotes the sheaf of morphisms between the underlying sheaves of sets of P and G. These conditions also mean that, for every object Q of $\underline{T}$, the map $u \longmapsto u \circ \mathrm{pr}_2$

$$\mathrm{Hom}_{\mathrm{sets}}(Q, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{sets}}(P \times Q, G) \tag{1.3.7.3}$$

is bijective. This property of P (with G fixed) is stable under Cartesian products; in particular it implies the same property for $P \times P$, $P \times P \times P$, and so forth. It follows immediately that it implies the hypotheses made in 1.3.5 and 1.3.6. It also implies that all the categories occurring in (1.3.4.1) are rigid, that is, every automorphism of an object of any one of them is the identity.

Moreover, since condition (1.3.7.1) on P, G is stable under every base change $S \longrightarrow e$, descent together with the uniqueness result 1.3.5 a) shows that a rigidified bitorsor E over P comes from an extension of P by G as soon as it does so locally over e. By 1.3.5 b), this may also be expressed by the vanishing of the section

$$\underline{\delta}(E) \in \Gamma R^1 h_*(G_{P \times P})$$

defined by $\delta(E)$, where

$$h : P \times P \longrightarrow e$$

is the structural morphism.

1.3.8. Let $p : P \longrightarrow S$ be a morphism of schemes which is coherent (that is, quasi-compact and quasi-separated) and flat, and suppose that

$$p_*(\mathcal{O}_P) \xleftarrow{\sim} \mathcal{O}_S. \tag{1.3.8.1}$$

Since the formation of $p_*(\mathcal{O}_P)$ commutes with every flat base change (by coherence of p), the stated conditions on the relative scheme P/S are stable under Cartesian products and hence hold, in particular, for the Cartesian powers $P \times P$, $P \times P \times P$, and so forth. They are also stable under every flat base change.

Now let G be an affine scheme over S. Relation (1.3.7.2) follows at once from (1.3.8.1), since

$$\mathrm{Hom}_S(P, G) = \mathrm{Hom}_{\mathrm{Alg}}(\underline{A}, p_*(\mathcal{O}_P)) \simeq \mathrm{Hom}_{\mathrm{Alg}}(\underline{A}, \mathcal{O}_S),$$

where $\underline{A}$ is the quasi-coherent algebra over S such that $G = \mathrm{Spec}(\underline{A})$. By the preceding observation, the same conclusion applies to $P \times P$, $P \times P \times P$, and so on. Likewise, if G is flat over S, the same conclusion applies to the relative scheme P_G over G with respect to the relative scheme G_G over G.

When P and G are group schemes over S, and one works with a topology on $(\mathrm{Sch})_{/S}$ coarser than the canonical topology, so that P and G may be identified with sheaves on this site, every morphism from P, $P \times P$, $P \times P \times P$, and so on to G which is compatible with the marked sections is trivial. Similarly, if G is flat, every S-morphism $P \times G \longrightarrow G$ compatible with the marked sections is trivial, as one sees by interpreting it as a G-morphism from P_G to G_G. Thus the hypotheses of 1.3.5 hold, as do those of 1.3.6 when G is flat. Moreover, the three categories occurring in (1.3.4.1) are rigid.

1.3.8.2. The most interesting case in which (1.3.8.1) holds, and continues to hold after every extension of the base, is that in which P is an abelian scheme over S; then the conditions of 1.3.7 themselves hold. In this case the conclusions of 1.3.5 and 1.3.6 are therefore valid for every affine group G over S. This is the case originally considered by J. P. Serre, and the preceding developments were plainly inspired by it. An interesting case in which (1.3.8.1) holds without holding universally is that in which P is the Néron model of an abelian scheme over the fraction field of a discrete valuation ring with spectrum S.

1.4. Extensions of a free commutative group

Let I be an object of $\underline{T}$, and put

$$P = \mathbb{Z}[I], \tag{1.4.1}$$

the free $\mathbb{Z}$-module generated by I (SGA 4, IV, 11). With every extension E of P by a given $\mathbb{Z}$-module G (from now on only commutative group extensions will occur), associate the corresponding torsor under G_P, and then restrict it to I along the canonical morphism

$$I \longrightarrow P = \mathbb{Z}[I].$$

This gives a canonical functor

$$\mathrm{EXT}(P, G) \longrightarrow \mathrm{TORS}(I, G_I). \tag{1.4.2}$$

I claim that this functor is an equivalence of categories. Equivalently, for every G, the maps

$$\mathrm{Hom}_{\mathbb{Z}}(P, G) \longrightarrow H^0(I, G_I) = G(I) \tag{1.4.3}$$

and

$$\mathrm{Ext}^1_{\mathbb{Z}}(P, G) \longrightarrow H^1(I, G_I) \tag{1.4.4}$$

are isomorphisms. More generally, if A is a ring (not necessarily commutative) of $\underline{T}$ and $P = A[I]$ is the free A-module generated by I, there are canonical isomorphisms, functorial in the variable A-module G,

$$\mathrm{Ext}^i_A(A[I], G) \xrightarrow{\sim} H^i(I, G_I). \tag{1.4.5}$$

For $A = \mathbb{Z}$ and $i = 0, 1$, these coincide with (1.4.3) and (1.4.4). To define (1.4.5), recall that its left-hand side is the derived functor of

$$G \longmapsto \mathrm{Hom}_A(A[I], G),$$

which, by the definition of $A[I]$, is isomorphic to $G \longmapsto G(I) = H^0(I, G_I)$. Since $G \longmapsto G_I$ carries injective modules to injective modules of $\underline{T}_{/I}$ (SGA 4, V), the assertion follows. We leave to the reader the verification of the stated compatibility with (1.4.3) and (1.4.4).

Finally, the functor (1.4.2) is compatible with inverse images under a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

2. <u>The notion of a biextension of abelian sheaves</u>

2.0. Throughout this section, P, Q, G denote commutative groups of the topos $\underline{T}$. We shall frequently use the Cartesian diagram

(2.0.1)
$$\begin{tikzcd} P & P \times Q \arrow[l] \arrow[d] \\ e \arrow[u, leftarrow] & Q \arrow[l] \end{tikzcd}$$

and the groups G_P (respectively G_Q, $G_{P\times Q}$) over P (respectively Q, $P \times Q$), obtained by inverse image from the group G over e. We identify $G_{P\times Q}$ either with the inverse image $(G_P)_{P\times Q}$ under pr_1, or with the inverse image $(G_Q)_{P\times Q}$ under pr_2. In this notation the group structures of $P, Q, P \times Q$ play no role; indeed, we shall almost never use the product-group structure on $P \times Q$. We shall instead use on $P \times Q$ the structure of a group over P obtained by inverse image from Q over e,

(2.0.2)
$$P \times Q = Q_P,$$

and, similarly, the structure of a group over Q obtained by inverse image from P over e,

(2.0.3)
$$P \times Q = P_Q.$$

Thus over P we have the two groups G_P, Q_P, and over Q the two groups G_Q, P_Q.

If E is a torsor over $P \times Q$ under $G_{P\times Q}$, it may be viewed either as a torsor over Q_P under $G_{P\times Q} = (G_P)_{Q_P}$, or as a torsor over P_Q under $G_{P\times Q} = (G_Q)_{P_Q}$. It therefore makes sense to equip E with a commutative extension structure in either variable. Applied in the induced topoi $\underline{T}_{/P}$ and $\underline{T}_{/Q}$, 1.1.6 gives a convenient explicit description, using the canonical isomorphisms

$$\underline{T}_{/P\times Q} \simeq (\underline{T}_{/P})_{/P\times Q}, \qquad \underline{T}_{/P\times Q} \simeq (\underline{T}_{/Q})_{/P\times Q}.$$

An extension structure in the P-direction—that is, of P_Q by G_Q—is equivalent to giving, for every object S of $\underline{T}$ and elements $p, p' \in P(S)$, $q \in Q(S)$, an isomorphism of torsors under G_S

(2.0.4)
$$\varphi_{p,p';q} : E_{p,q}E_{p',q} \xrightarrow{\sim} E_{pp',q}.$$

As usual (cf. 1.1.2.1), for $(p, q) \in (P \times Q)(S)$, $E_{p,q}$ denotes the torsor under G_S obtained from E by inverse image along $(p, q) : S \longrightarrow P \times Q$. The data (2.0.4) must be <u>functorial in S</u>, associative, and commutative. The associativity condition is the commutativity of

(2.0.5)
$$\begin{tikzcd} & E_{pp',q}E_{p'',q} \arrow[dr, "\varphi_{pp',p'';q}"] & \\ E_{p,q}E_{p',q}E_{p'',q} \arrow[ur, "\varphi_{p,p';q}\wedge\mathrm{id}"] \arrow[dr, "\mathrm{id}\wedge\varphi_{p',p'';q}"'] & & E_{pp'p'',q} \\ & E_{p,q}E_{p'p'',q} \arrow[ur, "\varphi_{p,p'p'';q}"'] & \end{tikzcd}$$

for $p, p', p'' \in P(S)$ and $q \in Q(S)$. Commutativity is expressed by

$$
\begin{tikzcd}
E_{p,q}E_{p',q} \arrow[r, "\varphi_{p,p';q}"] \arrow[d, "\mathrm{sym}"'] & E_{pp',q} \arrow[d, "\mathrm{id}"] \\
E_{p',q}E_{p,q} \arrow[r, "\varphi_{p',p;q}"'] & E_{p'p,q}.
\end{tikzcd}
\tag{2.0.6}
$$

Here $p, p' \in P(S)$ and $q \in Q(S)$.

Symmetrically, an extension structure in the Q-direction—that is, of Q_P by G_P—is described by isomorphisms of torsors under G_S

$$
\psi_{p;q,q'} : E_{p,q}E_{p,q'} \xrightarrow{\sim} E_{p,qq'},
\tag{2.0.7}
$$

for $p \in P(S)$ and $q, q' \in Q(S)$. These are functorial in S and satisfy the symmetric associativity and commutativity conditions

$$
\begin{tikzcd}
& E_{p,qq'}E_{p,q''} \arrow[dr, "\psi_{p;qq',q''}"] & \\
E_{p,q}E_{p,q'}E_{p,q''} \arrow[ur, "\psi_{p;q,q'}\wedge\mathrm{id}"] \arrow[dr, "\mathrm{id}\wedge\psi_{p;q',q''}"'] & & E_{p,qq'q''} \\
& E_{p,q}E_{p,q'q''} \arrow[ur, "\psi_{p;q,q'q''}"'] &
\end{tikzcd}
\tag{2.0.8}
$$

for $p \in P(S)$ and $q, q', q'' \in Q(S)$, and

$$
\begin{tikzcd}
E_{p,q}E_{p,q'} \arrow[r, "\psi_{p;q,q'}"] \arrow[d, "\mathrm{sym}"'] & E_{p,qq'} \arrow[d, "\mathrm{id}"] \\
E_{p,q'}E_{p,q} \arrow[r, "\psi_{p;q',q}"'] & E_{p,q'q}.
\end{tikzcd}
\tag{2.0.9}
$$

Here $p \in P(S)$ and $q, q' \in Q(S)$.

Definition 2.1. Let P, Q, G be abelian groups of $\underline{T}$, and let E be a torsor under $G_{P\times Q}$ equipped with an extension structure in the P-direction, defined by the isomorphisms (2.0.4), and an extension structure in the Q-direction, defined by the isomorphisms (2.0.7), each functorial in S and subject to the associativity and commutativity conditions above. The two structures are called compatible if, for every object S of $\underline{T}$ and $p, p' \in P(S)$, $q, q' \in Q(S)$, the following diagram commutes:

$$
\begin{tikzcd}
& E_{p,qq'}E_{p',qq'} \arrow[ddr, "\varphi_{p,p';qq'}"] & \\
E_{p,q}E_{p,q'}E_{p',q}E_{p',q'} \arrow[ur, "\psi_{p;q,q'}\wedge\psi_{p';q,q'}"] \arrow[d] & & \\
E_{p,q}E_{p',q}E_{p,q'}E_{p',q'} \arrow[dr, "\varphi_{p,p';q}\wedge\varphi_{p,p';q'}"'] & & E_{pp',qq'} \\
& E_{pp',q}E_{pp',q'} \arrow[ur, "\psi_{pp';q,q'}"'] &
\end{tikzcd}
\tag{2.1.1}
$$

where the vertical arrow is induced by the symmetry isomorphism

$$E_{p,q'}E_{p',q} \xrightarrow{\sim} E_{p',q}E_{p,q'}.$$

A *biextension* of (P,Q) by G is a torsor E under $G_{P\times Q}$ equipped with compatible extension structures in the P- and Q-directions in the preceding sense.

2.1.1. Equivalently, a biextension structure on the $G_{P\times Q}$-torsor E is given by two partially defined composition laws $\varphi(x,y)$ and $\psi(x,y)$ on E. The first is defined when $\mathrm{pr}_2\, j(x) = \mathrm{pr}_2\, j(y)$, and the second when $\mathrm{pr}_1\, j(x) = \mathrm{pr}_1\, j(y)$, where $j : E \longrightarrow P \times Q$ is the structural morphism of the torsor. Besides associativity relations of the type

$$\varphi(gx,y) = \varphi(x,gy) = g\varphi(x,y) \qquad (x,y \in E(S),\ g \in G(S)) \tag{2.1.1.1}$$

and the analogous relations for ψ, the axioms say that these partial composition laws are associative and commutative and satisfy

$$\varphi(\psi(x,y),\psi(z,t)) = \psi(\varphi(x,z),\varphi(y,t)), \tag{2.1.1.2}$$

where x,y,z,t lie respectively in $E_{p,q}(S), E_{p,q'}(S), E_{p',q}(S), E_{p',q'}(S)$.

2.2. Identity sections. Consider first the identity section of E regarded as an extension of Q_P by G_P,

$$e_{E/P} \in \Gamma(E/P) = \mathrm{Hom}_P(P,E), \tag{2.2.1}$$

and, symmetrically, the identity section of E regarded as an extension of P_Q by G_Q,

$$e_{E/Q} \in \Gamma(E/Q) = \mathrm{Hom}_Q(Q,E). \tag{2.2.2}$$

In particular, the torsor E over $P \times Q$ under $G_{P\times Q}$ is canonically trivialized over $P \times 1_Q$ by $e_{E/P}$, and over $1_P \times Q$ by $e_{E/Q}$. These trivializations coincide over the identity section $1_P \times 1_Q$ of $P \times Q$, and hence define the same section

$$e_E \in \Gamma(E), \tag{2.2.3}$$

which, by abuse of language, will be called the identity section (or neutral section) of E.

To prove the asserted equality, let e, e' be the two sections of E deduced respectively from $e_{E/P}$ and $e_{E/Q}$. They are characterized as the identity sections in the inverse image $E_{1_P,1_Q}$ of E along the identity section of $P \times Q$, for the two composition laws φ and ψ, respectively. Applying (2.1.1.2) to the four sections e, e', e', e of $E_{1,1}$ gives e on the left and e' on the right; hence $e = e'$. It follows moreover that the two composition laws φ and ψ on $E_{1,1}$ are identical. Indeed, under transport of structure by the morphisms $g \mapsto g.e$ and $g \mapsto g.e'$, respectively, both correspond to the group law of G, and $e = e'$.

The extension of Q_P by G_P over the base P, restricted over the identity section 1_P of P, gives an extension of Q by G on the inverse image of $1_P \times Q$ in E. The homomorphism $e_{E/Q}$ is immediately seen to be multiplicative for this extension structure, so the extension is canonically split. Symmetrically, the extension of P by G obtained by restricting the extension of P_Q by G_Q over Q to the identity section 1_Q is canonically split by $e_{E/P}$.

2.3. Morphisms of biextensions. Let E be a biextension of (P,Q) by G, and let E' likewise be a biextension of (P',Q') by G', where P', Q', G' are again commutative groups of $\underline{T}$. A morphism of biextensions from E to E' is a system (u,v,w,f), where

$$u : P \longrightarrow P', \qquad v : Q \longrightarrow Q', \qquad w : G \longrightarrow G' \tag{2.3.1}$$

are morphisms of groups, and

$$f : E \longrightarrow E' \tag{2.3.2}$$

is a morphism of the sheaves of sets underlying E and E', such that f is simultaneously a homomorphism of group extensions over P and P', respectively, associated with the relative group homomorphisms

$$u \times v : Q_P \longrightarrow Q'_{P'}, \qquad u \times w : G_P \longrightarrow G'_{P'},$$

and a homomorphism of group extensions over Q and Q', respectively, associated with the relative group homomorphisms

$$u \times v : P_Q \longrightarrow P'_{Q'}, \qquad v \times w : G_Q \longrightarrow G'_{Q'}.$$

Notice moreover that u, v, w are determined by f alone. Indeed, passing to the quotient recovers $u \times v$ from f, and then recovers u and v by restriction to $P \times 1_Q$ and $1_P \times Q$, respectively; restricting f to $G.e_E$ recovers w. Thus no ambiguity arises from identifying the morphism (u, v, w, f) with f.

After taking the usual precaution of adjoining their sources and targets to morphisms (the canonical genuflection), one sees that <u>the biextensions of the topos $\underline{T}$ are the objects of a category</u>, whose objects and morphisms are those defined in 2.1 and 2.3 and whose composition of morphisms is defined in the evident way. Moreover, $(P, Q, G, E) \mapsto (P, Q, G)$ gives a canonical functor

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}), \tag{2.3.3}$$

where $\text{BIEXT}(\underline{T})$ denotes the category of biextensions over $\underline{T}$, and $\text{GRAB}(\underline{T})$ the category of abelian groups over $\underline{T}$. It is useful to decompose this canonical functor into the two functors $E \mapsto (P, Q)$,

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}), \tag{2.3.4}$$

and $E \mapsto G$,

$$\text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}). \tag{2.3.5}$$

2.4. <u>The category $\text{BIEXT}(P, Q; G)$ for variable P, Q, G</u>.

I claim that <u>the canonical functor (2.3.4) is fibred</u>, and that <u>the canonical functor (2.3.5) is cofibred</u>. The first assertion means, in particular, that if E is a biextension of (P, Q) by G, and if

$$u : P' \longrightarrow P, \qquad v : Q' \longrightarrow Q$$

are morphisms of abelian groups, then there is a biextension E' of (P', Q') by G and a morphism of biextensions

$$f : E' \longrightarrow E$$

compatible with u, v, id_G, having the usual universal property for morphisms of biextensions into E compatible with u, v, id_G.

The second assertion is expressed similarly. In particular, if E is a biextension of (P, Q) by G, and if

$$w : G \longrightarrow G'$$

is a morphism of abelian groups, then there is a biextension E' of (P, Q) by G' and a morphism of biextensions

$$g : E \longrightarrow E'$$

compatible with $\mathrm{id}_P, \mathrm{id}_Q, w$, having the usual universal property for morphisms of biextensions from E compatible with $\mathrm{id}_P, \mathrm{id}_Q, w$. We leave it to the reader to spell out the two constructions whose existence has just been asserted; they are obtained easily by paraphrasing the familiar analogous constructions for extensions of ordinary groups.

Thus, for three abelian groups P, Q, G of $\underline{T}$, write

$$\text{BIEXT}(P, Q; G) \tag{2.4.1}$$

for the category of biextensions of (P, Q) by G, defined as the fibre category of (2.3.3) over (P, Q, G) (so its morphisms are the morphisms of biextensions compatible with the identity morphisms of P, Q, G). Then this category depends covariantly on G and contravariantly on P, Q. Moreover, with these variances, the category $\mathrm{BIEXT}(P, Q; G)$ depends additively on each of the three arguments P, Q, G. For example, for the argument G, this means precisely:

a) $\mathrm{BIEXT}(P, Q; 0)$ is equivalent to the one-point category;
b) for two abelian groups G, G' of $\underline{T}$, the functor

$$\mathrm{BIEXT}(P, Q; G \times G') \longrightarrow \mathrm{BIEXT}(P, Q; G) \times \mathrm{BIEXT}(P, Q; G') \tag{2.4.2}$$

deduced from the two projections $G \times G' \longrightarrow G$ and $G \times G' \longrightarrow G'$ is an equivalence of categories.

In the language of [4], one also says that, for fixed P, Q, the category of biextensions of (P, Q) by variable groups G is, via the functor $E \mapsto G$ induced by (2.3.5), an additive cofibred category over $\mathrm{GRAB}(\underline{T})$.

Remark 2.4.3. In the terminology of *loc. cit.*, the preceding cofibred category is not merely additive but even left exact [4, 2.2]. We shall not verify this property here, since it will reappear below in 3.6 as a consequence of more precise facts. We only point out that the general theory of *loc. cit.* would then describe the structure of this cofibred category, up to equivalence of fibred categories, in terms of a chain complex $L_\bullet$ in $\mathrm{GRAB}(\underline{T})$ depending only on P and Q. We shall also give below a direct construction of this complex as

$$P \overset{\mathbb{L}}{\otimes} Q,$$

the tensor product taken in a derived category.

2.5. Structure of the category $\mathrm{BIEXT}(P, Q; G)$.

The general formal considerations of [4, no. 1] give several useful consequences of the additive dependence of the categories $\mathrm{BIEXT}(P, Q; G)$ on G.

2.5.1. First, for fixed G, the category $\mathrm{BIEXT}(P, Q; G)$ has a composition law which assigns functorially to two biextensions E, E' of (P, Q) by G a product biextension, denoted $E.E'$. This construction is simply the analogue of the familiar product of two extensions of one commutative group by another. It is compatible with that construction whether E, E' are viewed as group extensions of Q_P by G_P, or as group extensions of P_Q by G_Q. In particular, as a torsor under $G_{P\times Q}$, EE' is the product of the torsors under $G_{P\times Q}$ defined by E and E'.

The formation of $E.E'$ is moreover associative and commutative up to canonical isomorphisms, and this composition law on $\mathrm{BIEXT}(P, Q; G)$ has a two-sided neutral object, called the trivial biextension of (P, Q) by G. It may be constructed, as in *loc. cit.*, from the unique biextension E_0 of (P, Q) by the zero group 0, up to isomorphism, by means of the homomorphism $0 \longrightarrow G$. From this description one immediately obtains an alternative description of the trivial biextension as the trivial torsor

$$E = P \times Q \times G.$$

Its two group structures over P and over Q are obtained, respectively, from the product group structures on $Q \times G$ and $P \times G$ by the base changes $P \longrightarrow e$ and $Q \longrightarrow e$.

Finally, for every biextension E of (P, Q) by G, one can define the inverse biextension, denoted E^{-1}, together with an isomorphism from EE^{-1} to the trivial biextension. This determines E^{-1} up to unique isomorphism and makes its dependence on E functorial.

In short, the category $\mathrm{BIEXT}(P, Q; G)$ behaves like a group object in the category **Cat** of categories, except that associativity, commutativity, unit, and inverse relations hold only up to specified isomorphisms. One may call it a pseudo-group in the 2-category **Cat**.

2.5.2. For fixed P, Q, G, the preceding properties imply the following facts. For every biextension E of (P, Q) by G, the monoid of endomorphisms of E is canonically isomorphic to the monoid of endomorphisms of the trivial biextension $\underline{1}$: an endomorphism u of $\underline{1}$ is sent to $u.\mathrm{id}_E$ on $\underline{1}.E \simeq E$. The endomorphism monoid of $\underline{1}$ is in fact a <u>commutative group</u>, denoted

$$\mathrm{Biext}^0(P, Q; G). \tag{2.5.2.1}$$

Its composition law is also given by $(u, v) \mapsto u.v \in \mathrm{End}(\underline{1}.\underline{1}) \simeq \mathrm{End}(\underline{1})$.

Likewise, the composition law $(E, E') \mapsto EE'$ equips the set

$$\mathrm{Biext}^1(P, Q; G) \tag{2.5.2.2}$$

of isomorphism classes of biextensions of (P, Q) by G with the structure of a <u>commutative group</u>; the inverse of the class of E is the class of E^{-1}.

2.5.3. Given a group homomorphism $G \longrightarrow G'$, the corresponding functor

$$\mathrm{BIEXT}(P, Q; G) \longrightarrow \mathrm{BIEXT}(P, Q; G') \tag{2.5.3.1}$$

is compatible, up to canonical isomorphisms, with the operations $E.E'$ and E^{-1} of 2.5.1. Consequently, the two evident maps

$$\mathrm{Biext}^i(P, Q; G) \longrightarrow \mathrm{Biext}^i(P, Q; G') \qquad (i = 0, 1) \tag{2.5.3.2}$$

are <u>group homomorphisms</u>. Moreover, if E is a biextension of (P, Q) by G and E' the corresponding biextension by G', the natural homomorphism

$$\mathrm{Hom}(E, E) = \mathrm{Aut}(E) \longrightarrow \mathrm{Hom}(E', E') = \mathrm{Aut}(E') \tag{2.5.3.3}$$

is precisely (2.5.3.2) for $i = 0$.

2.5.4. The group $\mathrm{Biext}^0(P, Q; G)$ admits a simple explicit description. If E is a biextension of (P, Q) by G, an automorphism u of E is first of all an automorphism of the underlying torsor under $G_{P\times Q}$, and hence is given by a morphism

$$u_0 : P \times Q \longrightarrow G$$

through the formula

$$u(x) = u_0(j(x))x.$$

Compatibility of u with the first, respectively the second, extension structure on E is equivalent to additivity of u_0 in the second, respectively the first, argument. Thus <u>u is an automorphism of the biextension structure of E if and only if u_0 is bilinear</u>. This gives a <u>canonical bijection</u>

$$\mathrm{Biext}^0(P, Q; G) \simeq \mathrm{Hom}(P \otimes Q, G). \tag{2.5.4.1}$$

It is readily checked that this bijection is independent of E, respects the group structures, and is functorial in G for the functoriality on the left described in (2.5.3.2).

Below we shall determine $\mathrm{Biext}^1(P, Q; G)$ and find that, canonically and functorially in G, it is isomorphic to

$$\mathrm{Ext}^1\big(P \overset{\mathbb{L}}{\otimes} Q, G\big).$$

2.6. In 2.5 only G was varied, while P and Q were held fixed. We now say a few words about variation in P, Q. The essential fact, from which the others follow formally, is that the contravariant dependence of $\mathrm{BIEXT}(P, Q; G)$ on P, Q <u>commutes</u> with its covariant dependence on G. Thus the categories in question may be viewed as the fibre categories of a <u>cofibred category</u>

$$\mathrm{BIEXT}'(\underline{T}) \longrightarrow \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T}), \tag{2.6.1}$$

where the superscript op denotes the opposite category and the structural functor is $E \mapsto (P, Q, G)$. This cofibred category is additive in each of its three arguments, or, as we shall say, <u>tri-additive</u>.

By paraphrasing standard arguments, which we shall not reproduce, it follows that the composition $E.E'$ in the fibre category $\mathrm{BIEXT}(P,Q;G)$ may be described using the additive structure of any one of the three arguments: the three resulting pseudo-group structures are canonically isomorphic. Consequently, the abelian groups

$$\mathrm{Biext}^i(P,Q;G), \qquad i=0,1, \tag{2.6.2}$$

may be regarded as two trilinear functors in P,Q,G, respectively in $\mathrm{GRAB}(\underline{T})^{\mathrm{op}}$, $\mathrm{GRAB}(\underline{T})^{\mathrm{op}}$, and $\mathrm{GRAB}(\underline{T})$. Equivalently, they are trilinear functors of variable P,Q,G in $\mathrm{GRAB}(\underline{T})$, contravariant in P,Q and covariant in G, with values in the category of abelian groups. For this variance, the isomorphism (2.5.4.1) is functorial in all of P,Q,G, and not merely in G. The analogous assertion will hold below for the description of Biext^1 by an Ext^1.

2.7. The symmetric biextension of a given biextension.

If E is a biextension of (P,Q) by G, define its symmetric biextension ${}^sE$, which is a biextension of (Q,P) by G. The underlying sheaf of sets of ${}^sE$ is that of E, and the actions of G on E and ${}^sE$ are the same. The structural morphism

$${}^sE \longrightarrow Q\times P$$

is the composite of $E\longrightarrow P\times Q$ with the symmetry $P\times Q\longrightarrow Q\times P$; finally, the left and right extension structures of ${}^sE$ are, respectively, the right and left extension structures of E. Evidently $E\mapsto {}^sE$ is an automorphism of the category $\mathrm{BIEXT}(\underline{T})$ of biextensions, and indeed an involutive automorphism:

$${}^s({}^sE)=E.$$

Every statement about bitorsors $E,\ldots$ therefore gives a symmetric statement upon being applied to ${}^sE,\ldots$. We may call it the statement obtained by *exchanging left and right*. In general we shall spell out only one of the two symmetric statements, choosing between left and right, for example, according to our political preferences.

2.8. Change of topos. Localization.

Let

$$f:\underline{T}'\longrightarrow \underline{T}$$

be a morphism of topoi. The inverse-image functor f^* transforms a biextension of (P,Q) by G into a biextension $f^*(E)$ of (f^*P,f^*Q) by f^*G, as always for a structure expressible in terms of finite projective limits and arbitrary inductive limits. To prove compatibility of the two extension structures on $f^*(E)$ induced by those on E, one expresses compatibility by commutativity of (2.1.1) in the universal case

$$S=P\times P\times Q\times Q,$$

where p,p',q,q' are the four projections. As expected, $E\mapsto f^*(E)$ gives a canonical functor

$$f^*:\mathrm{BIEXT}(\underline{T})\longrightarrow \mathrm{BIEXT}(\underline{T}').$$

In what follows we shall use without further mention the fact that every construction made with biextensions is compatible with inverse images under morphisms of topoi. This applies in particular to the product EE' and inverse E^{-1} considered in 2.5.

One may take for $\underline{T}'$ an induced topos $\underline{T}_{/S'}$, and for f the canonical morphism, called the localization morphism (SGA 4 IV 5),

$$f=i_{S'}:\underline{T}_{/S'}\longrightarrow \underline{T}.$$

Then $f^*(E)$ is called the biextension induced by E on S', and is also denoted, as usual, by $E|_{S'}$ or $E_{S'}$.

Since restriction has the usual transitivity properties, the categories $\mathrm{BIEXT}(\underline{T}_{/S})$, for variable $S\in\underline{T}$, form the fibres of a fibred category over $\underline{T}$ (SGA 1 VI 8). In fact, this category is even a <u>stack</u> over $\underline{T}$ in the sense of Giraud [2]: morphisms and objects in its fibres can be glued. For example, to give a biextension over S up to canonical isomorphism, it suffices to give a covering family (that is, an epimorphic family) $S_i \longrightarrow S$, a biextension E_i over each S_i (an object of $\underline{T}_{/S_i}$), and, for every pair (i,j), an isomorphism between $E_i|_{S_j}$ and $E_j|_{S_i}$, subject to the usual cocycle condition.

We call this the <u>stack of biextensions over $\underline{T}$</u>. Replacing $\underline{T}$ in (2.3.3), and hence also in (2.3.4) and (2.3.5), by $\underline{T}_{/S}$ for variable S gives morphisms of stacks over $\underline{T}$. The same applies to the operations $E.E'$ and E^{-1} of (2.5.1) in the categories $\mathrm{BIEXT}(P_S, Q_S; G_S)$ for variable S, where P, Q, G are fixed commutative groups over $\underline{T}$.

2.9. <u>Relations with birigidified torsors</u>.

In the terminology of 1.3.2, we saw in 2.2 that, for every biextension E of (P,Q) by G, the underlying torsor structure under $G_{P\times Q}$ is canonically birigidified relative to the pair of identity sections of P and Q. This gives a canonical functor

$$\mathrm{BIEXT}(P,Q;G) \xrightarrow{i} \mathrm{TORSBIRIG}(P,Q;G), \tag{2.9.1}$$

where the category on the right consists of birigidified torsors over $P\times Q$ under $G_{P\times Q}$.

Using the identity sections of P and Q, for every product of n objects each isomorphic to P or Q one obtains a canonical morphism inj_i from the i-th factor into that product. Thus, for a morphism from such a product to G, one may speak of its restriction to the i-th factor. Consider the products

$$\begin{cases} P\times Q, \\ P\times Q\times Q, \quad P\times Q\times Q\times Q, \\ P\times P\times Q, \quad P\times P\times P\times Q, \\ P\times P\times Q\times Q. \end{cases} \tag{2.9.2}$$

<u>Proposition 2.9.3. Suppose that every morphism from one of the products (2.9.2) to G which is trivial on each factor of that product (embedded by the identity sections as just explained) is trivial. Then:</u>

a) <u>The canonical functor (2.9.1) is fully faithful, and both sides of (2.9.1) are rigid categories. In particular, if E is a torsor under $G_{P\times Q}$ birigidified relative to (e_P, e_Q), there is at most one biextension structure on E compatible with its birigidified torsor structure.</u>

b) <u>Such a biextension structure exists on E if and only if E admits an extension structure of Q_P by G_P compatible with its torsor structure under G_P, and an extension structure of P_Q by G_Q compatible with its torsor structure under G_Q (cf. 1.3.5(b)). In that case there is a unique extension structure of Q_P by G_P compatible with the torsor structure under G_P and the given rigidification over $P\times e_Q$, and a unique extension structure of P_Q by G_Q compatible with the torsor structure under G_Q and the given rigidification over $e_P\times Q$; these two extension structures are compatible in the sense of 2.1.</u>

<u>Proof.</u> a) If E is a birigidified torsor over $P\times Q$, its automorphism group identifies with the group of morphisms $P\times Q \longrightarrow G$ trivial on the two partial factors. By the hypothesis applied to $P\times Q$, this group contains only the identity automorphism, proving rigidity. Now let E, E' be biextensions of (P,Q) by G. To see that every morphism $f : E \longrightarrow E'$ of birigidified torsors is a morphism of biextensions, apply 1.3.5(a) to the two extension structures defined by E, E' over P and over Q, respectively. This uses the hypothesis for $P\times Q\times Q$ and $P\times P\times Q$ in (2.9.2).

b) Necessity is clear. For sufficiency, 1.3.5(b) first shows that the stated condition gives unique extension structures of Q_P by G_P and of P_Q by G_Q, compatible with the torsor structure and the rigidifications. Here the hypothesis is used for $P\times Q\times Q\times Q$ and $P\times P\times P\times Q$. It remains to prove that the two extension structures are compatible, that is, that diagram (2.1.1) commutes. Its two composites are isomorphisms of torsors over $P\times P\times Q\times Q$, and hence differ by a morphism

$$u : P\times P\times Q\times Q \longrightarrow G.$$

Because the partial trivializations of E are identity sections for the laws φ and ψ, respectively, u is trivial on each of the four factors. The hypothesis for $P \times P \times Q \times Q$ therefore makes u trivial. This proves the proposition.

Corollary 2.9.4. Let P, Q, G be commutative groups satisfying

$$(2.9.4.1) \qquad \underline{\mathrm{Hom}}_{\mathrm{pt}}(P, G) = e, \qquad \underline{\mathrm{Hom}}_{\mathrm{pt}}(Q, G) = e$$

(cf. 1.3.7). Then the hypothesis of Proposition 2.9.3 is satisfied, and conclusions (a) and (b) of that proposition hold. Moreover, if E is a torsor under $G_{P\times Q}$ birigidified relative to (e_P, e_Q), it admits a biextension structure of (P, Q) by G compatible with its given structures if and only if the two morphisms of pointed sheaves which it defines,

$$(2.9.4.2) \qquad Q \longrightarrow R^1 p_*(G_P), \qquad P \longrightarrow R^1 q_*(G_Q),$$

where $p : P \longrightarrow e$ and $q : Q \longrightarrow e$ are the structural morphisms, are morphisms of groups.

The first assertion was proved in 1.3.7. For the second, necessity is clear. To prove sufficiency, apply Proposition 2.9.3(b) and 1.3.5(b). Multiplicativity of the first morphism in (2.9.4.2) says that the rigidified torsor E under G_P satisfies the condition of 1.3.5(b) locally over P. As observed in 1.3.7, that condition is then true globally over P, so E admits an extension structure of Q_P by G_P compatible with its rigidified torsor structure. Symmetrically, E admits an extension structure of P_Q by G_Q compatible with its rigidified torsor structure under G_Q. The conclusion follows from Proposition 2.9.3(b).

Example 2.9.5. Let S be a scheme, let P and Q be abelian schemes over S, and let $G = \underline{G}_{m,S}$ be the multiplicative group scheme over S. We noted in 1.3.8 that (2.9.4.1) then holds. A birigidified torsor E over $P \times Q$ is equivalently a birigidified invertible sheaf over $P \times Q$. Recall that a divisorial correspondence on P, Q is the isomorphism class of such a birigidified invertible sheaf (see, for example, [1]). In the fppf topology, the morphisms (2.9.4.2) are

$$(2.9.5.1) \qquad Q \longrightarrow \underline{\mathrm{Pic}}_{P/S}, \qquad P \longrightarrow \underline{\mathrm{Pic}}_{Q/S},$$

defined by the divisorial correspondence. The theorem of the square (*loc. cit.*) says that these morphisms necessarily respect the group structures. Thus 2.9.4 shows that the category of biextensions of (P, Q) by $G = \underline{G}_{m,S}$ is equivalent to the category of birigidified invertible sheaves on $P \times Q$; it is therefore a rigid category, and its set of isomorphism classes identifies with the set of divisorial correspondences on (P, Q). This identification plainly respects the natural group laws on the two sets, the first being that of (2.5.2.2).

Remark 2.9.6. Additivity of the first morphism in (2.9.4.2) is equivalent to factorization of the second through the subsheaf $\underline{\mathrm{Ext}}^1(Q, G)$; note that, by 1.3.5, the canonical morphism

$$\underline{\mathrm{Ext}}^1(Q, G) \longrightarrow R^1 q_*(G_Q)$$

is injective. Likewise, additivity of the second morphism in (2.9.4.2) means that the first factors through $\underline{\mathrm{Ext}}^1(P, G)$. This gives group isomorphisms

$$(2.9.6.1) \qquad \mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}\big(Q, \underline{\mathrm{Ext}}^1(P, G)\big) \simeq \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q, G)\big).$$

In the situation of 2.9.5 these reduce to the familiar isomorphisms

$$(2.9.6.2) \qquad \mathrm{Corr}(P, Q) \simeq \mathrm{Hom}(P, Q') \simeq \mathrm{Hom}(Q, P'),$$

where

$$P' = \underline{\mathrm{Ext}}^1(P, \underline{G}_m), \qquad Q' = \underline{\mathrm{Ext}}^1(Q, \underline{G}_m)$$

are the dual abelian schemes of P and Q, identified respectively with $\underline{\mathrm{Pic}}^0_{P/S}$ and $\underline{\mathrm{Pic}}^0_{Q/S}$. In both formulas, Hom denotes the group of group homomorphisms.

2.10. Various generalizations of the notion of biextension. We mention them for completeness, although we shall not use them.

2.10.1. One can define a biextension of (P,Q) by G without assuming P or Q commutative, only G being commutative. It is a torsor under $G_{P\times Q}$ equipped with central extension structures of Q_P by G_P and of P_Q by G_Q, each compatible with the torsor structure and compatible with one another, meaning that they make (2.1.1) commute. Most of the preceding developments remain valid, *mutatis mutandis*, for this more general notion. Even when P, Q are commutative, it is more general than the notion of 2.1, which is the case in which the two underlying relative group structures on E are commutative. In the noncommutative case, the category of biextensions of (P,Q) by G still depends additively on G, but not on P and Q.

2.10.2. Paraphrasing 2.1, for a fixed integer $n \geq 1$ one defines an n-extension of a sequence of n commutative groups $(P_1, P_2, \ldots, P_n)$ by a commutative group G. It consists of a torsor E under $G_{P_1\times\cdots\times P_n}$, equipped with n partial extension structures over the partial products of $n-1$ factors, with a compatibility relation of type (2.1.1) for every pair of distinct indices between 1 and n.

If $n \geq 3$, the P_i are abelian schemes over a scheme S, and $G = \underline{G}_{m,S}$, every n-extension of $(P_1, \ldots, P_n)$ by G is trivial.

2.10.3. The notion of biextension in 2.1 may be viewed as a variant of the notion of a commutative extension of commutative groups, that is, of an extension of $\mathbb{Z}$-modules. Let A be a ring of $\underline{T}$, assumed commutative for simplicity, and let P, Q, G be three A-modules. More generally one can define an A-module biextension of (P,Q) by G, a variant of an extension of A-modules.

Such a biextension is a biextension in the sense of 2.1 whose underlying partial extension structures are enriched to extension structures of A_P-modules and A_Q-modules, respectively, compatible with the relative module structures already present on Q_P, G_P and P_Q, G_Q. The two relative module structures on E may be interpreted as two actions of the underlying multiplicative monoid of A on the sheaf of sets underlying E, and these two actions are required to commute. This condition is added to the compatibility expressed by (2.1.1), and is itself expressed by an analogous diagram, which we leave to the reader. Finally, the first action of A on E is required to be distributive with respect not only to the first composition law on E but also to the second; the same is required of the second action.

The dictionary of 1.1.6 also has an A-linear variant. An extension of the A-module P by the A-module G amounts to a family of torsors E_p under G_p, parametrized by points $p \in P(S)$ for variable $S \in \operatorname{ob} \underline{T}$, with E_p varying A-linearly, rather than merely additively, in p. Besides the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1), this is expressed by isomorphisms

$$\mu_{a,p} : {}^aE_p \xrightarrow{\sim} E_{ap} \qquad (p \in P(S),\ a \in A(S)),$$

where ${}^aE_p$ denotes the image of the G_S-torsor E_p under extension of the structure group along

$$g \longmapsto ag : G_S \longrightarrow G_S.$$

We leave to the reader the conditions on φ and μ which make these data correspond to an extension of A-modules E of P by G (cf. 3.5.4(b)).

3. Supplements on homological algebra

The principal purpose of this section is to define the isomorphism

$$\operatorname{Biext}^1(P,Q;G) \simeq \operatorname{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

announced above. For this we shall need certain developments in homological algebra which are of independent interest and will occupy sections 3.1 through 3.4. The developments in 3.1 and 3.3 may be regarded as supplements to [4], especially to section 6.9 therein.

3.1. Construction of an additive cofibred category $\underline{\Psi}_{L_\bullet}$ in terms of a chain complex $L_\bullet$ in an abelian category $\underline{C}$

Let $\underline{C}$ be an abelian category and let $L_\bullet$ be a chain complex in $\underline{C}$,

$$L_2 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \tag{3.1.1}$$

written with lower-degree conventions, so that the differential has degree -1. Thus $L_i = 0$ for $i < 0$. We shall define a cofibred category

$$\underline{\Psi} = \underline{\Psi}_{L_\bullet}$$

over $\underline{C}$; up to canonical equivalence of cofibred categories it is the category of the same name in [4, 6.9]. The objects of $\underline{\Psi}_{L_\bullet}$ are pairs

$$(X, \alpha),$$

where X is an extension of L_0 by an object A of $\underline{C}$, depending on X,

$$0 \longrightarrow A \xrightarrow{i} X \xrightarrow{j} L_0 \longrightarrow 0, \tag{3.1.2}$$

and α is a trivialization of the inverse image $d_0^*(X)$ of this extension by $d_0 : L_1 \longrightarrow L_0$. It is required that the corresponding trivialization of the inverse image $d_1^* d_0^*(X)$ by $d_1 : L_2 \longrightarrow L_1$ be the evident one, obtained from the transitivity isomorphism

$$d_1^* d_0^*(X) \simeq (d_0 d_1)^*(X)$$

and the relation $d_0 d_1 = 0$.

Equivalently, α is a lifting of $d_0 : L_1 \longrightarrow L_0$ to a morphism $\alpha : L_1 \longrightarrow X$, subject to $\alpha d_1 = 0$:

$$\begin{tikzcd}
 & & 0 \arrow[d] \\
 & & A \arrow[d, "i"] \\
 & & X \arrow[d, "j"] \\
L_2 \arrow[r, "d_1"'] & L_1 \arrow[r, "d_0"'] \arrow[ur, "\alpha"] & L_0 \arrow[d] \\
 & & 0
\end{tikzcd} \quad , \qquad \alpha d_1 = 0. \quad (*) \tag{3.1.3}$$

Here is a simpler description, pointed out by Deligne: $\underline{\Psi}_{L_\bullet}(A)$ is the category of extensions of the complex $L_\bullet$ by A, regarded as a complex concentrated in degree zero.

In this form one sees that the description of the objects (X, α) depends only on the chain complex $[L_\bullet]$ <u>truncated in degree 1</u>, obtained from $L_\bullet$ by "killing $H_i(L_\bullet)$ for $i \geq 2$": its terms are zero in degrees at least 2, agree with those of $L_\bullet$ in degrees at most 0, and its degree-1 term is the cokernel of $d_1 : L_2 \longrightarrow L_1$.

A homomorphism from (X, α) to (X', α') is a homomorphism of extensions $u : X \longrightarrow X'$, compatible with the identity on L_0, such that $u\alpha = \alpha'$. Homomorphisms are composed by composing the corresponding homomorphisms of extensions. This gives the announced category $\underline{\Psi}_{L_\bullet}$. Up to canonical isomorphism, it depends only on the degree-1 truncation $[L_\bullet]$ of $L_\bullet$.

There is an evident functor $(X,\alpha) \longmapsto A$,

$$\pi : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{C}. \tag{3.1.4}$$

This functor is cofibred. Indeed, let $u : A \longrightarrow B$ be a morphism of $\underline{C}$, and let (X,α) be an object of the fibre $\underline{\Psi}_{L_\bullet}(A)$. Thus X is an extension of L_0 by A. Form the extension $u_*(X)$ of L_0 by B, the direct image of X under u, and equip it with the partial trivialization $\beta = f\alpha$, where $f : X \longrightarrow u_*(X)$ is the canonical morphism. It follows immediately that

$$f : (X,\alpha) \longrightarrow (u_*(X),\beta)$$

is cocartesian for (3.1.4): it has the usual universal property for u-morphisms in $\underline{\Psi}_{L_\bullet}$ with source (X,α). This is just the familiar and elementary corresponding universal property of $f : X \longrightarrow u_*(X)$.

Finally, the cofibred category $\underline{\Psi}_{L_\bullet}$ is additive; that is, it satisfies

$$\underline{\Psi}_{L_\bullet}(0) \quad \text{is equivalent to the one-point category,} \tag{3.1.5}$$

and, for arbitrary A, B in $\underline{C}$,

$$\underline{\Psi}_{L_\bullet}(A \times B) \longrightarrow \underline{\Psi}_{L_\bullet}(A) \times \underline{\Psi}_{L_\bullet}(B) \quad \text{is an equivalence.} \tag{3.1.6}$$

This follows immediately from the corresponding properties of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$. We shall freely use the consequences of the additivity of $\underline{\Psi}_{L_\bullet}$ established in [4, 1] and recalled above in 2.5.

3.1.7. Among these consequences, for every object A of $\underline{C}$ the category $\underline{\Psi}_{L_\bullet}(A)$ has a zero, or trivial, object (2.5.1). It is the trivial extension $X = L_0 + A$ of L_0 by A, equipped with the partial trivialization $\alpha : L_1 \longrightarrow X$ whose components are d_0 and 0.

Moreover, by 2.5.2, for every (X,α) in $\underline{\Psi}_{L_\bullet}(A)$ the automorphism group of this object is canonically isomorphic to the automorphism group of the trivial object. Denote the latter commutative group by $\Psi^0_{L_\bullet}(A)$. The explicit description above gives the canonical isomorphisms

$$\Psi^0_{L_\bullet}(A) = \mathrm{Ker}\big(\mathrm{Hom}(L_0,A) \longrightarrow \mathrm{Hom}(L_1,A)\big) \simeq \mathrm{Hom}(H_0(L_\bullet),A) \simeq \mathrm{Ext}^0(L_\bullet,A), \tag{3.1.8}$$

induced by the usual identification of the automorphism group of the extension X with $\mathrm{Hom}(L_0,A)$. The isomorphism (3.1.8) is plainly functorial in A.

3.1.9. Finally, recall from 2.5.2 that the natural multiplicative structure on the fibre category $\underline{\Psi}_{L_\bullet}(A)$ endows its set of isomorphism classes with a natural group structure. We denote this group by $\Psi^1_{L_\bullet}(A)$; it is an additive functor of A.

3.2. Determination of $\Psi^1_{L_\bullet}(A)$

Consider an object (X,α) (3.1.3) of $\underline{\Psi}_{L_\bullet}(A)$. Consider the cochain complex of length one

$$[X \longrightarrow L_0] : \quad 0 \longrightarrow X \longrightarrow L_0 \longrightarrow 0 \longrightarrow \cdots \tag{3.2.1}$$

defined by the morphism $j : X \longrightarrow L_0$, with X as its degree-zero component. It is a resolution of A, and hence may be identified with A as an object of the derived category $D(\underline{C})$ [5]. The diagram (3.1.3) then gives a degree-one morphism of complexes

$$L_\bullet \longrightarrow [X \longrightarrow L_0], \tag{3.2.2}$$

made explicit by the following diagram with anticommutative squares:

(3.2.3)
$$\begin{tikzcd} \cdots & L_2 \arrow[r, "d_1"] & L_1 \arrow[r, "d_0"] \arrow[d, "u^{-1}=-\alpha"'] & L_0 \arrow[r] \arrow[d, "u^0=\mathrm{id}"'] & 0 & \cdots \\ \cdots & 0 \arrow[r] & X \arrow[r, "j"] & L_0 \arrow[r] & 0 & \cdots \end{tikzcd}$$

(We use the usual convention which associates with the chain complex $L_\bullet$ a complex whose differential has degree 1 and whose degree-i component is L_{-i}.) We thus obtain a degree-one morphism in the derived category

(3.2.3) $$c(X,\alpha) : L_\bullet \longrightarrow A \qquad \big(\text{a degree-one morphism in } D(\underline{C})\big),$$

which plainly depends only on the isomorphism class of (X,α). We have thereby defined a canonical map

(3.2.4) $$\Psi^1_{L_\bullet}(A) \longrightarrow \mathrm{Ext}^1(L_\bullet, A) \overset{\mathrm{def}}{=} \mathrm{Hom}_{D(\underline{C})}(L_\bullet, A[1]).$$

Theorem 3.2.5. The map (3.2.4) is an isomorphism of groups, functorial in the object A of $\underline{C}$.

We leave to the reader the verification that (3.2.4) is additive and functorial in A, and shall confine ourselves to proving that it is bijective. For injectivity, let us first show that if the morphism of complexes u in (3.2.3) is zero in $D(\underline{C})$, then it is already zero in the homotopy category $K(\underline{C})$; that is, there exists a morphism

$$s : L_0 \longrightarrow X$$

such that

(3.2.5.1) $$js = u^0 = \mathrm{id}_{L_0}, \qquad sd_0 = -u^{-1} = \alpha.$$

Indeed, the hypothesis says that the conclusion becomes true after the morphism of complexes under consideration is composed with a suitable quasi-isomorphism v

(*)
$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r, "j"] \arrow[d] & L_0 \arrow[r] \arrow[d] & 0 & \\ 0 \arrow[r] & C^0 \arrow[r, "j'"] & C^1 \arrow[r] & C^2 \arrow[r] & \cdots \end{tikzcd}$$

(and, after replacing C^i by 0 for $i \geq 2$ and C^1 by $\mathrm{Im}(C^0 \longrightarrow C^1) = \mathrm{Ker}(C^1 \longrightarrow C^2)$ if necessary, we may suppose that $C^i = 0$ for $i \geq 2$). But the morphism (*) induces the identity on A, and therefore identifies X with the fibre product of L_0 and C^0 over C^1. A morphism $s' : L_0 \longrightarrow C^0$ defining a homotopy $vu \sim 0$, and hence satisfying $j's' = v^1u^0$, consequently factors through a morphism $s : L_0 \longrightarrow X$ satisfying $js = u^0 = \mathrm{id}_{L_0}$. Thus s splits the extension X of L_0 by A, so the complex $X \longrightarrow L_0$ is isomorphic in $K(\underline{C})$ to the complex $K^\bullet$ concentrated in A. For the latter complex it is clear that

$$\mathrm{Hom}_{K(\underline{C})}(L_\bullet, K^\bullet) \longrightarrow \mathrm{Hom}_{D(\underline{C})}(L_\bullet, K^\bullet)$$

is an isomorphism, proving the assertion.

Now the choice of a morphism $s : L_0 \longrightarrow X$ satisfying (3.2.5.1) is precisely equivalent to a trivialization of the extension X of L_0 by A compatible with the partial trivialization α. This proves the injectivity of (3.2.4).

For surjectivity, start with an element of the right-hand side. By definition it may be described by a degree-one morphism of complexes

$$
\begin{tikzcd}
\cdots & L_2 \arrow[r, "d_1"] & L_1 \arrow[r, "d_0"] \arrow[d, "u^{-1}"'] & L_0 \arrow[r] \arrow[d, "u^0"'] & 0 \\
& 0 \arrow[r] & C^0 \arrow[r] & C^1 \arrow[r] & \cdots
\end{tikzcd}
\tag{3.2.5.2}
$$

with anticommutative squares, where $C^\bullet$ is a resolution of A and, as above, we may suppose that $C^i = 0$ for $i \geq 2$. Thus C^0 is an extension of C^1 by A, and we may form the extension X obtained by pulling it back along $L_0 \longrightarrow C^1$. Anticommutativity in (3.2.5.2) shows that $-u^{-1}$ factors through a morphism $\alpha : L_1 \longrightarrow X$ which is a partial splitting, that is, $j\alpha = d_0$ (with j as in (3.1.2)). Moreover, the relation $u^{-1}d_1 = 0$ in (3.2.5.2) implies $\alpha d_1 = 0$. Hence (X, α) is indeed an object of $\underline{\Psi}_{L_\bullet}(A)$. By construction, the morphism (3.2.5.2) is the composite of the morphism (3.2.3) associated with (X, α) and the morphism

$$[X \longrightarrow L_0] \longrightarrow C^\bullet = [C^0 \longrightarrow C^1]$$

whose components are (f, u^0), where $f : X \longrightarrow C^0$ is the canonical morphism obtained from the definition of X as the pullback of the extension C^0.

The preceding morphism is a morphism of resolutions of A and induces the identity on A. It follows, by the definition $\mathrm{Ext}^1 = \mathrm{Hom}^1_{D(\underline{C})}$, that the given element of $\mathrm{Ext}^1(L_\bullet, A)$ described by (3.2.5.2) is precisely $c(X, \alpha)$. This proves the surjectivity of (3.2.4) and completes the proof of 3.2.5.

3.3. Properties of the cofibred category $\underline{\Psi}_{L_\bullet}$

In this section, which is not essential for understanding what follows, we freely use the notions of [4], our purpose being to compare the preceding results with those of that work.

Theorem 3.3.1. The additive cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ is left exact ([4, 2.2]), admits a quasi-maximal object ([4, 4.4]), and, for every object $E = (X, \alpha)$ of this category, there exists a typical complex $L^E_\bullet$ for E ([4, 3.1]). Moreover, the typical pro-complex of $\underline{\Psi}_{L_\bullet}$ ([4, 3.7 and 4.4]) is canonically isomorphic in $D(\underline{C})$ to the truncated complex $[L_\bullet]$ considered in 3.1.

We now spell out the meaning of the several assertions in 3.3.1.

3.3.2. To say that the cofibred category $\underline{\Psi}_{L_\bullet}$ is left exact means that it is additive and also satisfies the following condition:

3.3.2.1. For every exact sequence

$$0 \longrightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \longrightarrow 0$$

in $\underline{C}$, the natural functor (in the notation of [4, 1.8])

$$\underline{\Psi}_{L_\bullet}(A) \longrightarrow \underline{\Psi}_{L_\bullet}([A \longrightarrow A''])$$

is an equivalence of categories.

Recall that if $\underline{\Psi}$ is a cofibred category over $\underline{C}$ and $v : A \longrightarrow A''$ is a morphism in $\underline{C}$, then $\underline{\Psi}([A \longrightarrow A''])$ is the category of pairs (E, ρ) in which E is an object of $\underline{\Psi}(A)$ and ρ is a "trivialization" of the corresponding object $v_*(E)$ of $\underline{\Psi}(A'')$; that is, an isomorphism from it to the zero object (2.5.1) of $\underline{\Psi}(A'')$. The definition of the functor

$$\underline{\Psi}(A') \longrightarrow \underline{\Psi}([A \longrightarrow A''])$$

in 3.3.2.1 is evident. When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the objects of $\underline{\Psi}_{L_\bullet}([A \longrightarrow A''])$ are explicitly the triples (X, α, ρ), where (X, α) is as in (3.1.3) and $\rho : X \longrightarrow A''$ is a morphism in $\underline{C}$ satisfying

$$\rho i = v, \qquad \rho\alpha = 0. \tag{*}$$

If $v : A \longrightarrow A''$ belongs to a short exact sequence as in 3.3.2.1, then, as is well known, giving a morphism satisfying the first relation (*) is equivalent, by the left exactness of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$, to giving an extension X' of L_0 by $A' \simeq \operatorname{Ker} v$ which is a subextension of X: one takes $X' = \operatorname{Ker} \rho$. The second relation (*) says that α factors through a morphism $\alpha' : L_1 \longrightarrow X'$. Up to the compatibilities whose verification is immediate, this proves the left exactness stated in 3.3.2.1.

3.3.3. Recall that an object E_0 of a cofibred category $\underline{\Psi}$ over $\underline{C}$ is called quasi-maximal if, for every object E of $\underline{\Psi}$ lying over an object A of $\underline{C}$, there is a morphism $E \longrightarrow E'$ in $\underline{\Psi}$ inducing a monomorphism $A \longrightarrow A'$ in $\underline{C}$, such that E is dominated by E_0, meaning that there is a morphism $E_0 \longrightarrow E'$. If $\underline{C}$ has enough injective objects, this is equivalent to saying that every object of $\underline{\Psi}$ over an injective object of $\underline{C}$ is dominated by E_0.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ there is the following canonical quasi-maximal object. Put

$$L_1' = \operatorname{Coker}(d_1 : L_2 \longrightarrow L_1)$$

and let $\bar{d}_0 : L_1' \longrightarrow L_0$ be the morphism induced by $d_0 : L_1 \longrightarrow L_0$. Let $E_0 = (X_0, \alpha_0)$ be the object of $\underline{\Psi}_{L_\bullet}(L_1')$ for which $X_0 = L_0 \oplus L_1'$ is the trivial extension of L_0 by L_1' and α_0 is the composite

$$L_1 \twoheadrightarrow L_1' \xrightarrow{(\bar{d}_0, \mathrm{id}_{L_1'})} L_0 \oplus L_1'.$$

To see that E_0 is quasi-maximal, observe that for every object (X, α) of $\underline{\Psi}_{L_\bullet}$ one can find a morphism from it to an object (X', α'), inducing a monomorphism $A \longrightarrow A'$ in $\underline{C}$, such that the extension X' of L_0 by A' is split. Indeed, it suffices to take $A' = X$ and $(X', \alpha') = i_*(X, \alpha)$, with the notation of (3.1.2). On the other hand, it is immediate that any object (X', α') of $\underline{\Psi}$ for which the extension X' is split is dominated by E_0.

3.3.4. Let E be an object of an additive cofibred category $\underline{\Psi}$ over $\underline{C}$, lying above an object A of $\underline{C}$, and consider the following functor of the variable object B of $\underline{C}$:

$$B \longmapsto \operatorname{Hom}(E, \theta_B), \tag{3.3.4.1}$$

where θ_B denotes the trivial object of $\underline{\Psi}(B)$. We say that E admits a typical complex if this functor is representable by an object Ω_E of $\underline{C}$. Since the functor in question maps canonically to the functor $B \longmapsto \operatorname{Hom}(A, B)$ represented by A, one obtains a canonical morphism

$$A \longrightarrow \Omega_E, \tag{3.3.4.2}$$

and the corresponding length-one chain complex $A \longrightarrow \Omega_E$ in $\underline{C}$ is called the typical complex of E.

When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ and $E = (X, \alpha)$, one checks immediately that the functor (3.3.4.1) is represented by $\operatorname{Coker} \alpha$, and that the canonical morphism (3.3.4.2) is simply the composite

$$A \xrightarrow{i} X \xrightarrow{\text{can}} \operatorname{Coker} \alpha.$$

3.3.5. If an additive cofibred category $\underline{\Psi}$ over $\underline{C}$ admits a quasi-maximal object (3.3.3), and the typical complexes of its objects exist, then the typical complex $T_\bullet$ of a quasi-maximal object E_0, regarded as an object of the derived category $D(\underline{C})$, is independent, up to canonical isomorphism, of the choice of E_0 ([4, 4.4]). It is called the typical complex of the additive cofibred category $\underline{\Psi}$. If, in addition, $\underline{\Psi}$ is left exact (3.3.2), then $\underline{\Psi}$ can be reconstructed from the typical complex $T_\bullet$, up to an equivalence of fibred categories over $\underline{C}$ unique up to unique isomorphism ([4, 6.10]). There are also canonical isomorphisms, functorial in $A \in \underline{C}$ ([4, 5.4]),

$$\Psi^0(A) \xrightarrow{\sim} \operatorname{Ext}^0(T_\bullet, A), \qquad \Psi^1(A) \xrightarrow{\sim} \operatorname{Ext}^1(T_\bullet, A). \tag{3.3.5.1}$$

To complete the comparison with [4], we shall prove that when $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ its typical complex is the truncation $[L_\bullet]$. This gives another definition of the isomorphisms (3.1.8) and (3.2.4) in terms of (3.3.5.1), and one checks that it is compatible with the definitions in 3.1 and 3.2.

For this assertion we may plainly restrict to the case where $L_\bullet$ is already truncated in degree 1, since 3.1 noted that truncation does not change $\underline{\Psi}_{L_\bullet}$ up to an isomorphism of cofibred categories. The explicit construction in 3.3.4 then shows immediately that the typical complex of the quasi-maximal object described in 3.3.3 is canonically isomorphic to $L_\bullet$.

Remark 3.3.6. Let

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

be an exact sequence in $\underline{C}$. It gives an exact sequence of Ext^i groups

$$0 \longrightarrow \operatorname{Ext}^0(L_\bullet, A') \longrightarrow \operatorname{Ext}^0(L_\bullet, A) \longrightarrow \cdots \longrightarrow \operatorname{Ext}^1(L_\bullet, A) \longrightarrow \operatorname{Ext}^1(L_\bullet, A''),$$

which could of course be continued to the right by introducing Ext^i for $i \geq 2$. Using the isomorphisms (3.1.8) and (3.2.4), we obtain the exact sequence

$$\begin{aligned} 0 \longrightarrow \Psi^0_{L_\bullet}(A') \longrightarrow \Psi^0_{L_\bullet}(A) \longrightarrow \Psi^0_{L_\bullet}(A'') \xrightarrow{\partial} \Psi^1_{L_\bullet}(A') \\ \longrightarrow \Psi^1_{L_\bullet}(A) \longrightarrow \Psi^1_{L_\bullet}(A''). \end{aligned} \tag{3.3.6.1}$$

All arrows other than ∂ are defined simply by functoriality of $\Psi^0_{L_\bullet}$ and $\Psi^1_{L_\bullet}$. On the other hand, [4, 2.5.2] defines an exact sequence (3.3.6.1), for every left-exact cofibred category $\underline{\Psi}$ over $\underline{C}$, through a morphism

$$\partial : \Psi^0(A'') \longrightarrow \Psi^1(A'). \tag{3.3.6.2}$$

It associates with an element u of the left-hand side the class, in the right-hand side, of the object of $\underline{\Psi}(A')$, determined up to canonical isomorphism, whose image in $\underline{\Psi}([A \longrightarrow A''])$ (notation of 3.3.2) is the pair (X_0, u), where X_0 is the trivial object of $\underline{\Psi}(A)$.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the morphism (3.3.6.2) defined by this "geometric" construction is the negative of the boundary morphism ∂ in (3.3.6.1). Thus, after the identifications above and a change of sign in ∂, the latter exact sequence is precisely that of [4, 2.5.2]. This follows from the final assertion of [4, 5.4], in view of the compatibilities noted in 3.3.5.

Remark 3.3.6.3. We take this opportunity to clarify a point in [4, no. 3.11] concerning the fundamental isomorphisms (3.11.3) and (3.11.4), which play an essential role in the proof of [4, 5.4]. The diagram on page 35 of [4] is commutative, not anticommutative, and therefore does not as it stands define a morphism

$$u : [L_1^X \longrightarrow L_0^X] \longrightarrow K^\bullet[1].$$

It must be made anticommutative by replacing u^1 by $-u^1$. One could instead adopt the opposite convention and replace u^0 by $-u^0$; however, the stated convention, in which u^1 becomes $-u^1$, is the one for which (3.11.3) generalizes the isomorphism (3.5.3) of [4], as claimed there. With this clarification, the claim made in [4, 5.4] must also be corrected: the two boundary morphisms considered there differ by a sign.

3.3.7. In [4, 6.13] the variance of the cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ as $L_\bullet$ varies was also made explicit, in particular by reconstructing the category of cocartesian functors from $\underline{\Psi}_{L_\bullet}$ to $\underline{\Psi}_{L'_\bullet}$ in terms of the chain complexes $L_\bullet$ and $L'_\bullet$. The set of isomorphism classes of such cocartesian functors is canonically in bijection with

$$\operatorname{Hom}_{D(\underline{C})}(L'_\bullet, L_\bullet).$$

We shall use only a small part of these results, which we develop in the next section using the simple description of $\underline{\Psi}_{L_\bullet}$ given in 3.1.

3.4. Variance of $\underline{\Psi}_{L_\bullet}$ with $L_\bullet$

Consider a morphism of complexes

$$f : L'_\bullet \longrightarrow L_\bullet \tag{3.4.1}$$

given by a commutative diagram

$$\begin{tikzcd} {} \arrow[r] & L'_2 \arrow[r] \arrow[d, "f_2"'] & L'_1 \arrow[r] \arrow[d, "f_1"'] & L'_0 \arrow[r] \arrow[d, "f_0"'] & 0 \\ {} \arrow[r] & L_2 \arrow[r] & L_1 \arrow[r] & L_0 \arrow[r] & 0. \end{tikzcd} \tag{3.4.2}$$

We associate with it a canonical functor

$$f^* : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{\Psi}_{L'_\bullet}. \tag{3.4.3}$$

It is in fact a cocartesian functor of cofibred categories over $\underline{C}$: it is compatible with the structural functors to $\underline{C}$ and carries cocartesian morphisms to cocartesian morphisms; see SGA 1, Exposé VI.

If (X, α) is an object of $\underline{\Psi}_{L_\bullet}$, define $f^*(X, \alpha) = (X', \alpha')$ as follows. The extension X' of L'_0 by A is the pullback, along f_0, of the given extension X of L_0 by A, and α' is the partial trivialization deduced from α by transitivity in the second square of (3.4.2). The compatibility of α' with the relation $d'_0 d'_1 = 0$ follows immediately from the analogous compatibility for α. By construction, (3.4.3) commutes with the structural functors to $\underline{C}$. Its cocartesian character follows at once from the fact that, in the category of extensions of objects of $\underline{C}$, direct images of extensions commute with inverse images of extensions.

The functor (3.4.3) induces morphisms of functors

$$f^i : \Psi^i_{L_\bullet} \longrightarrow \Psi^i_{L'_\bullet}, \qquad i = 0, 1 \tag{3.4.4}$$

(compare 2.6). The morphism (3.4.1), on the other hand, induces morphisms

$$f^i : \operatorname{Ext}^i(L_\bullet, -) \longrightarrow \operatorname{Ext}^i(L'_\bullet, -), \qquad i \in \mathbb{Z}. \tag{3.4.5}$$

For $i = 0, 1$, the morphisms (3.4.4) and (3.4.5) are compatible with the isomorphisms (3.1.8) and (3.2.4). In other words, for $A \in \operatorname{Ob} \underline{C}$ and $i = 0, 1$, the following diagram is commutative:

$$\begin{tikzcd} \Psi^i_{L_\bullet}(A) \arrow[r, "\sim"] \arrow[d] & \operatorname{Ext}^i(L_\bullet, A) \arrow[d] \\ \Psi^i_{L'_\bullet}(A) \arrow[r, "\sim"] & \operatorname{Ext}^i(L'_\bullet, A). \end{tikzcd} \tag{3.4.6}$$

The verification of this compatibility is essentially immediate and is left to the reader.

Proposition 3.4.7. The functor (3.4.3) is faithful (respectively fully faithful, respectively an equivalence of categories) if and only if

$$H_0(f) : H_0(L'_\bullet) \longrightarrow H_0(L_\bullet)$$

is an epimorphism (respectively, if $H_0(f)$ is an isomorphism and $H_1(f)$ is an epimorphism; respectively, if $H_0(f)$ and $H_1(f)$ are isomorphisms).

Indeed, with f^i denoting the functorial morphism (3.4.4), the required condition is that f^0 be a monomorphism (respectively, that f^0 be an isomorphism and f^1 a monomorphism; respectively, that f^0 and f^1 be isomorphisms). By the commutativity of (3.4.6), these conditions may be expressed by the analogous conditions on the functors Ext^i, $i = 0, 1$. The proposition then reduces to a straightforward statement of homological algebra, which we leave to the reader as an exercise.

3.4.8. Consider now an exact sequence of chain complexes

$$0 \longrightarrow L_\bullet \xrightarrow{f} L'_\bullet \xrightarrow{g} L''_\bullet \longrightarrow 0. \tag{3.4.8.1}$$

It gives functors of the form (3.4.3),

$$\underline{\Psi}_{L''_\bullet} \xrightarrow{g^*} \underline{\Psi}_{L'_\bullet} \xrightarrow{f^*} \underline{\Psi}_{L_\bullet}.$$

The transitivity property implicit in functors of the form (3.4.3) gives an isomorphism of functors

$$f^*g^* \simeq (gf)^* = (0_{L_\bullet \to L''_\bullet})^*.$$

The last member is evidently canonically isomorphic to the cocartesian zero functor from $\underline{\Psi}_{L''_\bullet}$ to $\underline{\Psi}_{L_\bullet}$. The preceding isomorphism therefore defines a canonical functor

$$\underline{\Psi}_{L''_\bullet} \longrightarrow \underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}. \tag{3.4.8.2}$$

Here $\underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}$ denotes the category of pairs (X', t) in which X' is an object of $\underline{\Psi}_{L'_\bullet}$ and t is a trivialization of its inverse image $f^*(X')$, that is, an isomorphism

$$f^*(X') \simeq 0_{L_\bullet, A},$$

where $0_{L_\bullet, A}$ denotes the zero object of $\underline{\Psi}_{L_\bullet}(A)$ and A is the object of $\underline{C}$ underlying X'. We claim that the functor (3.4.8.2) is an equivalence of categories. Together with the left exactness in A established in 3.3.1, this completes the exactness property in $L_\bullet$ of the expression $\underline{\Psi}_{L_\bullet}(A)$.

Let us sketch the proof. To give an object of the target of (3.4.8.2) is to give an extension

$$0 \longrightarrow A \xrightarrow{i'} X' \xrightarrow{j'} L'_0 \longrightarrow 0$$

of L'_0 by an object A of $\underline{C}$, together with a morphism $\alpha' : L'_1 \longrightarrow X'$ satisfying $j'\alpha' = d'_0$ and $\alpha' d'_1 = 0$, and a morphism $t : L_0 \longrightarrow X'$ satisfying $j't = f_0$ and $td_0 = \alpha' f_1$. These data are displayed in the following diagram:

$$\begin{tikzcd}
 & & & & 0 \arrow[dl] & 0 \arrow[dl, dashed] \\
 & & & A \arrow[dl, "i'"'] & A \arrow[dl, dashed, "i''"'] & \\
 & & X' \arrow[d, shift right, "j'"'] & X'' \arrow[dl, dashed, "j''"'] & & \\
0 \arrow[r] & L_0 \arrow[ur, bend left, "t"] \arrow[r, "f_0"] & L'_0 \arrow[r, "g_0"] & L''_0 \arrow[r] & 0 & \\
0 \arrow[r] & L_1 \arrow[u, "d_0"] \arrow[r, "f_1"] & L'_1 \arrow[u, "d'_0"] \arrow[uu, bend right, "\alpha'"'] \arrow[r, "g_1"] & L''_1 \arrow[u, "d''_0"] \arrow[uu, dashed, bend right, "\alpha''"'] \arrow[r] & 0 & \\
0 \arrow[r] & L_2 \arrow[u] \arrow[r] & L'_2 \arrow[u, "d'_1"] \arrow[r] & L''_2 \arrow[u, "d''_1"] \arrow[r] & 0 &
\end{tikzcd}
\qquad
\begin{cases}
j'\alpha' = d'_0, \\
\alpha' d'_1 = 0, \\
j't = f_0, \\
td_0 = \alpha' f_1.
\end{cases}$$

The familiar left exactness, in the first argument, of extensions of an object L by an object A, applied to the first row of the diagram, shows that a morphism t satisfying $j't = f_0$ allows one to interpret X' as the inverse image of an extension X'' of L''_0 by A. Under this interpretation, t corresponds to the natural trivialization of the inverse image of that extension along $g_0 f_0 = 0$.

The morphism α' satisfying $j'\alpha' = d_0'$ and $\alpha' f_1 = td_0$ is interpreted by essentially the same dictionary, now applied to the second row of the diagram, as arising from a morphism $\alpha'' : L_1'' \longrightarrow X''$ satisfying $j''\alpha'' = d_0''$. Here α' and α'' are regarded as trivializations of the inverse images of X' and X'' along d_0' and d_0'', respectively. Finally, the relation $\alpha' d_1' = 0$ is then equivalent to $\alpha'' d_1'' = 0$. Thus the data (X', α', t), an object of $\underline{\Psi}_{[L_\bullet \longrightarrow L'_\bullet]}$, are equivalent to the data (X'', α''), an object of $\underline{\Psi}_{L''_\bullet}$. This proves, in substance, that (3.4.8.2) is indeed an equivalence of categories.

3.4.9. Continue to consider the exact sequence (3.4.8.1), and the exact sequence of Ext^i groups that it defines:

$$(3.4.9.1) \quad 0 \longrightarrow \mathrm{Ext}^0(L''_\bullet, A) \xrightarrow{g^0} \mathrm{Ext}^0(L'_\bullet, A) \xrightarrow{f^0} \mathrm{Ext}^0(L_\bullet, A) \xrightarrow{\partial^0} \mathrm{Ext}^1(L''_\bullet, A)$$
$$\xrightarrow{g^1} \mathrm{Ext}^1(L'_\bullet, A) \xrightarrow{f^1} \mathrm{Ext}^1(L_\bullet, A) \longrightarrow \cdots .$$

The canonical isomorphisms of the forms (3.1.8) and (3.2.4) give a geometric interpretation of the first six terms of this exact sequence in terms of the categories $\underline{\Psi}_{L_\bullet}$, $\underline{\Psi}_{L'_\bullet}$, and $\underline{\Psi}_{L''_\bullet}$. Compatibility (3.4.6) likewise gives a geometric interpretation of the arrows f^0, g^0, f^1, g^1. We shall now give such an interpretation of ∂^0, using the result just proved that (3.4.8.2) is an equivalence of categories.

Let

$$u \in \mathrm{Ext}^0(L_\bullet, A), \qquad \text{i.e.} \qquad u : L_0 \longrightarrow A, \quad ud_0 = 0.$$

Consider the object $(0, \overline{u})$ of $\underline{\Psi}_{[L_\bullet \to L'_\bullet]}$. Here 0 is the zero object of $\underline{\Psi}_{L'_\bullet}$, whose inverse image along f_0 is therefore identified with the zero object of $\underline{\Psi}_{L_\bullet}$, and $\overline{u}$ is the automorphism of the latter defined by u through the canonical isomorphism (3.1.8). Since (3.4.8.2) is an equivalence, $(0, \overline{u})$ comes from an object of $\underline{\Psi}_{L''_\bullet}(A)$, determined up to unique isomorphism. Its isomorphism class is an element of

$$\Psi^1_{L''_\bullet}(A) \simeq \mathrm{Ext}^1(L''_\bullet, A).$$

We claim that this element is precisely $\partial^0(u)$.

We prove this compatibility by reducing it to the analogous one in [4, Cor. 5.5], where the sign must be corrected as indicated in 3.3.6.3. The argument also gives another proof of the equivalence (3.4.8.2), by reducing it to [4, Prop. 2.7]. Consider the opposite category $\underline{C}^{\mathrm{o}}$ and interpret the construction of $\underline{\Psi}_{L_\bullet}(A)$, for fixed A, together with the constructions of 3.4.8 and the present section, in terms of $\underline{C}^{\mathrm{o}}$ and the category $\underline{E}$ cofibred over $\underline{C}^{\mathrm{o}}$ whose objects are extensions of the fixed object A of $\underline{C}^{\mathrm{o}}$ by a variable object L of $\underline{C}^{\mathrm{o}}$. This is an additive cofibred category over $\underline{C}^{\mathrm{o}}$, and is even left exact there (in the sense of [4, 2.2]).

A chain complex $L_\bullet$ of $\underline{C}$ may be interpreted as a cochain complex of $\underline{C}^{\mathrm{o}}$. Under this interpretation, the construction of $\underline{\Psi}_{L_\bullet}(A)$ coincides with the opposite of the category $\widehat{\underline{E}}(L_\bullet)$ in the sense of [4, 1.8]. The functor (3.4.8.2) is then identified with the opposite of the functor of the form [4, (2.1.2)], with E replaced by its extension $\widehat{\underline{E}}$ to the category of cochain complexes. Consequently, the assertion that this functor is an equivalence is the special case [4, 2.7, a)$\Rightarrow$d)].

On the other hand, the exact sequence (3.4.9.1) is identified with the exact sequence of the groups $\mathrm{Ext}^i(A, L''_\bullet)$, $\mathrm{Ext}^i(A, L'_\bullet)$, and $\mathrm{Ext}^i(A, L_\bullet)$, apart from a factor $(-1)^i$ on ∂^i (Verdier dixit). Similarly, the isomorphisms (3.1.8) and (3.2.4) are identified with the analogous isomorphisms (3.11.5) and (3.11.4). The asserted compatibility is therefore the special case of [4, Cor. 5.5] corrected as above.

Remark 3.4.9.2. The editor, being very afraid of signs, confesses that he does not have complete confidence in the preceding proof: he fears that signs may have been omitted in passing from C to C^{o}. He recommends that a courageous reader verify the asserted compatibility directly (or, respectively, its opposite). Fortunately, these sign questions will play no role in the following exposés.

3.5. A canonical partial flat resolution of an arbitrary $\mathbb{Z}$-module

We apply the preceding results when $\underline{C}$ is the category of commutative groups of the topos $\underline{T}$, that is, the category of $\mathbb{Z}$-modules of $\underline{T}$. Let P be an object of $\underline{C}$. We shall define, functorially in

P, a chain complex $L_\bullet(P)$ augmented to P:

(3.5.1)
$$\begin{array}{ccccccccc} 0 \longrightarrow & \overbrace{\mathbb{Z}[P\times P]\times\mathbb{Z}[P\times P\times P]}^{L_2} & \xrightarrow{d_1} & \overbrace{\mathbb{Z}[P\times P]}^{L_1} & \xrightarrow{d_0} & \overbrace{\mathbb{Z}[P]}^{L_0} & \longrightarrow 0 \\ & & & & & \Big\downarrow{\scriptstyle\varepsilon} & \\ & & & & & P & \end{array}$$

with $L_i(P)=0$ for $i\geq 3$. The components are displayed in (3.5.1), using the notation of 1.4. To describe the differentials d_0, d_1 and the augmentation ε, write $[p]$ for the section of $\mathbb{Z}[P](S)$ defined by $p\in P(S)$, and similarly write $[p,q]$ and $[p,q,r]$ for the corresponding sections of $\mathbb{Z}[P\times P](S)$ and $\mathbb{Z}[P\times P\times P](S)$. Then set

(3.5.2)
$$\begin{cases} \varepsilon[p] = p, \\ d_0[p,q] = [p+q]-[p]-[q], \\ d_1[p,q] = [p,q]-[q,p], \\ d_1[p,q,r] = [p+q,r]-[p,q+r]+[p,q]-[q,r]. \end{cases}$$

These homomorphisms of commutative groups do define a complex $L_\bullet(P)$ augmented to P:

$$\varepsilon d_0 = 0, \qquad d_0 d_1 = 0.$$

The first relation is immediate. The second splits into two relations according to the two factors of $L_2(P)$; they express, respectively, commutativity and associativity of the composition law on P.

The augmented complex depends functorially on P. A morphism $u : P \longrightarrow P'$ gives a commutative diagram

$$\begin{array}{ccc} L_\bullet(P) & \xrightarrow{L_\bullet(u)} & L_\bullet(P') \\ \Big\downarrow & & \Big\downarrow \\ P & \xrightarrow{u} & P'. \end{array}$$

Its components are flat, as is every free $\mathbb{Z}$-module.

Proposition 3.5.3. One has $H_1(L_\bullet(P))=0$, and the augmentation ε induces an isomorphism

$$H_0(L_\bullet(P)) \xrightarrow{\sim} P.$$

For the proof one can make the standard reduction to the case in which $\underline{T}$ is the category of sets and P is an ordinary $\mathbb{Z}$-module, where (3.5.1) is likely familiar. We instead give a direct proof that does not use this reduction, applying 3.4.7 to the augmentation

$$\varepsilon : L_\bullet(P) \longrightarrow P,$$

where P is regarded, as usual, as a chain complex concentrated in degree zero. It remains to prove that, for every object G of $\underline{C}$, the functor

(*)
$$\varepsilon^* : \underline{\Psi}_P(G) \longrightarrow \underline{\Psi}_{L_\bullet(P)}(G)$$

is an equivalence of categories.

The source of (*) is plainly isomorphic to the category of extensions of the $\mathbb{Z}$-module P by G. To describe its target, use 1.4 to identify the categories of extensions of L_0, L_1, L_2 by G, respectively, with the category of torsors under G_P, the category of torsors under $G_{P\times P}$, and the category of pairs consisting of a torsor under $G_{P\times P}$ and a torsor under $G_{P\times P\times P}$.

Thus an object of the target of (*) is a pair (E, φ), where E is a torsor under G_P and φ is a trivialization of the torsor E' on $P \times P$ corresponding to the inverse image along d_0 of the extension of $\mathbb{Z}[P]$ by G defined by E. The trivialization is subject to the compatibility with $d_0 d_1 = 0$ described below. Formula (3.5.2) immediately gives

$$E' = \pi^* E \ \mathrm{pr}_1^*(E)^{-1} \ \mathrm{pr}_2^*(E)^{-1},$$

and a trivialization of this torsor is the same as an isomorphism, also denoted by φ,

$$\varphi : \mathrm{pr}_1^* E \ \mathrm{pr}_2^* E \xrightarrow{\sim} \pi^* E.$$

Expanding the compatibility of φ with $d_0 d_1 = 0$ gives the commutativity condition (1.2.1) and the associativity condition (1.1.4.1), corresponding to the two factors of L_2. The assertion that (*) is an equivalence is therefore precisely the dictionary of 1.1.6. This proves 3.5.3.

Remark 3.5.4. a) It would be interesting to strengthen 3.5.3 by constructing a resolution $L_\bullet(P)$ of P, functorial in P, whose components are functorially isomorphic to sums of free $\mathbb{Z}$-modules generated by cartesian powers of P. In view of isomorphisms of the form (1.4.5), this would give a method for partially computing $\mathrm{Ext}^i(P, A)$ as the abutment of a spectral sequence whose initial term is expressed through groups

$$H^j(P \times P \times \cdots \times P, A).$$

Here we obtain only very partial information about Ext^1, and no information about the higher Ext^i. There is, in any case, a natural way to continue the partial resolution $L_\bullet(P)$ by one degree. [1]

b) There is a variant of (3.5.1) and 3.5.3 in the category of A-modules, where A is a fixed ring of $\underline{T}$. The form of $L_\bullet(P)$ is more complicated because the description 1.1.6 must be enlarged (compare 2.10.3). One obtains

$$\text{(3.5.5)} \qquad \begin{cases} L_0 = A[P], \\ L_1 = A[P \times P] + A[A \times P], \\ L_2 = A[P \times P \times P] + A[P \times P] + A[A \times A \times P] + A[A \times P \times P]. \end{cases}$$

The augmented-complex structure is given by (3.5.2), supplemented by

$$\text{(3.5.6)} \qquad \begin{cases} d_0[\lambda, p] = [\lambda p] - \lambda[p], \\ d_1[\lambda, \mu, p] = [\lambda, \mu p] - [\lambda\mu, p] + \lambda[\mu, p], \\ d_1[\lambda, p, q] = [\lambda, p + q] - [\lambda, p] - [\lambda, q] + \lambda[p, q] - [\lambda p, \lambda q]. \end{cases}$$

The relation $d_0 d_1 = 0$ follows from the analogous relation on the components appearing in (3.5.2), which, as we saw, is equivalent to associativity and commutativity of addition on P, together with the relations

$$d_0 d_1[\lambda, \mu, p] = 0, \qquad d_0 d_1[\lambda, p, q] = 0.$$

The latter express, respectively, associativity of the given action of A on P,

$$\lambda(\mu p) = (\lambda\mu) p,$$

and its distributivity over addition,

$$\lambda(p + q) = \lambda p + \lambda q.$$

For a torsor E under G_P, a trivialization of $d_0^* E$ amounts to an internal composition law on E (satisfying the compatibility conditions already described for the internal law on E and its torsor structure), together with an external composition law of A on E, compatible with the action of A

1. L. Breen constructed such a resolution (editor's note).

on P and on G. The compatibility of this trivialization with the evident trivialization of $d_1^* d_0^* E$ is equivalent to the conditions already described above, which say that the internal law on E is a commutative-group law extending P by G, together with associativity of the external law and distributivity of the latter with respect to the internal law. These conditions say exactly that E, with its two composition laws and its torsor structure, is an A-module extension of the A-modules P and G.

It follows, by essentially the same method as in 3.5.3, that $L_\bullet$ is a partial resolution of the module P, that is, $H_1(L_\bullet) = 0$. It would again be interesting to extend this partial resolution to an infinite resolution of the same kind, with components of the form

$$A[A \times A \times \cdots \times P \times P \times \cdots].$$

Compare the analogous question in a) for commutative groups.

3.6. Application to the description of the cofibred category of biextensions of a given pair (P, Q) of commutative groups

Let P and Q be two $\mathbb{Z}$-modules of the topos $\underline{T}$. Choose a flat resolution $L'_\bullet(P)$ of P that agrees in degrees at most 2 with the augmented complex $L_\bullet(P)$ of 3.5. This is possible by 3.5.3 and because every $\mathbb{Z}$-module on $\underline{T}$ is a quotient of a free, hence flat, $\mathbb{Z}$-module. Similarly choose a flat resolution $L'_\bullet(Q)$ of Q agreeing with $L_\bullet(Q)$ in degrees at most 2. We have canonical homomorphisms

$$L_\bullet(P) \longrightarrow L'_\bullet(P), \qquad L_\bullet(Q) \longrightarrow L'_\bullet(Q), \tag{3.6.1}$$

and hence a canonical homomorphism

$$\begin{aligned} L_\bullet(P, Q) &\overset{\text{def}}{=} L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow L'_\bullet(P, Q) \\ &\overset{\text{def}}{=} L'_\bullet(P) \otimes L'_\bullet(Q) \simeq P \overset{\mathbb{L}}{\otimes} Q. \end{aligned} \tag{3.6.2}$$

It is an isomorphism on components in degrees at most 2. Consequently, by definition,

$$\underline{\Psi}_{L_\bullet(P,Q)} = \underline{\Psi}_{L'_\bullet(P,Q)}. \tag{3.6.3}$$

Theorem 3.6.4. The additive cofibred category over the category

$$\underline{C} = \mathrm{GRAB}(\underline{T})$$

of commutative groups of $\underline{T}$, consisting of biextensions of (P, Q) by a variable abelian group G (2.4), is canonically equivalent to the additive cofibred category

$$\underline{\Psi}_{L_\bullet(P,Q)} = \underline{\Psi}_{L'_\bullet(P,Q)}$$

defined from $L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q)$ as in 3.1.

This and 3.2.5 give the principal result of the present section, announced above:

Corollary 3.6.5. For fixed P and Q there is a canonical isomorphism, functorial in G,

$$\mathrm{Biext}^1(P, Q; G) \xrightarrow{\sim} \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G). \tag{3.6.5.1}$$

To prove 3.6.5, first make explicit the components of degrees $0, 1, 2$ of $L_\bullet(P, Q)$:

$$\begin{cases} L_0(P, Q) = L_0(P) \otimes L_0(Q) = \mathbb{Z}[P \times Q], \\ L_1(P, Q) = L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \\ \qquad\qquad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times P \times Q], \\ L_2(P, Q) = L_0(P) \otimes L_2(Q) + L_1(P) \otimes L_1(Q) + L_2(P) \otimes L_0(Q) \\ \qquad\qquad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times Q \times Q \times Q] \\ \qquad\qquad\quad + \mathbb{Z}[P \times P \times Q \times Q] \\ \qquad\qquad\quad + \mathbb{Z}[P \times P \times Q] + \mathbb{Z}[P \times P \times P \times Q]. \end{cases} \tag{3.6.6}$$

To describe the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ through its fibres $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, use 1.4 to express extensions of $L_i(P,Q)$, for $i = 0,1,2$, by G in terms of torsors. An extension of $L_0(P,Q)$ by G is a torsor under $G_{P\times Q}$. An extension of $L_1(P,Q)$ by G is a pair consisting of a torsor under $G_{P\times Q\times Q}$ and one under $G_{P\times P\times Q}$. An extension of $L_2(P,Q)$ by G is similarly a system of five torsors, under the groups obtained from G by base change to

$$P \times Q \times Q, \quad P \times Q \times Q \times Q, \quad P \times P \times Q \times Q,$$
$$P \times P \times Q, \quad P \times P \times P \times Q.$$

Under this dictionary, a pair (X, α) consisting of an extension of $L_0(P,Q)$ by G and a trivialization of d_0^*X on $L_1(P,Q)$ becomes a torsor E on $P \times Q$ equipped with two trivializations (φ, ψ) of the torsors (E_1, E_2) on $P \times Q \times Q$ and $P \times P \times Q$, respectively, obtained from E by inverse image along

$$d_0 : L_1(P,Q) \longrightarrow L_0(P,Q).$$

Expanding d_0, as in the proof of 3.5.3, identifies φ and ψ with isomorphisms

$$\varphi : q_1^*E\, q_2^*E \xrightarrow{\sim} \pi_Q^*E, \qquad \psi : p_1^*E\, p_2^*E \xrightarrow{\sim} \pi_P^*E.$$

Here $\pi_Q, q_1, q_2 : P \times Q \times Q \longrightarrow P \times Q$ are induced by the composition law and the first and second projections $Q \times Q \longrightarrow Q$; the maps

$$\pi_P, p_1, p_2 : P \times P \times Q \longrightarrow P \times Q$$

are defined symmetrically. Thus φ and ψ are data of the forms (2.0.4) and (2.0.7) appearing in 2.1.

Finally, the condition that (X, α) be an object of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, namely compatibility of the partial trivialization α with $d_0 d_1 = 0$, becomes five compatibility relations for the structure (E, φ, ψ) on torsors over, respectively,

$$P \times Q \times Q, \quad P \times Q \times Q \times Q, \quad P \times P \times Q \times Q,$$
$$P \times P \times Q, \quad P \times P \times P \times Q.$$

They correspond to the five terms in the expression (3.6.6) for $L_2(P,Q)$. As in the proof of 3.5.3, the first two are precisely the commutativity relation (2.0.6) and the associativity relation (2.0.5); together they say that φ gives E the structure of a commutative extension of Q_P by G_P. Symmetrically, the last two say that ψ gives E the structure of a commutative extension of P_Q by G_Q. The middle relation, on torsors over $P \times P \times Q \times Q$, is exactly the commutativity of diagram (2.1.1). It says that φ and ψ are compatible, hence, in view of the other conditions, define on E a biextension structure of (P,Q) by G.

The verification of these interpretations is immediate and is left to the reader. The preceding discussion indeed defines an equivalence of categories cofibred over $\underline{C} = \mathrm{GRAB}(\underline{T})$, as asserted in 3.6.4.

Remark 3.6.7. Each of the three generalizations of biextensions mentioned in 2.10 has a corresponding variant of 3.6.4 and hence of 3.6.5. The generalization to any number of factors (2.10.2) is immediate, as is the generalization to biextensions of A-modules (2.10.3), once the variant of $L_\bullet(P)$ described in 3.5.4 b) has been constructed. From the viewpoint of homological algebra, the case in which P and Q are not assumed commutative (2.10.1) is more interesting.

Associate with a not necessarily commutative group P of $\underline{T}$ a complex $K_\bullet(P)$ truncated in degree 2. It is described as $L_\bullet(P)$ was in 3.5, except that

$$K_2(P) = \mathbb{Z}[P \times P \times P],$$

the first factor occurring in $L_2(P)$ being omitted; d_0 and d_1 are defined as in (3.5.2). Then

$$H_0(K_\bullet(P)) = P/[P,P] \simeq \underline{H}_1(P, \mathbb{Z}),$$

and similarly

$$H_1(K_\bullet(P)) \simeq \underline{H}_2(P, \mathbb{Z}).$$

Indeed, $K_\bullet(P)$ is precisely the truncation in degree 2 of the chain complex $C_\bullet(P, \mathbb{Z})$ of P with coefficients in $\mathbb{Z}$.

The argument proving 3.5.3 also shows that the cofibred category $\underline{\Psi}_{K_\bullet(P)}$ over $\underline{C} = \mathrm{GRAB}(\underline{T})$ is canonically equivalent to the category of central extensions of P by variable commutative groups G. Likewise, for two groups P, Q, the cofibred category of biextensions of (P, Q) in the sense of 2.10.1 is equivalent to $\underline{\Psi}_{K_\bullet(P,Q)}$, where

$$K_\bullet(P, Q) = K_\bullet(P) \overset{\mathbb{L}}{\otimes} K_\bullet(Q).$$

These considerations suggest that, when P is commutative, the canonical resolution sought in 3.5.4 a) might be obtained by a suitable modification of $C_\bullet(P, \mathbb{Z})$.

3.7. Various compatibilities

In this section we examine the behaviour, for variable P and Q, of the equivalence of cofibred categories in 3.6.4. The variance of the cofibred category of biextensions of (P, Q) by a variable G with respect to variable P, Q is defined in principle in 2.4; it may also be expressed by saying that one has a cofibred functor (2.6.1). Notice that the variance in P, Q is contravariant. The variance of the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ with respect to variable P, Q is also clear, since $L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q)$ depends functorially on the pair (P, Q), while $\underline{\Psi}_{L_\bullet}$ depends contravariantly on $L_\bullet$ (3.4). With this understood, a first compatibility is expressed, for every triple of morphisms

$$P' \longrightarrow P, \qquad Q' \longrightarrow Q, \qquad G \longrightarrow G', \tag{3.7.1}$$

by the fact that the diagram of functors

$$\begin{tikzcd} \underline{\Psi}_{L_\bullet(P,Q)}(G) \arrow[r] \arrow[d] & \underline{\Psi}_{L_\bullet(P',Q')}(G') \arrow[d] \\ \mathrm{BIEXT}(P, Q; G) \arrow[r] & \mathrm{BIEXT}(P', Q'; G') \end{tikzcd} \tag{3.7.2}$$

is commutative up to a canonical isomorphism. We shall admit this compatibility: making the commutativity isomorphism explicit could only be as obvious as it is tedious (as, indeed, could the complete explication it presupposes of the equivalence of cofibred categories in 3.6.4).

3.7.3. It follows in particular from the preceding compatibility that the isomorphism (3.6.5.1) is functorial not only in G, but also in P and Q. One may state, similarly and more trivially, the analogous functoriality of the isomorphism defined through (3.1.8), just as (3.6.5.1) was defined through (3.2.4):

$$\mathrm{Biext}^0(P, Q; G) \xrightarrow{\sim} \mathrm{Ext}^0(P \overset{\mathbb{L}}{\otimes} Q, G) \simeq \mathrm{Hom}(P \otimes Q, G). \tag{3.7.4}$$

But such an isomorphism was already obtained in (2.5.4.1), and was observed in 2.6 to be functorial in its three arguments. It is immediate to verify that the two isomorphisms (3.7.4) and (2.5.4.1) are equal.

3.7.5. The value of the isomorphism in 3.6.5 is that it expresses the remarkable group of geometric origin $\mathrm{Biext}^1(P, Q; G)$ in terms of the very simple homological-algebra invariant $\mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$. The latter belongs to the sequence of invariants $\mathrm{Ext}^i(P \overset{\mathbb{L}}{\otimes} Q, G)$, to which the usual formalism of homological algebra applies. In particular, the sequence of these Ext^i is a cohomological functor in the sense of [3] (or a connected sequence of functors in the terminology of [5]) in each of the three arguments P, Q, G. Thus an exact sequence

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \tag{3.7.5.1}$$

(respectively,

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0, \qquad \text{or} \qquad 0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0) \tag{3.7.5.2}$$

gives rise to an infinite exact sequence of the Ext^i. Its first six terms, which involve only Ext^0 and Ext^1, may be interpreted in terms of biextensions by 3.6.5 and 3.7.4. We have already indicated in 3.3.6 a geometric interpretation of the exact sequence deduced from (3.7.5.1), and especially of the boundary map

$$\partial : \mathrm{Biext}^0(P, Q; G'') \longrightarrow \mathrm{Biext}^1(P, Q; G'), \tag{3.7.5.3}$$

within the framework of a left exact cofibred category. One may seek an analogous interpretation of the six-term exact sequence associated with either short exact sequence in (3.7.5.2), say the first; equivalently, in view of 3.7.3, one may seek a geometric interpretation of the boundary homomorphism

$$\partial : \mathrm{Biext}^0(P', Q; G) \longrightarrow \mathrm{Biext}^1(P'', Q; G). \tag{3.7.5.4}$$

Indeed, we have the following result.

Proposition 3.7.6. The cofibred category over

$$\mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})^{\mathrm{op}} \times \mathrm{GRAB}(\underline{T})$$

of biextensions of (P, Q) by G, with P, Q, G variable (cf. (2.6.1)), is left exact not only in the argument G, but also in each of the arguments P and Q. Moreover, the boundary homomorphism (3.7.5.4) deduced from the exact sequence of the Ext^i associated with (3.7.5.2), by means of the isomorphisms 3.6.5 and 3.7.4, is the boundary homomorphism in the theory of left exact cofibred categories over $\underline{C} = \mathrm{GRAB}(\underline{T})^{\mathrm{op}}$ recalled in 3.3.6, applied to the cofibred category of biextensions of (P, Q) by G, where Q and G are fixed and P varies (subject to a possible sign error in 3.4.9; cf. 3.4.9.2).

We sketch the proof of 3.7.6. By 3.6.4 and compatibility (3.7.2), we interpret biextensions of (P, Q) by G as objects of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$. Consider the homomorphism

$$L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q)$$

defined by the augmentation $L_\bullet(P) \longrightarrow P$. It follows from 3.5.3 and the flatness of $L_\bullet(Q)$ that this homomorphism induces isomorphisms on H_0 and H_1, and hence by 3.4.7 an equivalence

$$\underline{\Psi}_{P\otimes L_\bullet(Q)} \longrightarrow \underline{\Psi}_{L_\bullet(P,Q)}.$$

We may therefore interpret biextensions of (P, Q) by G as objects of $\underline{\Psi}_{P\otimes L_\bullet(Q)}(G)$. The exact sequence (3.7.5.2) now gives an exact sequence of chain complexes

$$0 \longrightarrow P' \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q) \longrightarrow P'' \otimes L_\bullet(Q) \longrightarrow 0. \tag{3.7.6.1}$$

The asserted left exactness follows from 3.4.8 applied to this exact sequence of complexes. (A direct verification would in any event be immediate.) Moreover, the homomorphism (3.7.5.4) is precisely the boundary homomorphism $\mathrm{Ext}^0 \longrightarrow \mathrm{Ext}^1$ deduced from this exact sequence by taking the Ext^i with values in G. The last assertion of 3.7.6 is consequently a special case of 3.4.9.

3.8. Biextensions of (P, Q) trivialized over extensions of P and Q

First consider two exact sequences of abelian groups

$$0 \longrightarrow L_1 \xrightarrow{i} L_0 \xrightarrow{u} P \longrightarrow 0, \qquad 0 \longrightarrow M_1 \xrightarrow{j} M_0 \xrightarrow{v} Q \longrightarrow 0. \tag{3.8.1}$$

We shall describe the category of biextensions E of (P, Q) by an abelian group G for which the inverse image $(u, v)^*(E)$ is trivial; that is, those for which there exists a homomorphism s making the following triangle commutative:

$$\begin{tikzcd} E \arrow[d] & \\ P \times Q & L_0 \times M_0 \arrow[l, "u \times v"'] \arrow[ul, dash, "s"'] \end{tikzcd} \tag{3.8.2}$$

and satisfying the additivity condition required in each of its two arguments. The choice of such an s therefore identifies $(u, v)^*(E)$ with the trivial biextension $E_0 = L_0 \times M_0 \times G$ of (L_0, M_0) by G. The left exactness in P, Q of the cofibred category of biextensions of (P, Q) by fixed G (3.7.6) then shows that to give the biextension E of (P, Q), together with the trivialization s of its inverse image, is essentially equivalent to giving two partial trivializations of the trivial biextension E_0 of (L_0, M_0) by G, over (L_1, M_0) and (L_0, M_1) respectively, which agree over (L_1, M_1). Since the biextensions induced by E_0 over (L_1, M_0), (L_0, M_1), and (L_1, M_1) identify with the trivial biextensions, a partial trivialization over (L_1, M_0) (respectively over (L_0, M_1)) is equivalent to giving a homomorphism (or pairing)

$$f : L_1 \otimes M_0 \longrightarrow G \qquad \big(\text{respectively } g : L_0 \otimes M_1 \longrightarrow G\big). \tag{3.8.3}$$

The compatibility in question simply means that the two homomorphisms agree over $L_1 \otimes M_1$, namely

$$f(x_1 \otimes y_1) = g(x_1 \otimes y_1), \tag{3.8.4}$$

for sections x_1, y_1 of L_1, M_1 over an object of $\underline{T}$. In terms of the homomorphism s of (3.8.2), the forms f and g are characterized by

$$s(x_1, y_0) = f(x_1 \otimes y_0) e_{E/Q}(y_0), \qquad s(x_0, y_1) = g(x_0 \otimes y_1) e_{E/P}(x_0), \tag{3.8.5}$$

which again makes clear the necessity of condition (3.8.4).

Once the trivialization s has been chosen, the other trivializations are the morphisms $s' : L_0 \times M_0 \longrightarrow E$ of the form

$$s'(x_0, y_0) = s(x_0, y_0) + h(x_0 \otimes y_0), \tag{3.8.6}$$

where

$$h : L_0 \otimes M_0 \longrightarrow G$$

is an arbitrary form. It is then clear from (3.8.5) that the forms f, g defined by s are changed into

$$\begin{aligned} f'(x_1 \otimes y_0) &= f(x_1 \otimes y_0) + h(x_1 \otimes y_0), \\ g'(x_0 \otimes y_1) &= g(x_0 \otimes y_1) + h(x_0 \otimes y_1). \end{aligned} \tag{3.8.7}$$

Thus we obtain:

Proposition 3.8.8. The subgroup of $\mathrm{Biext}^1(P, Q; G)$ formed by the classes of biextensions whose inverse image under (u, v) is a trivial biextension of (L_0, M_0) (that is, the kernel of $\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(L_0, M_0; G)$) is canonically isomorphic, by the preceding description, to the quotient of the group of pairs of forms (f, g) in (3.8.3) satisfying compatibility condition (3.8.4), by the pairs of forms which arise by restriction from a form $h : L_0 \otimes M_0 \longrightarrow G$.

We now interpret this isomorphism in terms of homological algebra, taking into account the canonical isomorphism

$$\mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

of 3.6.5. To this end, interpret the data (3.8.1) as two complexes resolving P and Q, respectively:

$$\text{(3.8.9)} \qquad L_\bullet : \ 0 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0, \qquad M_\bullet : \ 0 \longrightarrow M_1 \longrightarrow M_0 \longrightarrow 0,$$

and consider the tensor-product complex $L_\bullet \otimes M_\bullet$

$$0 \longrightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \longrightarrow 0.$$

There is, moreover, a canonical homomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes} Q \longrightarrow L_\bullet \otimes M_\bullet,$$

and hence an equally canonical homomorphism

$$\text{(3.8.10)} \qquad H^1(\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, G)) \longrightarrow \mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G).$$

The group of 1-cocycles of $\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, G)$ identifies with the group of pairs (f, g) as in (3.8.3) which satisfy compatibility condition (3.8.4): to such a pair one associates the homomorphism $F : L_1 \otimes M_0 + L_0 \otimes M_1 \longrightarrow G$ with components (f, g),

$$\text{(3.8.11)} \qquad F(x_1 \otimes y_0 + x_0 \otimes y_1) = f(x_1 \otimes y_0) + g(x_0 \otimes y_1).$$

Moreover, the coboundary of the 0-cochain $h : L_0 \otimes M_0 \longrightarrow G$ is precisely the pair of its restrictions to $L_1 \otimes M_0$ and $L_0 \otimes M_1$. Consequently, the group of pairs (f, g) modulo the pairs (h, h) described in 3.8.8 is canonically isomorphic to the first term of (3.8.10). With this understood, one has the following compatibility.

Proposition 3.8.12. *If*

$$\xi \in \mathrm{Ker}\big(\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Biext}^1(L_0, M_0; G)\big)$$

is described by the pair (f, g) as in 3.8.8, then the image of ξ in $\mathrm{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$ *under the isomorphism (3.6.5.1) is the image under (3.8.10) of the element of its first term defined by the cocycle (3.8.11).*

The verification of this compatibility is left to the reader.

3.8.13. We shall apply this especially in the case

$$L_0 = \mathbb{Z}[P], \qquad M_0 = \mathbb{Z}[Q],$$

where L_1 and M_1 are respectively the kernels of the canonical homomorphisms $\mathbb{Z}[P] \longrightarrow P$ and $\mathbb{Z}[Q] \longrightarrow Q$. Notice the canonical isomorphisms

$$L_0 \overset{\mathbb{L}}{\otimes} M_0 \simeq L_0 \otimes M_0 \simeq \mathbb{Z}[P \times Q].$$

Since L_0 and M_0 are flat, (1.4.5) gives canonical isomorphisms

$$\mathrm{Ext}^i(L_0 \overset{\mathbb{L}}{\otimes} M_0, G) \simeq H^i(P \times Q, G_{P \times Q}).$$

It follows from this and 3.6.5, or directly from the definitions and 1.4, that the functor which assigns to each biextension of (L_0, M_0) by G the $G_{P\times Q}$-torsor over $P \times Q$ that it induces is an equivalence of categories. (This remains valid without using an additive structure on P, Q.) Consequently, if E is a biextension of (P, Q) by G, giving a trivialization of its inverse image over (L_0, M_0) is equivalent to giving a section of E considered only as an object over $P \times Q$, that is, a trivialization of E considered as a torsor under $G_{P\times Q}$.

Biextensions equipped with this additional structure are therefore described by the pairs (f, g) of (3.8.3) satisfying compatibility (3.8.4), and their classification up to isomorphism is the one given in 3.8.8.

With the notation of 3.5, one may also regard L_1 as the cokernel of $L_2(P) \longrightarrow L_1(P)$, and similarly for M_1. This makes it possible to express the data (f, g) in terms of a system of homomorphisms

$$\begin{cases} f' : L_1(P) \otimes L_0(Q) = \mathbb{Z}(P \times P \times Q) \longrightarrow G, \\ g' : L_0(P) \otimes L_1(Q) = \mathbb{Z}(P \times Q \times Q) \longrightarrow G, \end{cases}$$

that is, homomorphisms of sheaves of <u>sets</u>

$$P \times P \times Q \longrightarrow G, \qquad P \times Q \times Q \longrightarrow G,$$

satisfying five compatibility conditions. These express that f' and g' factor through f and g—which gives 2×2 equations, in view of the expression of $L_2(P)$ and $L_2(Q)$ as sums of two terms—together with condition (3.8.4). One then sees that f' and g' are precisely the partial multiplication data defining a biextension, and that the five conditions in question are exactly the five axioms for biextensions considered in 2.1 (cf. the proof of 3.6.4).

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SGA 7

Exposé VIII

COMPLEMENTS ON BIEXTENSIONS. GENERAL PROPERTIES OF BIEXTENSIONS OF GROUP SCHEMES

by A. Grothendieck

Contents

0. Introduction

0.1. We continue here the general study of biextensions begun in the preceding exposé. First we give some complements in the case of an arbitrary base topos, developing in particular, with the detail it deserves, the formalism of the pairings associated with biextensions (Section 2), whose importance in the theory of abelian schemes is well known. We then begin the study of biextensions of general group schemes, the case of greatest interest for us being that of biextensions by the multiplicative group $\underline{G}_m$. We shall nevertheless be obliged, as a technical intermediate step, also to study biextensions by other types of group schemes (Sections 5 and 6).

The most interesting result for what follows is given in 7.1, which gives conditions under which, for P and Q smooth over a trait with generic point η, a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$ can be prolonged to a biextension of (P, Q) by $\underline{G}_m$, determined up to a unique isomorphism. This result will be very convenient in the following exposé, devoted to the theory of the Néron model of abelian varieties. It is likely that one could manage there at less expense, without the imposing paperweight of Exposés VII and VIII (which will no longer be used in the remainder of the Seminar); it nevertheless seems likely to us that the developments given in these exposés are destined to be useful elsewhere as well.

0.2. When we work with group schemes over a scheme S, it will be understood that we work with the fppf topos of S. This is the topos associated with the $\underline{U}$-site of schemes locally of finite presentation over S (and belonging to $\underline{U}$), endowed with the fppf topology (SGA 3 IV 6.3). The reader will readily verify that most of the statements we shall give would also be valid (and essentially equivalent to the fppf statements) for other topologies on S, whose underlying site contains at least the schemes locally of finite presentation over S, such as the Zariski, étale, and fpqc topologies. One reason is that, by descent theory, the category of extensions of a group scheme P locally of finite presentation over S by the group scheme $\underline{G}_m$ does not depend, up to equivalence, on the chosen topology; the same statement holds for the notion of biextension.

The technical reason for preferring the fppf topology (or a finer topology, such as fpqc) to coarser topologies such as the Zariski and étale topologies is that, for every integer $n > 0$, the homomorphism $n \cdot \mathrm{id}$ from $\underline{G}_m$ to itself is an fppf epimorphism (because it is faithfully flat and of finite presentation), but in general is not a Zariski or an étale epimorphism. Notice that it is an étale epimorphism if and only if n is prime to the residue characteristics of S, and a Zariski epimorphism if and only if S is empty.

0.3. Finally, most of the statements concerning extensions or biextensions by $\underline{G}_m$ would remain valid if that group were replaced by a torus G. Since such a G is locally isomorphic (for the fppf, and even for the étale topology) to a group $\underline{G}_m^r$, this is an essentially trivial generalization.

1. Various special cases over an arbitrary topos

1.1. In this section and the next, as in Exposé VII, T denotes a topos, and P, Q, G denote three abelian groups of T. Thus there is a canonical isomorphism (VII 3.6.5)

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right). \tag{1.1.1}$$

Recall the canonical isomorphisms (called the Cartan isomorphisms)

$$\begin{aligned}\mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) &\simeq \mathrm{Ext}^1(P, R\underline{\mathrm{Hom}}(Q,G)) \\ &\simeq \mathrm{Ext}^1(Q, R\underline{\mathrm{Hom}}(P,G)).\end{aligned} \tag{1.1.2}$$

All these isomorphisms are functorial in the three arguments P, Q, G.

Taking the standard truncation of the derived tensor product, whose homology objects are the $\mathrm{Tor}_i(P,Q)$, one obtains from (1.1.1) the following infinite exact sequence:

$$\begin{aligned}&0 \longrightarrow \mathrm{Ext}^1(P\otimes Q, G) \longrightarrow \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}(\mathrm{Tor}_1(P,Q),G) \\ &\longrightarrow \mathrm{Ext}^2(P\otimes Q, G) \longrightarrow \mathrm{Ext}^2\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) \longrightarrow \cdots.\end{aligned} \tag{1.1.3}$$

On the other hand, since the cohomology objects of $R\underline{\mathrm{Hom}}(Q,G)$ are the $\underline{\mathrm{Ext}}^i(Q,G)$, the first equality in (1.1.2) gives rise to the five-term exact sequence

$$\begin{aligned}&0 \longrightarrow \mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}(P,\underline{\mathrm{Ext}}^1(Q,G)) \\ &\longrightarrow \mathrm{Ext}^2(P,\underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{Ext}^2(P, R\underline{\mathrm{Hom}}(Q,G)),\end{aligned} \tag{1.1.4}$$

associated with the usual spectral sequence

$$\mathrm{Ext}^*(P, R\underline{\mathrm{Hom}}(Q,G)) \Longleftarrow \mathrm{Ext}^p(P, \underline{\mathrm{Ext}}^q(Q,G)).$$

We leave it to the reader, should the need arise, to make the first arrow of (1.1.3), respectively of (1.1.4), explicit by means of the natural fully faithful functor

$$\mathrm{EXT}(P\otimes Q, G) \longrightarrow \mathrm{BIEXT}(P,Q;G), \tag{1.1.5}$$

respectively

$$\mathrm{EXT}(P, \underline{\mathrm{Hom}}(Q,G)) \longrightarrow \mathrm{BIEXT}(P,Q;G), \tag{1.1.6}$$

where EXT and BIEXT denote the evident categories of extensions and biextensions.

Of course, the isomorphism

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1(Q, R\underline{\mathrm{Hom}}(P,G))$$

gives rise to an exact sequence analogous to (1.1.3), with the roles of P and Q interchanged. Its first arrow can be made explicit by means of the fully faithful functor

$$\mathrm{EXT}(Q, \underline{\mathrm{Hom}}(P,G)) \longrightarrow \mathrm{BIEXT}(P,Q;G) \tag{1.1.7}$$

analogous to (1.1.6).

1.2. Suppose that

$$\operatorname{Tor}_1(P,Q)=0. \tag{1.2.1}$$

Then (1.1.3) supplies a canonical isomorphism

$$\operatorname{Biext}^1(P,Q;G)\simeq \operatorname{Ext}^1(P\otimes Q,G). \tag{1.2.2}$$

This holds, for example, if P or Q is a flat $\mathbb{Z}$-module. More precisely, under condition (1.2.1), the functor (1.1.5) is an equivalence of categories.

1.3. Suppose that

$$P\otimes Q=0. \tag{1.3.1}$$

Then (1.1.3) supplies a canonical isomorphism

$$\operatorname{Biext}^1(P,Q;G)\simeq \operatorname{Hom}(\operatorname{Tor}_1(P,Q),G). \tag{1.3.2}$$

The case of greatest interest to us in which (1.3.1) holds is that in which there is an integer $n>0$ such that

$$nP=P,\qquad Q=\varinjlim_i {}_{n^i}Q; \tag{1.3.3}$$

that is, P is divisible by n, and Q is the limit of subgroups annihilated by powers of n. Put, as usual,

$$T_n(P)=({}_{n^i}P)_{i\geq 0}\quad \text{(a projective system)},\qquad {}_{n^\infty}P=\varinjlim_i {}_{n^i}P.$$

From (1.3.3) one obtains the following formulas, more precise than (1.3.1):

$$\begin{aligned} P\otimes Q&=0,\\ \operatorname{Tor}_1(P,Q)&\simeq \varinjlim_i\big({}_{n^i}P\otimes{}_{n^i}Q\big)\simeq T_n(P)\otimes Q\\ &\simeq {}_{n^\infty}P\otimes T_n(Q). \end{aligned} \tag{1.3.4}$$

Thus the isomorphism (1.3.2) here assumes the more concrete form

$$\begin{aligned} \operatorname{Biext}^1(P,Q;G)&\simeq \operatorname{Hom}(T_n(P),D(Q))\\ &\simeq \operatorname{Hom}(Q,D(T_n(P)))\\ &\simeq \operatorname{Hom}(T_n(P)\otimes T_n(Q),T_n(G)). \end{aligned} \tag{1.3.5}$$

Let us specify the meaning of these formulas.

a) In the inductive system occurring in (1.3.4), the transition homomorphism

$${}_{n^i}P\otimes{}_{n^i}Q\longrightarrow {}_{n^j}P\otimes{}_{n^j}Q\qquad (i\leq j)$$

is described by identifying the first member with ${}_{n^j}P\otimes{}_{n^i}Q$ by means of $n^{j-i}\otimes\mathrm{id}$, and then using the homomorphism

$$\mathrm{id}\otimes\mathrm{inj}:{}_{n^j}P\otimes{}_{n^i}Q\longrightarrow{}_{n^j}P\otimes{}_{n^j}Q,$$

where $\mathrm{inj}:{}_{n^i}Q\longrightarrow{}_{n^j}Q$ denotes the inclusion.

b) By definition, put

$$T_n(P)\otimes Q\overset{\text{def}}{=}\varinjlim_i\big(T_n(P)\otimes{}_{n^i}Q\big),$$

where

$$T_n(P)\otimes M\overset{\text{def}}{=}{}_{n^i}P\otimes M\qquad \text{if } n^iM=0.$$

The meaning of the third term written in (1.3.4) is specified in the same way when $nQ=Q$ is also assumed, so that $T_n(Q)$, like $T_n(P)$, is a strict projective system. The last term of (1.3.5) is a Hom between projective systems of groups.

c) The letter D denotes $\underline{\mathrm{Hom}}(-,G)$. The object $D(Q)$ is defined as the projective system of groups

$$D(Q) = \big(D({}_{n^i}Q)\big)_{i\geq 0} = \big(\underline{\mathrm{Hom}}({}_{n^i}Q, G)\big)_{i\geq 0},$$

the second member of (1.3.5) being the group of homomorphisms of projective systems of groups. The following definition gives a simpler interpretation of the next member:

$$D(T_n(P)) = \varinjlim_i D({}_{n^i}P),$$

where the transition morphisms of this inductive system are obtained by transposition from the transition morphisms

$$n^{j-i} : {}_{n^j}P \longrightarrow {}_{n^i}P$$

of $T_n(P)$.

We leave it to the reader to make explicit the calculation of $\mathrm{Tor}_1(P,Q)$ indicated in (1.3.4), and to derive (1.3.5) from it. We note only that one uses the relations, deduced from the second hypothesis in (1.3.3),

$$\mathrm{Tor}_p(P,Q) \simeq \varinjlim_i \mathrm{Tor}_p(P, {}_{n^i}Q).$$

For $p=0$ this immediately gives $P \otimes Q = 0$, and it reduces the calculation of $\mathrm{Tor}_1(P,Q)$ to that of the $\mathrm{Tor}_1(P, {}_{n^i}Q)$ and their transition morphisms. Thus, when Q is annihilated by an integer $m = n^i$ such that $mP = P$, it remains to make explicit an isomorphism

$$\mathrm{Tor}_1(P,Q) \xleftarrow{\sim} {}_mP \otimes {}_mQ \qquad (mP = P,\ mQ = 0). \tag{1.3.6}$$

There are several ways to do this. One consists in using the exact Tor sequence associated with

$$0 \longrightarrow {}_mP \longrightarrow P \xrightarrow{m} P \longrightarrow 0.$$

One may also use the homomorphism (2.1.1) below; the fact that the transition morphisms are those specified in part (a) is then precisely 2.1.11. In any case, the choice of (1.3.6) amounts to the choice of a sign; see 2.1.10.

1.4. Suppose that

$$\underline{\mathrm{Hom}}(Q,G) = 0. \tag{1.4.1}$$

Then (1.1.4) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G) \xrightarrow{\sim} \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q,G)). \tag{1.4.2}$$

This case occurs in particular when Q is an abelian scheme and $G = \underline{G}_m$; see VII 1.3.8. Hence the functor in P,

$$P \longmapsto \mathrm{Biext}^1(P,Q;G),$$

is representable by $Q^* = \underline{\mathrm{Ext}}^1(Q,G)$. In particular, there is a canonical biextension E^Q of (Q^*,Q) by G, determined up to a unique isomorphism, since its automorphism group is

$$\mathrm{Hom}(Q^*, \underline{\mathrm{Hom}}(Q,G)) = 0.$$

This biextension, or its symmetric biextension ${}^sE^Q$ of (Q,Q^*) by G, may be called the Weil biextension of (Q^*,Q), respectively (Q,Q^*), by G.

1.5. Suppose that

$$\mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q,G)) = 0. \tag{1.5.1}$$

This holds in particular if $\underline{\mathrm{Ext}}^1(Q,G)=0$; for example, if Q is a finite locally free group scheme or a group of multiplicative type (SGA 3 IX), and $G=\underline{G}_m$; see 3.4 below. Then (1.1.4) supplies a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;G) \xleftarrow{\sim} \mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)), \tag{1.5.2}$$

where $\underline{\mathrm{Hom}}(Q,G)$ plays the role of a "dual group" of Q (relative to G). More precisely, it follows from (1.5.2) that the fully faithful functor (1.1.6) is an <u>equivalence of categories</u>.

1.6. Suppose that the two relations

$$\underline{\mathrm{Hom}}(P,G)=0, \qquad \mathrm{Hom}(P,\underline{\mathrm{Ext}}^1(Q,G))=0 \tag{1.6.1}$$

hold simultaneously. By 1.4 and 1.5, respectively, these two relations give isomorphisms

$$\mathrm{Biext}^1(P,Q;G) \xrightarrow{\sim} \mathrm{Hom}(Q,\underline{\mathrm{Ext}}^1(P,G)),$$

$$\mathrm{Biext}^1(P,Q;G) \xleftarrow{\sim} \mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)).$$

Their composite is the canonical isomorphism

$$\mathrm{Ext}^1(P,\underline{\mathrm{Hom}}(Q,G)) \xrightarrow{\sim} \mathrm{Hom}(Q,\underline{\mathrm{Ext}}^1(P,G)) \quad \big(\simeq \mathrm{Biext}^1(P,Q;G)\big), \tag{1.6.2}$$

which plays the role of a duality formula.

Notice also that the first relation in (1.6.1) plainly implies $\mathrm{Hom}(P\otimes Q,G)=0$. Hence the category $\mathrm{EXT}(P,\underline{\mathrm{Hom}}(Q,G))$ of commutative extensions of P by $\underline{\mathrm{Hom}}(Q,G)$ is rigid (every automorphism of an object is the identity), as is $\mathrm{BIEXT}(P,Q;G)$. The structure of these categories is therefore known once the group of isomorphism classes of objects is known; by (1.6.2), this group is canonically isomorphic to

$$\mathrm{Hom}(Q,\underline{\mathrm{Ext}}^1(P,G)).$$

The map (1.6.2) is in fact defined in all cases, without assuming (1.6.1), and can also be described directly, without passing through $\mathrm{Biext}^1(P,Q;G)$, in terms of the natural pairing

$$Q\times Q' \longrightarrow G, \qquad Q'=\underline{\mathrm{Hom}}(Q,G).$$

Interpreting it as a homomorphism $Q\longrightarrow \underline{\mathrm{Hom}}(Q',G)$ gives a composite

$$\begin{aligned}\mathrm{Ext}^1(P,Q')\times Q &\longrightarrow \mathrm{Ext}^1(P,Q')\times \mathrm{Hom}(Q',G)\\ &\longrightarrow \mathrm{Ext}^1(P,G),\end{aligned} \tag{1.6.3}$$

which recovers (1.6.2).

For a useful case in which the considerations of 1.6 apply, see 3.7 below.

2. Pairings defined by a biextension

2.1. Let P and Q be two abelian groups of T, and let $n>0$ be an integer. Recall that ${}_nP$ denotes the kernel of multiplication by $n\cdot \mathrm{id}_P$. We shall define a canonical homomorphism

$$\alpha=\alpha^n_{P,Q} : {}_nP\otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.1}$$

which will be the composite

$${}_nP\otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1({}_nP,{}_nQ) \longrightarrow \underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.2}$$

where the first arrow is $\alpha^n_{{}_nP,{}_nQ}$ and the second is induced by the inclusions ${}_nP\longrightarrow P$ and ${}_nQ\longrightarrow Q$. We are thus reduced to defining (2.1.1) when $nP=0$ and $nQ=0$, that is, when P and Q are $\mathbb{Z}/n\mathbb{Z}$-modules.

Suppose only that $nQ = 0$. Taking a flat resolution $L_\bullet$ of P (as a $\mathbb{Z}$-module), one obtains canonical isomorphisms

$$P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} Q \simeq L_\bullet \otimes_{\mathbb{Z}} Q \simeq (L_\bullet \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} Q,$$

which may be written as an isomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} Q \simeq \left(P \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}\right) \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}/n\mathbb{Z}} Q. \tag{2.1.3}$$

This gives rise to the homological Künneth-type spectral sequence (cf. SGA 1 IV 5.2)

$$\underline{\mathrm{Tor}}_*^{\mathbb{Z}}(P, Q) \Longleftarrow E^2_{p,q} = \underline{\mathrm{Tor}}_p^{\mathbb{Z}/n\mathbb{Z}}\big(\underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}), Q\big). \tag{2.1.4}$$

Recall also the canonical isomorphisms

$$\underline{\mathrm{Tor}}_0^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) = P_n := P/nP, \qquad \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) \simeq {}_nP,$$

and

$$\underline{\mathrm{Tor}}_i^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) = 0 \qquad \text{for } i \geq 2.$$

They follow by computing these objects from the canonical flat resolution of $(\mathbb{Z}/n\mathbb{Z})_T$:

$$\begin{tikzcd} 0 \arrow[r] & \mathbb{Z}_T \arrow[r, "n"] \arrow[d, equal] & \mathbb{Z}_T \arrow[r] \arrow[d, equal] & 0 \\ & L_1 & L_0 & \end{tikzcd}$$

whose augmentation $L_0 \longrightarrow (\mathbb{Z}/n\mathbb{Z})_T$ sends 1 to 1. Consequently, the spectral sequence (2.1.4) reduces essentially to the exact sequence

$$\begin{aligned} 0 \longrightarrow \underline{\mathrm{Tor}}_2^{\mathbb{Z}/n\mathbb{Z}}(P_n, Q) \longrightarrow {}_nP \otimes Q \xrightarrow{\alpha} \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q) \\ \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}/n\mathbb{Z}}(P_n, Q) \longrightarrow 0, \end{aligned} \tag{2.1.5}$$

where $P_n = P \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = P/nP$. The desired homomorphism α is the edge homomorphism $E^2_{0,1} \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}}$ in this exact sequence. Since the sequence is plainly functorial in P and Q, functoriality in P shows that the homomorphism in question is indeed the composite

$${}_nP \otimes Q \longrightarrow \underline{\mathrm{Tor}}_1({}_nP, Q) \longrightarrow \underline{\mathrm{Tor}}_1(P, Q),$$

where the first homomorphism is the α just defined, with (P, Q) replaced by $({}_nP, Q)$, and the second is induced by the inclusion ${}_nP \longrightarrow P$.

2.1.6. This therefore defines, for arbitrary P and Q, a homomorphism (2.1.1) satisfying the factorization condition (2.1.2). Moreover, when $nQ = 0$, the exact sequence (2.1.5) shows that α is an isomorphism whenever P_n is flat over $\mathbb{Z}/n\mathbb{Z}$. This holds in particular if $P_n = 0$, that is, if $nP = P$ and P is n-divisible, or if P itself is a flat $\mathbb{Z}/n\mathbb{Z}$-module. It also holds if Q is a flat $\mathbb{Z}/n\mathbb{Z}$-module.

2.1.7. Let us make the calculation of the homomorphism α in (2.1.5) explicit. To determine it, it suffices, for every section q of Q (over an object S of T; take the final object for simplicity), to determine the corresponding homomorphism

$${}_nP \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q).$$

The section q may be regarded as the image of the section 1 of $(\mathbb{Z}/n\mathbb{Z})_T$ under the uniquely determined homomorphism

$$u_q : (\mathbb{Z}/n\mathbb{Z})_T \longrightarrow Q.$$

Functoriality of α in Q therefore reduces us to the case $Q = (\mathbb{Z}/n\mathbb{Z})_T$, with q its unit section. In this case, identify

$$\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q) = \underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z})$$

with ${}_nP$ as explained after (2.1.4). The homomorphism in question is then the identity. Equivalently, the edge homomorphism

$$E^2_{0,q} = \underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \longrightarrow \underline{\mathrm{Tor}}_q^{\mathbb{Z}}(P, \mathbb{Z}/n\mathbb{Z})$$

is the identity. This follows immediately from the definitions. The analogous assertion remains valid when $\mathbb{Z}$ and its quotient $\mathbb{Z}/n\mathbb{Z}$ are replaced by any ring A on T and any quotient ring A/I.

2.1.8. The homomorphism α of (2.1.1) may also be determined by describing, for every pair of sections p of ${}_nP$ and q of ${}_nQ$ (over an object S of T, again taken to be final to simplify the notation), the section of $\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(P, Q)$ determined by (p, q). By functoriality in P and Q, as above, we are reduced to the case

$$P = Q = (\mathbb{Z}/n\mathbb{Z})_T,$$

with p and q the unit section. Thus we need only determine

$$\alpha : (\mathbb{Z}/n\mathbb{Z})_T \otimes (\mathbb{Z}/n\mathbb{Z})_T \longrightarrow \underline{\mathrm{Tor}}_1^{\mathbb{Z}}\big((\mathbb{Z}/n\mathbb{Z})_T, (\mathbb{Z}/n\mathbb{Z})_T\big).$$

The description in 2.1.7 shows that this is the isomorphism

(2.1.8.1) $$(\mathbb{Z}/n\mathbb{Z})_T \xrightarrow{\sim} \underline{\mathrm{Tor}}_1^{\mathbb{Z}}\big((\mathbb{Z}/n\mathbb{Z})_T, (\mathbb{Z}/n\mathbb{Z})_T\big)$$

deduced from the canonical flat resolution

(2.1.8.2) $$0 \longrightarrow \mathbb{Z}_T \xrightarrow{n} \mathbb{Z}_T \longrightarrow 0$$

of the <u>first</u> factor $P = (\mathbb{Z}/n\mathbb{Z})_T$ occurring in $\underline{\mathrm{Tor}}_1$.

2.1.9. The definition given in (2.1.1) is not symmetric. By interchanging the roles of P and Q in the preceding construction, one may define an analogous homomorphism

(2.1.9.1) $$\beta^n_{P,Q} : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P, Q),$$

first reducing to the case $nP = 0$ and then using a flat resolution <u>of</u> Q. Equivalently, β may be defined in terms of α by requiring the following diagram to commute:

(2.1.9.2)
$$\begin{tikzcd} {}_nP \otimes {}_nQ \arrow[r, "\sim"] \arrow[d, "\beta^n_{P,Q}"'] & {}_nQ \otimes {}_nP \arrow[d, "\alpha^n_{Q,P}"] \\ \underline{\mathrm{Tor}}_1(P, Q) \arrow[r, "\sim"] & \underline{\mathrm{Tor}}_1(Q, P). \end{tikzcd}$$

Here the horizontal arrows are the symmetry isomorphisms. With this convention, the relation between α and β is

(2.1.9.3) $$\beta^n_{P,Q} = -\alpha^n_{P,Q}.$$

In other words:

<u>Lemma 2.1.10.</u> <u>Let P and Q be two commutative groups and let $n > 0$ be an integer. Then the following diagram is anticommutative, where α is defined in (2.1.1):</u>

(2.1.10.1)
$$\begin{tikzcd} {}_nP \otimes {}_nQ \arrow[r, "\sim"] \arrow[d, "\alpha^n_{P,Q}"'] \arrow[dr, phantom, "-"] & {}_nQ \otimes {}_nP \arrow[d, "\alpha^n_{Q,P}"] \\ \underline{\mathrm{Tor}}_1(P, Q) \arrow[r, "\sim"] & \underline{\mathrm{Tor}}_1(Q, P). \end{tikzcd}$$

The horizontal arrows are the symmetry isomorphisms.

Proof. The procedure of 2.1.8 immediately reduces us to the case $P = Q = (\mathbb{Z}/n\mathbb{Z})_T$. We must then prove that the two isomorphisms (2.1.8.1), obtained by resolving respectively the first and second arguments of $\underline{\mathrm{Tor}}_1$ by (2.1.8.2), differ by a sign (equivalently, that the symmetry isomorphism on $\underline{\mathrm{Tor}}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ is $-\mathrm{id}$). We may plainly suppose that T is the punctual topos, so that we work with ordinary commutative groups. Let $L_\bullet$ be the complex

$$L_\bullet = \big(0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0\big),$$

regarded as a flat resolution of $P = Q = \mathbb{Z}/n\mathbb{Z}$. Use $L_\bullet \otimes L_\bullet$ to compute

$$\mathrm{Tor}_1(P, Q) = \mathrm{Tor}_1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}).$$

The isomorphisms with $H_1(L_\bullet \otimes \mathbb{Z}/n\mathbb{Z})$, which defines α, and with $H_1(\mathbb{Z}/n\mathbb{Z} \otimes L_\bullet)$, which defines β, are induced respectively by the augmentations $L_\bullet \longrightarrow \mathbb{Z}/n\mathbb{Z}$ of the second and first factors of $L_\bullet \otimes L_\bullet$. It follows at once that $\alpha(1,1)$ is represented by the cycle

$$-1_0 \otimes 1_1 + 1_1 \otimes 1_0 \in (L_\bullet \otimes L_\bullet)_1 = L_0 \otimes L_1 + L_1 \otimes L_0,$$

where 1_0 and 1_1 denote the element 1 of L_0 and L_1 respectively, whereas $\beta(1,1)$ is represented by $1_0 \otimes 1_1 - 1_1 \otimes 1_0$. This proves 2.1.10.

To finish these generalities on the pairings α, we must still specify the relations among the homomorphisms $\alpha^n_{P,Q}$ as n varies. The main result is the following.

Lemma 2.1.11. Let P, Q be two commutative groups of T, and let n, n' be positive integers such that n' is a multiple of n, say $n' = mn$. Then the following diagram is commutative:

(2.1.11.1)
$$\begin{tikzcd}
 & {}_{n'}P \otimes {}_nQ \arrow[dl, "P_m \otimes \mathrm{id}_{{}_nQ}"'] \arrow[dr, "\mathrm{id}_{{}_{n'}P} \otimes i_m"] & \\
{}_nP \otimes {}_nQ \arrow[dr, "\alpha^n_{P,Q}"'] & & {}_{n'}P \otimes {}_{n'}Q \arrow[dl, "\alpha^{n'}_{P,Q}"] \\
 & \underline{\mathrm{Tor}}_1(P,Q) &
\end{tikzcd}$$

Here $P_m : {}_{n'}P \longrightarrow {}_nP$ is the homomorphism induced by $m \cdot \mathrm{id}_P$, and $i_m : {}_nQ \longrightarrow {}_{n'}Q$ is the inclusion. Likewise, the diagram (2.1.11.2) symmetric to (2.1.11.1) is commutative; it has the same vertices except that the upper vertex ${}_{n'}P \otimes {}_nQ$ is replaced by ${}_nP \otimes {}_{n'}Q$.

It suffices to prove the first assertion; the second follows by symmetry in view of 2.1.10. By the procedure of 2.1.8, we are reduced to the punctual topos, where P and Q are ordinary commutative groups and

$$P = \mathbb{Z}/n'\mathbb{Z}, \qquad Q = \mathbb{Z}/n'\mathbb{Z}.$$

In fact, for P we use only the hypothesis $n'P = 0$. Using the resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n'} \mathbb{Z} \longrightarrow 0$$

of Q, identify $\mathrm{Tor}_1(P, Q)$ with ${}_{n'}P = P$. Likewise, using

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow 0$$

to resolve ${}_nQ \simeq \mathbb{Z}/n\mathbb{Z}$, identify $\mathrm{Tor}_1({}_nP, {}_nQ)$ with ${}_nP$. The augmentation in the latter resolution sends 1_0 to $m \cdot 1_Q$, where 1_Q denotes the element 1 of $Q = \mathbb{Z}/n'\mathbb{Z}$. Under these identifications the homomorphism

$$\begin{tikzcd}
\mathrm{Tor}_1({}_nP, {}_nQ) \arrow[r] \arrow[d, dash, "\wr"'] & \mathrm{Tor}_1(P,Q) \arrow[d, dash, "\wr"'] \\
{}_nP & {}_{n'}P = P
\end{tikzcd}$$

induced by the inclusions ${}_nP \hookrightarrow P$ and ${}_nQ \hookrightarrow Q$ is simply the natural inclusion ${}_nP \hookrightarrow {}_{n'}P$.

Consequently, identify the upper vertex of (2.1.11.1) with P/nP using the generator $m \cdot 1_Q$ of ${}_nQ$, which identifies ${}_nQ$ with $\mathbb{Z}/n\mathbb{Z}$. The diagram becomes

$$\begin{tikzcd}
& P/nP \arrow[dl, "x \mapsto mx"'] \arrow[dr, "x \mapsto mx"] & \\
{}_nP \arrow[dr, "x \mapsto x"'] & & P \arrow[dl, "x \mapsto x"] \\
& {}_{n'}P &
\end{tikzcd}$$

which is plainly commutative.

Corollary 2.1.12. With the notation of 2.1.11, the following diagram is commutative:

$$\begin{tikzcd}
{}_{n'}P \otimes {}_{n'}Q \arrow[r, "m \otimes m"] \arrow[d, "\alpha^{n'}_{P,Q}"'] & {}_nP \otimes {}_nQ \arrow[d, "\alpha^n_{P,Q}"] \\
{}_{n'}\underline{\mathrm{Tor}}_1(P,Q) \arrow[r, "m"] & {}_n\underline{\mathrm{Tor}}_1(P,Q).
\end{tikzcd} \tag{2.1.12.1}$$

In the description of the horizontal arrows, m denotes the homomorphisms ${}_{n'}E \longrightarrow {}_nE$ induced by $m \cdot \mathrm{id}_E$, for $E = P, Q$, and $\underline{\mathrm{Tor}}_1(P,Q)$ respectively.

Indeed, by 2.1.11, for $x \in {}_{n'}P(S)$ and $y \in {}_{n'}Q(S)$ one has

$$\alpha^n(mx \otimes my) = \alpha^{n'}(x \otimes my).$$

The right-hand side equals $m\alpha^{n'}(x \otimes y)$, since $x \otimes my = m(x \otimes y)$. This proves the commutativity of (2.1.12.1).

2.1.13. Let ℓ be a prime number. For every commutative group E, denote by $T_\ell(E)$ the inverse system

$$T_\ell(E) = ({}_{\ell^\nu}E)_{\nu \geq 0}, \tag{2.1.13.1}$$

which plainly depends functorially on E. The compatibilities of 2.1.12, for n, n' of the form $\ell^\nu, \ell^{\nu'}$ with $\nu' \geq \nu$, show that the homomorphisms $\alpha^{\ell^\nu}_{P,Q}$, as ν varies, define a homomorphism of inverse systems, functorial in P, Q:

$$\alpha^{(\ell)}_{P,Q} : T_\ell(P) \otimes T_\ell(Q) \longrightarrow T_\ell\big(\underline{\mathrm{Tor}}_1(P,Q)\big). \tag{2.1.13.2}$$

Knowledge of these homomorphisms for every prime ℓ is plainly equivalent to knowledge of the $\alpha^n_{P,Q}$ for every integer n. The antisymmetry relations of 2.1.10 then become the analogous ℓ-adic antisymmetry relations, for every ℓ, expressed by the anticommutativity of the diagrams

$$\begin{tikzcd}
T_\ell(P) \otimes T_\ell(Q) \arrow[r] \arrow[d, "\alpha^{(\ell)}_{P,Q}"'] \arrow[dr, phantom, "-"] & T_\ell(Q) \otimes T_\ell(P) \arrow[d, "\alpha^{(\ell)}_{Q,P}"] \\
T_\ell\big(\underline{\mathrm{Tor}}_1(P,Q)\big) \arrow[r] & T_\ell\big(\underline{\mathrm{Tor}}_1(Q,P)\big).
\end{tikzcd} \tag{2.1.13.3}$$

Here the horizontal arrows are the symmetry arrows.

Remark 2.1.14. The construction of the homomorphism (2.1.1), and the results 2.1.10 and 2.1.11 concerning it, may be generalized to the situation in which one is given a commutative ring $\underline{A}$ on T and two ideals $\underline{a}$ and $\underline{b}$ of $\underline{A}$. In the case treated above, both ideals are the ideal $n\mathbb{Z}_T$ of the constant ring $\mathbb{Z}_T$ on T. One then defines a homomorphism, functorial in the $\underline{A}$-modules P, Q,

$$ {}_{\underline{a}}P \otimes {}_{\underline{b}}Q \xrightarrow{\alpha} \underline{\mathrm{Hom}}_{\underline{A}}\left(\frac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1^{\underline{A}}(P,Q)\right), \tag{2.1.14.1} $$

as the composite of the homomorphisms

$$ \begin{aligned} {}_{\underline{a}}P \otimes {}_{\underline{b}}Q &\longrightarrow \underline{\mathrm{Hom}}(\underline{A}/\underline{a}, P) \otimes \underline{\mathrm{Hom}}(\underline{A}/\underline{b}, Q) \\ &\longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{b}), \underline{\mathrm{Tor}}_1(P,Q)), \end{aligned} $$

using the well-known canonical isomorphism

$$ \underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{b}) \simeq \frac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}. \tag{2.1.14.2} $$

In fact, the sign in 2.1.10 is "hidden" in the choice of this isomorphism, according as one works with the resolution $0 \longrightarrow \underline{a} \longrightarrow \underline{A} \longrightarrow 0$ of $\underline{A}/\underline{a}$, or with the resolution $0 \longrightarrow \underline{b} \longrightarrow \underline{A} \longrightarrow 0$ of $\underline{A}/\underline{b}$. Here it becomes the anticommutativity of the diagram

$$ \begin{tikzcd} {}_{\underline{a}}P \otimes {}_{\underline{b}}Q \arrow[rr] \arrow[d] \arrow[drr, phantom, "-1"] & & {}_{\underline{b}}Q \otimes {}_{\underline{a}}P \arrow[d] \\ \underline{\mathrm{Hom}}\left(\frac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1(P,Q)\right) \arrow[rr] & & \underline{\mathrm{Hom}}\left(\frac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1(Q,P)\right). \end{tikzcd} \tag{2.1.14.3} $$

The horizontal arrows are the symmetry arrows. When $\underline{a} = \underline{b}$, this anticommutativity may also be expressed by saying that the symmetry automorphism on $\underline{\mathrm{Tor}}_1(\underline{A}/\underline{a}, \underline{A}/\underline{a})$ is $-\mathrm{id}$.

In the special case where $\underline{a} = \underline{b} = n\underline{A}$, with n a section of $\underline{A}$, multiplication by n induces an epimorphism

$$ \underline{A}/\underline{a} \longrightarrow \frac{\underline{a}\cap\underline{a}}{\underline{a}\cdot\underline{a}} = \underline{a}/\underline{a}^2, $$

and hence a functorial injective homomorphism, in the $\underline{A}$-module T,

$$ \underline{\mathrm{Hom}}(\underline{a}/\underline{a}^2, T) \longrightarrow {}_{\underline{a}}T. $$

Thus in this case the homomorphism (2.1.14.1) may be interpreted as a homomorphism

$$ {}_nP \otimes {}_nQ \longrightarrow {}_n\underline{\mathrm{Tor}}_1(P,Q), \tag{2.1.14.4} $$

which is closer in form to (2.1.1).

Continue to suppose that $\underline{a} = \underline{b} = n\underline{A}$, and that n is a regular section of $\underline{A}$. The spectral sequence (2.1.4) still makes sense in the present setting and yields the following exact sequence of low-degree terms, generalizing (2.1.5), where Q is now an $\underline{A}$-module annihilated by $\underline{a}$:

$$ \begin{aligned} \underline{\mathrm{Tor}}_2^{\underline{A}}(P,Q) \longrightarrow \underline{\mathrm{Tor}}_2^{\underline{A}/n\underline{A}}(P_n,Q) \longrightarrow {}_nP \otimes Q \xrightarrow{\alpha} \underline{\mathrm{Tor}}_1^{\underline{A}}(P,Q) \\ \longrightarrow \underline{\mathrm{Tor}}_1^{\underline{A}/n\underline{A}}(P_n,Q) \longrightarrow 0. \end{aligned} \tag{2.1.14.5} $$

Return now to arbitrary $\underline{a}$ and $\underline{b}$, and let p and q be two sections of $\underline{A}$. We then obtain the commutative diagram below; (2.1.11.1) is essentially the special case obtained by taking $q = 1$:

$$\begin{tikzcd}[column sep=small]
& {}_{p\underline{a}}P \otimes {}_{q\underline{b}}Q \arrow[dl, "p\otimes q"'] \arrow[dr, "i\otimes j"] & \\
{}_{\underline{a}}P \otimes {}_{\underline{b}}Q \arrow[d, "\alpha"'] & & {}_{pq\underline{a}}P \otimes {}_{pq\underline{b}}Q \arrow[d, "\alpha"] \\
\underline{\mathrm{Hom}}\Big(\dfrac{\underline{a}\cap\underline{b}}{\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1(P,Q)\Big) \arrow[dr, "\text{incl.}"'] & & \underline{\mathrm{Hom}}\Big(\dfrac{pq\underline{a}\cap pq\underline{b}}{pq\underline{a}\cdot pq\underline{b}}, \underline{\mathrm{Tor}}_1(P,Q)\Big) \arrow[dl, "(pq)^*"] \\
& \underline{\mathrm{Hom}}\Big(\dfrac{\underline{a}\cap\underline{b}}{pq\underline{a}\cdot\underline{b}}, \underline{\mathrm{Tor}}_1(P,Q)\Big) &
\end{tikzcd} \tag{2.1.14.6}$$

Here

$$i : {}_{p\underline{a}}P \longrightarrow {}_{pq\underline{a}}P, \qquad j : {}_{q\underline{b}}Q \longrightarrow {}_{pq\underline{b}}Q$$

are the canonical inclusions, and $(pq)^*$ denotes the injective homomorphism obtained, by passing to $\underline{\mathrm{Hom}}$, from the surjective homomorphism

$$\frac{\underline{a}\cap\underline{b}}{pq\underline{a}\cdot\underline{b}} \longrightarrow \frac{pq\underline{a}\cap pq\underline{b}}{pq\underline{a}\cdot pq\underline{b}}$$

induced by multiplication by pq. When $\underline{a} = \underline{b} = n\underline{A}$, this diagram may be rewritten in a simpler form, given in (2.1.11.1) when $q = 1$.

2.2. Consider now, in addition to P and Q, a commutative group G. In (1.1.3) we defined a canonical homomorphism, functorial in P, Q, and G,

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(\mathrm{Tor}_1(P,Q),G\big). \tag{2.2.1}$$

On the other hand, if n is a positive integer (respectively, if ℓ is a prime number), the homomorphism $\alpha^n_{P,Q}$ of (2.1.1) (respectively, the homomorphism $\alpha^{(\ell)}_{P,Q}$ of (2.1.13.2)) gives a homomorphism

$$\mathrm{Hom}\big(\mathrm{Tor}_1(P,Q),G\big) \longrightarrow \mathrm{Hom}({}_nP\otimes{}_nQ,G)$$

(respectively, a homomorphism

$$\mathrm{Hom}\big(\mathrm{Tor}_1(P,Q),G\big) \longrightarrow \mathrm{Hom}\big(T_\ell(P)\otimes T_\ell(Q),T_\ell(G)\big)).$$

These last homomorphisms are also functorial in P, Q, G. Composing them with (2.2.1), we obtain canonical homomorphisms, functorial in P, Q, G,

$$\varphi^n = \varphi^n_{P,Q} : \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}({}_nP\otimes{}_nQ,G), \tag{2.2.2}$$

$$\varphi^{(\ell)} = \varphi^{(\ell)}_{P,Q} : \mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Hom}\big(T_\ell(P)\otimes T_\ell(Q),T_\ell(G)\big). \tag{2.2.3}$$

For

$$\xi \in \mathrm{Biext}^1(P,Q;G), \tag{2.2.4}$$

denote its image under (2.2.2) (respectively, under (2.2.3)) by φ^n_ξ (respectively, by $\varphi^{(\ell)}_\xi$):

$$\varphi^n_\xi : {}_nP\otimes{}_nQ \longrightarrow G \qquad \text{(or, equivalently, } {}_nG\text{)}, \tag{2.2.5}$$

$$\varphi_\xi^{(\ell)} : T_\ell(P) \otimes T_\ell(Q) \longrightarrow T_\ell(G). \tag{2.2.6}$$

2.2.7. Plainly, knowledge of $\varphi_\xi^{(\ell)}$ is equivalent to knowledge of the φ_ξ^n for all $n = \ell^\nu$, with $\nu \geq 0$. For fixed n, knowledge of φ_ξ^n is equivalent to knowledge, on the image of

$$\alpha_{P,Q}^n : {}_nP \otimes {}_nQ \longrightarrow \underline{\mathrm{Tor}}_1(P,Q),$$

of the homomorphism

$$\underline{\mathrm{Tor}}_1(P,Q) \longrightarrow G \tag{2.2.8}$$

which is the image of ξ under (2.2.1).

Observe that if $P = nP$ and $Q = nQ$, then the preceding image is ${}_n\underline{\mathrm{Tor}}_1(P,Q)$. This follows readily by considering the exact sequence (1.3.5) and the analogous exact sequence for Q. Furthermore, if P and Q are divisible, that is, if $nP = P$ and $nQ = Q$ for every positive integer n, one readily checks that $\underline{\mathrm{Tor}}_1(P,Q)$ is torsion, that is,

$$\underline{\mathrm{Tor}}_1(P,Q) = \varinjlim_n {}_n\underline{\mathrm{Tor}}_1(P,Q).$$

Thus, in this case, knowledge of the φ_ξ^n for all n is equivalent to knowledge of (2.2.8), whereas knowledge of $\varphi_\xi^{(\ell)}$ for a given prime ℓ is equivalent to knowledge of the restriction of (2.2.8) to the ℓ-primary component of $\underline{\mathrm{Tor}}_1(P,Q)$.

We now make explicit the symmetry properties of the pairings (2.2.5) and (2.2.6) associated with biextensions. First observe that there is a commutative diagram

$$\begin{tikzcd}
\mathrm{Biext}^1(P,Q;G) \arrow[r, "\sim"] \arrow[d] & \mathrm{Biext}^1(Q,P;G) \arrow[d] \\
\mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(P,Q),G\big) \arrow[r, "\sim"] & \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(Q,P),G\big),
\end{tikzcd} \tag{2.2.9}$$

where the horizontal arrows are the symmetry arrows (the first of which was defined in VII 2.7), and the vertical arrows are (2.2.1). The commutativity follows from that of the two constituent squares in the fuller diagram

$$\begin{tikzcd}
\mathrm{Biext}^1(P,Q;G) \arrow[r, "\sim"] \arrow[d] & \mathrm{Biext}^1(Q,P;G) \arrow[d] \\
\mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) \arrow[r, "\sim"] \arrow[d] & \mathrm{Ext}^1\left(Q \overset{\mathbb{L}}{\otimes} P, G\right) \arrow[d] \\
\mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(P,Q),G\big) \arrow[r, "\sim"] & \mathrm{Hom}\big(\underline{\mathrm{Tor}}_1(Q,P),G\big).
\end{tikzcd} \tag{2.2.10}$$

The first vertical arrow in each column is the canonical isomorphism of VII 3.6.5, and the second is the evident homomorphism. The second square in (2.2.10) is trivially commutative: by definition, the symmetry isomorphism

$$P \overset{\mathbb{L}}{\otimes} Q \xrightarrow{\sim} Q \overset{\mathbb{L}}{\otimes} P$$

induces the symmetry isomorphism $\underline{\mathrm{Tor}}_1(P,Q) \simeq \underline{\mathrm{Tor}}_1(Q,P)$, and, for a variable complex $L_\bullet$, the canonical isomorphism

$$\mathrm{Ext}^1(L_\bullet, G) \simeq \mathrm{Hom}\big(H_1(L_\bullet), G\big)$$

is functorial in $L_\bullet$.

As for the commutativity of the first square of (2.2.10), it follows by making explicit simultaneously the symmetry isomorphisms and the vertical isomorphisms defined in VII 3.6.5. We leave the details of the verification to the reader. One uses the fact that the symmetry isomorphism

$$L_\bullet(P) \otimes L_\bullet(Q) \simeq L_\bullet(Q) \otimes L_\bullet(P)$$

induces in degree 1 the isomorphism

$$L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \simeq L_0(Q) \otimes L_1(P) + L_1(Q) \otimes L_0(P),$$

which, on corresponding summands, is given by the ordinary symmetry isomorphism of the underlying ungraded modules (and hence introduces no sign).

In view of the commutativity of (2.2.9), the anticommutativity of (2.1.10.1), in terms of the φ^n or $\varphi^{(\ell)}$, becomes the following relation.

Proposition 2.2.11. The following diagrams are anticommutative:

(2.2.11.1)
$$\begin{tikzcd}
\mathrm{Biext}^1(P,Q;G) \arrow[rr, "\sim"] \arrow[d, "\varphi^n_{P,Q}"'] \arrow[drr, phantom, "-"] & & \mathrm{Biext}^1(Q,P;G) \arrow[d, "\varphi^n_{Q,P}"] \\
\mathrm{Hom}({}_nP \otimes {}_nQ, G) \arrow[rr, "\sim"] & & \mathrm{Hom}({}_nQ \otimes {}_nP, G),
\end{tikzcd}$$

(2.2.11.2)
$$\begin{tikzcd}
\mathrm{Biext}^1(P,Q;G) \arrow[rr, "\sim"] \arrow[d, "\varphi^{(\ell)}_{P,Q}"'] \arrow[drr, phantom, "-"] & & \mathrm{Biext}^1(Q,P;G) \arrow[d, "\varphi^{(\ell)}_{Q,P}"] \\
\mathrm{Hom}\big(T_\ell(P) \otimes T_\ell(Q), T_\ell(G)\big) \arrow[rr, "\sim"] & & \mathrm{Hom}\big(T_\ell(Q) \otimes T_\ell(P), T_\ell(G)\big),
\end{tikzcd}$$

where the horizontal arrows are the symmetry arrows.

We record the important corollary.

Corollary 2.2.12. Suppose that $P = Q$, and suppose that ξ is the class of a symmetric (respectively, antisymmetric) biextension of (P,P) by G; that is, suppose ${}^s\xi = \xi$ (respectively, ${}^s\xi = -\xi$), where ${}^s\xi$ denotes the image of ξ under symmetry. Then the bilinear maps φ^n_ξ, on ${}_nP \times {}_nP$, and $\varphi^{(\ell)}_\xi$, on $T_\ell(P) \times T_\ell(P)$, are antisymmetric (respectively, symmetric).

Remark 2.2.13. Suppose that ξ is symmetric. If n (respectively, ℓ) is odd, then 2.2.12 shows that φ^n_ξ (respectively, $\varphi^{(\ell)}_\xi$) is in fact an alternating form. This conclusion is not valid in general without the restriction that n (respectively, ℓ) be odd, as is already seen when P is the ordinary group $\mathbb{Z}/2\mathbb{Z}$ and $G = \mathbb{Z}/2\mathbb{Z}$.

Suppose, however, that $2P = P$. This is the case in particular when P is an abelian scheme, working as usual in the fppf or fpqc topology. The preceding forms are then still alternating. It is enough to prove that $\varphi^{2^\nu}_\xi(x,x) = 0$. Since $2P = P$, a given element $x \in {}_{2^\nu}P(S)$ may be written as $x = 2y$, with $y \in {}_{2^{\nu+1}}P(S)$. Hence

$$\varphi^{2^\nu}_\xi(x,x) = 2\varphi^{2^{\nu+1}}_\xi(y,y) = 0,$$

because $\varphi_{\xi}^{2^{\nu+1}}$ is antisymmetric.

2.3. <u>Relation with "Cartier duality"</u>. The description given in 2.2 of the pairings defined by a biextension, through the isomorphism theorem VII 3.6.5 and the homomorphism (2.2.1), is plainly not the definition formerly used for abelian schemes, at a time when the very notion of biextension did not yet exist. That definition proceeds through Cartier duality, and we now give its relation to the present one.

Consider again the homomorphism

$$\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q, G)\big) \tag{2.3.1}$$

arising from the exact sequence (1.4). It is in fact an isomorphism when

$$\underline{\mathrm{Hom}}(Q, G) = 0. \tag{2.3.2}$$

This condition holds in particular when Q is an abelian scheme over a scheme S and G is the multiplicative group $\underline{G}_m$ (VII 1.3.8).

There is also an evident "restriction to nQ" homomorphism

$$\underline{\mathrm{Ext}}^1(Q, G) \longrightarrow \underline{\mathrm{Ext}}^1(nQ, G),$$

and hence a composite homomorphism

$$\begin{aligned} \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(Q, G)\big) &\longrightarrow \mathrm{Hom}\big(P, \underline{\mathrm{Ext}}^1(nQ, G)\big) \\ &\longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ, G)\big), \end{aligned} \tag{2.3.4}$$

where the second arrow is restriction to ${}_nP$. Composing with (2.3.1), we obtain a homomorphism

$$\mathrm{Biext}^1(P, Q; G) \longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ, G)\big). \tag{2.3.5}$$

On the other hand, the exact sequence

$$0 \longrightarrow {}_nQ \longrightarrow Q \xrightarrow{\ n\ } nQ \longrightarrow 0 \tag{2.3.6}$$

gives rise to a long exact sequence of the $\underline{\mathrm{Ext}}^i(-, G)$, and hence to the short exact sequence

$$0 \longrightarrow \underline{\mathrm{Hom}}(Q, G)_n \longrightarrow \underline{\mathrm{Hom}}({}_nQ, G) \xrightarrow{\ \partial\ } {}_n\underline{\mathrm{Ext}}^1(nQ, G) \longrightarrow 0, \tag{2.3.7}$$

where, as usual, E_n denotes E/nE. We retain from this sequence the canonical surjective homomorphism

$$\underline{\mathrm{Hom}}({}_nQ, G) \xrightarrow{\qquad \partial \qquad} {}_n\underline{\mathrm{Ext}}^1(nQ, G), \tag{2.3.8}$$

which is an isomorphism if $\underline{\mathrm{Hom}}(Q, G)_n = 0$, and a fortiori if (2.3.2) holds. From (2.3.8) we deduce a homomorphism

$$\begin{aligned} \mathrm{Hom}\big({}_nP, \underline{\mathrm{Hom}}({}_nQ, G)\big) &\simeq \mathrm{Hom}({}_nP \otimes {}_nQ, G) \\ &\longrightarrow \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ, G)\big), \end{aligned} \tag{2.3.9}$$

which is an isomorphism when (2.3.2) holds.

All the homomorphisms under consideration are functorial in P, Q, G. The homomorphisms (2.3.5), (2.3.9), and (2.2.2) form the diagram

$$\begin{tikzcd} & \mathrm{Biext}^1(P, Q; G) \arrow[dl, "\varphi^n"'] \arrow[dr, "(2.3.5)"] \arrow[d, phantom, "-"] & \\ \mathrm{Hom}({}_nP \otimes {}_nQ, G) \arrow[rr, "(2.3.9)"] & & \mathrm{Hom}\big({}_nP, {}_n\underline{\mathrm{Ext}}^1(nQ, G)\big), \end{tikzcd} \tag{2.3.10}$$

where the horizontal arrow (2.3.9) is an isomorphism if (2.3.2) holds.

Proposition 2.3.11. The diagram (2.3.10) is anticommutative.

When $\underline{\mathrm{Hom}}(Q,G)=0$, this statement therefore recovers a characterization of the homomorphism φ^n of (2.2.2) in terms of the homomorphisms (2.3.5) and (2.3.9). This is the characterization formerly used in the theory of abelian schemes, where moreover $nQ=Q$.

Proof of 2.3.11.[2] Let ξ be an element of $\mathrm{Biext}^1(P,Q;G)$, let

$$\varphi_\xi : {}_nP\otimes {}_nQ \longrightarrow G$$

be the associated pairing, and let

$$u_\xi : {}_nP \longrightarrow {}_n\underline{\mathrm{Ext}}^1(nQ,G)$$

be the image of ξ under (2.3.5). We must prove that $-u_\xi$ is obtained from φ_ξ by the arrow (2.3.9), that is, that it is the composite

$$(*)\qquad {}_nP \xrightarrow{\varphi_\xi} \underline{\mathrm{Hom}}({}_nQ,G) \xrightarrow{(2.3.8)} {}_n\underline{\mathrm{Ext}}^1(nQ,G).$$

To prove this, we may take a section x of ${}_nP$ and prove that its image under $-u_\xi$ equals its image under the composite (*). If desired, this would allow us to reduce to the case $P=(\mathbb{Z}/n\mathbb{Z})_T$, but we shall not need that reduction.

a) Choose flat resolutions $L_\bullet$ and $M_\bullet$,

$$0\longrightarrow L_1\longrightarrow L_0\longrightarrow 0,\qquad 0\longrightarrow M_1\longrightarrow M_0\longrightarrow 0,$$

of P and Q, respectively. A flat resolution of P, for example, is obtained by writing P as a quotient of a flat $\mathbb{Z}_T$-module L_0 and taking L_1 to be the kernel of $L_0\longrightarrow P$; this kernel is flat because it is torsion-free. We obtain the complex $L_\bullet\otimes M_\bullet$

$$0\longrightarrow L_1\otimes M_1\xrightarrow{d_2} L_1\otimes M_0+L_0\otimes M_1\xrightarrow{d_1} L_0\otimes M_0\longrightarrow 0,$$

whose sheaf $\underline{H}_1$ will serve as $\underline{\mathrm{Tor}}_1(P,Q)$.

Let L_0' (respectively, M_0') be the subsheaf of L_0 (respectively, M_0) which is the inverse image of ${}_nP$ (respectively, ${}_nQ$). Thus

$$0\longrightarrow L_1\longrightarrow L_0'\longrightarrow 0\qquad \bigl(\text{respectively, } 0\longrightarrow M_1\longrightarrow M_0'\longrightarrow 0\bigr)$$

is a flat resolution of ${}_nP$ (respectively, of ${}_nQ$). The homomorphism

$${}_nP\otimes {}_nQ\longrightarrow \underline{\mathrm{Tor}}_1(P,Q)=\underline{H}_1(L_\bullet\otimes M_\bullet)$$

may then be described, by passage to the quotient, from the homomorphism

$$L_0'\otimes M_0'\longrightarrow \underline{H}_1(L_\bullet\otimes M_\bullet)$$

given by

$$x_0\otimes y_0\longmapsto \text{the class of the 1-cycle } [nx_0]\otimes y_0-x_0\otimes[ny_0].$$

Here the brackets indicate that the element in question is regarded as a section of the subsheaf L_1 of L_0 (respectively, M_1 of M_0). This description follows from the calculation made in the proof of 2.1.10.

b) Take an injective resolution

$$C^\bullet=\bigl(0\longrightarrow C^0\xrightarrow{d^0} C^1\longrightarrow\cdots\bigr)$$

2. For a more elegant proof, see 2.3.19 below.

of G, and identify

$$\mathrm{Biext}^1(P,Q;G) \simeq \mathrm{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right) = H^1(\mathrm{Hom}^{\cdot}(L_\bullet \otimes M_\bullet, C^{\cdot}))\,.$$

The element ξ may therefore be represented by a pair of homomorphisms f_1, f_0, giving the anti-commutative diagram

(2.3.12)
$$\begin{tikzcd} L_1 \otimes M_0 + L_0 \otimes M_1 \arrow[r, "d_1"] \arrow[d, "f_1"'] \arrow[dr, phantom, "-"] & L_0 \otimes M_0 \arrow[d, "f_0"] \\ C^0 \arrow[r, "d^0"] & C^1. \end{tikzcd}$$

By definition, the pairing

$$\varphi_\xi : {}_nP \otimes {}_nQ \longrightarrow G$$

is obtained, by passage to the quotient, from the homomorphism

$$\varphi : L_0' \otimes M_0' \longrightarrow G$$

given by

(2.3.13)
$$\varphi(x_0 \otimes y_0) = f_1\big([nx_0] \otimes y_0 - x_0 \otimes [ny_0]\big) \in G = \ker(C^0 \longrightarrow C^1).$$

Since assertion 2.3.11 is local on T, we may suppose that the given section x of ${}_nP$ lifts to a section x_0 of L_0'. Regarding y_0 as variable, the preceding formula then defines a homomorphism

(2.3.14)
$$\varphi' : M_0' \longrightarrow G, \qquad y_0 \longmapsto f_1\big([nx_0] \otimes y_0 - x_0 \otimes [ny_0]\big),$$

which passes to the quotient to give

$${}_nQ \longrightarrow G, \qquad y \longmapsto \varphi_\xi(x,y),$$

the homomorphism associated with φ_ξ and x.

c) We must take the image of this homomorphism under the connecting operator ∂ of (2.3.7), associated with the exact sequence (2.3.6). Now $nQ \simeq Q/{}_nQ$ is plainly also isomorphic to M_0/M_0', so that we have a homomorphism of exact sequences

$$\begin{tikzcd} 0 \arrow[r] & M_0' \arrow[r] \arrow[d] & M_0 \arrow[r] \arrow[d] & M_0/M_0' \arrow[r] \arrow[d, "\wr"'] & 0 \\ 0 \arrow[r] & {}_nQ \arrow[r] & Q \arrow[r] & nQ \arrow[r] & 0 \end{tikzcd} \quad ,$$

where the last vertical arrow is an isomorphism. This shows that the desired section of $\underline{\mathrm{Ext}}^1(nQ,G)$ may also be computed as the image of φ' in (2.3.14) under the connecting operator

$$\underline{\mathrm{Hom}}(M_0', G) \xrightarrow{\partial} \underline{\mathrm{Ext}}^1(M_0/M_0', G),$$

or, equivalently, as the section of the second term defined by the element of global Ext^1 which is the image of φ' under the connecting operator

$$\mathrm{Hom}(M_0', G) \longrightarrow \mathrm{Ext}^1(M_0/M_0', G).$$

The usual construction of the connecting operator gives the following recipe. Extend the composite (2.3.13)

$$M'_0 \xrightarrow{\varphi'} G \longrightarrow C^0$$

to a homomorphism

$$\psi : M_0 \longrightarrow C^0,$$

which is possible since C^0 is injective, and consider the unique homomorphism

$$\lambda : M_0/M'_0 \longrightarrow C^1 \tag{2.3.15}$$

which makes the diagram

$$\begin{tikzcd}
0 \arrow[r] & M'_0 \arrow[r] \arrow[d, "\varphi'"'] & M_0 \arrow[r] \arrow[d, "\psi"'] & M_0/M'_0 \arrow[r] \arrow[d, "\lambda"'] & 0 \\
0 \arrow[r] & G \arrow[r] & C^0 \arrow[r] & C^1 &
\end{tikzcd}$$

commutative. It is clear that (2.3.15) takes values in the 1-cocycles—that is, it is a 1-cocycle—and that this cocycle represents the desired element of

$$\mathrm{Ext}^1(nQ, G) = \mathrm{Ext}^1(M_0/M'_0, G).$$

d) It remains to describe explicitly the section of $\underline{\mathrm{Ext}}^1(nQ, G)$ obtained from ξ and x by means of arrow (2.3.5) in diagram (2.3.10). For this purpose we may disregard the fact that the section x of P is annihilated by n. In terms of the chosen lift x_0 of x to a section of L_0, the description is as follows. This lift defines an augmentation homomorphism

$$u : M_\bullet \longrightarrow L_\bullet \otimes M_\bullet, \qquad y_0 \longmapsto x_0 \otimes y_0, \quad y_1 \longmapsto x_0 \otimes y_1.$$

The section of $\underline{\mathrm{Ext}}^1(Q, G)$ obtained from (2.3.1) is therefore the one obtained from the element ξ of $\mathrm{Ext}^1(L_\bullet \otimes M_\bullet, G)$ by composition with the preceding homomorphism. Thus it is represented by the degree-one homomorphism

$$\begin{tikzcd}
0 \arrow[r] & M_1 \arrow[r, "d_1"] \arrow[d, "y_1 \mapsto f_1(x_0 \otimes y_1)"'] \arrow[dr, phantom, "-"] & M_0 \arrow[r] \arrow[d, "y_0 \mapsto f_0(x_0 \otimes y_0)"] & 0 \\
0 \arrow[r] & C^0 \arrow[r, "d^0"] & C^1 & \cdots
\end{tikzcd}$$

To restrict this class to $nQ \hookrightarrow Q$, use the isomorphism $nQ \simeq M_0/M'_0$, which gives a flat resolution

$$0 \longrightarrow M'_0 \longrightarrow M_0 \longrightarrow 0$$

of nQ. The inclusion $nQ \hookrightarrow Q$ lifts to the following homomorphism of resolutions:

$$\begin{tikzcd}
0 \arrow[r] & M'_0 \arrow[r] \arrow[d, "y'_0 \mapsto {[ny'_0]}"'] \arrow[dr, phantom, "+"] & M_0 \arrow[r] \arrow[d, "y_0 \mapsto ny_0"] & 0 \\
0 \arrow[r] & M_1 \arrow[r] & M_0 \arrow[r] & 0
\end{tikzcd}$$

By composition, the desired section of $\underline{\mathrm{Ext}}^1(nQ, G)$ is therefore associated with the element of $\mathrm{Ext}^1(nQ, G)$ represented by the degree-one homomorphism of complexes

$$\begin{tikzcd}
0 \arrow[r, dash] & M_0' \arrow[r, dash, "d_1"] \arrow[d, "g_1"'] & M_0 \arrow[r, dash] \arrow[d, "g_0"] & 0 \\
0 \arrow[r, dash] & C^0 \arrow[r, dash] & C^1 \arrow[r, dash] & \cdots
\end{tikzcd} ,$$

where

$$g_1(y_0') = f_1\big(x_0 \otimes [ny_0']\big), \qquad g_0(y_0) = nf_0(x_0 \otimes y_0).$$

e) Using now the fact that x is a section of ${}_nP$, we show that the preceding element of $\mathrm{Ext}^1(nQ, G)$ is the negative of the element defined by (2.3.15). For any homomorphism

$$h : M_0 \longrightarrow C^0,$$

the element of Ext^1 considered in d) is unchanged if g_1, g_0 are replaced, respectively, by

$$g_1' = g_1 + hd_1, \qquad g_0' = g_0 - d^0 h.$$

We have at our disposal the homomorphism $\psi : M_0 \longrightarrow C^0$ constructed in c), which by definition satisfies

$$\psi(y_0') = f_1\big([nx_0] \otimes y_0' - x_0 \otimes [ny_0']\big)$$

for every section y_0' of M_0'. Equivalently,

$$f_1\big(x_0 \otimes [ny_0']\big)\big(= g_1(y_0')\big) = f_1\big([nx_0] \otimes y_0'\big) - \psi(y_0').$$

Define $h : M_0 \longrightarrow C^0$ by

$$h(y_0) = -f_1\big([nx_0] \otimes y_0\big) + \psi(y_0).$$

By construction, then,

$$g_1' = 0, \qquad g_0'(y_0) = nf_0(x_0 \otimes y_0) + d^0 f_1\big([nx_0] \otimes y_0\big) - d^0\psi(y_0).$$

Since $d^0 f_1 = -f_0 d_1$, we have

$$d^0 f_1\big([nx_0] \otimes y_0\big) = -f_0(nx_0 \otimes y_0),$$

and hence

$$g_1' = 0, \qquad g_0' = -d^0\psi.$$

Thus the element in question is the one defined by the homomorphism

$$M_0/M_0' \simeq nQ \longrightarrow C^1$$

induced by $-d^0\psi$ upon passage to the quotient. We therefore obtain the negative of the element defined by (2.3.15).

<u>Remarks 2.3.16.</u> The preceding proof actually gives a slightly more precise compatibility than the one stated in 2.3.11, obtained by choosing a section x of ${}_nP$. This section then defines a homomorphism

$${}_nQ \longrightarrow G, \qquad y \longmapsto \varphi_\xi(x, y),$$

and hence, by the connecting homomorphism associated with (2.3.6), an element of $\mathrm{Ext}^1(nQ, G)$. On the other hand, if we forget that x is annihilated by n, we consider the homomorphism $\mathbb{Z}_T \longrightarrow P$ defined by x. Its inverse image is a biextension of $(\mathbb{Z}_T, Q)$ by G, whose class may be interpreted as an element of $\operatorname{Ext}^1(Q, G)$ and therefore induces an element of $\operatorname{Ext}^1(nQ, G)$. With these conventions, the two elements obtained in $\operatorname{Ext}^1(nQ, G)$ are opposite.

2.3.17. Let us use the explicit cocycle description of the pairing

$$\varphi_\xi^n : {}_nP \otimes {}_nQ \longrightarrow G$$

given in (2.3.12) to obtain another expression for it. First suppose that the biextension E of class ξ admits a section s over $P \times Q$. As in VII 3.8.13, take for the flat resolution of P the one with $L_0 = \mathbb{Z}[P]$ and L_1 the kernel of the canonical homomorphism $\mathbb{Z}[P] \longrightarrow P$, and define $M_\bullet$ similarly. There is then a unique biextension trivialization of the inverse image of E on (L_0, M_0) whose restriction to $P \times Q$ is given by s; denote this trivialization again by s (cf. VII 3.8.2). By VII 3.8.12, the class of ξ in

$$\operatorname{Ext}^1\left(P \overset{\mathbb{L}}{\otimes} Q, G\right)$$

is represented by the cocycle with components (f, g) characterized by

$$s(x_1, y_0) = f(x_1 \otimes y_0)e_{E/Q}(y_0), \qquad s(x_0, y_1) = g(x_0 \otimes y_1)e_{E/P}(x_0).$$

If x and y are sections of ${}_nP$ and ${}_nQ$, lift them to the corresponding sections $x_0 = [x]$ and $y_0 = [y]$ of L_0 and M_0. Applying formula (2.3.13), and putting $z = s(x, y)$, gives

$$\varphi_\xi^n(x, y) = {}^nz/e_{E/Q}(y) - z^n/e_{E/P}(x). \tag{2.3.18}$$

Here the exponent n written on the left, respectively on the right, of z denotes the nth power for the left, respectively right, multiplicative structure on E. It therefore gives a section of E over $(0, y)$, respectively $(x, 0)$; the meaning of the "quotient" of such an element by $e_{E/Q}(y)$, respectively $e_{E/P}(x)$, is clear.

The preceding formula (2.3.18) still makes sense whenever z is a global section of E lifting the sections x, y of ${}_nP$, ${}_nQ$. I claim that (2.3.18) is valid for every such section, without assuming that a section of E over $P \times Q$ exists. The section (x, y) of $P \times Q$ always lifts locally to a section of E, since $E \longrightarrow P \times Q$ is an epimorphism; thus the formula computes $\varphi_\xi^n(x, y)$ in every case. Moreover, if z is replaced by another section z' of E lifting (x, y), then $z' = z + g$ for a section g of G. The two terms on the right-hand side of (2.3.18) corresponding to z' are the terms corresponding to z, each increased by ng. Their difference is therefore independent of z, showing that (2.3.18) defines a well-determined homomorphism

$${}_nP \otimes {}_nQ \longrightarrow G.$$

It remains to show that this homomorphism is indeed φ_ξ^n. For this we reduce at once to the case

$$P = Q = (\mathbb{Z}/n\mathbb{Z})_T.$$

Then $P \times Q$ is of the form I_T, where $I = \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ is a finite set. In this case the torsor E under $G_{P\times Q}$—which identifies with a family of torsors E_i under G, for $i \in I$—plainly becomes trivial locally on T. We are thus reduced to the already treated case in which E has a global section over $P \times Q$.

2.3.19. Another proof of 2.3.11 (by Deligne).

a) Reduction to $P = \mathbb{Z}/n\mathbb{Z}$ and to $T = (\text{Sets})$.

b) Let ∂ be the following arrow from nQ to ${}_nQ[1]$:

$$nQ \underset{\text{“}n\text{”}}{\overset{\sim}{\longleftarrow}} [\,{}_nQ \longrightarrow Q\,] \longrightarrow {}_nQ[1].$$

The following diagram is anticommutative:

$$\begin{tikzcd}
nQ[0] \arrow[r, "1"] \arrow[d, "\partial"'] & {[Q \xrightarrow{n} Q]} \\
{}_nQ[1] \arrow[ur, "1"'] &
\end{tikzcd}.$$

Indeed, the arrows

$$\begin{tikzcd}
{[{}_nQ} \arrow[r] \arrow[d, "0"'] & {Q]} \arrow[d, "n"] \\
{[Q} \arrow[r, "n"] & {Q]}
\end{tikzcd}
\quad \text{and} \quad
\begin{tikzcd}
{[{}_nQ} \arrow[r] \arrow[d, "1"'] & {Q]} \arrow[d, "0"] \\
{[Q} \arrow[r, "n"] & {Q]}
\end{tikzcd}
\quad \text{are opposite by means of} \quad H \ : \quad
\begin{tikzcd}
{[{}_nQ} \arrow[r] & {Q]} \arrow[dl, "1"] \\
{[Q} \arrow[r, "n"] & {Q]}
\end{tikzcd}$$

c) Diagram 2.3.10 is

$$\begin{tikzcd}
 & & H^1\mathrm{RHom}(\mathbb{Z} \overset{\mathbb{L}}{\otimes} Q, G) \arrow[dd] \\
H^1\mathrm{RHom}(\mathbb{Z}/n\mathbb{Z} \overset{\mathbb{L}}{\otimes} Q, G) \arrow[r] \arrow[urr] \arrow[dd] & H^1\mathrm{RHom}(\mathbb{Z}/n\mathbb{Z}, \mathrm{RHom}(Q,G)) \arrow[dd] & \\
 & & \mathrm{Ext}^1(Q,G) \arrow[ddl] \\
\mathrm{Hom}(\mathrm{Tor}_1(\mathbb{Z}/n\mathbb{Z}, Q), G) \arrow[d] \arrow[r, phantom, "-?"] & \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathrm{Ext}^1(Q,G)) \arrow[d] & \\
\mathrm{Hom}({}_nQ, G) \arrow[r] & \mathrm{Ext}^1(nQ, G) &
\end{tikzcd}$$

and:

c1) We have an isomorphism

$$\mathbb{Z}/n\mathbb{Z} \overset{\mathbb{L}}{\otimes} Q \simeq [Q \xrightarrow{n} Q],$$

and, under this isomorphism, the first composite vertical arrow is induced by

$${}_nQ[1] \longrightarrow [Q \xrightarrow{n} Q].$$

c2) The second horizontal arrow is the transpose of the arrow ∂ displayed above in b).

c3) If the diagram is traversed clockwise, it is clear that one obtains the arrow induced by

$$nQ[0] \longrightarrow [Q \xrightarrow{n} Q].$$

This proves 2.3.11.

2.4. Let

$$f : \underline{T}' \longrightarrow \underline{T}$$

be a morphism of topoi. There is then a canonical homomorphism (VII 2.8)

$$\mathrm{Biext}^1(P,Q;G) \longrightarrow \mathrm{Biext}^1(P',Q';G'),$$

where primes denote inverse-image groups. Since, for every commutative group E of $\underline{T}$, there is plainly a canonical isomorphism

$$({}_nE)' \simeq {}_n(E'),$$

the homomorphisms (2.2.2) over $\underline{T}$ and $\underline{T}'$ give a canonical diagram

$$\begin{tikzcd} \mathrm{Biext}^1(P,Q;G) \arrow[r] \arrow[d, "\varphi^n_{P,Q}"'] & \mathrm{Biext}^1(P',Q';G') \arrow[d, "\varphi^n_{P',Q'}"] \\ \mathrm{Hom}({}_nP \otimes {}_nQ, G) \arrow[r] & \mathrm{Hom}({}_nP' \otimes {}_nQ', G'). \end{tikzcd} \tag{2.4.1}$$

Similarly, the homomorphisms (2.2.3) give an analogous diagram in terms of the T_ℓ. I claim that these diagrams are commutative. The verification is essentially immediate from the definitions and is left to the reader. More generally, notice that all the canonical homomorphisms considered in this section and the preceding one are compatible with change of topos, in the evident sense. We shall henceforth use compatibilities of this kind without further mention.

3. Biextensions of group schemes (P,Q) by $\underline{G}_m$: generalities

Throughout this section we fix a base scheme S and, as announced in Section 0, work in the fppf topos of S.

3.1. Let us first recall the most important example of a group $\mathrm{Biext}^1(P,Q;\underline{G}_m)$: the case where P and Q are two abelian schemes over S, and where Biext^1 is interpreted as the group of divisorial correspondences on (P,Q) (VII 2.9.4). The ℓ-adic pairings of the preceding section are then the familiar pairings [1] in the theory of abelian schemes.

3.2. More generally, suppose only that Q is an abelian scheme. It is well known (VII 1.3.8) that

$$\underline{\mathrm{Hom}}(Q,\underline{G}_m) = 0, \tag{3.2.1}$$

and hence (1.4) there is a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;\underline{G}_m) \simeq \mathrm{Hom}(P,Q^*), \tag{3.2.2}$$

where

$$Q^* = \underline{\mathrm{Ext}}^1(Q,\underline{G}_m) \tag{3.2.3}$$

is the dual abelian scheme of Q. (For the representability of Q^*, due to M. Raynaud in the general case, see [1].)

When P is also an abelian scheme, symmetrically to (3.2.2), there is a canonical isomorphism

$$\mathrm{Biext}^1(P,Q;\underline{G}_m) \simeq \mathrm{Hom}(Q,P^*). \tag{3.2.4}$$

Thus the datum of a biextension ξ of (P,Q) by $\underline{G}_m$ may be interpreted equivalently as the datum of a homomorphism

$$u : P \longrightarrow Q^*$$

or of a homomorphism

$$v : Q \longrightarrow P^*.$$

Recall that a homomorphism of abelian schemes $w : A \longrightarrow B$ is an isomorphism if and only if, for every integer $n > 0$, the induced homomorphism ${}_nw : {}_nA \longrightarrow {}_nB$ is an isomorphism. In the case of u and v above, ${}_nu : {}_nP \longrightarrow {}_nQ^*$ is identified with the homomorphism ${}_nP \longrightarrow D({}_nQ)$ defined by the pairing

$$\varphi_\xi^n : {}_nP \times {}_nQ \longrightarrow \underline{G}_m$$

(see 2.3.11), whereas ${}_nv : {}_nQ \longrightarrow {}_nP^*$ is identified with the homomorphism ${}_nQ \longrightarrow D({}_nP)$ defined by the pairing

$$\psi_\xi^n : {}_nP \times {}_nQ \longrightarrow \underline{G}_m.$$

Now $\varphi_\xi^n = -\psi_\xi^n$ (2.1.10), so, up to sign, ${}_nu$ and ${}_nv$ are transposes of one another. Since ${}_nP$ and ${}_nQ$ are finite and locally free over S, and therefore satisfy Cartier duality, it follows that ${}_nu$ is an isomorphism if and only if ${}_nv$ is an isomorphism. Consequently u is an isomorphism if and only if v is an isomorphism.

An equivalent, though slightly less symmetric, formulation is the following. Consider the canonical biextension (the Weil biextension) of (P, P^*) by $\underline{G}_m$, corresponding to the identity homomorphism $v : P^* \longrightarrow P^*$. Then the corresponding canonical homomorphism

$$u : P \longrightarrow P^{**}$$

is an isomorphism. This is the biduality theorem of Barsotti and Cartier.

3.3. We now turn to the case where Q is finite locally free over S, or is a group scheme of multiplicative type over S. The key lemma in this case is the following well-known result.

Proposition 3.3.1. Let Q be a commutative group scheme over S, and suppose that Q is finite locally free over S, or that Q is of finite type and of multiplicative type over S (SGA 3 IX). Then

$$\underline{\mathrm{Ext}}^1(Q, \underline{G}_m) = 0.$$

Proof. We distinguish the two cases.

a) Suppose that Q is finite locally free over S. We must prove that every extension E of Q by $\underline{G}_m$ splits locally for the fppf topology. We may plainly suppose S affine; there is then an integer $n > 0$ such that ${}_nQ = 0$ (SGA 3 IX 6.1). Moreover, $n \cdot \mathrm{id}$ is an epimorphism of $\underline{G}_m$ for the fppf topology, and hence the Kummer exact sequence

$$0 \longrightarrow \mu_n \longrightarrow \underline{G}_m \xrightarrow{n} \underline{G}_m \longrightarrow 0$$

gives an exact sequence of the $\mathrm{Ext}^i(Q, -)$ and, in particular, an epimorphism

$$\mathrm{Ext}^1(Q, \mu_n) \longrightarrow {}_n\mathrm{Ext}^1(Q, \underline{G}_m) = \mathrm{Ext}^1(Q, \underline{G}_m).$$

The last equality follows because multiplication by n, being zero on Q, is also zero on $\mathrm{Ext}^1(Q, \underline{G}_m)$. Thus the extension E comes from an extension F of Q by μ_n.

The group F is represented by a finite locally free scheme over S (fpqc descent theory, SGA 1 VII). By the known properties of Cartier duality for finite locally free groups (SGA 3 VII 3),

$$D(F) = \underline{\mathrm{Hom}}(F, \underline{G}_m)$$

is an extension of $D(\mu_n)$ by $D(Q)$. Here one uses the following fact: if $A \longrightarrow B$ is a monomorphism of commutative finite locally free groups over S, then $D(B) \longrightarrow D(A)$ is faithfully flat. Since $D(B)$ and $D(A)$ are themselves finite locally free over S, the fibrewise criterion for faithful flatness reduces us to the case where S is the spectrum of a field k. In that case the assertion says that the homomorphism of affine coordinate rings

$$k(D(A)) \longrightarrow k(D(B))$$

is injective (SGA 3 VI$_B$ 11.14). This homomorphism is the transpose of $k(B) \longrightarrow k(A)$, which is surjective because $A \longrightarrow B$ is a closed immersion (SGA 3 VI$_B$ 1.4.2). Hence $k(D(A)) \longrightarrow k(D(B))$ is indeed injective.

The inclusion $\mu_n \longrightarrow \underline{G}_m$ identifies with a section of

$$D(\mu_n) = \underline{\mathrm{Hom}}(\mu_n, \underline{G}_m),$$

and it remains only to prove that, locally for the fppf topology on S, this inclusion extends to a homomorphism $F \longrightarrow \underline{G}_m$; equivalently, that the section of $D(\mu_n)$ just considered lifts fppf-locally to a section of $D(F)$. This is clear, since $D(F) \longrightarrow D(\mu_n)$ is faithfully flat and of finite presentation.

b) By SGA 3 X 4.5, after replacing S by an ètale cover, we may suppose that Q is diagonalizable, say $Q = D(M)$, where M is a finitely generated $\mathbb{Z}$-module. Let T be the torsion submodule of M, and put $L = M/T$, so that $L \simeq \mathbb{Z}^r$. The exact sequence

$$0 \longrightarrow T \longrightarrow M \longrightarrow L \longrightarrow 0$$

then gives an exact sequence (SGA 3 IX 3.1)

$$0 \longrightarrow D(L) \longrightarrow D(M) \longrightarrow D(T) \longrightarrow 0.$$

It is therefore enough to prove separately that the $\underline{\mathrm{Ext}}^1$ of $D(L)$ and of $D(T)$ by $\underline{G}_m$ vanish. The case of $D(T)$ follows from a), and the case of $D(L)$ reduces to $L = \mathbb{Z}$. Thus it remains to check that

$$\underline{\mathrm{Ext}}^1(\underline{G}_m, \underline{G}_m) = 0,$$

that is, that every extension E of $\underline{G}_m$ by $\underline{G}_m$ is fppf-locally trivial.

Such an extension is represented by an affine flat group scheme over S (fpqc descent theory, SGA 1 VII) whose fibres are tori; it is therefore a torus (SGA 3 X 4.9). By SGA 3 X 4.5 we may suppose that E is a split torus

$$E = D(R) \simeq \underline{G}_m^{;2}.$$

After a Zariski localization, we may also suppose that the monomorphism $\underline{G}_m \hookrightarrow E = D(R)$ comes from a constant morphism $R \longrightarrow \mathbb{Z}_S$ (SGA 3 VIII 1.4), which is necessarily an epimorphism (SGA 3 VIII 3.2 b)). Choosing $r \in R$ above $1 \in \mathbb{Z}$ gives a homomorphism $E \longrightarrow \underline{G}_m$ splitting the extension, and proves the assertion.

Corollary 3.3.2. Let Q be as in 3.3.1. Then, for every P, there is a canonical isomorphism

$$\mathrm{Biext}^1(P, Q; \underline{G}_m) \xleftarrow{\sim} \mathrm{Ext}^1(P, D(Q)), \tag{3.3.2.1}$$

where $D(Q)$ denotes the Cartier dual $\underline{\mathrm{Hom}}(Q, \underline{G}_m)$ of Q. More precisely, the canonical functor (1.1.6) is an equivalence from the category of extensions of P by $D(Q)$ to the category of biextensions of (P, Q) by $\underline{G}_m$.

This is the special case of 1.5 obtained by applying 3.3.1.

Remarks 3.3.3. Under the hypotheses of 3.3.1, $D(Q)$ is represented by a separated group scheme, locally of finite presentation and locally quasi-finite over S. More precisely, if Q is finite locally free over S, then so is $D(Q)$, as is well known (SGA 3 VII 3). If Q is of multiplicative type, then $D(Q)$ is a finitely generated twisted constant group (SGA 3 X 5.1), as was seen in SGA 3 IX 4.5 and X 5.7. If P is a group scheme, every extension of P by $D(Q)$ is therefore representable, by a familiar lemma from fpqc descent theory (SGA 3 X 5.4).

3.3.4. It is false that $\underline{\mathrm{Ext}}^1(Q, \underline{G}_m) = 0$ for every affine flat group scheme Q of finite presentation over S. This already fails for

$$Q = \underline{G}_{a_S}, \qquad S = \operatorname{Spec} k[t]/(t^2),$$

where k is a field of characteristic $p > 0$.

Proposition 3.4. Let P be a smooth group scheme of finite presentation over S, with connected fibres, and let Q be a torus over S. Then the category of biextensions of (P, Q) by $\underline{G_m}$ is equivalent to the one-point category; that is,

$$\text{(3.4.1)} \qquad \mathrm{Biext}^0(P, Q; \underline{G_m}) = \mathrm{Biext}^1(P, Q; \underline{G_m}) = 0.$$

Proof. Put $M = D(Q)$. This is a twisted constant group over S whose fibres are isomorphic to groups $\mathbb{Z}^r$ (3.3.3). By 3.3.2, the category in question is equivalent to the category of extensions of P by M. This category is rigid: the automorphism group of any object is the trivial group. This is the first equality in (3.4.1), which may also be written

$$\mathrm{Hom}(P, M) = 0.$$

It follows at once from the fact that M is etale over S, whereas P has connected fibres. It remains to prove

$$\mathrm{Ext}^1(P, M) = 0,$$

that is, that every extension E of P by M is trivial.

By the uniqueness of a splitting when one exists, it is enough to prove that E is locally trivial. We may therefore reduce to the case $M = \mathbb{Z}^r_S$, and then to $M = \mathbb{Z}_S$. When S is the spectrum of a field, the conclusion follows from 5.5 (i) below. (Notice that we shall not use 3.4 before IX 6, so in particular there is no vicious circle.) We now reduce the general case to this special case.

It is enough to find a section f of E over P such that $f(0) = 0$, for such a section is necessarily multiplicative. Indeed,

$$f(x + y) - f(x) - f(y) = 0 :$$

the left-hand side defines a morphism $P \times_S P \longrightarrow \mathbb{Z}_S$ which vanishes on the zero section of $P \times_S P$. It therefore vanishes identically, since $P \times_S P$ has connected fibres (the geometric fibres of P being connected). The required conclusion now follows from the following lemma.

Lemma 3.4.2. Let S be a scheme, P a smooth S-scheme of finite presentation with geometrically connected fibres, Z a group, and E a Z_P-torsor (for the fppf, etale, or fpqc topology; these notions are equivalent here, cf. SGA 3 X 5.4). Let e be a section of P over S, and let e' be a section of E over S above e (that is, a section of $e^{-1}(E)$ over S).

The torsor E is trivial if and only if, for every $s \in S$, the induced torsor E_s over P_s is trivial. In that case there is a unique section f of E over P such that $f \circ e = e'$.

Proof. Uniqueness is clear. Two such sections f and g "differ" by a morphism

$$h : P \longrightarrow Z_S, \qquad h \circ e = 1,$$

and such a morphism is constantly equal to 1, since P has connected fibres. Moreover, if a section f of E over P exists, it can always be corrected by a morphism $P \longrightarrow Z_S$ coming from a morphism $z : S \longrightarrow Z_S$, so as to obtain a section f' satisfying $f' \circ e = e'$. One need only choose z so that

$$f(e(t)) = e'(t)z(t).$$

Applying this observation fibrewise, for every $s \in S$ we obtain a well-determined section f_s of E_s over P_s taking e_s to e'_s. It remains to prove that the f_s arise from a section f of E over P. Equivalently (EGA IV 17.9.3), we must prove that

$$U = \bigcup_{s \in S} f_s(P_s) \subset E$$

is open in E.

Suppose first that S is noetherian. Then U is constructible. To see this, for each $s \in S$ let T be the closure of s, endowed with its reduced induced structure. There is an open neighbourhood T' of s in T and a section of $E_{T'}$ over $P_{T'}$ extending f_s and compatible with the marked sections e and e' (as follows directly from EGA IV 8.8.2). By the criterion of EGA IV 1.10.1, it remains to show that U is stable under generization. The usual argument using EGA II 7.1.9 reduces this to the case in which S is the spectrum of a discrete valuation ring.

Since P is smooth over S, it is regular and hence normal; and since it has connected fibres, it is connected. The given section of E over the generic fibre of P over S therefore extends uniquely to a section f of E over P (EGA IV 18.10.19 and 18.10.20), necessarily compatible with e and e'. Thus in this case $U = f(P)$ is open. This proves 3.4 when S is noetherian. The general case reduces readily to this one by the usual limit arguments of EGA IV, 8 and 9; we leave the details to the reader.

Remark 3.4.3. Lemma 3.4.2 can be "crushed" under learned references, namely SGA 4 XV 1.15 (the case $n = 1$) and 4.1.

Corollary 3.5. Let P be a smooth group scheme of finite presentation over S, with connected fibres, and let

$$0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0$$

be an exact sequence of commutative groups over S, with Q' a torus. Then the inverse-image functor on biextensions

$$\mathrm{BIEXT}(P, Q''; \underline{G_m}) \xrightarrow{\sim} \mathrm{BIEXT}(P, Q; \underline{G_m}) \tag{3.5.1}$$

is an equivalence of categories. In particular, the following natural homomorphisms are isomorphisms:

$$\mathrm{Biext}^i(P, Q''; \underline{G_m}) \xrightarrow{\sim} \mathrm{Biext}^i(P, Q; \underline{G_m}), \qquad i = 0, 1. \tag{3.5.2}$$

Indeed, up to equivalence, $\mathrm{BIEXT}(P, Q''; \underline{G_m})$ identifies with the category of biextensions E of (P, Q) by $\underline{G_m}$ equipped with a trivialization of the induced biextension E' of (P, Q') by $\underline{G_m}$ (VII 3.7.6). By 3.4 there is exactly one such trivialization of E' for each given biextension E. One can of course also prove directly that (3.5.2) is an isomorphism for $i = 0, 1$ - which is equivalent to the conclusion of 3.5 - by using the exact sequence of the Biext^i in VII 3.7.5 associated with

$$0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0.$$

Remark 3.6. The symmetric statement obtained from 3.5 by interchanging left and right is of course also valid; it also follows formally from 3.5 by passing to symmetric biextensions (VII 2.7). Suppose, in addition to the data and hypotheses of 3.5, that the symmetric data and hypotheses are given: namely, that Q has connected fibres (equivalently, that Q'' has connected fibres, since Q' has connected fibres), and that there is an exact sequence

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0$$

with P' a torus. Successive application of 3.5 and of its symmetric form, applied to this last sequence and to Q'', shows that the functor

$$\mathrm{BIEXT}(P'', Q''; \underline{G_m}) \xrightarrow{\sim} \mathrm{BIEXT}(P, Q; \underline{G_m}) \tag{3.6.1}$$

is an equivalence of categories. In particular, the following homomorphism is an isomorphism:

$$\mathrm{Biext}^i(P'', Q''; \underline{G_m}) \xrightarrow{\sim} \mathrm{Biext}^i(P, Q; \underline{G_m}), \qquad i = 0, 1. \tag{3.6.2}$$

Proposition 3.7. Let A be an abelian scheme over S, let T be a group scheme of multiplicative type (SGA 3 IX) and of finite type over S, and let

$$\underline{M} = \underline{\mathrm{Hom}}(T, \underline{G_m})$$

be the Cartier dual of T (a finitely generated twisted constant group over S, by 3.3.3). Then the categories $\mathrm{EXT}(A, T)$ and $\mathrm{BIEXT}(A, \underline{M}; \underline{G_m})$, formed respectively by commutative extensions of A by T and by biextensions of $(A, \underline{M})$ by $\underline{G_m}$, are rigid and canonically equivalent. Their groups of isomorphism classes are canonically isomorphic to $\mathrm{Hom}(\underline{M}, A^*)$, where

$$A^* = \underline{\mathrm{Ext}}^1(A, \underline{G_m})$$

is the dual abelian scheme of A (3.2). In particular, there is a canonical isomorphism

$$\mathrm{Ext}^1(A, T) \xrightarrow{\sim} \mathrm{Hom}(\underline{M}, A^*). \tag{3.7.1}$$

This is the special case of 1.6 obtained by taking $P = A$ and $Q = \underline{M}$. By the biduality theorem (SGA 3 X 5.7), we do indeed have

$$T \simeq \underline{\mathrm{Hom}}(\underline{M}, \underline{G_m}),$$

and it remains only to verify the conditions (1.6.1), which here read

$$\underline{\mathrm{Hom}}(A, \underline{G_m}) = 0, \qquad \mathrm{Hom}\big(A, \underline{\mathrm{Ext}}^1(\underline{M}, \underline{G_m})\big) = 0.$$

The first equality was recalled in (3.2.1); the second may be strengthened to

$$\underline{\mathrm{Ext}}^1(\underline{M}, \underline{G_m}) = 0.$$

This assertion is local on S, so we may suppose that $\underline{M}$ is constant, $\underline{M} = M_S$. Decomposing M as a direct sum of cyclic $\mathbb{Z}$-modules reduces us to $M = \mathbb{Z}$, which is trivial, and to $M = \mathbb{Z}/n\mathbb{Z}$, which follows from the n-divisibility of $\underline{G_m}$ (or, more learnedly, from 3.3.1).

When $T = \underline{G_m}$, and hence $\underline{M} = \mathbb{Z}_S$, (3.7.1) reduces to the identity isomorphism (3.2.3). Of course, a direct proof of (3.7.1) would be essentially trivial, without introducing the group

$$\mathrm{Ext}^1\big(A \overset{\mathbb{L}}{\otimes} \underline{M}, \underline{G_m}\big) = \mathrm{Biext}^1(A, \underline{M}; \underline{G_m}).$$

4. Extensions and biextensions of smooth connected group schemes over a field

We recall the following lemma due to Rosenlicht [3 bis, Proposition 3].

Lemma 4.1. Let k be a field, and let X and Y be two separable, geometrically connected k-schemes of finite type, equipped respectively with k-rational points e_X and e_Y. Then every morphism

$$X \times Y \longrightarrow \underline{G_m}$$

which is trivial on the two subschemes $X \times e_Y$ and $e_X \times Y$ is trivial.

We shall use this result only when X and Y are group schemes over k; in that case the hypotheses imply that X and Y are in fact smooth over k. By induction, 4.1 also implies the analogous result for any number of factors. We may therefore apply the results of VII 1.3 and VII 2.9, and obtain the following two propositions.

Proposition 4.2. Let P be a smooth connected group over the field k. The natural functor from the category of group extensions of P by $\underline{G_m}$ to the category of line bundles on P rigidified at the identity element e_P is fully faithful. Its essential image consists of the rigidified line bundles $\mathbb{L}$ on P for which

$$\pi^*\mathbb{L} \simeq \mathrm{pr}_1^*\,\mathbb{L} \otimes \mathrm{pr}_2^*\,\mathbb{L},$$

where $\pi : P \times P \longrightarrow P$ is the group law of P, and $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections.

Proposition 4.3. Let P and Q be two smooth connected algebraic groups over the field k. The natural functor from the category of biextensions of (P, Q) by $\underline{G}_m$ to the category of line bundles on $P \times Q$ birigidified with respect to the identity sections e_P and e_Q of P and Q is fully faithful; moreover, both categories are rigid. A birigidified line bundle $\mathbb{L}$ on $P \times Q$ comes from a biextension if and only if its pullbacks to $P \times P \times Q$ and $P \times Q \times Q$ by $\pi_P \times \mathrm{id}_Q$ and $\mathrm{id}_P \times \pi_Q$ satisfy the "theorem of the cube" ([1] or [2]).

This last condition is simply the translation of the conditions in VII 2.9.3 b), in the explicit form given in VII 3.5 b).

4.4. I do not know whether the cube condition in 4.3 is automatically satisfied. At least it is known [2, IV 2.6] that, for a given $\mathbb{L}$, there is always an integer $n \geq 1$, a power of the characteristic exponent of k, such that $\mathbb{L}^{\otimes n}$ satisfies that condition and hence comes from a biextension of (P, Q) by $\underline{G}_m$. It is also always possible to find a purely inseparable extension k' of k such that the required condition becomes true after the base extension $k \longrightarrow k'$; after this extension, therefore, $\mathbb{L}$ is a birigidified line bundle arising from a biextension of (P, Q) by $\underline{G}_m$.

In particular, when k is perfect, the condition in 4.3 is always satisfied. This last fact, which formally implies the preceding statements by a trace argument, can also be seen by applying 4.7 below; cf. 4.9.

We recall, in the form of a lemma, the following result from EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.13.

Lemma 4.5. Let $f : X \longrightarrow Y$ be a smooth surjective morphism with integral fibres, where Y is normal and integral with generic point y. Let $\mathbb{L}$ be a line bundle on X whose restriction to X_y is trivial. Then there is a line bundle $\mathbb{M}$ on Y such that

$$\mathbb{L} \simeq f^* \mathbb{M}.$$

Proposition 4.6. Let P, Q be two smooth connected algebraic groups over the field k, with P affine. Suppose in addition either that P admits a composition series each of whose factors is a torus or is isomorphic to the additive group $\underline{G}_a$ (a condition satisfied in particular when k is perfect [SGA 3, XVII 7.2.1, 4.1.1 (i)$\Longleftrightarrow$ (ii), and 4.1.5]), or that Q is an extension of an abelian variety B by a smooth connected affine algebraic group satisfying the preceding condition. Under either hypothesis, every birigidified line bundle $\mathbb{L}$ on $P \times Q$ is trivial; a fortiori, by 4.3, every biextension of (P, Q) by $\underline{G}_m$ is trivial.

Proof. First observe that every automorphism of a birigidified line bundle is the identity (4.3). Descent theory therefore shows that it is enough to prove that $\mathbb{L}$ becomes trivial after a finite separable extension of k. Passing to the limit immediately reduces us to the case where k is separably closed.

Every torus over such a field is split. Hence, if P admits a composition series as in 4.6, it also admits one whose quotients are isomorphic to $\underline{G}_a$ or $\underline{G}_m$. Repeated application of 4.5, together with

$$\mathrm{Pic}(\underline{G}_{a_K}) = \mathrm{Pic}(\underline{G}_{m_K}) = 0$$

for every field K, gives

$$\mathrm{Pic}(P) = 0.$$

This equality remains true after every extension of the base field k. Applying 4.5 to

$$\mathrm{pr}_2 : P \times Q \longrightarrow Q$$

and to $\mathbb{L}$, we find a line bundle $\mathbb{M}$ on Q such that $\mathbb{L} \simeq \mathrm{pr}_2^* \mathbb{M}$. Since the restriction of $\mathbb{L}$ to $e_P \times Q$ is trivial, $\mathbb{M}$ is trivial, and so is $\mathbb{L}$. This proves the first case.

In the second case, let Q be an extension of an abelian scheme B by a group M admitting a composition series with quotients isomorphic to $\underline{G}_m$ or $\underline{G}_a$. Apply 4.5 to

$$f : X = P \times Q \longrightarrow Y = P \times B.$$

The fibre over the generic point y of $P\times B$ is an M_y-torsor. It is necessarily trivial by the particular structure of M and the classical equalities

$$H^1(K,\underline{G}_a)=0, \qquad H^1(K,\underline{G}_m)=0,$$

valid for every field K, here taken to be $k(y)$. Since $\mathrm{Pic}(M)=0$, we have $\mathrm{Pic}(X_y)=0$; thus 4.5 applies and shows that $\mathbb{L}$ is the pullback of a line bundle $\mathbb{N}$ on $Y=P\times B$.

Let $\mathbb{N}_1$ and $\mathbb{N}_2$ be the restrictions of $\mathbb{N}$ to $P\times e_B$ and $e_P\times B$, respectively. Replacing $\mathbb{N}$ by

$$\mathbb{N}\otimes \mathrm{pr}_1^*(\mathbb{N}_1)^{-1}\otimes \mathrm{pr}_2^*(\mathbb{N}_2)^{-1},$$

we may suppose that $\mathbb{N}$ is birigidified and that $\mathbb{L}$ is its birigidified pullback. We are thereby reduced to the case in which Q itself is an abelian scheme.

Then $\mathbb{L}$ defines a canonical morphism

$$(*) \qquad P\longrightarrow \underline{\mathrm{Pic}}_{Q/k}$$

taking the identity to the identity. Since P is connected, this morphism factors through the dual abelian scheme

$$Q'=\underline{\mathrm{Pic}}^0_{Q/k}.$$

It is well known that every morphism of schemes from a smooth connected affine algebraic group to an abelian scheme is constant. This is a geometric assertion, so we may suppose k algebraically closed; a devissage then reduces it immediately to $P=\underline{G}_a$ or $P=\underline{G}_m$, and every rational map from $\mathbb{P}^1_k$ to an abelian scheme over k is constant.

The constancy, hence vanishing, of (*) means that $\mathbb{L}$ is the pullback of a line bundle $\mathbb{M}$ on P. Since $\mathbb{L}$ is trivial on $P\times e_Q$, $\mathbb{M}$ is trivial, and therefore so is $\mathbb{L}$. This completes the proof.

Proposition 4.7. Let P and Q be two smooth connected algebraic groups over the field k. Suppose that P is an extension of an abelian scheme A by a smooth connected affine algebraic group L admitting a composition series each of whose factors is isomorphic to a torus or to $\underline{G}_a$ (a condition always satisfied when k is perfect). Then the functor considered in 1.3 is an equivalence of categories; that is, every birigidified line bundle $\mathbb{L}$ on $P\times Q$ comes from a biextension of (P,Q). Moreover, the functor between the rigid categories

$$\mathrm{BIEXT}(A,Q;\underline{G}_m)\longrightarrow \mathrm{BIEXT}(P,Q;\underline{G}_m)$$

is an equivalence of categories; in other words, it induces an isomorphism of groups

$$(4.7.1) \qquad \mathrm{Biext}^1(A,Q;\underline{G}_m)\xrightarrow{\sim}\mathrm{Biext}^1(P,Q;\underline{G}_m).$$

For the first assertion, descent and 4.3 again reduce us to the case where k is separably closed. Proceeding as in the proof of 4.6, we deduce from 4.5 and the hypothesis on L that $\mathbb{L}$ is isomorphic to the pullback of a birigidified line bundle $\mathbb{M}$ on $A\times Q$. For the first assertion of 4.7 we are thus reduced to the case $A=P$. The line bundle $\mathbb{L}$ then defines a morphism of pointed schemes

$$f: Q\longrightarrow \underline{\mathrm{Pic}}_{P/k},$$

which necessarily factors through the dual abelian scheme P' of P. Let $\mathbb{L}'$ be the pullback of the Weil sheaf on $P\times P'$ along

$$\mathrm{id}_P\times f: P\times Q\longrightarrow P\times P'.$$

This is a birigidified line bundle which, by the definition of f, differs from $\mathbb{L}$ only by the pullback of a line bundle $\mathbb{M}$ on P. Since both $\mathbb{L}'$ and $\mathbb{L}$ have trivial restriction to $P\times e_Q$, it follows that $\mathbb{M}$ is trivial and hence that $\mathbb{L}\simeq\mathbb{L}'$. We noted in VII 2.9.4 that divisorial correspondences on a product of two abelian schemes arise from biextensions. Consequently $\mathbb{L}'$, and therefore $\mathbb{L}$, comes from a biextension.

Returning to the general case, where P is an extension of A by L, it remains to prove that (4.7.1) is an isomorphism. This follows from the exact sequence of the Biext^i, for $i = 0, 1$, associated with

$$0 \longrightarrow L \longrightarrow P \longrightarrow A \longrightarrow 0,$$

and from the relations

$$\mathrm{Biext}^0(L, Q; \underline{G}_m) = 0, \qquad \mathrm{Biext}^1(L, Q; \underline{G}_m) = 0.$$

The first is a special case of the rigidity assertion in 4.3, and the second is precisely the first case of 4.6.

Corollary 4.8. Under the hypotheses of 4.7, suppose that Q is an extension of an algebraic group B (necessarily smooth and connected) by a smooth connected affine algebraic group M. Then the functor between the rigid categories

$$\mathrm{BIEXT}(A, B; \underline{G}_m) \longrightarrow \mathrm{BIEXT}(P, Q; \underline{G}_m)$$

is an equivalence of categories; in other words, it induces an isomorphism of groups

(4.8.1) $$\mathrm{Biext}^1(A, B; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(P, Q; \underline{G}_m).$$

In view of (4.7.1), it remains to prove that

$$\mathrm{Biext}^1(A, B; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(A, Q; \underline{G}_m)$$

is an isomorphism. As in the proof of (4.7.1), this follows from the exact sequence of the Biext^i associated with

$$0 \longrightarrow M \longrightarrow Q \longrightarrow B \longrightarrow 0,$$

together with the relations

$$\mathrm{Biext}^0(P, M; \underline{G}_m) = 0, \qquad \mathrm{Biext}^1(P, M; \underline{G}_m) = 0.$$

Indeed, (4.7.1), applied with M in place of Q, identifies these groups with the corresponding groups having A in the first variable, as required by the preceding exact sequence. The second vanishing displayed above is a special case of the second case of 4.6, after interchanging the two factors.

Remark 4.9. Let P and Q be two smooth connected algebraic groups over a perfect field k. It is then known [3] that each is an extension of an abelian scheme by a smooth connected affine group, the latter being the product of a torus with an iterated extension of copies of $\underline{G}_a$ (cf. the reference given in 4.6):

$$0 \longrightarrow L \longrightarrow P \longrightarrow A \longrightarrow 0, \qquad 0 \longrightarrow M \longrightarrow Q \longrightarrow B \longrightarrow 0.$$

Thus 4.8 applies and shows that the theory of biextensions of (P, Q) by $\underline{G}_m$ reduces to the analogous (and much more classical!) theory for the pair (A, B) of abelian varieties.

Corollary 4.10. Let P and Q be two smooth connected algebraic groups over k, let L be a smooth connected affine algebraic subgroup of P, and let E be a biextension of (P, Q) by $\underline{G}_m$. By 2.2, E induces a pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\underline{G}_m).$$

Then the induced pairing

$$T_\ell(L) \times T_\ell(Q) \longrightarrow T_\ell(\underline{G}_m)$$

is zero.

Indeed, we may plainly suppose that k is algebraically closed. We are then under the hypotheses of 4.6, which imply that the restriction of E to (L, Q) is the trivial biextension. The asserted conclusion follows, since the induced pairing in question is the pairing defined by this restriction.

4.11. Let P and Q be algebraic groups smooth over k, let E be a biextension of (P, Q) by $\underline{G}_m$, and let T be the largest subtorus of P. We shall determine the left annihilator of the pairing

$$T_\ell(P) \times T_\ell(Q) \longrightarrow T_\ell(\underline{G}_{m,k}), \tag{4.11.1}$$

where ℓ is again a fixed prime number. By 4.10 this annihilator always contains the subgroup $T_\ell(T)$ of $T_\ell(P)$.

Choose an extension k' of k over which P and Q can each be written as an extension of an abelian scheme by an affine group scheme that is an iterated extension of tori and copies of $\underline{G}_a$. If P and Q have unipotent rank zero, we may take $k' = k$; in general, we may take for k' the perfect closure of k (or a finite radicial extension). Let A' and B' be the abelian parts of $P_{k'}$ and $Q_{k'}$, respectively. By 4.7, the biextension $E_{k'}$ of $(P_{k'}, Q_{k'})$ comes from a biextension F' of (A', B'), uniquely determined up to unique isomorphism. Consequently the pairing

$$T_\ell(P_{k'}) \times T_\ell(Q_{k'}) \longrightarrow T_\ell(\underline{G}_{m,k'}), \tag{4.11.2}$$

obtained from (4.11.1) by the base-field extension $k \to k'$, and also defined by the biextension $E_{k'}$, is obtained from the pairing

$$T_\ell(A') \times T_\ell(B') \longrightarrow T_\ell(\underline{G}_{m,k'}) \tag{4.11.3}$$

defined by F', by composing with the canonical homomorphisms (which are in fact epimorphisms if P and Q are ℓ-divisible)

$$T_\ell(P_{k'}) \longrightarrow T_\ell(A'), \qquad T_\ell(Q_{k'}) \longrightarrow T_\ell(B'). \tag{4.11.4}$$

It follows that the left annihilator of (4.11.2), which is obtained from the left annihilator of (4.11.1) by base change, is the inverse image under the first homomorphism in (4.11.4) of the left annihilator of (4.11.3). Let C' be the largest abelian subscheme of A' such that the restriction of F' to (C', B') is the trivial biextension. Equivalently, C' is the reduced identity component of the kernel of the homomorphism

$$A' \longrightarrow (B')^* \tag{4.11.5}$$

defined by F', where $(B')^*$ is the dual abelian scheme of B'. The left annihilator of (4.11.3) identifies with $T_\ell(C')$; therefore the left annihilator of (4.11.1) is precisely the inverse image of

$$T_\ell(C') \hookrightarrow T_\ell(A')$$

under the first morphism in (4.11.4).

In particular, to say that the left annihilator of (4.11.1) is exactly $T_\ell(T)$ – equivalently, that the left annihilator of (4.11.2) is the kernel $T_\ell(T')$ of the first morphism in (4.11.4) – is to say that (4.11.3) is nondegenerate on the left, that is, that $C' = 0$, or equivalently that (4.11.5) has finite kernel. In this last form the condition is visibly independent of ℓ, and in any of its forms it is plainly independent of the choice of k'. When it holds, we shall simply say that the given biextension E of (P, Q) by $\underline{G}_m$ is <u>nondegenerate on the left</u>. By duality, this also means that the homomorphism

$$B' \longrightarrow (A')^* \tag{4.11.6}$$

defined by F' is an epimorphism, that is, is surjective.

The notion of a biextension nondegenerate on the right is defined symmetrically: (4.11.6) must have finite kernel, or, equivalently, (4.11.5) must be an epimorphism. The biextension E of (P, Q) by $\underline{G}_{m,k}$ is called nondegenerate if it is nondegenerate on both the left and the right. This is equivalent to saying that (4.11.5), or equivalently (4.11.6), is an isogeny.

It is immediate from the definition that these notions are invariant under every extension of the base field k.

4.12. Let P be a smooth connected group scheme, let $\mathbb{L}$ be a line bundle on P, and put

$$\delta(\mathbb{L}) = \pi^*\mathbb{L} \otimes \mathrm{pr}_1^*(\mathbb{L})^{-1} \otimes \mathrm{pr}_2^*(\mathbb{L})^{-1}, \tag{4.12.1}$$

with the notation of 4.2. As already observed in VII 1.3.4, this is a birigidified line bundle on $P \times P$. By 4.3, if $\delta(\mathbb{L})$ comes from a biextension, that biextension is determined up to unique isomorphism. In particular, by 4.7, it is defined when P is an extension of an abelian scheme A by an iterated extension of tori and copies of $\underline{G}_a$. Whenever it is defined, this biextension is plainly symmetric.

In the preceding favourable case, and assuming that the maximal torus of P is split, the construction of $\delta(\mathbb{L})$ can be reduced to the case where P is an abelian scheme. Indeed, the construction is functorial in P, while the homomorphism

$$\mathrm{Pic}(A) \longrightarrow \mathrm{Pic}(P) \tag{4.12.2}$$

induced by the canonical quotient homomorphism $f : P \longrightarrow A$ is surjective by 4.5 (cf. the beginning of the proof of 4.6). We can therefore find an isomorphism

$$\mathbb{L} \simeq f^*\mathbb{M},$$

where $\mathbb{M}$ is a line bundle on A. Then $\delta(\mathbb{L})$ is the pullback of the biextension $\delta(\mathbb{M})$ of (A, A) by $\underline{G}_{m,k}$.

Proposition 4.13. Under the hypotheses of 4.12, suppose that $\mathbb{L}$ is an ample line bundle on P. If the biextension $\delta(\mathbb{L})$ of (P, P) by $\underline{G}_{m,k}$ is defined, then it is nondegenerate in the sense of 4.11. If P is an extension of an abelian scheme A by a smooth connected affine group L satisfying the condition of 4.7, then the biextension of (A, A) from which $\delta(\mathbb{L})$ comes by 4.8 is of the form $\delta(\mathbb{M})$, where $\mathbb{M}$ is an ample line bundle on A.

We reduce to the case where k is algebraically closed. By the definition, it is then enough, with the notation of 4.12, to prove that the biextension $\delta(\mathbb{M})$ of (A, A) is nondegenerate. M. Raynaud proves in his thesis [2, XI 1.11] that if $\mathbb{L} = f^*\mathbb{M}$ is ample, then $\mathbb{M}$ is ample as well. The theory of A. Weil then gives the familiar conclusion that $\delta(\mathbb{M})$ is nondegenerate [1, 2], that is, that the homomorphism from A to its dual abelian scheme is an isogeny.

Remark 4.14. Suppose we are in the favourable case where P is an extension of an abelian scheme A by a group L which is an iterated extension of tori and copies of $\underline{G}_a$, and consider the canonical composite

$$\mathrm{NS}(A) \xrightarrow{\delta} \mathrm{Biext}^1(A, A; \underline{G}_m) \xrightarrow{\sim} \mathrm{Biext}^1(P, P; \underline{G}_m). \tag{4.14.1}$$

By 4.8 the second arrow is bijective. On the other hand, the theory of abelian varieties shows that the first arrow identifies the Neron–Severi group $\mathrm{NS}(A)$ of A with the subgroup of

$$\mathrm{Biext}^1(A, A; \underline{G}_m) = \mathrm{Corr}(A, A)$$

consisting of the classes of symmetric biextensions, or equivalently symmetric divisorial correspondences. Thus (4.14.1) identifies $\mathrm{NS}(A)$ with the group of symmetric biextensions of (P, P) by $\underline{G}_m$. By 4.7 this group can equally be described as the group of classes of symmetric divisorial correspondences on (P, P).

It is natural to call a divisorial correspondence on (P, P) ample, and likewise to call the corresponding biextension of (P, P) by $\underline{G}_m$ ample, when it comes from an ample element of $\mathrm{NS}(A)$. This condition is plainly invariant under extension of the base field; the notion therefore still makes sense without the preceding restrictive hypothesis on P, assuming only that P is smooth and connected.

Suppose further that L is "split solvable over k", in the sense that it admits a devissage by copies of $\underline{G}_a$ and $\underline{G}_m$. The preceding observation and the surjectivity of (4.12.2) show that the image of

$$\mathrm{Pic}(P) \xrightarrow{\delta} \mathrm{Biext}^1(P, P; \underline{G}_m) \tag{4.14.2}$$

consists of the symmetric biextensions, and that its kernel is the subgroup $\mathrm{Pic}^0(P)$ of $\mathrm{Pic}(P)$ which is the image of $\mathrm{Pic}^0(A)$. Consequently, if we put

$$\mathrm{NS}(P) = \mathrm{Pic}(P)/\mathrm{Pic}^0(P) \simeq \mathrm{NS}(A),$$

we may also interpret (4.14.1) as a canonical isomorphism

$$\mathrm{NS}(P) \xrightarrow{\sim} \mathrm{Biext}^1(P, P; \underline{G}_m)^{\mathrm{sym}}. \tag{4.14.3}$$

Finally, the theorem of Raynaud cited in 4.13 says that an element ξ of $\mathrm{Pic}(P)$ is ample if and only if its image $\delta(\xi)$ under (4.14.2) is the class of an ample biextension.

5. Extensions and biextensions by torsion-free truncated constant group schemes.

In this section and the next we give some auxiliary results that will be used in section 7 in the case $Z = \mathbb{Z}$.

Proposition 5.1. Let S be an irreducible geometrically unibranch scheme with generic point η, and let Z be a torsion-free group, hence giving a constant group scheme Z_S over S. Then every torsor E under Z_S is trivial, and the functor

$$E \longmapsto E(\eta)$$

from the category of torsors under Z_S to the category of torsors under $Z = Z_S(\eta)$ is an equivalence of categories.

A familiar lemma of descent theory (SGA 3 X 5.4) shows that every torsor E under Z_S is representable. In fact, that result applies to every group scheme over S which is separated, locally of finite presentation, and locally quasi-finite over S. Since Z_S is etale over S, fpqc descent shows that the same is true of every torsor E under Z_S; such a torsor is therefore locally trivial for the etale topology (and not merely for the fppf topology). To prove that E is trivial, it is consequently enough to prove

$$H^1(S_{\text{ét}}, Z_S) = e.$$

Likewise, the full-faithfulness assertion of 5.1 reduces to the relation

$$Z \xrightarrow{\sim} H^0(S_{\text{ét}}, Z_S).$$

The latter is immediate. To prove the former, consider the inclusion

$$i : \eta \longrightarrow S.$$

Because S is geometrically unibranch, we have

$$Z_S \xrightarrow{\sim} i_*(Z_\eta)$$

(SGA 4 IX 2.14.1). The noncommutative low-degree exact sequence that replaces the Leray spectral sequence for $i : \eta \to S$ (SGA 4 XII 3.2) therefore begins

$$e \longrightarrow H^1(S, i_*(Z_\eta)) \longrightarrow H^1(\eta, Z_\eta) \longrightarrow \cdots,$$

and shows that

$$H^1(S, Z_S) \longrightarrow H^1(\eta, Z_\eta)$$

is injective. We are thus reduced to the case in which S is the spectrum of a field k. If π is the Galois group of a separable closure of k, then

$$H^1(k, Z) = \mathrm{Hom}_{\mathrm{cont}}(\pi, Z)/Z = e,$$

because Z is torsion-free. This completes the proof.

<u>Corollary 5.2.</u> <u>With the notation of 5.1, let P be a smooth scheme over S. Then every torsor E under Z_P is trivial, and the restriction functor</u>

$$E \longmapsto E_\eta$$

<u>from the category of torsors on P under Z_P to the category of torsors on P_η under Z_{P_η} is an equivalence of categories.</u>

Indeed, P is geometrically unibranch (EGA IV 17.5.7). Moreover, every irreducible component of P dominates S, because P is flat over S, and the set of irreducible components of P is locally finite. Thus P is the disjoint sum of irreducible geometrically unibranch schemes P_i. The assertion now follows immediately from 5.1 applied to the various P_i and their generic fibres $P_{i\eta}$, noting that the generic point p_i of P_i is also the generic point of $P_{i\eta}$.

5.2.1. The preceding corollary immediately gives the analogous statement for bitorsors on P under (Z_P, Z_P): restriction to the generic fibre induces an equivalence from their category to the category of bitorsors on P_η under (Z_{P_η}, Z_{P_η}). Indeed, it is enough to express the bitorsor structure on E in terms of its structure as a right torsor under Z_P, together with an isomorphism from Z_P to the commutant of Z_P for its action on E. Since E is necessarily trivial, that commutant is itself isomorphic to Z_P. We are thus reduced, on each irreducible component X of P, to the following immediate observation: if X is a nonempty connected scheme, the functor $Z \mapsto Z_X$ from sets to X-schemes is fully faithful.

<u>Proposition 5.3.</u> <u>Let S and Z be as in 5.1, and let P be a smooth group scheme over S. Consider extensions E of group schemes of P by Z_S. Then restriction to the generic fibre</u>

$$E \longmapsto E_\eta$$

<u>is an equivalence from the category of these extensions to the category of extensions of P_η by Z_η. If P and Z are commutative, it also induces an equivalence between the corresponding categories of commutative extensions.</u>

First observe that every extension of sheaves of groups of P by Z_S (and similarly of P_η by Z_η) gives, by 5.2, a trivial torsor, hence in particular a representable one. The categories in 5.3 may therefore equally well be interpreted as categories of extensions of sheaves of groups. The description in VII 1.1.6 and 1.2 expresses these in terms of bitorsors on P, $P \times P$, and $P \times P \times P$ (respectively on P_η, $P_\eta \times P_\eta$, and $P_\eta \times P_\eta \times P_\eta$), under the pairs of groups obtained from (Z, Z) by base change. Applying 5.2 in the commutative case and 5.2.1 in the general case to the smooth S-schemes P, $P \times P$, and $P \times P \times P$ gives the asserted conclusion. The same argument gives the following result.

<u>Corollary 5.4.</u> <u>Let S, Z be as in 5.1, and let P, Q be two smooth group schemes over S. Suppose that P, Q, and Z are commutative, and consider biextensions E of (P, Q) by Z_S. Then restriction to the generic fibre</u>

$$E \longmapsto E_\eta$$

<u>is an equivalence from the category of these biextensions to the category of biextensions of (P_η, Q_η) by Z_η.</u>

The preceding results reduce the determination of categories of extensions or biextensions by a torsion-free constant group to the case in which the base scheme is the spectrum of a field. In that case we obtain the following.

Proposition 5.5. Let k be a field, Z a torsion-free group, P a k-group scheme locally of finite type, P° its identity component, and $\Phi = P/P^\circ$ its group of connected components (SGA 3 VI_A 4). Then:

(i) The natural functor from the category of group schemes which are extensions (respectively commutative extensions) of Φ by Z_k to the category of group schemes which are extensions (respectively commutative extensions) of P by Z_k is an equivalence. In particular, if P is connected, every extension of P by Z_k is trivial, and the category of such extensions is rigid.

(ii) Suppose that Φ is a torsion group and that P and Z are commutative. Then the category of extensions of P by Z_k is rigid: every automorphism of an object of this category is trivial. If, in addition, there is an integer $n > 0$ such that $n\mathrm{id}_\Phi = 0$—for example, if Φ is finite, as happens when P is of finite type—then the group of classes of commutative extensions of P by Z_k is given by the canonical isomorphism

$$\mathrm{Ext}^1(P, Z_k) \xleftarrow{\sim} \mathrm{Hom}\big(\Phi, (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z\big), \tag{5.5.1}$$

defined by the boundary operator in the exact sequence of Ext groups associated with the exact sequence

$$0 \longrightarrow Z \longrightarrow Z \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z \longrightarrow 0. \tag{5.5.2}$$

The canonical morphism

$$P \longrightarrow \Phi$$

plainly induces an isomorphism on the sets of irreducible components, since it is flat with irreducible fibres. By 5.1, the induced functor from torsors under Z_Φ to torsors under Z_P is therefore an equivalence. (Recall that a group scheme locally of finite type over k is geometrically unibranch.) The same conclusion applies to bitorsors and to the morphisms

$$P \times P \longrightarrow \Phi \times \Phi, \qquad P \times P \times P \longrightarrow \Phi \times \Phi \times \Phi.$$

The description in VII 1.1.6 of extensions of P by Z_k and of Φ by Z_k (and VII 1.3 in the commutative case) now gives the assertions in (i).

We restrict attention to commutative extensions. The automorphism group of an extension of P by Z_k is

$$\mathrm{Hom}(P, Z_k) \simeq \mathrm{Hom}(\Phi, Z_k),$$

which is plainly zero when Φ is a torsion group. Likewise, if $V = Z \otimes_{\mathbb{Z}} \mathbb{Q}$ and Φ is annihilated by the integer $n > 0$, then

$$\mathrm{Ext}^i(\Phi, V_k) = 0 \qquad \text{for every } i \in \mathbb{Z},$$

because the groups in question are $\mathbb{Q}$-vector spaces annihilated by n. The isomorphism (5.5.1) follows at once.

The same method proves, essentially, the following.

Corollary 5.6. Let k be a field, Z a torsion-free commutative group, P and Q two commutative group schemes locally of finite type over k, and let $\Phi = P/P^\circ$ and $\Psi = Q/Q^\circ$ be their groups of connected components. Then:

(i) The natural functor from the category of biextensions of (Φ, Ψ) by Z_k to the category of biextensions of (P, Q) by Z_k is an equivalence. In particular, if P or Q is connected, the category of biextensions of (P, Q) by Z_k is equivalent to the terminal category.

(ii) Suppose that Φ or Ψ is a torsion group. Then the category of biextensions of (P, Q) by Z_k is rigid. If, in addition, there is an integer $n > 0$ such that $n\mathrm{id}_\Phi = 0$ or $n\mathrm{id}_\Psi = 0$, then the group of classes of biextensions of (P, Q) by Z_k is given by the canonical isomorphism

$$\operatorname{Biext}^1(P, Q; Z_k) \xleftarrow{\sim} \operatorname{Hom}\big(\Phi \otimes \Psi, (Z \otimes_{\mathbb{Z}} \mathbb{Q})/Z\big), \tag{5.6.1}$$

defined by the boundary operator in the exact sequence of Biext^i groups ($i = 0, 1$) associated with (5.5.2).

5.7. Let S be a scheme, T a closed subscheme, and Z a set. The constant scheme over S, truncated along T, with value Z, denoted $Z_{S,T}$, is the scheme obtained by taking the sum of copies S_z of S, one for each $z \in Z$, and gluing all these copies along the open subscheme $U = S \setminus T$. Thus

$$Z_{S,T} \times_S U \simeq U, \qquad Z_{S,T} \times_S T \simeq Z_T, \tag{5.7.1}$$

and $Z_{S,T}$ is etale over S. For a variable S-scheme S' there is an isomorphism, functorial in S',

$$\operatorname{Hom}_S(S', Z_{S,T}) \simeq \operatorname{Hom}_T(S'_T, Z_T). \tag{5.7.2}$$

The map is restriction to S'_T, using the second isomorphism in (5.7.1); we leave it to the reader to check that the map is bijective. If

$$i : T \longrightarrow S$$

denotes the inclusion, the functorial isomorphism (5.7.2) may also be written as an isomorphism of sheaves

$$Z_{S,T} \simeq i_*(Z_T). \tag{5.7.3}$$

This formula, or the definition directly, shows that as Z varies, $Z_{S,T}$ depends functorially on Z, and that this functor commutes with finite inverse limits. In particular, it takes groups to group schemes and commutative groups to commutative group schemes, as is also clear from the form (5.7.2) of the functor represented by $Z_{S,T}$.

It is also immediate from the definition, or from (5.7.3), that the formation of $Z_{S,T}$ commutes with every base change $f : S' \to S$. This gives an isomorphism, functorial in Z,

$$Z_{S,T} \times_S S' \simeq Z_{S',T'}, \qquad T' = T \times_S S' = T_{S'}. \tag{5.7.4}$$

5.8. We shall study torsors E under a group scheme of the form $Z_{S,T}$, where Z is a group, assumed commutative here for simplicity. There is a natural functor

$$E \longmapsto E_T \tag{5.8.0}$$

from the category of torsors under $Z_{S,T}$ to the category of torsors under Z_T. We claim that this functor is an equivalence of categories. Its full faithfulness means that for two torsors E, F under $Z_{S,T}$, the map

$$\operatorname{Hom}(E, F) \longrightarrow \operatorname{Hom}(E_T, F_T)$$

is bijective. By fpqc descent this reduces immediately to the case in which E and F are trivial, hence to $E = F = Z_{S,T}$, where the assertion is precisely the bijectivity of (5.7.2) for $S' = S$.

It remains to prove that (5.8.0) is essentially surjective. Let E_T be a torsor under Z_T. We already observed in the proof of 5.1 that E_T is representable and locally trivial for the etale topology. It therefore suffices to prove that

$$H^1(S_{\text{ét}}, Z_{S,T}) \longrightarrow H^1(T_{\text{ét}}, Z_T)$$

is bijective. This follows from (5.7.3) and the relation

$$R^1 i_{\text{ét},*}(Z_T) = 0,$$

which is the special case SGA 4 VIII 5.6.

5.8.1. Let now P be a commutative group scheme over S. We wish to determine the category of extensions, as sheaves of commutative groups, of P by $Z_{S,T}$. For this we use the description in VII 1.1.6 and 1.3 in terms of torsors on P, $P \times P$, and $P \times P \times P$, under the inverse images of $Z_{S,T}$. By (5.7.4), these inverse images are again of the same form as $Z_{S,T}$. Applying the preceding result to those torsors shows that they correspond to torsors on P_T, $P_T \times_T P_T$, and $P_T \times_T P_T \times_T P_T$ under the inverse images of Z_T. We therefore obtain the following.

Proposition 5.9. Let S be a scheme, T a closed subscheme, Z a commutative group, $Z_{S,T}$ the corresponding truncated constant group scheme over S defined in 5.7, and P a commutative group scheme over S. Then restriction to T induces an equivalence between the category of sheaves of commutative groups which are extensions of P by $Z_{S,T}$ and the category of sheaves of commutative groups which are extensions of P_T by Z_T.

An essentially identical argument gives the following.

Corollary 5.10. With the notation of 5.9, let Q be another commutative group scheme over S. Then restriction to T induces an equivalence between the category of biextensions of (P, Q) by $Z_{S,T}$ and the category of biextensions of (P_T, Q_T) by Z_T.

6. Extensions and biextensions by the Neron model of G_m.

6.1. Let S be a scheme, T a closed subscheme, and Z a set, giving a truncated constant scheme $Z_{S,T}$ over S (5.7). There is a canonical homomorphism, functorial in Z, of etale schemes over S,

$$Z_S \longrightarrow Z_{S,T}, \tag{6.1.1}$$

which is a surjective morphism, as follows at once from (5.7.1). Its functoriality in Z shows that when Z is a group, (6.1.1) is a homomorphism of etale group schemes over S. Its kernel is therefore an etale group scheme over S, denoted $Z_{S,U}$, and there is a canonical exact sequence, functorial in Z,

$$0 \longrightarrow Z_{S,U} \longrightarrow Z_S \longrightarrow Z_{S,T} \longrightarrow 0. \tag{6.1.2}$$

It follows immediately from (5.7.1) that $Z_{S,U}$ is the sum of a family of S-schemes S_n, $n \in Z$, where $S_n = S$ for $n = 0$ and $S_n = U$ for $n \neq 0$. This description of $Z_{S,U}$ readily shows that for every group F over S, restriction to U induces a bijection

$$\mathrm{Hom}_{\mathrm{gr}}(Z_{S,U}, F) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr}}(Z_U, F|U). \tag{6.1.3}$$

6.1.4. Henceforth we suppose that Z is commutative, and we wish to examine extensions of commutative groups of $Z_{S,T}$ by a given commutative group G over S. By the exact sequence (6.1.2), the category of these extensions is equivalent to the category of extensions F of Z_S by G, furnished with a trivialization of the inverse image of the extension along $Z_{S,U} \to Z_S$. By (6.1.3), such a trivialization amounts to a trivialization of the extension $F|U$ of Z_U by $G|U$.

For what follows we take Z to be the group $\mathbb{Z}$ of integers. Thus $\mathbb{Z}_S$ is the free group on one generator, and VII 1.4 gives an equivalence between the category of extensions of $\mathbb{Z}_S$ by G and the category of torsors under G. Applying the same result to $\mathbb{Z}_U$ and $G|U$ gives the following.

Lemma 6.2. Let S be a scheme, T a closed subscheme, $U = S \setminus T$, and G a commutative group over S. The category of commutative groups over S which are extensions of $\mathbb{Z}_{S,T}$ (defined in (5.7)) by G is equivalent to the category of pairs (F, s) consisting of a torsor F under G and a section s of $F|U$. If $\alpha : S \to \mathbb{Z}_{S,T}$ is the section of $\mathbb{Z}_{S,T}$ which is the image of the section 1 of $\mathbb{Z}_S$ over S, then the torsor corresponding to an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by G is $F = \alpha^*(\overline{G})$ under G, furnished with the section of $F|U$ given by restricting the unit section of $\overline{G}$ to U.

6.2.1. If $\overline{G}$ corresponds to the pair (F, s), then the inverse image $f_n^*(\overline{G})$ in $\overline{G}$ of the section f_n of $\mathbb{Z}_{S,T}$ defined by the constant section of $\mathbb{Z}_S$ with value $n \in \mathbb{Z}$ is canonically identified, as a torsor under G, with the n-th power $F^{(n)}$ of the G-torsor F. The canonical isomorphism of G_U-torsors

$$f_n^*(\overline{G})|U \simeq G_U,$$

defined by the first relation in (5.7.1), is determined by the section $s^{(n)}$ of $F_U^{(n)}$, the n-th power of the given section s of F_U. This makes the following description of the sections of $\overline{G}$ over S explicit. A section $\overline{g}$ of $\overline{G}$ over S is determined by:

a) a section $\underline{n}$ of $\mathbb{Z}_{S,T}$ over S, which may be interpreted as a section of $\mathbb{Z}_T$, that is, as a locally constant function on T with values in $\mathbb{Z}$, or equivalently as a partition

$$T = \coprod_{i \in \mathbb{Z}} T_i$$

of T into a disjoint sum of open subsets indexed by $i \in \mathbb{Z}$;

b) a family of sections

$$\overline{g}_i \in \Gamma(F^{(i)}/S_i'), \qquad i \in \mathbb{Z},$$
$$S_i' = S \setminus T_i', \qquad T_i' = \bigcup_{\substack{j \in \mathbb{Z} \\ j \neq i}} T_j,$$

subject to the following compatibility condition. For $i, j \in \mathbb{Z}$, $\overline{g}_i$ and $\overline{g}_j$ agree on $S_i' \cap S_j' = U$ after $F^{(i)}|U$ and $F^{(j)}|U$ have been identified with G_U by means of the sections $s^{(i)}$ and $s^{(j)}$, respectively. In other words, there must exist a necessarily unique section $h_{ij} \in \Gamma(G/U)$ such that

$$\overline{g}_i = h_{ij} s^{(i)}, \qquad \overline{g}_j = h_{ij} s^{(j)}.$$

Since the description in 6.2 of an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by G is plainly compatible with every base change $S' \to S$, the preceding gives a complete description of the sheaf $\overline{G}$ as a functor of the argument $S' \in \mathrm{Ob}(\mathrm{Sch}_{/S})$.

6.2.2. If T is a scheme which is a sum of nonempty connected schemes T_i, $i \in I$, the description in 6.2.1 of the sections of $\overline{G}$ over S may be simplified. Such a section is determined by a family of integers $(n_i)_{i \in I}$ indexed by I, and a family of sections

$$\overline{g}_i \in \Gamma(F^{(n_i)}/S_i'), \qquad i \in I,$$

where

$$S_i' = S \setminus T_i', \qquad T_i' = \coprod_{\substack{j \in I \\ j \neq i}} T_j.$$

The $\overline{g}_i$ satisfy the following compatibility condition: for every pair $i, j \in I$, the restrictions of $\overline{g}_i$ and $\overline{g}_j$ to $S_i' \cap S_j' = U$ agree under the identifications of $F^{(n_i)}|U$ and $F^{(n_j)}|U$ deduced from the respective sections $s^{(n_i)}$ and $s^{(n_j)}$ of these torsors.

6.2.3. Finally, if T is assumed nonempty and connected, giving a section $\overline{g}$ of $\overline{G}$ over S amounts to giving an integer n and a section $\overline{g}$ of $F^{(n)}$ over S. There is no longer any compatibility condition to impose, so the section s of $F|U$ does not occur in the description of the group of sections of $\overline{G}$ over S. (Of course, s occurs again as soon as one seeks to describe the points of $\overline{G}$ with values in a general S-scheme S'.)

6.2.4. Since $\overline{G}$ is the union of the "open subschemes" $f_n^*(\overline{G})$, $n \in \mathbb{Z}$, described in 6.2.1, it follows that $\overline{G}$ is representable if and only if the torsors $F^{(n)}$ are representable. For example, it suffices that G be representable and affine over S, by the descent theory of SGA 1 VIII 2.1.

6.3. In what follows we suppose that $G = \underline{G}_{mS}$. Every extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by $G = \underline{G}_{mS}$ is then representable (6.2.4), and is of course smooth over S, being an extension of the etale group scheme $\mathbb{Z}_{S,T}$ by the smooth group scheme $\underline{G}_{mS}$. Moreover, giving a torsor F under $\underline{G}_m$ is equivalent to giving an invertible Module $\mathbb{L}$ over S (by associating to the latter the torsor of invertible sections of $\mathbb{L}$). Thus $\overline{G}$ is described by a pair $(\mathbb{L}, s)$, where $\mathbb{L}$ is an invertible Module over S and s is a trivialization of $\mathbb{L}$ over $U = S \setminus T$, that is, an invertible section of $\mathbb{L}|U$.

When U is schematically dense in S, and S is locally noetherian, or is reduced with a locally finite set of irreducible components, giving such a pair is equivalent to giving a <u>divisor D on S such that</u>

$$\operatorname{supp} D \subset T, \tag{6.3.1}$$

by associating to such a divisor the pair

$$\big(\underline{O}_S(-D), s_{-D}|U\big) \tag{6.3.2}$$

formed by the invertible Module $\underline{O}_S(-D)$ associated with $-D$ (EGA IV 21.2.8.1) and the restriction to U of its canonical rational section s_{-D} (EGA IV 21.2.11 and 20.2.11(ii)). In any event, without any hypothesis on S, a divisor D on S satisfying (6.3.1) defines by (6.3.2) a pair $(\mathbb{L}, s)$ and hence an extension $\overline{G}$ of $\mathbb{Z}_{S,T}$ by $\underline{G}_m$. We call it the <u>Neron–Raynaud model of $\underline{G}_m$</u> associated with the divisor D and the closed subscheme T of S satisfying (6.3.1). If $T = \operatorname{supp} D$, we omit T from the terminology.

It is immediate that the formation of this extension $\overline{G}$ commutes with every base change $f : S' \to S$ for which the inverse-image divisor $f^*(D)$ is defined (EGA IV 21.4.2), and in particular with every <u>flat</u> base change.

6.3.3. The general description in 6.2.1 of the sections of $\overline{G}$ immediately shows that they may be identified with the sections f of the sheaf $\mathcal{M}_S^*$ of regular meromorphic sections on S (EGA IV 20.1.8) whose divisor $\operatorname{div}(f)$ is locally, for the Zariski topology, an integral multiple of D. If V is an open subset of S on which the restriction of $\operatorname{div}(f)$ equals that of nD, where $n \in \mathbb{Z}$, then the section of $\mathbb{Z}_{S,T}$ defined by f is simply the image in (6.1.2) of the constant section of $\mathbb{Z}_S$ with value n. The preceding description of the group of sections of $\overline{G}$ over S remains valid over every open subset of S, and more generally for the group of points of $\overline{G}$ with values in any flat S-scheme S'. Since $\overline{G}$ itself is flat (indeed smooth) over S, this description again gives a complete characterization of the Neron–Raynaud model of $\underline{G}_m$ defined by D and $T \supset \operatorname{supp} D$.

6.4. Suppose now that S is <u>normal</u>, that its set of irreducible components is locally finite (so that S is a sum of normal integral schemes), and that D is a positive divisor with all multiplicities equal to 1 (EGA IV 21.6), that $T = \operatorname{supp} D$, and that T is locally irreducible, that is, its irreducible components are disjoint. Then for every rational function f on S which is regular on $U = S \setminus T$, the divisor $\operatorname{div}(f)$ is locally a multiple of D. Thus in this case there is a canonical isomorphism of groups

$$\Gamma(S, \overline{G}) \xrightarrow{\sim} \Gamma(U, \underline{G}_{mU}). \tag{6.4.1}$$

Suppose further that T is geometrically unibranch and locally integral, a hypothesis stable under smooth base change (EGA IV 17.5.7). Since all the hypotheses are then stable under base change, (6.4.1) implies that <u>for every S' smooth over S there is a canonical group isomorphism, functorial in S',</u>

$$\operatorname{Hom}_S(S', \overline{G}) \xrightarrow{\sim} \operatorname{Hom}_U(S'_U, \underline{G}_{mU}) \simeq \Gamma(S'_U, \underline{O}^*_{S'_U}). \tag{6.4.2}$$

Since $\overline{G}$ is itself smooth over S, this isomorphism characterizes the group scheme $\overline{G}$ over S up to unique isomorphism. It also shows that $\overline{G}$ plays the role of a "Neron model" for the group scheme

$\underline{G}_{mU}$ over the open subset U of S. Together with the fact that it was first introduced and used by M. Raynaud, this justifies the terminology adopted in 6.3 for $\overline{G}$.

Theorem 6.5. Let S be a normal locally integral scheme and let D be a positive divisor on S, without multiple components, whose support is geometrically unibranch and locally integral. Let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$ over S relative to D (6.3), and consider the category of torsors E over S under the group scheme $\overline{G}$. The restriction functor

$$E \longmapsto E_U = E|U$$

from this category to the category of torsors over U under the group scheme $\underline{G}_{mU}$ is fully faithful. If S is regular (more generally, if its local rings are factorial), this functor is an equivalence of categories.

Let us prove that the functor in question is fully faithful, that is, that for two torsors E, F under $\overline{G}$, the restriction map

$$(*) \qquad \operatorname{Hom}(E, F) \longrightarrow \operatorname{Hom}(E_U, F_U)$$

is bijective, where Hom denotes the set of homomorphisms of torsors. First observe that this question is plainly local on S for the Zariski topology. Indeed, suppose that S is the union of open subsets S_i such that for every open subset S' of S contained in some S_i (and hence in particular for $S' = S_i$ or $S_i \cap S_j$), the map

$$\operatorname{Hom}(E_{S'}, F_{S'}) \longrightarrow \operatorname{Hom}(E_{S'\cap U}, F_{S'\cap U})$$

is bijective. Then (*) is bijective. On the other hand, it follows at once from 5.8 applied to $S, T, \mathbb{Z}$, and from 5.1 applied to $T, \mathbb{Z}$, that every torsor under $\overline{G}$ is locally trivial for the Zariski topology. These two observations reduce us to proving the bijectivity of (*) when $E = F = \overline{G}$. In that case the map is precisely (6.4.1), whose bijectivity has already been established.

We now prove that the functor $E \mapsto E_U$ is essentially surjective when the local rings of S are factorial. Since $\underline{G}_{mS}$ is identified with an open subgroup scheme of $\overline{G}$, it is enough to show that every torsor under $\underline{G}_{mU}$ is isomorphic to the restriction to U of a torsor under $\underline{G}_{mS}$. Equivalently, every invertible Module over U must be isomorphic to the restriction of an invertible Module over S. This follows from the hypothesis on S (EGA IV 21.6.11, where only the locally noetherian case is treated; that is the only case needed below). This proves 6.5.

6.5.1. Suppose now, in addition, that S is locally noetherian and regular. Since S is then locally factorial, giving D is equivalent to giving a reduced, purely 1-codimensional closed subscheme, again denoted D, which is geometrically unibranch. Notice also that the hypotheses on S and D are stable under every smooth base change $S' \to S$, so that the conclusion of 6.5 applies to S' with the subscheme $D' = D \times_S S'$. The description in VII 1.1.6 of extension groups now immediately gives the following consequence.

Corollary 6.6. Let S be a locally noetherian regular scheme, let D be a reduced, geometrically unibranch, purely 1-codimensional closed subscheme (thus defining a divisor on S), let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$ relative to D, and let P be a smooth commutative group scheme over S. Consider the category of commutative extensions, as sheaves of groups, of P by $\overline{G}$. Restriction to U, $E \mapsto E|U$, induces an equivalence between this category and the category of extensions of P_U by $\underline{G}_{mU}$.

6.6.1. The same method gives the analogous statement for extensions which are not necessarily commutative (with P no longer assumed commutative). We leave the details of the proof to the reader; they are slightly more involved because bitorsors must be considered in place of torsors.

By essentially the same proof as for 6.6, one likewise obtains:

Corollary 6.7. Let $S, D, \overline{G}$ be as in 6.6, and let P, Q be two smooth commutative group schemes over S. Consider the category of biextensions of (P, Q) by $\overline{G}$. Then the restriction functor to U,

$E \mapsto E|U$, from this category to the category of biextensions of (P_U, Q_U) by $\underline{G}_{mU}$ is an equivalence of categories.

Remarks 6.8. a) We shall apply 6.6 and 6.7 when S is a trait and D is the divisor defined by the inclusion of the closed point s of S into S. In this case U is the spectrum of the fraction field K of S, and $\overline{G}$ is the usual Neron model of the multiplicative group over K, introduced by Raynaud. Notice that even in this special case, the proof of 6.6 and 6.7 uses 6.5 over bases more general than traits (such as P and its Cartesian powers). Nevertheless, in this case it is enough to know the existence of the Neron model $\overline{G}$ in the special situation under consideration: 6.5 for base schemes smooth over S, and hence 6.6 and 6.7, then follow formally. For us, the introduction of $\overline{G}$ will chiefly serve as a convenient intermediate step in studying, in the next section, the problem of extending extensions and biextensions by the multiplicative group.

b) A Neron–Raynaud model $\overline{G}$ of $\underline{G}_{mU}$, satisfying the fundamental condition (6.4.2) for every S' smooth over S, can be defined under much more general conditions than those considered above, with U merely a schematically dense open subset of a normal locally noetherian scheme S. In this case $\overline{G}$ is an fppf sheaf, not necessarily representable. Let $\underline{D}$ denote the sheaf, on the small etale site of S, of divisors on S whose support is contained in $T = S \setminus U$, and, by abuse of language, also denote by the same letter its canonical extension to a sheaf on $(\mathrm{Sch})_{/S}$. Its sections over a relative scheme $f : S' \to S$ are the sections of $f^*_{\text{ét}}(\underline{D})$ over S'. The desired model is an extension of groups

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow \overline{G} \longrightarrow \underline{D} \longrightarrow 0. \tag{6.8.1}$$

The construction of a canonical extension (6.8.1), without any normality hypothesis on S, presents no difficulty when one uses the general description VII 1.1.6; at the same time one obtains (6.4.2) for S' etale over S. For the same relation to hold for every S' smooth over S, it is equivalent to verify that the formation of the sheaf $\underline{D}$ commutes with every smooth base change $f : S' \to S$. For normal S, this follows from EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.9.

6.9. "Self-criticism" (observations of P. Deligne).[3]

Sections 5 and 6 are really rather uninspiring. One periodically has the impression that the formalism of i_*, i^* is being redone "by hand", without seeing what is geometric and what is "general nonsense".

I propose extracting the "geometric" statements a), b), c):

a) (cf. 6.8 b)) Let $j : U \hookrightarrow S$ be a schematically dense open subset of a normal noetherian scheme S. The etale sheaf $\underline{G}_{mS}$ is then identified with a subsheaf of the etale sheaf $j_*\underline{G}_{mU}$. Let $\underline{D}$ be the quotient etale sheaf in the exact sequence

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow j_*\underline{G}_{mU} \longrightarrow \underline{D} \longrightarrow 0.$$

Then the formation of $\underline{D}$ commutes with every smooth base change (EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.9); moreover, $R^1 j_*\underline{G}_m = 0$ (Theorem 90) if the strictly local rings of S are factorial.

b) If $U = S \setminus Y$, where Y is a geometrically unibranch divisor, and if i is the inclusion of Y into S, then

$$\underline{D} = i_*\mathbb{Z}_Y.$$

c) If Y is a closed subscheme of a scheme X, and E is a set, then the etale sheaf $i_*\underline{E}_Y$ is represented by an etale scheme over X, whose formation commutes with every base change.

Corollary. Let S be a normal noetherian scheme, let Y be a geometrically unibranch divisor, and let $U = S \setminus Y$:

3. The author of the present Expose having admitted defeat, the reader is invited, if necessary, to rewrite sections 5 and 6 according to the accompanying directions.

$$U \underset{j}{\hookrightarrow} S \xleftarrow{i} Y \qquad .$$

<u>There exists a unique smooth group scheme $\mathcal{G}_m$ over S such that after every smooth base change one has $\mathcal{G}_m|S_{\text{ét}} = j_*(\underline{G_m})_U$.</u>

Indeed, the preceding requirements determine $\mathcal{G}_m$ as a sheaf, for the etale topology, on the category of schemes smooth over S. By a) and b), the sequence

$$0 \longrightarrow \underline{G}_m \longrightarrow \mathcal{G}_m \xrightarrow{q} i_*\mathbb{Z}_Y \longrightarrow 0$$

is exact. By c), $i_*\mathbb{Z}_Y$ is representable. Finally, the morphism q is representable.

Use of the preceding "small smooth site" should make it possible to deduce the statements of section 6 formally from a).

<u>7. Canonical extensions of extensions and biextensions by $\underline{G}_m$.</u>

7.0. We now reap the fruit of our preparations in sections 5 and 6. First let the base scheme S be a <u>trait</u>, that is, the spectrum of a discrete valuation ring; let η be its generic point, s its closed point, and $k = k(s)$. Let P be a smooth commutative group scheme over S, and write $P_0 = P_s$ for its special fibre. In the present section we consider only commutative groups; in particular, all group extensions under consideration are commutative.

<u>Theorem 7.1.</u> <u>The notation is that recalled in 7.0.</u>

a) <u>Suppose that the group of connected components of the special fibre of P,</u>

$$\Phi = P_0/P_0^\circ,$$

<u>is a torsion group—as is the case, in particular, if P is of finite type over S. Then the functor $E \mapsto E_\eta$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_η by $\underline{G}_{m\eta}$ is fully faithful. Suppose that there is an integer $n > 0$ such that $n\Phi = 0$. Then every extension E_η of P_η by $\underline{G}_{m\eta}$ has an associated canonical homomorphism</u>

$$d(E_\eta) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{7.1.1}$$

<u>whose vanishing is necessary and sufficient for E_η to be isomorphic to the generic fibre of an extension of P by $\underline{G}_{mS}$. In particular, if P_0 is connected, the preceding functor $E \mapsto E_\eta$ is an equivalence of categories.</u>

b) <u>Let P and Q be two smooth group schemes over S, put</u>

$$\Phi = P_0/P_0^\circ, \qquad \Psi = Q_0/Q_0^\circ,$$

<u>and suppose that either Φ or Ψ is a torsion group (for example, that either P or Q is of finite type over S). Then the functor $E \mapsto E_\eta$ from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of biextensions of (P_η, Q_η) by $\underline{G}_{m\eta}$ is fully faithful. Suppose that there is an integer $n > 0$ such that $n\Phi = 0$ or $n\Psi = 0$. Then every biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$ has an associated canonical pairing</u>

$$d(E_\eta) : \Phi \otimes_{\mathbb{Z}} \Psi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{7.1.2}$$

<u>whose vanishing is necessary and sufficient for E_η to be isomorphic to the generic fibre of a biextension E of (P, Q) by $\underline{G}_{mS}$. In particular, if either P_0 or Q_0 is connected, the preceding functor $E \mapsto E_\eta$ is an equivalence of categories.</u>

<u>Proof.</u> a) Let $\overline{G}$ be the Neron–Raynaud model of $\underline{G}_m$. The functor under consideration is the composite of the functor "extension of the kernel $\underline{G}_{mS} \to \overline{G}$" with the functor "restriction to η" defined on the category of extensions by $\overline{G}$. The latter functor is an equivalence of categories by 6.6, so it remains to study the first. For this purpose we use the exact sequence of group schemes

$$0 \longrightarrow \underline{G}_{mS} \longrightarrow \overline{G} \longrightarrow \mathbb{Z}_{S,s} \longrightarrow 0, \tag{7.1.3}$$

where the notation $\mathbb{Z}_{S,s}$ is that of 5.7. The exact sequence of Ext^i, $i = 0, 1$, associated with (7.1.3) shows that the functor in question is fully faithful if and only if

$$(7.1.4) \qquad \mathrm{Hom}(P, \mathbb{Z}_{S,s}) = 0,$$

and that the obstruction for an extension $\overline{E}$ of P by $\overline{G}$ to come from an extension E of P by $\underline{G}_{mS}$ lies in

$$(7.1.5) \qquad \mathrm{Ext}^1(P, \mathbb{Z}_{S,s}).$$

By (5.7.2), the first group in (7.1.4) is isomorphic to $\mathrm{Hom}(P_0, \mathbb{Z}_k)$, which in turn is isomorphic to $\mathrm{Hom}(\Phi, \mathbb{Z}_k)$. Thus (7.1.4) is equivalent to

$$(7.1.4\text{ bis}) \qquad \mathrm{Hom}(\Phi, \mathbb{Z}_k) = 0.$$

This relation certainly holds if Φ is a torsion group. Moreover, if $n\Phi = 0$, then 5.9 and 5.5 give canonical isomorphisms

$$(7.1.6) \qquad \mathrm{Ext}^1(P, \mathbb{Z}_{S,s}) \simeq \mathrm{Ext}^1(P_0, \mathbb{Z}_k) \simeq \mathrm{Hom}\big(\Phi, (\mathbb{Q}/\mathbb{Z})_k\big),$$

which completes the proof of 7.1(a).

b) The proof is essentially the same. By 6.7, we reduce to studying the functor "extension of the kernel $\underline{G}_{mS} \to \overline{G}$" from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of biextensions of (P, Q) by $\overline{G}$. We use the exact sequence (7.1.3) and the associated exact sequence of Biext^i, $i = 0, 1$ (VII 3.7.5). Full faithfulness of the functor is equivalent to

$$(7.1.7) \qquad \mathrm{Hom}(P \otimes Q, \mathbb{Z}_{S,s}) = 0.$$

By (5.7.2), the left side is isomorphic to $\mathrm{Hom}(P_0 \otimes Q_0, \mathbb{Z}_k)$ and hence to $\mathrm{Hom}(\Phi \otimes \Psi, \mathbb{Z}_k)$; this group is indeed zero if either Φ or Ψ is a torsion group. On the other hand, for a given biextension $\overline{E}$ of (P, Q) by $\overline{G}$, the obstruction to its coming from a biextension E of (P, Q) by $\underline{G}_{mS}$ lies in

$$\mathrm{Biext}^1(P, Q; \mathbb{Z}_{S,s}).$$

By 5.10 this is isomorphic to $\mathrm{Biext}^1(P_0, Q_0; \mathbb{Z}_k)$, which, when $n\Phi = 0$ or $n\Psi = 0$, is itself isomorphic by 5.6 to

$$\mathrm{Hom}\big(\Phi \otimes \Psi, (\mathbb{Q}/\mathbb{Z})_k\big).$$

This completes the proof of 7.1.

Remark 7.2. Given a smooth group scheme P over a connected regular locally noetherian scheme S with generic point η, one may ask to study the functor $E \mapsto E_\eta$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_η by $\underline{G}_{m\eta}$. This problem is readily reduced to the one treated in 7.1(a). For simplicity, suppose that P is of finite type over S. It follows easily from 7.1 that this functor is fully faithful.[4] One readily deduces that if E_η is an extension of P_η by $\underline{G}_{m\eta}$, then it is isomorphic to the generic fibre of an extension E of P by $\underline{G}_{mS}$ if and only if, for every $s \in S$ such that

$$\dim \mathrm{Spec}(\underline{O}_{S,s}) = 1,$$

the analogous extension problem obtained by base change $\mathrm{Spec}(\underline{O}_{S,s}) \to S$ has a solution. By 7.1(a), this is expressed by the vanishing of a certain homomorphism

$$(7.2.1) \qquad d(E_\eta, s) : P_s/P_s^\circ \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k(s)}.$$

The preceding assertion plainly reduces to the following one. Let U be an open subset of S whose complement T satisfies

$$\mathrm{codim}(T, S) \geq 2.$$

4. For this it is enough that S be normal rather than regular.

Then the functor $E \mapsto E|U$ from the category of extensions of P by $\underline{G}_{mS}$ to the category of extensions of P_U by $\underline{G}_{mU}$ is an equivalence of categories. In terms of the description VII 1.1.6 this assertion is essentially trivial: for every smooth S-scheme X (such as P, $P \times P$, $P \times P \times P$, and so on), X is locally factorial and $X_T = X \setminus X_U$ has codimension at least 2 in X. Consequently, the functor $F \mapsto F|X_U$ from the category of torsors under $\underline{G}_{mX}$ to the category of torsors under $\underline{G}_{mX_U}$ is an equivalence of categories.

There are entirely analogous results for the problem of extending biextensions; we leave their formulation to the reader.

7.3. We shall make explicit some functorial properties of the obstructions defined in (7.1.1) and (7.1.2), with P, Q, S variable, and some relations among them. In what follows, we shall suppose that the conditions allowing these obstructions to be defined are satisfied. Since the verifications present no difficulty, we leave them to the reader.

7.3.1. Let

$$u : P' \longrightarrow P$$

be a morphism of smooth group schemes over S, and let E_η be an extension of P_η by $\underline{G}_{m\eta}$. Pullback by $u_\eta : P'_\eta \to P_\eta$ gives an extension E'_η of P'_η by $\underline{G}_{m\eta}$. With this notation, the following diagram is commutative:

(7.3.1.1)
$$\begin{tikzcd} P'_0/P'^{\circ}_0 \arrow[d, "u_0"'] \arrow[dr, "d(E'_\eta)"] & \\ & (\mathbb{Q}/\mathbb{Z})_k \\ P_0/P^{\circ}_0 \arrow[ur, "d(E_\eta)"'] & \end{tikzcd} .$$

Similarly, if we have morphisms of smooth group schemes over S

$$u : P' \longrightarrow P, \qquad v : Q' \longrightarrow Q,$$

and a biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$, whose pullback by (u_η, v_η) is a biextension E'_η of (P'_η, Q'_η) by $\underline{G}_{m\eta}$, then the following diagram is commutative:

(7.3.1.2)
$$\begin{tikzcd} P'_0/P'^{\circ}_0 \otimes Q'_0/Q'^{\circ}_0 \arrow[d, "u_0 \otimes v_0"'] \arrow[dr, "d(E'_\eta)"] & \\ & (\mathbb{Q}/\mathbb{Z})_k \\ P_0/P^{\circ}_0 \otimes Q_0/Q^{\circ}_0 \arrow[ur, "d(E_\eta)"'] & \end{tikzcd}$$

The essentially mechanical verification of these compatibilities is left to the reader.

7.3.2. It follows from the compatibility (7.3.1.1) that, if E_η is an extension of P_η by $\underline{G}_{m\eta}$, there is a largest group subscheme V of P such that the restriction of E_η to V_η extends to an extension of V by $\underline{G}_{mS}$. It is the open subgroup V of P containing P_η and defined by the condition

(7.3.2.1)
$$V_0/V^{\circ}_0 = \ker d(E_\eta).$$

If $u : P' \to P$ is then a morphism of smooth group schemes over S, the extension E'_η, obtained by pulling E_η back along u_η, extends to an extension of P' by $\underline{G}_{mS}$ if and only if

(7.3.2.2)
$$u(P') \subset V.$$

In the case of a biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$, there is likewise a largest group subscheme V of P such that the restriction of E_η to (V_η, Q_η) extends to a biextension of (V, Q) by $\underline{G}_{mS}$. It is the open subgroup of P containing the generic fibre P_η and characterized by the fact that V_0/V_0° is the left annihilator in Φ of the pairing (7.1.2), that is, the kernel of the corresponding homomorphism from Φ to $\underline{\mathrm{Hom}}(\Psi, (\mathbb{Q}/\mathbb{Z})_k)$:

$$V_0/V_0^\circ = \ker\left(\Phi \xrightarrow{d(E_\eta)} \underline{\mathrm{Hom}}(\Psi, (\mathbb{Q}/\mathbb{Z})_k)\right). \tag{7.3.2.3}$$

If $u : P' \to P$ is a morphism of smooth group schemes over S, the biextension of (P'_η, Q_η) by $\underline{G}_{m\eta}$ obtained by pulling E_η back along u_η extends to a biextension of (P', Q) by $\underline{G}_{mS}$ if and only if (7.3.2.2) holds. Symmetric results hold upon interchanging the roles of P and Q.

Of course, in general there is no largest pair of subgroups $P' \to P$, $Q' \to Q$ for which the restriction E'_η of E_η to a biextension of (P'_η, Q'_η) by $\underline{G}_{m\eta}$ extends to a biextension of (P', Q') by $\underline{G}_{mS}$. One can only say that, for a given pair (P', Q'), or more generally for a given pair of morphisms $P' \to P$, $Q' \to Q$ of smooth group schemes, the desired extension exists if and only if the restriction of the form $d(E_\eta)$ to

$$P'_0/P'^\circ_0 \otimes Q'_0/Q'^\circ_0$$

is zero.

7.3.3. Symmetry. Let E_η be a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$, and consider the transposed biextension ${}^sE_\eta$ (VII 2.7), which is a biextension of (Q_η, P_η) by $\underline{G}_{m\eta}$. The following diagram is commutative:

$$\begin{tikzcd} \Phi \otimes \Psi \arrow[dd, "\mathrm{sym}"'] \arrow[dr, "d(E_\eta)"] & \\ & (\mathbb{Q}/\mathbb{Z})_k \\ \Psi \otimes \Phi \arrow[ur, "d({}^sE_\eta)"'] & \end{tikzcd}, \tag{7.3.3.1}$$

where the vertical arrow is the canonical symmetry. (Compare this statement with the antisymmetry statement of 2.2.10.) It follows in particular that, if $P = Q$ and E_η is a symmetric biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$, that is, if E_η is isomorphic to ${}^sE_\eta$, then the bilinear form $d(E_\eta)$ on $\Phi \times \Phi$ with values in $(\mathbb{Q}/\mathbb{Z})_k$ is symmetric (and not alternating, as in 2.2.12!). If it is zero, that is, if E_η extends to a biextension E of (P, P) by $\underline{G}_{mS}$, the latter is necessarily symmetric as well: by 7.1(a), every isomorphism ${}^sE_\eta \to E_\eta$ extends uniquely to an isomorphism ${}^sE \to E$.

7.3.4. Reduction of obstructions of type (7.1.2) to those of type (7.1.1).

Suppose, for definiteness, that there is an integer $n > 0$ such that $n\Phi = 0$, and let

$$q \in Q(S)$$

be a section of Q. By an abuse of notation, write

$$q_0 \in \Psi(k)$$

for the image in $\Psi(k)$ of the value of q at s. Regarding E_η as an extension of $P_\eta \times_\eta Q_\eta$ by $\underline{G}_{mQ_\eta}$ over Q_η, pullback gives an extension

$$q^*_\eta(E_\eta), \qquad \text{an extension of } P_\eta \text{ by } \underline{G}_{m\eta},$$

and hence an obstruction

$$d_q \overset{\text{dfn}}{=} d\big(q_\eta^*(E_\eta)\big) : \Phi \longrightarrow (\mathbb{Q}/\mathbb{Z})_k. \tag{7.3.4.1}$$

With $d = d(E_\eta)$, one has the relation

$$d_q(x) = d(x, q_0), \tag{7.3.4.2}$$

valid for every point x of Φ with values in a k-scheme T.

7.3.4.3. For every $q \in Q(S)$, this formula determines d_q in terms of the pairing d; conversely, it plainly recovers d from the invariants d_q, at least when every point of Ψ comes from a section of Q. By Hensel's lemma, this condition holds automatically when S is strictly local. The pairing d is known once it is known over the separable closure of k, and its formation commutes with etale localization on S by 7.3.5.3 below. Thus, after replacing S by its strict localization if necessary, the determination of d reduces to that of the invariants d_q.

In particular, when every point of Ψ comes from a section q of Q, the pairing d is zero—that is, E_η extends to a biextension E of (P, Q) by $\underline{G}_m$—if and only if d_q is zero for every section q of Q, that is, if and only if $q_\eta^*(E_\eta)$ extends to an extension of P by $\underline{G}_m$. With no hypothesis on Ψ, the analogous criterion remains valid provided that the q are allowed to be points with values in local etale S-schemes S'.

7.3.4.4. Conversely, starting from the situation of 7.1(a), the obstruction (7.1.1) may be interpreted as an obstruction of type (7.1.2): take $Q = \mathbb{Z}_S$ and the biextension of $(P_\eta, \mathbb{Z}_\eta)$ deduced from E_η. Indeed, it suffices to apply the preceding construction with q equal to the section 1 of $\mathbb{Z}_S$.

7.3.5. <u>Effect of a base change $S' \to S$.</u> Suppose given a dominant local morphism of traits

$$f : S' \longrightarrow S,$$

and denote by η' the generic point of S', by s' its closed point, and put $k' = k(s')$. If $S = \operatorname{Spec}(V)$ and $S' = \operatorname{Spec}(V')$, then V' is a discrete valuation ring dominating the discrete valuation ring V. Put

$$e(S'/S) = e(V'/V) = \operatorname{length}_{V'}(V'/\mathfrak{m}V'), \tag{7.3.5.1}$$

where $\mathfrak{m}$ is the maximal ideal of V.

Let now P be as in 7.1(a), let E_η be an extension of P_η by $\underline{G}_{m\eta}$, put $P' = P \times_S S'$, and let $E'_{\eta'}$ be the extension of $P'_{\eta'}$ by $\underline{G}_{m\eta'}$ obtained by pulling back E_η. With the notation of 4.1(a), there is a canonical isomorphism

$$\Phi' \overset{\text{dfn}}{=} P'_0/P'^{\circ}_0 \simeq \Phi \otimes_k k' \simeq \Phi_{k'},$$

and hence one may consider

$$d(E_\eta)_{k'} : \Phi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k'}.$$

Then

$$d(E'_{\eta'}) = e(S'/S)\, d(E_\eta)_{k'}. \tag{7.3.5.2}$$

Likewise, under the hypotheses of 7.1(b) and with the evident notation, (7.3.5.2) holds with $d(E_\eta)$ now denoting the pairing (7.1.2), $d(E_\eta)_{k'}$ the pairing

$$\Phi' \otimes \Psi' \longrightarrow (\mathbb{Q}/\mathbb{Z})_{k'}$$

obtained from it by base change $k \to k'$, and $E'_{\eta'}$ the biextension of $(P'_{\eta'}, Q'_{\eta'})$ by $\underline{G}_{m\eta'}$ obtained from the biextension E_η of (P_η, Q_η) by $\underline{G}_{m\eta}$ through the base change $\eta' \to \eta$.

7.3.5.3. An interesting case is $e(S'/S) = 1$, that is,

$$\mathfrak{m}' = \mathfrak{m}V',$$

where $\mathfrak{m}'$ denotes the maximal ideal of V'. This holds in particular if V' is etale over V, or if V' is the henselization, strict henselization, or completion of V. In this case (7.3.5.2) simply says that the formation of obstruction (7.1.1), respectively (7.1.2), commutes with the base change under consideration $S' \to S$.

7.3.5.4. On the other hand, (7.3.5.2) in the general case shows that, for every extension E_η as in 7.1(a) (respectively every biextension E_η as in 7.1(b)), there is a finite morphism of traits $S' \to S$ such that $k(\eta')$ is a separable extension of $k(\eta)$ and such that the extension (respectively biextension) $E'_{\eta'}$ obtained from E_η by base change $\eta' \to \eta$ extends to an extension (respectively biextension) E' over all of S'. Indeed, it is enough to choose S'/S so that $e(S'/S)$ is a multiple of the integer $n > 0$ occurring in 7.1, and such an extension can always be found with $k(\eta')$ separable over $k(\eta)$.

Proposition 7.4. Let S be a reduced scheme.

a) Let P be a smooth group scheme over S whose fibres at the maximal points of S are connected. Then the natural functor

$$\mathrm{EXT}(P, \underline{G}_{mS}) \longrightarrow \mathrm{TORSRIG}(P, e_P; \underline{G}_{mS}) \tag{7.4.1}$$

from the category of extensions of P by $\underline{G}_{mS}$ to the category of torsors under $\underline{G}_{mP}$ rigidified along the identity section of P is fully faithful. If S is normal and P has connected fibres, an object E of the target of (7.4.1) belongs to the essential image of this functor if and only if, for every maximal point η of S, the rigidified torsor E_η on P_η comes from an extension of P_η by $\underline{G}_{m\eta}$.

b) Let P, Q be two smooth group schemes over S whose fibres at the maximal points of S are connected. Then the natural functor

$$\mathrm{BIEXT}(P, Q; \underline{G}_{mS}) \longrightarrow \mathrm{TORSBIRIG}(P, Q; \underline{G}_{mS}) \tag{7.4.2}$$

from the category of biextensions of (P, Q) by $\underline{G}_{mS}$ to the category of torsors under $\underline{G}_{m,P\times_S Q}$ birigidified along the pair of identity sections of P and Q is fully faithful. If S is normal and either P or Q has connected fibres, an object E of the target of (7.4.2) belongs to the essential image of the functor if and only if, for every maximal point η of S, E_η comes from a biextension of (P_η, Q_η) by $\underline{G}_{m\eta}$.

We prove (a); the essentially identical proof of (b) is left to the reader. For the full-faithfulness assertion, note that 4.1 remains valid if X and Y are flat schemes over a reduced scheme S whose fibres at the maximal points satisfy the conditions stated in 4.1. Indeed, the family of maximal fibres of $X \times_S Y$ (which is flat over the reduced scheme S) is schematically dense in $X \times_S Y$. Hence a morphism from $X \times_S Y$ to the separated S-scheme $\underline{G}_{mS}$ is trivial as soon as it is trivial on the maximal fibres. Thus the full faithfulness of (7.4.1) is proved as in 4.2, and the characterization of its essential image given there remains valid.

It remains to prove that, if the rigidified torsor E on P is such that the E_η come from extensions, then so does E, assuming that S is normal and P has connected fibres. The latter condition implies that P is of finite presentation over S (SGA 3 VI_B 5.5). Full faithfulness also shows that the question is local on S, so that we may suppose S affine and local. The usual limit argument, using Nagata's theorem on the finiteness of the integral closure of a reduced finitely generated $\mathbb{Z}$-algebra (EGA IV 7.8), reduces us further to the case in which S is noetherian.

Then 7.1(a) shows that there is an open subset $U \subset S$ whose complement has codimension at least 2 and such that E_U comes from an extension of P_U by $\underline{G}_{mU}$. In other words, the desired isomorphism

$$\pi^*(E) \simeq \mathrm{pr}_1^*(E)\,\mathrm{pr}_2^*(E)$$

can be constructed over $(P \times_S P)_U$. This is an open subset of the normal scheme $P \times_S P$ whose complement has codimension at least 2. The isomorphism on $(P \times_S P)_U$ therefore extends over all of $P \times_S P$, completing the proof.

Corollary 7.5. Let S be a normal scheme and let P and Q be two smooth group schemes over S, with connected fibres at the maximal points, either P or Q having connected fibres. Suppose in addition that, for every maximal point η of S, either P_η or Q_η is an extension of an abelian scheme by a torus (or, more generally, admits a composition series whose factors are abelian schemes, tori, or groups $\underline{G_a}$). Then the functor (7.4.2) is an equivalence of categories.

This is an immediate consequence of 7.4(b) and 4.7.

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Exposé IX

NÉRON MODELS AND MONODROMY

by A. Grothendieck
(with an appendix by M. Raynaud)

Contents

0. Introduction

0.1. In the present exposé we apply the results of the preceding exposés to the study of the Néron model (1.1) A of an abelian scheme A_K defined over the function field K of the Henselian trait S. We shall see that the most important properties of the Néron model can be interpreted in terms of the action of the local monodromy group I on the Tate module $T_\ell(A_K(\overline{K}))$ (where ℓ is a prime number distinct from p). Since the dual of the latter can be interpreted as the ℓ-adic cohomology group $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$, and since, for every smooth projective scheme X over K, $H^1(X_{\overline{K}}, \mathbb{Z}_\ell)$ is identified with $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$, where $A_{\overline{K}}$ is the Albanese variety of $X_{\overline{K}}$, the present exposé may also be viewed as a detailed study of local monodromy phenomena for ℓ-adic H^1 (or, better still, for "motivic" H^1) of smooth projective varieties over K. From this point of view, it seems clear that the principal results of the present exposé are destined to be subsumed within a "Néron theory" for motives of arbitrary weight, that is, for H^i (ℓ-adic, de Rham, Hodge, and so forth) with arbitrary i, of which at present we are only beginning to catch a glimpse. (See [9] in this connection, and more particularly Deligne's Conjectures 9.8 through 9.13 in the cited report.)

0.2. The most important result of the present exposé is the "semistable reduction theorem" 3.6, which asserts that, after a finite separable extension K' of K, the Néron model of $A_{K'}$ has a special fiber whose identity component is an extension of an abelian scheme by a torus. This result was conjectured by J.-P. Serre in 1964, and the principle of the proof given here [5] goes back to the same year. It appears in 3.5 that this property of "semistable reduction" is equivalent to the action of the monodromy group I on $T_\ell(A_K(\overline{K}))$ being "essentially unipotent of step 2" (where ℓ is a fixed prime distinct from the residual characteristic p of S). In this form it was proved, in terms of ℓ-adic H^1, in Exposé III as a special case of the "monodromy theorem", using resolution of singularities for excellent schemes of dimension 2 (due to S. Abhyankar). To prove the equivalence of the two properties, we shall also need an "orthogonality theorem" 2.4, due to Igusa [11] in the special case in which A_K is the Jacobian of a smooth curve X_K over K, the generic fiber of a regular connected projective scheme X over S whose geometric special fiber has no singular points other than ordinary quadratic singularities. This is indeed a semistable-reduction case, but it would not suffice for our purposes; it is chiefly in order to establish the orthogonality theorem in its general form that we use the convenient tool supplied by the theory of biextensions developed in the two preceding exposés. Let us also point out that D. Mumford has found a very different proof of the semistable reduction theorem, one which does not use resolution of singularities and instead uses his theory of theta functions (unpublished); unfortunately, he must restrict himself to residual characteristic different from 2.

0.3. The semistable reduction theorem is the principal tool used in the subsequent sections to study Néron models, which we undertake here from the "algebraic" point of view. It also seems indispensable for studying abelian varieties over K from the "rigid-analytic" point of view, or from the point of view of "theta functions" (these terms apparently being synonymous when abelian varieties over complete valued fields are in question), as developed recently by D. Mumford and M. Raynaud following work of Tate and Morikawa. (In Sections 13 and 14 we shall say a few words about the relations between the algebraic and transcendental points of view.) Finally, the same theorem is a key ingredient in the proof of the "semistable reduction theorem for algebraic curves" of Artin–Deligne–Mumford [3], [6] which we have already had occasion to use in the study of the action of monodromy on the fundamental group (the "essentially tame action theorem" of Exposé V), and which is used again in the proof of the integrality and positivity theorem 10.4 in the present exposé. [7]

0.4. The present exposé differs from the other exposés of the Seminar in that, in studying the Tate module $T_\ell(A_K)$, we have systematically taken into account also the case in which ℓ equals the residual characteristic p. It turns out that the language of Barsotti–Tate groups [33] [8] makes it possible to extend to this case all the most important phenomena established for $\ell \neq p$, subject on occasion to having available Tate's fundamental theorem [33, Theorem 4], which at present has been established only when K has characteristic zero. Of course, in the latter case the pro-p Barsotti–Tate group $T_p(A_K)$ may still be interpreted in terms of the action of the monodromy group I on the p-adic module $T_p(A_K(\overline{K}))$, but one should bear in mind that most of the statements suggested by this superficial analogy with the case $\ell \neq p$ are grossly false. (Thus it is no longer true that the monodromy action is essentially unipotent when $\ell = p$.) By using more judicious translations, however (such as 5.13 below), it is already clear that the study of pro-p Barsotti–

5. It is developed in a letter from the lecturer to J.-P. Serre dated 30 October 1964.

6. P. Deligne gave the proof of this theorem in the oral Seminar, but it was not considered useful to reproduce it in the Seminar texts.

7. There is another proof of the latter theorem, by way of rigid-analytic theory, which does not use the semistable reduction theorem.

8. Or "p-divisible groups" in Tate's terminology.

Tate groups defined geometrically over the function field of a trait of residual characteristic $p > 0$, whether or not the trait has mixed characteristic, must be regarded as part of the theory of "local monodromy", in which these groups play the role of p-adic local systems on $\mathrm{Spec}(K)$. In fact, these are essentially the "local systems" that enter into the study, from the "p-adic" point of view, of H^1 of smooth projective varieties over K. For the still-to-be-developed study of higher H^i from the p-adic point of view, it appears that the category of pro-p Barsotti–Tate groups over K, respectively over S, will have to be suitably enlarged, taking inspiration from the viewpoints provided by "crystalline cohomology".

0.5. The exposé given here follows the oral lectures very closely up to Section 4, but is markedly more detailed and complete from the following sections onward, which had been only partially sketched. It will not be used in the subsequent exposés. Some results of the present exposé (10.4 and 12.5), on the other hand, use the Picard–Lefschetz formula, which will be established only later.

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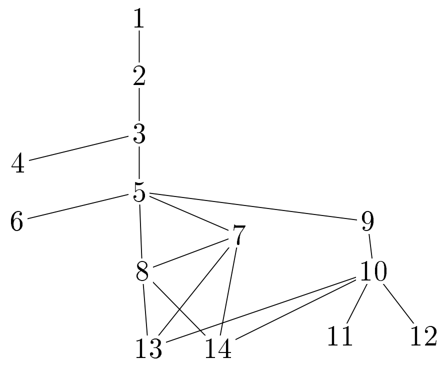

1. Neron models of abelian schemes: notation; the canonical pairing

$$\Phi_o \times \Phi'_o \longrightarrow (\mathbb{Q}/\mathbb{Z})_k,$$

and the canonical biextension W^o of (A^o, A'^o).

1.0. Throughout this exposé we fix a trait S, for which we use the usual notation s, η, k, K, and so forth, of I 1. Let A_K be an abelian scheme over K, and denote its dual abelian scheme [5] by A'_K. The duality between A_K and A'_K is given by a divisorial correspondence W_K on (A_K, A'_K), that is, an invertible sheaf on $A_K \times A'_K$, birigidified with respect to the identity sections of A_K and A'_K. It is determined up to unique isomorphism by its class (cf. VIII 3.1)

$$w_K \in \mathrm{Biext}^1(A_K, A'_K; \mathbb{G}_{mK}). \tag{1.0.1}$$

An important role will be played by the corresponding pairing (VIII 2.2)

$$\varphi : T_\ell(A_K) \times T_\ell(A'_K) \longrightarrow \mathbb{Z}_\ell(1)_K \overset{\text{def}}{=} T_\ell(\mathbb{G}_{mK}), \tag{1.0.2}$$

where ℓ is an arbitrary prime number. When $\ell \neq p = \mathrm{char}\, k$, giving φ is equivalent to giving a pairing of finite free $\mathbb{Z}_\ell$-modules

$$\varphi : T_\ell(A_K(\overline{K})) \times T_\ell(A'_K(\overline{K})) \longrightarrow T_\ell(\overline{K}^*), \tag{1.0.3}$$

compatible with the action of

$$\pi = \mathrm{Gal}(\overline{K}/K)$$

on the three modules in question, where $\overline{K}$ is a separable closure of K. It is known (VIII 3.2) that this pairing is a perfect duality: it identifies each of the modules $T_\ell(A_K(\overline{K}))$ and $T_\ell(A'_K(\overline{K}))$ with the dual of the other, with values in the invertible module $\mathbb{Z}_\ell(1) = T_\ell(\overline{K}^*)$.

At times we shall suppose that a polarization of A_K has been chosen; thus we are given an element

$$\xi \in \mathrm{NS}(A_K/K) = \underline{\mathrm{NS}}_{A_K/K}(K), \tag{1.0.4}$$

where

$$\underline{\mathrm{NS}}_{A_K/K} \stackrel{\mathrm{def}}{=} \underline{\mathrm{Pic}}_{A_K/K}/\big(\underline{\mathrm{Pic}}^o_{A_K/K} = A'_K\big).$$

In the usual way it defines a homomorphism

$$\psi_\xi : A_K \longrightarrow A'_K, \tag{1.0.5}$$

hence a homomorphism

$$T_\ell(\psi_\xi), \quad \text{or simply} \quad \psi_\xi : T_\ell(A_K) \longrightarrow T_\ell(A'_K), \tag{1.0.6}$$

and a bilinear form φ_ξ

$$\begin{gathered} \varphi_\xi(x,y) = \varphi(x, \psi_\xi(y)), \\ \varphi_\xi : T_\ell(A_K) \times T_\ell(A_K) \longrightarrow \mathbb{Z}_\ell(1)_K, \end{gathered} \tag{1.0.7}$$

which, for $\ell \neq p$, may also be interpreted as a pairing of finite free $\mathbb{Z}_\ell$-modules with operators:

$$\varphi_\xi : T_\ell(A_K(\overline{K})) \times T_\ell(A_K(\overline{K})) \longrightarrow T_\ell(\overline{K}^*). \tag{1.0.8}$$

These constructions make sense for every element of the Neron–Severi group of A_K, and yield an <u>alternating</u> form φ_ξ, in either interpretation (1.0.7) or (1.0.8) (VIII 2.2.11). The fact that ξ is a <u>polarization</u> implies that ψ_ξ in (1.0.5) is an isogeny; equivalently, since $\dim A_K = \dim A'_K$, it is an epimorphism of groups. This is also equivalent to the injectivity of $T_\ell(\psi_\xi)$ in (1.0.6) (or to its having finite cokernel), and finally to the form φ_ξ in (1.0.7) being <u>separating</u>.

1.1. Recall that there exists a <u>smooth scheme</u> A over S, called the <u>Neron model of A_K</u>, which represents the functor

$$S' \longmapsto \mathrm{Hom}(S'_\eta, A_K) \tag{1.1.1}$$

on the category of smooth S-schemes. Thus, by definition, there is an isomorphism, functorial in the smooth relative S-scheme S',

$$\mathrm{Hom}_S(S', A) \simeq \mathrm{Hom}_K(S'_\eta, A_K). \tag{1.1.2}$$

Neron [19] constructed A when the residue field k of S is perfect. Raynaud extended the construction to the general case in an unwritten seminar held at IHES in 1966–67 (see [21] for the statement of Raynaud's principal results, which complete those of Neron in loc. cit.). We shall admit this existence result here, together with the fact that A is a quasi-projective <u>scheme</u> over S. Applying (1.1.2) when S' is smooth <u>over</u> η, one also finds that the following canonical morphism is an isomorphism:

$$A \times_S \mathrm{Spec}(K) \xrightarrow{\sim} A_K. \tag{1.1.3}$$

This justifies the paired notation A, A_K; when convenient we shall also write A_η in place of A_K. Using (1.1.3) as an identifying isomorphism, the canonical bijection (1.1.2) is simply the map given by <u>restriction to the generic fibers</u>.

Since the functor (1.1.1) plainly has the structure of a commutative group, the Neron model A representing it itself has the structure of a <u>smooth commutative group scheme over S</u>. The remainder of this exposé is devoted to the study of this group scheme.

An important invariant arising from the Neron model is, of course, the smooth commutative group scheme of finite type over k

$$A_o = A \times_S s = A_k. \tag{1.1.4}$$

Following the general notation, we denote its identity component by A_o^o and also consider the group scheme

$$\Phi_o = A_o / A_o^o \tag{1.1.5}$$

of connected components of A_o, which is finite and etale over k. In terms of a separable closure $\overline{k}$ of k, it is therefore determined by the ordinary finite group

$$\Phi_o(\overline{k}), \tag{1.1.6}$$

together with the natural action of

$$\pi_o = \mathrm{Gal}(\overline{k}/k)$$

on this group.

Along with the Neron model A of A_K, we shall study

$$A' = \text{the Neron model of } A'_K, \tag{1.1.7}$$

which is likewise a smooth commutative group scheme of finite type over S. We shall denote by

$$\Phi'_o = A'_o / A'^o_o \tag{1.1.8}$$

the group of connected components of its special fiber A'_o.

1.2. Apply VIII 7.1(b) to the pair (A, A') and to the biextension W_K of (A_K, A'_K) by $\mathbb{G}_{mK}$ in (1.0.1). We thereby obtain a <u>canonical pairing</u>

$$\Phi_o \times \Phi'_o \longrightarrow (\mathbb{Q}/\mathbb{Z})_k, \tag{1.2.1}$$

that is, a pairing of π_o-modules

$$\Phi_o(\overline{k}) \times \Phi'_o(\overline{k}) \longrightarrow \mathbb{Q}/\mathbb{Z}. \tag{1.2.2}$$

Here the pairing appears as the obstruction to extending W_K to a biextension W of (A, A') by $\mathbb{G}_{mS}$; the meaning of this point of view is explained in VIII 7.3.2. We record at once the following conjecture concerning this pairing.

<u>Conjecture 1.3.</u> <u>The pairing</u> (1.2.1) <u>is a perfect duality</u>: <u>it identifies each of the component groups</u> Φ_o <u>and</u> Φ'_o <u>with the dual of the other, with values in</u> $(\mathbb{Q}/\mathbb{Z})_k$.

1.3.1. The formation of (1.2.1) commutes with formally etale base change (VIII 7.3.5.3), and the formation of Neron models has the analogous property. Consequently, to prove Conjecture 1.3 it would suffice to treat the case in which S is complete and has separably closed residue field. Artin and Mazur proved the conjecture when A_K is the Jacobian of a proper smooth curve over K, using their general, unpublished theory of the self-duality of the relative Jacobian (in the sense of derived categories) of a proper curve scheme over S. They also proved Conjecture 1.3 when the residue field is <u>finite</u>, using Tate duality [32]. We shall likewise see in 11.4 and 11.5 (subject to compatibility checks) that 1.3 holds on the ℓ-primary components for $\ell \neq p$, and that it holds in the case of "semistable reduction". These results make 1.3 extremely plausible. Moreover, 1.3 may also be regarded as a by-product of a general duality theory for group schemes over fraction fields of complete discrete valuation rings, formally analogous to SGA 5 I. Such a theory remains heuristic at present; it would also contain local class field theory in the form given to it by Serre [27].

1.3.2. For a given abelian scheme A_K, the validity of 1.3 means that the pairing (1.2.1) is separating on both the left and the right. By VIII 7.3.2, separation on the left says precisely that the only open subgroup scheme V of A for which W_K can be extended to a biextension of (V, A') by $\mathbb{G}_{mS}$ is the identity component A^o of A. There is a symmetric interpretation of separation on the right.

1.4. More generally, suppose that

$$U \subset A \qquad \text{and} \qquad U' \subset A' \tag{1.4.1}$$

are open subgroup schemes such that W_K extends to a biextension of (U, U') by $\mathbb{G}_{mS}$. By VIII 7.3.2, this says explicitly that the subgroups

$$F_o = U_o/U_o^o \subset \Phi_o, \qquad F'_o = U'_o/{U'}^o_o \subset \Phi'_o \tag{1.4.2}$$

defined by U and U' are orthogonal under (1.2.1). [9] We denote this extension (unique up to unique isomorphism) of W_K by W. It is determined up to unique isomorphism by its class

$$w \in \mathrm{Biext}^1(U, U'; \mathbb{G}_{mS}), \tag{1.4.3}$$

which is itself characterized by the condition that its image in $\mathrm{Biext}^1(A_K, A'_K; \mathbb{G}_{mK})$ be the class w_K of (1.0.1).

For example, one may take for U and U' the identity components A^o and A'^o of A and A'. One thereby obtains a canonical biextension W^o, or W, of (A^o, A'^o) by $\mathbb{G}_{mS}$, giving rise to a class

$$w = w^o \in \mathrm{Biext}^1(A^o, A'^o; \mathbb{G}_{mS}), \tag{1.4.4}$$

with which we shall work chiefly. Evidently, the restrictions to (A^o, A'^o) of the classes w in (1.4.3), for the various pairs (U, U'), are all equal to w^o in (1.4.4).

Proposition 1.5. Consider the special fiber W_s of the canonical biextension W of (A^o, A'^o) by $\mathbb{G}_{mS}$ in (1.4.4), so that W_s is a biextension of $(A_o^o, {A'}^o_o)$ by $\mathbb{G}_{mk}$. This biextension is separating (VIII 4.11).

Indeed, choose an ample invertible sheaf $\mathbb{L}$ on A^o. Then $\mathbb{L}_\eta$ defines a homomorphism (cf. 1.0.5)

$$A_K \longrightarrow A'_K,$$

which extends to a homomorphism $A \longrightarrow A'$, and hence gives

$$\psi : A^o \longrightarrow A'^o.$$

We may consider the biextension $W_{\mathbb{L}}$ of (A^o, A^o) by $\mathbb{G}_{mS}$ obtained by pulling back W along $(\mathrm{id}_{A^o}, \psi)$. Its restriction to the generic fibers is precisely the biextension $\delta(\mathbb{L}_\eta)$ of VIII 4.12. Hence, by VIII 7.4, $W_{\mathbb{L}}$ is associated with the birigidified invertible sheaf $\delta(\mathbb{L})$, and consequently $(W_{\mathbb{L}})_s$ is defined by $\delta(\mathbb{L}_s)$. On the other hand, it may also be described as the pullback of W_s by the morphisms

$$\mathrm{id}_{A_o^o} : A_o^o \longrightarrow A_o^o, \qquad \psi_o : A_o^o \longrightarrow {A'}^o_o.$$

Since $\mathbb{L}$, and hence $\mathbb{L}_s$, is ample, VIII 4.13 shows that $W_{\mathbb{L}_s}$ is separating, in particular separating on the left. A fortiori W_s is therefore separating on the left. By symmetry it is also separating on the right, which proves 1.5.

We shall substantially sharpen 1.5 when A_o^o has unipotent rank zero (5.4).

2. Fixed and toric parts of $T_\ell(A_K)$. Criteria for good reduction. Orthogonality theorem for $\ell \neq p$.

9. The printed formula, confirmed in a direct high-resolution check, reuses Φ'_o for the first subgroup and writes $\Psi'_o \subset \Psi_o$ for the second, colliding with the established notation for the component group of A'. The neutral local names F_o, F'_o above make the intended two subgroups unambiguous without changing the assertion.

2.1. Like every commutative algebraic group over k, A_o has a largest subtorus, whose formation commutes with every extension of the base field [SGA 3 XVII 7.2.1, XII 1.12]:

$$T_o \subset A_o. \tag{2.1.1}$$

When k is perfect, T_o is characterized by the condition that A_o^o/T_o be an extension of an abelian scheme B_o over k by a smooth connected unipotent group [25]. The latter necessarily has a composition series all of whose quotients are isomorphic to $\mathbb{G}_{ak}$ (SGA 3 XVII 4.1.3), and one has an exact sequence

$$0 \longrightarrow U_o \longrightarrow A_o^o/T_o \longrightarrow B_o \longrightarrow 0. \tag{2.1.2}$$

In any case, whether or not k is perfect, <u>if</u> A_o^o is an extension of an abelian scheme B_o by a smooth connected affine scheme L_o,

$$0 \longrightarrow L_o \longrightarrow A_o^o \longrightarrow B_o \longrightarrow 0, \tag{2.1.3}$$

this extension structure is known to be unique [25]. Moreover, $T_o \subset L_o$, and T_o is the largest subtorus of L_o. Thus one has the exact sequence (2.1.2), with

$$U_o = L_o/T_o. \tag{2.1.4}$$

2.1.5. Let us note that A_o^o has <u>unipotent rank zero</u> (that is, after passage to an algebraic closure of k, it has no subgroup isomorphic to $\mathbb{G}_a$) if and only if A_o^o/T_o is an abelian scheme B_o. In that case the preceding hypotheses apply with $U_o = 0$.

Let ℓ be a prime different from the characteristic of k, so that A_o^o is ℓ-divisible. The inclusion (2.1.1) then gives an exact sequence of pro-group schemes:

$$\begin{tikzcd}
0 \arrow[r] & T_\ell(T_o) \arrow[r] & T_\ell(A_o^o) \arrow[r] \arrow[d, "\wr"] & T_\ell(A_o^o/T_o) \arrow[r] \arrow[d, "\wr"] & 0 \\
& & T_\ell(A_o) & T_\ell(A_o/T_o) & ,
\end{tikzcd} \tag{2.1.6}$$

whose terms may also be interpreted as twisted constant free ℓ-adic sheaves. When an exact sequence (2.1.2) is available—for example, when k is perfect or when A_o^o has unipotent rank zero—one has

$$T_\ell(A_o^o/T_o) \xrightarrow{\sim} T_\ell(B_o), \tag{2.1.7}$$

so that the exact sequence (2.1.6) becomes

$$0 \longrightarrow T_\ell(T_o) \longrightarrow T_\ell(A_o^o) \longrightarrow T_\ell(B_o) \longrightarrow 0. \tag{2.1.8}$$

Put

$$n = \dim A_K = \dim A_o, \qquad \mu = \dim T_o, \qquad \lambda = \dim U_o, \qquad \alpha = \dim B_o. \tag{2.1.9}$$

Then

$$n = \mu + \lambda + \alpha, \tag{2.1.10}$$

and

$$\begin{cases} \operatorname{rank} T_\ell(T_o) = \mu, & \operatorname{rank} T_\ell(A_o^o/T_o) = \operatorname{rank} T_\ell(B_o) = 2\alpha, \\ \operatorname{rank} T_\ell(A_o^o) = \mu + 2\alpha. \end{cases} \tag{2.1.11}$$

The second relation is a well-known theorem of Weil [5, 15].

2.1.12. The integers n, μ, λ, α are plainly defined, after passing to an algebraic closure of k, even without assuming the existence of an exact sequence (2.1.2). They are respectively called the dimension of A_o, the reductive rank of A_o (equal to the dimension of T_o), the unipotent rank of A_o, and the abelian rank of A_o. They are still related by (2.1.10) and (2.1.11), with the term $\operatorname{rank} T_\ell(B_o)$ omitted from the latter.

2.1.13. Suppose that k has characteristic $p > 0$ and that A_o^o is p-divisible, or equivalently that it has unipotent rank zero (cf. 2.1.5). One should then also consider the pro-group $T_p(A_o^o)$, which is a Barsotti–Tate group (or a p-divisible group, in the terminology of [33]). The exact sequence (2.1.8) remains valid for $\ell = p$ and is then an exact sequence of Barsotti–Tate groups. Here $\lambda = 0$, and hence

$$n = \mu + \alpha. \tag{2.1.13.1}$$

The formulas (2.1.11) also remain valid, provided that rank is interpreted as the rank (or "height") of the Barsotti–Tate groups in question.

2.1.14. For A'^o_o we likewise introduce the invariants

$$T'_o \subset A'^o_o, \qquad U'_o, \qquad B'_o,$$

where T'_o is the toric part of A'^o_o, B'_o its abelian part (defined in particular when k is perfect or when A'^o_o has unipotent rank zero), and U'_o its unipotent part (defined, for example, when k is perfect, or trivially when A'^o_o has unipotent rank zero). There is a corresponding exact sequence of the form (2.1.6), or (2.1.8), which we shall not write again. One could introduce integers μ', λ', α' for A'^o_o as in (2.1.9), but we shall see below (2.2.7) that they are the same as those for A_o^o; this spares us from introducing new notation for them.

2.2. We propose to make more precise the relations between $T_\ell(A_K)$ and $T_\ell(A_o) = T_\ell(A_o^o)$ by bringing $T_\ell(A^o)$ into play. First recall the following.

Lemma 2.2.1. Let $m > 0$ be an integer. Then the following conditions are equivalent:
(i) $m \operatorname{id}_{A^o}$ is flat.
(ii) $m \operatorname{id}_{A^o}$ is surjective.
(iii) $m \operatorname{id}_{A^o}$ is quasi-finite.
(iv) m is prime to the characteristic of k, or A_o^o has unipotent rank zero (cf. 2.1.5).
Moreover, if these conditions hold for m, they hold for m^ν for every $\nu > 0$; the groups

$${}_{m^\nu}A^o \underset{\text{dfn}}{=} \ker(m^\nu \operatorname{id}_{A^o})$$

are quasi-finite, separated, and flat over S, and, as ν varies, they form an inverse system of group schemes whose transition morphisms

$$x \longmapsto m^{\nu'-\nu}x : {}_{m^{\nu'}}A^o \longrightarrow {}_{m^\nu}A^o \qquad (\nu' \geq \nu)$$

are faithfully flat.

Indeed, the fiberwise criterion for flatness (EGA IV 11.3.5) shows that (i), like (ii) and (iii), is a condition on the fibers; these three conditions are equivalent by SGA 3 VI_A 5.6, since the fibers in question are smooth and connected. In the form (ii), a classical theorem of Weil [5, 15] shows that the conditions hold on the generic fiber, which is an abelian scheme; it is therefore enough to state them on the special fiber. Their equivalence with (iv) is then also well known from the structure theory. In any of these forms, stability of the condition under $m \mapsto m^\nu$ is immediate. It follows that the ${}_{m^\nu}A^o$ are quasi-finite flat group schemes over S, and they are plainly separated over S because A^o is. Finally, faithful flatness of the transition morphisms—equivalently, the fact that they are epimorphisms for the fppf topology—follows formally from the fact that $m\,\mathrm{id}_{A^o}$ is faithfully flat, hence an epimorphism for that topology.

2.2.2. Consequently, we shall take m to be a prime ℓ satisfying the conditions of 2.2.1. We then denote by

$$T_\ell(A^o) = ({}_{\ell^\nu}A^o)_{\nu>0} \tag{2.2.2.1}$$

the inverse system described in 2.2.1. Under the two base changes $\eta \longrightarrow S$ and $s \longrightarrow S$, it recovers the inverse systems $T_\ell(A_K)$ and $T_\ell(A_o^o) = T_\ell(A_o)$ considered in 1.0 and 2.1 respectively.

2.2.3. Suppose now that S is henselian. Every quasi-finite separated scheme X over S then has a canonical decomposition as a disjoint union

$$X = X^f \coprod X', \tag{2.2.3.1}$$

where X^f is finite over S and $X'_o = \varnothing$, that is, X' is confined to its generic fiber X'_η; thus X^f has the same special fiber as X. This decomposition is plainly functorial in X. In particular, if X is a group scheme over S, then X^f is a subgroup scheme. If X is flat, respectively étale, over S, then so is X^f.

We apply this to the components ${}_{\ell^\nu}A^o$ of $T_\ell(A^o)$. This gives a projective subsystem of groups

$$\big(({}_{\ell^\nu}A^o)^f\big)_{\nu>0} \underset{\text{dfn}}{=} T_\ell(A^o)^f \subset T_\ell(A^o), \tag{2.2.3.2}$$

called the fixed part of the Tate module $T_\ell(A^o)$. By construction, there are canonical isomorphisms, functorial in A,

$$T_\ell(A^o)^f \times_S s \xrightarrow{\sim} T_\ell(A^o) \times_S s \xrightarrow{\sim} T_\ell(A_o^o). \tag{2.2.3.3}$$

On the other hand, the generic fiber of $T_\ell(A^o)^f$ defines a canonical sub-pro-group of $T_\ell(A_K)$, again called the "fixed part" of this Tate module:

$$T_\ell(A_K)^f \subset T_\ell(A_K). \tag{2.2.3.4}$$

2.2.4. When $\ell \neq p = \operatorname{char} k$, the fixed part $T_\ell(A^o)^f$ may be interpreted as a twisted constant ℓ-adic sheaf on S (the $({}_{\ell^\nu}A^o)^f$ being finite étale over S). Since the functor

$$X \longmapsto X_o = X \times_S s$$

from finite étale schemes over S to finite étale schemes over k is an equivalence of categories when S is henselian, there is an analogous equivalence for the categories of twisted constant ℓ-adic sheaves over S and over k. Hence (2.2.3.3), which gives the special fiber of $T_\ell(A^o)^f$, characterizes this fixed part up to unique isomorphism. On the other hand, as an ℓ-adic subsheaf of $T_\ell(A^o)$ it is plainly characterized by its generic fiber (2.2.3.4), or equivalently by the corresponding $\mathbb{Z}_\ell$-submodule, stable under $\pi = \operatorname{Gal}(\overline{K}/K)$:

$$T_\ell(A_K)^f(\overline{K}) \underset{\text{dfn}}{=} T_\ell(A(\overline{K}))^f \subset T_\ell(A(\overline{K})). \tag{2.2.4.1}$$

In fact, one has the following result.

Proposition 2.2.5. Suppose, as above, that $\ell \neq p$ and that S is henselian. Then the submodule (2.2.4.1) of $T_\ell(A(\overline{K})) = T_\ell(A_K)(\overline{K})$ corresponding to the fixed part $T_\ell(A_K)^f$ in (2.2.3.4) is the submodule $T_\ell(A(\overline{K}))^I$ of invariants under the inertia group $I \subset \pi$.

This follows, by passage to the inverse limit, from the analogous relation valid for every integer $m > 0$ prime to $p = \operatorname{char} k$:

$$({}_mA)^f(\overline{K}) = ({}_mA(\overline{K}))^I. \tag{2.2.5.1}$$

To prove this relation, introduce the strict localization $\widetilde{S}$ of S, whose fraction field $\widetilde{K}$ is the maximal unramified subextension of $\overline{K}/K$. Since $({}_mA)^f$ is finite étale over S, its points with values in $\overline{K}$ already have values in $\widetilde{K}$. Thus the left-hand side of (2.2.5.1) is $\mathrm{Hom}_S(\widetilde{S}, ({}_mA)^f)$, whereas the right-hand side is, by definition of $\widetilde{K}$, equal to ${}_mA(\widetilde{K})$. By the definition of $({}_mA)^f$, and since $\widetilde{S}$ is an inverse limit of finite étale S-schemes S_i, the former is also equal to

$$\mathrm{Hom}_S(\widetilde{S}, {}_mA) = {}_m\,\mathrm{Hom}_S(\widetilde{S}, A).$$

By the defining property of the Néron model, $\mathrm{Hom}_S(\widetilde{S}, A) = A(\widetilde{K})$, which proves (2.2.5.1).

To deduce 2.2.5, observe that (2.2.5.1) gives

$$(*) \qquad T_\ell(A(\overline{K}))^I = \varprojlim_\nu ({}_{\ell^\nu}A)^f(\overline{K}) \supset \varprojlim_\nu ({}_{\ell^\nu}A^o)^f(\overline{K}).$$

If (x_ν) belongs to the rightmost term, then for every $\nu \geq 0$, x_ν is a section of A that is infinitely divisible by ℓ. Its value at s is therefore an element of $A_o(k)$ infinitely divisible by ℓ, and consequently lies in $A_o^o(k)$. This proves equality of the last two terms of (*) and completes the proof of 2.2.5.

Proposition 2.2.6. Let C_K be a second abelian scheme over K, with Néron model C, and consider a homomorphism $u_K : C_K \longrightarrow A_K$, which therefore extends to a homomorphism $u : C \longrightarrow A$. Suppose that u_K is an isogeny. Then, for every prime ℓ, $T_\ell(u_K) : T_\ell(C_K) \longrightarrow T_\ell(A_K)$ induces an isogeny of fixed parts

$$(2.2.6.1) \qquad T_\ell(C_K)^f \longrightarrow T_\ell(A_K)^f,$$

and likewise the map induced by $u_o : C_o \longrightarrow A_o$

$$(2.2.6.2) \qquad T_\ell(u_o^o) : T_\ell(C_o^o) \longrightarrow T_\ell(A_o^o)$$

is an isogeny. Finally, $u_o : C_o \longrightarrow A_o$ induces an isogeny from the largest subtorus of C_o to the largest subtorus T_o of A_o; and over an extension k'/k on which the "abelian parts" of C_o and A_o are defined (for example, over the perfect closure of k), u_o induces an isogeny between the abelian parts of $C_{ok'}$ and $A_{ok'}$.

This statement follows formally from the fact that $T_\ell(A_K)^f$, $T_\ell(A_o^o)$, the largest subtorus of A_o, and the "abelian part" of $A_{ok'}$ are additive functors of the varying A_K, and from the fact that u_K is an isogeny precisely in the sense that there is a morphism $v_K : A_K \longrightarrow C_K$ such that

$$u_K v_K = n\,\mathrm{id}, \qquad v_K u_K = n\,\mathrm{id}$$

for some integer $n > 0$.

Corollary 2.2.7. If A_K and C_K are isogenous abelian schemes over K, then the corresponding invariants μ, λ, α in (2.1.11) are the same for A_K and C_K. In particular, they are the same for A_K and its dual A'_K.

Remarks 2.2.8. In the two preceding statements, 2.2.6 and 2.2.7, we assumed that S is henselian. It is immediate, however, that 2.2.7, as well as the assertions of 2.2.6 concerning the properties of the induced homomorphism $u_o : C_o \longrightarrow A_o$, holds without this hypothesis. Indeed, one reduces to the case already treated by the base change $S^h \longrightarrow S$, where S^h is the henselization of S, using the fact—immediate from the definitions—that formation of Néron models commutes with this base change.

Corollary 2.2.9. (Good reduction.) Let S be a trait, let A_K be an abelian scheme over its fraction field K, let A be the Néron model of A_K, and let ℓ be a prime different from $\mathrm{char}\,k$. The following conditions are equivalent.

(i) There exists an abelian scheme over S whose generic fiber is isomorphic to A_K.

(ii) A is an abelian scheme.
(ii^o) A^o is an abelian scheme.
(iii) A_o is an abelian scheme.
(iii^o) A_o^o is an abelian scheme, that is, with the notation of (2.1.11), $\lambda = \mu = 0$.
(iv) A is proper over S.
(iv^o) A^o is proper over S.
(v) $T_\ell(A_K)$ is "unramified over S", that is, it extends to a twisted constant ℓ-adic sheaf over S, or equivalently the inertia group I acts trivially on $T_\ell(A_K(\overline{K}))$.
(v bis) For every integer $\nu \geq 0$, ${}_{\ell^\nu}A_K$ is "unramified over S", that is, it extends to an étale covering of S, or equivalently the inertia group I acts trivially on ${}_{\ell^\nu}A_K(\overline{K})$.

Finally, an abelian scheme over S as in (i) is a Néron model of A_K, and in particular is determined up to unique isomorphism.

2.2.9.1. We say that A_K has good reduction over S if it satisfies condition (i) above. Criterion (v), or (v bis), is known as the Néron–Ogg–Shafarevich criterion for good reduction.

Let us prove 2.2.9. It is well known (cf., for example, [5]) that an abelian scheme over S satisfies the universal property of a Néron model; this reduces to the fact that a rational section of an abelian scheme over a regular base is defined everywhere. This proves the equivalence *(i)* $\iff$ *(ii)* $\iff$ *(ii^o)*, as well as the last assertion of 2.2.9. These conditions plainly imply (iii), (iv), and (iv^o), each of which implies (iii^o). Let us prove that (iii^o)$\Longrightarrow$(ii^o), which will establish the equivalence of (i) through (iv^o). This follows from the fact that a smooth group scheme over S whose fibers are abelian schemes is proper over S (and hence is an abelian scheme), by EGA IV 15.7.10.

On the other hand, in view of 2.2.5 and the fact that

$$T_\ell(A_K(\overline{K}))/T_\ell(A_K(\overline{K}))^I$$

is plainly a torsion-free $\mathbb{Z}_\ell$-module, condition (v) says that $T_\ell(A_K(\overline{K}))$ and its fixed part have the same rank. By (2.1.11) and (2.2.3.3), this is the condition

$$\mu + 2\alpha = 2(\lambda + \mu + \alpha),$$

that is, $2\lambda + \mu = 0$, or $\lambda = \mu = 0$, which is precisely (iii^o). Finally, the equivalence of (v) and (v bis) is clear from the definitions. This completes the proof of 2.2.9.

2.3. Suppose once again that S is henselian and that $\ell \neq p = \operatorname{char} k$. We now use the inclusion

$$T_\ell(T_o) \hookrightarrow T_\ell(A_o^o) \tag{2.3.1}$$

of (2.1.6), together with the equivalence of categories recalled in 2.2.4. We obtain a canonical twisted constant ℓ-adic subsheaf of $T_\ell(A^o)^f$, called the toric part of $T_\ell(A^o)$,

$$T_\ell(A^o)^t \hookrightarrow T_\ell(A^o)^f \hookrightarrow T_\ell(A^o), \tag{2.3.2}$$

characterized by the condition that its special fiber is the inclusion (2.3.1):

$$T_\ell(A^o)^t \times_S s = T_\ell(T_o) \hookrightarrow T_\ell(A_o^o). \tag{2.3.3}$$

Later we shall give an interpretation of the toric part of $T_\ell(A^o)$ as the Tate module $T_\ell(T)$ of a torus T over S lifting the torus T_o over k. That interpretation will allow the definition of the toric part to be extended to the case $\ell = p$. For the moment, and to fix ideas, we confine ourselves to the case $\ell \neq p$. In this case the toric part of $T_\ell(A^o)$ may be described by the corresponding $\mathbb{Z}_\ell$-submodule of $T_\ell(A(\overline{K}))$:

$$T_\ell(A(\overline{K}))^t \hookrightarrow T_\ell(A(\overline{K}))^f \hookrightarrow T_\ell(A(\overline{K})). \tag{2.3.4}$$

It is stable under the action of π; its first term is defined as $T_\ell(A^o)^t(\overline{K})$ and is, as one would expect, again called the toric submodule of the Tate module $T_\ell(A(\overline{K}))$. Thus this Tate module carries the canonical three-step filtration (2.3.4). Applying the same considerations to the dual abelian scheme A'_K, we obtain the analogous three-step filtration

$$T_\ell(A'(\overline{K}))^t \hookrightarrow T_\ell(A'(\overline{K}))^f \hookrightarrow T_\ell(A'(\overline{K})). \tag{2.3.5}$$

The middle terms in (2.3.4) and (2.3.5), the "fixed parts," are described in terms of the π-module structures of the Tate modules of $A(\overline{K})$ and $A'(\overline{K})$ as their submodules of invariants under the inertia group I (2.2.5). The toric parts are then completely described in terms of the fixed parts and the pairing φ of (1.0.2), by the following theorem (a special case of which already appears in [11]).

Theorem 2.4. (Orthogonality theorem.) Let S be a henselian trait, let A_K be an abelian scheme over its fraction field K, let A'_K be the dual abelian scheme, and let ℓ be a prime different from the residual characteristic of S. The Tate modules $T_\ell(A(\overline{K}))$ and $T_\ell(A'(\overline{K}))$ then carry the canonical filtrations (2.3.4) and (2.3.5), respectively, by the "fixed part" and the "toric part." The toric part $T_\ell(A(\overline{K}))^t$ is the intersection of the fixed part $T_\ell(A(\overline{K}))^f$ with the orthogonal of the fixed part $T_\ell(A'(\overline{K}))^f = T_\ell(A'(\overline{K}))^I$ with respect to the canonical pairing φ of (1.0.3). Symmetrically, the toric part $T_\ell(A'(\overline{K}))^t$ is the intersection of $T_\ell(A'(\overline{K}))^f$ with the orthogonal of the fixed part $T_\ell(A(\overline{K}))^f = T_\ell(A(\overline{K}))^I$. (Here the superscript I denotes the submodule of invariants under the action of the inertia group I.)

Proof. a) Let us prove that $T_\ell(A(\overline{K}))^t$ is orthogonal to $T_\ell(A'(\overline{K}))^f$. To do so, we interpret the restriction of φ to the fixed parts of the Tate modules using the canonical biextension $W^o = W$ of (1.4.4). This is the canonical extension of the biextension W_K of (A_K, A'_K) by $\mathbb{G}_{mK}$ to a biextension of (A^o, A'^o) by $\mathbb{G}_{mS}$. By VIII 2.2, W gives rise to a pairing

$$\varphi_W : T_\ell(A^o) \times T_\ell(A'^o) \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.1}$$

from which the pairing φ of (1.0.3) is obtained by passing to $\overline{K}$-valued points. It follows that the restriction of φ to the fixed parts considered in 2.4 is obtained, by passing to $\overline{K}$-valued points, from the pairing

$$T_\ell(A^o)^f \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.2}$$

the restriction of (2.4.1). Since the two factors on the left of (2.4.2) are now twisted constant ℓ-adic sheaves, this pairing may be studied on the special fibers. The fibers of the two factors are precisely the Tate modules $T_\ell(A^o_o)$ and $T_\ell(A'^o_o)$ (2.2.3.3), and the pairing induced on special fibers by (2.4.2) is

$$T_\ell(A^o_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.3}$$

obtained from the biextension W_o of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$, the restriction of W to the special fibers.

Recall from 2.3 that the toric part of $T_\ell(A(\overline{K}))$ is obtained by passing to $\overline{K}$-valued points in the twisted constant ℓ-adic subsheaf $T_\ell(A^o)^t$ of $T_\ell(A^o)^f$. Thus the orthogonality relation to be proved says that the pairing induced by (2.4.2),

$$T_\ell(A^o)^t \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}), \tag{2.4.4}$$

is identically zero. It is enough to verify this for the corresponding pairing on the special fibers, which by the definition of the toric part is

$$T_\ell(T_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{2.4.5}$$

induced by (2.4.3). By VIII 4.10 this pairing is necessarily zero, which proves the asserted orthogonality.

b) Let us prove that the inclusion

$$T_\ell(A(\overline{K}))^t \subset T_\ell(A(\overline{K}))^f \cap \big(T_\ell(A'(\overline{K}))^f\big)^\perp \tag{2.4.6}$$

is an equality, where $\perp$ denotes the orthogonal. Equivalently, in the pairing (2.4.2), the annihilator of the second factor—which, as we have just seen, contains the toric part $T_\ell(A^o)^t$—is equal to that toric part. Since these are again relations between twisted constant ℓ-adic sheaves, it is enough to check the assertion on the special fibers: namely, that the inclusion

$$T_\ell(T_o) \subset T_\ell(A'^o_o)^\perp \tag{2.4.7}$$

is an equality. In other words, we must verify that the biextension W_o of (A^o_o, A'^o_o) is <u>nondegenerate on the left</u> (VIII 4.11). But this is precisely what we already observed in 1.5, as a consequence of a result from Raynaud's thesis.

2.5. The orthogonality theorem has an equivalent formulation using a polarization ξ of A_K and the alternating bilinear form (1.0.8), compatible with the operations of π,

$$\varphi_\xi : T_\ell(A(\overline{K})) \times T_\ell(A(\overline{K})) \longrightarrow T_\ell(\overline{K}^*), \tag{2.5.1}$$

obtained from the form φ of (1.0.3) by transporting structure in the second argument along the isogeny of $\mathbb{Z}_\ell$-modules

$$T_\ell(A(\overline{K})) \longrightarrow T_\ell(A'(\overline{K}))$$

induced by the isogeny $\psi_\xi : A_K \longrightarrow A'_K$ defined by ξ. The equality in (2.4.6) then takes the simple form

$$W = V \cap V^\perp, \tag{2.5.2}$$

where we have set

$$\left\{\begin{aligned} U = T_\ell(A(\overline{K})),\qquad V = U^I = \text{the "fixed part" of } T_\ell(A(\overline{K})),\\ W = \text{the toric part of } T_\ell(A(\overline{K})),\end{aligned}\right. \tag{2.5.3}$$

and $\perp$ denotes the orthogonal with respect to the polarization form (2.5.1). These modules give the following diagram of inclusions:

$$\begin{tikzcd} & & V \arrow[dl, hook, "2\lambda"'] & & \\ U & W^\perp = V + V^\perp \arrow[l, hook', "\mu"'] & & W = V \cap V^\perp \arrow[ul, hook, "2\alpha"'] \arrow[dl, hook, "2\lambda"] & 0 \arrow[l, hook', "\mu"'] \\ & & V^\perp \arrow[ul, hook, "2\alpha"] & & \end{tikzcd} \quad, \tag{2.5.4}$$

where the integers 2α, 2λ, and μ are the ranks of the quotients of consecutive modules in the diagram, with the notation of (2.1.9). (That W has rank μ is the definition of μ; that V/W has rank 2α follows from the second relation in (2.1.11) together with the exact sequence (2.1.6). Hence, by orthogonality, $\operatorname{rank}(U/W^\perp) = \mu$ and $\operatorname{rank}(W^\perp/V^\perp) = 2\alpha$. Finally, the formulas

$$\operatorname{rank}\big(V/(V \cap V^\perp)\big) = \operatorname{rank}(V/W) = 2\alpha \tag{2.5.5}$$

and the dual formula follow from the preceding ones, in view of (2.1.10) and $\operatorname{rank} U = n$.)

<u>Remarks</u> 2.6. a) Here we obtained the orthogonality theorem 2.4 as an easy consequence of the theory of biextensions developed in the preceding sections. The <u>inclusion</u> (2.4.6) can be proved

without using the theory of biextensions, by an arithmetic argument in the style of the proof of the "monodromy theorem" given in Exposé I. One reduces to a situation in which S has finite residue field, where the orthogonality result follows from a simple consideration of weights for the Frobenius operation (using A. Weil's result on the eigenvalues of that Frobenius). There is no need to invoke resolution of singularities here: one may state the desired result for group schemes A over schemes S under sufficiently general conditions. Unfortunately, it does not appear possible to obtain by this method equality in the preceding inclusion. That equality will apparently be used in an essential way in the proof of the "semistable reduction theorem" in the next section.

b) The orthogonality theorem 2.4 allows the most important properties of the Néron model of an abelian variety A_K to be expressed in terms of the representation of the inertia group I on the Tate module $T_\ell(A(\overline{K}))$, where ℓ is a fixed prime different from the residual characteristic. (The case $\ell = p$ will be considered separately later, in section 5.) As a first statement of this kind, though one that does not use the orthogonality theorem, recall the good-reduction criterion 2.2.9. We shall give many further examples below, beginning in the next section (3.5).

3. The case of semistable reduction. The semistable reduction theorem.

We first give an auxiliary result in the style of SGA 3.

Proposition 3.1. Let S be a scheme, let G be a commutative S-group scheme that is flat, separated, and of finite presentation over S, let H be a quasi-separated group scheme locally of finite type over S, let $u : G \longrightarrow H$ be a homomorphism of S-group schemes, and put $K = \ker u$. For every $s \in S$, let $\bar{s}$ denote the spectrum of an algebraic closure of $k(s)$, and denote by $\mu(s)$ (respectively, $\alpha(s)$) the reductive rank (respectively, the abelian rank) of K_s, that is, the dimension of the largest subtorus (respectively, of the largest abelian-variety quotient) of $(K_{\bar{s}})^o_{\mathrm{red}}$. Let $s \in S$, let $t \in S$ be a generalization of s, and let p be the characteristic exponent of $k(s)$.

a) For every integer $n > 0$ prime to p, ${}_nK$ is etale, of finite type, and separated over S in a neighborhood of s, and one has the divisibility relation

$$\text{(3.1.1)} \qquad \operatorname{rank}({}_nK_s) \mid \operatorname{rank}({}_nK_t).$$

b) One has

$$\text{(3.1.2)} \qquad \mu(s) + 2\alpha(s) \leq \mu(t) + 2\alpha(t).$$

c) If $(K_{\bar{t}})^o_{\mathrm{red}}$ is unipotent, then so is $(K_{\bar{s}})^o_{\mathrm{red}}$.

d) Suppose that G_s has unipotent rank zero, that H is smooth over S, and that u_t is an isogeny. Then u is flat and quasi-finite over an open neighborhood U of s; a fortiori, $K|_U$ is flat and quasi-finite.

e) Under the hypotheses of d), suppose in addition that u_t is a monomorphism (and hence an isomorphism from G_t onto an open subgroup of H_t). Then u_s is an isomorphism from G_s onto an open subgroup of H_s. If, moreover, S is irreducible with generic point t, then there is an open neighborhood U of s such that u induces an isomorphism from $G|_U$ onto an open subgroup of $H|_U$.

Proof.

a) We may suppose that n is invertible on S. Then $n\,\mathrm{id}_G$ is etale on every fibre (SGA 3, XV, 1.3); since G is flat and of finite presentation over S, the fibrewise criterion for etaleness (EGA IV, 17.8.2) shows that $n\,\mathrm{id}_G$ is etale. Its kernel ${}_nG$ is therefore etale; it is also separated and of finite presentation over S, since G is. Similarly, ${}_nH$ is locally of finite type, quasi-separated, and formally unramified over S,[10] so its unit section is an open immersion (EGA IV, 17.4.2 b')). Consequently, the kernel of

$$ {}_nu : {}_nG \longrightarrow {}_nH $$

10. At least when H is commutative. In the general case, ${}_nH$ is still unramified over S along its unit section, which is all that is needed here.

is an open subscheme of ${}_nG$, hence is etale over S; it is quasi-compact in ${}_nG$ and therefore of finite presentation over S, and it is separated over S because G is. This kernel is plainly ${}_nK$.

Formula (3.1.1) follows from these properties of ${}_nK$. Indeed, after a base change we may reduce to the case where S is henselian local with closed point s, and then use the decomposition of ${}_nK$ into ${}_nK^f$ and a scheme with empty special fibre (cf. 2.2.3). More precisely, the same argument gives a monomorphism on geometric fibres

$$ {}_nK(\bar{s}) \hookrightarrow {}_nK(\bar{t}). $$

b) Choose an integer $n > 0$ prime to the characteristic exponent of $k(s)$ and to the ranks of the quotients of the geometric fibres of K at s and t by their reduced identity components. The two sides of (3.1.1) are then obtained by raising n, respectively, to the powers whose exponents are the two sides of (3.1.2), as follows from the known structure of commutative algebraic groups. Thus b) follows from a).

c) The hypothesis that $(K_{\bar{t}})^o_{\mathrm{red}}$ is unipotent is equivalent to the equalities $\mu(t) = \alpha(t) = 0$, and likewise for $K_{\bar{s}}$. Assertion c) therefore follows formally from b).

d) Since u_t is an isogeny, K_t is finite. We are therefore under the hypotheses of c), and $(K_{\bar{s}})^o_{\mathrm{red}}$ is unipotent. As G_s has unipotent rank zero, it follows that K_s is finite. Moreover, the fact that u_t is an isogeny gives $\dim G_t = \dim H_t$. Since G and H are flat and locally of finite presentation, one has

$$ \dim G_s = \dim G_t, \qquad \dim H_s = \dim H_t \quad (\text{SGA 3, VI}_B, 4.3), $$

and hence $\dim G_s = \dim H_s$. Since $\dim \ker u_s = 0$, it follows that $\dim H_s = \dim u_s(G_s)$ (SGA 3, VI_B, 1.2), which, because H_s is smooth, is equivalent to u_s being flat (SGA 3, VI_A, 5.6). There is therefore an open neighborhood U of s such that u is flat and quasi-finite at the points of the unit section over U (SGA 3, VI_B, 2.2(i) and 2.5(ii)). As for u_s, it follows that $u_{s'}$ is flat for every $s' \in U$; hence $u|_{G_U}$ is flat and quasi-finite, since it has these properties on every fibre.

e) For the first assertion, a base change immediately reduces us to the case where S is integral with generic point t, and hence to proving the second assertion of e). That assertion follows at once from d) and the following lemma.

Lemma 3.1.3. Let S be a scheme, let K be a quasi-finite S-scheme of finite presentation that is separated and flat over S, let $s \in S$, and let $t \in S$ be a generalization of s. Then

$$ \operatorname{rank} K_s \leq \operatorname{rank} K_t. $$

If $\operatorname{rank} K_\eta \leq 1$ for every maximal point η of S, then $K \longrightarrow S$ is an open immersion (and hence an isomorphism if it admits a section).

For the first assertion, strict localization reduces us to the case where S is strictly local. Decomposing K as the disjoint union of a finite scheme K^f and a scheme with empty special fibre, we obtain

$$ \operatorname{rank} K_s = \operatorname{rank} K^f_s = \operatorname{rank} K^f_t \leq \operatorname{rank} K_t. $$

The second equality follows because K^f is flat, finite, and of finite presentation over S, and therefore locally free over S.

For the second assertion, the preceding argument already shows that $\operatorname{rank} K_s \leq 1$ for every $s \in S$. Thus K_s is empty or isomorphic to $\operatorname{Spec} k(s)$. Consequently, $K \longrightarrow S$ is a monomorphism (SGA 3, VI_B, 2.11), and, being flat and of finite presentation, it is an open immersion (EGA IV, 17.9.1). This proves the lemma.

Remark 3.1.4. Part 3.1 a), and the remainder of Proposition 3.1, give no information about ${}_nK_s$ when n is a power of p (for $p > 1$). At least when G is smooth (and S is locally noetherian or H is of finite presentation over S), essentially the same method as in the proof of a) gives such information. Indeed, reducing to the case where S is complete local noetherian and using SGA 3, XV, 1.6 c), X, 3.2, IX, 3.6, and IX, 5.2, one obtains

$$\mathrm{rank}_{\mathrm{mult}}({}_nK_s) \mid \mathrm{rank}_{\mathrm{mult}}({}_nK_t), \tag{3.1.4.1}$$

where $\mathrm{rank}_{\mathrm{mult}}$ denotes the rank of the largest subgroup of multiplicative type of the commutative algebraic group under consideration (SGA 3, XVII, 7.2.1). Notice that a torsion algebraic group of multiplicative type is finite. Formula (3.1.4.1) is valid for every integer $n \geq 1$. It follows from SGA 3, XVII, 4.6.1(vi) that, in c), if K_t is unipotent (SGA 3, VII, 1.1), then K_s is unipotent as well.

Proposition 3.2. Let S be a regular integral locally noetherian scheme of dimension 1, let K be its field of rational functions, let A_K be an abelian scheme over K, and let A be its Neron model over S, which is a smooth separated group scheme over S (and is quasi-projective over S if S is noetherian [23, VIII, 2, 3°]). The following conditions are equivalent:

(i) For every $s \in S$, A^o_s has unipotent rank zero; that is, it is an extension of an abelian scheme by a torus (2.1.5).

(ii) There exists a smooth separated group scheme G of finite type over S, extending A_K, whose fibres have unipotent rank zero.

Moreover, if G is as in (ii), then the unique morphism $G \longrightarrow A$ inducing the given isomorphism on the generic fibres is an isomorphism from G onto an open subgroup of A; it therefore induces an isomorphism of identity components

$$G^o \xrightarrow{\sim} A^o.$$

3.2.1. Let us note at once that the definition of the Neron model A of A_K is the same as in the local case recalled in 1.1. The existence of this Neron model, and the fact that it is separated and of finite type over S, are well known and in fact reduce immediately to the local case. Of course, for every closed point $s \in S$,

$$S' = \mathrm{Spec}(\mathcal{O}_{S,s})$$

is a trait, and $A \times_S S'$ is precisely the Neron model of A_K over S' in the sense considered in 1.1.

Let us prove 3.2. It is trivial that (i) implies (ii), since it suffices to take $G = A$. It remains to prove that if G is as in (ii), then $G \longrightarrow A$ is an open immersion. This is indeed a special case of 3.1 e).

Corollary 3.3. With the notation of 3.2, suppose that the identity components of the fibres of the Neron model have unipotent rank zero. Let S' be a scheme satisfying the same hypotheses as S, let K' be its field of rational functions, let $f : S' \longrightarrow S$ be a dominant morphism, let

$$A_{K'} \simeq A_K \otimes_K K' \simeq \text{the generic fibre of } A \times_S S',$$

let A' be the Neron model of $A_{K'}$, and consider the unique morphism

$$u : A \times_S S' \longrightarrow A'$$

inducing the identity on the generic fibres. Then u is an open immersion and therefore induces an isomorphism on identity components

$$A^o \times_S S' \xrightarrow{\sim} A'^o. \tag{3.3.1}$$

Indeed, it suffices to apply 3.2 over S', taking $G \simeq A \times_S S'$.

3.3.2. Thus the equivalent properties of 3.2 imply a partial "stability" property for Neron models. One should bear in mind that when A_K does not have good reduction, that is, when A is not an abelian scheme, it is not true that, for every S' as in 3.3 (even for S' finite over S and K'/K separable), the morphism

$$A \times_S S' \longrightarrow A'$$

itself is an isomorphism, or that it induces isomorphisms on the groups of connected components of the fibres (see 11.10 below for the behavior of these groups of connected components under extension of the base). These observations, reinforced by 3.9 below, justify the following terminology, introduced by Mumford (and which also appears to agree with the notions of the same name studied in his book [17]).

Definition 3.4. With the notation of 3.2, the abelian scheme A_K over K is said to have semistable reduction over S if the equivalent conditions (i) and (ii) of 3.2 are satisfied.

3.4.0. Notice that in the form (ii), this condition does not invoke the notion of a Neron model. Moreover, by localization it reduces immediately to the case where S is a trait, as follows for example from 3.2.1. Furthermore, since formation of the Neron model commutes with base change from S to its henselization or its strict henselization, the notion even reduces to the case where S is a strictly local trait.

Proposition 3.5. (Galois criterion for semistable reduction). Suppose that S is a trait. The following conditions on the abelian scheme A_K are equivalent (the notation being that of (2.5.3), applied to the situation deduced from (A_K, S) by passing to the henselization of S):

(i) A_K has semistable reduction over S (3.4).

(ii) With the notation of (2.5.3) and (2.5.4), one has

$$V^{\perp} \subset V.$$

(iii) For the toric part W of the Tate module one has

$$W = V^{\perp}.$$

(iv) The action of the inertia group I on $U = T_\ell(A(\overline{K}))$ is unipotent of level 2 (that is, there is an I-stable submodule U' of U such that I acts trivially on U' and on U/U').

We may suppose that S is henselian (3.4.0). With the notation of 2.1.9, (i) means that

$$\lambda = 0.$$

By the rank formulas following (2.5.4), this is equivalent to $V^{\perp} = V \cap V^{\perp}$, that is, to (ii). Since $W = V \cap V^{\perp}$ by the orthogonality theorem 2.4, conditions (ii) and (iii) are equivalent. Moreover, (iv) also says that I acts trivially on $U/U^I = U/V$. By duality this means that it acts trivially on $V^{\perp}$, that is, that $V^{\perp} \subset U^I = V$, which is (ii). This completes the proof.

Remarks 3.5.1. Conditions (ii) and (iv) plainly depend only on the action of I on the $\mathbb{Q}_\ell$-vector space $U \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and could have been formulated in terms of that action. Let us also note that these conditions are manifestly stable under every base change by a morphism of traits $S' \longrightarrow S$. Finally, if (iv) holds for A_K, it plainly holds for every abelian subscheme and every quotient abelian scheme of A_K, and in particular for every abelian scheme isogenous to A_K (for example, for A'_K).

Corollary 3.5.2. Let g be a topological generator of $\mathbb{Z}_\ell(1) = T_\ell(\overline{k}^*)$. The equivalent conditions of 3.5 are also equivalent to the following condition:

(v) The action of the inertia group I on $U = T_\ell(A(\overline{K}))$ factors through the maximal pro-ℓ quotient $I(\ell)$ of I, which is canonically isomorphic to $\mathbb{Z}_\ell(1)$, and the action g_U of the generator g on U satisfies

$$(1 - g_U)^2 = 0. \qquad (3.5.3)$$

Indeed, for every continuous unipotent action of a profinite group I on an ℓ-adic vector space, one sees at once that the image of I is a pro-ℓ group. It follows that (v) is equivalent to (iv).

We can now prove the central result of the present expose:

Theorem 3.6. (Semistable reduction theorem). Let S be a connected regular noetherian scheme of dimension 1, let K be the field of rational functions of S, and let A_K be an abelian scheme over K. Then there is a finite Galois extension K' of K such that the abelian scheme $A_{K'} = A_K \otimes_K K'$ has semistable reduction (3.4) over the normalization S' of S in K'.

Notice that S' is connected, regular, noetherian, and of dimension 1, so that the stated conclusion makes sense. To prove 3.6, choose a separable closure $\overline{K}$ of K and write $\overline{K}$ as the filtered union of its finite Galois subextensions K_i. Let S_i be the normalization of S in K_i, and let U be the largest open subset of S over which A_K has good reduction. Since S is connected, noetherian, and one-dimensional, and $U \neq \varnothing$, the complement $T = S - U$ is a finite set of closed points of S, and $\mathcal{O}_{S,t}$ is a discrete valuation ring for every $t \in T$. It is enough to find, for every $t \in T$, an index $i = i(t)$ such that A_{K_i} has semistable reduction at the points of S_i above t: one then takes for K' the compositum of the extensions $K_{i(t)}$, for $t \in T$. Thus we are reduced to the case where S is a trait.

Moreover, since the Galois group permutes the points of the normalization lying above the closed point s of S, it is enough to find K' such that $A_{K'}$ has semistable reduction at *one* point of the normalization S'. Here it is unnecessary at first to require that K'/K be Galois: it is enough to find a finite etale extension K'/K satisfying the preceding condition, since its Galois closure satisfies the same condition. Using criterion 3.5(iv), we see that Theorem 3.6 reduces to the following assertion.

Corollary 3.7. Suppose again that S is a trait. Then there is an open subgroup π' of π such that the action of $I' = I \cap \pi'$ on $U = T_\ell(A(\overline{K}))$ is unipotent of level 2; equivalently, there is an open subgroup I' of I whose action on U is unipotent of level 2.

There is a canonical isomorphism, compatible with the actions of π,

$$(*) \qquad H^1(A_{\overline{K}}, \mathbb{Z}_\ell) \simeq \operatorname{Hom}\big(T_\ell(A(\overline{K})), \mathbb{Z}_\ell\big),$$

which shows that assertion 3.7 is equivalent to the same assertion for the actions of I on the left-hand side of (*). But that assertion was proved in Expose III (using resolution of singularities for excellent schemes of dimension 2).

Corollary 3.8. The equivalent conditions (i)–(iv) of 3.5 are also equivalent to the following condition:

(vi) The action of I on $U = T_\ell(A(\overline{K}))$ is unipotent.

Indeed, to prove that (vi) implies (iv), observe that if a group I acting unipotently on a finite-dimensional vector space E over a field of characteristic zero (here $\mathbb{Q}_\ell$) has a subgroup I' of finite index acting with level m (here $m = 2$), then I itself acts with level m. In fact, since I' has finite index in I, the algebraic group $G' \subset \mathrm{GL}(E)$ that it defines has finite index in the group G defined by I; but G is unipotent and hence connected (in characteristic zero), so $G = G'$. Thus I and I' have the same "level of unipotence."

We also obtain a converse to 3.3:

Corollary 3.9. The notation is that of 3.2. The conditions (i) and (ii) of 3.2 are also equivalent to the following condition:

(iii) For every finite etale extension K' of K, let S' denote the normalization of S in K', and let A' denote the Neron model of $A_{K'} = A \otimes_S K' = A_K \otimes_K K'$. Then the canonical morphism

$$A \times_S S' \longrightarrow A'$$

induces an isomorphism on identity components.

The necessity of the condition was proved, in a stronger form, in 3.3. For sufficiency, choose K' as in 3.6. Since the fibres of A'^o have unipotent rank zero, the same is true of the fibres of $(A \times_S S')^o = A^o \times_S S'$, and hence also of the fibres of A^o, from which the former are obtained by extension of the base field.

4. Application to a conjecture of Serre–Tate and to the conductor

4.1. We suppose that S is henselian, and retain the notation of (2.5.3), with $\ell \neq p$.[11] To study the action of $\pi = \mathrm{Gal}(\overline{K}/K)$ and of its inertia subgroup I on $U = T_\ell(A(\overline{K}))$, we shall use the fact that there exists an open subgroup I' of I, which we may moreover take to be normal in π, such that the action of I' on U is unipotent of level two; that is,

$$U^{I'} \supset (U^{I'})^\perp.$$

It is then clear that $U^{I'}$ does not depend on the particular choice of I'. We might call it the essentially fixed submodule of U, and shall denote it by U^{ef}. Its orthogonal will be called the essentially toric submodule of U, and will be denoted by U^{et}. We thus have a three-step filtration

$$U \supset U^{\mathrm{ef}} \supset U^{\mathrm{et}} \supset 0, \tag{4.1.1}$$

which is plainly invariant under the action of π.

Let us also note that there is a largest subgroup of I acting unipotently (or, equivalently here, unipotently of level two) on U: it is the subgroup I' consisting of the elements acting unipotently. It is also the kernel of the representation of I on the graded module associated with the filtration (4.1.1), or, equivalently in the present case, the kernel of the induced representation of I on U^{ef}. By the Galois criterion for semistable reduction (3.5), this subgroup I' corresponds to the smallest Galois extension $\widetilde{K}'$ of $\widetilde{K}$ (the maximal unramified extension of K) such that $A_{\widetilde{K}'}$ has semistable reduction. This description also shows that the subgroup I' of I is independent of the choice of the prime $\ell \neq p$.

Notice that, once the representation of π on U^{ef}, and hence on U^{et}, is known, the representation on U/U^{ef} is known up to isogeny, by means of the isogeny induced by the polarization pairing (2.5.1):

$$U/U^{\mathrm{ef}} \longrightarrow (U^{\mathrm{et}})^\vee(1) \qquad \text{an isogeny,} \tag{4.1.2}$$

where (1) denotes the Tate twist by $\mathbb{Z}_\ell(1) = T_\ell(\overline{K}^*)$; this homomorphism is plainly compatible with the operations of π. More precisely, there is a canonical isomorphism

$$U/U^{\mathrm{ef}} \simeq ((U')^{\mathrm{et}})^\vee(1), \qquad U' = T_\ell(A'(\overline{K})).$$

It therefore remains to study more closely the action of π on U^{ef}.

By construction this action is trivial on the normal subgroup I' of I, and hence factors through a representation of the quotient group

$$\pi' = \pi/I', \tag{4.1.3}$$

an action of which we shall now give a description "independent of ℓ."

4.2. Let K' be a Galois extension of K corresponding to an open normal subgroup V of π such that $V \cap I = I'$. The Galois group G of K'/K then occurs in an exact sequence

$$1 \longrightarrow \Gamma \longrightarrow G \longrightarrow G_o \longrightarrow 1, \tag{4.2.1}$$

where

$$\Gamma = I/I' \tag{4.2.2}$$

11. The case $\ell = p$ will be studied in §§5 and 6 below.

is the inertia subgroup, and G_o is the Galois group of the largest unramified subextension K'' of K'/K. Thus (4.2.1) is the image of the analogous exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \pi' \longrightarrow \pi_o \longrightarrow 1, \tag{4.2.3}$$

where π' is the group (4.1.3) acting on U^{ef}. We therefore have the following diagram of field inclusions:

$$\begin{tikzcd}
K \arrow[r, "G_o"] \arrow[dr, dash, "G"'] \arrow[rr, bend left, dash, "\pi_o"] & K'' = \widetilde{K} \cap K' \arrow[r] \arrow[d, "\Gamma"] & \widetilde{K} \arrow[d, "\Gamma"] \arrow[dd, bend left, dash, "I"] \\
& K' \arrow[r] & \widetilde{K}' \arrow[d, "I'"] \\
& & \overline{K}
\end{tikzcd} \tag{4.2.4}$$

Let S' be the normalization of S in K', and let A^* be the Neron model of $A_{K'}$. Then, by the Galois criterion for semistable reduction (3.5), $A_{K'}$ has semistable reduction over S'; that is, A_o^{*o} is an extension of an abelian scheme by a torus. On the other hand, by transport of structure, G acts on A^*, and hence on A^{*o}, by automorphisms in a manner compatible with the given action of G on the base scheme S' of A^*. It is then clear that the action of π' on

$$U^{\mathrm{ef}} = T_\ell(A^{*o})^f(\overline{K})$$

is obtained by transport of structure from the action of π' on $(A^{*o}/K', \widetilde{K}')$, where π' acts on A^* through G. Taking into account the canonical isomorphism

$$U^{\mathrm{ef}} \simeq T_\ell(A^{*o})^f(\overline{K}) \simeq T_\ell(A_o^{*o})(\overline{k}) \tag{4.2.6}$$

(valid for every twisted constant ℓ-adic sheaf over S'; cf. Exposé I [reference omitted in the source]), we obtain the following description of the action of π' on U^{ef}.

Consider the smooth connected group scheme A_o^{*o} over the residue field $k' \hookrightarrow \overline{k}$ of S', which is a quasi-Galois extension of k with Galois group G_o. The quotient group G of π' acts on A_o^{*o} compatibly with its action on k' through G_o. Moreover, π' acts on $\overline{k}$ through its quotient π_o, and hence acts on the situation $(A_o^{*o}/k', \overline{k})$. Thus, by transport of structure, π' acts on

$$U^{\mathrm{ef}} = T_\ell(A_o^{*o})(\overline{k}),$$

and this is the desired action of π'. In particular, the subgroup Γ of G, which acts trivially on k', acts on A_o^{*o} by k'-automorphisms, and the corresponding representation of $\Gamma = I/I'$ on $T_\ell(A_o^{*o})(\overline{k})$ is the representation deduced from the natural representation of the inertia group I on U^{ef}.

On the other hand, the essentially toric part U^{et} of U corresponds simply to the submodule

$$U^{\mathrm{et}} \simeq T_\ell(T_o^*)(\overline{k}) \subset U^{\mathrm{ef}} \simeq T_\ell(A_o^{*o})(\overline{k}), \tag{4.2.7}$$

where T_o^* is the largest subtorus of A_o^{*o}. Likewise, put

$$B_o^* = A_o^{*o}/T_o^* \tag{4.2.8}$$

for the abelian part of A_o^{*o}. The action of G on A_o^{*o} induces separately actions of G on T_o^* and on B_o^*, and the action of π' on U^{et} and on $U^{\text{ef}}/U^{\text{et}}$ is obtained from these actions by means of the isomorphism (4.2.7) and the corresponding isomorphism

$$U^{\text{ef}}/U^{\text{et}} \simeq T_\ell(B_o^*)(\overline{k}). \tag{4.2.9}$$

It follows from this and the isogeny (4.1.2) that the character $g \longmapsto \operatorname{Tr} g_U$ of the action of π' (or of π) on U is given by

$$\operatorname{Tr} g_{U_\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + \chi_\ell(g) \operatorname{Tr} g^{-1}_{T_o^*,\ell}. \tag{4.2.10}$$

Here U_ℓ has been written for U on the left to recall the chosen prime $\ell \neq p$; $g_{B_o^*,\ell}$ and $g_{T_o^*,\ell}$ denote, respectively, the actions of g on the Tate modules $T_\ell(B_o^*)(\overline{k})$ and $T_\ell(T_o^*)(\overline{k})$ induced by its actions on B_o^* and T_o^* just described; and

$$\chi_\ell : \pi' \longrightarrow \pi_o \longrightarrow \mathbb{Z}_\ell^* \tag{4.2.11}$$

is the character of π' (or of π_o) induced by the action on $\mathbb{Z}_\ell(1) = T_\ell(\overline{k}^*)$—a character that is trivial on Γ.

We are now in a position to prove a special case of a conjecture of Serre–Tate [29, Appendix]. [12]

Theorem 4.3. *Let S be a henselian trait with residue field k and function field K, and let A_K be an abelian scheme over K. Consider the representation of $\pi = \operatorname{Gal}(\overline{K}/K)$ on the Tate module $U_\ell = T_\ell(A(\overline{K}))$, where ℓ is a prime distinct from the residual characteristic.*

a) *There exists an open subgroup I' of the inertia group that acts unipotently on U_ℓ, so that the character of the representation of π on U_ℓ is in fact a function on π/I'. Moreover, on $\Gamma = I/I'$ this function takes integral values and is independent of ℓ. More generally, for every $g \in I$, the coefficients of the characteristic polynomial of g_{U_ℓ} are rational integers independent of ℓ.*

b) *Suppose that k is a finite field with q elements, so that*

$$\pi_o = \pi/I \simeq \operatorname{Gal}(\overline{k}/k)$$

is identified with the profinite completion $\widehat{\mathbb{Z}}$ of $\mathbb{Z}$, the generator 1 of $\mathbb{Z}$ corresponding to the Frobenius automorphism

$$f_q : x \longmapsto x^q$$

of $\overline{k}/k$. Then, for every $g \in \pi$ whose image in π_o lies in $\mathbb{Z}$, the trace (and, more generally, every coefficient of the characteristic polynomial) of g_{U_ℓ} is a rational number independent of ℓ. It is even an integer if the image of g in π_o is of the form f_q^n, with $n \geq 0$.

Let us prove, for example, the assertions concerning traces. The assertions concerning the coefficients of the characteristic polynomial could be proved in the same way. One may also note that they follow formally from the assertions about traces, by expressing these coefficients in terms of the traces of the iterates (although those formulas introduce denominators and therefore do not give the asserted integrality), together with a lemma of Dwork–Fatou, for which a pleasant proof is given by Kleiman in [14, 2.8].

a) We use formula (4.2.10), which here becomes

$$\operatorname{Tr} g_{U_\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + \operatorname{Tr} g^{-1}_{T_o^*,\ell}. \tag{4.3.0}$$

Here $g_{B_o^*}$ and $g_{T_o^*}$ are now k'-automorphisms of B_o^* and T_o^*, respectively. It therefore suffices to show that the two corresponding traces are rational integers independent of ℓ, which is a general "geometric" assertion about algebraic groups. For $g_{B_o^*}$ this is a well-known result of Weil in the theory of abelian varieties [5] [15]. For $g_{T_o^*}$ it is practically immediate: one describes the

12. This is essentially the case $i = 1$ of those conjectures for the H^i.

representation of Γ on $T = T_o^*$ by means of the corresponding representation on the dual group $M' \simeq \mathbb{Z}^\mu$. The required trace is that of the contragredient of the latter representation, by the canonical isomorphism

$$T_\ell(T) \simeq \check{M}_\ell(1) \overset{\text{def}}{=} \check{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1).$$

b) If the image of g in π_o is $f_q^{;n}$, formula (4.2.10) now becomes

$$\text{(4.3.1)} \qquad \operatorname{Tr} g_{U,\ell} = \operatorname{Tr} g_{B_o^*,\ell} + \operatorname{Tr} g_{T_o^*,\ell} + q^n \operatorname{Tr} g^{-1}_{T_o^*,\ell}.$$

We are thus reduced to proving the following lemma, in which k is to be replaced by an algebraic closure of k', and A by $A_o^{*o} \otimes_{k'} \overline{k}$.

Lemma 4.3.2. Let k be the algebraic closure of a finite field, let A be a commutative group scheme of finite type over k, and let g be an automorphism of this group scheme compatible with the automorphism f of $\operatorname{Spec}(k)$ obtained by transport of structure from an iterated Frobenius automorphism

$$x \longmapsto x^q, \qquad q = p^r, \quad r \geq 0,$$

where q is a power of the characteristic. Thus the action of g on functions φ satisfies

$$g^*(\lambda\varphi) = f^*(\lambda) g^*(\varphi) = \lambda^{q^{-1}} g^*(\varphi) = \lambda^{p^{-r}} g^*(\varphi) \qquad (\lambda \in k).$$

Let g_ℓ be the automorphism induced by g on $T_\ell(A(k))$ by "transport of structure," where ℓ is a prime different from p. Then the trace of g_ℓ (and, more generally, every coefficient of its characteristic polynomial) is a rational integer independent of ℓ, and its eigenvalues are algebraic integers of absolute value $q^{1/2}$ or q. If A is an abelian scheme, their absolute values are $q^{1/2}$; if A is affine, the eigenvalues are of the form $q\varepsilon_i$, where the ε_i are roots of unity.

Indeed, consider the "q-th power" endomorphism f_q^A of the underlying absolute scheme of A, defined as the identity on points and as the q-th power map on the structure sheaf. It is an endomorphism of the group scheme A compatible with the endomorphism $f_q^{\operatorname{Spec}(k)} = f^{-1}$ of the base $\operatorname{Spec}(k)$. Consequently the composite

$$\text{(4.3.3)} \qquad F = f_q^A g$$

is an endomorphism of the group scheme A compatible with the identity on $\operatorname{Spec}(k)$; in other words, it is a k-endomorphism of the group scheme A. I claim that the automorphism Γ_g of $A(k)$ defined by g through transport of structure, namely

$$\text{(4.3.4)} \qquad \Gamma_g(u) = g \circ u \circ f^{-1} \qquad \text{for } u \in A(k),$$

may also be described by

$$\text{(4.3.5)} \qquad \Gamma_g(u) = F(u) \qquad \text{for } u \in A(k).$$

To verify this, it is enough to observe that the underlying set maps of the morphisms $\operatorname{Spec}(k) \longrightarrow A$ represented by the two sides are the same. This reduces us to the equality obtained from (4.3.5) by composing its left-hand member on the left with f_q^A and its right-hand member on the right with f^{-1}, since f_q^A and f^{-1} induce the identity on the underlying sets of the source schemes. The resulting formula follows immediately from (4.3.3) and (4.3.4).

Applying (4.3.5) to $u \in {}_{\ell^\nu}A(k)$ and passing to the inverse limit over ν, we find that g_ℓ may also be interpreted as the endomorphism $T_\ell(F)$ of $T_\ell(A(k))$ associated with the endomorphism F of A. It is known, on the other hand, that $T_\ell(F)$ has a characteristic polynomial with integral coefficients independent of ℓ. By devissage, this reduces to the case in which A is either an abelian scheme or a torus, where the required result was already used in part a) of the proof. The same argument also proves the assertions about the eigenvalues, in view, in the abelian-scheme case, of the classical results of Weil [15].

This proves 4.3.2 and completes the proof of 4.3. At the same time it proves the following result.

Corollary 4.4. Suppose that the hypotheses of 4.3 b) hold. Then, for every $g \in \pi'$ whose image in π_o is of the form f_q^n with $n \geq 0$, the eigenvalues of the induced automorphisms of

$$U^{\mathrm{et}}, \qquad U^{\mathrm{ef}}/U^{\mathrm{et}}, \qquad U/U^{\mathrm{ef}}$$

are algebraic integers whose absolute values are, respectively, q^n, $q^{n/2}$, and 1. For the two extreme pieces one may say more precisely that the corresponding eigenvalues are, respectively,

$$(q^n\varepsilon_i)_{1\leq i\leq\mu} \qquad \text{and} \qquad (\varepsilon_i)_{1\leq i\leq\mu},$$

where the ε_i $(1 \leq i \leq \mu)$ are roots of unity.

Indeed, formula (4.1.2) gives for the last piece the eigenvalues

$$\frac{q^n}{q^n\varepsilon_i} = \varepsilon_i^{-1}.$$

But since ε_i is a root of unity, one has $\varepsilon_i^{-1} = \overline{\varepsilon_i}$ (complex conjugate), and since the family of the ε_i is stable under complex conjugation (being the family of roots of an equation with integral coefficients), the family of the ε_i^{-1} is also the family of the ε_i. This proves the last assertion of 4.4.

4.5. We shall give an application of 4.3 a) to the theory of the local conductor of abelian schemes. Suppose that k is algebraically closed. [13] Recall (SGA 5 X 6.2, or [20]) that, for every commutative etale ℓ-group scheme F_K over K $(\ell \neq p)$, one defines an integer

$$\alpha_s(F_K) = \alpha_s(i_*F_K), \tag{4.5.1}$$

where $i : \eta \longrightarrow S$ denotes the inclusion, as follows. One chooses a finite Galois extension K' of K, with group Γ, that splits F_K, so that F_K is represented by a finite ℓ-group M on which Γ acts. On the other hand, one associates with the extension K'/K a certain projective $\mathbb{Z}_\ell[\Gamma]$-module, defined up to isomorphism,

$$\mathrm{Sw} = \mathrm{Sw}(K'/K, \mathbb{Z}_\ell), \tag{4.5.2}$$

and sets

$$\alpha_s(F_K) = \operatorname{length} \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(\mathrm{Sw}, M). \tag{4.5.3}$$

One checks (loc. cit.) that this expression is independent of the choice of the extension K'/K. Moreover, the projectivity of Sw as a $\mathbb{Z}_\ell[\Gamma]$-module implies that $\alpha_s(F_K)$ is additive in exact sequences.

We apply this definition to the sheaf ${}_\ell A_K$, where A_K is our abelian scheme over K. In this case one therefore obtains an integer

$$\alpha_s(A_K;\ell) = \operatorname{length} \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}\big(\mathrm{Sw}, U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell\big), \qquad U_\ell = T_\ell(A(\overline{K})), \tag{4.5.4}$$

where Γ is chosen as above (a choice that a priori depends on ℓ). The integer (4.5.4) is called the *wild conductor exponent* of the abelian scheme A_K relative to the discrete valuation ring $V \subset K$. [14]

13. It would in fact suffice for k to be perfect, but, after passing to the strict henselization of S, one is immediately reduced to the case considered here.

14. One should also define the *tame conductor exponent* of A_K relative to V as the difference between the lengths over $\mathbb{Z}/\ell\mathbb{Z}$ of the geometric generic and special fibres of ${}_\ell A^o$. It is independent of ℓ and equals $2\lambda+\mu$ in the notation of (1.1.11). The (total) conductor exponent of A_K is the sum of the tame and wild exponents; it is the local term occurring in the Euler–Poincare formula for ${}_\ell A^o$ in SGA 5 X 6.2 and [20].

After passage to the henselization it is of course defined without assuming that V is henselian, and one is immediately reduced to the case in which S is strictly local. We intend to prove the following result.

Corollary 4.6. The wild conductor exponent (4.5.4) is independent of the choice of the prime $\ell \neq p$.

We may suppose that S is strictly local. We compute (4.5.4), replacing Sw by any other projective $\mathbb{Z}_\ell(\Gamma)$-module L. Observe that the filtration (4.1.1) of U_ℓ consists of $\mathbb{Z}_\ell$-submodules that are manifestly direct summands. It therefore induces a corresponding filtration of

$$U(\ell) = U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell,$$

whose successive factors are obtained, by tensoring with $\mathbb{F}_\ell$, from the factors V_i $(i = 1, 2, 3)$

$$U^{\mathrm{et}}, \qquad U^{\mathrm{ef}}/U^{\mathrm{et}}, \qquad U/U^{\mathrm{ef}}$$

appearing in (4.1.1). This filtration is, of course, stable under Γ. Additivity thus decomposes

$$\operatorname{length} \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell))$$

as a sum of three terms, the lengths of the groups

$$(*) \qquad \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell) \simeq \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i) \otimes_{\mathbb{Z}_\ell} \mathbb{F}_\ell.$$

The isomorphism in (*) again follows from the assumed projectivity of L; it is only necessary to choose K' large enough for Γ to act on the V_i. Now $\operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, V_i)$ is plainly a torsion-free $\mathbb{Z}_\ell$-module, so its rank is the length of the right-hand side of (*). If φ_i is the character of the representation of Γ on V_i, and ψ the character of the representation of Γ on $\check{L}$, we obtain

$$(4.6.1) \qquad \operatorname{length} \operatorname{Hom}_{\mathbb{Z}_\ell(\Gamma)}(L, U(\ell)) = \sum_{1 \le i \le 3} N^{-1} \sum_{g \in \Gamma} \psi(g) \varphi_i(g)$$

$$= N^{-1} \sum_{g \in \Gamma} \psi(g) \varphi(g), \qquad N = \operatorname{card} \Gamma,$$

where $\varphi(g) = \sum_i \varphi_i(g)$ is also the function on Γ obtained, by passage to the quotient, from the character of the monodromy representation of the inertia group I on U. By 4.3 a), this character is an integral-valued rational function independent of ℓ. Taking $L = \mathrm{Sw}$, the same is true of the character $\psi(g)$ of L, which is

$$a(g) - N\delta(g) + 1,$$

where δ is the Dirac function on Γ and a is the Artin conductor of K'/K [26]. This proves 4.6 and at the same time gives the following expression, independent of ℓ, for the wild conductor exponent:

$$(4.6.2) \qquad \alpha_s(A_K) = N^{-1} \sum_{g \in \Gamma} \big(a(g) - N\delta(g) + 1\big) \varphi(g).$$

Here Γ is the Galois group of a finite Galois extension K' of K that splits U^{ef} (for example the smallest such extension; cf. 4.1), $a(g)$ is the Artin conductor on Γ, and $\varphi(g)$ is the trace function on Γ described in (4.3.0). In the semistable-reduction case one may take $K' = K$, hence $\Gamma = \{e\}$, and obtains

$$(4.6.3) \qquad \alpha_s(A_K) = 0 \qquad \text{(semistable-reduction case)}.$$

To conclude, let us state explicitly a criterion for semistable reduction that follows from the general considerations of 4.2.

Proposition 4.7. (M. Raynaud's criterion for semistable reduction.) Let S be a trait, let A_K be an abelian scheme over its function field K, and let $n > 1$ be an integer prime to the residual characteristic p. Suppose that ${}_nA_K$ is unramified relative to S, that is, that the inertia group acts trivially on ${}_nA(\overline{K})$. Then either A_K has semistable reduction over S, or $n = 2$; in the latter case, if S is strictly local, A_K acquires semistable reduction over a finite Galois extension K' of K whose Galois group is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$.[15]

We may plainly suppose that S is strictly local. With the notation of 4.1 and 4.2, take K' as small as possible; in other words, suppose that Γ acts faithfully on A_o^{*o}. By hypothesis Γ acts trivially on ${}_nA(\overline{K})$, and hence a fortiori on

$${}_nA(\overline{K})^{\mathrm{ef}} \simeq {}_nA_o^*(k) \supset {}_nA_o^{*o}(k).$$

The conclusion follows from the next two lemmas.

Lemma 4.7.1. (J.-P. Serre.) Let G be a commutative group scheme over a field k, which is an extension of an abelian scheme by a torus, and let $n \geq 2$ be an integer. Every finite-order automorphism of G that induces the identity on ${}_nG$ is the identity if $n \neq 2$, and is either the identity or has order two if $n = 2$.

Lemma 4.7.2. A finite group in which every element distinct from 1 has order 2 is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^r$ for a suitable $r \geq 0$.

4.7.3. Let us recall the well-known proof of 4.7.1. Since $n\,\mathrm{id}_G$ is an epimorphism, the hypothesis that u induces the identity on ${}_nG$, or equivalently that $1 - u$ induces zero on ${}_nG$, is expressed by the condition

$$1 - u = nv$$

in the endomorphism ring of G. To prove the asserted conclusion, we reduce, by decomposing the finite cyclic group generated by u into its p-primary components for the various primes p dividing the order of u, to the case in which u has order p^N.

Let E be the subring of $\mathrm{End}(G)$ generated by v. Since $u = 1 - nv$ and $u^{p^N} = 1$, it is a quotient of

$$\mathbb{Z}[T]/\big((1 - nT)^{p^N} - 1\big);$$

it is therefore of finite type over $\mathbb{Z}$. It is moreover torsion-free (a formal consequence of the divisibility of G), and thus embeds in

$$E_{\mathbb{Q}} = E \otimes_{\mathbb{Z}} \mathbb{Q}.$$

The latter is a quotient, after the change of variable $u = 1 - nT$, of $\mathbb{Q}[u]/(u^{p^N} - 1)$, which is a direct product of finite field extensions of $\mathbb{Q}$; the same is therefore true of $E_{\mathbb{Q}}$. Considering the images of E, u, and v in the component fields K_i of $E_{\mathbb{Q}}$, it remains to prove the following assertion. Let $K = \mathbb{Q}(u)$ be generated by a primitive p^m-th root of unity u, and let $n \geq 2$ be an integer such that

$$u - 1 \equiv 0 \pmod{n}$$

in the ring of integers of K. Then $u = 1$, or $u = -1$ and $n = 2$.

This follows at once from the familiar fact that, if $u \neq 1$, that is, $m \geq 1$, the ideal $(u-1)$ is prime and (p) is a power of it, with exponent greater than 1 except when $p = 2$ and $m = 1$, that is, when $u = -1$. Since $(u-1)$ is prime and divides p, an integer $n \geq 2$ can divide $u - 1$ only if $n = p$; and p can divide $u - 1$ only if $(u-1) = (p)$, that is, if $u = -1$. Hence $n = 2$.

4.7.4. Let us prove 4.7.2. It is enough to prove that Γ is commutative. If $x, y \in \Gamma$, then $(xy)(xy) = 1$; since $x^2 = 1$ and $y^2 = 1$, this may also be written

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx,$$

15. As follows from an example communicated to me by J. Tate, 4.7 is no longer valid when n equals the residual characteristic p, even if K has characteristic zero and A_K is an elliptic curve with complex multiplication.

as required.

Remark 4.7.5. The proof given in 4.7.3, together with 4.7.2, shows that if G is a divisible commutative group in a topos, $n \geq 2$ is an integer, and Γ is a finite group acting faithfully on G while inducing the identity on ${}_nG$, then Γ is the trivial group if $n \geq 3$, and is isomorphic to a group $(\mathbb{Z}/2\mathbb{Z})^r$ if $n = 2$.

5. The orthogonality theorem in the case $\ell = p$ (semistable case). Duality of the abelian schemes B_o and B'_o. Characterization of the fixed part. Criteria for good reduction, essentially good reduction, and semistable reduction.

5.1. Let us place ourselves in the situation of 2.2.3, where S is henselian and A^o is ℓ-divisible, so that the fixed part

$$T_\ell(A^o)^f \hookrightarrow T_\ell(A)$$

is defined as the pro-ℓ Barsotti–Tate group formed by the $({}_{\ell^\nu}A^o)^f$. We shall now give a definition of the toric part (2.3.2) of $T_\ell(A^o)$ which is independent of the hypothesis $\ell \neq p$ made in 2.3, but which instead assumes that S is complete.[16] For this purpose, for every integer $n \geq 0$ introduce the local Artinian scheme

$$S_n = \operatorname{Spec}(V/\mathfrak{m}^{n+1}), \tag{5.1.1}$$

the n-th infinitesimal neighbourhood of s in S, and, for every S-scheme X, put $X_n = X \times_S S_n$. Since $(A^o)_n = (A_n)^o$ is smooth over S_n, it is known (SGA 3 IX 3.6 bis) that T_o is the special fibre of a unique subtorus

$$T_n \subset A^o_n. \tag{5.1.2}$$

By the uniqueness of the T_n, they are obtained from one another by the base changes $S_{n'} \longrightarrow S_n$. In other words, they define a formal subtorus $\widehat{T}$ of the group $\widehat{A}^o$ (in the category of formal schemes over S), the formal completion of A^o along the special fibre:

$$\widehat{T} \subset \widehat{A}^o. \tag{5.1.3}$$

Since the functor $X \longmapsto \widehat{X}$ (formal completion along the special fibre) induces an equivalence between the category of finite schemes over S and that of finite formal schemes over S (EGA III 4.8), we shall simply identify a finite scheme over S with the corresponding formal scheme. With this identification, the definition of the *fixed parts* immediately gives, for every $\nu \geq 0$,

$$({}_{\ell^\nu}A^o)^f = {}_{\ell^\nu}(\widehat{A}^o), \tag{5.1.4}$$

which may also be written as an identity of projective systems:

$$T_\ell(A^o)^f = T_\ell(\widehat{A}^o). \tag{5.1.5}$$

We therefore obtain a Barsotti–Tate pro-subgroup of the fixed part of $T_\ell(A^o)$ by taking $T_\ell(\widehat{T}) \subset T_\ell(\widehat{A}^o)$. We accordingly define the toric part of $T_\ell(A^o)$ by

$$T_\ell(A^o)^t \overset{\text{def}}{=} T_\ell(\widehat{T}) \subset T_\ell(\widehat{A}^o) = T_\ell(A^o)^f. \tag{5.1.6}$$

Thus, by definition,

$$T_\ell(A^o)^t = ({}_{\ell^\nu}\widehat{T})_{\nu \geq 0}, \tag{5.1.7}$$

the finite flat S-group ${}_{\ell^\nu}\widehat{T}$ itself being described as the inductive limit of the finite flat S_n-groups

$$ {}_{\ell^\nu}(T_n), \qquad n \text{ variable}; \tag{5.1.8}$$

16. This hypothesis is in fact not essential, as is shown in 6.1 below.

that is, its affine algebra is described as the projective limit of the affine algebras of the groups (5.1.8). Of course,

$$\big(T_\ell(A^o)^t\big)_n = T_\ell(T_n). \tag{5.1.9}$$

For $n = 0$ this is precisely formula (2.3.3), by which we characterized the toric part (as a profinite etale group) in the case $\ell \neq p$. This shows that definition (5.1.6) is compatible with that given in 2.3 when $\ell \neq p$.

Passing to generic fibres, the fixed and toric parts of $T_\ell(A^o)$ define fixed and toric parts in $T_\ell(A_K)$:

$$T_\ell(A_K)^t \subset T_\ell(A_K)^f \subset T_\ell(A_K). \tag{5.1.10}$$

These are inclusions of pro-ℓ Barsotti–Tate groups. Consider the analogous filtration corresponding to the dual abelian scheme A'_K:

$$T_\ell(A'_K)^t \subset T_\ell(A'_K)^f \subset T_\ell(A'_K). \tag{5.1.11}$$

We then have the following result, which generalizes 2.4 (at least in the case where S is complete, to which one could easily have reduced in 2.4).

Theorem 5.2. Suppose that S is a complete trait, and let A_K be an abelian scheme over K, A'_K the dual abelian scheme, and ℓ a prime such that the connected Neron model A^o is ℓ-divisible (cf. 2.2.1), so that one then has the filtrations (5.1.10) and (5.1.11) of the pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$. Then the toric parts are recovered from the fixed parts by the formula

$$T_\ell(A_K)^t = T_\ell(A_K)^f \cap \big(T_\ell(A'_K)^f\big)^\perp, \tag{5.2.1}$$

and by the analogous formula obtained by interchanging the roles of A_K and A'_K, where $\perp$ denotes the orthogonal complement with respect to the canonical pairing (1.0.2).

Finally, when A_K has semistable reduction over S (which follows from the other hypotheses in the case $\ell = p$), one even has the formula

$$T_\ell(A_K)^t = \big(T_\ell(A'_K)^f\big)^\perp. \tag{5.2.2}$$

5.2.3. The meaning of formula (5.2.1) is quite clear when ℓ is prime to the characteristic of K: as in 2.4, it can then be interpreted by passing to the ordinary $\mathbb{Z}_\ell$-modules, with an action of $\pi = \mathrm{Gal}(\overline{K}/K)$, formed by the $\overline{K}$-points of the pro-groups involved. Let us spell it out in the general case, using the notion of the dual

$$D(G) = \mathrm{Hom}\big(G, T_\ell(\mathbb{G}_m)\big) \tag{5.2.3.1}$$

of a pro-ℓ Barsotti–Tate group $G = (G(\nu))$. By definition, this is the projective system of Cartier duals $D(G(\nu))$ (SGA 3 VII 3.3.1 and VIII 1), whose transition morphisms are defined in the evident way (for example, by transposition from the inclusion morphisms among the $G(\nu)$). The duality theory of abelian varieties tells us that the pairing mentioned in 5.2 actually defines an isomorphism

$$T_\ell(A'_K) \simeq D\big(T_\ell(A_K)\big), \tag{5.2.3.2}$$

and hence, by composition, a morphism

$$T_\ell(A_K)^f \longrightarrow D\big(T_\ell(A'_K)^f\big). \tag{5.2.3.3}$$

Formula (5.2.1) says that the kernel of the preceding homomorphism (which a priori is a pro-subgroup of its first term) is precisely the toric part $T_\ell(A_K)^t$. Notice that this is an isomorphism of pro-objects, and not of projective systems: we do not assert (and it would in general be false) that, for every integer ν, $T_\ell(A_K)^t(\nu)$ equals the kernel of

$$T_\ell(A_K)^f(\nu) \longrightarrow D\big(T_\ell(A'_K)^f\big)(\nu);$$

the latter can a priori be larger. We shall see below, however, that it is not larger if A_K has semistable reduction; by the hypotheses made in 5.2, this is in particular the case when $\ell = p$.

5.2.4. The proof of (5.2.1) merely repeats, with practically no change, the proof of 2.4. Observe that the induced pairing

$$T_\ell(A_K)^f \times T_\ell(A'_K)^f \longrightarrow T_\ell(\mathbb{G}_{mK}) \tag{5.2.4.1}$$

which gives rise to (5.2.3.3) is induced by a pairing

$$T_\ell(A^o)^f \times T_\ell(A'^o)^f \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.2.4.2}$$

of pro-Barsotti–Tate groups over S, itself induced by the pairing of pro-groups

$$T_\ell(A^o) \times T_\ell(A'^o) \longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{5.2.4.3}$$

deduced from the biextension W of (A^o, A'^o) by $\mathbb{G}_{mS}$. Consequently, the pairing

$$T_\ell(A^o_o) \times T_\ell(A'^o_o) \longrightarrow T_\ell(\mathbb{G}_{mk}), \tag{5.2.4.4}$$

obtained from (5.2.4.2) by passing to the special fibres, is precisely the pairing deduced from the biextension W_o of (A^o_o, A'^o_o) by $\mathbb{G}_{mk}$. By VIII 4.11 and 1.5, the left annihilator of the pairing (5.2.4.4) is therefore $T_\ell(T_o)$, which is precisely the special fibre of the toric part $T_\ell(A^o)^t$. Returning to the definition of left annihilators given in 5.2.3, we conclude by means of the following lemma.

Lemma 5.2.5. Let $u : G \longrightarrow H$ be a homomorphism of pro-ℓ Barsotti–Tate groups over S. If $u_o : G_o \longrightarrow H_o$ is zero, then u is zero. If $u_o : G_o \longrightarrow H_o$ is an isogeny (respectively an isomorphism), then u is an isogeny (respectively an isomorphism).

Indeed, apply the first assertion of 5.2.5 to the composite morphism

$$T_\ell(A^o)^t \longrightarrow T_\ell(A^o)^f \longrightarrow D\big(T_\ell(A'^o)^f\big),$$

where the second morphism is deduced from the pairing (5.2.4.2). The composite is zero, so the pairing under consideration vanishes on the toric part on the left. By symmetry it also vanishes on the toric part on the right, and hence descends to a pairing on the quotients, or *abelian pieces*,

$$T_\ell(A^o)^{ab} \times T_\ell(A'^o)^{ab} \longrightarrow T_\ell(\mathbb{G}_m).$$

These quotients are again pro-Barsotti–Tate groups, as one checks readily. It remains to prove that the morphism corresponding to this pairing,

$$T_\ell(A^o)^{ab} \longrightarrow D\big(T_\ell(A'^o)^{ab}\big),$$

is an isogeny, which again follows from 5.2.5.

We leave to the reader the easy proof of 5.2.5, a result valid over any connected Noetherian base. Moreover, below in 7.4.7, by a somewhat more precise analysis of the biextensions W_n of (A^o_n, A'^o_n), we shall find a quite evident direct proof of the two special cases in which we had to use 5.2.5.

5.2.6. Suppose now that A_K has semistable reduction over S, so that ℓ may be any prime whatsoever (equal or unequal to p). Moreover, by the Galois criterion for semistable reduction 3.5, for $\ell \neq p$ one has

$$\big(T_\ell(A'_K)^f\big)^\perp \ \big(= T_\ell(A_K)^t\big) \subset T_\ell(A_K)^f. \tag{5.2.7}$$

To finish the proof of 5.2, it remains to show that this inclusion is valid without restriction on ℓ.

Indeed, the left-hand term is canonically isomorphic to

$$D\big(T_\ell(A'_K)/T_\ell(A'_K)^f\big)$$

and therefore has *rank* (or *height*) $2n-2\alpha-\mu=\mu$, in the notation of (2.1.9), since by hypothesis $\lambda=0$. This is also the rank of $T_\ell(A_K)^t$. Since the inclusion

$$(*)\qquad T_\ell(A_K)^t\subset T_\ell(A_K)$$

is strict, in the sense that it already induces a monomorphism on every term of the projective systems under consideration, it follows readily that the left-hand term of (*) is maximal among the pro-Barsotti–Tate subgroups of $T_\ell(A_K)$ having the same rank. Thus equality holds in (5.2.7); by the orthogonality theorem this is equivalent to the inclusion between the two extreme terms of (5.2.7).

5.3. Using the orthogonality correspondence between strict pro-Barsotti–Tate subgroups (in the sense specified above) of the dual pro-groups $T_\ell(A_K)$ and $T_\ell(A'_K)$, one formally deduces from (5.2.2) that the pairing (1.0.2) between these pro-groups induces a pairing

$$(5.3.2)\qquad T_\ell(A_K)^{ab}\times T_\ell(A'_K)^{ab}\longrightarrow T_\ell(\mathbb{G}_{mK})$$

which is a perfect duality, that is, it identifies each of the two factors on the left with the dual of the other (in the sense of 5.2.3).[17] It is also clear that (5.3.2) is induced by the pairing

$$(5.3.3)\qquad T_\ell(A^o)^{ab}\times T_\ell(A'^o)^{ab}\longrightarrow T_\ell(\mathbb{G}_{mS})$$

obtained from the pairing (5.2.4.2) by passing to the quotients. Since this is a perfect duality on the generic fibres, that is, since the corresponding homomorphism

$$(5.3.4)\qquad T_\ell(A^o)^{ab}\longrightarrow D\big(T_\ell(A'^o)^{ab}\big)$$

is an isomorphism on the generic fibres, one naturally wishes to deduce that it is so over the whole of S, or equivalently on the special fibres. This deduction is trivially valid if $\ell\neq p$. It is also valid, at least in unequal characteristics when $\ell=p$, by a deep theorem of Tate [33, Th. 4]. We shall nevertheless establish it directly below in 7.4.6, without using Tate's theorem.

To make precise the geometric meaning of the result we wish to prove, observe that the pairing obtained from (5.3.3) by passage to the special fibres is precisely the pairing

$$(5.3.5)\qquad T_\ell(B_o)\times T_\ell(B'_o)\longrightarrow T_\ell(\mathbb{G}_{mk})$$

deduced from the biextension W_o of (A^o_o,A'^o_o) by passage to the quotient, where B_o and B'_o denote respectively the abelian parts of A^o_o and A'^o_o. By VIII 4.8, the biextension W_o comes from a biextension W^{ab}_o of (B_o,B'_o) by $\mathbb{G}_{mk}$, unique up to a unique isomorphism. Thus the pairing (5.3.5) may also be interpreted as the one deduced from the divisorial correspondence W^{ab}_o, and the corresponding homomorphism

$$(5.3.6)\qquad T_\ell(B_o)\longrightarrow D\big(T_\ell(B'_o)\big)\simeq T_\ell\big(D(B'_o)\big)$$

(where, in the last term, D denotes the dual abelian scheme) is precisely the morphism obtained by applying T_ℓ functorially to the homomorphism

$$(5.3.7)\qquad B_o\longrightarrow D(B'_o)$$

defined by W^{ab}_o. Consequently, by the standard dictionary (cf. [5]), saying that (5.3.5) is a perfect duality is also saying that the kernel of (5.3.7), which is a finite group over k by 1.5, has zero

17. The superscripts ab denote the *abelian parts*, the quotient of the fixed part by the toric part.

ℓ-primary component. Saying that this holds for every ℓ therefore means that the isogeny (5.3.7) is an isomorphism, that is, that the divisorial correspondence W_o^{ab} is a perfect duality between B_o and B'_o. The preceding considerations show only that the kernel of (5.3.7) is a p-group. We therefore state the more complete result.

Theorem 5.4. Let S be a trait, and let A_K be an abelian scheme over K having semistable reduction over S. Let B_o and B'_o denote respectively the abelian parts of the Neron models of A_K and of the dual abelian scheme A'_K. Then the canonical divisorial correspondence W_o^{ab} on (B_o, B'_o), deduced by VIII 4.8 from the canonical biextension W_o of $(A_o^o, A_o'^o)$ (the special fibre of the biextension W of 1.4), is a perfect duality between B_o and B'_o; that is, it defines an isomorphism of each of these abelian schemes with the dual of the other.

The proof of 5.4 in the general case will be given in section 7, as a consequence of a very natural and important construction due to Raynaud. Let us merely observe here that, in proving 5.4, one may assume that S is complete, by 3.3.

5.5. To express conveniently the extreme factors in the filtrations (5.1.10) and (5.1.11), in the case of semistable reduction, let us introduce the twisted constant group schemes over S

$$\underline{M}'_o = D(T_o), \qquad \underline{M}_o = D(T'_o), \tag{5.5.1}$$

dual to the maximal tori T_o, T'_o of the special fibres of A, A' (SGA 3, X 5.7). Since S is complete local, these group schemes may be regarded as the special fibres of twisted constant schemes

$$\underline{M}', \qquad \underline{M} \quad \text{over } S. \tag{5.5.2}$$

Thus, if the torus T_o is split (for example, if k is separably closed), that is, if $\underline{M}'_o$ is the constant group defined by the ordinary group

$$M' = \mathrm{Hom}_{k\text{-gr}}(T_o, \mathbb{G}_{mk}), \tag{5.5.3}$$

then $\underline{M}'$ is the constant group defined by M':

$$\underline{M}' = M'_S.$$

One may moreover observe that, since T_o and T'_o are isogenous (2.2.6), T'_o is split whenever T_o is. In this case one should therefore also introduce

$$M = \mathrm{Hom}_{k\text{-gr}}(T'_o, \mathbb{G}_{mk}), \tag{5.5.4}$$

and then

$$\underline{M} = M_S.$$

With this notation, for every integer n there are evidently isomorphisms

$$\underline{M}'_n \simeq D(T_n), \qquad \underline{M}_n \simeq D(T'_n), \tag{5.5.5}$$

where T'_n is defined in terms of A' as T_n is in terms of A (5.1.2). Hence there follow canonical isomorphisms (where $\mathbb{Z}_\ell(1) = T_\ell(\mathbb{G}_{mS})$)

$$T_\ell(T_n) \simeq \check{\underline{M}}'_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1), \qquad T_\ell(T'_n) \simeq \check{\underline{M}}_n \otimes_{\mathbb{Z}} \mathbb{Z}_\ell(1), \tag{5.5.6}$$

and, on passing to the limit over n, canonical isomorphisms

$$\begin{cases} T_\ell(A^o)^t = T_\ell(\widehat{T}) \simeq \check{\underline{M}}' \otimes \mathbb{Z}_\ell(1), \\ T_\ell(A'^o)^t = T_\ell(\widehat{T}') \simeq \check{\underline{M}} \otimes \mathbb{Z}_\ell(1). \end{cases} \tag{5.5.7}$$

In the case of semistable reduction, duality, together with the orthogonality theorem 5.2 and relation (5.2.2), then gives canonical isomorphisms

$$(5.5.8)\qquad \begin{cases} T_\ell(A_K)/T_\ell(A_K)^f \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell, \\ T_\ell(A'_K)/T_\ell(A'_K)^f \simeq \underline{M}'_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell. \end{cases}$$

Likewise, the same formula (5.2.2) gives canonical isomorphisms

$$(5.5.9)\qquad \begin{cases} T_\ell(A_K)/T_\ell(A_K)^t \simeq D\big(T_\ell(A'_K)^f\big), \\ T_\ell(A'_K)/T_\ell(A'_K)^t \simeq D\big(T_\ell(A_K)^f\big), \end{cases}$$

where one should recall that the groups $T_\ell(A_K)^f$ and $T_\ell(A'_K)^f$ occurring on the right are the restrictions to η of the pro-ℓ Barsotti–Tate groups $T_\ell(A^o)^f$ and $T_\ell(A'^o)^f$, respectively, defined over S. Here D denotes duality in the sense of pro-Barsotti–Tate groups, namely $\underline{\mathrm{Hom}}(-, \mathbb{Z}_\ell(1))$. Thus we have:

Proposition 5.6. Suppose that S is complete and that A_K has semistable reduction over S. Then, for every prime number ℓ (which may equal the residue characteristic of S), the pro-ℓ Barsotti–Tate group $T_\ell(A_K)^t$ is the generic fibre of a pro-Barsotti–Tate group over S "of toric type", given by (5.5.7), whereas $T_\ell(A_K)/T_\ell(A_K)^f$ is the generic fibre of a pro-Barsotti–Tate group over S of pro-étale type, given by (5.5.8). Finally, the groups $T_\ell(A_K)^f$ and $T_\ell(A_K)/T_\ell(A_K)^t$ are the generic fibres of pro-Barsotti–Tate groups defined over S, equal respectively to $T_\ell(A^o)^f$ and to $D(T_\ell(A'^o)^f) = \underline{\mathrm{Hom}}(T_\ell(A'^o)^f, \mathbb{Z}_\ell(1))$.

5.6.1. Observe that, for an étale covering S' of S, splitting $T_\ell(A_K)^t$ is equivalent to splitting $T_\ell(A_K)/T_\ell(A_K)^f$: these conditions say respectively that S' splits the twisted constant groups $\underline{M}$ and $\underline{M}'$, and these are isogenous because T_o and T'_o are. At the same time one sees that there is a smallest étale extension S' of S which does the job.

We also record the following easy consequence of the definitions in 2.2.3.

Proposition 5.7. Suppose that A^o is ℓ-divisible and that S is henselian. Let F_K be a constructible ℓ-adic sheaf over K which is "unramified", that is, which extends to a twisted constant ℓ-adic sheaf F over S, and let $u : F_K \longrightarrow T_\ell(A_K)$ be a homomorphism of pro-groups. Then u factors through a homomorphism from F to $T_\ell(A_K)^f$.

Indeed, write

$$(5.7.1)\qquad u = (u(\nu))_{\nu \geqslant 0}, \qquad u(\nu) : F_K(\nu) \longrightarrow T_\ell(A_K)(\nu) = {}_{\ell^\nu}A_K.$$

By the universal property of A, since $F(\nu)$ is étale and hence smooth over S, the morphism $u(\nu)$ comes from a homomorphism

$$v(\nu) : F(\nu) \longrightarrow {}_{\ell^\nu}A.$$

By definition of the fixed part, this homomorphism factors as

$$v(\nu) : F(\nu) \longrightarrow ({}_{\ell^\nu}A)^f.$$

Since the image of the induced morphism on special fibres lies in the infinitely ℓ-divisible part of A_o, it follows that $v(\nu)$ in fact factors through $({}_{\ell^\nu}A^o)^f$. Passing to the limit over ν gives a homomorphism

$$v : F \longrightarrow T_\ell(A^o)^f,$$

and the induced morphism on generic fibres plainly gives the required factorization.

Remarks 5.7.2. When $\ell \neq p$, property 5.7 uniquely characterizes the sub-pro-group $T_\ell(A_K)^f$ of $T_\ell(A_K)$ as its largest unramified ℓ-adic subsheaf. When $\ell = p$, the statement of 5.7 ceases to be very satisfactory. We shall give a considerably stronger property, which can serve as an intrinsic characterization of the fixed part of $T_p(A_K)$ (in terms of the pro-Barsotti–Tate group structure of $T_p(A_K)$ alone), at least in unequal characteristics.

<u>Theorem</u> 5.8 (M. Raynaud). <u>Let A_K be an abelian scheme with semistable reduction over the function field K of a henselian trait S. Let ℓ be a prime number (the interesting case being $\ell = p$), and consider the pro-ℓ Barsotti–Tate group $X_\eta = T_\ell(A_K)$ and its "fixed part" X_η^f, which is the generic fibre of the pro-ℓ Barsotti–Tate group $X^f = T_\ell(A^o)^f$. If $\ell = p$, suppose that $\operatorname{char} K = 0$ (so that Tate's theorem [33, Th. 4] is available over S). Let F be a pro-ℓ Barsotti–Tate group over S. Then every homomorphism $F_\eta \longrightarrow X_\eta$ factors through X_η^f.</u>

Passing to the ind-ℓ groups associated with the pro-ℓ Barsotti–Tate groups under consideration, the preceding statement becomes equivalent, by the cited theorem of Tate, to the following.

<u>Corollary</u> 5.9. <u>If F is an ind-ℓ Barsotti–Tate group, then every homomorphism $u_\eta : F_\eta \longrightarrow A_\eta$ extends uniquely to a homomorphism $u : F \longrightarrow A$.</u>

In other words, the characteristic Neron property (1.1.2) of the Neron model remains valid for homomorphisms from ind-group schemes of Barsotti–Tate type, with no smoothness hypothesis on them. (Only the case of ind-étale groups would follow directly from the definition.)

<u>Remark</u> 5.9.1. In fact, M. Raynaud proves 5.9 without assuming that A_K has semistable reduction, and even for every Neron group scheme A over S (whose generic fibre need not be an abelian scheme). Of course, the assumption that S is henselian is also unnecessary in the formulation of 5.9. Raynaud's proof of this more general case reduces it to the special case considered in 5.8, using an unpublished property in the construction of Neron models. Here we shall give only Raynaud's proof of 5.9.

<u>Proof of 5.9.</u> It is known [33] that F is an extension of an ind-étale Barsotti–Tate group F' by a Barsotti–Tate group F'' with connected special fibre, so that for every integer n there is an exact sequence

$$0 \longrightarrow F''(n) \longrightarrow F(n) \longrightarrow F'(n) \longrightarrow 0,$$

with $F'(n)$ étale and $F''(n)_o$ purely infinitesimal. The given homomorphism u_η is defined by a sequence of homomorphisms

$$u_\eta(n) : F(n)_\eta \longrightarrow A_\eta,$$

and it is enough to extend these separately to homomorphisms $u(n) : F(n) \longrightarrow A$. Lemma 5.9.2 below then reduces us to proving that the homomorphism $u''_\eta : F''_\eta \longrightarrow A_\eta$ induced by u_η extends to $u'' : F'' \longrightarrow A$. Consequently, <u>we may suppose that F has connected special fibre</u>.

We may regard u_η as a homomorphism from F_η to the ind-ℓ Barsotti–Tate group Y_η associated with $X_\eta = T_\ell(A_K)$, and then consider the composite

$$v_\eta : F_\eta \longrightarrow Y_\eta \longrightarrow Y_\eta / Y_\eta^f. \tag{*}$$

By 5.6, the latter group extends to an ind-étale Barsotti–Tate group Z over S, namely

$$\underline{M} \otimes_{\mathbb{Z}} (\mathbb{Q}_\ell / \mathbb{Z}_\ell).$$

By Tate's theorem cited in 5.8, the homomorphism (*) then extends to a homomorphism

$$v : F \longrightarrow Z.$$

Since F has connected special fibre and Z is ind-étale, this morphism is zero. Hence v_η is zero as well, and u_η therefore factors through Y_η^f, as required.

It remains to prove the following lemma.

<u>Lemma</u> 5.9.2. Let S be a trait, let A be a Neron group scheme over S (1.1.2), let

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1$$

be an extension of group schemes over S, with G'' smooth, let $u : G' \longrightarrow A$ be a group homomorphism, and let $v_\eta : G_\eta \longrightarrow A_\eta$ be a group homomorphism extending u_η. Suppose that the extension of G'' by A obtained from the extension G by pushout along u is representable [18]—as is the case, for example, if $G \longrightarrow G''$ admits a section locally for the "finite locally free" topology (cf. SGA 3, IV 6.3) and if A is quasi-projective over S. Then there exists a unique group homomorphism $v : G \longrightarrow A$ extending both v_η and u.

Indeed, the last hypothesis reduces us to the case in which $G' = A$, u is the identity, and $v_\eta : G_\eta \longrightarrow A_\eta$ is a projection which we seek to extend to a projection $v : G \longrightarrow A$. But G is smooth, being an extension of two smooth group schemes, and the conclusion follows from the definition of the Neron property of A.

Corollary 5.10. Let A_K be an abelian scheme over the function field K of a trait S, and let ℓ be a prime number. When $\ell = p$, suppose that $\operatorname{char} K = 0$, so that Tate's theorem already invoked in 5.8 is available. Then A_K has good reduction over S if and only if the pro-ℓ Barsotti–Tate group $T_\ell(A_K)$ "has good reduction over S", that is, extends to a Barsotti–Tate group over S.

(This is the result conjectured in [8, 4.9].)

Necessity being clear, it remains to prove sufficiency; observe that the case $\ell \neq p$ has already been treated in 2.2.9. We may suppose that S is henselian (2.2.9). First suppose that A_K has semistable reduction over S. Applying 5.8 with $F = X$ gives

$$T_\ell(A_K) = T_\ell(A_K)^f.$$

This also says that the maximal torus of A_o is zero, that is, that A_o^o is an abelian scheme; the conclusion then follows from 2.2.9 (iii^o).

We now treat the general case. By 2.2.9 we may plainly assume that S is strictly local. Let K'/K be a finite Galois extension such that $A_{K'}$ has semistable reduction over the normalization S' of S in K' (3.6). If A^* denotes the corresponding Neron model, $G = \operatorname{Gal}(K'/K)$ acts on the situation (A^*, S'), and hence on A_o^{*o}. By 4.1 it is enough to prove that G acts trivially on the latter; this indeed implies that A_K has semistable reduction, reducing us to the case already treated.

Let X be a Barsotti–Tate group over S extending $X_\eta = T_\ell(A_K)$. By Tate's uniqueness theorem already invoked, there is a unique isomorphism

$$X \times_S S' \simeq T_\ell(A^{*o})$$

which induces the identity on generic fibres. By transport of structure, this isomorphism is compatible with the actions of G. Now G plainly acts trivially on the special fibre of $X \times_S S'$, and hence on $T_\ell(A_o^{*o})$. It follows at once that G acts trivially on A_o^{*o}, completing the proof.

Remark 5.11. Once Tate's theorem has been proved without any restrictive hypothesis on $\operatorname{char} K$, the preceding results 5.8, 5.9, and 5.10, as well as 5.13 below, will plainly hold without such a restriction.

We record a definition which will also be useful elsewhere.

Definition 5.12. Let S be a henselian trait, let ℓ be a prime number, and let X_η be a pro-ℓ Barsotti–Tate group over the generic point η of S. We say that X_η has *good reduction over S* if it extends to a Barsotti–Tate group over S (cf. 5.10). We say that it has *virtual good reduction over S* if there exists a filtration

$$\left(\operatorname{Fil}^i(X_\eta)\right)_{i\geq 0}$$

by strict Barsotti–Tate subgroups [19] such that the Barsotti–Tate groups

$$\operatorname{Gr}^i(X_\eta) = \operatorname{Fil}^i(X_\eta)/\operatorname{Fil}^{i+1}(X_\eta)$$

18. Unpublished results of M. Raynaud imply, moreover, that this condition is always satisfied.
19. That is, subobjects as projective systems, or equivalently subobjects term by term.

have good reduction over S. If the filtration can be chosen to have length at most n (that is, if at most n of the Gr^i are nonzero), we say that X_η has *virtual good reduction of level n over S*.

Finally, we say that X_η has *essential good reduction over S* (respectively, *essential virtual good reduction*, respectively, *essential virtual good reduction of level n*) if there exists a finite extension K'/K such that $X_{K'}$ has good reduction (respectively, virtual good reduction, respectively, virtual good reduction of level n) over the normalization S' of S in K'.

5.12.1. When $\ell \neq p$, the categories of pro-ℓ Barsotti–Tate groups over η and over S may be interpreted in terms of ℓ-adic representations of

$$\pi = \pi_1(\eta), \qquad \pi_o = \pi_1(S),$$

respectively. The six notions just introduced can then be expressed in terms of the natural representation of the inertia group I on $X_\eta(\overline{K})$: the representation is, respectively, trivial, unipotent, unipotent of level n, essentially trivial, essentially unipotent, or essentially unipotent of level n.

5.12.2. Take $X_\eta = T_\ell(A_K)$, where A_K is an abelian scheme over K. We saw in 5.6 that, if A_K has semistable reduction over S, then X_η has virtual good reduction of level 2 (and we shall see the converse in a moment). Consequently, by the semistable reduction theorem 3.6, X_η has essential virtual good reduction of level 2. This statement should be regarded as the "right" formulation of the geometric monodromy theorem in Exposé III [20] for H^1 of a proper smooth scheme over K, without excluding the case $\ell = p$ (which was not considered there). The geometric monodromy theorem—or, for the present, conjecture—for the higher H^i given in Exposé III should likewise be reformulated so as to include "p-adic" coefficients, as was observed in 0.4.

We now summarize the relations between the reduction properties of abelian schemes over K and those of the associated Barsotti–Tate groups.

Proposition 5.13. Let S be a henselian trait, let A_K be an abelian scheme over its fraction field, and let ℓ be a prime number. When $\ell = p$, assume that K has characteristic zero (cf. 5.11).

a) The abelian scheme A_K has good reduction over S (2.2.9) if and only if $T_\ell(A_K)$ has good reduction over S (5.12).

b) The abelian scheme A_K has essential good reduction over S—that is, there exists a finite extension K'/K such that $A_{K'}$ has good reduction over the normalization S' of S in K'—if and only if $T_\ell(A_K)$ has essential good reduction over S (5.12). In that case K' may be chosen to be a finite separable extension of K.

c) The following conditions are equivalent:

1) A_K has semistable reduction over S (3.4);
2) $T_\ell(A_K)$ has virtual good reduction of level 2 over S (5.12);
3) $T_\ell(A_K)$ has virtual good reduction over S.

Proof. Part a) is just 5.10 and has been included for reference.

For b), the first assertion follows immediately from a) and the definitions. To prove the final assertion, we may suppose that $\ell \neq p$. In view of a), it then suffices to take for K' the finite separable extension of K corresponding to an open subgroup π' of

$$\pi = \mathrm{Gal}(\overline{K}/K)$$

whose intersection with the inertia group I is contained in the kernel of the representation of I on $T_\ell(A_K(\overline{K}))$; this kernel is open by hypothesis (cf. 5.12.1).

For c), we have already noted that 1) implies 2) (5.12.2), and it is immediate that 2) implies 3). It remains to prove that 3) implies 1). For $\ell \neq p$, this is precisely the Galois criterion for semistable reduction 3.5. For $\ell = p$, argue as follows. Choose a prime $\ell' \neq p$ and apply

20. The source leaves the precise internal reference blank.

the considerations of 4.1 to it. We may suppose that S is strictly local (3.4.0), and choose a finite Galois extension K'/K, with group G, such that $A_{K'}$ has semistable reduction. We must prove that the representation of G on each of the three quotients occurring in (4.1.1) is trivial. Equivalently, their direct-sum representation is trivial, or again its trace is the "trivial" trace, that is, is constant. We saw in 4.3(a) that this trace function is independent of the choice of $\ell' \neq p$, and in 6.4 below we indicate a method of calculating it in terms of $T_p(A_K)$ as well. That calculation plainly gives a constant function when $T_p(A_K)$ has virtual good reduction, and this completes the proof.

Remark 5.13.1. It seems plausible that, if X_η is a pro-ℓ Barsotti–Tate group over η having virtual good reduction over S and essential virtual good reduction of level n, then it has virtual good reduction of level n. (This is true when $\ell \neq p$.)

The following result, an easy consequence of 5.7 and of the semistable reduction theorem 3.6, was pointed out to me by J.-P. Serre.

Proposition 5.14 (Criterion for essential good reduction). Let S be a trait with function field K, let A_K be an abelian scheme over K, and let ℓ be a prime number distinct from $\operatorname{char} K$.

a) Suppose that $\ell \neq p = \operatorname{char} k$. Then the following conditions are equivalent:

(i) A_K has essential good reduction over S; that is, there exists a finite separable extension K'/K such that $A_{K'}$ has good reduction over the normalization S' of S in K';

(ii) the action of the inertia group I on $U_\ell = T_\ell(A(\overline{K}))$ is essentially trivial; that is, it is trivial on an open subgroup of I, or equivalently factors through a finite quotient Γ of I;

(iii) the action of I on $U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ is semisimple.

b) Suppose that $\ell = p$. If $n = \dim A_K \neq 0$, then the image of I in the automorphism group of U_ℓ is not finite; more precisely, its image in the group $\mathbb{Z}_\ell^\times$ of automorphisms of $\bigwedge^{2n} U_\ell$ has finite index. A sufficient (but by no means necessary) condition for A_K to have essential good reduction is that the action of I on $U_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ be semisimple.

Proof. When $\ell \neq p$, the equivalence of (i) and (ii) follows immediately from the good-reduction criterion 2.2.9, and (ii) implies (iii) because $\mathbb{Q}_\ell$ has characteristic zero. The homomorphism

$$\pi = \operatorname{Gal}(\overline{K}/K) \longrightarrow \mathbb{Z}_\ell^\times \simeq \operatorname{Aut}\left(\bigwedge^{2n} U_\ell\right)$$

is χ^n, where χ is induced by the character

$$\mathbb{Z}_\ell^\times = \operatorname{Aut}(\mathbb{Z}_\ell(1))$$

arising from the action on the ℓ^ν-th roots of unity. It is known that, when $\ell = p$, the image of I under this character has finite index [33]. The fact that the action of I on $U_p \otimes \mathbb{Q}_p$ need not be semisimple even when A_K is an elliptic curve with good reduction is exhibited in [28, IV A 2.4].

It remains to prove that (iii) implies (i), for every ℓ. We use the semistable reduction theorem 3.6, which immediately reduces us to the case where A_K has semistable reduction over S: hypothesis (iii) is preserved upon replacing K by a finite separable extension K', or equivalently I by an open subgroup I'. When $\ell \neq p$, the action of I on $U_\ell \otimes \mathbb{Q}_\ell$ is unipotent (3.5(v)) and semisimple, hence trivial, and the good-reduction criterion 2.2.9 gives the result. We now give an argument valid with no restriction on ℓ, which yields a slightly sharper statement.

Corollary 5.15. With the notation of 5.14, suppose that A_K has semistable reduction over S, and that the fixed part

$$(U_{\ell\mathbb{Q}_\ell})^f \overset{\text{def}}{=} (U_\ell^f)_{\mathbb{Q}_\ell}$$

admits an I-stable complement in $U_{\ell\mathbb{Q}_\ell}$. Then A_K has good reduction.

It is enough to prove under these hypotheses that $U_\ell = U_\ell^f$. By 5.6 this says precisely that the toric part T_o of A_o^o is zero, so that A_o^o is an abelian scheme; hence A is an abelian scheme by 2.2.9.

We may plainly suppose that k is separably closed. The intersection of the asserted complement in 5.15 with U_ℓ then defines a pro-ℓ subgroup F_K of $T_\ell(A_K)$ which is isogenous to the cotoric part

$$T_\ell(A_K)/T_\ell(A_K)^f.$$

The latter is pro-étale and unramified by 5.6, and the same is therefore true of F_K. It follows from 5.7 that F_K is contained in $T_\ell(A_K)^f$; thus $F_K = 0$, proving that U_ℓ is equal to its fixed part.

6. Remarks on the construction of the toric part and the fixed part of $T_p(A)$ in the case of non-semistable reduction.

6.1. The construction of the toric part of $T_\ell(A_K)$ given in 5.1 still makes sense as soon as S is complete, without assuming that A^o is ℓ-divisible; that is (when $p = \ell$), without assuming semistable reduction. That hypothesis had been used only in order to formulate the orthogonality theorem 5.2. Moreover, one can construct $T_p(A_K)^t$ while assuming only that S is henselian: one shows that the toric part $T_p(A_{\widehat{K}})^t$ over the completion, constructed by the method of 5.1, comes from a Barsotti–Tate pro-subgroup of $T_p(A_K)$, necessarily determined uniquely.

More precisely, one can construct a pro-p Barsotti–Tate group $T_p(A)^t$ over S and a homomorphism

$$T_p(A)^t \longrightarrow T_p(A)$$

of projective systems which is a monomorphism term by term and whose pullback to $\widehat{S}$ gives the analogous objects constructed by the method of 5.1. This follows by paraphrasing the construction of 5.1, except that, instead of lifting the whole maximal torus T_o of A_o, for each integer $\nu \geq 0$ one merely lifts the subgroup ${}_{p^\nu}T_o$ to a multiplicative-type subgroup $({}_{p^\nu}A^o)^t$ of ${}_{p^\nu}A^o$.

For the existence and uniqueness of such a lifting, note that the functor

$$(\mathrm{Sch})^o_{/S} \longrightarrow \mathrm{Sets}, \qquad S' \longmapsto \{\text{finite multiplicative-type subgroups of } A^o_{S'}\}$$

is represented by an S-scheme X of finite type (cf. the proof of SGA 3 XI 3.12(a)), which is étale over S by SGA 3 IX 3.6 bis. Consequently, there is exactly one section of X over S extending a given section of X_s over s, which proves the assertion.

The same argument shows more generally that, for every finite multiplicative-type subgroup H_o of A_o, there is a unique finite multiplicative-type subgroup H of A extending it. Consequently, for every toric Barsotti–Tate p-subgroup H_o of A_o (that is, one whose components are of multiplicative type), there is a unique Barsotti–Tate p-subgroup H of A extending it. This in turn gives a Barsotti–Tate p-subgroup of ${}_{p^\infty}A_K$, or, equivalently, a pro-p Barsotti–Tate subgroup of $T_p(A_K)$.

In particular, this procedure applies when H_o is the largest toric Barsotti–Tate p-subgroup of A_o. It is obtained by taking the largest p-divisible subgroup $\overline{A}_o$ of A^o_o ($\overline{A}_o = p^N A^o_o$ for N large) and the "toric part" of ${}_{p^\infty}\overline{A}_o$ (defined, for example, by SGA 3 XVII 7.2.1). We thus obtain a Barsotti–Tate subgroup of ${}_{p^\infty}A_K$ which might be called the toric part of the latter.

6.2. Again let $\overline{A}_o$ be the largest p-divisible subgroup scheme of A^o_o, and consider the Barsotti–Tate p-group

$$F_o = {}_{p^\infty}(\overline{A}_o).$$

It is immediate that F_o may be characterized as the largest Barsotti–Tate p-subgroup of A_o (or of A^o_o). It is natural to ask whether this Barsotti–Tate group can be extended to a Barsotti–Tate subgroup F of A, which would then play the role of the fixed part $T_\ell(A)^f$ of (2.2.3.2) in the case in which A^o is ℓ-divisible.

In the semistable case, where $\overline{A}_o = A^o_o$, the method of 2.2 answers the question affirmatively. In the general case, whenever the problem has a solution, it is readily seen to be unique. The corresponding pro-subgroups of $T_p(A_K)$ and $T_p(A'_K)$ (which again deserve to be called the fixed parts of the pro-groups in question) still satisfy the orthogonality theorem 5.2, and the fixed part $T_p(A_K)^f$ may still be characterized as in 5.8.

6.2.1. We shall nevertheless show that in general the question has no affirmative answer; that is, the fixed part $T_p(A)^f$ cannot in general be defined in the case of non-semistable reduction. Indeed, by a lifting theorem of Serre–Tate [30], when S is complete, the existence of the fixed part is equivalent to the existence of infinitesimal liftings $\overline{A}_n$ of $\overline{A}_o$ in the A_n, or again to the existence of a smooth group scheme $\overline{A}$ over S, an extension of an abelian scheme by a torus T, having special fibre $\overline{A}_o$, together with a formal homomorphism

$$\widehat{\overline{A}} \longrightarrow \widehat{A}$$

extending the inclusion on the special fibres.

Thus, when $\overline{A}_o$ is an abelian scheme, that is, when $T_o = 0$, this is equivalent to the existence of an abelian scheme $\overline{A}$ extending $\overline{A}_o$ and a homomorphism $\overline{A} \longrightarrow A$ extending the inclusion of the special fibres. It will therefore suffice to give an example in which $T_o = 0$, A_o^o is neither unipotent nor proper, and A_K is simple over K.

6.2.2. Here is an example, due to M. Raynaud. It is based on the observation that, if S is a trait, S' is the normalization of S in a finite separable extension K' of K, and A' is an abelian scheme over S', then the group scheme

$$A = \prod_{S'/S} A'/S'$$

over S is a Néron model: it identifies with the Néron model of its generic fibre, which is indeed an abelian scheme. This follows immediately from the definitions and from the fact that A' itself is a Néron model over S'. Then

$$A_o = \prod_{S'_s/s} (A' \times_{S'} S'_s)/S'_s$$

is an extension of an abelian scheme, nonzero if $A' \neq 0$, by a connected unipotent scheme, nonzero if $A' \neq 0$ and S' is ramified over S.

It therefore suffices to give an example in which $A' \neq 0$, S' is ramified over S (a complete local trait), and

$$A_K = \prod_{K'/K} A'_{K'}/K'$$

is simple over K. Take K'/K to be a ramified quadratic extension (such an extension always exists), and take A' to be an elliptic curve over S'. Then A has relative dimension two, and saying that A_K is simple over K means that there is no nonzero homomorphism

$$B_K \longrightarrow A_K,$$

where B_K is an abelian scheme of dimension one over K. Giving such a homomorphism is equivalent to giving a nonzero homomorphism

$$B_{K'} \longrightarrow A'_{K'},$$

which would necessarily be an isogeny. Thus A_K is simple over K precisely when $A'_{K'}$ is not isogenous to an elliptic curve coming from K.

If the contrary held, the j-invariant of A' would be algebraically dependent (over the prime field) on its conjugate $\overline{j}$ under the nontrivial K-automorphism of K'. If j is chosen so that

$$a = j + \overline{j} = \operatorname{Tr}_{K'/K} j$$

is transcendental over the prime field, this is equivalent to saying that j is algebraically dependent on a. By cardinality considerations, however, there exists an element $a \in V$ transcendental over the prime subfield P of K, and then an element $b \in V'$ (the normalization of V in K') which is transcendental over $P(a)$ and satisfies

$$\operatorname{Tr}_{V'/V} b = 0.$$

It is enough to take $j = a + b \in V'$. We may moreover suppose that $b \in \mathfrak{r}(V')$, and that the image of a in $k = V/\mathfrak{m}$ is a prescribed element of k, chosen distinct from the exceptional values $j_o = 0$ and $j_o = 12^3$. This is possible even when k has only two elements, because then these two contemplated values of j_o coincide. Thus the image j_o of j in $k(s')$ is different from 0 and 12^3, and it is known that there then exists an elliptic curve over S' with j-invariant j.

6.3. Return to the case in which S is merely assumed henselian. In analogy with the definitions of 4.1 for $\ell \neq p$, one can then define a filtration

$$T_p(A_K) \supset T_p(A_K)^{\mathrm{ef}} \supset T_p(A_K)^{\mathrm{et}} \supset 0 \tag{6.3.1}$$

by strict Barsotti–Tate pro-subgroups, called respectively the essentially fixed part and the essentially toric part of $T_p(A_K)$.

To do so, choose a finite Galois extension K' of K such that $A_{K'}$ has semistable reduction over the normalization S' of S in K'. The fixed part $T_p(A_{K'})^f$ of $T_p(A_{K'}) = (T_p(A_K))_{K'}$ is then defined (2.2.3), as is the toric part (6.1). The latter is moreover the orthogonal complement of the fixed part $T_p(A'_{K'})^f$ in $T_p(A'_{K'})$ (5.2). Plainly

$$T_p(A_{K'})^t \subset T_p(A_{K'})^f$$

(this follows from 5.3, or directly from the construction of the toric part in 6.1), and the Galois group of K'/K preserves these two strict pro-subgroups of $(T_p(A_K))_{K'}$. Descent therefore gives the desired filtration (6.3.1). It is independent of the choice of K', by the semistable-reduction property 3.3. Descent also plainly gives the orthogonality property

$$\begin{aligned} T_p(A_K)^{\mathrm{et}} &= \big(T_p(A'_{K'})^{\mathrm{ef}}\big)^{\perp}, \\ T_p(A_K)^{\mathrm{ef}} &= \big(T_p(A'_{K'})^{\mathrm{et}}\big)^{\perp}. \end{aligned} \tag{6.3.2}$$

One could give an intrinsic characterization of the filtration (6.3.1) in terms of the Barsotti–Tate pro-group $X = T_p(A_K)$ over the fraction field K of S, in the spirit of 5.8—at least, for the moment, in unequal characteristics. The essentially fixed part is the largest Barsotti–Tate pro-subgroup having "essentially good reduction over S": after a finite separable extension K' of K, it gives a Barsotti–Tate pro-group $X_{K'}$ having good reduction over the normalization S' of S in K', that is, extending to a Barsotti–Tate pro-group over S'. Such an extension is necessarily unique up to a unique isomorphism, by the already-cited theorem of Tate [33, Th. 4]. The essentially toric part is then defined in terms of the essentially fixed part by orthogonality (6.3.2).

Let us also record the following immediate consequence of 5.8 and the definitions; for $\ell = p$ it reduces to 5.12(a).

Proposition 6.3.3. Let ℓ be a prime number. If $\ell = p$, assume that $\operatorname{char} K = 0$.[21] Then A_K has essential good reduction over S (5.12(a)) if and only if its essentially fixed part $T_\ell(A_K)^{\mathrm{ef}}$ (4.1.1 and 6.3.1) is all of $T_\ell(A_K)$.

6.4. Recall that, when $\ell \neq p$, in 4.3(a) we exhibited a remarkable integer-valued function on the inertia group I of S, independent of the choice of ℓ. When the residue field is perfect, knowledge of this function suffices to reconstruct the conductor exponent of A (4.5): one takes, on a sufficiently large quotient $\Gamma = I/I'$ of I, the inner product of this function with the Artin conductor on Γ (cf. (4.6.2)). It is useful to observe that this integer-valued function on I can also be recovered from the knowledge of $T_\ell(A_K)$ for $\ell = p$—at least, for the time being, in unequal characteristics.

When the residue field k is finite, an analogous observation applies to the $\mathbb{Q}$-valued function considered in 4.3(b), defined on the subgroup π' of π consisting of the elements whose image in π_o has the form f_q^n, $n \in \mathbb{Z}$. We indicate the principle of the verification. Using the filtration (6.3.1), one is reduced to attaching a suitable function (on I, respectively on π') to a pro-p Barsotti–Tate

21. This hypothesis is probably unnecessary; cf. 5.11.

group Y over K having "essentially good reduction over S" in the sense of 5.12. Here Y will be one of the three quotients arising from the filtration (6.3.1).

In the case of 4.3(a), a suitable residue-field extension within S (EGA 0_{III} 10.3.1) reduces us to the case of a perfect residue field, or even an algebraically closed one if desired. Choose a finite Galois extension K' of K such that $Y_{K'}$ has good reduction over S'. We then have a Barsotti–Tate group $Y_{S'}$ over S' and an action of G on $Y_{S'}$ compatible with its action on S'. If k' is the residue field of S', then G acts, not necessarily faithfully, on k' (whose invariant field is k), and on $Y_{k'}$ compatibly with its action on k'.

By Dieudonné theory [16 bis], the Barsotti–Tate pro-p group $Y_{k'}$ has a canonically associated finite free module M (of rank equal to the rank of $Y_{k'}$) over the ring $W(k')$ of Witt vectors of k'. The group G plainly acts on M compatibly with its actions on $W(k')$, through its actions on k'. Hence the inertia subgroup Γ of G acts $W(k')$-linearly on M; since Γ is a quotient of the inertia group I, this gives a trace function on Γ, and hence on I, which is the desired function.

A priori this function takes values in $W(k')$,[22] but it agrees with the analogous function defined in terms of the T_ℓ for $\ell \neq p$ whenever the action of G on $Y_{S'}$ is obtained from an action of G on a smooth p-divisible group scheme C' over S' by taking $Y = T_p(C')$ (cf. [25 bis, Th. 7, p. 737]). It is therefore integer-valued in this case. The case of 4.3(b) is handled similarly: to define a trace function on π', one uses the method of the proof of 4.3.2 to "linearize" on M the action defined by an element $g \in \pi$ lying above an f_q^n, with $n \geq 0$.

7. The Raynaud extension $G^\natural$ over S associated with the abelian scheme A_K having semistable reduction. Duality of the abelian schemes B, B' over S.

7.1. Let S be a Noetherian scheme which is the spectrum of a ring V, separated and complete for the topology defined by an ideal $\underline{m}$ of V. Let A be a smooth group scheme over S, let A^o be its identity component, and suppose that A_o/A_o^o is finite (hence finite etale) over S_o, and that A_o^o is an extension of an abelian scheme by a torus. As usual, we have put $A_n = A \times_S S_n$, where $S_n = \mathrm{Spec}(V/\underline{m}^{n+1})$. We shall be interested chiefly in the case where S is a trait, the generic fiber A_K is an abelian scheme having semistable reduction over S, and A is the Neron model of A_K.

Let T_o be the maximal torus of A_o, so that $B_o = A_o^o/T_o$ is an abelian scheme. The construction of 5.1 then applies and gives a formal subtorus $\widehat{T}$ of the formal completion $\widehat{A}$. For every n, the quotient

$$B_n = A_n^o/T_n \tag{7.1.1}$$

is representable (SGA 3 VI$_A$ 3.2); more precisely, it is a flat scheme of finite type over S_n whose pullback to S_o is the abelian scheme B_o. Consequently B_n is an abelian scheme over S_n. As n varies, the B_n are obtained from one another by base change and therefore define a formal abelian scheme

$$\widehat{B} = (B_n)_{n \geqslant 0}\,. \tag{7.1.2}$$

We shall give conditions implying that this formal abelian scheme is algebraizable (EGA III 5.4.1), more precisely, that it is the formal completion of a projective abelian scheme over S, which we shall then denote by B. By loc. cit., B is determined up to a unique isomorphism (as an S-scheme; the group structure of $\widehat{B}$ then comes uniquely from a group structure on B, again by loc. cit.). This abelian scheme B will then be called the abelian part of the scheme A, and will sometimes be denoted by A^{ab}.

7.1.3. By EGA III 5.4.5, projectivity of $\widehat{B}$ means that one can find an invertible sheaf $\widehat{\mathbb{M}}$ on $\widehat{B}$, that is, a sequence of invertible sheaves $\mathbb{M}_n$ on the B_n which glue as n varies, and which is relatively ample over S, that is, such that $\mathbb{M}_o$ is ample. The natural idea would be to start with an ample invertible sheaf $\mathbb{L}$ on A and show that, for every n, its restriction $\mathbb{L}_n$ to A_n is the pullback

22. In fact it is immediate that this function always takes values in $W(k)$, since it is invariant under $\mathrm{Gal}(k'/k)$.

of an invertible sheaf $\mathbb{M}_n$ on B_n.[23] I do not know whether such $\mathbb{M}_n$ necessarily exist (and strongly doubt it). We shall follow a slightly different method, constructing a polarization of $\widehat{B}$, that is, a projective system of polarizations ρ_n of the B_n which glue. It is known [5] that the doubles of these polarizations canonically define (symmetric) invertible sheaves $\mathbb{M}_n$ on the B_n. By their canonical nature these sheaves necessarily glue and furnish the desired $\widehat{\mathbb{M}}$.

We shall interpret an element of $\mathrm{NS}(B_n/S_n)$ as a symmetric divisorial correspondence on (B_n, B_n) [5], that is, as a biextension of (B_n, B_n) by $\mathbb{G}_{mS_n}$. By VIII 3.5, these biextensions correspond bijectively to biextensions of (A_n, A_n) by $\mathbb{G}_{mS_n}$. To obtain a coherent system of such biextensions W_n, it suffices to start with

(7.1.4) $\qquad W,\quad$ a symmetric biextension of (A, A) by $\mathbb{G}_{mS}$,

and take the W_n obtained from W by the base changes $S_n \longrightarrow S$. We must in addition suppose that the element of $\mathrm{NS}(B_o)$ defined by W_o is a polarization of B_o; we thereby obtain a polarization of $\widehat{B}$.

Proposition 7.1.5. With the preceding notation for $S, A, \widehat{B}$, suppose that S is normal. Then $\widehat{B}$ is algebraizable, and the abelian scheme B over S from which it comes is projective over S.

We may plainly suppose that $A = A^o$ and that S is integral. By Raynaud [23, XI 1.13], the hypothesis that S is normal implies that A is quasi-projective over S. Choose a relatively ample invertible sheaf $\mathbb{L}$ on A, and form the birigidified invertible sheaf

$$\delta(\mathbb{L}) = \pi^*(\mathbb{L})\, \mathrm{pr}_1^*(\mathbb{L})^{-1}\, \mathrm{pr}_2^*(\mathbb{L})^{-1}$$

on $A \times_S A$, as in VIII 4.12. Since the fibers of A_o have unipotent rank zero, the same is true of the fibers of A (this follows easily from 3.1). Therefore, by VIII 7.5, $\delta(\mathbb{L})$ defines a biextension W of (A, A), plainly symmetric, which by VIII 4.13 (that is, essentially by Raynaud [23, XI 1.11]) satisfies the required ampleness condition at the points of S_o.

Remark 7.1.6. Without assuming that S is normal, one can prove that $\widehat{B}$ is always algebraizable by reducing to the normal case as in 13.5.3(a).

7.2. We now record the following lemma.

Lemma 7.2.1. Let S be the spectrum of a Noetherian ring V, separated and complete for the topology defined by an ideal $\underline{m}$ of V. Let C be the category of group schemes Λ over S, smooth and of finite type over S, such that Λ/Λ^o is a finite etale group scheme over S and Λ^o is an extension of an abelian scheme B by an isotrivial torus T (SGA 3 IX 1.1). Let C' be the category of group objects in the category of formal schemes of finite type (EGA I 10.3.3) over $\widehat{S} = \mathrm{Spf}(V)$, and consider the functor

$$A \longmapsto \widehat{A}$$

from C to C' which takes the formal completion along $A_o = A \times_S S_o$, where $S_o = \mathrm{Spec}(V/\underline{m})$.[24] Then:

(a) The preceding functor is fully faithful.

(b) Let $\mathcal{A}$ be an object of C', smooth over $\widehat{S}$; that is, suppose that the ordinary schemes

$$A_n = \mathcal{A} \times_{\widehat{S}} S_n, \qquad S_n = \mathrm{Spec}(V/\underline{m}^{n+1}),$$

are smooth over S_n. Suppose that A_o/A_o^o is a finite etale group scheme over S_o, and that A_o^o is an extension of an abelian scheme B_o by an isotrivial torus T_o. Put

$$\mathcal{A}^o = (A_n^o)_{n\geq 0}, \qquad \mathcal{A}/\mathcal{A}^o = (A_n/A_n^o)_{n\geq 0}.$$

23. The printed text changes incoherently here from an invertible sheaf $\mathbb{M}$ on G to the restriction $\mathbb{L}_n$ of $\mathbb{L}$ to G_n. The notation above is the type-correct reading fixed by the surrounding construction and by Proposition 7.1.5.

24. At both occurrences in this statement the print writes $\mathrm{Specf}(A)$, and it also writes $\mathrm{Spec}(A/\underline{m}^{n+1})$ below. Since V is the adic base ring and A denotes the varying group scheme, the type-correct base is $\mathrm{Spf}(V)$ and $S_n = \mathrm{Spec}(V/\underline{m}^{n+1})$.

Then $\mathcal{A}/\mathcal{A}^o$ is finite etale over $\widehat{S}$, and $\mathcal{A}^o$ is, in an essentially unique manner and functorially in $\mathcal{A}$, an extension of a formal abelian scheme $\mathcal{B}$ over $\widehat{S}$ by a formal torus $\mathcal{T}$ over $\widehat{S}$, reducing respectively to B_o and T_o. The formal group scheme $\mathcal{A}$ belongs to the essential image of $C \longrightarrow C'$—or, as we shall also say, is "algebraizable"—if and only if $\mathcal{B}$ is algebraizable.

7.2.1.1. First observe, over an arbitrary scheme S, that if A is a smooth group scheme such that A^o is an extension of an abelian scheme by a torus, this extension structure is essentially unique and is functorial in A in the evident sense. Indeed, the assertion reduces at once to the fact that every morphism from a torus T to an abelian scheme B is zero. This is well known on geometric fibers [15], and SGA 3 IX 5.3 then gives the conclusion. Applying the result to the compatible systems of smooth group schemes A_n over the S_n occurring in 7.2.1 gives the analogous result for formal group schemes over S. Moreover, we already observed in 7.1 that, if A_o^o is an extension of an abelian scheme by a torus, the same is true of A_n^o. This proves the first assertion of (b).

7.2.1.2. Suppose next that the formal group scheme $\mathcal{A}$ over S is of the form $\widehat{A}$, with $A \in \operatorname{ob} C$. If A^o is an extension of an abelian scheme B by a torus T, then plainly $\mathcal{B} \simeq \widehat{B}$. Thus algebraizability of $\mathcal{A}$ implies algebraizability of $\mathcal{B}$.

Conversely, suppose that $\mathcal{B}$ is algebraizable, that is, that it comes from an abelian scheme B, and let us prove that $\mathcal{A}$ is algebraizable. We first prove that $\mathcal{A}^o = (A_n^o)$ is algebraizable. By VIII 3.7, the extensions A_n^o of B_n by T_n may be interpreted as homomorphisms

$$(*) \qquad \underline{M}_n \longrightarrow B'_n,$$

where $\underline{M}_n$ is the twisted constant group over S_n dual to T_n (SGA 3 X 5.7), and B'_n is the abelian scheme dual to B_n. Restriction to S_o, or to any S_n, induces an equivalence from the category of isotrivial tori over S to the corresponding category over S_o (SGA 3 X 3.2). Hence the T_n come from a torus T over S, determined up to a unique isomorphism. Denote its dual group by $\underline{M}$; its restriction to S_n is canonically isomorphic to $\underline{M}_n$.

The isotriviality of T means that $\underline{M}$ is a disjoint union of S-schemes $\underline{M}^i$ which are finite over S (SGA 3 X 5.11). It follows at once that the homomorphisms $(*)$ come from a unique homomorphism

$$(**) \qquad \underline{M} \longrightarrow B',$$

where B' is the abelian scheme dual to B. In turn, this homomorphism is interpreted as an extension of B by T, say A^o, which plainly supplies an algebraization of $\mathcal{A}^o$.

7.2.1.3. On the other hand, by hypothesis A_o/A_o^o is a finite group scheme Φ_o over S_o, and the A_n/A_n^o are likewise finite group schemes Φ_n over S_n; they are necessarily etale over S_n.[25] There is therefore an etale group scheme Φ over S, determined up to a unique isomorphism, whose formal completion is the formal group scheme $\mathcal{A}/\mathcal{A}^o$ defined by the Φ_n.

Since $G = A^o$ is divisible, the category of extensions, as fppf sheaves, of a finite locally constant group over S such as Φ by the group scheme A^o is canonically equivalent to the category of torsors under the sheaf

$$H = \underline{\mathrm{Hom}}_{\mathrm{gr}}(\Phi, G),$$

and this equivalence is compatible with every base change. Applying the same description to extensions of Φ_n by G_n in terms of torsors under

$$H_n \simeq \underline{\mathrm{Hom}}_{\mathrm{gr}}(\Phi_n, G_n),$$

the A_n define a coherent system of torsors P_n under the H_n. Because G is an extension of an abelian scheme by a torus, it follows readily that H is a group scheme finite and locally free over S. Consequently, the functor

$$P \longmapsto \widehat{P} = (P_n)$$

25. The print writes S rather than the fiber bases S_o and S_n in these sentences, and leaves blank the parenthetical citation from which representability of Φ_n is said to follow. The bases are restored from the statement of the lemma; the missing citation is not guessed.

is an equivalence between the category of torsors over S under H and the category of "formal" torsors $\widehat{P}$ under $\widehat{H}$, that is, coherent systems of torsors P_n under the H_n.

Thus the P_n defined by the extensions A_n come from a single torsor P over S under H. We obtain an extension A of Φ by $G = A^o$ which algebraizes $\mathcal{A}$, except that its representability remains to be checked. Regard A as a torsor over Φ under A^o_Φ. This torsor becomes trivial after base change by the finite, surjective, locally free morphism

$$P \times_S \Phi \longrightarrow \Phi,$$

since the base change $P \longrightarrow S$ splits the extension A of Φ by A^o. One may therefore conclude by one of Raynaud's many criteria for effectiveness of descent [23, XI 3.1(6)].

7.2.1.4. This completes the proof of (b), and only the full faithfulness assertion (a) remains. We leave it to the reader to verify that this assertion is essentially contained in the method of proof of (b), or, if preferred, to find an independent direct proof.

7.2.1.5. Notice that the appeal to Raynaud's delicate descent result becomes unnecessary if A^o is assumed quasi-projective over S, or, equivalently, if B is projective over S. This condition holds in particular when one starts from the formal group scheme $\widehat{A}$ over S, with A as in 7.1.5. This is the only case in which we shall use 7.2.1.

7.2.2. We apply 7.2.1 to the formal group scheme $\widehat{A}$ considered in 7.1, in the favorable case treated there in which the formal abelian scheme $\widehat{B}$ is algebraizable (cf. 7.1.5), and we suppose in addition that the torus T_o over S_o is isotrivial. We thereby obtain a group scheme $A^\natural$ over S, called the <u>Raynaud group</u> of A over S, characterized up to isomorphism by a canonical isomorphism of formal group schemes over $\mathrm{Spf}(V) = \widehat{S}$:

$$(A^\natural)^\wedge \simeq \widehat{A}. \tag{7.2.3}$$

It follows from 7.2.1 that

$$\Phi = A^\natural / A^{\natural o} \tag{7.2.4}$$

is a finite etale group over S, and that the identity component

$$A^{\natural o} = A^{o\natural}$$

is endowed with the structure of an extension

$$0 \longrightarrow T \longrightarrow A^{\natural o} \longrightarrow B \longrightarrow 0, \tag{7.2.5}$$

where B is the abelian scheme over S whose formal completion is the formal abelian scheme $\widehat{B}$ constructed in 7.1, and T is characterized, up to a unique isomorphism, as the isotrivial torus over S lifting T_o. On restriction to the S_n, the extension structure (7.2.5) gives the extension structures on $A^{\natural o}_n = A^o_n$ defined by (7.1.1).

7.2.6. It also follows immediately from 7.2.1 that the Raynaud group $A^\natural$, as well as the exact sequence (7.2.5), depends functorially on A (satisfying the conditions under consideration), and that its formation commutes with every base change $S' \longrightarrow S$ associated with a homomorphism of adic topological rings $V \longrightarrow V'$.

7.3. Let us now consider, for a fixed prime number ℓ, the pro-ℓ group $T_\ell(A^o)$. Identifying a finite scheme over S with the corresponding formal scheme, and defining the "<u>fixed part</u>" $T_\ell(A^o)^f$ as usual, we obtain an isomorphism as in (5.1.5):

$$T_\ell(A^o)^f \simeq T_\ell(\widehat{A}^o) \quad , \tag{7.3.1}$$

and, defining the "<u>toric part</u>" as in (5.1.6),

$$(7.3.2) \qquad T_\ell(A^o)^t = T_\ell(\widehat{T}) \simeq T_\ell(T) \subset T_\ell(A^o)^f \simeq T_\ell(\widehat{A}^o) \quad ,$$

we immediately obtain, for the "<u>abelian part</u>," an isomorphism

$$(7.3.3) \qquad T_\ell(A^o)^{ab} \overset{\text{def}}{=} T_\ell(A^o)^f / T_\ell(A^o)^t \simeq T_\ell(\widehat{B}) \quad .$$

We have not yet used here the hypothesis that $\widehat{B}$ is algebraizable, the hypothesis that allowed us to construct the abelian scheme B and the Raynaud extension of B by T. Since there is, of course, a canonical isomorphism (the special case of (7.3.1) applied to the abelian scheme B!) $T_\ell(\widehat{B}) \simeq T_\ell(B)$, the isomorphism (7.3.3) also gives us

$$(7.3.4) \qquad T_\ell(A^o)^{ab} \simeq T_\ell(B) \quad .$$

Let us also note in passing that (7.2.3) and (7.3.1) imply a canonical isomorphism

$$(7.3.5) \qquad T_\ell(A^o)^f \simeq T_\ell(A^{\natural o}) \quad ,$$

since the right-hand side is already a profinite group, that is, identical to its fixed part. The canonical exact sequence

$$(7.3.6) \qquad 0 \longrightarrow T_\ell(A^o)^t \longrightarrow T_\ell(A^o)^f \longrightarrow T_\ell(B) \longrightarrow 0 \quad ,$$

which can in any case (without the algebraizability hypothesis on $\widehat{B}$) be interpreted as the exact sequence obtained by applying the functor T_ℓ to the exact sequence

$$0 \longrightarrow \widehat{T} \longrightarrow \widehat{A}^o \longrightarrow \widehat{B} \longrightarrow 0 \quad ,$$

can also be regarded as obtained by applying the functor T_ℓ to the canonical exact sequence of ordinary group schemes

$$0 \longrightarrow T \longrightarrow A^{\natural o} \longrightarrow B \longrightarrow 0 \qquad .$$

7.4. Suppose now that a second group scheme A' over S is given, satisfying the same hypotheses as A, and suppose that A and A' have connected fibres. Let W be a biextension of (A, A') by $\mathbb{G}_{mS}$. We thereby obtain a pairing (VIII 2.2)

$$(7.4.1) \qquad T_\ell(A) \times T_\ell(A') \longrightarrow T_\ell(\mathbb{G}_{mS}) = \mathbb{Z}_\ell(1)$$

and, by restriction to the fixed parts, a pairing

$$(7.4.2) \qquad T_\ell(A)^f \times T_\ell(A')^f = T_\ell(\widehat{A}) \times T_\ell(\widehat{A'}) \longrightarrow \mathbb{Z}_\ell(1) \qquad .$$

This can also be interpreted as the pairing defined by the biextension $\widehat{W}$ of $(\widehat{A}, \widehat{A'})$ by $\widehat{\mathbb{G}}_{mS}$ obtained from W by completion (here, for simplicity, defining a biextension of formal group schemes as a coherent family of biextensions W_n of (A_n, A'_n) by $\mathbb{G}_{mS_n}$). Now we know (VIII 3.5) that W_n comes from a biextension W_n^{ab} of (B_n, B'_n), where B'_n is the abelian part of A'_n, so that (7.4.2) factors through

$$T_\ell(A)^{ab}\times T_\ell(A')^{ab}=T_\ell(\widehat{B})\times T_\ell(\widehat{B'})\longrightarrow \mathbb{Z}_\ell(1) \quad . \tag{7.4.3}$$

This is also the pairing defined by the biextension

$$\widehat{W}^{ab}=(W_n^{ab})$$

of $(\widehat{B},\widehat{B'})$ by $\widehat{\mathbb{G}}_{mS}$. When $\widehat{B}$ and $\widehat{B'}$ are algebraizable, so that abelian schemes B,B' over S are available, EGA IV 5.1.4 shows that $\widehat{W}^{ab}$ comes from a biextension, defined up to a unique isomorphism,

$$W^{ab} \quad \text{of} \quad (B,B') \quad \text{by} \quad \mathbb{G}_{mS},$$

and the corresponding pairing

$$T_\ell(B)\times T_\ell(B')\longrightarrow T_\ell(\mathbb{G}_{mS}) \tag{7.4.4}$$

is canonically isomorphic to (7.4.3).

Writing $D(B')$ for the abelian scheme dual to B', W^{ab} defines a homomorphism

$$u: B\longrightarrow D(B') \quad , \tag{7.4.5}$$

whose properties reflect those of the pairings (7.4.4), that is, (7.4.3). For example, the pairing (7.4.3) is a perfect duality at a point $t\in S$ if and only if $u_t : B_t \longrightarrow D(B')_t$ is an isogeny whose kernel has rank prime to ℓ. This implies that u itself is an isogeny (since S is local, hence connected), and $\ker u$ is flat and finite over S, so that $\ker u$ has the same rank at every point. Thus the condition just considered is independent of the chosen point t.

7.4.6. This completes in particular the proof of 5.4. Indeed, under the conditions of that theorem (where S may be assumed complete), applying the preceding discussion to A^o and A'^o shows that, for every ℓ, the pairing (7.4.3) is a perfect duality at the generic point η, and hence also at the point s.

7.4.7. The preceding considerations likewise make the orthogonality theorem 5.2 clear, without recourse to Lemma 5.2.4. Indeed, along the way in 7.4 we established that the pairing (5.2.3.1) = (7.4.2) vanishes on the toric parts on both sides, and therefore factors through the pairing (7.4.3). It remains only to establish that this pairing is nondegenerate on the generic fibres of the pro-groups under consideration, knowing that it is so on the special fibres. This follows from the fact that if (7.4.5) induces an isogeny on the special fibres, then it is an isogeny.

7.4.8. Return to the general conditions of 7.4, and suppose that an open subset $U\subset S-S_o$ is given such that A_U and A'_U are <u>abelian schemes</u>, dual to one another through the biextension W_U. Restricting (7.3.6), and the analogous exact sequence for A', to U, we obtain canonical filtrations on the Barsotti–Tate pro-ℓ groups $T_\ell(A_U)$ and $T_\ell(A'_U)$:

$$\begin{cases} T_\ell(A_U)\supset T_\ell(A_U)^f\supset T_\ell(A_U)^t\supset 0, \\ \\ T_\ell(A'_U)\supset T_\ell(A'_U)^f\supset T_\ell(A'_U)^t\supset 0 \end{cases} \quad . \tag{7.4.9}$$

I claim that <u>these two filtrations are dual to one another</u> with respect to the pairing (7.4.1), which here is a perfect duality (VIII 3.2); that is, relation (5.2.2) and its symmetric counterpart still hold, with A_K, A'_K replaced by A_U, A'_U. We saw above that this orthogonality assertion is also equivalent to saying that the biextension W^{ab} of (B, B') by $\mathbb{G}_{mS}$ is a perfect duality over U (or, equivalently when U is dense in S, over S). It is enough to verify this property at every point η of U; the usual argument then reduces us to the already known case in which S is a trait with generic point η.

The orthogonality relation also allows us to extend formulas (5.5.7), (5.5.8), and (5.5.9) to the present case, replacing restriction to the generic point η by restriction to the open subset U. Consequently, 5.6 likewise extends to the present case. We refrain from rewriting the preceding formulas and statement here in this context.

Let us summarize, from the point of view of Néron models, the results obtained concerning the abelian parts and their pairings.

Proposition 7.5. Let S be a complete trait with function field K. To every abelian scheme A_K over K having semistable reduction over S, the procedure of 7.1, applied to the identity component A^o of the Néron model A of A_K, associates its abelian part $B = A^{ab}$, which is an abelian scheme over S depending functorially on A_K. For every other abelian scheme A'_K with semistable reduction, one can associate to every divisorial correspondence W_K on (A_K, A'_K) a unique divisorial correspondence W^{ab} on (A^{ab}, A'^{ab}) such that, for every integer $n \geq 0$, the pullback of W^{ab}_n along $A^o_n \longrightarrow B_n = A^{oab}_n$ and $A'^o_n \longrightarrow B'_n = A'^{oab}_n$ is the biextension W_n of (A^o_n, A'^o_n) obtained from W by reduction, where W is the canonical extension (1.4) of W_K to (A^o, A'^o). The formation of W^{ab} is compatible with the formation of pullbacks of correspondences by morphisms of abelian schemes $\overline{A}_K \longrightarrow A_K$ and $\overline{A'}_K \longrightarrow A'_K$. If W_K is a perfect duality, then so is W^{ab}. If W_K is the divisorial correspondence on (A_K, A_K) associated with a polarization of A_K, then W^{ab} is associated with a polarization of A^{ab}.

The compatibility of the formation of the W^{ab} with pullbacks is immediate from the indicated characterization of the W^{ab}. The additional assertion concerning polarizations was established in 7.1.5 at the same time as the algebraizability of $\widehat{B}$.

7.6. Continue to suppose that S is a complete trait, with A the Néron model of an abelian scheme A_K over K. Set

$$A^{\natural}_K = (A^{\natural})_K \quad , \tag{7.6.1}$$

where $A^{\natural}$ is the Raynaud group scheme (7.2.2): it is an extension of the finite etale group Φ by $A^{\natural o}$, while $A^{\natural o}$ is an extension of the abelian scheme B by the torus T. Thus $A^{\natural}_K$ is itself an extension of Φ_K by $(A^{\natural}_K)^o = (A^{\natural o})_K = (A^{o\natural})_K$, which is an extension of the abelian scheme B_K by the torus T_K. For every prime number ℓ, (7.3.5) and (7.3.2) give

$$T_\ell(A^{\natural}_K) \simeq T_\ell(A^{\natural o}_K) \simeq T_\ell(A_K)^f \ , \qquad T_\ell(T_K) \simeq T_\ell(A_K)^t \quad . \tag{7.6.2}$$

7.6.3. By 3.3, the formation of $A^{\natural o}_K$ commutes with every base change associated with a morphism of complete traits $S' \longrightarrow S$.

8. The Raynaud extension $A^{\natural o}_K$ in the non-semistable case, and the ind-group $A^{\flat}_K$

8.1. Continue to suppose that the trait S is complete, but no longer suppose necessarily that A_K has semistable reduction. As usual, let K' be a finite Galois extension of K such that $A_{K'}$ has semistable reduction relative to the normalization S' of S in K'. Consider the Raynaud extension $A^{\natural}_{K'}$ associated with $A_{K'}$ (7.6.1). It is clear that the Galois group of K'/K acts by transport of structure on this group scheme; hence, by Galois descent, it gives a group scheme over K, denoted $A^{\natural}_{K,K'}$, which depends on the choice of the extension K'. Its identity component, however, is independent of this choice up to canonical isomorphism, by 7.6.3, and will simply be denoted $A^{\natural o}_K$. Descent shows that $A^{\natural o}_K$ is an extension

$$0 \longrightarrow T_K \longrightarrow A_K^{\natural o} \longrightarrow B_K \longrightarrow 0 \quad , \tag{8.1.1}$$

where T_K is a torus and B_K an abelian scheme over K. Likewise, by the definition of the essentially fixed and essentially toric parts of $T_\ell(A_K)$ (4.1 and 6.5), the isomorphisms (7.6.2) give canonical isomorphisms

$$T_\ell(A_K^\natural) \simeq T_\ell(A_K)^{ef} \quad , \qquad T_\ell(T_K) \simeq T_\ell(A_K)^{et} \quad , \tag{8.1.2}$$

and hence, on passing to the quotient,

$$T_\ell(B_K) \simeq T_\ell(A_K)^{ef}/T_\ell(A_K)^{et} \quad . \tag{8.1.3}$$

This therefore determines two of the quotients occurring in the filtration

$$T_\ell(A_K) \supset T_\ell(A_K)^{ef} \supset T_\ell(A_K)^{et} \supset 0 \tag{8.1.4}$$

Thus the extension of group schemes (8.1.1) describes two quotients in this filtration. For the third quotient $T_\ell(A_K)/T_\ell(A_K)^{ef}$, (6.5.2) gives a canonical isomorphism

$$T_\ell(A_K)/T_\ell(A_K)^{ef} \simeq D(T_\ell(A'_K)^{et}) \quad , \tag{8.1.5}$$

where $D = \underline{\mathrm{Hom}}(-, T_\ell(\mathbb{G}_m))$ denotes Cartier duality (in the sense of Barsotti–Tate pro-groups), and A'_K denotes the abelian scheme dual to A_K. Consider then the exact sequence of type (8.1.1)

$$0 \longrightarrow T'_K \longrightarrow {A'_K}^{\natural o} \longrightarrow B'_K \longrightarrow 0 \tag{8.1.6}$$

associated with the abelian scheme A'_K over K, together with the corresponding isomorphisms of type (8.1.2). Formula (8.1.5) then gives

$$T_\ell(A_K)/T_\ell(A_K)^{ef} \simeq D(T_\ell(T'_K)) \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad , \tag{8.1.7}$$

where, generalizing the notation introduced in (5.5.2), we have set

$$\underline{M}_K = D(T'_K) \quad , \qquad \underline{M}'_K = D(T_K) \quad , \tag{8.1.8}$$

and where D now denotes Cartier duality $\underline{\mathrm{Hom}}(-, \mathbb{G}_{mK})$ in the sense of ordinary group schemes. Thus $\underline{M}_K$ and $\underline{M}'_K$ are twisted constant groups over K which become constant over K' and isomorphic to groups of the form $\mathbb{Z}^r$, where $r = \dim T = \dim T'$.

8.2. The hypotheses are those of 8.1. It is clear that the extension (8.1.1) depends functorially on the abelian scheme A_K, and that its formation commutes with every morphism of traits $\widetilde{S} \longrightarrow S$. Choose a polarization of A_K, hence an isogeny $A_K \longrightarrow A'_K$. It induces a homomorphism from (8.1.1) to (8.1.6), which is immediately seen, by looking at the T_ℓ, to be an isogeny term by term. Likewise, using 7.5 and descent, one sees that every biextension by $\mathbb{G}_{mK}$ of abelian schemes A_K and $\overline{A}_K$ over K defines a biextension of $(B_K, \overline{B}_K)$ by $\mathbb{G}_{mK}$, where B_K and $\overline{B}_K$ denote the abelian quotients of the Raynaud extensions corresponding to A_K and $\overline{A}_K$, respectively. When W is a perfect duality, so is the corresponding biextension of $(B_K, \overline{B}_K)$. In particular, with the notation of 7.6, the abelian schemes B_K and B'_K are canonically dual to one another.

The correspondence between biextensions on $(A_K, \overline{A}_K)$ and biextensions on $(B_K, \overline{B}_K)$ has evident functoriality properties as A_K and $\overline{A}_K$ vary. These follow by descent from the analogous properties stated in 7.5, and we leave their formulation to the reader. We also leave to the reader the compatibility of the preceding construction with every change of traits $\widetilde{S} \longrightarrow S$.

8.3. Return to the group $A^{\natural}_{K,K'}$ and study its variation with K'. It is clear that two isomorphic extensions K' and K'_1 of K give canonically isomorphic groups, so it is enough to consider the subextensions K' of the separable closure $\overline{K}$ of K. If $K' \hookrightarrow K''$ are two such finite Galois subextensions of $\overline{K}$, it follows immediately from 3.3 that there is a canonical <u>open immersion</u>

$$A^{\natural}_{K,K'} \hookrightarrow A^{\natural}_{K,K''} \quad , \tag{8.3.1}$$

which identifies the first group with an open subgroup of the second. As K' varies, the $A^{\natural}_{K,K'}$ thus form an inductive system of smooth groups of finite type over K, whose transition morphisms are open immersions. If we set

$$\Phi_{K,K'} = A^{\natural}_{K,K'}/{A^{\natural}_{K,K'}}^{o} \quad , \tag{8.3.2}$$

the $\Phi_{K,K'}$ themselves form an inductive system of finite etale groups over K, which will be studied in greater detail below.

We shall set

$$A^{\flat}_K = \varinjlim_{K'} A^{\natural}_{K,K'} \quad , \tag{8.3.3}$$

where the limit is taken in the category of fppf sheaves over K. Since the transition morphisms in this inductive system are open immersions, it is clear moreover that this limit is represented by a group scheme over K (in general non-quasi-compact), which we shall naturally denote by the same symbol $A^{\flat}_K$. For this group scheme one has

$${A^{\flat}_K}^{o} = A^{\natural o}_K \quad , \qquad A^{\flat}_K/{A^{\flat}_K}^{o} \simeq \varinjlim_{K'} \Phi_{K,K'} \quad . \tag{8.3.4}$$

The latter group will be determined in 11.10: it is canonically isomorphic to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$, with the notation of (8.1.8).

9. <u>Definition of the monodromy pairing $\underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_{\ell S}$</u>.

9.1. In this section, suppose that S is henselian and that A_K has semistable reduction over S, or, equivalently, that A'_K has semistable reduction over S (3.5.1). We then have over S two twisted constant $\mathbb{Z}$-modules $\underline{M}$ and $\underline{M}'$ (5.5.2). Over a suitable finite etale covering of S (and hence over S itself if S is strictly local, that is, if k is separably closed), they split into constant groups with values M and M', respectively, noncanonically isomorphic to $\mathbb{Z}^r$, where r is the common dimension of the maximal tori of A_o and A'_o. For every prime number ℓ, we shall consider the corresponding ℓ-adic sheaves

$$\underline{M}_\ell = \underline{M} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad , \qquad \underline{M}'_\ell = \underline{M}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \quad , \tag{9.1.1}$$

whose geometric meaning is made explicit by formulas (5.5.7) and (5.5.8). In this section we shall define a canonical pairing, called the <u>monodromy pairing</u>,

$$u_\ell : \underline{M}_\ell \otimes \underline{M}'_\ell \longrightarrow \mathbb{Z}_\ell \quad , \tag{9.1.2}$$

which will be defined in terms of the Barsotti–Tate pro-ℓ-groups $T_\ell(A^o)$ and $T_\ell(A'^o)$, and even in terms of the generic fiber $T_\ell(A_K)$ of the first of these groups (subject, when $\ell = p$, to the availability of Tate's theorem [33, Th. 4], that is, for the time being, provided we are then in unequal characteristic). When $\ell \neq p$, this is equivalent to saying that the pairing (9.1.2) is described in terms of the Galois π-module

$$U = T_\ell(A_K)(\overline{K}) = T_\ell(A_{\overline{K}}) \tag{9.1.3}$$

considered in (2.5.3). More precisely, we shall see that knowing the pairing is essentially equivalent to knowing the action of the monodromy group I on U.

9.2. The case $\ell \neq p$. Begin with the simpler case $\ell \neq p$, using the notation introduced in (2.5.3). Note that, up to torsion, the relation $V = U^I$ also means that $V^\perp$ is the module generated by the elements $gx - x$, where $x \in U$ and $g \in I$, and in any case one has

$$gx - x \in V^\perp \quad \text{for } x \in U,\ g \in I \qquad . \tag{9.2.1}$$

Thus, for $g \in I$,

$$gx = x + u(g) \,.\, x \qquad , \tag{9.2.2}$$

where $g \longmapsto u(g)$ is a homomorphism from U to $V^\perp$ which vanishes on $V = U^I$, or equivalently

$$u \in \mathrm{Hom}(U/V, V^\perp) \simeq (U/V)^\vee \otimes V^\perp \qquad . \tag{9.2.3}$$

In the case of semistable reduction under consideration, that is, when $V^\perp \subset V$, formula (9.2.2) says that the action of g on U is a transvection parallel to $V^\perp = W$, and composition of these transvections is simply addition of the corresponding homomorphisms u in (9.2.3). Consequently, the action of the inertia group I on U is completely described by a group homomorphism

$$I \longrightarrow (U/V)^\vee \otimes V^\perp \qquad . \tag{9.2.4}$$

The orthogonality theorem, applied both to A_K and to A'_K, gives

$$V^\perp = W \quad , \qquad (U/V)^\vee \simeq W'(-1) \qquad ,$$

where W' is the toric part of $U' = T_\ell(A'(\overline{K}))$, and (-1) denotes the Tate twist. Moreover, the homomorphism (9.2.4) factors through $I(\ell) \simeq \mathbb{Z}_\ell(1)$, so that (9.2.4) may be interpreted as a canonical homomorphism

$$u : \mathbb{Z}_\ell(1) \longrightarrow W' \otimes W(-1) \qquad , \tag{9.2.5}$$

or, equivalently, as an element

$$u \in W' \otimes W(-2) \qquad . \tag{9.2.6}$$

Recall that W and W' are the groups of $\overline{K}$-valued points of the two twisted constant ℓ-adic sheaves $T_\ell(A^o)^t$ and $T_\ell(A'^o)^t$, whose special fibers are $T_\ell(T_o)$ and $T_\ell(T'_o)$, respectively. We obtain the canonical isomorphisms

$$\text{(9.2.7)} \qquad \begin{cases} W \simeq T_\ell(T_o)(\overline{k}) \simeq \check{M} \otimes \mathbb{Z}_\ell(1), \\ \\ W' \simeq T_\ell(T'_o)(\overline{k}) \simeq \check{M}' \otimes \mathbb{Z}_\ell(1), \end{cases}$$

where M and M' denote the character groups over $\overline{k}$ of T_o and T'_o, respectively:

$$\text{(9.2.8)} \qquad T_o = D(M) \quad , \qquad T'_o = D(M') \qquad .$$

It follows that the element u of (9.2.6), which describes the action of the inertia group I on the Tate module U of A_K, is interpreted as a canonical element of $(\check{M}' \otimes_{\mathbb{Z}} \check{M}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. It is moreover invariant under $\pi_o = \pi/I = \mathrm{Gal}(\overline{k}/k)$, since by definition it is invariant under transport of structure:

$$\text{(9.2.9)} \qquad u \in \left((\check{M}' \otimes_{\mathbb{Z}} \check{M}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell\right)^I \simeq \mathrm{Hom}_{\mathbb{Z}}(\underline{M} \otimes \underline{M}', \mathbb{Z}_\ell) \simeq \mathrm{Hom}_{\mathbb{Z}_\ell}(\underline{M}_\ell \otimes \underline{M}'_\ell, \mathbb{Z}_\ell) \quad .$$

Up to writing u in place of u_ℓ, this is the element (9.1.2) that we wished to define.

To define (9.1.2) in the general case, we shall need a few preliminaries, which will be developed in the next two sections.

9.3. Mixed extensions.

Let $\underline{C}$ be an abelian category, and let

$$\text{(9.3.1)} \qquad P,\ R,\ Q$$

be three objects of $\underline{C}$. We are interested in classifying the objects X of $\underline{C}$ endowed with a three-step filtration

$$(*) \qquad X^0 = X \supset X^1 \supset X^2 \supset X^3 = 0$$

and with given isomorphisms [26]

$$X^0/X^1 \xrightarrow{\sim} Q, \qquad X^1/X^2 \xrightarrow{\sim} R, \qquad X^2/X^3 = X^2 \xrightarrow{\sim} P.$$

Putting

$$E = X^0/X^2, \qquad F = X^1/X^3 = X^1,$$

we see that X gives rise to two ordinary extensions

$$\text{(9.3.2)} \qquad 0 \longrightarrow P \longrightarrow F \longrightarrow R \longrightarrow 0, \qquad 0 \longrightarrow R \longrightarrow E \longrightarrow Q \longrightarrow 0,$$

of R by P and of Q by R, respectively. Our point of view here will be to start, in addition to the objects (9.3.1), with the extension structures (9.3.2), which are likewise assumed to be given, and to consider the corresponding three-step filtered structures X.

More precisely, a mixed extension of the extension E by the extension F will mean an object X of $\underline{C}$, endowed with a three-step filtration (*) and with given isomorphisms

26. The printed source assigns P, R, Q, in that order, to these three graded pieces. The immediately following exact sequences and every subsequent construction require Q, R, P; the evident interchange of P and Q has therefore been corrected here.

$$E \simeq X^0/X^2, \qquad F \simeq X^1/X^3 = X^1, \tag{9.3.3}$$

compatible with the filtrations: they carry X^1/X^2 to $\mathrm{Im}(R \longrightarrow E)$ and X^2 to $\mathrm{Im}(P \longrightarrow F)$, respectively, and the isomorphism $R \simeq F/P$ (and hence $R \simeq X^1/X^2$) deduced from (9.3.3) is required to equal the one deduced from the first exact sequence in (9.3.2).

For fixed P, Q, R and fixed extensions E and F, the mixed extensions of E by F form a category in the evident way; it is in fact a groupoid (every arrow is invertible). The automorphism group of a mixed extension X of E by F is canonically isomorphic to $\mathrm{Hom}(Q, P)$ through

$$\mathrm{Hom}(Q,P) \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{extpan}}(X), \qquad u \longmapsto \mathrm{id}_X + \bar{u}, \tag{9.3.4}$$

where $\bar{u}$ is the endomorphism of the object of $\underline{C}$ underlying X obtained by composing $X \longrightarrow Q \xrightarrow{u} P \longrightarrow X$; the outer arrows are respectively the canonical epimorphism and the canonical monomorphism coming from the given mixed-extension structure on X.

One could make the structure of the category $\mathrm{EXTPAN}(E, F)$ of mixed extensions of E by F precise by regarding it as a "torsor" under the "group-category" (VII 2.5) $\mathrm{EXT}(Q, P)$ of extensions of Q by P, using a natural functor analogous to the Brauer composition of ordinary extensions:

$$\begin{gathered} \omega : \mathrm{EXT}(Q,P) \times \mathrm{EXTPAN}(E,F) \longrightarrow \mathrm{EXTPAN}(E,F), \\ \omega(G,X) = G \wedge X. \end{gathered} \tag{9.3.5}$$

For every object X of $\mathrm{EXTPAN}(E, F)$, the partial functor $G \longmapsto G \wedge X$ is an equivalence from $\mathrm{EXT}(Q, P)$ to $\mathrm{EXTPAN}(E, F)$.

We shall give only a brief description of ω. Regard a mixed extension X of E by F as defining an ordinary extension, also denoted X, of Q by F (viewed as an object of $\underline{C}$). For every extension G of Q by P, let $\bar{G}$ denote the extension of Q by F induced by the inclusion $P \longrightarrow F$. Set

$$G \wedge X = \omega(G,X) \overset{\mathrm{dfn}}{=} \bar{G} \wedge X, \tag{9.3.6}$$

where $\wedge$ denotes Brauer composition in the category of extensions of Q by F. The three-step filtration of the right-hand side X' of (9.3.6) is clear, as is the second isomorphism of type (9.3.3) attached to X'.

To define the first isomorphism of type (9.3.3) for X', observe that the functor

$$Y \longmapsto \widetilde{Y}, \qquad \mathrm{EXT}(Q,F) \longrightarrow \mathrm{EXT}(Q,R),$$

induced by the given morphism $F \longrightarrow R$ in (9.3.2) is compatible with Brauer composition. It therefore sends X' to an extension $\widetilde{X}'$ canonically isomorphic to $\widetilde{\bar{G}} \wedge \widetilde{X}$. The first isomorphism (9.3.3) gives an isomorphism $\widetilde{X} \simeq E$, while $\widetilde{\bar{G}}$—the extension induced from G by the composite $P \longrightarrow F \longrightarrow R$—is canonically isomorphic to the trivial extension, since that composite $P \longrightarrow R$ is zero. Consequently there is a canonical isomorphism $\widetilde{X}' \simeq E$, which specifies on the right-hand side of (9.3.6) its mixed-extension structure of E by F.[27]

We leave to the reader the definition of ω on the arrows of $\mathrm{EXT}(Q, P) \times \mathrm{EXTPAN}(E, F)$, the verification that (9.3.5) has the asserted property—namely, for every <u>fixed</u> mixed extension X, the functor $G \longmapsto G \wedge X$ is an equivalence from $\mathrm{EXT}(Q, P)$ to $\mathrm{EXTPAN}(E, F)$—and the proof of the associativity formula

27. At both occurrences in this paragraph the printed source has F where the objects involved are extensions of Q by R and hence must be identified with E.

$$G_1 \wedge (G_2 \wedge X) \simeq (G_1 \wedge G_2) \wedge X, \tag{9.3.7}$$

where $G_1 \wedge G_2$ denotes Brauer composition in $\mathrm{EXT}(Q,P)$.

The preceding discussion immediately gives parts (a) and (b) of the following result.

Proposition 9.3.8. Given two extensions E and F as in (9.3.2) in the abelian category $\underline{C}$, consider the category $\mathrm{EXTPAN}(E,F)$ of mixed extensions of E by F. Then:

a) $\mathrm{EXTPAN}(E,F)$ is a groupoid. For every one of its objects the automorphism group is canonically isomorphic to $\mathrm{Ext}^0(Q,P) = \mathrm{Hom}(Q,P)$.

b) The set $\mathrm{Extpan}(E,F)$ of isomorphism classes of objects of $\mathrm{EXTPAN}(E,F)$ is empty, or is naturally a torsor under the group $\mathrm{Ext}^1(Q,P)$.

c) One can canonically define a class

$$c(E,F) \in \mathrm{Ext}^2(Q,P), \tag{9.3.9}$$

whose vanishing is necessary and sufficient for $\mathrm{Extpan}(E,F) \neq \varnothing$, that is, for the existence of a mixed extension of E by F.

It remains to prove (c).[28] A mixed extension of E by F may also be described as an extension X of E by P together with an isomorphism of extensions of R by P

$$X^1 \simeq F,$$

where X^1 denotes the image of the extension X under the morphism $R \longrightarrow E$ in (9.3.2). Such a mixed extension therefore exists if and only if there is an element of $\mathrm{Ext}^1(E,P)$ whose image in $\mathrm{Ext}^1(R,P)$ is the class ξ of F. Consider the exact sequence of the $\mathrm{Ext}^i(-,P)$ associated with the second exact sequence in (9.3.2)[29]:

$$\cdots \longrightarrow \mathrm{Ext}^1(E,P) \longrightarrow \mathrm{Ext}^1(R,P) \xrightarrow{\partial} \mathrm{Ext}^2(Q,P) \longrightarrow \cdots .$$

It is enough to define $c(E,F)$ by

$$c(E,F) = \partial\xi, \qquad \xi = \mathrm{cl}(F) \in \mathrm{Ext}^1(R,P). \tag{9.3.10}$$

Now suppose that we are given an exact functor

$$r : \underline{C} \longrightarrow \underline{C}', \tag{9.3.11}$$

where $\underline{C}'$ is another abelian category. Write P', Q', R', E', F' for the objects of $\underline{C}'$ obtained from P, Q, R, E, F by applying r, so that E' is an extension of Q' by R' and F' is an extension of R' by P'. Up to canonical isomorphism, the following diagram of functors is commutative:

$$\begin{tikzcd} \mathrm{EXT}(Q,P) \times \mathrm{EXTPAN}(E,F) \arrow[r, "\omega"] \arrow[d] & \mathrm{EXTPAN}(E,F) \arrow[d] \\ \mathrm{EXT}(Q',P') \times \mathrm{EXTPAN}(E',F') \arrow[r, "\omega'"'] & \mathrm{EXTPAN}(E',F'). \end{tikzcd} \tag{9.3.12}$$

28. The printed source says “prove (d)”, although Proposition 9.3.8 has only parts (a), (b), and (c).

29. The printed final term is $\mathrm{Ext}^2(R,Q)$; applying $\mathrm{Ext}(-,P)$ to $0 \to R \to E \to Q \to 0$ gives the displayed $\mathrm{Ext}^2(Q,P)$, which is also the group specified in (9.3.9).

The vertical arrows are the evident functors induced by the exact functor r, and ω' is the arrow analogous to ω in (9.3.5), with $\underline{C}$ replaced by $\underline{C}'$. The lower arrow in the printed diagram is labelled ω without a prime; the explanatory sentence identifies it as ω', which is the notation used here. We obtain analogous compatibilities, with the functor r, of the structures considered in 9.3.8; their explicit formulation is left to the reader.

We shall use the following consequence of this commutativity. Suppose that $\mathrm{EXTPAN}(E, F)$ is nonempty and choose an object X in it. Then, by means of $G' \longmapsto G' \wedge X'$, where $X' = r(X)$, the category $\mathrm{EXTPAN}(E', F')$ is equivalent to $\mathrm{EXT}(Q', P')$. Thus an object Y' of $\mathrm{EXTPAN}(E', F')$ is known up to isomorphism when the isomorphism class of the corresponding object of $\mathrm{EXT}(Q', P')$ is known, that is, when the corresponding element

$$c_X(Y') \in \mathrm{Ext}^1(Q', P')$$

is known.

If X is replaced by another object $X_1 = H \wedge X$, where H is an extension of Q by P, then one immediately obtains [30]

$$c_{X_1}(Y') = c_X(Y') - \mathrm{cl}(H')$$

from the associativity formula (9.3.7). Hence, for fixed Y' and varying X, the elements $c_X(Y')$ run through exactly one coset

$$c(Y') \in \mathrm{Ext}^1(Q', P') / \operatorname{Im} \mathrm{Ext}^1(Q, P), \tag{9.3.13}$$

which is therefore canonically associated with the mixed extension Y' of E' by F'. By construction, its vanishing is necessary and sufficient for Y' to be isomorphic to the transform under r of a mixed extension X of E by F.

9.4. Let S be a henselian trait with generic point η, let n be an integer > 0, put $\Lambda = \mathbb{Z}/n\mathbb{Z}$, and let $\underline{C}$ be the category of Λ-modules on S (for the fppf site on S; cf. VIII 0.2) and $\underline{C}'$ the category of Λ-modules on η. Let P, R, Q be objects of $\underline{C}$, and let E, F be extensions of Q by R and of R by P, respectively (9.3.2). Suppose that Q is a locally free Λ_S-module of finite type and that P is represented by a finite group of multiplicative type over S (SGA 3 IX), so that it is of the form

$$P = D(Q^*),$$

where $D = \underline{\mathrm{Hom}}(-, \mathbb{G}_{mS})$ denotes Cartier duality. Let $P_\eta, R_\eta, Q_\eta, E_\eta, F_\eta$ denote the restrictions of P, R, Q, E, F to η, and suppose that a mixed extension X_η of E_η by F_η is given. Under these conditions we shall canonically define a pairing

$$c(X_\eta) : Q \otimes_\Lambda Q^* \longrightarrow \Lambda_S \qquad (\Lambda = \mathbb{Z}/n\mathbb{Z}). \tag{9.4.1}$$

For this purpose, first <u>suppose</u> that S is <u>strictly local</u>. Then Q and Q^* are constant, say $Q = Q_{oS}$ and $Q^* = Q^*_{oS}$, and giving (9.4.1) amounts to giving an element

$$c(X_\eta) \in \check{Q}_o \otimes \check{Q}^*_o, \tag{9.4.2}$$

where the check denotes the ordinary dual $\mathrm{Hom}(-, \Lambda)$. We apply the considerations of 9.3 with the restriction functor $r : \underline{C} \longrightarrow \underline{C}'$. First observe that a mixed extension of E by F exists. Indeed, the obstruction to its existence lies in $\mathrm{Ext}^2(Q, P)$ (9.3.8(c)), and it is enough to prove

$$\mathrm{Ext}^2(Q, P) = 0.$$

30. The printed right-hand side of the following formula has $c_X(Y)$; the fixed object in this construction is Y'.

This is a special case of the following lemma.

Lemma 9.4.3. Under the preceding hypotheses, with S strictly local, one has:

a) canonical isomorphisms

$$\mathrm{Ext}^i_\Lambda(Q,P) \simeq H^i(S,\check{Q}\otimes P), \qquad \mathrm{Ext}^i_\Lambda(Q_\eta,P_\eta) \simeq H^i(\eta,\check{Q}\otimes P);$$

b) a canonical isomorphism

$$\check{Q}\otimes P \simeq D(Q\otimes Q^*); \tag{b)}$$

c) for every finite constant group $H = H_{oS}$ over S,

$$H^i(S,D(H)) \simeq H^i(\eta,D(H)_\eta) = 0 \qquad \text{for } i \geq 2; \tag{c)}$$

d) canonical isomorphisms

$$\begin{aligned} H^1(S,D(H)) &\simeq \mathbb{G}_m(S)\otimes\check{H}_o = V^*\otimes\check{H}_o, \\ H^1(\eta,D(H)_\eta) &\simeq \mathbb{G}_m(\eta)\otimes\check{H}_o = K^*\otimes\check{H}_o, \end{aligned} \tag{d)}$$

where the check in (d) denotes $\mathrm{Hom}(-,\mathbb{Q}/\mathbb{Z})$.

The formulas in (a) and (b) follow immediately from the fact that Q is a locally free Λ_S-module of finite type. To prove (c), we plainly reduce to the case $H_o = \mathbb{Z}/m\mathbb{Z}$, that is, $D(H) = \mu_m$. The cohomology sequence associated with the Kummer exact sequence

$$0 \longrightarrow \mu_m \longrightarrow \mathbb{G}_m \xrightarrow{m\cdot\mathrm{id}} \mathbb{G}_m \longrightarrow 0$$

then reduces us to proving

$$H^i(S,\mathbb{G}_m) = H^i(\eta,\mathbb{G}_m) = 0 \qquad \text{for } i \geq 1.$$

These groups are the groups $H^i_{\text{ét}}$ computed on the étale site [6, Theorem 11.7], and therefore vanish for S, since S is strictly local. Their vanishing for η is likewise well known [6, Theorem 1.1].

It remains to describe the isomorphisms in (d). Over an arbitrary base U there is a canonical homomorphism

$$\mathbb{G}_m(U)\otimes\check{H}_o \longrightarrow H^1(U,D(H_{oU})). \tag{9.4.3.1}$$

If H_o is annihilated by an integer $m > 0$, this homomorphism is described using the boundary map $\mathbb{G}_m(U) \xrightarrow{\partial} H^1(U,\mu_m)$ associated with the Kummer sequence: it is the composite

$$\begin{aligned} \mathbb{G}_m(U)\otimes\check{H}_o &\longrightarrow H^1(U,\mu_m)\otimes\check{H}_o \\ &\longrightarrow H^1(U,\mu_m\otimes\check{H}_o) \longrightarrow H^1(U,D(H_{oU})). \end{aligned}$$

Here the second arrow is deduced from the evident homomorphism $\check{H}_o \longrightarrow \mathrm{Hom}(\mu_m,\mu_m\otimes\check{H}_o)$, and the third is a special case of (b), with m in place of n. We leave it to the reader to check that (9.4.3.1) is independent of the choice of the integer m such that $mH_o = 0$. We also claim that (9.4.3.1) is always injective and that its cokernel vanishes if $\mathrm{Pic}(U) = 0$. (In fact, the cokernel is always isomorphic to $\mathrm{Hom}(H_o,\mathrm{Pic}(U))$, but this is immaterial here.) Indeed, we reduce to the case $H_o = \mathbb{Z}/n\mathbb{Z}$, and the assertion then follows immediately from the cohomology sequence of the Kummer sequence above. This proves (d), and immediately yields:

Corollary 9.4.4. With the notation of 9.4.3, there is a canonical isomorphism
e)

$$H^1(\eta, D(H)_\eta)/\operatorname{Im} H^1(S, D(H)) \simeq \check{H}_o,$$

and hence, on combining (a) and (b), a canonical isomorphism

$$\operatorname{Ext}^1(Q_\eta, P_\eta)/\operatorname{Im}\operatorname{Ext}^1(Q, P) \simeq \check{Q}_o \otimes \check{Q}_o^*. \tag{9.4.5}$$

Returning to the definition of (9.4.2) when S is strictly local, the relation $\operatorname{Ext}^2(Q, P) = 0$ makes it possible to define the canonical element (9.3.13), taking for Y' the given mixed extension X_η of E_η by F_η. Formula (9.4.5) identifies this element $c(X_\eta)$ with an element of $\check{Q}_o \otimes \check{Q}_o^*$, thereby completing the definition of (9.4.2). The general considerations of 9.3 also show that this element is the obstruction to extending X_η to a mixed extension X of E by F.

9.4.6. The definition of (9.4.1) when S is henselian now reduces immediately to the strictly local case by Galois descent from the strict henselization of S.

Proposition 9.4.7. Let P, Q, R, E, F be as above, except that the trait S need not be henselian. Consider the functor

$$X \longmapsto X_\eta \tag{9.4.7.1}$$

from the category of mixed extensions of E by F to the category of mixed extensions of E_η by F_η. For every mixed extension X_η of E_η by F_η, consider the pairing

$$c_o(X_\eta) : Q_o \otimes Q_o^* \longrightarrow \Lambda_s \tag{9.4.7.2}$$

induced by the pairing of type (9.4.1) defined over the henselization S^h of S. Then the functor (9.4.7.1) is fully faithful, and a mixed extension X_η of E_η by F_η belongs to its essential image if and only if the pairing (9.4.7.2) is zero.

The printed target of (9.4.7.2) is Λ_s; the French transcription's $\wedge_s$ is a character-recognition error.

9.4.7.3. This statement still makes sense and remains valid if one assumes only that S is a connected regular noetherian scheme of dimension 1. Instead of the single pairing (9.4.7.2), one must then consider a pairing $Q_s \otimes Q_s^* \longrightarrow \Lambda_s$ for each closed point s of S. It will be most convenient to prove 9.4.7 in this more general form.

a) For every mixed extension X, the map

$$\operatorname{End}(X) \longrightarrow \operatorname{End}(X_\eta)$$

is bijective. The two sides are respectively the sections over S and η of $\underline{\operatorname{Hom}}(Q, P) = D(Q \otimes Q^*)$; hence the assertion follows from the normality of S and the finiteness of $D(Q \otimes Q^*)$ over S.[31] It follows at once that if X and Y are two mixed extensions already known to be isomorphic, then

$$\operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(X_\eta, Y_\eta)$$

is bijective.

b) If X and Y are two mixed extensions, every isomorphism $X_\eta \xrightarrow{\sim} Y_\eta$ comes from an isomorphism $X \longrightarrow Y$. By (a), the question is local for the étale topology, so we may reduce to the case in which S is strictly local. In that case 9.4.3 shows that the map

31. The printed proof has $\underline{\operatorname{Hom}}(P, Q)$ here. Proposition 9.3.8(a), formula (9.3.4), and the displayed identification with $D(Q \otimes Q^*)$ all require $\underline{\operatorname{Hom}}(Q, P)$.

$$\mathrm{Ext}^1_\Lambda(Q,P) \longrightarrow \mathrm{Ext}^1_\Lambda(Q_\eta,P_\eta)$$

identifies with

$$V^*\otimes \check{H}_o \longrightarrow K^*\otimes \check{H}_o, \qquad H_o = Q_o\otimes Q_o^*.$$

The latter is injective, since the exact sequence $0 \longrightarrow V^* \longrightarrow K^* \longrightarrow \mathbb{Z} \longrightarrow 0$ is split. By the discussion of 9.3 this implies that if X_η and Y_η are isomorphic, then so are X and Y; the conclusion now follows from (a).[32]

Thus (a) and (b) show that $X \longmapsto X_\eta$ is fully faithful.

c) Let X_η be a mixed extension over η. We prove that it extends to S if and only if the pairings (9.4.7.2) vanish. Here again, the uniqueness already proved shows that the question is local for the étale topology, reducing us once more to the strictly local case. The conclusion then holds by construction, as observed immediately before 9.4.6.[33]

This completes the proof of 9.4.7.

9.5. Fix a prime number ℓ, pro-ℓ Barsotti–Tate groups over S

$$P = (P(n))_{n\geq 0}, \qquad R = (R(n))_{n\geq 0}, \qquad Q = (Q(n))_{n\geq 0}, \tag{9.5.1}$$

and extensions of pro-ℓ Barsotti–Tate groups

$$0 \longrightarrow P \longrightarrow F \longrightarrow R \longrightarrow 0, \qquad 0 \longrightarrow R \longrightarrow E \longrightarrow Q \longrightarrow 0. \tag{9.5.2}$$

Thus, for every integer $n\geq 0$, these define extensions of Λ_n-modules, where $\Lambda_n = \mathbb{Z}/\ell^{n+1}\mathbb{Z}$:

$$\begin{gathered} 0 \longrightarrow P(n) \longrightarrow F(n) \longrightarrow R(n) \longrightarrow 0,\\ 0 \longrightarrow R(n) \longrightarrow E(n) \longrightarrow Q(n) \longrightarrow 0. \end{gathered} \tag{9.5.3}$$

We assume that Q is of étale type and P of multiplicative type. Finally, suppose that a pro-ℓ Barsotti–Tate group X_η is given over η which is a mixed extension of E_η by F_η. By this we mean that it is endowed, as a pro-ℓ Barsotti–Tate group, with a three-step filtration whose partial extensions are supplied with isomorphisms to E_η and F_η, respectively. It follows that, for every n, $X_\eta(n)$ is endowed with the structure of a mixed extension of $E(n)_\eta$ by $F(n)_\eta$. We may therefore associate with it a pairing of type (9.4.1):

$$c(X(n)_\eta) : Q(n)\otimes Q^*(n) \longrightarrow \Lambda_n = \mathbb{Z}/\ell^{n+1}\mathbb{Z}, \tag{9.5.4}$$

where the $Q^*(n)$ are the finite étale groups over S defined by

$$P(n) \simeq D(Q^*(n)), \qquad \text{i.e.} \qquad Q^*(n) = D(P(n)),$$

and D again denotes Cartier duality. The $Q^*(n)$ form a pro-ℓ group in the evident way, denoted by Q^*. We leave it to the reader to verify that, as n varies, the homomorphisms (9.5.4) define a homomorphism of projective systems, that is, a homomorphism of twisted constant ℓ-adic sheaves over S:

$$c(X_\eta) : Q\otimes Q^* \longrightarrow \mathbb{Z}_{\ell S}. \tag{9.5.5}$$

32. The printed Ext groups in this paragraph have the arguments (P,Q). The action of $\mathrm{Ext}^1(Q,P)$ on mixed extensions in Proposition 9.3.8(b), and Lemma 9.4.3 itself, require (Q,P).

33. The printed proof cites the nonexistent pairings "(9.4.6.2)"; these are the pairings (9.4.7.2).

We mention in passing the following result, which will not be used below.

Proposition 9.5.6. Let P, Q, R, E, F be as above, except that the trait S need not be henselian. Consider the functor

$$X \longmapsto X_\eta \tag{9.5.6.1}$$

from the category of Barsotti–Tate mixed extensions of E by F to the category of mixed extensions of E_η by F_η. For every Barsotti–Tate group X_η which is a mixed extension of E_η by F_η, consider the corresponding pairing

$$Q_o \otimes Q_o^* \longrightarrow \mathbb{Z}_{\ell s} \tag{9.5.6.2}$$

induced by the pairing of type (9.5.5) defined over the henselization of S. Then the functor (9.5.6.1) is fully faithful, and its essential image consists of those Barsotti–Tate groups X_η which are mixed extensions of E_η by F_η and for which the corresponding pairing (9.5.6.2) is zero. Moreover, when $\operatorname{char} K = 0$ (so that Tate's theorem [33, Theorem 4] applies), X_η belongs to the essential image if and only if it extends to a Barsotti–Tate group over S.

9.5.6.3. We leave it to the reader to extend this statement to the case in which S is assumed only to be a connected regular noetherian scheme of dimension 1.

The first assertion of 9.5.6 follows immediately from 9.4.7, applied to the components of the Barsotti–Tate groups under consideration.[34] It remains to prove that if a Barsotti–Tate group X_η, which is a mixed extension of E_η by F_η, extends to a Barsotti–Tate group X over S, then it also extends as a mixed extension.[35] By Tate's theorem [33], the homomorphism $X_\eta \longrightarrow Q_\eta$ extends uniquely to a homomorphism

$$X \longrightarrow Q.$$

For every integer n, the corresponding homomorphism $X(n) \longrightarrow Q(n)$ is surjective on the generic fibres and hence is surjective. Since $Q(n)$ is étale over S, the map $X(n) \longrightarrow Q(n)$ is flat, and consequently it is an epimorphism. Thus $X \longrightarrow Q$ is an epimorphism component by component.[36] Let $\overline{F}$ be the kernel of $X \longrightarrow Q$. Then $\overline{F}$ is a Barsotti–Tate group and X is an extension of Q by $\overline{F}$. The isomorphism $F_\eta \xrightarrow{\sim} \overline{F}_\eta$ extends, by Tate's theorem, to an isomorphism $F \simeq \overline{F}$. Hence X is an extension of Q by F. Consider now the quotient $\overline{E} = X/P$. A further application of Tate's theorem gives an isomorphism $E \simeq \overline{E}$. It follows that X is a mixed extension of E by F extending the mixed extension X_η, as required.

Remarks 9.5.6.4.

a) Of course, the last assertion of 9.5.6 should hold without restriction on K, since the same should be true of the invoked theorem of Tate.

b) One can give another proof of the good-reduction criterion 5.10. As in 5.10, reduce to the case where A_K has semistable reduction, and then apply 9.5.6 to $T_\ell(A_K)$ regarded as a mixed extension (cf. 9.6). This implies $u_\ell = 0$; using 11.6.2(a) below then gives $T_\ell(A_K)^t = 0$.

9.5.8. When $\ell \neq p$, the Barsotti–Tate groups under consideration over S and over K, respectively, may simply be interpreted in terms of twisted constant ℓ-adic sheaves over S, or equivalently in terms of representations of $\pi_o = \operatorname{Gal}(\overline{k}/k)$ and $\pi = \operatorname{Gal}(\overline{K}/K)$, respectively, on free $\mathbb{Z}_\ell$-modules

34. The printed proof cites 9.4.6, which only performs the henselian descent construction; the full-faithfulness and essential-image statement being applied componentwise is Proposition 9.4.7.

35. At this point the printed source gives the extension over S the name X_η as well. We reserve X_η for the generic fibre and write X for its extension over S.

36. The printed sentence says that $Q(n)$ is étale over η. The group $Q(n)$ was defined over S and is assumed of étale type there; the base required for this flatness argument is S.

of finite type. One then sees that the homomorphism (9.5.5) is the canonical homomorphism expressing the action of the inertia subgroup $I \subset \pi$ on $X_\eta(\overline{K})$, as defined by the method of 9.2.[37] To verify this, one reduces immediately to the situation of 9.4, taking n there to be a power of ℓ and R to be a locally constant Λ-module (as are Q and P); the same is then true of E, F, and X_η. Verification that the two definitions agree is left to the reader.

9.6. We are now in a position to define the announced canonical pairing (9.1.2), without any restriction on ℓ, in such a way that, for $\ell \neq p$, this definition agrees with the one given in 9.2. For this purpose, we apply the considerations of 9.5 to the following situation:

$$(9.6.1)\qquad \begin{cases} P = T_\ell(A^o)^t = \check{\underline{M}}'_\ell(1), \quad Q = \underline{M}_\ell, & R = T_\ell(A^o)^f/T_\ell(A^o)^t, \\ F = T_\ell(A^o)^f, \qquad E = D(F') = D\big(T_\ell(A'^o)^f\big). \end{cases}$$

In the last formula, D denotes Cartier duality in the sense of pro-ℓ Barsotti–Tate groups; that is, $D(Y)$ is the <u>projective</u> system of the $D(Y(n))$, with the usual transition morphisms. The structure of F as an extension of R by P is clear. Likewise,

$$F' = T_\ell(A'^o)^f$$

has the structure of an extension of R' by $P' = T_\ell(A'^o)^t = \check{\underline{M}}_\ell(1)$. Applying Cartier duality, we conclude that $E = D(F')$ is endowed with the structure of an extension of $D(P') = \underline{M}_\ell$ by $D(R')$. Now $D(R')$ is canonically isomorphic to R by 5.5 (proved in 7.4.6); this specifies the structure of E as an extension of Q by R.

As the mixed extension X_η of E_η by F_η, we take

$$(9.6.2)\qquad X_\eta = T_\ell(A_K) = X_\eta^0 \supset X_\eta^1 = T_\ell(A_K)^f \supset X_\eta^2 = T_\ell(A_K)^t \supset 0.$$

The isomorphism of extensions of R_η by P_η

$$X_\eta^1 \simeq F_\eta$$

is clear. The isomorphism of extensions of Q_η by R_η

$$X_\eta/X_\eta^2 \simeq E_\eta = D(F'_\eta)$$

is clear from the orthogonality theorem 5.2.[38]

We therefore have the data required to define a pairing (9.5.5), which here, since $Q^* = \underline{M}'_\ell$, takes the form (9.1.2).[39] Its compatibility with the definition of 9.2 follows from 9.5.8. The considerations of 5.8 also show, subject to the validity of Tate's theorem [33] (hence at least when $\operatorname{char} K = 0$), that the pairing in question can be described in terms of the single pro-ℓ Barsotti–Tate group $X_\eta = T_\ell(A_K)$ over η: its three-step filtration, and the extensions F and E extending X_η^1 and X_η/X_η^2, can be defined canonically in terms of X_η alone.[40]

<u>Remark 9.7.</u> We no longer necessarily assume that A_K has semistable reduction over the henselian trait S. Let K' be a finite Galois extension of K such that $A_{K'}$ has semistable reduction over

37. The printed source refers here to "the homomorphism (9.5.6)"; the homomorphism constructed in this section is (9.5.5), whereas 9.5.6 is the proposition.

38. The printed formula has $F_\eta = D(F'_\eta)$. Since $E = D(F')$ and the displayed quotient is the extension of Q_η by R_η, the required term is E_η.

39. The printed source calls this a pairing "(9.5.6)"; the pairing is the homomorphism (9.5.5), whereas 9.5.6 is the subsequent proposition.

40. The printed sentence says that X_η is "over S"; its notation and definition make it the generic-fibre group over η. It also lacks its closing parenthesis.

the normalization S' of S in K' (3.6). With the notation of (8.1.8), Galois descent from the monodromy pairing for $A_{K'}$ then defines a pairing

$$(*)\qquad v_\ell^{K'} : \underline{M}_K \otimes \underline{M}'_K \longrightarrow \mathbb{Z}_{\ell K}.$$

A priori this pairing depends on the choice of K', but it follows from (10.4.3) below that the pairing

$$(9.7.1)\qquad u_\ell : \underline{M}_K \otimes \underline{M}'_K \longrightarrow \mathbb{Q}_{\ell K}$$

given by

$$u_\ell = \frac{1}{e(S'/S)}\, v_\ell^{K'}$$

is independent of it, where

$$e(S'/S) = \text{length}(V'/\mathfrak{m}V').$$

We may again call (9.7.1) the monodromy pairing for the abelian scheme A_K over K and the prime number ℓ under consideration: when A_K itself has semistable reduction, it plainly coincides with the restriction to η of the monodromy pairing (9.1.2). It follows from 10.4 below that this pairing in fact always has values in $\mathbb{Q}_K$ and is independent of ℓ.

One may ask whether it has values in $\mathbb{Z}_K$, which is equivalent to saying that (9.7.1) has values in $\mathbb{Z}_{\ell K}$ for every prime number ℓ. This is so at least when A_K is an elliptic curve. The only case requiring verification is then the one in which $\underline{M}_K \simeq \underline{M}'_K$ has rank 1, so that $\underline{M}_K \otimes \underline{M}'_K$ is canonically isomorphic to $\mathbb{Z}_K$ and the monodromy pairing (9.7.1) can simply be interpreted as an ordinary rational number. It follows from the results of Néron [19] that this number is $v(j)$, where j is the absolute invariant of A_K and v is the normalized valuation of K.

10. Properties of the monodromy pairing. Integrality and positivity theorem

In this section we shall first give certain essentially trivial formal properties of the monodromy pairings, leaving the (somewhat tedious) details of the verifications to the reader. We shall also state a deeper theorem (10.4) concerning these pairings and indicate how its proof may be reduced to a special case of Jacobians, to be treated (by transcendental methods) in section 12, which is in fact logically independent of section 11.

10.1. Transport of structure. It is clear from the definition of the monodromy pairing (9.1.2) that this pairing is compatible with transport of structure in the situation (S, A_K, A'_K).

10.2. Symmetry. Starting with the dual scheme A'_K of A_K and applying the construction of section 9 to it, we must introduce the dual scheme $(A'_K)'$ of A'_K, which is canonically isomorphic to A_K by the biduality theorem ([5] or VIII 3.2). Denote by $u_\ell^{A_K}$ the monodromy pairing (9.1.2) relative to A_K. In the same way we obtain a pairing

$$(10.2.1)\qquad u_\ell^{A'_K} : \underline{M}'_\ell \otimes \underline{M}_\ell \longrightarrow \mathbb{Z}_\ell.$$

I claim that the latter is obtained from $u_\ell^{A_K}$ by the symmetry

$$s : \underline{M}'_\ell \otimes \underline{M}_\ell \xrightarrow{\sim} \underline{M}_\ell \otimes \underline{M}'_\ell;$$

that is,

$$(10.2.2)\qquad u_\ell^{A'_K} = u_\ell^{A_K} \circ s.$$

Taking account of the known anticommutativity property of the pairings φ_ξ ([5] or VIII 2.2.11), the symmetry formula (10.2.2) may be reformulated as follows in terms of a fixed polarization ξ of A_K. It defines a homomorphism

$$\widetilde{\xi} : A_K \longrightarrow A'_K$$

and hence a corresponding homomorphism (t standing for "toric")

$$\xi_t : \underline{M} \longrightarrow \underline{M}'. \tag{10.2.3}$$

We may therefore form the pairing

$$\begin{aligned} \varphi^t_{\xi,\ell} : \underline{M}_\ell \otimes \underline{M}_\ell &\longrightarrow \mathbb{Z}_\ell, \\ \varphi^t_{\xi,\ell}(x,y) &= u^{A_K}_\ell\big(x, \xi_t(y)\big). \end{aligned} \tag{10.2.4}$$

The preceding form $\varphi^t_{\xi,\ell}$ is symmetric:

$$\varphi^t_{\xi,\ell}(x,y) = \varphi^t_{\xi,\ell}(y,x). \tag{10.2.5}$$

In fact, this assertion is valid for every element ξ of the Néron–Severi group of A_K, not necessarily a polarization. But only when ξ is nondegenerate—that is, defines an isogeny $A_K \longrightarrow A'_K$, and hence also an isogeny (10.2.3)—is the symmetry relation (10.2.5) equivalent to (10.2.2). When ξ is a polarization, we shall moreover sharpen (10.2.5) considerably in 10.4(b) below.

Let us finally mention a third interpretation of the symmetry relation. Consider a biextension (that is, a divisorial correspondence) ξ on a pair $(A_K, \overline{A}_K)$ of abelian schemes with semistable reduction over K. It defines a homomorphism

$$A_K \longrightarrow \overline{A}'_K.$$

Passing to the Néron models, this defines a homomorphism on the maximal tori of the special fibres, and therefore, in the reverse direction, a homomorphism on their character groups:

$$\xi_t : \overline{\underline{M}} \longrightarrow \underline{M}', \tag{10.2.6}$$

where $\overline{\underline{M}}$ is defined from $\overline{A}_K$ as $\underline{M}$ is defined from A_K. We can therefore form the pairing

$$\begin{aligned} \varphi^t_{\xi} : \underline{M}_\ell \otimes \overline{\underline{M}}_\ell &\longrightarrow \mathbb{Z}_\ell, \\ \varphi^t_{\xi}(x,y) &= u^{A_K}_\ell\big(x, \xi_t(y)\big). \end{aligned} \tag{10.2.7}$$

[41] Notice in passing that, for a variable abelian scheme A_K with semistable reduction, the twisted constant group $\underline{M}$ is a covariant functor of A_K. Formation of the pairings (10.2.7) associated with biextensions of abelian schemes with semistable reduction over (K,S) is then compatible, in the evident sense, with taking inverse images of biextensions.

When ξ is the canonical ("Weil") biextension of (A_K, A'_K), the pairing (10.2.7) is of course precisely the monodromy pairing (9.1.2).[42] Now consider the transpose ${}^s\xi$ of ξ, a biextension of $(\overline{A}_K, A_K)$, and hence the pairing

41. In the printed right-hand side the underlying pairing is denoted only by φ^t, without a definition. It is the monodromy pairing $u^{A_K}_\ell$ used in the identical construction (10.2.4).

42. The printed source refers here to (9.2.1), which is the inertia orthogonality formula $gx - x \in V^\perp$, not the monodromy pairing. The pairing is numbered (9.1.2).

$$\varphi^t_{s\xi} : \overline{M}_\ell \otimes \underline{M}_\ell \longrightarrow \mathbb{Z}_\ell.$$

The symmetry relation is

(10.2.8) $$\varphi^t_{s\xi} = \varphi^t_\xi \circ s, \qquad s : \overline{M}_\ell \otimes \underline{M}_\ell \xrightarrow{\sim} \underline{M}_\ell \otimes \overline{M}_\ell,$$

where s is again the symmetry isomorphism. Compare this relation with the antisymmetry relation of VIII 2.2.11.

10.3. Change of trait. Let

(10.3.1) $$g : S' \longrightarrow S$$

be a dominant morphism of henselian traits, associated with an injective local homomorphism of discrete valuation rings

$$V \longrightarrow V'.$$

As usual, put

(10.3.2) $$e(S'/S) = \mathrm{length}_V\big(V'/\underline{m}V'\big),$$

where $\underline{m}$ is the maximal ideal of V. It is then known that the homomorphism induced by (10.3.1) on the "tame" parts of the inertia groups,

(10.3.3) $$g_* : I'_t \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell(1)(k') \longrightarrow I_t = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)(k),$$

is given by

(10.3.4) $$g_*(x) = e(S'/S)x,$$

after the two sides of (10.3.3) have been identified by means of the evident isomorphisms $\mathbb{Z}_\ell(1)(k) \xrightarrow{\sim} \mathbb{Z}_\ell(1)(k')$. [43]

On the other hand, if A_K is an abelian scheme over K with semistable reduction over S, formation of the twisted lattices $\underline{M}$ and $\underline{M}'$ commutes with every base change $S' \longrightarrow S$ as above, as follows immediately from the definitions (especially 3.4). We may therefore regard the pairing $u_\ell^{A_{K'}}$ defined by $A_{K'}$, where K' is the fraction field of V', as a pairing

$$u_\ell^{A_{K'}} : \underline{M}_{\ell S'} \otimes \underline{M}'_{\ell S'} \longrightarrow \mathbb{Z}_{\ell S'},$$

and compare it with the pairing $u_{\ell S'}^{A_K}$ obtained from $u_\ell^{A_K}$ by base change. When $\ell \neq p$, the construction of the pairings u_ℓ given in 9.2, together with formula (10.3.4), shows at once that [44]

(10.3.5) $$u_\ell^{A_{K'}} = e(S'/S)u_\ell^{A_K}.$$

In fact this formula remains valid for $\ell = p$. To see this, return to the general construction of 9.3–9.6 and reduce to the analogous compatibility assertion for the isomorphism 9.4.4(e), under a change of strictly local traits $S' \longrightarrow S$. Notice that the isomorphism in question is induced by the normalized valuation $K^*/V^* \simeq \mathbb{Z}$, and therefore gives the commutative diagram [45]

43. The printed source leaves the internal reference following "Exposé I" blank.

44. The printed source cites (10.3.3), the displayed map g_*; the multiplication formula it uses is (10.3.4).

45. The printed source says that $K^*/V^* \simeq \mathbb{Z}$ is deduced from the "norm." The displayed map is the normalized discrete valuation.

$$\begin{array}{ccc} K^*/V^* & \xrightarrow{\sim} & \mathbb{Z} \\ \downarrow & & \downarrow{\scriptstyle e(S'/S)} \\ K'^*/V'^* & \xrightarrow{\sim} & \mathbb{Z}. \end{array}$$

Here is a deeper result concerning the monodromy pairings. We shall prove it by transcendental methods in section 12, relying on results that will be proved later in the Seminar. (We shall not, in fact, use 10.4 in the remainder of the Seminar, any more than we shall use the other results of the present exposé.)

Theorem 10.4. Let S be a henselian trait, and let A_K be an abelian scheme over K with semistable reduction over S. Consider the lattices $\underline{M}$ and $\underline{M}'$ associated as in 9.1 (cf. 5.5.2). Then the following assertions hold:

a) There exists a unique pairing

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S \tag{10.4.1}$$

such that, for every prime number ℓ, $u \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ is the monodromy pairing u_ℓ of (9.1.2). This pairing is separating.

b) Let ξ be a polarization of A_K, thus defining a homomorphism (10.2.3)

$$\xi_t : \underline{M} \longrightarrow \underline{M}',$$

and consider the form

$$\begin{aligned} \varphi_\xi^t : \underline{M} \otimes \underline{M} &\longrightarrow \mathbb{Z}_S, \\ \varphi_\xi^t(x,y) &= u\big(x, \xi_t(y)\big). \end{aligned} \tag{10.4.2}$$

This form is symmetric, positive, and nondegenerate.

Of course, in terms of the geometric fibre M_o of $\underline{M}$ at a given geometric point of S, the preceding assertion says that the corresponding form

$$M_o \otimes M_o \longrightarrow \mathbb{Z}$$

is symmetric, positive, and nondegenerate. The pairing (10.4.1) will again be called the monodromy pairing.

10.5. Reduction of 10.4 to a particular case

10.5.1. The uniqueness of a u satisfying the stated condition (even for a single ℓ) is clear. On the other hand, existence simply means that every u_ℓ in fact maps $\underline{M} \otimes \underline{M}'$ into $\mathbb{Z}_S$, and that the homomorphism thereby obtained

$$\underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S$$

is independent of ℓ. In fact it even suffices to prove the analogous assertion for the $u_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and for a homomorphism

$$u_{\mathbb{Q}} : \underline{M}_{\mathbb{Q}} \otimes \underline{M}'_{\mathbb{Q}} \longrightarrow \mathbb{Q}_S.$$

Suppose, then, that existence has been established after a base change $S' \longrightarrow S$ as in 9.3. It follows immediately from (10.3.5) that there exists a homomorphism

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \frac{1}{e(S'/S)} \mathbb{Z}_S \subset \mathbb{Q}_S$$

compatible with the $u_\ell \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$; this proves a). As for b), it is then plain that, if it holds for $A_{K'}$, it also holds for A_K.

Consequently, in proving 10.4 we may make any dominant change of traits $S' \longrightarrow S$. In particular, this allows us to suppose that S is complete with algebraically closed residue field. In that case $\underline{M}$ and $\underline{M}'$ are constant group schemes, with values M and M', finitely generated free $\mathbb{Z}$-modules, and a) and b) may be reformulated more conveniently in terms of the latter.

10.5.2. Suppose now that A_K is isomorphic to a direct factor of an abelian scheme $\overline{A}_K$ over K with semistable reduction, and that 10.4 has been proved for the latter. It follows at once that 10.4 also holds for A_K, in view of the evident functorial behaviour of the lattices $\underline{M}, \underline{M}'$ and of the monodromy pairings with respect to direct sums of abelian schemes with semistable reduction. [46]

It is equally immediate from 10.5.1 that the validity of 10.4 depends only on the isogeny class of A_K. Together with the preceding remark and the "Poincaré reducibility theorem" [15], this shows that, if A_K is a quotient of an abelian scheme $\overline{A}_K$ for which 10.4 holds, then 10.4 also holds for A_K. [47]

10.5.3. We now use the well-known fact [31, Th. 11 (p. 20)] that, after making a finite extension of K if necessary (which is permissible by 10.5.1), every abelian scheme A_K is a quotient of the Jacobian

$$\underline{\mathrm{Pic}}^0_{X_K/K}$$

of a proper, smooth, geometrically connected curve X_K over K. Thus, by 10.5.2, we are reduced to proving 10.4 for such a Jacobian. [48]

10.5.4. By a theorem of Artin–Mumford [3] (already cited and used in Exposé V), after making a further finite extension K' of K if necessary, we may suppose that X_K extends to a projective and flat scheme over S which is regular and whose geometric special fibre $X_{\bar{s}}$ has no singularities other than ordinary quadratic singularities (that is, here, normal crossings of two branches). [49] In this case we shall prove 10.4 a) in 12.5 below, by describing the monodromy pairing (10.4.1) completely in terms of the geometric data of the situation; at the same time we shall verify 10.4 b) for the canonical polarization of the Jacobian.

10.5.5. It remains to see that, if for a given A_K with semistable reduction assertion 10.4 a) holds and assertion 10.4 b) holds for one given polarization ξ_o of A_K, then the same conclusion holds for every polarization ξ of A_K. Indeed, let

$$\widetilde{\xi}_o, \widetilde{\xi} : A_K \longrightarrow A'_K$$

be the homomorphisms defined by ξ_o and ξ; these are isogenies. Then

$$\widetilde{\xi} = \widetilde{\xi}_o \alpha = {}^t(\alpha^*)\widetilde{\xi}_o, \qquad \alpha \in \mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q},$$

where $\alpha \longmapsto \alpha^*$ denotes the canonical involution of the algebra $\mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q}$ defined by the polarization ξ_o, and the superscript t in ${}^t(\alpha^*)$ denotes transpose (associating with an endomorphism of A_K an endomorphism of A'_K). On the other hand, it is known [15, Chap. V, Th. 3] that

$$\alpha = \sum_i \beta_i^* \beta_i, \qquad \beta_i \in \mathrm{End}(A_K) \otimes_{\mathbb{Z}} \mathbb{Q},$$

and hence

$$\widetilde{\xi} = \sum_i {}^t\beta_i\, \widetilde{\xi}_o\, \beta_i.$$

46. The printed source says that $\overline{A}_K$ is "over k"; the notation and the direct-factor argument require "over K."

47. The printed source leaves the pinpoint following citation [15] blank.

48. The printed source says that the curve X_K is "over X"; the displayed relative Picard scheme and the argument require "over K."

49. The printed source leaves the internal reference following "Exposé V" blank.

Equivalently, in terms of the biextensions (that is, divisorial correspondences) associated with the polarizations under consideration (VIII 4.14),

(10.5.5.1) $$\delta(\xi) = \sum_i \delta(\xi_o) \circ (\beta_i, \beta_i).$$

By what was said after (10.2.7), this formula implies (with the notation of (10.4.2))

(10.5.5.2) $$\varphi_\xi^t = \sum_i \varphi_{\xi_o}^t(\beta_i, \beta_i) \in \mathrm{Hom}\big(M_\mathbb{Q} \otimes_\mathbb{Q} M_\mathbb{Q}, \mathbb{Q}\big).$$

It indeed follows that, if $\varphi_{\xi_o}^t$ is symmetric and positive, then so is φ_ξ^t.

Finally, the nondegeneracy of $\varphi_{\xi_o}^t$ is plainly equivalent to the assertion that the pairing (10.4.1) is separating (since $\xi_{ot} : M \otimes_\mathbb{Z} \mathbb{Q} \longrightarrow M' \otimes_\mathbb{Z} \mathbb{Q}$ is an isomorphism, $\widetilde{\xi}_o$ being an isogeny), and is therefore likewise independent of the chosen polarization. One may also observe that, in order to verify that (10.4.1) is separating, it suffices to verify the same assertion for the monodromy pairing u_ℓ of (9.1.2) for one prime number ℓ. Taking $\ell \neq p$, one may use the description of u_ℓ in 9.2; the fact that u_ℓ is separating on the left (and hence on both sides, for reasons of rank) is then immediate.

11. Connected components of the Néron model: duality relation and asymptotic behaviour

11.0. The purpose of the present section is to study the finite étale k-group

(11.0.1) $$\Phi_o = A_o / A_o^{;o}$$

already considered in section 1. For every prime number ℓ, denote by

$$\Phi_o(\ell)$$

its ℓ-primary component. Thus knowledge of Φ_o is equivalent to knowledge of the $\Phi_o(\ell)$, by the formula

(11.0.2) $$\Phi_o = \coprod_\ell \Phi_o(\ell).$$

Notice that the groups Φ_o and $\Phi_o(\ell)$ are determined, respectively, by the ordinary finite groups $\Phi_o(\bar{k})$ and $\Phi_o(\ell)(\bar{k})$ together with the natural action of

$$\pi_o = \mathrm{Gal}(\bar{k}/k)$$

on them. On the other hand, it is clear that $\Phi_o(\bar{k})$ is also the group of connected components of the Néron model of $A_{\widetilde{K}}$, where $\widetilde{K}$ is the fraction field of the strict henselization of S relative to the chosen separable closure $\bar{k}$ of k. Thus, in studying Φ_o, we may reduce to the case where S is strictly local; the actions of

$$\pi_o \simeq \mathrm{Gal}(\widetilde{K}/K)$$

on $\Phi_o(\bar{k})$ are then recovered automatically by transport of structure. We could likewise reduce further to the case where S is complete, using the base change $\widehat{S} \longrightarrow S$ and the fact that formation of the Néron model commutes with it [21, Th. 3.4 b)].

11.1. Suppose, then, that S is strictly local, so that Φ_o may be identified with the ordinary finite group $\Phi_o(k)$. We obtain

$$\Phi_o(k) \simeq A_o(k)/A_o^o(k) \simeq A(S)/A^o(S) \simeq A(K)/A(K)^o, \tag{11.1.1}$$

where the first equality follows from the fact that A_o is smooth over k, the second from Hensel's lemma since A is smooth over S and S is henselian, and the third from the relation $A(S) \xrightarrow{\sim} A(K)$ and from the convention

$$A(K)^o := A^o(S). \tag{11.1.2}$$

Thus $A(K)^o$ appears as a natural subgroup of finite index in $A(K)$ (and is not to be confused with $A^o(K) = A(K)$). The isomorphisms (11.1.1) are plainly functorial in A and compatible with transport of structure in (S, A_K), and in particular with the action of any Galois groups.

Now let ℓ be a prime number different from p. Then A^o is ℓ-divisible; more precisely, $\ell\,\mathrm{id}_{A^o}$ is an étale surjective morphism. It consequently induces a surjective endomorphism of $A^o(S) = A(K)^o$, which is therefore an <u>ℓ-divisible group</u>. It follows at once that, for every integer $n \geq 0$ prime to $p = \mathrm{char}(k)$,

$$_n\Phi_o(k) \simeq {}_nA(K)/{}_nA(K)^o, \qquad (n,p) = 1. \tag{11.1.3}$$

By the definition (2.2.3) of the fixed part of $_nA_K$, we have

$$_nA(K)^o = ({}_nA_K)^f(\bar{K}) \simeq T_n(A_K)^f(\bar{K}) \otimes_{\mathbb{Z}_n} \mathbb{Z}/n\mathbb{Z}. \tag{11.1.4}$$

Taking n to be a power of ℓ, and putting as in (2.5.3)

$$U = T_\ell\big(A(\bar{K})\big), \qquad T_\ell\big(A(\bar{K})\big)^f = U^I, \tag{11.1.5}$$

where U^I denotes the submodule of U consisting of invariants under the inertia group I, we may write (11.1.3) in the form

$$_n\Phi_o(k) \simeq (U_n)^I/(U^I)_n, \qquad n = \ell^{r+1}, \tag{11.1.6}$$

where the subscript n denotes reduction modulo n, that is, tensor product over $\mathbb{Z}_\ell$ with $\mathbb{Z}_\ell/n\mathbb{Z}_\ell$. The transition morphisms between the groups occurring on the two sides of (11.1.6) being evident, passage to the inductive limit over the powers of ℓ gives the following result.

<u>Proposition 11.2.</u> <u>Suppose that S is strictly local, and let ℓ be a prime number different from p. Then there is a canonical isomorphism</u>

$$\Phi_o(\ell) \simeq \frac{(U \otimes D_\ell)^I}{U^I \otimes D_\ell}, \tag{11.2.1}$$

<u>where U is defined in (11.1.5) and $D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell$.</u>

Thus the group $\Phi_o(\ell)$ can be described solely in terms of the Tate module $U = T_\ell(A(\bar{K}))$, regarded as a Galois module under the action of the inertia group I.

11.3. We shall use the explicit expression (11.2.1) to sketch the definition of a canonical pairing

$$\Phi_o(\ell) \otimes \Phi_o'(\ell) \longrightarrow \mathbb{Q}/\mathbb{Z} \tag{11.3.1}$$

which is a perfect duality; here Φ'_o denotes, as in section 1, the group of connected components associated with the Néron model of the dual A'_K of A_K. There is little doubt (although the author has dispensed with the verification) that this pairing is none other than, possibly up to sign, the one defined in (1.2.1), and the reader is invited, as an exercise, to establish the required compatibility (with the correct sign!).

Put

$$U' = T_\ell\big(A'(\bar K)\big), \qquad T = T_\ell(1)(\bar K) = T_\ell(\bar K^*).$$

There is a perfect pairing of finitely generated free $\mathbb{Z}_\ell$-modules, compatible with the actions of I,

$$\varphi : U \otimes U' \longrightarrow T, \qquad (T \simeq \mathbb{Z}_\ell \text{ noncanonically}), \tag{11.3.2}$$

and, by the known structure of I, an exact sequence [50]

$$1 \longrightarrow R \longrightarrow I \longrightarrow T \longrightarrow 1, \tag{11.3.3}$$

where R is a profinite group prime to ℓ. Starting now from the data (11.3.2) and (11.3.3), introduce the groups

$$\Phi = \frac{(U \otimes D_\ell)^I}{U^I \otimes D_\ell}, \qquad \Phi' = \frac{(U' \otimes D_\ell)^I}{U'^I \otimes D_\ell}, \qquad D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell. \tag{11.3.4}$$

We shall define a perfect duality

$$\Phi \otimes \Phi' \longrightarrow \mathbb{Q}/\mathbb{Z}, \tag{11.3.5}$$

which will in particular give the announced pairing (11.3.1). Consider the exact sequence of I-modules

$$0 \longrightarrow U \xrightarrow{i} U \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \xrightarrow{j} U \otimes D_\ell \longrightarrow 0 \tag{11.3.6}$$

and the associated exact cohomology sequence, where the cohomology of I is defined by continuous cochains with coefficients in the topological groups under consideration:

(11.3.7)

$$\begin{array}{ccccccccc}
0 \to H^o(I,U) & \xrightarrow{i^o} & H^o(I, U\otimes \mathbb{Q}_\ell) & \xrightarrow{j^o} & H^o(I, U \otimes D_\ell) & \xrightarrow{\partial} & H^1(I,U) & \xrightarrow{i^1} & H^1(I, U\otimes\mathbb{Q}_\ell) \\
\Big\| \wr & & \Big\| \wr & & \Big\| \wr & & & & \Big\| \wr \\
U^I & & (U \otimes \mathbb{Q}_\ell)^I \simeq U^I \otimes_{\mathbb{Z}} \mathbb{Q} & & (U\otimes D_\ell)^I & & & & H^1(I,U)\otimes_{\mathbb{Z}} \mathbb{Q}.
\end{array}$$

It is immediate that the group Φ defined by (11.3.4) is precisely $\operatorname{coker} j^o$, and is therefore canonically isomorphic to $\ker i^1$. Thus we obtain a canonical exact sequence

$$0 \longrightarrow \Phi \longrightarrow H^1(I,U) \longrightarrow H^1(I,U) \otimes_{\mathbb{Z}} \mathbb{Q}, \tag{11.3.8}$$

which shows that Φ is canonically isomorphic to the torsion subgroup of $H^1(I,U)$.

50. The printed source leaves the internal reference following the first occurrence of I blank.

On the other hand, from the exact sequence (11.3.3) we deduce, for every $n = \ell^r$, an isomorphism

$$H^1(I, T_n) \simeq \mathrm{Hom}(I, T_n) \xleftarrow{\sim} \mathrm{Hom}(T, T_n) \xleftarrow{\sim} \mathbb{Z}/n\mathbb{Z}. \tag{11.3.9}$$

(Notice that I acts trivially on T, and hence on $T_n = T/nT$.) It follows that, for every finite I-$\mathbb{Z}_\ell$-module M, if

$$M' = \mathrm{Hom}(M, T \otimes D_\ell) \tag{11.3.10}$$

denotes its "Cartier dual," there is a canonical pairing

$$H^i(I, M) \otimes H^{1-i}(I, M') \longrightarrow H^1(I, T \otimes D_\ell) \simeq D_\ell, \tag{11.3.11}$$

where the last isomorphism is deduced from the isomorphisms (11.3.9) by passage to the inductive limit over the powers n of ℓ. It is well known that these pairings are <u>perfect dualities</u> (see, for example, SGA 5 I 5). Taking M in these pairings to be of the form U_n, so that $M' = U'_n$, and passing to the limit, we obtain a perfect pairing

$$H^o(I, U \otimes D_\ell) \times H^1(I, U') \longrightarrow D_\ell, \tag{11.3.12}$$

where the second factor on the left is a finitely generated $\mathbb{Z}_\ell$-module. A further passage to the limit in the pairings (11.3.12), for transition morphisms of the form ℓ^r id, gives a second perfect pairing

$$H^o(I, U \otimes \mathbb{Q}_\ell) \times H^1(I, U' \otimes \mathbb{Q}_\ell) \longrightarrow D_\ell \tag{11.3.13}$$

of finite-dimensional vector spaces over $\mathbb{Q}_\ell$. Let i' and j' denote the morphisms defined as in (11.3.6), with U replaced by U'. It is then easy to see that the following homomorphisms

$$H^o(I, U \otimes \mathbb{Q}_\ell) \xrightarrow{\quad j^o \quad} H^o(I, U \otimes D_\ell)$$

$$H^1(I, U' \otimes \mathbb{Q}_\ell) \xleftarrow{\quad i'^1 \quad} H^1(I, U')$$

are <u>transposes</u> of one another under the pairings (11.3.12) and (11.3.13). It follows immediately that $\operatorname{coker} j^o$ and $\ker i'^1$ are dual to one another by a canonical pairing with values in D_ℓ; this is the announced pairing (11.3.5).

11.4. The preceding description of a pairing (11.3.1), by means of the expression (11.2.1) for the ℓ-primary component of the group of connected components, is valid only when $\ell \neq p$. Suppose now that A_K has <u>semistable reduction</u> over S. We shall again define pairings that are perfect dualities

$$\Phi_o(\ell) \otimes \Phi'_o(\ell) \longrightarrow \mathbb{Q}/\mathbb{Z} \tag{11.4.1}$$

without any restriction on ℓ. As in 11.3,[51] one should establish that this recovers (up to a sign, perhaps, which ought to be specified) the pairing defined in (1.2.1). This would therefore establish Conjecture 1.3 in the case of semistable reduction, just as it does when one restricts to the ℓ-primary components for $\ell \neq p$.

To define (11.4.1), we shall give, essentially, an expression for $\Phi_o(\ell)$ in terms of the Barsotti–Tate group $T_\ell(A_K)$ over K; in the present case this will play the same role as 11.2 did in the case treated above.

Theorem 11.5. Suppose that S is henselian and that A_K has semistable reduction over S. Let ℓ be a prime and consider the monodromy pairing (9.1.2), and hence the canonical homomorphism

$$\widetilde{u}_\ell : \underline{M}_\ell \longrightarrow \underline{\check{M}}'_\ell.$$

Put $\Phi(\ell) = \operatorname{coker} \widetilde{u}_\ell$. Then there is a canonical isomorphism

$$\Phi_o(\ell) \xrightarrow{\sim} \Phi(\ell) \times_S s. \tag{11.5.1}$$

This statement immediately gives the definition of a perfect pairing (11.4.1). Indeed, applying 11.5 to A'_K in place of A_K and using (10.2.2), we also obtain a canonical isomorphism

$$\Phi'_o(\ell) \xrightarrow{\sim} \Phi'(\ell) \times_S s, \qquad \Phi'(\ell) = \operatorname{coker}\big({}^t\widetilde{u}_\ell : \underline{M}'_\ell \longrightarrow \underline{\check{M}}_\ell\big).$$

Remarks 11.5.2. a) Theorem 11.5 implies in particular that $\widetilde{u}_\ell$ is an isogeny, that is, that u_ℓ is separating, including in the case $\ell = p$, where this assertion is not immediate from the definition in 9.6. Notice that we do not use here the (considerably deeper) Theorem 10.4, which will be proved in the next section.

b) Using 10.4, which supplies us with a monodromy pairing

$$u : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_S,$$

and denoting by Φ the cokernel of the corresponding homomorphism

$$\widetilde{u} : \underline{M} \longrightarrow \underline{\check{M}}',$$

we obtain an equivalent formulation of 11.5, independent of the choice of ℓ, in the canonical isomorphism

$$\Phi_o \xrightarrow{\sim} \Phi \times_S s. \tag{11.5.3}$$

11.6. To define the isomorphism (11.5.1), we may restrict ourselves to the case in which S is strictly local (11.0), so that $\underline{M}$ and $\underline{M}'$ are constant, with values

$$M = D(T'_o), \qquad M' = D(T_o), \tag{11.6.1}$$

where T_o and T'_o are respectively the maximal tori of A_o and A'_o. Likewise identifying Φ_o with an ordinary finite group, we must define a canonical isomorphism between its component $\Phi_o(\ell)$ and

$$\Phi(\ell) = \operatorname{coker}\big(\widetilde{u}_\ell : M_\ell \longrightarrow \check{M}'_\ell\big).$$

In fact, we shall directly define a canonical isomorphism

51. The printed source reads “as in 11.4” here, within section 11.4 itself. The reference is corrected to the immediately preceding construction in 11.3.

$$\Phi_o(\ell) \xrightarrow{\sim} \ker\bigl(\widetilde{u}_\ell \otimes D_\ell : M_\ell \otimes D_\ell \longrightarrow \check{M}'_\ell \otimes D_\ell\bigr). \tag{11.6.2}$$

Since $\Phi_o(\ell)$ is finite and M_ℓ and $\check{M}'_\ell$ have the same rank, it will first follow that $\widetilde{u}_\ell : M_\ell \to \check{M}'_\ell$ is an isogeny. It then follows that the right-hand side of (11.6.2) is canonically isomorphic to $\Phi(\ell)$:

$$\Phi(\ell) = \operatorname{coker} \widetilde{u}_\ell \xrightarrow{\sim} \ker(\widetilde{u}_\ell \otimes D_\ell). \tag{11.6.3}$$

Indeed, identifying M_ℓ by means of $\widetilde{u}_\ell$ with a finite-index submodule of $\check{M}'_\ell$, we may write the source and target of $\widetilde{u}_\ell \otimes D_\ell$ as[52]

$$M_\ell \otimes D_\ell = V/M_\ell, \qquad \check{M}'_\ell \otimes D_\ell = V/\check{M}'_\ell,$$

where

$$V = M_\ell \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \check{M}'_\ell \otimes \mathbb{Q}_\ell.$$

Composing (11.6.2) with (11.6.3) now immediately gives the isomorphism (11.5.1) to be defined, so everything reduces to defining the canonical isomorphism (11.6.2).

Writing $M_\ell \otimes D_\ell$ (respectively, $\check{M}'_\ell \otimes D_\ell$) as the inductive limit of the M_n (respectively, the $\check{M}'_n$), where n ranges over the powers of ℓ, it is equivalent to define canonical isomorphisms

$${}_n\Phi_o \xrightarrow{\sim} \ker\bigl(\widetilde{u}_n : M_n \longrightarrow \check{M}'_n\bigr), \qquad n = \ell^{r+1}, \tag{11.6.4}$$

compatible with the evident transition morphisms as n varies. Here, by an abuse of notation, $\widetilde{u}_n$ now denotes the homomorphism obtained by reducing $\widetilde{u}_\ell : M_\ell \to \check{M}'_\ell$ modulo n.

For this purpose, consider the group scheme ${}_nA$ over S. By the semistable-reduction hypothesis it is flat and quasi-finite over S (2.2.1). There is a canonical filtration

$${}_nA \supset ({}_nA^o)^f \supset ({}_nA^o)^t \supset 0, \tag{11.6.5}$$

where the superscript f (respectively, t) denotes the "fixed part" (2.2.3) (respectively, the "toric part" (5.1)) of the group scheme to which it is applied. In the definition of the toric part in 5.1, we had assumed S complete; this is permissible here (11.0). (We saw moreover in 6.1 that the construction generalizes to the henselian case, if one insists.) Put

$${}_n\Psi = {}_nA/({}_nA^o)^f. \tag{11.6.6}$$

This is a flat, separated, quasi-finite group scheme over S (SGA 3 VI_A 3.2 and VI_B 9.2), with generic fibre

$${}_n\Psi_K = ({}_nA_K)/({}_nA_K)^f,$$

where, by definition, $({}_nA_K)^f = ({}_nA^o)^f_K$ denotes the r-th constituent ($n = \ell^{r+1}$) of the Barsotti–Tate pro-group $T_\ell(A_K)^f$. The first of relations (5.5.8) therefore gives a canonical isomorphism

$${}_n\Psi_K \xrightarrow{\sim} \underline{M}_{nK}, \qquad \underline{M}_n := \underline{M} \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}). \tag{11.6.7}$$

52. At this point the printed source labels V/M_ℓ as $\Phi(\ell)$ and $V/\check{M}'_\ell$ as $\Phi'(\ell)$. Those quotients are instead the source and target of the map whose kernel is the finite cokernel $\check{M}'_\ell/M_\ell$; the contradictory labels are not reproduced.

In fact, we have:

Lemma 11.6.8. The group scheme ${}_n\Psi$ over S defined in (11.6.6) is *étale* over S, and the isomorphism (11.6.7) extends uniquely to a morphism

$$(11.6.8.1) \qquad {}_n\Psi \hookrightarrow \underline{M}_n$$

which is an open immersion.

To see that ${}_n\Psi$ is étale, in view of (11.6.7) it remains only to verify this at the points of its special fibre, or equivalently to verify that this fibre is étale. This is immediate, since the fibre in question is

$${}_nA_o/{}_nA_o^o \subset A_o/A_o^o = \Phi_o.$$

It follows that ${}_n\Psi$ is normal; and since $\underline{M}_n$ is finite over S, the existence of the extension (11.6.8.1) follows at once. This is a morphism of étale group schemes over S, so its kernel is étale over S. Since ${}_n\Psi$, and hence this kernel, is separated, while its generic fibre is the unit group by (11.6.7), the kernel is the unit group (3.1.3). Thus (11.6.8.1) is an étale monomorphism, hence an open immersion (EGA IV 17.9.1), proving the lemma.

Now put

$$(11.6.9) \qquad {}_n\Phi = ({}_nA)^f/({}_nA^o)^f \subset {}_n\Psi.$$

Thus ${}_n\Phi$ is a finite open subscheme of ${}_n\Psi$, since $({}_nA)^f$ is a finite open subgroup scheme of ${}_nA$. More precisely, it is immediate that

$$(11.6.10) \qquad {}_n\Phi \xrightarrow{\sim} ({}_n\Psi)^f.$$

Composing (11.6.9) with (11.6.8.1), we obtain an open immersion

$$(11.6.11) \qquad {}_n\Phi \hookrightarrow \underline{M}_n.$$

On the other hand,

$${}_n\Phi \times_S s = {}_n\Psi \times_S s = {}_nA_o/{}_nA_o^o,$$

and, since A_o^o is n-divisible (2.2.1), the last member is immediately identified with ${}_n\Phi_o$. We therefore obtain an isomorphism

$$(11.6.12) \qquad {}_n\Phi \times_S s \xrightarrow{\sim} {}_n\Phi_o,$$

which shows that ${}_n\Phi$ is characterized, up to unique isomorphism, as the finite étale scheme over S whose special fibre is ${}_n\Phi_o$. Thus, to define the desired isomorphism (11.6.4) by means of the open immersion (11.6.11), it is enough to prove the following.

Lemma 11.6.13. The image of the canonical homomorphism (11.6.11) is equal to the kernel of

$$\widetilde{u}_n : \underline{M}_n \longrightarrow \underline{\check{M}}'_n.$$

The required compatibility of the homomorphisms (11.6.11) with the transition homomorphisms as n varies is immediate from the definitions. Consequently Lemma 11.6.13 will complete the proof of Theorem 11.5.

To prove the lemma, note that

$$K_n = \ker \widetilde{u}_n$$

is also the left annihilator of the canonical pairing

$$u_n : \underline{M}_n \times \underline{M}'_n \longrightarrow (\mathbb{Z}/n\mathbb{Z})_S,$$

whose explicit definition is given in 9.4 and 9.6 in terms of the extensions

(11.6.14)

$$\begin{tikzcd} 0 \arrow[r] & \check{\underline{M}}'_n(1) \arrow[r] & F_n \arrow[r] \arrow[d, equal] & R_n \arrow[r] & 0 \\ & & ({}_nA^o)^f & & \end{tikzcd} \quad , \quad \begin{tikzcd} 0 \arrow[r] & \check{\underline{M}}_n(1) \arrow[r] & F'_n \arrow[r] \arrow[d, equal] & R'_n \arrow[r] & 0 \\ & & ({}_nA'^o)^f & & \end{tikzcd}$$

(where R_n and R'_n are Cartier dual to one another), and in terms of the mixed extension $X_{n\eta}$ of $E_{n\eta}$ by $F_{n\eta}$, which is the generic fibre of (11.6.5),[53] where we put

(11.6.15)

$$E_n = D(F_n), \qquad \begin{tikzcd} 0 \arrow[r] & R_n \arrow[r] \arrow[d, dash, "\wr"'] & E_n \arrow[r] & \underline{M}_n \arrow[r] & 0 \\ & D(R'_n) & & & \end{tikzcd} \quad .$$

It follows immediately from Proposition 9.4.7[54] that K_n is the largest among the finite étale subgroups K'_n of $\underline{M}_n$ for which the inverse image $\overline{X}_{n\eta}$ of K'_n in the mixed extension $X_{n\eta}$, regarded as a mixed extension of $\overline{E}_{n\eta}$ by $F_{n\eta}$, extends to a mixed extension of $\overline{E}_n$ by F_n. Here $\overline{E}_n$ is the evident subgroup extension of K'_n by R_n.[55]

This condition is plainly satisfied for the subgroup ${}_n\Phi$ of $\underline{M}_n$ in (11.6.11), since the corresponding mixed extension $({}_nA)^f$ over S is available. Hence ${}_n\Phi \subset K_n$. To prove the opposite inclusion, consider the mixed extension $\overline{X}_n$ of $\overline{E}_n$ by F_n extending $\overline{X}_{n\eta}$, defined as above by taking $K'_n = K_n$. Thus it is an extension

$$0 \longrightarrow F_n \longrightarrow \overline{X}_n \longrightarrow K_n \longrightarrow 0,$$

and there is a canonical homomorphism

$$(*) \qquad F_n = ({}_nA^o)^f \hookrightarrow A,$$

which on the generic fibre is induced by the canonical homomorphism

$$(**) \qquad \overline{X}_{n\eta} \hookrightarrow X_{n\eta} = {}_nA_K \hookrightarrow A_K.$$

It follows immediately from (5.9.2) that the homomorphism $(**)$ extends to a unique homomorphism

53. The printed source refers here to nonexistent formula (11.7.5); the filtration whose generic fibre is being used is (11.6.5).

54. The printed source cites 9.4.6, but the required extension criterion is Proposition 9.4.7.

55. The printed source says "of K_n" in this general criterion, although the subgroup being varied at this point is K'_n; the specialization $K'_n = K_n$ is made only below.

(***) $$\overline{X}_n \longrightarrow A$$

inducing $(*)$. Since this last homomorphism maps the generic fibre into ${}_nA$, which is a closed subscheme of A, it factors through $\overline{X}_n \to {}_nA$; and since $\overline{X}_n$ is finite, it factors through $({}_nA)^f$. Looking at the induced homomorphism on generic fibres, we conclude that

$$\overline{X}_{n\eta} \subset ({}_nA)^f_\eta, \quad \text{i.e.} \quad K_{n\eta} \subset {}_n\Phi_\eta, \quad \text{i.e.} \quad K_n \subset {}_n\Phi.$$

This proves the lemma.

Remarks 11.7. The construction made in 11.6 [56] shows that, from the extensions (11.6.14) over S and the mixed extension $X_{n\eta} = {}_nA_K$ of E_n by F_n, one can reconstruct, up to unique isomorphism, the mixed extension (11.6.5), and hence also the subgroup

$$({}_nA)^f \supset ({}_nA^o)^f$$

of ${}_nA$. Indeed, by 9.4 these data determine the pairing

$$u_n : \underline{M}_n \times \underline{M}'_n \longrightarrow (\mathbb{Z}/n\mathbb{Z})_S,$$

and the preceding construction gives $({}_nA)^f$ as the mixed extension $\overline{X}_n$ of $\overline{E}_n$ by F_n, extending the mixed extension $\overline{X}_{n\eta}$ induced by $X_{n\eta}$. Here $\overline{E}_n$ is the inverse image of $\ker \widetilde{u}_n$ under $E_n \to \underline{M}_n$. By (9.4.8), this mixed extension $\overline{X}_n$ is thus determined up to unique isomorphism. We then recover ${}_nA$ by gluing $\overline{X}_n$ and $X_{n\eta}$ along their common open subscheme $\overline{X}_{n\eta}$.

Thus, if the structural data (9.6.1) and the mixed extension

$$X_\eta = T_\ell(A_K)$$

of E_η by F_η are known—equivalently, if the extensions (11.6.14) and $X_{n\eta}$ are known for every $n = \ell^{r+1}$, $r \geq 0$—then one can reconstruct the inductive (or, if preferred, projective) system of mixed extensions (11.6.5).

11.8. As in 8.3, regard the separable closure $\overline{K}$ of K as the inductive limit of its finite subextensions K'. For every such K', let $A[K']$ denote the Néron model of

$$A_{K'} = A_K \otimes_K K'.$$

It gives a finite étale group scheme

(11.8.1) $$\Phi_o[K'] = A[K']_o / A[K']^o_o$$

of connected components, defined over the residue field of the normalization $S[K']$ of S in K'. For simplicity, suppose that the residue field of S is separably closed, so that the same is true of the residue fields of the $S[K']$. The groups (11.8.1) may then be identified with ordinary finite groups $\Phi[K']$, and it is immediate that, as K' varies, these form an inductive system, which we now propose to study. We shall be interested in particular in the group

(11.8.2) $$\Psi = \varinjlim_{K'} \Phi[K'].$$

We know from 3.6 that, for K' sufficiently large, say $K' \supset K'_o$, $A[K']$ has semistable reduction. Hence for $K' \supset K'_o$ the transition morphisms are injective (3.3). Moreover, with the notation of 5.5, retained in the present section, there is for $K' \supset K'_o$ a canonical isomorphism

56. The printed source says "the construction made in 11.7" at the opening of 11.7; the reference is to the construction just completed in 11.6.

$$(11.8.3) \qquad \Phi[K'](\ell) \simeq \ker\bigl(\widetilde{u}[K']_\ell \otimes D_\ell : M_\ell \otimes D_\ell \longrightarrow \check{M}'_\ell \otimes D_\ell\bigr),$$

coming from (11.6.2) applied to $A[K']$. One also verifies immediately that the corresponding immersions

$$\Phi[K'](\ell) \hookrightarrow M_\ell \otimes D_\ell$$

are compatible with the transition morphisms as K' varies. Passing to the limit therefore gives a canonical monomorphism

$$(11.8.4) \qquad \Psi(\ell) \overset{\text{def}}{=} \varinjlim_{K'} \Phi[K'](\ell) \hookrightarrow M_\ell \otimes D_\ell$$

from the ℓ-primary component of Ψ; summing over all ℓ gives

$$(11.8.5) \qquad \Psi \overset{\text{def}}{=} \varinjlim_{K'} \Phi[K'] \longrightarrow M \otimes \mathbb{Q}/\mathbb{Z}.$$

Theorem 11.9. The canonical homomorphism (11.8.5) is an isomorphism.

We already know that it is a monomorphism. To prove that it is an isomorphism, we must use the characterization (11.8.3) of the image of $\Phi[K'](\ell)$ and the behaviour of $\widetilde{u}[K']$ as K' varies. That behaviour is given by (10.3.4).

We may reduce to the case in which S is strictly local. To prove 11.9 we are then reduced to the following familiar assertion, applied to $S[K'_o]$: if S is a strictly local trait with fraction field K, then for every integer $n \geq 1$ there is a separable extension K' of K whose degree is divisible by n.[57]

Corollary 11.10. Under the hypotheses (S complete) and with the notation of 8.3, the ind-finite group scheme over K of connected components of the group scheme $A^\flat_K$, namely $A^\flat_K/A^{\flat\,o}_K$, is canonically isomorphic to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$, where $\underline{M}_K$ is the twisted constant group over K defined in (8.1.8).

This follows immediately from (8.2.4) and 11.9, in view of the definition of $\underline{M}_K$.

Remarks 11.11. a) If one restricts to the ℓ-primary components for $\ell \neq p$, Theorem 11.9 is immediate in terms of 11.2 and does not require the more delicate considerations of 11.6 (via 9.4). Likewise, if K is assumed to be of characteristic zero, 11.9 can be proved without using (11.8.3): it is immediate that the image of ${}_n\Phi[K'](\ell)$ in

$$M_n = {}_n(M_\ell \otimes D_\ell)$$

contains the image of the group ${}_nA(K')$ of K'-valued points of ${}_nA$. For K' sufficiently large this group is equal to ${}_nA(\overline{K})$, and therefore maps onto M_n. If $\operatorname{char} K = p$, on the other hand, it appears that there is no "trivial" proof—that is, no proof avoiding 9.4—of 11.10 for the p-primary component.

b) The existence of an isomorphism

$$\Psi \simeq (\mathbb{Q}/\mathbb{Z})^r,$$

together with the correct value of r, was conjectured in 1964 by J.-P. Serre. 12. Néron model of a Jacobian and the Picard–Lefschetz formula

57. The printed source appends "cf. I" followed by a blank internal reference. No target is supplied, so none is invented here.

Let us first recall the following general results of M. Raynaud [22, Ths. 5 and 7; 22 bis].

Theorem 12.1 (M. Raynaud). Let S be a strictly local trait, and let $f : X \to S$ be a scheme proper and flat over S, with one-dimensional fibres, such that

$$f_*\underline{\mathcal{O}}_X \simeq \underline{\mathcal{O}}_S, \tag{12.1.1}$$

and such that the local rings of X are factorial. For every maximal point x_i of X_o, put

$$d_i = \operatorname{length} \underline{\mathcal{O}}_{X_o,x_i}, \qquad d_i^t = \mu_i d_i, \tag{12.1.2}$$

where μ_i is the radicial multiplicity of $k(x_i)$ over k (EGA IV, 4.7.4) (thus a power of the characteristic exponent of k). Let d (respectively d^t) be the gcd of the integers d_i (respectively d_i^t); hence $d = d^t$ if k is perfect. Put

$$P = \underline{\mathrm{Pic}}_{X/S} : (\mathrm{Sch})^{;o}_{/S} \longrightarrow (\mathrm{Ab}). \tag{12.1.3}$$

a) Consider the following conditions:

1) $d^t = 1$.

2) X has a zero-cycle of degree 1.

3) There exists an invertible sheaf L on $X \times P^o$ such that the corresponding homomorphism $P^o_\eta \to \underline{\mathrm{Pic}}_{X_\eta/\eta} = P_\eta$ is the canonical inclusion. (It is known, by J.-P. Murre, that P_η is representable by a group scheme locally of finite type over η; see, for example, SGA 6, XII, 1.5.)

4) X is cohomologically flat over S (EGA III, 7.8.1) (equivalently, cohomologically flat in dimension zero, or equivalently $k \xrightarrow{\sim} H^o(X_o, \underline{\mathcal{O}}_{X_o})$), and $d = 1$.

5) P is representable (equivalently, representable by a group scheme smooth over S).

6) P^o is representable (equivalently, representable by a group scheme smooth over S). (For the definition of P^o, see SGA 3, VI_B, 3.1.)

7) P^o is separated over S, that is, it satisfies the valuative criterion of separatedness of EGA II, 7.2.3 (equivalently, every section s of P^o over S for which $s(\eta) = 0$ is zero).

8) $d = 1$.

Then one has the implications

$$1) \Longleftrightarrow 2) \Longrightarrow 3) \Longrightarrow 4) \Longleftrightarrow 5) \Longleftrightarrow 6) \Longleftrightarrow 7) \Longrightarrow 8). \tag{12.1.4}$$

In particular, if $d = d^t$, for example if k is perfect, the conditions 1) through 8) are equivalent.

b) Suppose that $d^t = 1$, so that P is representable by a group scheme smooth over S. Then the schematic closure of the identity section of P is a group subscheme E which is ètale over S. If I is the set of reduced irreducible components X_o^i of X_o, one obtains an isomorphism

$$\mathbb{Z}^I \simeq \Gamma(S, E) \tag{12.1.5}$$

by associating to every $i \in I$ the section S_i of P over S (zero over η) defined by the invertible sheaf $\underline{\mathcal{O}}_X(X_o^i)$ on X [58]. (Because the local rings of X are factorial by hypothesis, the cycles X_o^i are indeed divisors; thus $\underline{\mathcal{O}}_X(X_o^i)$ has its stated meaning as the invertible sheaf associated with that divisor; see EGA IV, 21.2.8.1.) One has

58. The printed source says "on S" here and later writes X_i^o for the same divisor. The sheaf is the invertible sheaf on X associated with the component X_o^i; the English text keeps that notation consistently.

$$P^o \cap E = \text{the zero section of } P. \tag{12.1.6}$$

c) Under the conditions of b), the quotient

$$Q = P/E \tag{12.1.7}$$

is representable by a smooth separated group scheme over S, and this group scheme is Néron, that is, it satisfies the universal property (1.1.2) in terms of its special fibre.

d) Under the conditions of c), suppose in addition that the geometric generic fibre of X is integral. Then there is a canonical exact sequence

$$0 \longrightarrow A \longrightarrow Q \xrightarrow{\deg} \mathbb{Z}_S \longrightarrow 0, \tag{12.1.8}$$

where deg is induced by the "degree" homomorphism $P \to \mathbb{Z}_S$, and where A is a smooth separated group scheme of finite type over S, which is likewise Néron, that is, a Néron model of

$$A_\eta = P_\eta^o = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}. \tag{12.1.9}$$

Moreover, the composite homomorphism $P^o \to P \to Q$ induces an isomorphism

$$P^o \xrightarrow{\sim} A^o. \tag{12.1.10}$$

12.1.11. The case of interest to us is that in which X_η is smooth over η, so that, in view of (12.1.1), the geometric generic fibre of X is smooth and connected (hence integral), and in which X_η has a zero-cycle of degree one. All the conditions considered in 12.1 (and in particular conditions 1) through 8) of a)) then hold. Moreover, $A_\eta = P_\eta^o$ is then an abelian scheme, and d) gives us a very explicit construction of its Néron model A. In particular, (12.1.10) gives an isomorphism

$$A_o^{;o} \simeq \underline{\mathrm{Pic}}^o_{X_o/k}, \tag{12.1.12}$$

which shows in particular that A has semistable reduction over S if and only if $\underline{\mathrm{Pic}}^o_{X_o/k}$ has unipotent rank zero. When X is regular and X_o is separable over k, this is known to hold if and only if the geometric special fibre $\overline{X}_o$ has no singular points other than ordinary double points (that is, points whose completed local ring is the same as that of the union of the two coordinate axes of affine 2-space at the origin). Verification of this fact, together with the "well-known" description of the Picard scheme of a proper separable curve over a field k, using EGA IV, 21.8.5, is left to the reader as an exercise. Notice also that the implication "$\overline{X}_o$ has ordinary double points" $\Longrightarrow$ "$A_\eta = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}$ has semistable reduction" follows immediately from the "geometric monodromy theorem" III [59], in view of the Galois criterion for semistable reduction in 3.5. This argument, however, does not give the important isomorphism (12.1.10) and its consequence (12.1.12).

12.2. For the proof of 12.1, we refer to [22 bis]; in fact we shall need only the case where X_η is smooth over η. Let us also note that [22 bis] proves relations between Picard schemes and Néron models under conditions substantially more general than those of 12.1 c) and d). As their statements are more technical, we have chosen not to include them in this review. We mention only that b), c), and d) are easy consequences of a), and that in a) the only delicate points are the

59. The printed source leaves the internal locator following "III" blank; no target is invented here.

implications 3) $\Longrightarrow$ 4) $\Longrightarrow$ 5) and 7) $\Longrightarrow$ 4), the first being the most difficult. This point is trivial, however, if one assumes that X_o is separable over k (which implies that X has a section over S, since S is strictly local, and hence that condition 1), $d^t = 1$, holds). We advise the reader to give the proof in this particularly simple case, assuming in addition, if necessary, that X_η is smooth, so that P^o_η is an abelian scheme. One may then use its Néron model A, and A^o represents $\underline{\mathrm{Pic}}^o_{X/S}$. We shall use 12.1 only in this particular case, and under the further assumption that $\overline{X}_o$ has no singularities other than ordinary double points, that is, by (12.1.11), that A_η has semistable reduction. This is the case considered in 12.5.4, to which we restrict ourselves from now on. 12.3.

Let C be a proper separable curve over a field k, which for simplicity we suppose separably closed (or even algebraically closed, if the reader prefers). If $\widetilde{C}$ is the normalization of C, it is known that $\underline{\mathrm{Pic}}^o_{\widetilde{C}/k}$ has "reductive rank" zero (that is, its maximal torus is zero). Consequently the maximal torus of $\underline{\mathrm{Pic}}^o_{C/k}$ is contained in

$$N = \ker\big(\underline{\mathrm{Pic}}_{C/k} \longrightarrow \underline{\mathrm{Pic}}_{\widetilde{C}/k}\big),$$

and is therefore canonically isomorphic to the maximal torus of this last group. The structure of N is easily made explicit by EGA IV, 21.8.5: N is canonically an extension of a torus T, which we shall describe, by a connected unipotent group U (which itself has, over k, a composition series whose quotients are isomorphic to the additive group $\mathbb{G}_a$). Moreover, if T' is the maximal torus of N, then $T' \cap U$ is both unipotent and of multiplicative type, and is therefore zero (SGA 3, XVII, 2.4), while the image of T' in T is all of T (SGA 3, XII, 7.1 e)). Thus $T' \to T$ is an isomorphism. Hence T is canonically isomorphic to the maximal torus of N, and therefore to that of $\underline{\mathrm{Pic}}^o_{C/k}$.[60]

To make the structure of N explicit, it is convenient to suppose that k is separably closed. The irreducible components C_i of C are then geometrically irreducible, and the nonnormal points of C are radicial over k. Let J be the set of irreducible components of $\widetilde{C}$, and for every $x \in C$ let $J(x)$ be the set of "branches" of C through x (that is, the points of $\widetilde{C}$ above x). Finally put

$$R(x) = \mathbb{Z}^{J(x)}/\operatorname{Im}\mathbb{Z}, \tag{12.3.1}$$

where $\mathbb{Z} \to \mathbb{Z}^{J(x)}$ is the diagonal map, and

$$R = \left(\bigoplus_x R(x)\right)/\operatorname{Im}\mathbb{Z}^J. \tag{12.3.2}$$

Clearly it is enough to take the sum over the finite set of points of C which belong to at least two branches of C. Thus R is a finitely generated $\mathbb{Z}$-module. There is then a canonical isomorphism

$$T\ \big(\simeq \text{ the maximal torus of } \underline{\mathrm{Pic}}^o_{C/k}\big) \simeq R \otimes \mathbb{G}_{m,k}. \tag{12.3.3}$$

In other words, the character group M of T is

$$M \overset{\text{def}}{=} D(T) = \check{R} \simeq \ker\left(\bigoplus_x \check{R}(x) \longrightarrow \mathbb{Z}^J\right), \tag{12.3.4}$$

where $\check{R}(x)$ may also be interpreted as the set of integer-valued functions on $J(x)$ whose values have sum zero.[61]

60. The printed source has $\underline{\mathrm{Pic}}^o_{X/k}$ at the end of this argument. The curve throughout 12.3 is C, and the preceding and following occurrences both read $\underline{\mathrm{Pic}}^o_{C/k}$; the English text restores C.

61. The printed source numbers these two displays (12.2.3) and (12.2.4). They belong to section 12.3 and are cited below as (12.3.3) and (12.3.4); those coherent numbers are used here.

12.3.5. We shall need only the case in which at most two branches of C pass through any point x (as happens, in particular, for ordinary double points). If $\operatorname{card} J(x) = 2$, then $R(x)$ is isomorphic to $\mathbb{Z}$, but the isomorphism depends on an ordering of the two branches (C', C'') through x, or equivalently on the choice of one branch C': to C' one associates the element of $R(x)$ which is the canonical image of $0_{C'} + 1_{C''}$. One then obtains an isomorphism

$$(12.3.6) \qquad R = \mathbb{Z}^I / \operatorname{Im} \mathbb{Z}^J, \qquad T \simeq \mathbb{G}_m^I / \operatorname{Im} \mathbb{G}_m^J, \qquad M = \ker\big(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J\big),$$

where I is the set of points $x \in C$ which belong to two branches of C, after a choice of signs as just explained.

Remarks 12.3.7. A sometimes convenient way to describe the character group M of the maximal torus T of $\underline{\mathrm{Pic}}^o_{C/k}$ is the following. Consider the "unoriented one-dimensional graph" $\Gamma(C)$ (N. Bourbaki, *Groupes et algèbres de Lie*, Chapter IV, Appendix), whose vertices are the irreducible components of $\widetilde{C}$. Two vertices $\widetilde{C}_i$ and $\widetilde{C}_j$ are joined by a set of edges in bijective correspondence with the pairs of distinct branches of C, contained respectively in $\widetilde{C}_i$ and $\widetilde{C}_j$, which pass through the same point of C. This set is $\widetilde{C}_i \times_C \widetilde{C}_j$ when $i \neq j$; when $i = j$, it is

$$\big(\widetilde{C}_i \times_C \widetilde{C}_i - \Delta(\widetilde{C}_i)\big)/(\text{symmetry}),$$

where $\Delta : \widetilde{C} \to \widetilde{C} \times_C \widetilde{C}$ is the diagonal. (It is therefore empty if C_i is geometrically unibranch.) One then has canonical isomorphisms

$$(12.3.7) \qquad M = D(T) \xleftarrow{\ \sim\ } H_1(\Gamma(C), \mathbb{Z}), \qquad \check{M} = R \simeq H^1(\Gamma(C), \mathbb{Z}).$$

We shall not need this combinatorial interpretation of M, and shall confine ourselves to defining the canonical homomorphism

$$(12.3.8) \qquad H_1(\Gamma(C), \mathbb{Z}) \longrightarrow M,$$

leaving it to the reader to prove that this is an isomorphism. Let I be the set of points of C which belong to at least two branches of C, let J, as above, be the set of irreducible components of $\widetilde{C}$ (or of C, which amounts to the same thing), and let K be the set of "branches" of C at the points of I, that is, the inverse image of I in $\widetilde{C}$. There is a canonical map

$$(12.3.9) \qquad h : K \longrightarrow I \times J, \qquad h = (f, g),$$

and the description (12.3.4) of M is expressed in terms of h as

$$(12.3.10) \qquad M \simeq \ker\left(\mathbb{Z}^{(h)} : \mathbb{Z}^{(K)} \longrightarrow \mathbb{Z}^{(I \times J)}\right).$$

Indeed, the expression (12.3.3) for $R = \check{M}$ may plainly be interpreted as the cokernel of the homomorphism $\mathbb{Z}^{I \times J} \to \mathbb{Z}^K$ induced by h, which is the transpose of $\mathbb{Z}^{(h)}$. The graph $\Gamma(C)$ is likewise described in terms of h by

$$\Gamma_0 = J, \qquad \Gamma_1 = (K \times_I K - \Delta(K))/(\text{symmetry}).$$

Choose an orientation on each edge of $\Gamma(C)$, and denote by Γ_1 the set of these oriented edges (x, C', C''). Then

(12.3.11)
$$H_1(\Gamma(C), \mathbb{Z}) = \ker\left(d : \mathbb{Z}^{(\Gamma_1)} \longrightarrow \mathbb{Z}^{(J)}\right),$$

where the differential is

(12.3.12)
$$d(x, C', C'') = g(C'') - g(C').$$

The homomorphism (12.3.8) is induced by the following homomorphism v of chain complexes: [62]

(12.3.13)
$$\begin{aligned} v = (v_0, v_1) : C_\bullet(\Gamma(C)) = &\left[\mathbb{Z}^{(\Gamma_1)} \longrightarrow \mathbb{Z}^{(J)}\right] \\ &\longrightarrow \left[\mathbb{Z}^{(K)} \longrightarrow \mathbb{Z}^{(I\times J)}\right], \end{aligned}$$

given by

(12.3.14)
$$v_1(x, C', C'') = (x, C'') - (x, C'), \qquad v_0(\widetilde{C}_j) = (0, \widetilde{C}_j).$$

12.4. We now place ourselves in the case considered at the end of 12.2. By (12.1.12) and (12.3.6), applied with $C = X_o$, the character group M of the maximal torus T_o of the special fibre A_o of the Néron model A of $A = \underline{\mathrm{Pic}}^o_{X_\eta/\eta}$ is given by

(12.4.1)
$$M = \ker\left(\mathbb{Z}^I \xrightarrow{d} \mathbb{Z}^J\right),$$

where I is the set of singular points of X_o, and J is the set of its irreducible components, the homomorphism $\mathbb{Z}^I \to \mathbb{Z}^J$ being given by

(12.4.2)
$$d(x_i) = C'_i - C''_i.$$

Here C'_i and C''_i are the two irreducible components of $\widetilde{C}$ corresponding to the branches through the double point x_i. The operator d depends on the choice of an order (C'_i, C''_i) on these two branches, and the isomorphism (12.4.1) plainly depends on this system of choices. Nevertheless, if, for $i \in I$, we denote by

(12.4.3)
$$\varphi_{x_i} : M \longrightarrow \mathbb{Z}$$

the composite

$$M \longrightarrow \mathbb{Z}^I \xrightarrow{\mathrm{pr}_i} \mathbb{Z},$$

where the first homomorphism comes from (12.4.1), then, on returning to the expression (12.3.4), one sees at once that φ_{x_i} is well-defined up to sign. More precisely, it depends only on the order chosen between the two branches C'_i, C''_i through x_i, and changes sign when that order is reversed. It follows in particular that the bilinear form

62. The printed display defining (12.3.8) carries no number even though the following text cites it, and (12.3.13) writes $\mathbb{Z}^{(\Gamma)}$ where the edge chain group is $\mathbb{Z}^{(\Gamma_1)}$. The English text supplies the cited label and the required subscript. The working transcription's $\widetilde{C}_i \times_C \widetilde{C}_1$ in the loop-edge description is also restored from the scan as $\widetilde{C}_i \times_C \widetilde{C}_i$.

$$\varphi_{x_i} \otimes \varphi_{x_i} \in \check{M} \otimes_{\mathbb{Z}} \check{M} \simeq \mathrm{Hom}(M \otimes M, \mathbb{Z}) \tag{12.4.4}$$

is determined canonically by the singular point x_i, without any sign ambiguity. We may therefore put

$$u = \sum_i \varphi_{x_i} \otimes \varphi_{x_i} : M \otimes_{\mathbb{Z}} M \longrightarrow \mathbb{Z}. \tag{12.4.5}$$

This gives a well-defined pairing, independent of every choice of signs. It is plainly positive definite, since it is the restriction to $M \hookrightarrow \mathbb{Z}^I$ of the standard quadratic form $\sum_i X_i^2$ on $\mathbb{Z}^I$.

The notation M for the character group of T_o is not, a priori, consistent with that of the preceding sections, where this group was written M', while M denoted the analogous group associated with the dual abelian scheme A'_K of A_K. But since A_K is a Jacobian, it has a canonical polarization ξ [5] [63], which induces a canonical isomorphism

$$\psi_\xi : A_K \xrightarrow{\sim} A'_K. \tag{12.4.6}$$

By means of this isomorphism we shall identify A_K and A'_K, so that there is no longer any need to distinguish M from M'. In particular, for every prime number ℓ, the monodromy pairing (9.1.2) may be regarded as a pairing

$$u_\ell : M_\ell \otimes_{\mathbb{Z}_\ell} M_\ell \longrightarrow \mathbb{Z}_\ell. \tag{12.4.7}$$

With this understood, we can state the principal result of the present section:

Theorem 12.5 (Picard–Lefschetz formula). Let S be a strictly local trait, and let X be a regular scheme, projective and flat over S, with one-dimensional fibres, whose geometric generic fibre is smooth and connected and whose geometric special fibre has no singularities other than ordinary quadratic singularities. With the notation introduced in 12.4, for every prime number ℓ, the monodromy pairing (12.4.7) is equal to the pairing obtained from (12.4.5) by extension of scalars from $\mathbb{Z}$ to $\mathbb{Z}_\ell$. 12.6. Notice that this statement proves 10.4 in the particular case under consideration when, in 10.4 b), one takes for ξ the canonical polarization considered there. As we showed in 10.5, this suffices to imply 10.4 in the general case.

12.7. Proof of 12.5. The case $\ell \neq p$. It relies essentially on the cohomological Picard–Lefschetz formula in the situation considered in 12.5, a formula giving the action of the monodromy group I on the following group: [64]

$$H^1(X_{\bar\eta}, \mathbb{Z}_\ell) \simeq H^1(A_{\bar\eta}, \mathbb{Z}_\ell) \simeq \mathrm{Hom}\big(T_\ell(A_{\bar\eta}), \mathbb{Z}_\ell\big) \tag{12.7.1}$$

when $\ell \neq p$. This formula, which will be proved in a later exposé (Exposé XV), and indeed in a substantially more general case, is essentially identical to assertion 12.5 for the prime ℓ in question, in view of the definition in 9.2 of u_ℓ for $\ell \neq p$. It therefore remains to treat the case $\ell = p$. We shall do this by a method which reduces it to characteristic zero (and, moreover, to the case where X_o has only one singular point), which we now explain.

63. The printed citation following the canonical polarization is "[5]" followed by a blank pinpoint. The English text preserves the reference without inventing a missing locator.

64. The working transcription adds a dual superscript to $T_\ell(A_{\bar\eta})$ in (12.7.1). A direct 5,000-dpi inspection of the printed formula shows no such superscript, in agreement with the standard identification $H^1(A_{\bar\eta}, \mathbb{Z}_\ell) = \mathrm{Hom}(T_\ell(A_{\bar\eta}), \mathbb{Z}_\ell)$.

Remarks 12.7.2. Notice also that, once the announced reduction has been made, the usual arguments would allow us to reduce to the case in which the base field is the field $\mathbb{C}$ of complex numbers, and then to invoke the Picard–Lefschetz formula in its classical form—except that there does not seem to be a satisfactory reference for that formula. The proof of the Picard–Lefschetz formula given in our Seminar will proceed similarly, by reduction to the transcendental case, which is amenable to properly transcendental methods. One may therefore say that, in any event, the proof of 12.5 (and hence of 10.4) proposed here is transcendental in nature. Let us also point out that the only other known proof of 10.4 comes from rigid-analytic theory [24], and must for that reason likewise be regarded as essentially "transcendental." It would be interesting to find purely algebraic proofs of these results (although it is by no means evident that this is possible!).

12.8. Proof of 12.5: reduction to the case in which S has characteristic zero. We shall use an argument based on the same principle as the proof of the "theorem on the essentially tame action of monodromy on π_1" in Exposé V.[65] Notice that we may assume S complete, by 10.3. Let W be a Cohen p-ring with residue field k, and choose a homomorphism $W \longrightarrow V$ inducing the identity on residue fields (EGA 0_{IV}, 19.8.6). Let $(x_\lambda)_{\lambda \in L}$ be the set of singular points of X_o, and consider the formal moduli scheme $\underline{S}$ of X_o over W, together with the "universal curve"

$$\underline{f} : \underline{X} \longrightarrow \underline{S}$$

whose special fibre is X_o (Exposé VI).[66] We saw in the cited place that $\underline{S}$ is isomorphic to the formal spectrum of a power-series ring $W[[T_1, \dots, T_N]]$, and that in $\underline{S}$ one can find a normal-crossings divisor

$$D = \sum_{\lambda \in L} D_\lambda, \tag{12.8.1}$$

whose irreducible components D_λ are indexed by L, each D_λ being described by an equation of the form $T_{i_\lambda} = 0$. (If desired, one may suppose $i_\lambda = \lambda$ after indexing L by a set of integers $[1, r]$.) The component D_λ is the image under $\underline{f} : \underline{X} \longrightarrow \underline{S}$ of the component D'_λ through the singular point x_λ of X_o in the non-smoothness locus

$$D' = \bigcup_{\lambda \in L} D'_\lambda$$

of $\underline{f}$. By the construction itself, $f : X \longrightarrow S$ is isomorphic to the pullback of $\underline{f} : \underline{X} \longrightarrow \underline{S}$ by a suitable W-morphism $g : S \longrightarrow \underline{S}$:

$$\begin{tikzcd} \underline{X} \arrow[d, "\underline{f}"'] & X \arrow[l] \arrow[d, "f"] \\ \underline{S} & S \arrow[l, "g"'] \end{tikzcd} \tag{12.8.2}$$

Set

$$P = \underline{\mathrm{Pic}}^o_{\underline{X}/\underline{S}}. \tag{12.8.3}$$

Since the special fibre of $\underline{f}$ is geometrically reduced of dimension 1, it follows that $\underline{f}$ is cohomologically flat in dimension zero,[67] by EGA III, 7.8.1, and that P is formally smooth over $\underline{S}$ (the

65. The printed source leaves the internal pinpoint following "V" blank; no section target is supplied.

66. The printed source likewise leaves the internal pinpoint following "VI" blank.

67. The working transcription reads $\underline{F}$ here. Direct inspection of the printed page shows $\underline{f}$, as also required by the sentence's reference to the special fibre of the universal curve.

latter fact follows from $H^2(X_o, \underline{\mathcal{O}}_{X_o}) = 0$, since X_o has dimension 1). For simplicity, we shall assume that P is representable by a group scheme over $\underline{S}$; it will necessarily be a smooth group scheme with connected fibres. This is a special case of an unpublished result of Mumford cited in [22] (and considerably generalized by Raynaud in the note cited there, when the base is a trait). Since no proof of either Mumford's or Raynaud's result has been published, the reader may prefer to use the general results of M. Artin [1]. They give the representability of P—under the sole assumption that $\underline{f}$ is proper, flat, and cohomologically flat in dimension zero—by an "algebraic space" in Artin's sense, which would suffice for our purpose. Put

$$U = \underline{S} - \operatorname{supp} D. \tag{12.8.4}$$

Thus U is the largest open subset of $\underline{S}$ over which $\underline{X}$ is smooth, and hence

$$P|_U = \underline{\mathrm{Pic}}^o_{\underline{X}_U/U}$$

is an abelian scheme over U. Its pullback to η by $g : \eta \longrightarrow U$ is the abelian scheme A_η over η that we wish to study. More precisely, there is a canonical isomorphism over S (12.1.10):

$$P \times_{\underline{S}} S \simeq A^o. \tag{12.8.5}$$

Since the special fibre $P_o = A^o_o$ is ℓ-divisible for every ℓ, the same holds for P itself; all its fibres are therefore extensions of an abelian variety by a torus. Consider the pro-ℓ group $T_\ell(P)$ over $\underline{S}$, whose pullback by g is precisely $T_\ell(A^o)$. Applying the constructions of 7.3, one defines a filtration of this pro-group by a "fixed part" and a "toric part":

$$T_\ell(P) \supset T_\ell(P)^f = T_\ell(\widehat{P}) \supset T_\ell(P)^t = T_\ell(T) \supset 0. \tag{12.8.6}$$

Its pullback by g is the analogous filtration of $T_\ell(A^o)$ studied in sections 2 and 5.

Consider the restriction of this filtration over the open subset U of $\underline{S}$. There we have the canonical self-duality of P_U, as the Jacobian of a smooth curve scheme over U, and hence a corresponding pairing

$$\varphi : T_\ell(P_U) \otimes T_\ell(P_U) \longrightarrow T_\ell(\mathbb{G}_{mU}). \tag{12.8.7}$$

This is a self-duality, in the sense of Barsotti–Tate groups, of $T_\ell(P_U)$.[68] We saw in 7.4.8 that, for this pairing, $T_\ell(P_U)^f$ and $T_\ell(P_U)^t$ are each other's orthogonal complements. It follows that

$$E_U = T_\ell(P_U)/T_\ell(P_U)^t$$

is canonically isomorphic to the Barsotti–Tate pro-group dual to $T_\ell(P_U)^f = F_U$. It therefore extends canonically to a Barsotti–Tate group E over $\underline{S}$, namely

$$E = D(F), \qquad \text{where} \qquad F = T_\ell(P)^f. \tag{12.8.8}$$

We are now in a typical situation of mixed extensions (9.3), and can repeat the constructions of 9.4–9.6 (for whose having been written in the stale and piss-poor setting of a base scheme

68. The printed source reads $T_\ell(B_U)$ here, although no B_U has been introduced and the pairing and every adjacent term involve P_U. The English text restores the internally forced P_U.

reduced to a trait we apologize!): $T_\ell(P_U)$ is a mixed extension of E_U by F_U, where E and F are Barsotti–Tate groups over $\underline{S}$ which are extensions, dual to one another:

$$
\begin{array}{lccccccccc}
 & 0 & \longrightarrow & P & \longrightarrow & F & \longrightarrow & R & \longrightarrow & 0 \\
 & & & \wr\| & & \wr\| & & & & \\
 & & & T_\ell(P)^t \simeq \underline{\mathrm{Hom}}\big(M, T_\ell(\mathbb{G}_{m\underline{S}})\big) & & T_\ell(P)^f & & & & \\
(12.8.9) & & & & & & & & & \\
 & 0 & \longrightarrow & R & \longrightarrow & E & \longrightarrow & Q & \longrightarrow & 0 \\
 & & & \wr\| & & & & \wr\| & & \\
 & & & D(R) & & & & M \otimes \mathbb{Z}_{\ell\,\underline{S}} & & .
\end{array}
$$

The printed upper identification in (12.8.9) omits the outer closing parenthesis after $\underline{\mathrm{Hom}}(M, T_\ell(\mathbb{G}_{m\underline{S}}))$; the balanced expression has been supplied here. Notice that R is canonically self-dual, as follows from the self-duality (12.8.7) and the orthogonality statement noted above. The only difference from the cited construction is that the pairing obtained on $M_\ell \times M_\ell$ takes values in the group

$$\big(\mathbb{G}_m(U)/\mathbb{G}_m(\underline{S})\big) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

which is no longer canonically isomorphic to $\mathbb{Z}_\ell$, but rather to $\mathbb{Z}_\ell^L$, by means of the isomorphism

$$\mathbb{Z}^L \simeq \mathbb{G}_m(U)/\mathbb{G}_m(\underline{S}). \tag{12.8.10}$$

It maps the element indexed by λ in the canonical basis of $\mathbb{Z}^L$ to the class of T_{i_λ} in the right-hand side, where T_{i_λ} is an equation for the divisor D_λ. We thus obtain a canonical pairing

$$\underline{u_\ell} : M_\ell \otimes M_\ell \longrightarrow \mathbb{Z}_\ell^L \quad . \tag{12.8.11}$$

It is essentially immediate that the monodromy pairing u_ℓ (12.4.7) over S which we propose to study is the composite of $\underline{u_\ell}$ with the homomorphism $\mathbb{Z}_\ell^L \longrightarrow \mathbb{Z}_\ell$ obtained, by tensoring with $\mathbb{Z}_\ell$ over $\mathbb{Z}$, from the homomorphism

$$
\begin{array}{ccc}
\mathbb{G}_m(U)/\mathbb{G}_m(\underline{S}) & \longrightarrow & \mathbb{G}_m(\eta)/\mathbb{G}_m(S) \\
\wr\| & & \wr\| \\
\mathbb{Z}^L & & \mathbb{Z}
\end{array}
$$

induced by $g : S \longrightarrow \underline{S}$ (which sends η into U). This homomorphism is given by an element

$$(d_\lambda)_{\lambda\in L} \in \mathbb{Z}^L, \qquad d_\lambda \geqslant 1 \quad \text{for every } \lambda \in L, \tag{12.8.12}$$

where, for every $\lambda \in L$, d_λ is the multiplicity of the inverse image in S of the divisor D_λ on $\underline{S}$. It in fact depends only on the completion $\widehat{\mathcal{O}}_{X,x_\lambda}$ as a V-algebra (Exposé VI).[69] Moreover, it is known (Exposé VI) that, for a given λ, one has $d_\lambda = 1$ if and only if X is <u>regular</u> at x_λ.[70] Since in 12.5 we assumed X regular, that is, all the d_λ equal to 1, assertion 12.5 will therefore follow from the following lemma, in which X, S, g no longer occur.

<u>Lemma</u> 12.8.13. <u>If $\lambda \in L$, the composite of (12.8.11) with pr_λ is equal to the form $\varphi_{x_\lambda} \otimes \varphi_{x_\lambda}$ of (12.4.4).</u>

69. Both printed citations to Exposé VI in this sentence leave their internal section pinpoints blank.
70. The printed source has the lowercase letter x as the subject of "is regular." The following sentence and the mathematical statement require the scheme X.

To prove this last assertion, consider the completed strict localization $\underline{S}^\lambda$ of $\underline{S}$ at the generic point of D_λ,[71] and put

$$\underline{X}^\lambda = \underline{X} \times_{\underline{S}} \underline{S}^\lambda.$$

The special fibre $\underline{X}_o^\lambda$ then has only one singular point y_λ, which is again an ordinary quadratic singularity. In the corresponding monodromy pairing

$$u_\ell^\lambda : M_\ell^\lambda \otimes M_\ell^\lambda \longrightarrow \mathbb{Z}_\ell \qquad , \tag{12.8.14}$$

M^λ consequently has rank 1 or 0. We leave to the reader the verification of the following easy facts, which make explicit, in the special case in which we need them, the behaviour under localization of the preceding constructions.

a) There is a unique subtorus T_λ of $P \times_{\underline{S}} D_\lambda$ whose restriction to the generic point s_λ of D_λ is the maximal torus of P_{s_λ} (which has dimension 0 or 1).

b) Consider the transpose homomorphism $M \longrightarrow M^\lambda$ on character groups of the inclusion $T_{\lambda o} \hookrightarrow T_o$, where T_o is the maximal torus of P_o, and the form $\varphi_{y_\lambda} : M^\lambda \longrightarrow \mathbb{Z}$, defined up to sign as in 12.4.3 (with X replaced by $\underline{X}^\lambda$). Then, up to sign, the following diagram is commutative:

$$\begin{tikzcd} M \arrow[rr] \arrow[dr, "\varphi_{x_\lambda}"'] & & M^\lambda \arrow[dl, "\varphi_{y_\lambda}"] \\ & \mathbb{Z} & \end{tikzcd} \qquad . \tag{12.8.15}$$

The working transcription misreads the right diagonal label in (12.8.15) as $\varphi_{\lambda_\lambda}$. The printed diagram, the immediately preceding definition, and the formula below all give φ_{y_λ}. (In fact, choosing an ordering of the two branches of X_o through x_λ determines an analogous choice at the point y_λ of $\underline{X}_o^\lambda$, and for choices corresponding in this way the preceding diagram is commutative.)

c) The following diagram is commutative:

$$\begin{tikzcd} M_\ell \otimes M_\ell \arrow[rr] \arrow[dr, "\mathrm{pr}_\lambda\, \underline{u}_\ell"'] & & M_\ell^\lambda \otimes M_\ell^\lambda \arrow[dl, "u_\ell^\lambda"] \\ & \mathbb{Z}_\ell & \end{tikzcd} \qquad ,$$

where the notation is that of 12.8.13 and (12.8.14). These relations show at once that, to prove 12.8.13, it is enough to prove the analogous relation for $\underline{X}^\lambda$:

$$u_\ell^\lambda = (\varphi_{y_\lambda} \otimes \varphi_{y_\lambda}) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell,$$

which is nothing other than 12.5 for $\underline{X}^\lambda$ over $\underline{S}^\lambda$. Now $\underline{S}^\lambda$ is a trait of <u>characteristic zero</u>.[72] This completes the announced reduction.

<u>Remarks 12.9.</u> Notice that this reduction simultaneously reduces us to the case in which X_o has only one singular point, the case classically considered in the Picard–Lefschetz formula. Notice also that, in this case, X_o has one or two irreducible components, corresponding respectively to

71. In this reduction the printed source drops the subscript λ from D_λ and prints an un-underlined X as the right-hand factor of the following fibre product. Both marks are restored: the localization is at the component D_λ, and the fibre product is formed from the universal curve $\underline{X}$ over $\underline{S}$.

72. At this final step the printed source writes $\underline{S}$ rather than the localized trait $\underline{S}^\lambda$ used throughout the reduction. The English text restores the mathematically required superscript.

rank $M = 1$ and rank $M = 0$ (the latter being the case in which the Jacobian of the generic fibre has good reduction).

12.10. The proof given in 12.8 yields a result more general than 12.5, valid without assuming that X is regular, by introducing the integers $d_\lambda \geqslant 1$ of (12.8.12): all the monodromy pairings are then deduced from the pairing

$$u = \sum_{\lambda \in L} d_\lambda \, \varphi_{x_\lambda} \otimes \varphi_{x_\lambda} : M \otimes M \longrightarrow \mathbb{Z} \quad . \tag{12.10.1}$$

13. Links with transcendental theory: the complex analytic case.

For the general notions concerning analytic spaces, we refer to [2].

13.1. Review of analytic groups over S. Let S be an analytic space, and let A be an analytic space over S, endowed with the structure of a commutative group over S (that is, A is a commutative group in the category of analytic spaces over S). We assume that A is smooth over S, and denote by

$$\underline{E} = \underline{\mathrm{Lie}}(A/S) \tag{13.1.1}$$

the locally free module on S obtained by pulling the relative tangent sheaf of A over S back along the zero section. We shall admit here the definition of the exponential homomorphism

$$\exp : E \longrightarrow A \quad , \tag{13.1.2}$$

where E is the analytic vector bundle on S whose module of holomorphic sections is $\underline{E}$. In a neighbourhood of a point $s \in S$, this homomorphism is first defined near the zero section of E by the familiar methods of the theory of formal groups in characteristic zero, and is then extended in the evident way, by multiplicativity, to all of E. It is a homomorphism of relative analytic groups over S; it is étale, and its image is the open subspace A^o of A which is the union of the identity components of the fibres of A over S. If R is the analytic subspace which is the kernel of exp, we therefore have an exact sequence of relative analytic groups

$$0 \longrightarrow R \longrightarrow E \longrightarrow A^o \longrightarrow 0 \quad . \tag{13.1.3}$$

Here the morphism $E \longrightarrow A^o$ is surjective and étale, so R is a relative analytic group étale over S. It is an analytic subspace of E, and is closed in E if and only if A^o is separated over S, that is, if and only if the zero section of A^o is closed in A^o.

13.1.4. Of course, the exact sequence (13.1.3) is functorial in A, and commutes with every base change $S' \longrightarrow S$. In this way one obtains an equivalence of categories, compatible with base changes, between the category of relative analytic groups over S which are smooth and have connected fibres, and the category of pairs (E, R) consisting of a locally free vector bundle E over S and an analytic subgroup R of E which is étale over S.

13.1.5. When A is an "abeloid" group over S, by which we mean that it is smooth, proper, and separated over S, with connected fibres—so that in particular $A = A^o$—then R is a locally constant group over S, whose rank at each point s (as a $\mathbb{Z}$-module) is equal to $2\,\mathrm{rank}_s(E)$, and conversely. In this way one obtains the familiar interpretation of an abeloid space over S in terms of a "local system" R over S and a complex analytic vector-bundle structure on the real analytic bundle $R \otimes_{\mathbb{Z}} \mathbb{R}$ defined by R.

13.2. Formal completion along a fibre. We now fix a point $s \in S$, and shall work locally in a neighbourhood of s. For every integer $n \geqslant 0$, let S_n denote the n-th infinitesimal neighbourhood of s in S, and, for every analytic space X over S, put likewise $X_n = X \times_S S_n$. As n varies, the system of the X_n then forms an inductive system of analytic spaces over S, which we denote by $\widehat{X}$ and call the formal completion of X along X_s. It clearly depends functorially on X, and its formation commutes with finite projective limits; hence $X \longmapsto \widehat{X}$ takes groups to groups. Notice that $\widehat{X}$ in fact depends only on the restriction of X to an arbitrary open neighbourhood U of s in S.

Lemma 13.2.1. The data are those of 13.1, and suppose that the following conditions hold: a) $A = A^o$; b) R is constant in a neighbourhood of s; and c) R_s spans the complex vector space E_s. Let A' be a second smooth analytic group over S, and consider the map

$$u \longmapsto \widehat{u} : \mathrm{Hom}_{S\text{-gr}}(A, A') \longrightarrow \mathrm{Hom}_{\widehat{S}\text{-gr}}(\widehat{A}, \widehat{A}') \quad ,$$

where the notation is that of 13.2, and the analogous map obtained by restricting over a variable open neighbourhood U of s and passing to the inductive limit:

$$u \longmapsto \widehat{u} : \mathrm{Hom}_{(S,s)\text{-gr}}(A, A') \longrightarrow \mathrm{Hom}_{\widehat{S}\text{-gr}}(\widehat{A}, \widehat{A}') \quad . \tag{13.2.1.1}$$

(Here (S, s) denotes the germ of the analytic space S at s.) Then the latter map is bijective.

Indeed, consider the exact sequence

$$0 \longrightarrow R' \longrightarrow E' \longrightarrow A'^o \longrightarrow C$$

analogous to (13.1.3) and defined by A'. Since the homomorphisms $A \longrightarrow A'$ necessarily factor through A'^o, and the same holds after the base changes $U \longrightarrow S$ and $S_n \longrightarrow S$, we may assume that $A' = A'^o$. By the dictionary of 13.1.4, it remains to prove that giving a germ of a vector bundle homomorphism $E \longrightarrow E'$ which takes R into R' is equivalent to giving a coherent system of homomorphisms $E_n \longrightarrow E'_n$ which take R_n into R'_n.

If $\underline{\mathcal{O}}_s$ denotes the local ring of S at s, giving a coherent system of homomorphisms $E_n \longrightarrow E'_n$ is equivalent to giving a homomorphism of $\widehat{\underline{\mathcal{O}}}_s$-modules

$$\widehat{F} : \widehat{\underline{E}}_s \longrightarrow \widehat{\underline{E}}'_s,$$

where $\underline{E}_s$ and $\underline{E}'_s$ are the free $\underline{\mathcal{O}}_s$-modules obtained at s from the $\underline{\mathcal{O}}_s$-modules $\underline{E}$ and $\underline{E}'$, corresponding to E and E'. On the other hand, a germ of a vector bundle homomorphism $E \longrightarrow E'$ is equivalent to giving a homomorphism of $\underline{\mathcal{O}}_s$-modules

$$F : \underline{E}_s \longrightarrow \underline{E}'_s \quad .$$

This already shows that (13.2.1.1) is injective. For surjectivity, it is enough to prove that if $\widehat{F}$ is a homomorphism as above such that, for every n, the induced homomorphism $F_n : E_n \longrightarrow E'_n$ takes R_n into R'_n, then $\widehat{F}$ comes from an F—that is, it takes $\underline{E}_s$ into $\underline{E}'_s$—and the corresponding germ of homomorphisms $E \longrightarrow E'$ takes R into R'.

By hypothesis c), $R_s = R_o$ spans E_s; it follows from Nakayama's lemma that the group R_o (identified either with the group of sections of $\widehat{R}$ over $\widehat{S}$, or with the group of germs of sections of R over S) generates $\widehat{E}_s$. Thus, to prove that $\widehat{F}(\underline{E}_s) \subset \underline{E}'_s$, it is enough to prove that $\widehat{F}(R_o) \subset \underline{E}'_s$. But $\widehat{F}(R_o) \subset R'_o$, where R'_o is itself interpreted as the group of germs of sections of $R' \subset E'$ in a neighbourhood of s. This gives the asserted inclusion, and hence the existence of F. Moreover, we see that $F(R_o) \subset R'_o$, where R_o and R'_o are regarded as groups of germs of sections of E and E', respectively. By b), R_o generates R in a neighbourhood of s; it follows that $F(R) \subset R'$ in a neighbourhood of s, which completes the proof.

Corollary 13.2.2. If A satisfies conditions a), b), and c) of 13.2.1, it is determined up to unique isomorphism in a neighbourhood of s by the knowledge of the $\widehat{S}$-group $\widehat{A}$.[73] 13.3. The analytic Raynaud extension.

We are now in a position to paraphrase, in analytic geometry, the construction of the Raynaud extension given in 7. Suppose that A is as in 13.1, with $A = A^o$,[74] and consider $R_o = R_s$, which is a free $\mathbb{Z}$-module of finite type. After restricting S to a sufficiently small open neighbourhood of s, and interpreting R_o as the group of germs of sections of R near s, one obtains a homomorphism

$$R_{oS} \longrightarrow R \subset E, \tag{13.3.1}$$

where R_{oS} denotes the constant analytic group over S with value R_o. On choosing the neighbourhood of s small enough that every section of $\underline{\mathcal{O}}_S$ whose germ at s is zero is itself zero, one sees that the preceding homomorphism is injective. Its image is therefore a constant analytic subgroup of R; it is plainly the largest constant subgroup of R, and is called the fixed part of R and denoted by R^f:

$$R_{oS} \simeq R^f \hookrightarrow R, \qquad \text{inducing an isomorphism } R^f_s \simeq R_s. \tag{13.3.2}$$

We may then consider the quotient group

$$A^\natural = E/R^f, \tag{13.3.3}$$

and, in view of (13.1.3), obtain a canonical exact sequence

$$0 \longrightarrow R/R^f \xrightarrow{i} A^\natural \longrightarrow A \longrightarrow 0. \tag{13.3.4}$$

Notice that, after restricting S to a neighbourhood of s, R^f is always closed in E, and hence $A^\natural$ is separated over S. Notice also that the homomorphism $A^\natural \longrightarrow A$ is étale, and therefore induces an isomorphism

$$\underline{\mathrm{Lie}}(A^\natural) \xrightarrow{\sim} \underline{\mathrm{Lie}}(A) = \underline{E}. \tag{13.3.5}$$

Moreover, for every $n \geqslant 0$, the canonical homomorphism $A^\natural \longrightarrow A$ induces an isomorphism $A^\natural_n \simeq A_n$ (since $R^f_n \xrightarrow{\sim} R_n$), and consequently an isomorphism

$$\widehat{A^\natural} \xrightarrow{\sim} \widehat{A}. \tag{13.3.6}$$

This last fact shows that $A^\natural$ does indeed play the role of the group, likewise denoted by $A^\natural$, that we considered in 7. Here it has been constructed without any hypothesis on the fibre $A_s = A_o$ of the form "an extension of an abeloid group by a torus". Such a hypothesis will nevertheless be necessary if we wish to recover on $A^\natural$ the important extension structure (7.2.3). By contrast, the homomorphism $A^\natural \longrightarrow A$ in (13.3.4) is essentially transcendental in nature and has no counterpart in the purely algebraic theory of § 7. In § 14 we shall return to the question of a purely algebraic interpretation of the homomorphism $i : R/R^f \longrightarrow A^\natural$ in (13.3.4).

Suppose first that we are given a complex subtorus

73. The printed source refers here to the conditions of 13.2.2 itself; the three conditions are introduced in Lemma 13.2.1.

74. The printed source has $A = A_o$ here. The required hypothesis, and the one used below, is $A = A^o$; compare 13.1 and Lemma 13.2.1.

$$T_o \subset A_o, \tag{13.3.7}$$

where by a "<u>complex torus</u>" we mean an analytic group isomorphic to a Cartesian power of $\mathbb{C}^*$. Let

$$T = T_o \times S \tag{13.3.8}$$

be the constant torus over S with value T_o. Proceeding as in 5.1, one obtains a unique homomorphism (indeed, an immersion) of inductive systems of analytic groups

$$\widehat{T} \longrightarrow \widehat{A^\natural},$$

which over S_o reduces to (13.3.7). By 13.2.2, this homomorphism comes from a unique germ of homomorphisms

$$T \longrightarrow A^\natural, \tag{13.3.9}$$

and, after taking S small enough around s, we may suppose that it is defined over all of S. The corresponding homomorphism

$$\underline{\mathrm{Lie}}(T) \longrightarrow \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{E}$$

is injective over each S_n, hence over $\widehat{S}$, and therefore over $\underline{\mathcal{O}}_s$ by flat descent. Thus there is a neighbourhood of s in S, which we may take to be all of S, on which $\underline{\mathrm{Lie}}(T)$ identifies with a locally direct summand $\underline{E}^t$ of $\underline{E}$; denote the corresponding vector subbundle of E by E^t:

$$\underline{\mathrm{Lie}}(T) \xrightarrow{\sim} \underline{E}^t \subset \underline{E} \simeq \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{\mathrm{Lie}}(A), \qquad E^t \subset E. \tag{13.3.10}$$

Likewise, let

$$R_o^t \subset R_o, \qquad R^t = R_{oS}^t \subset R_{oS} \simeq R^f \tag{13.3.11}$$

denote respectively the sublattice of R_o defined by the subtorus T_o of A_o, and the corresponding constant analytic group over S. Then the homomorphism (13.3.9) corresponds simply to the homomorphism

$$E^t/R_{oS}^t = E^t/R^t \longrightarrow E/R_{oS} = E/R^f$$

induced by the inclusions

$$E^t \hookrightarrow E, \qquad R^t \hookrightarrow R^f.$$

Using the fact that (13.3.9) induces an immersion on the fibre over s, we conclude that the homomorphism

$$R_o/R_o^t \longrightarrow E_o/E_o^t = (E/E^t)_o$$

is injective. It follows at once that

$$R_o/R_o^t \longrightarrow (E/E^t)_{s'}$$

is still injective for s' near s. Thus, after shrinking S around s, we may suppose that (13.3.9) is a closed immersion.

We are therefore led to introduce

$$B = A^{\natural}/T \simeq (E/E^t)/(R^f/R^t), \tag{13.3.12}$$

so that $A^{\natural}$ acquires an extension structure

$$0 \longrightarrow T \longrightarrow A^{\natural} \longrightarrow B \longrightarrow 0, \tag{13.3.13}$$

where B is a relative analytic group over S, smooth and separated over S, with connected fibres. By construction we therefore have

$$\underline{\mathrm{Lie}}(B) \simeq \underline{E}/\underline{E}^t, \tag{13.3.14}$$

and the exponential exact sequence for B is

$$0 \longrightarrow R^f/R^t \longrightarrow E/E^t \longrightarrow B \longrightarrow 0. \tag{13.3.15}$$

Notice that

$$R^f/R^t = (R_o/R_o^t)_S \tag{13.3.16}$$

is a constant analytic group over S. It follows from 13.1.5 that B is an abeloid group over S in a neighbourhood of s if and only if its fibre at s,

$$B_o = A_o/T_o,$$

is an abeloid analytic group; in that case A_o appears as an extension of such a group B_o by a torus T_o. Every way of writing A_o as an extension of an abeloid group by a complex torus $(*)$ therefore gives rise on A, in a neighbourhood of s, to a unique extension structure (13.3.13) of an abeloid group B by a torus T, inducing the given extension structure at s.

13.4. Interpretation in terms of Deligne's "mixed Hodge structures".

Let us first return to the dictionary of 13.1.4, which we shall reformulate in a slightly different way, better suited to introducing the "Hodge structures" relevant to the present situation. For this purpose, let $\underline{R}$ be the sheaf of sections of R over S, and introduce the $\underline{\mathcal{O}}_S$-module

$$\mathcal{E}(A) = \mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \quad . \tag{13.4.1}$$

It is locally free of finite type when R is locally constant. (Note that $\mathcal{E}$ need not be coherent when R is not locally constant.) The homomorphism (13.1.3) $\underline{R} \longrightarrow \underline{E}$ then defines a canonical homomorphism

$$\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S \longrightarrow \underline{E} \quad . \tag{13.4.2}$$

We denote its kernel by $\underline{F}(A)$, or simply by $\underline{F}$; thus there is an exact sequence

$$0 \longrightarrow \underline{F} \longrightarrow \mathcal{E} \longrightarrow \underline{E} \quad . \tag{13.4.3}$$

The most interesting case is that in which (13.4.2) is an epimorphism, that is, in which R_s spans the complex vector space E_s for every $s \in S$. We then have an exact sequence

$$(13.4.4) \qquad 0 \longrightarrow \underline{F} \longrightarrow \mathcal{E} \longrightarrow \underline{E} \longrightarrow 0 \quad .$$

If R is locally constant, this is a short exact sequence of locally free modules over S. Knowledge of the pair (E, R) is then equivalent to knowledge of the pair $(R, \underline{F})$, where R is a locally constant analytic group over S whose fibres are free $\mathbb{Z}$-modules of finite type, and $\underline{F}$ is a locally direct-summand submodule of $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S}$, subject in addition to

$$(13.4.5) \qquad \underline{F} \cap \underline{R} = 0 \quad .$$

From the standpoint of Hodge structures, $\mathcal{E}$ should be regarded as filtered by the "<u>Hodge filtration</u>"

$$(13.4.6) \qquad \mathrm{Fil}^1(\mathcal{E}) = \mathcal{E}, \qquad \mathrm{Fil}^0(\mathcal{E}) = \underline{F}, \qquad \mathrm{Fil}^{-1}(\mathcal{E}) = 0 \quad .$$

Notice that the exact sequence (13.4.3) still depends functorially on the analytic group A over S, and commutes with arbitrary base change.

Returning now to the conditions of 13.3, put

$$(13.4.7) \qquad \mathcal{E}^f = \underline{R}^f \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S}, \qquad \mathcal{E}^t = \underline{R}^t \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S} \quad .$$

These modules are defined on a suitable neighbourhood of s, which we may suppose to be all of S, and define a filtration of $\mathcal{E}$, called the <u>weight filtration</u>,

$$(13.4.8) \qquad \mathcal{E} \supset \mathcal{E}^f \supset \mathcal{E}^t \supset 0 \quad .$$

Here $\mathcal{E}^t$, $\mathcal{E}^f$, and $\mathcal{E}$ are to be regarded as having weights -2, -1, and 0, respectively; thus this is an increasing filtration. Like R^t, the module $\mathcal{E}^t$ depends on the choice of a subtorus T_o of $A_o = A_s$. Since R^f and R^t are locally constant, <u>$\mathcal{E}^f$ and $\mathcal{E}^t$ are locally free</u>; they may also be interpreted as

$$(13.4.9) \qquad \mathcal{E}^f = \mathcal{E}(A^\natural), \qquad \mathcal{E}^t = \mathcal{E}(T) \quad .$$

Applying the functoriality of (13.4.3), as A varies, to the homomorphisms

$$T \longrightarrow A^\natural \longrightarrow A$$

shows at once that $\underline{F}(A^\natural)$ and $\underline{F}(T)$ identify respectively with

$$(13.4.10) \qquad \underline{F}(A^\natural) \simeq \underline{F}^f \stackrel{\mathrm{def}}{=} \underline{F} \cap \mathcal{E}^f, \qquad \underline{F}(T) = \underline{F}^t = \underline{F} \cap \mathcal{E}^t = 0 \quad .$$

The last equality follows immediately from the fact that $\underline{F}(T) = 0$ for a torus T: in this case the exact sequence of type (13.4.3) is $0 \longrightarrow 0 \longrightarrow \mathcal{E}(T) \longrightarrow \underline{\mathrm{Lie}}(T) \longrightarrow 0$. At the same time we see that the composite homomorphism

$$\mathcal{E}^t \longrightarrow \mathcal{E} \longrightarrow \underline{E}$$

induces an isomorphism

(13.4.11) $$\mathcal{E}^t \xrightarrow{\sim} \underline{E}^t \simeq \underline{\mathrm{Lie}}(T) \qquad .$$

Suppose now that R_s spans the complex vector space E_s, as happens in particular in the favourable case 13.3.16, where A_s is an extension of an abeloid group by a complex torus T_o. After restricting S to a suitable neighbourhood of s, the canonical homomorphism of type (13.4.2), with A replaced by $A^\natural$,

$$\mathcal{E}^f = \mathcal{E}(A^\natural) \longrightarrow \underline{\mathrm{Lie}}(A^\natural) \simeq \underline{E}$$

is an epimorphism; a fortiori, so is (13.4.2). We therefore have the exact sequences (13.4.4) and

(13.4.12) $$0 \longrightarrow \underline{F}^f \longrightarrow \mathcal{E}^f \longrightarrow \underline{E} \longrightarrow 0 \qquad .$$

In particular,

(13.4.13) $$\underline{F} + \mathcal{E}^f = \mathcal{E} \qquad ,$$

since $\mathcal{E}/\underline{F} \simeq \underline{E}$. Moreover, since $B = A^\natural/T$ is a quotient of $A^\natural$,[75] it satisfies the analogous surjectivity condition, and hence gives an exact sequence of type (13.4.4):

(13.4.14) $$0 \longrightarrow \underline{F}(B) \longrightarrow \mathcal{E}(B) \longrightarrow \underline{\mathrm{Lie}}(B) \longrightarrow 0 \qquad .$$

Its interpretation in terms of the modules already introduced is immediate. Indeed, by (13.3.12),

(13.4.15) $$\begin{cases} \mathcal{E}(B) \simeq (\underline{R}^f/\underline{R}^t) \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S} \simeq \mathcal{E}^f/\mathcal{E}^t, \\ \underline{\mathrm{Lie}}(B) \simeq \underline{E}/\underline{E}^t. \end{cases}$$

It follows from (13.4.11) and the relation $\underline{F} \cap \mathcal{E}^t = 0$ in (13.4.10) that $\underline{F}(B) = \ker(\mathcal{E}^f/\mathcal{E}^t \longrightarrow \underline{E}/\underline{E}^t)$ is isomorphic to $\underline{F}^f$ through the homomorphism

(13.4.16) $$\underline{F}^f \simeq \underline{F}(A^\natural) \xrightarrow{\sim} \underline{F}(B) \qquad .$$

Let us summarize the properties just obtained for the weight filtration (13.4.8), and its relation with the Hodge filtration (13.4.6) of $\mathcal{E}$ defined by the submodule $\underline{F}$, in the favourable case 13.3.16 where A_s is an extension of an abeloid analytic group B_o by a torus T_o. We use this torus to define R^t, so that B is an abeloid analytic group over S.

Proposition 13.4.17. Under the preceding conditions, the Hodge and weight filtrations on $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}_S}$,

(13.4.17.1) $$\mathcal{E} \supset \underline{F}, \qquad \mathcal{E} \supset \mathcal{E}^f \supset \mathcal{E}^t \supset 0,$$

satisfy the following conditions:

a) The weight filtration is obtained, by tensoring with $\underline{\mathcal{O}_S}$ over $\mathbb{Z}$, from the analogous filtration on $\underline{R}$

75. The printed source says here that B is a quotient of A; the displayed definition $B = A^\natural/T$ gives the intended group.

$$\underline{R} \supset \underline{R}^f \supset \underline{R}^t \supset 0,$$

where $\underline{R}^f$ and $\underline{R}^t$ are locally constant analytic groups over S, with values free $\mathbb{Z}$-modules of finite type, namely $R_o^f = R_s = \pi_1(A_s)$ and $R_o^t = \pi_1(T_o)$.

b) One has the relations

$$(*) \qquad \underline{F} \cap \mathcal{E}^t = 0, \qquad \underline{F} + \mathcal{E}^f = \mathcal{E} \quad .$$

In other words, the filtration on $\mathcal{E}^t$ induced by the Hodge filtration, with the degree conventions of (13.4.6), is concentrated in degree -1, whereas the filtration on $\mathcal{E}/\mathcal{E}^f$ induced by the Hodge filtration is concentrated in degree 0.

c) Consider on

$$\mathcal{E}^{\mathrm{ab}} \stackrel{\mathrm{def}}{=} \mathcal{E}^f/\mathcal{E}^t \simeq \underline{R}^{\mathrm{ab}} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S, \qquad \underline{R}^{\mathrm{ab}} = \underline{R}^f/\underline{R}^t \quad \text{(cf. a))},$$

the filtration induced by the Hodge filtration on $\mathcal{E}$. It is therefore defined by the submodule $\underline{F}^{\mathrm{ab}}$ of $\mathcal{E}^{\mathrm{ab}}$ which is the image of

$$\underline{F}^f \stackrel{\mathrm{def}}{=} \underline{F} \cap \mathcal{E}^f \longrightarrow \mathcal{E}^{\mathrm{ab}}.$$

The arrow is a monomorphism by the first relation $(*)$. Then the pair

$$\left(\underline{R}^{\mathrm{ab}}, \underline{F}^{\mathrm{ab}} \subset \underline{R}^{\mathrm{ab}} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S = \mathcal{E}^{\mathrm{ab}}\right)$$

is isomorphic to the pair $(\underline{R}(B), \underline{F}(B))$ associated with an abeloid group over S, as explained after (13.4.4). Thus

$$B \simeq \mathcal{E}^{\mathrm{ab}}/(F^{\mathrm{ab}} + R^{\mathrm{ab}}),$$

where the symbols without underlining denote the corresponding analytic groups over S.

Remark 13.4.18. In Deligne's terminology [4], the conclusion of the preceding proposition, after restriction to an open subset U of S on which R is locally constant, says that R, endowed with the two filtrations (13.4.17.1) on $\mathcal{E} = \underline{R} \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S$, is an "analytic family of mixed Hodge structures over U". The situation just described served as the model for Deligne's conjectures on Néron theory in higher dimensions, stated in [9, 9.8–9.13].

There is, however, an important feature of the algebraic theory studied in § 7 which is missing from the description in 13.4.17. Suppose that A is an abeloid group over an open subset U complementary to a nowhere-dense analytic subset Z of S—say $Z = \{s\}$ when $\dim S = 1$—so that, by (13.1.5), $R|U$ is a local system over U. Then the local system R/R^t over U ought to be constant; more precisely, it ought to be dual to the local system R'^f, where R' is the analytic group, ètale over S, which would correspond (as R corresponds to A) to a suitable analytic group A' over S such that $A'|U$ is the abeloid group dual to $A|U$. In the notation of § 5, the algebraic analogue is the duality between $T_\ell(A_K)/T_\ell(A_K)^t$ and $T_\ell(A'_K)^f$ in (5.5.9), which follows from the duality theorem 5.2.

I do not know whether such a statement holds under only the hypotheses of 13.4.17, or even whether the action of $\pi_1(U,t)$ on R_t is unipotent for $t \in U$. Nor do the naive methods used in the present paragraph seem capable of proving such a result, even under the additional hypothesis that $A|U$ is endowed with a "polarization", that is, with an invertible sheaf ample on every fibre. This is, however, something that the "semi-simple group people" apparently know how to prove, at least when S is nonsingular of dimension 1; see the report [9]. In the next subsection we shall show that everything works under relative algebraicity hypotheses on A, and at the same time relate the result to the construction of § 7. 13.5. The case of a relatively algebraic situation. We now suppose that A comes from a relative group scheme over S which is smooth and separated, and whose fibres are extensions of abelian schemes by tori. We denote this relative scheme by the same letter A. (We allow ourselves this abuse of notation because the functor $G \longmapsto G^{\mathrm{an}}$, from the category of relative group schemes of the type under consideration to the category of analytic groups over S, is fully faithful, as would be easy to prove using results of relative GAGA type [10].) For everything concerning the notion of a relative scheme and the relations between relative schemes over S and analytic spaces over S, we refer to [10] (see also [2, Exp. 15]).

If $\underline{O}$ denotes the local ring of S at s, then the relative scheme A is determined near s by the ordinary scheme $A_{\underline{O}}$ over $\mathrm{Spec}(\underline{O})$ obtained from A by localization at s. It is then clear that the formal completion of $A_{\underline{O}}$ along its special fibre (isomorphic to $A_s = A_o$) is isomorphic to the formal completion of the analytic group A along A_s (13.2), where, by the abuse of notation introduced above, the $A_n = A \times_S S_n$ are identified with algebraic schemes over S. In the favourable case in which $\hat{A}$, viewed as a formal group scheme over $\hat{\underline{O}}$, is algebraizable (7.2.1), as is the case in particular when S is normal at s and A is quasi-projective over S (7.1.5), we therefore obtain a group scheme $A^{\natural}_{\hat{\underline{O}}}$ over $\mathrm{Spec}(\hat{\underline{O}})$, which is an extension of an abelian scheme $B_{\hat{\underline{O}}}$ by a torus. For simplicity, we suppose that $B_{\hat{\underline{O}}}$ is projective over $\hat{\underline{O}}$ (as is indeed the case, we know, if $\underline{O}$, and hence $\hat{\underline{O}}$, is normal). We shall then prove:

<u>Proposition 13.5.1.</u> <u>Under the preceding conditions, the analytic group $A^{\natural}$ of (13.3.3) comes, in a neighbourhood of s, from a relative group scheme determined up to a unique isomorphism, which is an extension of a projective relative abelian scheme over S by a torus. If $A^{\natural}_{\underline{O}}$ denotes the group scheme it defines over $\underline{O} = \underline{O}_{S,s}$, then the group scheme</u>

$$A^{\natural}_{\underline{O}} \otimes_{\underline{O}} \hat{\underline{O}}$$

<u>is canonically isomorphic, with its extension structure, to the Raynaud extension $A^{\natural}_{\hat{\underline{O}}}$ introduced above.</u>

The latter assertion follows immediately from the first, in view of 7.2.1, since, by (13.3.6), the formal completion of $A^{\natural}_{\underline{O}} \otimes_{\underline{O}} \hat{\underline{O}}$ along its special fibre is $\hat{A}_{\underline{O}}$. To prove the algebraicity of $A^{\natural}$ near s, proceeding as in 7.2.1, we are immediately reduced to proving that of the abelian quotient B of $A^{\natural}$, which follows from the next lemma.

<u>Lemma 13.5.2.</u> <u>Let B be an abeloid analytic group over the analytic space S, and let s be a point of S such that the completion $\hat{B}$ along the fibre B_s is algebraizable; that is, the $B_n = B \times_S S_n$ are algebraizable and $\hat{B}$, now viewed as a formal scheme over $\hat{\underline{O}}_{S,s}$, is algebraizable, or equivalently comes from an abelian scheme $B_{\hat{\underline{O}}}$ over $\hat{\underline{O}}$. Suppose moreover that $B_{\hat{\underline{O}}}$ is projective over $\hat{\underline{O}}$. Then B is projective over S in a neighbourhood of s.</u>

The hypothesis means that there is a polarization of the formal analytic space $\hat{B}$, that is, a sequence of polarizations p_n of the B_n which "glue together". The conclusion means that there is a polarization p of B over a neighbourhood of s. Let P be the analytic space over S of relative polarizations of B over S, which represents the functor

$$S' \longmapsto \{\text{polarizations of } B_{S'} \text{ over } S'\}$$

on the category of analytic spaces S' over S. This is an open subfunctor of the Néron–Severi functor $\underline{\mathrm{NS}}_{B/S}$ over S. The simplest standard arguments, which we omit here, show that $\underline{\mathrm{NS}}_{B/S}$, and hence also P, is represented by an <u>unramified</u> analytic space over S. The hypothesis then says that there is a <u>formal</u> section p of P over S at s. Since P is unramified and therefore locally quasi-finite over S, it follows from [2, Exp. 18, no. 1] that every formal section of P over S at s comes from an analytic section near s, that is, from a polarization of B over a neighbourhood of s. This proves the assertion.

<u>Remarks 13.5.3.</u>

a) If in 13.5.2 we omit the hypothesis that $B_{\underline{\hat{O}}}$ is projective over $\underline{\hat{O}}$,[76] we can still conclude that B is algebraizable near s. For this one uses an unpublished result (due independently to M. Artin and P. Deligne, using the theory of the sheaves $\underline{T}^i$ of VI 2) which asserts that, if X is proper over S and X_{red} is algebraizable, then X is algebraizable. This reduces us to the case where S is reduced. On the other hand, when S is normal, so that $\underline{\hat{O}}$ is normal, every abelian scheme over $\underline{\hat{O}}$ is known to be projective over $\underline{\hat{O}}$ (see, for example, [23]), and 13.5.2 applies. To obtain the case of an arbitrary reduced S, one uses non-flat descent from the normalization S' of S, by means of the following (delicate) result, which we shall not prove here: a proper morphism of noetherian schemes $f : S' \longrightarrow S$ which is an epimorphism (that is, such that $\underline{O}_S \longrightarrow f_*(\underline{O}_{S'})$ is injective) is an <u>effective descent morphism</u> for the fibred category of relative abelian schemes.

b) It appears that the proof of 13.5.2 should also prove the analogous result for every analytic space B proper over S. Probably more delicate is the question whether an analytic space X proper over S whose formal completion $\hat{X}$ along X_s is algebraizable is itself algebraizable over S near s. We merely point out that here the notion of relative algebraizability of X over S should be taken in a broader sense than in [10]: near s, X should be required to come from an <u>algebraic space</u> over $\underline{O}_{S,s}$ in the sense of M. Artin [1].

13.5.4. Let us place ourselves under the conditions of 13.5.1. For every integer $n > 0$, choose also the standard isomorphism of groups over $\mathbb{C}$

$$\mu_{n\mathbb{C}} \simeq (\mathbb{Z}/n\mathbb{Z})_{\mathbb{C}}, \qquad m \longmapsto \exp(2\pi i m/n) \qquad .$$

By passing to the inverse limit over the powers of a fixed prime ℓ, this gives a canonical isomorphism of ℓ-adic sheaves over $\mathbb{C}$,

$$\mathbb{Z}_\ell(1)_{\mathbb{C}} \simeq \mathbb{Z}_{\ell\mathbb{C}} \qquad ,$$

which we shall use to identify the two sides.

Applying the exact sequence of $\underline{\mathrm{Ext}}^i_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -)$ to the exact sequence (13.1.3), we obtain a canonical isomorphism

$$(13.5.5) \qquad {}_nA^{\circ} \simeq \underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \qquad ,$$

which, as n runs through the powers of a given prime ℓ, gives an isomorphism of inverse systems of groups over S,

$$(13.5.6) \qquad T_\ell(A^{\circ}) \simeq \underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad ,$$

where $\mathbb{Z}_\ell$ is regarded as the inverse system of the $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$. In the situation considered here, in which A comes from a separated group scheme over S, it is then immediately clear that the $\underline{R} \otimes \mathbb{Z}/n\mathbb{Z}$ may be regarded as relative schemes. This gives a meaning to the symbol $(\underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}}$ (whereas $R_{\underline{O}}$ would in general have none!), and yields a canonical isomorphism

$$(13.5.7) \qquad T_\ell(A_{\underline{O}}) \simeq (\underline{R} \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}} \qquad .$$

It follows immediately from the constructions of paragraph 2 and 13.3 that the preceding isomorphism carries the fixed part to the fixed part and the toric part to the toric part:

$$(13.5.8) \qquad T_\ell(A_{\underline{O}})^f \simeq (\underline{R}^f \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}}, \qquad T_\ell(A_{\underline{O}})^t \simeq (\underline{R}^t \otimes_{\mathbb{Z}} \mathbb{Z}_\ell)_{\underline{O}} \qquad .$$

76. The printed source says "projective over S" here; the hypothesis of Lemma 13.5.2 and the ensuing normality argument require "projective over $\underline{\hat{O}}$".

If A' is a relative analytic group over S satisfying the same hypotheses as A, and if $W_{\underline{O}}$ is a biextension of $(A_{\underline{O}}, A'_{\underline{O}})$ by $\mathbb{G}_{m\underline{O}}$, hence defines over a neighbourhood of s a biextension of (A, A') by $\mathbb{G}_{mS}$ (now using the notion in the complex-analytic sense, obtained by working in the site of analytic spaces over S for a suitable topology), then, by VIII 2.2, W defines pairings

$$(13.5.9) \qquad (\underline{R} \otimes \mathbb{Z}_\ell) \times (\underline{R}' \otimes_{\mathbb{Z}} \mathbb{Z}_\ell) \simeq T_\ell(A) \times T_\ell(A') \longrightarrow \mathbb{Z}_\ell(1)_S \simeq \mathbb{Z}_{\ell S} \qquad ,$$

and it is again clear that the corresponding pairing

$$(13.5.10) \qquad T_\ell(A_{\underline{O}}) \times T_\ell(A'_{\underline{O}}) \longrightarrow \mathbb{Z}_\ell(1)_S$$

is precisely the one associated, by loc. cit., with the biextension $W_{\underline{O}}$ of $(A_{\underline{O}}, A'_{\underline{O}})$. It is also not difficult to see that, after restricting to a neighbourhood of s, the pairings (13.5.9) arise from a unique pairing (no longer depending on ℓ)

$$(13.5.11) \qquad \underline{R} \otimes \underline{R}' \longrightarrow \mathbb{Z} \qquad .$$

With these preliminaries, the algebraic results concerning the $T_\ell(A_{\underline{O}})$ can be used to draw conclusions about the sheaf $\underline{R}$. Thus the pairing (13.5.11) is zero on $\underline{R}^t \times \underline{R}'^f$ and on $\underline{R}^f \times \underline{R}'^t$ (7.4.8). Moreover, if S is nonsingular of dimension 1 and, over $U = S - \{s\}$, A and A' are relative abelian schemes dual to one another through W, then $\underline{R}^t_U$ and $\underline{R}'^f_U$, as well as $\underline{R}^f_U$ and $\underline{R}'^t_U$, are exact annihilators of one another over a sufficiently small neighbourhood of s (7.4.8). Since (13.5.11) is a perfect duality over U, it follows that there is a duality over U between $\underline{R}/\underline{R}^t$ and $\underline{R}'^f$. In particular, $\underline{R}/\underline{R}^t$ is constant, and a fortiori the action of $\pi_1(U, t)$ on $\underline{R}_t$ is unipotent of level 2. Similarly, over U there is a perfect duality between $\underline{R}/\underline{R}^f$ and $\underline{R}'^t$. Since the latter is also isomorphic to the dual torus of the lattice M_S over S, where M is the group defined in (5.5.4), we obtain a canonical isomorphism

$$(13.5.12) \qquad \underline{R}_U/\underline{R}^f_U \simeq M_U \qquad ,$$

and the homomorphism i of (13.3.4) may be written as a homomorphism

$$(13.5.13) \qquad i : M_U \longrightarrow A^\natural \qquad ,$$

which we shall encounter again in a slightly different context in the next paragraph. 14. <u>Links with transcendental theory: the rigid-analytic case. The canonical homomorphism</u> $\underline{M}_K \longrightarrow A^{\natural\circ}_K$.

14.1. We assume here the foundations of the theory of rigid-analytic spaces, for which one may consult [12], [13], and [34].

Let S be a complete trait, and let A_K be an abelian scheme over its field of fractions. In 8.1, by descent from the Raynaud extension in the semistable case, we constructed an extension $G_K = A^{\natural\circ}_K$ (8.1.1)

$$(14.1.1) \qquad 0 \longrightarrow T_K \longrightarrow G_K \longrightarrow B_K \longrightarrow 0$$

of an abelian scheme B_K by a torus T_K. Considering the dual abelian scheme A'_K, we likewise have an extension $G'_K = A'^{\natural\circ}_K$

$$(14.1.2) \qquad 0 \longrightarrow T'_K \longrightarrow G'_K \longrightarrow B'_K \longrightarrow 0 \qquad ,$$

where B'_K is moreover identified with the abelian scheme dual to B_K (8.2). As in (8.1.8), let $\underline{M}_K$ and $\underline{M}'_K$ denote the lattices over K dual to T'_K and T_K, respectively:

$$\underline{M}_K = D(T'_K), \qquad \underline{M}'_K = D(T_K) \quad . \tag{14.1.3}$$

Thus the extension structures (14.1.2) and (14.1.1) may respectively be interpreted (VIII 3.7) as equivalent to the data of homomorphisms

$$\underline{M}_K \xrightarrow{p_K} B_K, \qquad \underline{M}'_K \xrightarrow{p'_K} B'_K \quad . \tag{14.1.4}$$

Let $W^{\natural}_K$ be the biextension of (B_K, B'_K) by $\mathbb{G}_{mK}$ which defines the canonical duality between these abelian schemes (8.2), and let E_K be the biextension of $(\underline{M}_K, \underline{M}'_K)$ by $\mathbb{G}_{mK}$ obtained by pulling $W^{\natural}_K$ back along the homomorphisms (14.1.4). One checks immediately that trivializations of E_K correspond canonically and bijectively to homomorphisms

$$i : \underline{M}_K \longrightarrow G_K \tag{14.1.5}$$

lifting $p_K : \underline{M}_K \longrightarrow B_K$, or, symmetrically, to homomorphisms

$$i' : \underline{M}'_K \longrightarrow G'_K \tag{14.1.6}$$

lifting $p'_K : \underline{M}'_K \longrightarrow B'_K$.

With this understood, write a superscript $^{\mathrm{an}}$ for the rigid-analytic groups over K associated with the algebraic groups under consideration [13]. In [24] one defines a canonical exact sequence

$$0 \longrightarrow \underline{M}^{\mathrm{an}}_K \xrightarrow{i^{\mathrm{an}}} G^{\mathrm{an}}_K \xrightarrow{e} A^{\mathrm{an}}_K \longrightarrow 0 \quad . \tag{14.1.7}$$

Here i^{an}, which may also be interpreted as a homomorphism $i : \underline{M}_K \longrightarrow G_K$ of ordinary group schemes, lifts the homomorphism p in (14.1.4), that is, it is of the form considered in (14.1.5). This exact sequence is the rigid-analytic analogue of the exact sequence (13.3.4). It depends functorially on A_K, and its formation is compatible with every change of traits $S' \longrightarrow S$.

The latter fact also shows that, in order to define and study (14.1.7), finite descent reduces us to the case where A_K has semistable reduction. In this case the group schemes $T_K, G_K, B_K, \underline{M}_K, \dots$ are the generic fibres of group schemes $T, G, B, M, \dots$ over S, with G an extension of the abelian scheme B by the torus T (paragraph 7). We may then consider the formal completion $\hat{G}$ of G along its special fibre, and by construction there is a canonical isomorphism (7.2.6)

$$\hat{G} \xrightarrow{\sim} \widehat{A^{\mathrm{o}}} \quad . \tag{14.1.8}$$

Passing to the "generic fibres", which are rigid-analytic groups over K, gives a canonical isomorphism

$$\hat{G}^{\mathrm{an}} \xrightarrow{\sim} \widehat{A^{\mathrm{o}}}^{\,\mathrm{an}} \quad . \tag{14.1.9}$$

On the other hand, there are canonical homomorphisms (rigid-analytic open immersions, but not in general isomorphisms)

$$\hat{G}^{\mathrm{an}} \hookrightarrow G^{\mathrm{an}}_K, \qquad \widehat{A^{\mathrm{o}}}^{\,\mathrm{an}} \hookrightarrow A^{\mathrm{an}}_K \quad . \tag{14.1.10}$$

The homomorphism e in (14.1.7) is uniquely characterized by the commutativity of the diagram

$$
\begin{array}{ll}
(14.1.11) &
\begin{array}{ccc}
\hat{G}^{\mathrm{an}} & \longrightarrow & \widehat{A^{\mathrm{o}}}^{\mathrm{an}} \\
\downarrow & & \downarrow \\
G_K^{\mathrm{an}} & \longrightarrow & A_K^{\mathrm{an}}
\end{array}
\end{array}
$$

where the unnamed arrows are those of (14.1.9) and (14.1.10).

Remarks 14.1.12.

a) Notice that the uniqueness of e making the preceding diagram commute is immediate, since the rigid-analytic space G_K^{an} is connected. The existence of e, on the other hand, is by no means trivial, and on this point we merely refer to the cited note of M. Raynaud. Moreover, the definition of i^{an} given in loc. cit. is likewise transcendental in nature, although it may be interpreted as an object of an essentially algebraic nature (14.1.5), once the construction of G itself, and hence of G_K, has been obtained through the constructions of formal geometry in paragraph 7. I know of no purely algebro-geometric description of it, but below (14.4) I shall propose an "arithmetic" characterization, obtained by reducing to a situation of finite type over the ring $\mathbb{Z}$. Thus, from the standpoint of algebraic geometry, the homomorphism (14.1.5) seems to be a more hidden piece of structure than those encountered in the preceding paragraphs.

b) As explained above, the homomorphism i is canonically associated with a well-determined trivialization of the biextension E_K of $(\underline{M}_K, \underline{M}'_K)$, or equivalently with a well-determined homomorphism (14.1.6) lifting p'_K. One then proves that the latter is precisely the homomorphism occurring in the exact sequence

$$(14.1.13) \qquad 0 \longrightarrow \underline{M}'^{\mathrm{an}}_K \xrightarrow{i'^{\mathrm{an}}} G'^{\mathrm{an}}_K \longrightarrow A'^{\mathrm{an}}_K \longrightarrow 0$$

analogous to (14.1.7), associated with the abelian scheme A'_K dual to A_K (unless it is the opposite???).

14.2. Relations of the homomorphism $i : \underline{M}_K \longrightarrow G_K$ with monodromy.

Let n be an integer > 0. From the exact sequence (14.1.7), by means of the exact sequence of the $\underline{\mathrm{Ext}}^i(\mathbb{Z}/n\mathbb{Z}, -)$ on the site of rigid-analytic spaces over K, one then obtains a canonical exact sequence (taking account of the fact that $\underline{M}_K^{\mathrm{an}}$ is torsion-free and G_K^{an} is n-divisible):

$$0 \longrightarrow {}_nG_K^{\mathrm{an}} \longrightarrow {}_nA_K^{\mathrm{an}} \longrightarrow \underline{M}_{K,n}^{\mathrm{an}} \longrightarrow 0 \qquad ,$$

where $\underline{M}_{K,n} \overset{\mathrm{dfn}}{=} \underline{M}_K / n\underline{M}_K$. Since all the terms occurring in this exact sequence come from finite group schemes over K, it may be rewritten as arising from an exact sequence

$$(14.2.1) \qquad 0 \longrightarrow {}_nG_K \longrightarrow {}_nA_K \longrightarrow \underline{M}_{K,n} \longrightarrow 0 \qquad .$$

As n varies, these exact sequences behave in the expected way, and, on passing to the inverse limit over the powers of a prime number ℓ, one obtains the analogous exact sequence

$$(14.2.2) \qquad 0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longrightarrow 0 \qquad .$$

But one already has such an exact sequence by purely algebraic means, thanks to the first formula (8.1.2) and to (8.1.7), which give

$$(14.2.3) \qquad T_\ell(G_K) \xrightarrow{\sim} T_\ell(A_K)^{\mathrm{ef}} \subset T_\ell(A_K), \qquad T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}} \simeq \underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \qquad .$$

With this understood, I claim that the homomorphisms (14.2.3) arising from the algebraic theory are precisely those deduced from the exact sequence (14.2.2) arising from the transcendental theory. For the first homomorphism, which corresponds to the first homomorphism of (14.2.2), our assertion follows at once, after reduction to the semistable situation, from the commutativity of (14.1.11), since the vertical arrows there induce isomorphisms on the T_ℓ, while applying T_ℓ to the horizontal arrows gives, respectively, the arrow under consideration in (14.2.3) and that in (14.2.2). We shall admit the asserted compatibility for the two isomorphisms obtained between $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}}$ and $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. Notice, moreover, that this compatibility (for a given ℓ) already suffices to characterize uniquely, by transcendental means, the homomorphism (14.1.5) occurring in (14.1.7), interpreted as an isomorphism from $\underline{M}_K^{\mathrm{an}}$ onto $\operatorname{Ker} e$.

One may also say that the homomorphism e of (14.1.7) identifies the $T_\ell(A_K)/T_\ell(A_K)^{\mathrm{ef}}$ with the ℓ-adic completions $\underline{R}_K \otimes \mathbb{Z}_\ell$ of a single lattice $\underline{R}_K = \operatorname{Ker} e$ over K, and that this "integral structure" on the preceding system of pro-ℓ groups is compatible, in the evident sense, with the integral structure arising from formulas (8.1.7) (orthogonality theorem 5.2). This then defines canonically an isomorphism

$$i : \underline{M}_K \xrightarrow{\sim} \operatorname{Ker} e \qquad .$$

Since the homomorphism i (14.1.5) enables one to reconstruct the usual extension structure on $T_\ell(A_K)$, namely (14.2.2), and hence also its structure as a blended extension, described in 9.6, by means of the exact sequence deduced from (14.1.1),

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T_\ell(T_K) & \longrightarrow & T_\ell(G_K) & \longrightarrow & T_\ell(B_K) & \longrightarrow & 0 \\ & & & & & & & & \\ & & \underline{M}'_K \otimes \mathbb{Z}_\ell(1) & & & & & & \end{array}$$

it also enables one, in the semistable case, to reconstruct the monodromy pairings (9.1.2)

$$u_\ell : \underline{M} \otimes \underline{M}' \longrightarrow \mathbb{Z}_\ell \qquad , \tag{14.2.4}$$

these having been defined precisely in terms of blended extensions in 9.6. Let us merely indicate the result of translating the construction of 9.6. For this purpose, observe that in the semistable case p_K and p'_K arise from homomorphisms

$$p : \underline{M} \longrightarrow B, \qquad p' : \underline{M}' \longrightarrow B' \qquad ,$$

where B, B' are abelian schemes over S, dual to one another. Pulling back by (p, p') therefore gives a biextension E of $(\underline{M}, \underline{M}')$ by $\mathbb{G}_{mS}$ whose generic fibre is the biextension E_K already considered in 4.1.

With this said, suppose for simplicity that $\underline{M}$ and $\underline{M}'$ are constant, with values M and M'. For every pair $(m, m') \in M \times M'$, $E_{m,m'}$ is a torsor under $\mathbb{G}_{mS}$, and hence corresponds to an invertible module $L_{m,m'}$ over the ring V of S. It therefore admits a basis u_0, determined up to multiplication by a unit of V. Consequently one obtains a canonical "valuation" map

$$\underline{v} : E_{m,m'}(K) \longrightarrow \mathbb{Z}$$

which associates with every basis u of $L_{m,m'} \otimes_V K$, written $u = \lambda u_0$ ($\lambda \in K^*$), the element

$$\underline{v}(u) = \underline{v}(\lambda) \in \mathbb{Z} \qquad ,$$

where $\underline{v}$ on the right denotes the valuation associated with V (normalized in the usual way, so that its value group is $\mathbb{Z}$); the resulting integer is plainly independent of the choice of u_0. If φ is now a trivialization of E, one may consider the corresponding map

$$(14.2.5) \qquad (m,m') \longmapsto \underline{v}\big(\varphi(m,m')\big) : M \times M' \longrightarrow \mathbb{Z} \qquad .$$

With this understood, one finds that the monodromy homomorphism (14.2.4) takes its values in $\mathbb{Z}$ and is precisely the homomorphism (14.2.5), in which φ is the splitting of E_K associated with the homomorphism i (14.1.5).

Remarks 14.2.6. In particular, one recovers (by means of the transcendental rigid-analytic theory) the fact that the monodromy pairing is integral and independent of ℓ (10.4 a)), proved by a different (and likewise transcendental) method in 12.5. According to M. Raynaud, rigid-analytic theory also recovers the positivity assertion 10.4 b) (which is likewise contained in 12.5).

Remarks 14.2.7. Regard $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Q}$ as an extension of $\underline{M}_K \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$ by $\underline{M}_K$. Using the homomorphism (14.1.5) $i : \underline{M}_K \longrightarrow G_K$, one deduces an extension A^b_K

$$(14.2.8) \qquad 0 \longrightarrow G_K \longrightarrow A^b_K \longrightarrow \underline{M}_K \otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0 \qquad .$$

Observe moreover that, since G_K is a divisible group, the datum of such an extension is equivalent to that of a family of extensions of the form (14.2.2), namely those deduced from (14.2.8) by applying the T_ℓ. Since i is a monomorphism, the extensions in question which define (14.2.8) are precisely the extensions (14.2.2) arising from the exact sequence (14.2.1). As these latter extensions are defined by purely algebraic means, the same is true of the extension (14.2.8). In fact, the characterization just given shows at once that this extension is canonically isomorphic to the group, likewise denoted A^b_K, introduced in (8.2.3), in view of the isomorphism 11.11 from the group of connected components of the latter to $\underline{M}_K \otimes \mathbb{Q}/\mathbb{Z}$.

14.3. Application to an arithmetic characterization of $i : \underline{M}_K \longrightarrow G_K$.

The characterization we have in mind rests on the observation that, for every prime number ℓ, the extension (14.2.2) of $\underline{M}_K \otimes \mathbb{Z}_\ell$ by $T_\ell(G_K)$ deduced from i is known by purely algebraic means. In particular, the image of i under the canonical homomorphism

$$(14.3.1) \qquad \mathrm{Hom}(\underline{M}_K, G_K) \longrightarrow \prod_\ell \mathrm{Ext}^1\big(\underline{M}_K \otimes \mathbb{Z}_\ell, T_\ell(G_K)\big) \qquad .$$

is known. In the case (to which one may always reduce by descent) in which $\underline{M}_K$ is constant, with value M, the preceding homomorphism may be interpreted as

$$(14.3.2) \qquad \mathrm{Hom}\big(M, G_K(K)\big) \longrightarrow \prod_\ell \mathrm{Hom}\big(M, H^1(K, T_\ell(G_K))\big),$$

deduced from the canonical homomorphism

$$(14.3.3) \qquad G_K(K) \longrightarrow \prod_\ell H^1(K, T_\ell(G_K)) \qquad ,$$

which comes from the connecting homomorphisms in the Kummer exact sequences

$$0 \longrightarrow {}_nG_K \longrightarrow G_K \xrightarrow{\ n\ } G_K \longrightarrow 0 \qquad .$$

(The homomorphism (14.3.1) could in fact always be interpreted as the connecting homomorphism analogous to (14.3.3) for the twisted group $\check{\underline{M}}_K \otimes G_K = \underline{\mathrm{Hom}}(\underline{M}_K, G_K)$.) The kernel of (14.3.3) plainly consists of the elements of $G_K(K)$ which are divisible by n for every integer $n \geq 1$, or, as we shall say, which are infinitely divisible. Thus this already determines i modulo an element of $\mathrm{Hom}(M, g)$, where g is the subgroup of $G_K(K)$ formed by its infinitely divisible elements:

$$g = \bigcap_{n \geq 1} nG_K(K) \qquad .$$

We now have the following result.

Lemma 14.3.4. Let V be a discrete valuation ring whose residue field k has characteristic $p > 0$ and is of finite type over its prime field, and let G_K be a group scheme of finite type over K, of unipotent rank zero if $\mathrm{char}\, K = 0$. Then the group g of infinitely divisible elements of $G_K(K)$ is zero; that is, (if G_K is divisible) the homomorphism (14.3.3) is injective.

After making a finite extension K'/K and replacing V by its normalization in K', we may suppose that G_K admits a composition series whose successive quotients are of one of the following types: a constant finite group, an abelian variety, $\mathbb{G}_{mK}$, or $\mathbb{G}_{aK}$. We are therefore immediately reduced to the case where G_K itself is of one of these types. The case of a constant finite group, or of $\mathbb{G}_{aK}$, is trivial. In the abelian case, G_K may be regarded as the generic fibre of its Néron model G, so we are reduced to showing that, if G is a scheme of finite type over S, the subgroup of $G(S)$ formed by the infinitely divisible points is zero.

This is true of the group $G(k)$, as follows at once from the usual dévissage of G_o and from the Mordell–Weil theorem [16] (which says that $G_o(k)$ is a finitely generated $\mathbb{Z}$-module if G_o is an abelian scheme over the field of finite type k). Consequently g is contained in the kernel $G(S)^{(1)}$ of $G(S) \longrightarrow G(k)$. The kernel of $G(S)^{(1)} \longrightarrow G(V/\underline{m}^{n+1})$ has a filtration whose quotients are isomorphic to vector groups over k (after supposing, as we may, that S is complete), and these are killed by p. It follows that g is contained in the kernel of the preceding homomorphism for every n, and hence that it is zero. The case $G_K = \mathbb{G}_{mK}$ reduces to the preceding one, since $G_K(K) = K^*$ is an extension of $\mathbb{Z}$ (which has no nonzero infinitely divisible elements) by $V^* = \mathbb{G}_m(S)$, and the latter satisfies the same condition by what we have just proved.

Remarks 14.3.4.1. Notice that in 14.3.4 it is essential to require divisibility by all integers ≥ 1, and not only by the powers of a fixed prime ℓ; equivalently, one must consider the kernel of (14.3.3), rather than restricting the right-hand side to a single factor $H^1(K, T_\ell(G_K))$. Indeed, if G_K comes from a smooth group scheme G over S and S is henselian, every element of the kernel of $G(S) \longrightarrow G(k)$ is infinitely ℓ-divisible for every $\ell \neq p$! The argument proving 14.3.4 shows that the subgroup of $G_K(K)$ formed by the infinitely p-divisible elements (that is, the kernel of $G_K(K) \longrightarrow H^1(K, T_p(G_K))$ when G_K is p-divisible) is a finite group.

14.3.5. The preceding considerations show that, when the residue field k has characteristic $p > 0$ and is of finite type in the absolute sense, the homomorphism i (14.1.5) is completely determined by its image under (14.3.1), that is, by the extensions (14.2.2). To deduce from this a principle characterizing i in the general case, with no restriction on k, observe that in paragraph 7 we were able to construct the scheme G over S under conditions considerably more general than those in which S is a complete trait and one has an abelian variety with semistable reduction over its field of fractions.

More generally, let S denote the spectrum of a normal noetherian ring V, separated and complete for an $\underline{m}$-adic topology, and consider a smooth quasi-projective group scheme A over S, with connected fibres. Suppose that $A_o = A \times_S S_o$, where $S_o = \mathrm{Spec}(V/\underline{m})$, is an extension of an abelian scheme B_o by an isotrivial torus T_o, and that A_U is an abelian scheme, where $U \subset S - S_o$ is a nonempty open subset. In 7.22 we then constructed a scheme G over S, an extension of an abelian scheme B by a torus T, characterized up to unique isomorphism by the isomorphism (14.1.8)

$$\hat{G} \xrightarrow{\sim} \hat{A} \tag{14.3.5.1}$$

on the formal completions along G_o and A_o. For every prime number ℓ one deduces an isomorphism (7.3.5)

$$T_\ell(G) \xrightarrow{\sim} T_\ell(A)^f \qquad , \tag{14.3.5.2}$$

and hence, on restricting to U, the structure of an extension of pro-ℓ Barsotti–Tate groups

$$\begin{tikzcd} 0 \arrow[r] & T_\ell(G_U) \arrow[r] \arrow[d, equal] & T_\ell(A_U) \arrow[r] & T_\ell(A_U)/T_\ell(A_U)^f \arrow[r] & 0 \\ & T_\ell(A_U)^f & & & \end{tikzcd} \quad . \tag{14.3.5.3}$$

Suppose now that a second group scheme A' over S is given, satisfying the same hypotheses as A, together with a biextension W of (A, A') by $\mathbb{G}_m$ which over U induces a duality between the abelian schemes A_U and A'_U. (In fact, by VIII 7.1, giving W is equivalent to giving such a duality between A_U and A'_U.) Then W defines pairings

$$T_\ell(A_U) \times T_\ell(A'_U) \longrightarrow \mathbb{Z}_\ell(1) \qquad , \tag{14.3.5.4}$$

which give rise to the orthogonality theorem (7.4.8): $T_\ell(A_U)^f$ is the orthogonal of $T_\ell(A'_U)^t = T_\ell(T'_U)$, where T' is the toric part of the Raynaud extension $G' = A'^\natural$ associated with A'. Thus one again obtains an isomorphism

$$T_\ell(A_U)/T_\ell(A_U)^f \simeq \underline{M}_U \otimes \mathbb{Z}_\ell \qquad , \tag{14.3.5.5}$$

where $\underline{M} = D(T')$ is the lattice over S dual to the torus T'. Consequently (14.3.5.3) and (14.3.5.5) give an extension

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow \underline{M}_U \otimes \mathbb{Z}_\ell \longrightarrow 0 \qquad . \tag{14.3.5.6}$$

With this understood, we may state the following conjecture.

<u>Conjecture 14.4.</u> The hypotheses being those just set out in 14.3.5, there exists a homomorphism

$$i : \underline{M}_U \longrightarrow G_U \tag{14.4.1}$$

such that, for every prime number ℓ, the extension (14.3.5.6) is isomorphic to the extension deduced from i as explained for (14.2.2). More precisely, there is one and only one way of associating with every situation of the type considered in 14.3.5 a homomorphism (14.4.1) satisfying the preceding condition, in such a way that the formation of i is compatible with every base change $S' \longrightarrow S$ associated with a continuous homomorphism $V \longrightarrow V'$ of noetherian adic rings.

14.4.2. It seems that the existence of i, compatibly with base change, follows from M. Raynaud's rigid-analytic theory, in view of the fact that he is developing it in a framework more general than that stated in 14.1, working, as here, over general noetherian adic rings. Moreover, to prove the existence and uniqueness asserted in 14.4, one is immediately reduced to considering only those S for which S_o is a scheme of finite type over $\mathbb{Z}$. Indeed, V is the filtered inductive limit of subrings V_i of finite type over $\mathbb{Z}$; these may be supposed normal by Nagata's theorem (EGA IV 7.8), and, for i sufficiently large, the situation (A, A', W) comes from an analogous situation over

V_i, or alternatively over the separated completion of V_i for the topology defined by $m_i = \underline{m}V_i$. This gives the asserted reduction.

14.4.3. Observe now that $S - S_o$ is a Jacobson scheme (EGA IV 10.5.8). Thus, to see that two solutions i, i' of the problem, for a given situation (A, A', W), are equal, it is enough to prove that these homomorphisms are equal at every closed point η of U. Let S' be the normalization of the closure of η in S. Then S' is a complete trait, and the uniqueness result of 4.3 may be applied to the pullback to S' of the situation over S, since the residue field of S' is finite. Once uniqueness is thus established, it implies compatibility of the formation of i with base change, subject to existence.

Remarks 14.5. It would remain to formulate with the desired generality, and to establish, a compatibility between the homomorphism i introduced in 14.4 by a rigid-analytic method and the analogous homomorphism (13.5.13) of complex-analytic theory—or, better still, to find a common generalization of them!

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